Formalization Blueprint for “An Introduction to Algebraic Combinatorics” by Darij Grinberg

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Chapter 1

Formal Power Series

1.1 Notations and elementary facts

We will use the following notations and conventions:

- The symbol $\mathbb{N}$ will denote the set $\{0, 1, 2, 3, \ldots\}$ of nonnegative integers.
- The size (i.e., cardinality) of a set A will be denoted by $|A|$.

We will need some basics from enumerative combinatorics:

- **Addition principle (sum rule):** If A and B are two disjoint sets, then $|A \cup B| = |A| + |B|$.
- **Multiplication principle (product rule):** If A and B are any two sets, then $|A \times B| = |A| \cdot |B|$.
- **Bijection principle:** There is a bijection between two sets X and Y if and only if $|X| = |Y|$.
- A set with n elements has 2^n subsets, and has $\dbinom{n}{k}$ size-k subsets for any $k \in \mathbb{R}$.
- A set with n elements has $n!$ permutations.

1.1.1 Binomial coefficients

Definition 1.1. For any numbers n and k, we set

$$\binom{n}{k} = \begin{cases} \dfrac{n(n-1)(n-2)\cdots(n-k+1)}{k!}, & \text{if } k \in \mathbb{N}; \\ 0, & \text{else.} \end{cases} \tag{1.1}$$

Note that "numbers" is to be understood fairly liberally here. In particular, n can be any integer, rational, real or complex number (or, more generally, any element in a $\mathbb{Q}$-algebra), whereas k can be anything (although the only nonzero values of $\dbinom{n}{k}$ will be achieved for $k \in \mathbb{N}$, by the above definition).

Lemma 1.2. *For any $k \in \mathbb{N}$, we have* $\binom{-1}{k} = (-1)^k$.

Proof. We have

$$\begin{aligned}\binom{-1}{k} &= \frac{(-1)(-1-1)(-1-2)\cdots(-1-k+1)}{k!} \\ &= \frac{(-1)(-2)(-3)\cdots(-k)}{k!} = \frac{(-1)^k k!}{k!} = (-1)^k.\end{aligned}$$

$\square$

If $n, k \in \mathbb{N}$ and $n \geq k$, then

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}. \tag{1.2}$$

But this formula only applies to the case when $n, k \in \mathbb{N}$ and $n \geq k$. The above definition is more general than it.

Lemma 1.3. *For $n, k \in \mathbb{N}$ with $k \leq n$, we have* $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$.

Proof. This follows directly from the definition by cancellation. $\square$

Lemma 1.4. *Let $n \in \mathbb{N}$. Then,* $\binom{2n}{n} = \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)}{n!} \cdot 2^n$.

Proof. We have

$$\begin{aligned}(2n)! &= 1 \cdot 2 \cdot \cdots \cdot (2n) \\ &= (1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)) \cdot \underbrace{(2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n))}_{=2^n(1\cdot 2\cdot\cdots\cdot n)} \\ &= (1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)) \cdot 2^n \underbrace{(1 \cdot 2 \cdot \cdots \cdot n)}_{=n!} \\ &= (1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)) \cdot 2^n n!.\end{aligned}$$

Now, (1.2) yields

$$\begin{aligned}\binom{2n}{n} &= \frac{(2n)!}{n!n!} = \frac{(1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)) \cdot 2^n n!}{n! \cdot n!} \\ &= \frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)}{n!} \cdot 2^n.\end{aligned}$$

$\square$

Proposition 1.5 (*Pascal's identity*, aka *recurrence of the binomial coefficients*)**.** *We have*

$$\binom{m}{n} = \binom{m-1}{n-1} + \binom{m-1}{n} \tag{1.3}$$

for any numbers m and n.

Proof. This follows from the identity $\binom{r+1}{k+1} = \binom{r}{k} + \binom{r}{k+1}$ (the successor form of Pascal's identity) by substituting $r = m-1$ and $k = n-1$. $\square$

Lemma 1.6. *For natural numbers m, n with $m > 0$ and $n > 0$, we have* $\binom{m}{n} = \binom{m-1}{n-1} + \binom{m-1}{n}$.

Proof. This is the natural number specialization of Proposition 1.5. $\square$

Lemma 1.7. *For any element r in a binomial ring and $k \in \mathbb{N}$,* $\binom{r+1}{k+1} = \binom{r}{k} + \binom{r}{k+1}$.

Proof. This is the successor form of Pascal's identity, which is a standard result in Mathlib. $\square$

Proposition 1.8. *Let $m, n \in \mathbb{N}$ satisfy $m < n$. Then,* $\binom{m}{n} = 0$.

Proof. If $m < n$, then in the product $m(m-1)(m-2)\cdots(m-n+1)$ in the numerator of $\binom{m}{n}$, one of the factors is $m - m = 0$ (since $m - n + 1 \leq 0 \leq m$ when $m < n$ and $m \in \mathbb{N}$). Hence the entire product is 0, and so $\binom{m}{n} = 0$.

Note that this really requires $m \in \mathbb{N}$. For example, $1.5 < 2$ but $\binom{1.5}{2} = 0.375 \neq 0$. $\square$

Theorem 1.9 (*Symmetry of the binomial coefficients*)**.** *Let $n \in \mathbb{N}$ and $k \in \mathbb{R}$. Then,*

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof. For $k \notin \{0, 1, \ldots, n\}$, both sides are 0 (if $k < 0$ or $k > n$, then both $\binom{n}{k} = 0$ and $\binom{n}{n-k} = 0$ by Proposition 1.8 or by the definition). For $k \in \{0, 1, \ldots, n\}$, we use the factorial formula (1.2):

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k}.$$

Note the $n \in \mathbb{N}$ requirement. Theorem 1.9 would fail for $n = -1$ and $k = 0$, since $\binom{-1}{0} = 1$ but $\binom{-1}{-1} = 0$. $\square$

Lemma 1.10. *For any $a, b \in \mathbb{N}$, we have* $\binom{a+b}{a} = \binom{a+b}{b}$.

Proof. This follows from Theorem 1.9 since $(a+b) - a = b$. $\square$

Lemma 1.11. *For any binomial ring R, natural numbers n, k with $k \leq n$,* $\binom{n}{k} = \binom{n}{n-k}$ *where n is viewed as an element of R.*

Proof. This follows from the natural-number symmetry (Theorem 1.9) and the fact that the generalized binomial coefficient agrees with the natural-number binomial coefficient on natural numbers (Lemma 1.16). $\square$

Lemma 1.12. *For any r, we have* $\binom{r}{0} = 1$.

Proof. The empty product in the numerator of $\binom{r}{0}$ equals 1, and $0! = 1$. $\square$

Lemma 1.13. *For any r, we have* $\binom{r}{1} = r$.

Proof. We have $\binom{r}{1} = \frac{r}{1!} = r$. $\square$

Lemma 1.14. *For $k > 0$, we have* $\binom{0}{k} = 0$.

Proof. The numerator of $\binom{0}{k}$ contains the factor 0. $\square$

Lemma 1.15. *For any element r in a binomial ring and $n \in \mathbb{N}$, $n! \cdot \binom{r}{n} = r(r-1)(r-2)\cdots(r-n+1)$.*

Proof. This is the defining property of generalized binomial coefficients in a binomial ring, expressed using the descending Pochhammer symbol. $\square$

Lemma 1.16. *For natural numbers n, k, the generalized binomial coefficient $\binom{n}{k}$ (computed in any binomial ring) agrees with the natural-number binomial coefficient $\binom{n}{k}$ as computed from Definition 1.1.*

Proof. This follows from the fact that the generalized binomial coefficient agrees with the natural-number binomial coefficient when n and k are natural numbers. $\square$

1.1.2 Helpers for Lemma 1.4

Lemma 1.17. *For any $n \in \mathbb{N}$, $\prod_{i=0}^{n-1}(2i+1) = (2n-1)!!$. That is, the product of odd numbers $1 \cdot 3 \cdot 5 \cdots (2n-1)$ equals the double factorial $(2n-1)!!$.*

Proof. By induction on n, using the double factorial recurrence $(2m+3)!! = (2m+3) \cdot (2m+1)!!$. $\square$

Lemma 1.18. *For any $n \in \mathbb{N}$, $n!$ divides $\left(\prod_{i=0}^{n-1}(2i+1)\right) \cdot 2^n$.*

Proof. Uses the factorization $(2n)! = 2^n \cdot n! \cdot (2n-1)!!$ together with the divisibility $n! \cdot n! \mid (2n)!$. $\square$

1.1.3 The Fibonacci sequence (Example 1)

Definition 1.19. The *Fibonacci sequence* $(f_0, f_1, f_2, \ldots)$ is defined by $f_0 = 0$, $f_1 = 1$, and $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$.

Definition 1.20. The *golden ratio* $\phi_+ = \frac{1+\sqrt{5}}{2}$.

Definition 1.21. The *conjugate golden ratio* $\phi_- = \frac{1-\sqrt{5}}{2}$.

Theorem 1.22. *The generating function of the Fibonacci sequence is*

$$F(x) = \sum_{n \geq 0} f_n x^n = \frac{x}{1 - x - x^2}.$$

Proof. Let $D = 1 - x - x^2$. We verify $F \cdot D = x$ by checking that for each n, $[x^n](F \cdot D) = f_n - f_{n-1} - f_{n-2} = 0$ for $n \geq 2$, $[x^1](F \cdot D) = 1$, and $[x^0](F \cdot D) = 0$. Then $F = x \cdot D^{-1}$ since the constant term of D is nonzero. □

Theorem 1.23 (Binet's formula)**.** *For any $n \in \mathbb{N}$,*

$$f_n = \frac{\phi_+^n - \phi_-^n}{\sqrt{5}}.$$

Proof. This follows from Binet's formula as established in Mathlib. □

1.1.4 Dyck words and Catalan numbers (Example 2)

Definition 1.24. A *Dyck word* of length $2n$ is a list of 0s and 1s with equal numbers of each, such that every prefix has at least as many 1s as 0s.

Definition 1.25. The n-th *Catalan number* is $c_n = \frac{1}{n+1}\binom{2n}{n}$.

Theorem 1.26. *For $n \geq 1$, the Catalan numbers satisfy the recurrence*

$$c_n = \sum_{k=0}^{n-1} c_k \, c_{n-1-k}.$$

Proof. Follows from Mathlib's Catalan number recurrence, converting between Mathlib's recursive definition and the explicit formula. □

Lemma 1.27. *The explicit formula for Catalan numbers: $c_n = \frac{1}{n+1}\binom{2n}{n}$.*

Proof. This is immediate from the definition of c_n. □

Lemma 1.28. *For $n \geq 1$, $c_n = \binom{2n}{n} - \binom{2n}{n-1}$.*

Proof. Uses the identity $\binom{2n}{n-1} \cdot (n+1) = \binom{2n}{n} \cdot n$ (from the recurrence for binomial coefficients) and the divisibility $(n+1) \mid \binom{2n}{n}$. □

Theorem 1.29. *The generating function $C(x) = \sum_{n \geq 0} c_n x^n$ satisfies the functional equation $C(x) = 1 + x\, C(x)^2$.*

Proof. Follows from Mathlib's identity $C(x)^2 \cdot x + 1 = C(x)$ for the Catalan generating function, after showing that $C(x)$ equals the image of Mathlib's Catalan series under the natural map. □

Helpers for the Catalan generating function

Lemma 1.30. *For $n \in \mathbb{N}$, $(n+1)\binom{1/2}{n+1} = \left(\frac{1}{2} - n\right)\binom{1/2}{n}$.*

Proof. Follows from the descending Pochhammer recurrence applied to $\binom{1/2}{n}$. □

Lemma 1.31. *For $n \in \mathbb{N}$, $\binom{1/2}{n+1} \cdot (-4)^{n+1} \cdot (n+1) = -2 \cdot \binom{2n}{n}$.*

Proof. By induction on n, using Lemma 1.30 and the central binomial coefficient recurrence $(n+1)\binom{2(n+1)}{n+1} = 2(2n+1)\binom{2n}{n}$. □

Lemma 1.32. *For $n \in \mathbb{N}$, $\binom{1/2}{n+1} \cdot (-4)^{n+1} = -2\,c_n$.*

Proof. Follows from Lemma 1.31 by dividing both sides by $(n+1)$, using $(n+1) \mid \binom{2n}{n}$ and $c_n = \binom{2n}{n}/(n+1)$. □

Theorem 1.33. *The generating function of the Catalan numbers satisfies*

$$2x\,C(x) = 1 - \sqrt{1-4x},$$

where $\sqrt{1-4x} = \sum_{n\geq 0}\binom{1/2}{n}(-4)^n x^n$ is the binomial series.

Proof. We check coefficient by coefficient. For $n = 0$, both sides give 0. For $n \geq 1$, the n-th coefficient of the left side is $2c_{n-1}$, and the n-th coefficient of $1 - \sqrt{1-4x}$ is $-\binom{1/2}{n}(-4)^n$, which equals $2c_{n-1}$ by Lemma 1.32. □

1.1.5 The Vandermonde convolution (Example 3)

Theorem 1.34 (Vandermonde convolution)**.** *For $a, b, n \in \mathbb{N}$,*

$$\binom{a+b}{n} = \sum_{k=0}^{n} \binom{a}{k}\binom{b}{n-k}.$$

Proof. Follows from the Vandermonde convolution identity for natural-number binomial coefficients from Mathlib, with the antidiagonal sum rewritten as a range sum. □

1.1.6 Solving a recurrence (Example 4)

Definition 1.35. The sequence (a_n) defined by $a_0 = 1$ and $a_{n+1} = 2a_n + n$ for all $n \geq 0$.

Theorem 1.36. *For all $n \in \mathbb{N}$, $a_n = 2^{n+1} - (n+1)$.*

Proof. By induction on n. The base case $a_0 = 1 = 2^1 - 1$ is immediate. For the induction step, $a_{n+1} = 2a_n + n = 2(2^{n+1} - (n+1)) + n = 2^{n+2} - (n+2)$. □

1.1.7 Auxiliary generating function results

Lemma 1.37. *The geometric series identity:* $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots = \sum_{n\geq 0} x^n$.

Proof. We verify $(1-x) \cdot \sum_{n\geq 0} x^n = 1$ and invert, using that the constant term of $(1-x)$ is nonzero. □

Lemma 1.38. *For any* $\alpha \in \mathbb{Q}$, $\frac{1}{1-\alpha x} = \sum_{n\geq 0} \alpha^n x^n$.

Proof. We verify $(1-\alpha x) \cdot \sum_{n\geq 0} \alpha^n x^n = 1$ by checking coefficients, then invert. □

Lemma 1.39. *The formal derivative of* $\frac{1}{1-x}$ *is* $\frac{1}{(1-x)^2}$.

Proof. Follows from the derivative of an inverse formula in Mathlib. □

Lemma 1.40. *The power series identity* $\sum_{n\geq 0}(n+1)x^n = \frac{1}{(1-x)^2}$.

Proof. Uses the identity $\sum_{n\geq 0}(n+1)x^n = \frac{1}{(1-x)^2}$ from Mathlib. □

Lemma 1.41. *The power series identity* $\sum_{n\geq 0} n\, x^n = \frac{x}{(1-x)^2}$.

Proof. For $n = 0$ both sides give 0; for $n \geq 1$, $[x^n]\left(\frac{x}{(1-x)^2}\right) = [x^{n-1}]\frac{1}{(1-x)^2} = n$, using Lemma 1.40. □

1.2 Commutative rings and modules

1.2.1 Reminder: Commutative rings

Definition 1.42. A *commutative ring* means a set K equipped with three maps

$$\begin{aligned} \oplus &: K \times K \to K, \\ \ominus &: K \times K \to K, \\ \odot &: K \times K \to K \end{aligned}$$

and two elements $\mathbf{0} \in K$ and $\mathbf{1} \in K$ satisfying the following axioms:

1. *Commutativity of addition:* We have $a \oplus b = b \oplus a$ for all $a, b \in K$.
2. *Associativity of addition:* We have $a \oplus (b \oplus c) = (a \oplus b) \oplus c$ for all $a, b, c \in K$.
3. *Neutrality of zero:* We have $a \oplus \mathbf{0} = \mathbf{0} \oplus a = a$ for all $a \in K$.
4. *Subtraction undoes addition:* Let $a, b, c \in K$. We have $a \oplus b = c$ if and only if $a = c \ominus b$.
5. *Commutativity of multiplication:* We have $a \odot b = b \odot a$ for all $a, b \in K$.
6. *Associativity of multiplication:* We have $a \odot (b \odot c) = (a \odot b) \odot c$ for all $a, b, c \in K$.
7. *Distributivity:* We have

   $$a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c) \qquad \text{and} \qquad (a \oplus b) \odot c = (a \odot c) \oplus (b \odot c)$$

   for all $a, b, c \in K$.

8. *Neutrality of one:* We have $a \odot \mathbf{1} = \mathbf{1} \odot a = a$ for all $a \in K$.

9. *Annihilation:* We have $a \odot \mathbf{0} = \mathbf{0} \odot a = \mathbf{0}$ for all $a \in K$.

The operations $\oplus$, $\ominus$ and $\odot$ are called the *addition*, the *subtraction* and the *multiplication* of the ring K. When confusion is unlikely, we will denote these three operations $\oplus$, $\ominus$ and $\odot$ by $+$, $-$ and $\cdot$, respectively, and we will abbreviate $a \odot b = a \cdot b$ by ab.

The elements $\mathbf{0}$ and $\mathbf{1}$ are called the *zero* and the *unity* (or the *one*) of the ring K. We will simply call these elements 0 and 1 when confusion with the corresponding numbers is unlikely.

We will use *PEMDAS conventions* for the three operations $\oplus$, $\ominus$ and $\odot$. These imply that the operation $\odot$ has higher precedence than $\oplus$ and $\ominus$, while the operations $\oplus$ and $\ominus$ are left-associative.

1.2.2 Standard rules in commutative rings

Lemma 1.43. *Let K be a commutative ring. For any $a, b \in K$, we have $-(a+b) = (-a)+(-b)$.*

Proof. Standard ring theory. $\square$

Lemma 1.44. *Let K be a commutative ring. For any $a \in K$, we have $-(-a) = a$.*

Proof. Standard ring theory. $\square$

Lemma 1.45. *Let K be a commutative ring. For any $a \in K$ and $n, m \in \mathbb{Z}$, we have $(n+m)\,a = na + ma$.*

Proof. Standard ring theory. $\square$

Lemma 1.46. *Let K be a commutative ring. For any $a \in K$ and $n, m \in \mathbb{Z}$, we have $(nm)\,a = n\,(ma)$.*

Proof. Standard ring theory. $\square$

Lemma 1.47. *Let K be a commutative ring. For any $a, b, c \in K$, we have $a\,(b-c) = (ab)-(ac)$.*

Proof. Standard ring theory. $\square$

Lemma 1.48. *Let K be a commutative ring. For any $a, b \in K$ and $n \in \mathbb{N}$, we have $(ab)^n = a^n b^n$.*

Proof. Standard ring theory (induction on n). $\square$

Lemma 1.49. *Let K be a commutative ring. For any $a \in K$ and $n, m \in \mathbb{N}$, we have $a^{n+m} = a^n a^m$.*

Proof. Standard ring theory (induction on m). $\square$

Lemma 1.50. *Let K be a commutative ring. For any $a \in K$ and $n, m \in \mathbb{N}$, we have $a^{nm} = (a^n)^m$.*

Proof. Standard ring theory (induction on m). $\square$

Theorem 1.51. ***(Binomial Theorem.)*** *Let K be a commutative ring. For any $a, b \in K$ and $n \in \mathbb{N}$,*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

Proof. Standard; this is the binomial theorem from Mathlib. □

Lemma 1.52. *Let K be a commutative ring. For any $a, b \in K$ and $n \in \mathbb{N}$,*

$$(a-b)^n = \sum_{m=0}^{n} (-1)^{m+n} \cdot a^m \cdot b^{n-m} \cdot \binom{n}{m}.$$

Proof. Write $a - b = a + (-b)$ and apply Theorem 1.51. □

1.2.3 Finite sums and products

Lemma 1.53. *Let K be a commutative ring. Let S be a finite type, and let X, Y be disjoint finite subsets of S. For any family $(a_s)_{s\in S}$ of elements of K, we have $\sum_{s\in X\cup Y} a_s = \sum_{s\in X} a_s + \sum_{s\in Y} a_s$.*

Proof. Standard. □

Lemma 1.54. *Let K be a commutative ring. Let T be a finite set and $(a_s)_{s\in T}$, $(b_s)_{s\in T}$ be two families of elements of K. Then $\sum_{s\in T} (a_s + b_s) = \sum_{s\in T} a_s + \sum_{s\in T} b_s$.*

Proof. Standard. □

Lemma 1.55. *Let K be a commutative ring. If $S = \varnothing$, then $\sum_{s\in S} a_s = 0$.*

Proof. By definition. □

Lemma 1.56. *Let K be a commutative ring. If $S = \varnothing$, then $\prod_{s\in S} a_s = 1$.*

Proof. By definition. □

1.2.4 K-modules

Definition 1.57. Let K be a commutative ring.

A *K-module* means a set M equipped with three maps

$$\oplus : M \times M \to M,$$
$$\ominus : M \times M \to M,$$
$$\rightharpoonup : K \times M \to M$$

(notice that the third map has domain $K \times M$, not $M \times M$) and an element $\overrightarrow{0} \in M$ satisfying the following axioms:

1. *Commutativity of addition:* We have $a \oplus b = b \oplus a$ for all $a, b \in M$.
2. *Associativity of addition:* We have $a \oplus (b \oplus c) = (a \oplus b) \oplus c$ for all $a, b, c \in M$.
3. *Neutrality of zero:* We have $a \oplus \overrightarrow{0} = \overrightarrow{0} \oplus a = a$ for all $a \in M$.
4. *Subtraction undoes addition:* Let $a, b, c \in M$. We have $a \oplus b = c$ if and only if $a = c \ominus b$.
5. *Associativity of scaling:* We have $u \rightharpoonup (v \rightharpoonup a) = (uv) \rightharpoonup a$ for all $u, v \in K$ and $a \in M$.
6. *Left distributivity:* We have $u \rightharpoonup (a \oplus b) = (u \rightharpoonup a) \oplus (u \rightharpoonup b)$ for all $u \in K$ and $a, b \in M$.

7. *Right distributivity:* We have $(u+v) \rightharpoonup a = (u \rightharpoonup a) \oplus (v \rightharpoonup a)$ for all $u, v \in K$ and $a \in M$.
8. *Neutrality of one:* We have $1 \rightharpoonup a = a$ for all $a \in M$.
9. *Left annihilation:* We have $0 \rightharpoonup a = \overrightarrow{0}$ for all $a \in M$.
10. *Right annihilation:* We have $u \rightharpoonup \overrightarrow{0} = \overrightarrow{0}$ for all $u \in K$.

The operations $\oplus$, $\ominus$ and $\rightharpoonup$ are called the *addition*, the *subtraction* and the *scaling* (or the *K-action*) of the K-module M. When confusion is unlikely, we will denote these three operations $\oplus$, $\ominus$ and $\rightharpoonup$ by $+$, $-$ and $\cdot$, respectively, and we will abbreviate $a \rightharpoonup b = a \cdot b$ by ab.

The element $\overrightarrow{0}$ is called the *zero* (or the *zero vector*) of the K-module M. We will usually just call it 0.

When M is a K-module, the elements of M are called *vectors*, while the elements of K are called *scalars*.

We will use *PEMDAS conventions* for the three operations $\oplus$, $\ominus$ and $\rightharpoonup$, with the operation $\rightharpoonup$ having higher precedence than $\oplus$ and $\ominus$.

1.2.5 Additive inverses in modules

Lemma 1.58. *Let K be a commutative ring and M a K-module. For any $a \in M$, the additive inverse of a can be constructed as $(-1) \cdot a$, i.e., $-a = (-1) \cdot a$.*

Proof. Standard module theory. □

Lemma 1.59. *Let K be a commutative ring and M a K-module. For any $a, b \in M$, we have $a - b = a + (-b)$.*

Proof. Standard module theory. □

Lemma 1.60. *Let K be a commutative ring and M a K-module. For any $a, b \in M$, we have $a - b = a + (-1) \cdot b$.*

Proof. Combine the fact that $-b = (-1) \cdot b$ with $a - b = a + (-b)$. □

Lemma 1.61. *Let K be a commutative ring and M a K-module. For any $a, b \in M$, we have $-(a+b) = (-a) + (-b)$.*

Proof. Standard module theory. □

Lemma 1.62. *Let K be a commutative ring and M a K-module. For any $a \in M$, we have $-(-a) = a$.*

Proof. Standard module theory. □

1.2.6 Inverses in commutative rings

Definition 1.63. Let L be a commutative ring and $a, b \in L$.

(a) We say that b is an *inverse* (or *multiplicative inverse*) of a if $a \cdot b = 1$.

(b) We say that a is *invertible* in L (or a *unit* of L) if a has an inverse.

Lemma 1.64. *Let L be a commutative ring. If b and c are both inverses of a, then $b = c$. In other words, the inverse of an element (if it exists) is unique.*

Proof. We have $b = b \cdot 1 = b \cdot (a \cdot c) = (b \cdot a) \cdot c = 1 \cdot c = c$. □

Definition 1.65. An element $a \in L$ is *invertible* if there exists $b \in L$ such that $a \cdot b = 1$.

Lemma 1.66. *The definition of invertibility from Definition 1.107 is equivalent to the definition of a unit in Mathlib.*

Proof. Both express the existence of a multiplicative inverse; the equivalence is immediate from the definitions. □

Lemma 1.67. *If a and b are invertible in L, then $a \cdot b$ is invertible.*

Proof. Convert to the equivalent notion of a unit and use the fact that the product of two units is a unit. □

Lemma 1.68. *In a field F, an element a is invertible if and only if $a \neq 0$.*

Proof. Every nonzero element of a field has a multiplicative inverse. □

1.2.7 Fractions and integer powers

Definition 1.69. For an invertible element a and any $b \in L$, the *fraction* b/a is defined as $b \cdot a^{-1}$.

Lemma 1.70. *For an invertible element a and elements $b, c \in L$, we have $b/a = c$ if and only if $b = c \cdot a$.*

Proof. Multiply both sides by a (or a^{-1}) and use $a \cdot a^{-1} = 1$. □

Lemma 1.71. *For an invertible element a and a natural number n, we have $a^{-n} = (a^{-1})^n$. This defines negative integer powers.*

Proof. By the definition of integer powers on units. □

Lemma 1.72. *For an invertible element a and integers $m, n \in \mathbb{Z}$, we have $a^{m+n} = a^m \cdot a^n$.*

Proof. Standard property of integer powers. □

1.3 Formal power series: definition and basic properties

1.3.1 Convention

Fix a commutative ring K. (For example, K can be $\mathbb{Z}$ or $\mathbb{Q}$ or $\mathbb{C}$.)

1.3.2 The definition of formal power series

Definition 1.73. A *formal power series* (or, short, *FPS*) in 1 indeterminate over K means a sequence $(a_0, a_1, a_2, \ldots) = (a_n)_{n \in \mathbb{N}} \in K^{\mathbb{N}}$ of elements of K.

Definition 1.74. (a) The *sum* of two FPSs $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ and $\mathbf{b} = (b_0, b_1, b_2, \ldots)$ is defined to be the FPS

$$(a_0 + b_0, \quad a_1 + b_1, \quad a_2 + b_2, \quad \ldots) .$$

It is denoted by $\mathbf{a} + \mathbf{b}$.

(b) The *difference* of two FPSs $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ and $\mathbf{b} = (b_0, b_1, b_2, \ldots)$ is defined to be the FPS

$$(a_0 - b_0, \quad a_1 - b_1, \quad a_2 - b_2, \quad \ldots).$$

It is denoted by $\mathbf{a} - \mathbf{b}$.

(c) If $\lambda \in K$ and if $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ is an FPS, then we define an FPS

$$\lambda \mathbf{a} := (\lambda a_0, \lambda a_1, \lambda a_2, \ldots).$$

(d) The *product* of two FPSs $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ and $\mathbf{b} = (b_0, b_1, b_2, \ldots)$ is defined to be the FPS $(c_0, c_1, c_2, \ldots)$, where

$$\begin{aligned} c_n &= \sum_{i=0}^{n} a_i b_{n-i} = \sum_{\substack{(i,j) \in \mathbb{N}^2; \\ i+j=n}} a_i b_j \\ &= a_0 b_n + a_1 b_{n-1} + a_2 b_{n-2} + \cdots + a_n b_0 \qquad \text{for each } n \in \mathbb{N}. \end{aligned}$$

This product is denoted by $\mathbf{a} \cdot \mathbf{b}$ or just by $\mathbf{ab}$.

(e) For each $a \in K$, we define $\underline{a}$ to be the FPS $(a, 0, 0, 0, \ldots)$. An FPS of the form $\underline{a}$ for some $a \in K$ (that is, an FPS $(a_0, a_1, a_2, \ldots)$ satisfying $a_1 = a_2 = a_3 = \cdots = 0$) is said to be *constant*.

(f) The set of all FPSs (in 1 indeterminate over K) is denoted $K[[x]]$.

Definition 1.75. If $n \in \mathbb{N}$, and if $\mathbf{a} = (a_0, a_1, a_2, \ldots) \in K[[x]]$ is an FPS, then we define an element $[x^n]\, \mathbf{a} \in K$ by

$$[x^n]\, \mathbf{a} := a_n.$$

This is called the *coefficient of* x^n *in* $\mathbf{a}$, or the *n-th coefficient* of $\mathbf{a}$, or the *x^n-coefficient* of $\mathbf{a}$.

Theorem 1.76. ***(a)*** *The set $K[[x]]$ is a commutative ring (with its operations $+$, $-$ and $\cdot$ defined in Definition 1.74). Its zero and its unity are the FPSs $\underline{0} = (0, 0, 0, \ldots)$ and $\underline{1} = (1, 0, 0, 0, \ldots)$. This means, concretely, that the following facts hold:*

1. Commutativity of addition: *We have $\mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}$ for all $\mathbf{a}, \mathbf{b} \in K[[x]]$.*
2. Associativity of addition: *We have $\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}$ for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in K[[x]]$.*
3. Neutrality of zero: *We have $\mathbf{a} + \underline{0} = \underline{0} + \mathbf{a} = \mathbf{a}$ for all $\mathbf{a} \in K[[x]]$.*
4. Subtraction undoes addition: *Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in K[[x]]$. We have $\mathbf{a} + \mathbf{b} = \mathbf{c}$ if and only if $\mathbf{a} = \mathbf{c} - \mathbf{b}$.*
5. Commutativity of multiplication: *We have $\mathbf{ab} = \mathbf{ba}$ for all $\mathbf{a}, \mathbf{b} \in K[[x]]$.*
6. Associativity of multiplication: *We have $\mathbf{a}(\mathbf{bc}) = (\mathbf{ab})\,\mathbf{c}$ for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in K[[x]]$.*
7. Distributivity: *We have*

   $$\mathbf{a}(\mathbf{b} + \mathbf{c}) = \mathbf{ab} + \mathbf{ac} \qquad \text{and} \qquad (\mathbf{a} + \mathbf{b})\,\mathbf{c} = \mathbf{ac} + \mathbf{bc}$$

   for all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in K[[x]]$.
8. Neutrality of one: *We have $\mathbf{a} \cdot \underline{1} = \underline{1} \cdot \mathbf{a} = \mathbf{a}$ for all $\mathbf{a} \in K[[x]]$.*

9. Annihilation: *We have* $\mathbf{a} \cdot \underline{0} = \underline{0} \cdot \mathbf{a} = \underline{0}$ *for all* $\mathbf{a} \in K[[x]]$*.*

(b) *Furthermore,* $K[[x]]$ *is a* K*-module (with its scaling being the map that sends each* $(\lambda, \mathbf{a}) \in K \times K[[x]]$ *to the FPS* $\lambda\mathbf{a}$ *defined in Definition 1.74* ***(c)****). Its zero vector is* $\underline{0}$*. Concretely, this means that:*

10. Associativity of scaling: *We have* $\lambda(\mu\mathbf{a}) = (\lambda\mu)\mathbf{a}$ *for all* $\lambda, \mu \in K$ *and* $\mathbf{a} \in K[[x]]$*.*

11. Left distributivity: *We have* $\lambda(\mathbf{a} + \mathbf{b}) = \lambda\mathbf{a} + \lambda\mathbf{b}$ *for all* $\lambda \in K$ *and* $\mathbf{a}, \mathbf{b} \in K[[x]]$*.*

12. Right distributivity: *We have* $(\lambda + \mu)\mathbf{a} = \lambda\mathbf{a} + \mu\mathbf{a}$ *for all* $\lambda, \mu \in K$ *and* $\mathbf{a} \in K[[x]]$*.*

13. Neutrality of one: *We have* $1\mathbf{a} = \mathbf{a}$ *for all* $\mathbf{a} \in K[[x]]$*.*

14. Left annihilation: *We have* $0\mathbf{a} = \underline{0}$ *for all* $\mathbf{a} \in K[[x]]$*.*

15. Right annihilation: *We have* $\lambda\underline{0} = \underline{0}$ *for all* $\lambda \in K$*.*

(c) *We have* $\lambda(\mathbf{a} \cdot \mathbf{b}) = (\lambda\mathbf{a}) \cdot \mathbf{b} = \mathbf{a} \cdot (\lambda\mathbf{b})$ *for all* $\lambda \in K$ *and* $\mathbf{a}, \mathbf{b} \in K[[x]]$*.*

(d) *Finally, we have* $\lambda\mathbf{a} = \underline{\lambda} \cdot \mathbf{a}$ *for all* $\lambda \in K$ *and* $\mathbf{a} \in K[[x]]$*.*

Proof. Most parts of Theorem 1.76 are straightforward to verify. Let us check the associativity of multiplication.

Let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in K[[x]]$. We must prove that $\mathbf{a}(\mathbf{bc}) = (\mathbf{ab})\mathbf{c}$. Let $n \in \mathbb{N}$. Consider the two equalities

$$\begin{aligned}[x^n]((\mathbf{ab})\mathbf{c}) &= \sum_{j=0}^{n} \underbrace{[x^{n-j}](\mathbf{ab})}_{=\sum_{i=0}^{n-j}[x^i]\mathbf{a}\cdot[x^{n-j-i}]\mathbf{b}} \cdot [x^j]\mathbf{c} \\ &= \sum_{j=0}^{n} \sum_{i=0}^{n-j} [x^i]\mathbf{a} \cdot [x^{n-j-i}]\mathbf{b} \cdot [x^j]\mathbf{c}\end{aligned}$$

and

$$\begin{aligned}[x^n](\mathbf{a}(\mathbf{bc})) &= \sum_{i=0}^{n} [x^i]\mathbf{a} \cdot \underbrace{[x^{n-i}](\mathbf{bc})}_{=\sum_{j=0}^{n-i}[x^{n-i-j}]\mathbf{b}\cdot[x^j]\mathbf{c}} \\ &= \sum_{i=0}^{n} \sum_{j=0}^{n-i} [x^i]\mathbf{a} \cdot [x^{n-i-j}]\mathbf{b} \cdot [x^j]\mathbf{c}.\end{aligned}$$

The right hand sides are equal since

$$\sum_{j=0}^{n} \sum_{i=0}^{n-j} = \sum_{\substack{(i,j)\in\mathbb{N}^2;\\ i+j\leq n}} = \sum_{i=0}^{n} \sum_{j=0}^{n-i}$$

and $n - j - i = n - i - j$. Thus $[x^n]((\mathbf{ab})\mathbf{c}) = [x^n](\mathbf{a}(\mathbf{bc}))$ for each $n \in \mathbb{N}$, which entails $(\mathbf{ab})\mathbf{c} = \mathbf{a}(\mathbf{bc})$.

The remaining claims of Theorem 1.76 follow by similar (and easier) coefficient-wise verifications. In the Lean formalization, the commutative ring and module structures are provided by Mathlib's existing instances on power series. $\square$

Lemma 1.77. *The product formula can be written as a sum over a range:*

$$[x^n]\,(\mathbf{ab}) = \sum_{i=0}^{n} \left[x^i\right] \mathbf{a} \cdot \left[x^{n-i}\right] \mathbf{b}.$$

Proof. This rewrites the antidiagonal sum from Definition 1.74 as a range sum using the bijection $(i, n-i) \leftrightarrow i \in \{0, \ldots, n\}$. ☐

1.3.3 Essentially finite sums and summable families

Definition 1.78. **(a)** A family $(a_i)_{i\in I} \in K^I$ of elements of K is said to be *essentially finite* if all but finitely many $i \in I$ satisfy $a_i = 0$ (in other words, if the set $\{i \in I \mid a_i \neq 0\}$ is finite).

(b) Let $(a_i)_{i\in I} \in K^I$ be an essentially finite family of elements of K. Then, the infinite sum $\sum_{i\in I} a_i$ is defined to equal the finite sum $\sum_{\substack{i\in I;\\ a_i\neq 0}} a_i$. Such an infinite sum is said to be *essentially finite.*

Definition 1.79. A (possibly infinite) family $(\mathbf{a}_i)_{i\in I}$ of FPSs is said to be *summable* (or *entrywise essentially finite*) if

$$\text{for each } n \in \mathbb{N}, \text{ all but finitely many } i \in I \text{ satisfy } [x^n]\,\mathbf{a}_i = 0.$$

In this case, the sum $\sum_{i\in I} \mathbf{a}_i$ is defined to be the FPS with

$$[x^n]\left(\sum_{i\in I} \mathbf{a}_i\right) = \underbrace{\sum_{i\in I} [x^n]\,\mathbf{a}_i}_{\substack{\text{an essentially}\\ \text{finite sum}}} \qquad \text{for all } n \in \mathbb{N}.$$

Proposition 1.80. *Let $(\mathbf{a}_i)_{i\in I}$ be a summable family of FPSs. Then, any subfamily of $(\mathbf{a}_i)_{i\in I}$ is summable as well.*

Proof. Let J be a subset of I. We must prove that the subfamily $(\mathbf{a}_i)_{i\in J}$ is summable.

Let $n \in \mathbb{N}$. Then, all but finitely many $i \in I$ satisfy $[x^n]\,\mathbf{a}_i = 0$ (since the family $(\mathbf{a}_i)_{i\in I}$ is summable). Hence, all but finitely many $i \in J$ satisfy $[x^n]\,\mathbf{a}_i = 0$ (since J is a subset of I). Since we have proved this for each $n \in \mathbb{N}$, we thus conclude that the family $(\mathbf{a}_i)_{i\in J}$ is summable. ☐

Proposition 1.81. *Sums of summable families of FPSs satisfy the usual rules for sums (such as the breaking-apart rule $\sum_{i\in S} a_s = \sum_{i\in X} a_s + \sum_{i\in Y} a_s$ when a set S is the union of two disjoint sets X and Y). See [19s, Proposition 7.2.11] for details. Again, the only* ***caveat*** *is about interchange of summation signs: The equality*

$$\sum_{i\in I}\ \sum_{j\in J} \mathbf{a}_{i,j} = \sum_{j\in J}\ \sum_{i\in I} \mathbf{a}_{i,j}$$

holds when the family $(\mathbf{a}_{i,j})_{(i,j)\in I\times J}$ is summable (i.e., when for each $n \in \mathbb{N}$, all but finitely many ***pairs*** *$(i,j) \in I \times J$ satisfy $[x^n]\,\mathbf{a}_{i,j} = 0$); it does not generally hold if we merely assume that the sums $\sum_{i\in I}\ \sum_{j\in J} \mathbf{a}_{i,j}$ and $\sum_{j\in J}\ \sum_{i\in I} \mathbf{a}_{i,j}$ are summable.*

Proof. The proof is tedious (as there are many rules to check), but fairly straightforward (the idea is always to focus on a single coefficient, and then to reduce the infinite sums to finite sums). For example, consider the discrete Fubini rule, which says that

$$\sum_{i\in I}\ \sum_{j\in J}\mathbf{a}_{i,j} = \sum_{(i,j)\in I\times J}\mathbf{a}_{i,j} = \sum_{j\in J}\ \sum_{i\in I}\mathbf{a}_{i,j}$$

whenever $(\mathbf{a}_{i,j})_{(i,j)\in I\times J}$ is a summable family of FPSs. In order to prove this rule, we fix a summable family $(\mathbf{a}_{i,j})_{(i,j)\in I\times J}$ of FPSs. It is easy to see that the families $(\mathbf{a}_{i,j})_{j\in J}$ for all $i \in I$ are summable as well, as are the families $(\mathbf{a}_{i,j})_{i\in I}$ for all $j \in J$, and the families $\left(\sum_{j\in J}\mathbf{a}_{i,j}\right)_{i\in I}$ and $\left(\sum_{i\in I}\mathbf{a}_{i,j}\right)_{j\in J}$.

To verify the Fubini identity, it suffices to check that

$$[x^n]\left(\sum_{i\in I}\ \sum_{j\in J}\mathbf{a}_{i,j}\right) = [x^n]\left(\sum_{(i,j)\in I\times J}\mathbf{a}_{i,j}\right) = [x^n]\left(\sum_{j\in J}\ \sum_{i\in I}\mathbf{a}_{i,j}\right)$$

for each $n \in \mathbb{N}$. Fix $n \in \mathbb{N}$; then, we have $[x^n](\mathbf{a}_{i,j}) = 0$ for all but finitely many $(i,j) \in I \times J$ (since the family $(\mathbf{a}_{i,j})_{(i,j)\in I\times J}$ is summable). That is, the set of all pairs $(i,j) \in I \times J$ satisfying $[x^n](\mathbf{a}_{i,j}) \neq 0$ is finite. Hence, the set I' of the first entries and the set J' of the second entries of all these pairs are also finite. The three sums reduce to finite sums over $I' \times J'$, which are equal by the usual Fubini rule for finite sums. See [19s, proof of Proposition 7.2.11] for more details. $\square$

1.3.4 The indeterminate x and powers

Definition 1.82. Let x denote the FPS $(0, 1, 0, 0, 0, \ldots)$. In other words, let x denote the FPS with $\left[x^1\right] x = 1$ and $\left[x^i\right] x = 0$ for all $i \neq 1$.

Lemma 1.83. *Let* $\mathbf{a} = (a_0, a_1, a_2, \ldots)$ *be an FPS. Then,* $x \cdot \mathbf{a} = (0, a_0, a_1, a_2, \ldots)$.

Proof. If n is a positive integer, then

$$\begin{aligned}
[x^n](x\cdot\mathbf{a}) &= \sum_{i=0}^{n}\left[x^i\right]x\cdot a_{n-i} \\
&= \underbrace{\left[x^0\right]x}_{=0}\cdot a_{n-0} + \underbrace{\left[x^1\right]x}_{=1}\cdot a_{n-1} + \sum_{i=2}^{n}\underbrace{\left[x^i\right]x}_{\substack{=0\\ (\text{since } i\geq 2>1)}}\cdot a_{n-i} \\
&= a_{n-1}.
\end{aligned}$$

A similar argument for $n = 0$ yields $\left[x^0\right](x \cdot \mathbf{a}) = 0$. Thus, for each $n \in \mathbb{N}$,

$$[x^n](x\cdot\mathbf{a}) = \begin{cases} a_{n-1}, & \text{if } n > 0; \\ 0, & \text{if } n = 0. \end{cases}$$

In other words, $x \cdot \mathbf{a} = (0, a_0, a_1, a_2, \ldots)$. $\square$

Proposition 1.84. *We have*

$$x^k = \left(\underbrace{0, 0, \ldots, 0}_{k \text{ times}}, 1, 0, 0, 0, \ldots\right) \qquad \text{for each } k \in \mathbb{N}.$$

Proof. We induct on k.

Induction base: We have $x^0 = \underline{1} = (1, 0, 0, 0, 0, \ldots) = \left(\underbrace{0, 0, \ldots, 0}_{0 \text{ times}}, 1, 0, 0, 0, \ldots \right)$.

Induction step: Let $m \in \mathbb{N}$. Assume that Proposition 1.84 holds for $k = m$. We have $x^m = \left(\underbrace{0, 0, \ldots, 0}_{m \text{ times}}, 1, 0, 0, 0, \ldots \right)$. Thus, Lemma 1.83 (applied to $\mathbf{a} = x^m$) yields

$$x \cdot x^m = \left(0, \underbrace{0, 0, \ldots, 0}_{m \text{ times}}, 1, 0, 0, 0, \ldots \right) = \left(\underbrace{0, 0, \ldots, 0}_{m+1 \text{ times}}, 1, 0, 0, 0, \ldots \right).$$

Since $x \cdot x^m = x^{m+1}$, this completes the induction step. □

Corollary 1.85. *Any FPS* $(a_0, a_1, a_2, \ldots) \in K[[x]]$ *satisfies*

$$(a_0, a_1, a_2, \ldots) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots = \sum_{n \in \mathbb{N}} a_n x^n.$$

In particular, the right hand side here is well-defined, i.e., the family $(a_n x^n)_{n \in \mathbb{N}}$ *is summable.*

Proof. By Proposition 1.84, we have

$$\begin{aligned}
& a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots \\
= \quad & a_0 (1, 0, 0, 0, \ldots) \\
& + a_1 (0, 1, 0, 0, \ldots) \\
& + a_2 (0, 0, 1, 0, \ldots) \\
& + a_3 (0, 0, 0, 1, \ldots) \\
& + \cdots \\
= \quad & (a_0, 0, 0, 0, \ldots) \\
& + (0, a_1, 0, 0, \ldots) \\
& + (0, 0, a_2, 0, \ldots) \\
& + (0, 0, 0, a_3, \ldots) \\
& + \cdots \\
= \; & (a_0, a_1, a_2, a_3, \ldots).
\end{aligned}$$

□

1.3.5 Helper lemmas from the formalization

Lemma 1.86. *Two FPSs* f *and* g *are equal if and only if* $[x^n] f = [x^n] g$ *for all* $n \in \mathbb{N}$.

Proof. Immediate from the definition of FPS as a sequence. □

Lemma 1.87. *For any* $\mathbf{a}, \mathbf{b} \in K[[x]]$, *we have* $[x^0](\mathbf{ab}) = [x^0]\mathbf{a} \cdot [x^0]\mathbf{b}$. *That is, the constant term of a product is the product of the constant terms.*

Proof. Follows from the product formula with $n = 0$: $[x^0](\mathbf{ab}) = \sum_{i=0}^{0} [x^i]\mathbf{a} \cdot [x^{0-i}]\mathbf{b} = [x^0]\mathbf{a} \cdot [x^0]\mathbf{b}$. □

Lemma 1.88. *For any $k \in \mathbb{N}$ and any FPS $\mathbf{a}$,*

$$[x^n]\left(x^k \cdot \mathbf{a}\right) = \begin{cases} 0, & \text{if } n < k; \\ \left[x^{n-k}\right]\mathbf{a}, & \text{if } n \geq k. \end{cases}$$

That is, multiplication by x^k shifts coefficients by k positions.

Proof. By induction on k, using Lemma 1.83 for the induction step. □

Lemma 1.89. *If $(\mathbf{a}_i)_{i\in I}$ and $(\mathbf{b}_i)_{i\in I}$ are summable families of FPSs, then so is $(\mathbf{a}_i + \mathbf{b}_i)_{i\in I}$.*

Proof. For each $n \in \mathbb{N}$, $\{i \mid [x^n](\mathbf{a}_i + \mathbf{b}_i) \neq 0\} \subseteq \{i \mid [x^n]\mathbf{a}_i \neq 0\} \cup \{i \mid [x^n]\mathbf{b}_i \neq 0\}$, and the union of two finite sets is finite. □

Lemma 1.90. *If $(\mathbf{a}_i)_{i\in I}$ is a summable family of FPSs, then so is $(-\mathbf{a}_i)_{i\in I}$.*

Proof. For each n, $[x^n](-\mathbf{a}_i) = -[x^n]\mathbf{a}_i$, so $\{i \mid [x^n](-\mathbf{a}_i) \neq 0\} = \{i \mid [x^n]\mathbf{a}_i \neq 0\}$. □

Lemma 1.91. *The sum of two essentially finite families is essentially finite.*

Proof. $\{i \mid a_i + b_i \neq 0\} \subseteq \{i \mid a_i \neq 0\} \cup \{i \mid b_i \neq 0\}$, and the union of two finite sets is finite. □

Lemma 1.92. *The negation of an essentially finite family is essentially finite.*

Proof. $\{i \mid -a_i \neq 0\} = \{i \mid a_i \neq 0\}$. □

Lemma 1.93. *The difference of two essentially finite families is essentially finite.*

Proof. Write $a_i - b_i = a_i + (-b_i)$ and apply Lemma 1.91 and Lemma 1.92. □

Definition 1.94. For an essentially finite family $(a_i)_{i\in I}$, the *essentially finite sum* $\sum_{i\in I} a_i$ is defined as $\sum_{i\in S} a_i$ where $S = \{i \in I \mid a_i \neq 0\}$.

Lemma 1.95. *The essentially finite sum distributes over addition: if $(a_i)_{i\in I}$ and $(b_i)_{i\in I}$ are essentially finite families, then $\sum_{i\in I}(a_i + b_i) = \sum_{i\in I} a_i + \sum_{i\in I} b_i$.*

Proof. All three sums can be computed over the finite set $S = \{i \mid a_i \neq 0\} \cup \{i \mid b_i \neq 0\}$, so this reduces to the finite sum identity $\sum_{i\in S}(a_i + b_i) = \sum_{i\in S} a_i + \sum_{i\in S} b_i$. □

Lemma 1.96. *For a scalar $c \in K$ and an essentially finite family $(a_i)_{i\in I}$, $\sum_{i\in I} c\, a_i = c \cdot \sum_{i\in I} a_i$.*

Proof. The support of $(c\, a_i)$ is contained in the support of (a_i), so this reduces to the finite identity $\sum_{i\in S} c\, a_i = c \sum_{i\in S} a_i$. □

Lemma 1.97. *If $(\mathbf{a}_i)_{i\in I}$ and $(\mathbf{b}_i)_{i\in I}$ are summable families of FPSs, then so is $(\mathbf{a}_i - \mathbf{b}_i)_{i\in I}$.*

Proof. Write $\mathbf{a}_i - \mathbf{b}_i = \mathbf{a}_i + (-\mathbf{b}_i)$ and apply Lemma 1.89 and Lemma 1.90. □

Lemma 1.98. *If $(\mathbf{a}_i)_{i\in I}$ is a family of FPSs such that $\{i \mid \mathbf{a}_i \neq 0\}$ is finite, then the family is summable.*

Proof. For each n, $\{i \mid [x^n]\mathbf{a}_i \neq 0\} \subseteq \{i \mid \mathbf{a}_i \neq 0\}$, which is finite by hypothesis. □

Lemma 1.99. *For $a \in K$ and $k, n \in \mathbb{N}$, $[x^n](\underline{a} \cdot x^k) = \begin{cases} a & \text{if } n = k, \\ 0 & \text{otherwise.} \end{cases}$*

Proof. Follows from $[x^n]x^k = \delta_{n,k}$ (Proposition 1.84) and $[x^n](\underline{a} \cdot f) = a \cdot [x^n]f$. $\square$

Lemma 1.100. *Multiplying on the right by x also shifts coefficients: $[x^{n+1}](\mathbf{a} \cdot x) = [x^n]\mathbf{a}$.*

Proof. By commutativity of multiplication, $\mathbf{a} \cdot x = x \cdot \mathbf{a}$, so this follows from Lemma 1.83. $\square$

Lemma 1.101. *For any $k \in \mathbb{N}$ and any FPS $\mathbf{a}$, $[x^n](\mathbf{a} \cdot x^k) = \begin{cases} 0 & \text{if } n < k, \\ [x^{n-k}]\mathbf{a} & \text{if } n \geq k. \end{cases}$*

Proof. By commutativity: $\mathbf{a} \cdot x^k = x^k \cdot \mathbf{a}$, so this follows from Lemma 1.88. $\square$

1.3.6 Generating functions

Definition 1.102. The *(ordinary) generating function* of a sequence $(a_0, a_1, a_2, \ldots)$ is the FPS $(a_0, a_1, a_2, \ldots) = \sum_{n \geq 0} a_n x^n$.

Lemma 1.103. *The n-th coefficient of the generating function of a sequence (a_n) is a_n.*

Proof. Immediate from the definitions. $\square$

1.3.7 The Chu–Vandermonde identity

Proposition 1.104. *Let $a, b \in \mathbb{N}$, and let $n \in \mathbb{N}$. Then,*

$$\binom{a+b}{n} = \sum_{k=0}^{n} \binom{a}{k} \binom{b}{n-k}.$$

Proof. Multiply out the identity $(1+x)^{a+b} = (1+x)^a (1+x)^b$ using the binomial formula and compare the coefficient of x^n on both sides. The left side gives $\binom{a+b}{n}$, and the right side gives $\sum_{k=0}^{n} \binom{a}{k} \binom{b}{n-k}$ by the product formula for FPSs. $\square$

Lemma 1.105. *The Vandermonde identity in range-sum form: for $a, b, n \in \mathbb{N}$, $\binom{a+b}{n} = \sum_{k=0}^{n} \binom{a}{k} \binom{b}{n-k}$.*

Proof. Rewrites the antidiagonal sum from Proposition 1.104 as a range sum. $\square$

Theorem 1.106 (Vandermonde convolution identity, aka Chu–Vandermonde identity)**.** *Let $a, b \in \mathbb{C}$, and let $n \in \mathbb{N}$. Then,*

$$\binom{a+b}{n} = \sum_{k=0}^{n} \binom{a}{k} \binom{b}{n-k}.$$

Proof. Fix $n \in \mathbb{N}$ and $b \in \mathbb{N}$. By Proposition 1.104, the equality

$$\binom{a+b}{n} = \sum_{k=0}^{n} \binom{a}{k} \binom{b}{n-k}$$

holds for each $a \in \mathbb{N}$. However, both sides are polynomials in a:

$$\binom{a+b}{n} = \frac{(a+b)(a+b-1)\cdots(a+b-n+1)}{n!},$$
$$\sum_{k=0}^{n} \binom{a}{k}\binom{b}{n-k} = \sum_{k=0}^{n} \frac{a(a-1)\cdots(a-k+1)}{k!}\binom{b}{n-k}.$$

Two univariate polynomials that agree on all of $\mathbb{N}$ (infinitely many points) must be identical. Hence the equality holds for all $a \in \mathbb{C}$.

Now fix $a \in \mathbb{C}$ and repeat the argument with b as the variable: both sides are polynomials in b that agree on $\mathbb{N}$, hence they agree on $\mathbb{C}$. (See [20f, proofs of Lemma 7.5.8 and Theorem 7.5.3] or [19s, §2.17.3] for this proof in more detail. Alternatively, see [Grinbe15, §3.3.2 and §3.3.3] for two other proofs.) $\square$

1.4 Dividing FPSs

1.4.1 Inverses in commutative rings

Definition 1.107. Let L be a commutative ring. Let $a \in L$. Then:

(a) An *inverse* (or *multiplicative inverse*) of a means an element $b \in L$ such that $ab = ba = 1$.

(b) We say that a is *invertible* in L (or a *unit* of L) if a has an inverse.

Theorem 1.108. *Let L be a commutative ring. Let $a \in L$. Then, there is **at most one** inverse of a.*

Proof. Let b and c be two inverses of a. Since b is an inverse of a, we have $ab = ba = 1$. Since c is an inverse of a, we have $ac = ca = 1$. Now, $b(ac) = b \cdot 1 = b$ and $(ba)c = 1 \cdot c = c$. However, $b(ac) = (ba)c$ by associativity. Hence, $b = b(ac) = (ba)c = c$. $\square$

Definition 1.109. Let L be a commutative ring. Let $a \in L$. Assume that a is invertible. Then:

(a) The inverse of a is called a^{-1}.

(b) For any $b \in L$, the product $b \cdot a^{-1}$ is called $\dfrac{b}{a}$ (or b/a).

(c) For any negative integer n, we define a^n to be $\left(a^{-1}\right)^{-n}$. Thus, the n-th power a^n is defined for each $n \in \mathbb{Z}$.

Proposition 1.110. *Let L be a commutative ring. Then:*

(a) *Any invertible element $a \in L$ satisfies $a^{-1} = 1/a$.*

(b) *For any invertible elements $a, b \in L$, the element ab is invertible as well, and satisfies $(ab)^{-1} = b^{-1}a^{-1} = a^{-1}b^{-1}$.*

(c) *If $a \in L$ is invertible, and if $n \in \mathbb{Z}$ is arbitrary, then $a^{-n} = \left(a^{-1}\right)^n = (a^n)^{-1}$.*

(d) *Laws of exponents hold for negative exponents as well: If a and b are invertible elements of L, then*

$$\begin{aligned} a^{n+m} &= a^n a^m && \text{for all } n, m \in \mathbb{Z}; \\ (ab)^n &= a^n b^n && \text{for all } n \in \mathbb{Z}; \\ (a^n)^m &= a^{nm} && \text{for all } n, m \in \mathbb{Z}. \end{aligned}$$

(e) Laws of fractions hold: If a and c are two invertible elements of L, and if b and d are any two elements of L, then

$$\frac{b}{a} + \frac{d}{c} = \frac{bc + ad}{ac} \qquad \text{and} \qquad \frac{b}{a} \cdot \frac{d}{c} = \frac{bd}{ac}.$$

(f) Division undoes multiplication: If a, b, c are three elements of L with a being invertible, then the equality $\frac{c}{a} = b$ is equivalent to $c = ab$.

Proof. Exercise (see, e.g., [19s, solution to Exercise 4.1.1] for a proof of parts **(c)** and **(d)** in the special case where $L = \mathbb{C}$; essentially the same argument works in the general case). □

1.4.2 Inverses in $K[[x]]$

Proposition 1.111. *Let $a \in K[[x]]$. Then, the FPS a is invertible in $K[[x]]$ if and only if its constant term $[x^0]a$ is invertible in K.*

Proof. $\Longrightarrow$: If $ab = 1$, then $[x^0]a \cdot [x^0]b = [x^0](ab) = 1$ (by Lemma 1.87).

$\Longleftarrow$: If $[x^0]a = a_0$ is invertible, define a sequence $(b_0, b_1, b_2, \ldots)$ recursively by $b_0 = a_0^{-1}$ (Lemma 1.113) and $b_n = -a_0^{-1} \sum_{k=1}^{n} a_k b_{n-k}$ for $n \geq 1$ (Lemma 1.114). Then $ab = 1$. □

Corollary 1.112. *Assume that K is a field. Let $a \in K[[x]]$. Then, the FPS a is invertible in $K[[x]]$ if and only if $[x^0]a \neq 0$.*

Proof. An element of K is invertible in K if and only if it is nonzero (since K is a field). Hence, Corollary 1.112 follows from Proposition 1.111. □

1.4.3 Coefficient formulas for inverses

Lemma 1.113. *Assume that K is a field. Let $f \in K[[x]]$. Then the constant term of f^{-1} equals the inverse of the constant term of f:*

$$\left[x^0\right] f^{-1} = \left(\left[x^0\right] f\right)^{-1}.$$

Proof. Direct consequence of the definition of the inverse of an FPS. □

Lemma 1.114. *Assume that K is a field. Let $f \in K[[x]]$. For each $n \in \mathbb{N}$,*

$$\left[x^{n+1}\right] f^{-1} = -\left(\left[x^0\right] f\right)^{-1} \sum_{k=0}^{n} \left[x^{k+1}\right] f \cdot \left[x^{n-k}\right] f^{-1}.$$

This recurrence allows computing the coefficients of f^{-1} one by one.

Proof. Follows from the recurrence relation for the coefficients of f^{-1} obtained by expanding the equation $f \cdot f^{-1} = 1$ and solving for $[x^{n+1}]f^{-1}$. □

Lemma 1.115. *Assume that K is a field. Let $f \in K[[x]]$ be an invertible FPS. Then the inverse of f obtained from the invertibility proof equals f^{-1} (the inverse as defined in the power series ring).*

Proof. Both satisfy $f \cdot g = 1$, and inverses are unique (Theorem 1.108). □

Lemma 1.116. *Assume that K is a field. Let $f \in K[[x]]$ be an invertible FPS. Then*

$$\left[x^0\right] f^{-1} = \left(\left[x^0\right] f\right)^{-1}.$$

Proof. By Lemma 1.115, the inverse obtained from the invertibility proof equals f^{-1}, so the result follows from Lemma 1.113. □

Lemma 1.117. *Assume that K is a field. Let $f \in K[[x]]$ be an invertible FPS. For each $n \in \mathbb{N}$,*

$$\left[x^{n+1}\right] f^{-1} = -\left(\left[x^0\right] f\right)^{-1} \sum_{k=0}^{n} \left[x^{k+1}\right] f \cdot \left[x^{n-k}\right] f^{-1}.$$

Proof. By Lemma 1.115, the inverse obtained from the invertibility proof equals f^{-1}, so the result follows from Lemma 1.114. □

1.4.4 Newton's binomial formula

Proposition 1.118. *The FPS $1 + x \in K[[x]]$ is invertible, and its inverse is*

$$(1+x)^{-1} = \sum_{n \in \mathbb{N}} (-1)^n x^n.$$

Proof. Direct computation: $(1+x) \cdot \sum_{n \geq 0} (-1)^n x^n = 1$ by telescoping (all powers of x other than 1 cancel out). □

Theorem 1.119. *For each $n \in \mathbb{Z}$, we have*

$$(1+x)^n = \sum_{k \in \mathbb{N}} \binom{n}{k} x^k.$$

Proof. For $n \geq 0$, this is the standard binomial theorem (the sum $\sum_{k \in \mathbb{N}} \binom{n}{k} x^k$ reduces to $\sum_{k=0}^{n} \binom{n}{k} x^k$ since $\binom{n}{k} = 0$ for $k > n$ by Proposition 1.8).

For $n < 0$, we have $-n \in \mathbb{N}$, so Corollary 1.122 (applied to $-n$ instead of n) yields $(1+x)^{-(-n)} = \sum_{k \in \mathbb{N}} \binom{-(-n)}{k} x^k$, which rewrites as $(1+x)^n = \sum_{k \in \mathbb{N}} \binom{n}{k} x^k$. □

Theorem 1.120. *Let $n \in \mathbb{C}$ and $k \in \mathbb{Z}$. Then,*

$$\binom{-n}{k} = (-1)^k \binom{k+n-1}{k}.$$

Proof. If $k < 0$, then both $\binom{-n}{k}$ and $\binom{k+n-1}{k}$ are 0. For $k \geq 0$, expand using the definition and observe the numerators differ by a factor of $(-1)^k$. □

Proposition 1.121. *For each $n \in \mathbb{N}$, we have*

$$(1+x)^{-n} = \sum_{k \in \mathbb{N}} (-1)^k \binom{n+k-1}{k} x^k.$$

Proof. By induction on n.

Induction base: $(1+x)^{-0} = 1$, and $\sum_{k\in\mathbb{N}}(-1)^k\binom{0+k-1}{k}x^k = 1$ since $\binom{k-1}{k} = 0$ for all $k > 0$ (by Proposition 1.8).

Induction step: Assume the formula holds for $n = j$. We must prove it for $n = j+1$. We have $(1+x)^{-(j+1)} = (1+x)^{-j}\cdot(1+x)^{-1}$. Using the induction hypothesis and multiplying by $1+x$, it suffices to show

$$(1+x)^{-j} = \left(\sum_{k\in\mathbb{N}}(-1)^k\binom{j+k}{k}x^k\right)\cdot(1+x)\,.$$

Expanding the right hand side, using Pascal's identity $\binom{j+k}{k} = \binom{j+k-1}{k-1} + \binom{j+k-1}{k}$ (Proposition 1.5), and collecting terms (the $\binom{j+k-1}{k-1}$ terms telescope), we obtain $\sum_{k\in\mathbb{N}}(-1)^k\binom{j+k-1}{k}x^k = (1+x)^{-j}$ by the induction hypothesis. □

Corollary 1.122. *For each $n \in \mathbb{N}$, we have*

$$(1+x)^{-n} = \sum_{k\in\mathbb{N}}\binom{-n}{k}x^k.$$

Proof. Proposition 1.121 yields $(1+x)^{-n} = \sum_{k\in\mathbb{N}}(-1)^k\binom{n+k-1}{k}x^k$. By Theorem 1.120, we have $(-1)^k\binom{n+k-1}{k} = (-1)^k\binom{k+n-1}{k} = \binom{-n}{k}$. □

1.4.5 Dividing by x

Definition 1.123. Let $a = (a_0, a_1, a_2, \ldots)$ be an FPS whose constant term a_0 is 0. Then, $\dfrac{a}{x}$ is defined to be the FPS $(a_1, a_2, a_3, \ldots)$.

Lemma 1.124. *Let $a \in K[[x]]$ with $[x^0]a = 0$. Then for each $n \in \mathbb{N}$,*

$$[x^n]\,\frac{a}{x} = \left[x^{n+1}\right]a.$$

Proof. By definition, $\frac{a}{x} = (a_1, a_2, a_3, \ldots)$, so its n-th coefficient is $a_{n+1} = [x^{n+1}]a$. □

Proposition 1.125. *Let $a, b \in K[[x]]$ be two FPSs. Then, $a = xb$ if and only if $\left[x^0\right]a = 0$ and $b = \dfrac{a}{x}$.*

Proof. Follows directly from the definitions. □

Helpers for division by x

Lemma 1.126. *Let $a \in K[[x]]$ with $[x^0]a = 0$. Then $a = x\cdot\frac{a}{x}$.*

Proof. Follows from Proposition 1.125: setting $b = \frac{a}{x}$, the right-to-left direction gives $a = xb$. □

Lemma 1.127. *For any FPS $b \in K[[x]]$, we have $\frac{xb}{x} = b$.*

Proof. For each $n \in \mathbb{N}$, $[x^n]\frac{xb}{x} = [x^{n+1}](xb) = [x^n]b$ (by Lemma 1.124 and the coefficient formula for xb). □

1.4.6 A few lemmas

Lemma 1.128. *Let $a \in K[[x]]$ be an FPS with $\left[x^0\right] a = 0$. Then, there exists an $h \in K[[x]]$ such that $a = xh$.*

Proof. The FPS $\dfrac{a}{x}$ is well-defined (since $a_0 = 0$). Set $h = \dfrac{a}{x}$; then $a = x \cdot \dfrac{a}{x} = xh$ (by Lemma 1.126). ☐

Lemma 1.129. *Let $k \in \mathbb{N}$. Let $a \in K[[x]]$ be any FPS. Then, the first k coefficients of the FPS $x^k a$ are 0.*

Proof. For $m < k$, we have $[x^m](x^k a) = \sum_{i=0}^{m} [x^i](x^k) \cdot [x^{m-i}]a = 0$ since $[x^i](x^k) = 0$ for all $i \leq m < k$. ☐

Lemma 1.130. *Let $k \in \mathbb{N}$. Let $f \in K[[x]]$ be any FPS. Then, the first k coefficients of f are 0 if and only if f is a multiple of x^k.*

Proof. $\Longleftarrow$: By Lemma 1.129.

$\Longrightarrow$: If $f_n = 0$ for each $n \in \{0, 1, \ldots, k-1\}$, then $f = \sum_{n \geq k} f_n x^n = x^k \sum_{n \geq k} f_n x^{n-k}$. ☐

Lemma 1.131. *Let $a, f, g \in K[[x]]$ be three FPSs. Let $n \in \mathbb{N}$. Assume that $[x^m]f = [x^m]g$ for each $m \in \{0, 1, \ldots, n\}$. Then, $[x^m](af) = [x^m](ag)$ for each $m \in \{0, 1, \ldots, n\}$.*

Proof. Let $m \in \{0, 1, \ldots, n\}$. Each $j \in \{0, 1, \ldots, m\}$ satisfies $j \leq m \leq n$ and thus $[x^j]f = [x^j]g$. Hence, $[x^m](af) = \sum_{j=0}^{m} [x^{m-j}]a \cdot [x^j]f = \sum_{j=0}^{m} [x^{m-j}]a \cdot [x^j]g = [x^m](ag)$. ☐

Lemma 1.132. *Let $u, v \in K[[x]]$ be two FPSs such that v is a multiple of u. Let $n \in \mathbb{N}$. Assume $[x^m]u = 0$ for each $m \in \{0, 1, \ldots, n\}$. Then $[x^m]v = 0$ for each $m \in \{0, 1, \ldots, n\}$.*

Proof. Write $v = ua$. We have $[x^m]u = 0 = [x^m]0$ for each $m \in \{0, 1, \ldots, n\}$. Lemma 1.131 (applied to $f = u$ and $g = 0$) yields $[x^m](au) = [x^m](a \cdot 0) = 0$ for each $m \in \{0, 1, \ldots, n\}$. Since $v = ua = au$, we get $[x^m]v = 0$. ☐

Lemma 1.133. *Let $a, b, c, d \in K[[x]]$ be four FPSs. Let $n \in \mathbb{N}$. Assume $[x^m]a = [x^m]b$ and $[x^m]c = [x^m]d$ for each $m \in \{0, 1, \ldots, n\}$. Then $[x^m](ac) = [x^m](bd)$ for each $m \in \{0, 1, \ldots, n\}$.*

Proof. The FPS $ac - bc = (a - b)c$ is a multiple of $a - b$, and $[x^m](a - b) = 0$ for each $m \in \{0, 1, \ldots, n\}$. By Lemma 1.132, $[x^m](ac - bc) = 0$, so $[x^m](ac) = [x^m](bc)$.

Similarly, $bc - bd = b(c - d) = (c - d)b$ is a multiple of $c - d$, and $[x^m](c - d) = 0$. By Lemma 1.132, $[x^m](bc - bd) = 0$, so $[x^m](bc) = [x^m](bd)$.

Combining: $[x^m](ac) = [x^m](bc) = [x^m](bd)$. ☐

1.5 Polynomials

1.5.1 Definition

Definition 1.134. **(a)** An FPS $a \in K[[x]]$ is said to be a *polynomial* if all but finitely many $n \in \mathbb{N}$ satisfy $[x^n]\, a = 0$ (that is, if all but finitely many coefficients of a are 0).

(b) We let $K[x]$ be the set of all polynomials $a \in K[[x]]$. This set $K[x]$ is a subring of $K[[x]]$ (according to Theorem 1.148 below), and is called the *univariate polynomial ring* over K.

Lemma 1.135. *An FPS $f \in K[[x]]$ is a polynomial if and only if there exists an $N \in \mathbb{N}$ such that $[x^n]\, f = 0$ for all $n \geq N$.*

Proof. $\Longrightarrow$: If the set of nonzero-coefficient indices is finite, it is bounded above by some N, so all coefficients beyond N vanish.

$\Longleftarrow$: If all coefficients beyond N vanish, then the set of nonzero-coefficient indices is contained in $\{0, 1, \ldots, N-1\}$, which is finite. $\square$

Lemma 1.136. *An FPS $f \in K[[x]]$ is a polynomial (in the sense of Definition 1.134) if and only if f equals the image of some element of $K[x]$ under the canonical embedding $K[x] \hookrightarrow K[[x]]$.*

Proof. $\Longrightarrow$: Use the degree bound from Lemma 1.135 and truncation.

$\Longleftarrow$: The image of any polynomial has finite support (Lemma 1.137). $\square$

Lemma 1.137. *The image of any polynomial $p \in K[X]$ under the canonical embedding $K[X] \hookrightarrow K[[x]]$ is a polynomial in the FPS sense.*

Proof. The set of indices with nonzero coefficients is contained in the support of p, which is a finite set. $\square$

Converting between polynomial FPSs and polynomials

Definition 1.138. Let $f \in K[[x]]$ be a polynomial FPS (in the sense of Definition 1.134). We define $\operatorname{toPolynomial}(f) \in K[X]$ as the truncation of f at the degree bound given by Lemma 1.135.

Lemma 1.139. *Let $f \in K[[x]]$ be a polynomial FPS. Then the coercion of* $\operatorname{toPolynomial}(f)$ *back into $K[[x]]$ equals f:*

$$\iota\big(\operatorname{toPolynomial}(f)\big) = f,$$

where $\iota \colon K[X] \hookrightarrow K[[x]]$ is the canonical embedding.

Proof. By the degree bound, all coefficients of f beyond the bound are zero, so truncation and coercion recover every coefficient of f. $\square$

Lemma 1.140. *For any polynomial $p \in K[X]$, we have* $\operatorname{toPolynomial}(\iota(p)) = p$*, where ι is the canonical embedding.*

Proof. By Lemma 1.139, both sides have the same image under ι, and ι is injective. $\square$

1.5.2 Polynomials form a subring of power series

Lemma 1.141. *The zero power series $\underline{0}$ is a polynomial.*

Proof. All coefficients of $\underline{0}$ are 0, so the set of nonzero-coefficient indices is empty, hence finite. $\square$

Lemma 1.142. *The unity $\underline{1}$ is a polynomial.*

Proof. The unity $\underline{1}$ equals the image of $1 \in K[X]$ under the canonical embedding. $\square$

Lemma 1.143. *The sum of two polynomial power series is a polynomial.*

Proof. If f and g are polynomials, then the set of indices where $(f+g)$ has a nonzero coefficient is contained in the union of the corresponding sets for f and g. A finite union of finite sets is finite. $\square$

Lemma 1.144. *The negation of a polynomial power series is a polynomial.*

Proof. Negation does not change which coefficients are nonzero. $\square$

Lemma 1.145. *The difference of two polynomial power series is a polynomial.*

Proof. Write $f - g = f + (-g)$ and apply Lemma 1.143 and Lemma 1.144. $\square$

Lemma 1.146. *The product of two polynomial power series is a polynomial.*

Proof. Let $a, b \in K[x]$. Since a is a polynomial, there exists a finite subset I of $\mathbb{N}$ such that $[x^i]\, a = 0$ for all $i \in \mathbb{N} \setminus I$. Similarly, there exists a finite subset J of $\mathbb{N}$ such that $[x^j]\, b = 0$ for all $j \in \mathbb{N} \setminus J$.

Let $S = \{i + j \mid i \in I \text{ and } j \in J\}$. This set S is again finite (since I and J are finite), and we have

$$[x^n]\,(ab) = \sum_{i=0}^{n} [x^i]\, a \cdot [x^{n-i}]\, b = 0 \text{ for all } n \in \mathbb{N} \setminus S,$$

because for each $i \in \{0, 1, \ldots, n\}$, letting $j = n - i$, we cannot have both $i \in I$ and $j \in J$ (otherwise $n = i + j \in S$, contradicting $n \notin S$), so at least one of $[x^i]\, a$ and $[x^j]\, b$ is 0, making each summand 0. Thus, all but finitely many $n \in \mathbb{N}$ satisfy $[x^n]\,(ab) = 0$, so $ab \in K[x]$. $\square$

Lemma 1.147. *Scalar multiplication of a polynomial power series by an element of K is a polynomial.*

Proof. If $[x^n]\, f = 0$, then $[x^n]\,(cf) = c \cdot [x^n]\, f = 0$. So the nonzero-coefficient set of cf is contained in that of f, which is finite. $\square$

Theorem 1.148. *The set $K[x]$ is a subring of $K[[x]]$ (that is, it is closed under addition, subtraction and multiplication, and contains the zero $\underline{0}$ and the unity $\underline{1}$) and is a K-submodule of $K[[x]]$ (that is, it is closed under addition and scaling by elements of K).*

Proof. Follows from the preceding lemmas: $K[x]$ contains $\underline{0}$ (Lemma 1.141) and $\underline{1}$ (Lemma 1.142), is closed under addition (Lemma 1.143), multiplication (Lemma 1.146), and scalar multiplication (Lemma 1.147). $\square$

Lemma 1.149. *The polynomial $x \in K[[x]]$ is a polynomial.*

Proof. The only nonzero coefficient of x is the coefficient of x^1, which is 1. Thus the set of nonzero-coefficient indices is contained in $\{1\}$, which is finite. $\square$

Lemma 1.150. *For any $c \in K$, the constant FPS $\underline{c}$ is a polynomial.*

Proof. The only possibly nonzero coefficient of $\underline{c}$ is the coefficient of x^0. Thus the set of nonzero-coefficient indices is contained in $\{0\}$, which is finite. $\square$

1.5.3 Reminders on rings and K-algebras

Definition 1.151. The notion of a *ring* (also known as a *noncommutative ring*) is defined in the exact same way as we defined the notion of a commutative ring in Definition 1.42, except that the "Commutativity of multiplication" axiom is removed.

Definition 1.152. A K-*algebra* is a set A equipped with four maps

$$\begin{aligned} \oplus &: A \times A \to A, \\ \ominus &: A \times A \to A, \\ \odot &: A \times A \to A, \\ \rightharpoonup &: K \times A \to A \end{aligned}$$

and two elements $\overrightarrow{0} \in A$ and $\overrightarrow{1} \in A$ satisfying the following properties:

1. The set A, equipped with the maps $\oplus$, $\ominus$ and $\odot$ and the two elements $\overrightarrow{0}$ and $\overrightarrow{1}$, is a (noncommutative) ring.
2. The set A, equipped with the maps $\oplus$, $\ominus$ and $\rightharpoonup$ and the element $\overrightarrow{0}$, is a K-module.
3. We have
$$\lambda \rightharpoonup (a \odot b) = (\lambda \rightharpoonup a) \odot b = a \odot (\lambda \rightharpoonup b) \tag{1.4}$$
for all $\lambda \in K$ and $a, b \in A$.

(Thus, in a nutshell, a K-algebra is a set A that is simultaneously a ring and a K-module, with the property that the ring A and the K-module A have the same addition, the same subtraction and the same zero, and satisfy the additional compatibility property (1.4).)

1.5.4 Evaluation aka substitution into polynomials

Definition 1.153. Let $f \in K[x]$ be a polynomial. Let A be any K-algebra. Let $a \in A$ be any element. We then define an element $f[a] \in A$ as follows:

Write f in the form $f = \sum_{n \in \mathbb{N}} f_n x^n$ with $f_0, f_1, f_2, \ldots \in K$. (That is, $f_n = [x^n] f$ for each $n \in \mathbb{N}$.) Then, set
$$f[a] := \sum_{n \in \mathbb{N}} f_n a^n.$$
(This sum is essentially finite, since f is a polynomial.)

The element $f[a]$ is also denoted by $f \circ a$ or by $f(a)$, and is called the *value* of f at a (or the *evaluation* of f at a, or the *result of substituting* a for x in f).

Lemma 1.154. *The evaluation $f[a]$ can be computed as $f[a] = \sum_n f_n \cdot a^n$, where the sum ranges over all n in the support of f.*

Proof. Immediate from the definitions. $\square$

Lemma 1.155. *The evaluation $f[a]$ equals $\sum_{n \in \operatorname{supp}(f)} f_n \cdot a^n$, making the finiteness of the sum explicit.*

Proof. This follows from the definition by restricting the sum to the support. $\square$

Lemma 1.156. *The evaluation $f[a]$ equals $\sum_{n=0}^{\deg f} f_n \cdot a^n$, where the sum is taken up to the degree of f.*

Proof. Since all coefficients beyond the degree of f are zero, extending or restricting the sum to this range does not change the result. $\square$

Theorem 1.157. *Let A be a K-algebra. Let $a \in A$. Then:*

(a) *Any $f, g \in K[x]$ satisfy*
$$(f+g)[a] = f[a] + g[a] \qquad \text{and} \qquad (fg)[a] = f[a] \cdot g[a].$$

(b) *Any $\lambda \in K$ and $f \in K[x]$ satisfy*
$$(\lambda f)[a] = \lambda \cdot f[a].$$

(c) *Any $\lambda \in K$ satisfies $\underline{\lambda}[a] = \lambda \cdot 1_A$, where 1_A is the unity of the ring A.*

(d) *We have $x[a] = a$.*

(e) *We have $x^i[a] = a^i$ for each $i \in \mathbb{N}$.*

(f) *Any $f, g \in K[x]$ satisfy $f[g[a]] = (f[g])[a]$.*

Proof. See [19s, Theorem 7.6.3] for parts **(a)**, **(b)**, **(c)**, **(d)** and **(e)**. Part **(f)** is [19s, Proposition 7.6.14]. ☐

1.5.5 Special evaluation results

Lemma 1.158. *For any polynomial $f \in K[x]$, we have $f[x] = f$.*

Proof. The evaluation map at x is the identity on $K[x]$. ☐

Lemma 1.159. *For any polynomial $f \in K[x]$ and any K-algebra A, we have $f[0] = \left[x^0\right] f$ (the constant term of f, coerced into A).*

Proof. Setting $a = 0$: all terms $f_n \cdot 0^n$ for $n \geq 1$ vanish, leaving only $f_0 \cdot 0^0 = f_0 \cdot 1 = f_0$. ☐

Lemma 1.160. *For any polynomial $f \in K[x]$ and any K-algebra A, we have $f[1] = \left[x^0\right] f + \left[x^1\right] f + \left[x^2\right] f + \cdots$ (the sum of all coefficients of f, coerced into A).*

Proof. Setting $a = 1$: each term $f_n \cdot 1^n = f_n$, so $f[1] = \sum_n f_n$. ☐

1.6 Substitution and evaluation of power series

1.6.1 Defining substitution

Definition 1.161. Let f and g be two FPSs in $K[[x]]$. Assume that $[x^0]g = 0$ (that is, $g = g_1x + g_2x^2 + g_3x^3 + \cdots$ for some $g_1, g_2, g_3, \ldots \in K$).

We then define an FPS $f[g] \in K[[x]]$ as follows:

Write f in the form $f = \sum_{n \in \mathbb{N}} f_n x^n$ with $f_0, f_1, f_2, \ldots \in K$. (That is, $f_n = [x^n]f$ for each $n \in \mathbb{N}$.) Then, set

$$f[g] := \sum_{n \in \mathbb{N}} f_n g^n.$$

(This sum is well-defined, as we will see in Proposition 1.163 **(b)** below.)

This FPS $f[g]$ is also denoted by $f \circ g$, and is called the *composition* of f with g, or the result of *substituting* g for x in f.

Equivalently, the n-th coefficient of $f[g]$ is the finite sum

$$[x^n](f[g]) = \sum_{d=0}^{n} f_d \cdot [x^n](g^d).$$

Lemma 1.162. *The composition $f[g]$ defined above equals the standard substitution operation on formal power series.*

Proof. By definition. ☐

1.6.2 Well-definedness of substitution

Proposition 1.163. *Let f and g be two FPSs in $K[[x]]$. Assume that $[x^0]g = 0$. Write f in the form $f = \sum_{n \in \mathbb{N}} f_n x^n$ with $f_0, f_1, f_2, \ldots \in K$. Then:*

(a) *For each $n \in \mathbb{N}$, the first n coefficients of the FPS g^n are 0.*

(b) *The sum $\sum_{n \in \mathbb{N}} f_n g^n$ is well-defined, i.e., the family $(f_n g^n)_{n \in \mathbb{N}}$ is summable.*

(c) *We have $[x^0](\sum_{n \in \mathbb{N}} f_n g^n) = f_0$.*

Proof. **(a)** The FPS g has constant term $[x^0]g = 0$. Hence, by Lemma 1.128 (applied to $a = g$), there exists an $h \in K[[x]]$ such that $g = xh$. Now, let $n \in \mathbb{N}$. From $g = xh$, we obtain $g^n = (xh)^n = x^n h^n$. However, Lemma 1.129 (applied to $k = n$ and $a = h^n$) yields that the first n coefficients of $x^n h^n$ are 0. In other words, the first n coefficients of g^n are 0.

(b) This follows from part **(a)**. We must prove that for each $n \in \mathbb{N}$, all but finitely many $i \in \mathbb{N}$ satisfy $[x^n](f_i g^i) = 0$. Fix $n \in \mathbb{N}$. Let $i \in \mathbb{N}$ satisfy $i > n$. Then $n < i$, and by part **(a)** (applied to i instead of n), the first i coefficients of g^i are 0. Since $n < i$, the coefficient $[x^n](g^i) = 0$. Thus $[x^n](f_i g^i) = f_i \cdot 0 = 0$.

(c) Let n be a positive integer. By part **(a)**, the first n coefficients of g^n are 0, so $[x^0](g^n) = 0$ and thus $[x^0](f_n g^n) = 0$. Therefore,

$$[x^0]\left(\sum_{n\in\mathbb{N}} f_n g^n\right) = [x^0](f_0 \underbrace{g^0}_{=1}) + \sum_{n>0} \underbrace{[x^0](f_n g^n)}_{=0} = f_0.$$

□

Lemma 1.164. *Let $g \in K[[x]]$ with $[x^0]g = 0$. Then $0[g] = 0$.*

Proof. Immediate from the definitions: $0[g] = \sum_{n\in\mathbb{N}} 0 \cdot g^n = 0$. □

Lemma 1.165. *Let $g \in K[[x]]$ with $[x^0]g = 0$. Then $\underline{1}[g] = \underline{1}$.*

Proof. Immediate from the definitions: $\underline{1}[g] = 1 \cdot g^0 + \sum_{n>0} 0 \cdot g^n = 1$. □

1.6.3 Lemma for first k coefficients

Lemma 1.166. *Let $f, g \in K[[x]]$ satisfy $[x^0]g = 0$. Let $k \in \mathbb{N}$ be such that the first k coefficients of f are 0. Then, the first k coefficients of $f \circ g$ are 0.*

Proof. We have $[x^0]g = 0$. Hence, by Lemma 1.128 (applied to $a = g$), there exists an $h \in K[[x]]$ such that $g = xh$.

Write $f = (f_0, f_1, f_2, \ldots)$. The first k coefficients of f are 0, so $f_n = 0$ for each $n < k$. Now,

$$f \circ g = \sum_{n\in\mathbb{N}} f_n g^n = \sum_{\substack{n\in\mathbb{N};\\ n<k}} \underbrace{f_n}_{=0} g^n + \sum_{\substack{n\in\mathbb{N};\\ n\geq k}} f_n \underbrace{g^n}_{=x^n h^n} = \sum_{\substack{n\in\mathbb{N};\\ n\geq k}} f_n x^n h^n.$$

For any $m < k$, each term $f_n x^n h^n$ with $n \geq k$ satisfies $[x^m](x^n h^n) = 0$ (since $m < k \leq n$). Thus $[x^m](f \circ g) = 0$ for each $m < k$. □

1.6.4 Kronecker delta notation

Definition 1.167. If i and j are any objects, then $\delta_{i,j}$ means the element

$$\delta_{i,j} = \begin{cases} 1, & \text{if } i = j; \\ 0, & \text{if } i \neq j \end{cases}$$

of K.

Lemma 1.168. $\delta_{i,i} = 1$.

Proof. Immediate from the definition. □

Lemma 1.169. *If $i \neq j$, then $\delta_{i,j} = 0$.*

Proof. Immediate from the definition. □

Lemma 1.170. $\delta_{i,j} = \delta_{j,i}$.

Proof. Case split on whether $i = j$. □

Lemma 1.171. $\delta_{i,j} \cdot a = \begin{cases} a & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$

Proof. Case split on whether $i = j$. □

Lemma 1.172. $a \cdot \delta_{i,j} = \begin{cases} a & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$

Proof. Case split on whether $i = j$. □

Lemma 1.173. *If α is a finite type, then $\sum_{i \in \alpha} \delta_{i,j} = 1$.*

Proof. All terms vanish except $i = j$, which contributes 1. □

Lemma 1.174. *If α is a finite type, then $\sum_{j \in \alpha} \delta_{i,j} = 1$.*

Proof. By symmetry ($\delta_{i,j} = \delta_{j,i}$), this reduces to Lemma 1.173. □

Lemma 1.175. *If α is a finite type, then $\sum_{i \in \alpha} \delta_{i,j} \cdot f(i) = f(j)$.*

Proof. By the multiplication property, $\delta_{i,j} \cdot f(i) = 0$ for $i \neq j$ and $f(j)$ for $i = j$. The sum collapses to the single term $f(j)$. □

Lemma 1.176. *If α is a finite type, then $\sum_{j \in \alpha} f(j) \cdot \delta_{i,j} = f(i)$.*

Proof. Analogous to Lemma 1.175. □

Lemma 1.177. *For natural numbers $n, m \in \mathbb{N}$, we have $\delta_{n,m} = \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$*

Proof. This is an immediate specialization of the definition to natural numbers. □

Lemma 1.178. *For integers $n, m \in \mathbb{Z}$, we have $\delta_{n,m} = \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m. \end{cases}$*

Proof. This is an immediate specialization of the definition to integers. □

Lemma 1.179. *For a nontrivial ring K and any i, j, we have $\delta_{i,j} = 0$ if and only if $i \neq j$.*

Proof. If $i \neq j$, then $\delta_{i,j} = 0$ by definition. Conversely, if $i = j$, then $\delta_{i,j} = 1 \neq 0$ (since K is nontrivial). □

Lemma 1.180. *For a nontrivial ring K and any i, j, we have $\delta_{i,j} = 1$ if and only if $i = j$.*

Proof. If $i = j$, then $\delta_{i,j} = 1$ by definition. Conversely, if $i \neq j$, then $\delta_{i,j} = 0 \neq 1$ (since K is nontrivial). □

1.6.5 Laws of substitution

Proposition 1.181. *Composition of FPSs satisfies the rules you would expect it to satisfy:*

(a) *If $f_1, f_2, g \in K[[x]]$ satisfy $[x^0]g = 0$, then $(f_1 + f_2) \circ g = f_1 \circ g + f_2 \circ g$.*

(b) *If $f_1, f_2, g \in K[[x]]$ satisfy $[x^0]g = 0$, then $(f_1 \cdot f_2) \circ g = (f_1 \circ g) \cdot (f_2 \circ g)$.*

(c) *If $f_1, f_2, g \in K[[x]]$ satisfy $[x^0]g = 0$, then $\dfrac{f_1}{f_2} \circ g = \dfrac{f_1 \circ g}{f_2 \circ g}$, as long as f_2 is invertible. (In particular, $f_2 \circ g$ is automatically invertible under these assumptions.)*

(d) *If $f, g \in K[[x]]$ satisfy $[x^0]g = 0$, then $f^k \circ g = (f \circ g)^k$ for each $k \in \mathbb{N}$.*

(e) *If $f, g, h \in K[[x]]$ satisfy $[x^0]g = 0$ and $[x^0]h = 0$, then $[x^0](g \circ h) = 0$ and $(f \circ g) \circ h = f \circ (g \circ h)$.*

(f) *We have $\underline{a} \circ g = \underline{a}$ for each $a \in K$ and $g \in K[[x]]$.*

(g) *We have $x \circ g = g \circ x = g$ for each $g \in K[[x]]$.*

(h) *If $(f_i)_{i \in I} \in K[[x]]^I$ is a summable family of FPSs, and if $g \in K[[x]]$ is an FPS satisfying $[x^0]g = 0$, then the family $(f_i \circ g)_{i \in I} \in K[[x]]^I$ is summable as well and we have $\left(\sum_{i \in I} f_i\right) \circ g = \sum_{i \in I} f_i \circ g$.*

Proof. **(a)** Write $f_1 = \sum_{n \in \mathbb{N}} f_{1,n} x^n$ and $f_2 = \sum_{n \in \mathbb{N}} f_{2,n} x^n$. Then $f_1 + f_2 = \sum_{n \in \mathbb{N}} (f_{1,n} + f_{2,n}) x^n$, so

$$(f_1 + f_2)[g] = \sum_{n \in \mathbb{N}} (f_{1,n} + f_{2,n}) g^n = \sum_{n \in \mathbb{N}} f_{1,n} g^n + \sum_{n \in \mathbb{N}} f_{2,n} g^n = f_1[g] + f_2[g].$$

(f) We have $\underline{a} = \sum_{n \in \mathbb{N}} a \delta_{n,0} x^n$. Hence $\underline{a}[g] = \sum_{n \in \mathbb{N}} a \delta_{n,0} g^n = a \cdot 1 + 0 = \underline{a}$.

(g) We have $x = \sum_{n \in \mathbb{N}} \delta_{n,1} x^n$, so $x[g] = \sum_{n \in \mathbb{N}} \delta_{n,1} g^n = g^1 = g$. Also, writing $g = \sum_{n \in \mathbb{N}} g_n x^n$, we get $g[x] = \sum_{n \in \mathbb{N}} g_n x^n = g$.

(b) Write $f_1 = \sum_{i \in \mathbb{N}} f_{1,i} x^i$ and $f_2 = \sum_{j \in \mathbb{N}} f_{2,j} x^j$. Then

$$f_1[g] \cdot f_2[g] = \left(\sum_{i \in \mathbb{N}} f_{1,i} g^i\right) \left(\sum_{j \in \mathbb{N}} f_{2,j} g^j\right) = \sum_{n \in \mathbb{N}} \left(\sum_{\substack{(i,j) \in \mathbb{N}^2; \\ i+j=n}} f_{1,i} f_{2,j} \right) g^n.$$

The same computation with g replaced by x gives $f_1 \cdot f_2 = \sum_{n \in \mathbb{N}} c_n x^n$ where $c_n = \sum_{\substack{(i,j) \in \mathbb{N}^2; \\ i+j=n}} f_{1,i} f_{2,j}$. Thus $(f_1 \cdot f_2)[g] = \sum_{n \in \mathbb{N}} c_n g^n = f_1[g] \cdot f_2[g]$.

The interchange of infinite summation signs (discrete Fubini) is justified because the family $(f_{1,i} f_{2,j} g^{i+j})_{(i,j) \in \mathbb{N}^2}$ is summable: for any $m \in \mathbb{N}$ and any (i,j) with $m < i + j$, we have $[x^m](g^{i+j}) = 0$ by Proposition 1.163 **(a)**.

(c) Assume f_2 is invertible. By part **(b)** applied to f_2^{-1}, $(f_2^{-1} \cdot f_2) \circ g = (f_2^{-1} \circ g) \cdot (f_2 \circ g)$, so $(f_2^{-1} \circ g) \cdot (f_2 \circ g) = \underline{1} \circ g = \underline{1}$ by part **(f)**. Hence $f_2 \circ g$ is invertible. Then by part **(b)** applied to f_1/f_2, $(f_1/f_2 \cdot f_2) \circ g = (f_1/f_2 \circ g) \cdot (f_2 \circ g)$, i.e., $f_1 \circ g = (f_1/f_2 \circ g) \cdot (f_2 \circ g)$. Dividing by $f_2 \circ g$ gives $f_1/f_2 \circ g = (f_1 \circ g)/(f_2 \circ g)$.

(d) By induction on k. Base case: $f^0 \circ g = \underline{1} \circ g = \underline{1} = (f \circ g)^0$ by part **(f)**. Induction step: $f^{m+1} \circ g = (f \cdot f^m) \circ g = (f \circ g) \cdot (f^m \circ g) = (f \circ g) \cdot (f \circ g)^m = (f \circ g)^{m+1}$ by part **(b)**.

(h) First, we show $(f_i \circ g)_{i \in I}$ is summable. Since $(f_i)_{i \in I}$ is summable, for each $n \in \mathbb{N}$ there is a finite $I_n \subseteq I$ such that $[x^n] f_i = 0$ for all $i \in I \setminus I_n$. For $i \in I \setminus (I_0 \cup I_1 \cup \cdots \cup I_n)$, the first $n+1$ coefficients of f_i are all 0, so by Lemma 1.166, $[x^n](f_i \circ g) = 0$. Thus the family $(f_i \circ g)_{i \in I}$ is summable.

The equality $\left(\sum_{i \in I} f_i\right) \circ g = \sum_{i \in I} f_i \circ g$ follows by writing each $f_i = \sum_{n \in \mathbb{N}} f_{i,n} x^n$ and interchanging summation signs (justified by summability of the family $(f_{i,n} g^n)_{(i,n) \in I \times \mathbb{N}}$).

(e) Write $g = \sum_{n \in \mathbb{N}} g_n x^n$. By Proposition 1.163 **(c)** (applied to g, h), $[x^0](g \circ h) = g_0 = [x^0]g = 0$.

For associativity: $f \circ g = \sum_{n \in \mathbb{N}} f_n g^n$, and the family $(f_n g^n)_{n \in \mathbb{N}}$ is summable by Proposition 1.163 **(b)**. By part **(h)**,

$$(f \circ g) \circ h = \left(\sum_{n \in \mathbb{N}} f_n g^n \right) \circ h = \sum_{n \in \mathbb{N}} (f_n g^n) \circ h.$$

By parts **(b)**, **(d)**, and **(f)**, $(f_n g^n) \circ h = (\underline{f_n} \cdot g^n) \circ h = (\underline{f_n} \circ h) \cdot (g^n \circ h) = \underline{f_n} \cdot (g \circ h)^n = f_n (g \circ h)^n$. Hence $(f \circ g) \circ h = \sum_{n \in \mathbb{N}} f_n (g \circ h)^n = f \circ (g \circ h)$. $\square$

1.6.6 Example: Fibonacci generating function

Definition 1.182. The *geometric series* is the FPS $\frac{1}{1-x} = (1-x)^{-1} \in K[[x]]$ (where K is a field).

Lemma 1.183. *Substituting $x + x^2$ into the geometric series $\frac{1}{1-x}$ yields the generating function for shifted Fibonacci numbers:*

$$\frac{1}{1-x}\left[x + x^2\right] = \frac{1}{1 - x - x^2}.$$

Proof. By Proposition 1.181 **(c)**, substitution preserves inverses when the original has invertible constant coefficient. Since $[x^0](1-x) = 1 \neq 0$, we have $\frac{1}{1-x}\left[x + x^2\right] = \frac{1}{(1-x)[x+x^2]}$. Now $(1-x)[x+x^2] = 1 - (x + x^2) = 1 - x - x^2$ by the linearity of substitution, so the result follows. $\square$

1.7 Derivatives of FPSs

Our definition of the derivative of a FPS copycats the well-known formula for the derivative of a power series in analysis:

Definition 1.184. Let $f \in K[[x]]$ be an FPS. Then, the *derivative* f' of f is an FPS defined as follows: Write f as $f = \sum_{n \in \mathbb{N}} f_n x^n$ (with $f_0, f_1, f_2, \ldots \in K$), and set

$$f' := \sum_{n > 0} n f_n x^{n-1}.$$

Lemma 1.185. *Let $f \in K[[x]]$ be an FPS. Then f' is the power series with coefficient function $n \mapsto (n+1) \cdot f_{n+1}$.*

Proof. Immediate from the definition by reindexing $n \mapsto n+1$. $\square$

Lemma 1.186. *The derivative operation $f \mapsto f'$ agrees with the standard derivative operation on formal power series.*

Proof. By definition. $\square$

Lemma 1.187. *Let $f \in K[[x]]$ be an FPS. Then for each $n \in \mathbb{N}$, the n-th coefficient of f' is $(n+1) \cdot f_{n+1}$.*

Proof. Follows directly from the definition of the derivative. $\square$

Lemma 1.188. *We have $0' = 0$.*

Proof. All coefficients of 0 are 0, so the derivative is 0. □

Lemma 1.189. *We have* $1' = 0$.

Proof. 1 is constant, so $1' = 0$. □

Lemma 1.190. *For any* $c \in K$, *we have* $(c)' = 0$ *(where* c *is viewed as a constant FPS).*

Proof. All coefficients of c beyond the constant term are 0, so the derivative is 0. □

Lemma 1.191. *We have* $x' = 1$.

Proof. The 0-th coefficient of x' is $(0+1) \cdot [x^1]x = 1$, and all higher coefficients are 0. □

Lemma 1.192. *For any* $n \in \mathbb{N}$, *we have* $\left(x^{n+1}\right)' = (n+1) \cdot x^n$.

Proof. Direct coefficient comparison: $[x^m](x^{n+1})' = (m+1) \cdot [x^{m+1}](x^{n+1})$. This is $(n+1)$ if $m = n$ and 0 otherwise. □

Lemma 1.193. *For any* $n \in \mathbb{N}$ *with* $n > 0$, *we have* $(x^n)' = n \cdot x^{n-1}$.

Proof. Write $n = m + 1$ and apply Lemma 1.192. □

Lemma 1.194. *Let* p *be a polynomial. Then the derivative of* p *viewed as a power series equals the polynomial derivative of* p *viewed as a power series.*

Proof. Both derivatives are defined by the same coefficient formula. □

1.7.1 Derivative rules

Theorem 1.195. ***(a)*** *We have* $(f+g)' = f' + g'$ *for any* $f, g \in K[[x]]$.

(b) *If* $(f_i)_{i \in I}$ *is a summable family of FPSs, then the family* $(f_i')_{i \in I}$ *is summable as well, and we have*

$$\left(\sum_{i \in I} f_i\right)' = \sum_{i \in I} f_i'.$$

(c) *We have* $(cf)' = cf'$ *for any* $c \in K$ *and* $f \in K[[x]]$.

(d) *We have* $(fg)' = f'g + fg'$ *for any* $f, g \in K[[x]]$. *(This is known as the* Leibniz rule.*)*

(e) *If* $f, g \in K[[x]]$ *are two FPSs such that* g *is invertible, then*

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

(This is known as the quotient rule.*)*

(f) *If* $g \in K[[x]]$ *is an FPS, then* $(g^n)' = ng'g^{n-1}$ *for any* $n \in \mathbb{N}$ *(where the expression* $ng'g^{n-1}$ *is to be understood as* 0 *if* $n = 0$*).*

(g) *Given two FPSs* $f, g \in K[[x]]$, *we have*

$$(f \circ g)' = (f' \circ g) \cdot g'$$

if f *is a polynomial or if* $\left[x^0\right] g = 0$. *(This is known as the* chain rule.*)*

(h) *If* K *is a* $\mathbb{Q}$*-algebra, and if two FPSs* $f, g \in K[[x]]$ *satisfy* $f' = g'$, *then* $f - g$ *is constant.*

Proof. This combines parts **(a)** through **(h)**, proved individually below. $\square$

Lemma 1.196 (Theorem 1.195 **(a)**)**.** *We have $(f+g)' = f' + g'$ for any $f, g \in K[[x]]$.*

Proof. This is part of [19s-mt3s, Exercise 5 **(b)**] (specifically, it is Statement 1 in [19s-mt3s, solution to Exercise 5 **(b)**]). $\square$

Lemma 1.197. *We have $(-f)' = -f'$ for any $f \in K[[x]]$ (where K is a commutative ring).*

Proof. The derivative is an additive map, so it preserves negation. $\square$

Lemma 1.198. *We have $(f-g)' = f' - g'$ for any $f, g \in K[[x]]$ (where K is a commutative ring).*

Proof. The derivative is an additive map, so it preserves subtraction. $\square$

Lemma 1.199 (Theorem 1.195 **(b)**, finite version)**.** *Let I be a finite index set and $(f_i)_{i\in I}$ a family of FPSs. Then $\left(\sum_{i\in I} f_i\right)' = \sum_{i\in I} f_i'$.*

Proof. Follows from additivity (part **(a)**) by induction on $|I|$. $\square$

Lemma 1.200. *If $(f_i)_{i\in I}$ is a summable family of FPSs, then $(f_i')_{i\in I}$ is also summable.*

Proof. For each coefficient index n, the set of i with $[x^n]f_i' \neq 0$ is a subset of the set of i with $[x^{n+1}]f_i \neq 0$, which is finite by summability of (f_i). $\square$

Lemma 1.201 (Theorem 1.195 **(b)**, infinite version)**.** *If $(f_i)_{i\in I}$ is a summable family of FPSs, then the family $(f_i')_{i\in I}$ is summable as well, and we have*

$$\left(\sum_{i\in I} f_i\right)' = \sum_{i\in I} f_i'.$$

Proof. This is the natural generalization of Theorem 1.195 **(a)** to (potentially) infinite sums. The proof works coefficient-by-coefficient: for each n,

$$\begin{aligned}
[x^n]\left(\sum_{i\in I} f_i\right)' &= \left(\sum_{i\in I}[x^{n+1}]f_i\right)\cdot(n+1) \\
&= \sum_{i\in I}[x^{n+1}]f_i\cdot(n+1) = \sum_{i\in I}[x^n]f_i',
\end{aligned}$$

where the interchange of summation and multiplication by $(n+1)$ is justified because the sum has finite support. $\square$

Lemma 1.202 (Theorem 1.195 **(c)**)**.** *We have $(cf)' = cf'$ for any $c \in K$ and $f \in K[[x]]$.*

Proof. This is part of [19s-mt3s, Exercise 5 **(b)**] (specifically, it is Statement 3 in [19s-mt3s, solution to Exercise 5 **(b)**]). $\square$

Lemma 1.203. *We have $(C(c)\cdot f)' = C(c)\cdot f'$ for any $c \in K$ and $f \in K[[x]]$, where $C(c)$ denotes the constant power series with value c.*

Proof. This follows from the scalar multiplication rule (Lemma 1.202), since $C(c)\cdot f = c\cdot f$ in the power series ring. $\square$

Lemma 1.204 (Theorem 1.195 **(d)**)**.** *We have* $(fg)' = f'g + fg'$ *for any* $f, g \in K[[x]]$. *(This is known as the* Leibniz rule.*)*

Proof. This is [19s-mt3s, Exercise 5 **(c)**] and [Grinbe17, Proposition 0.2 **(c)**]. ☐

Lemma 1.205 (Theorem 1.195 **(e)**)**.** *If* $f, g \in K[[x]]$ *are two FPSs such that* g *is invertible, then*

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

(This is known as the quotient rule.*)*

Proof. Let $f, g \in K[[x]]$ be two FPSs such that g is invertible. Then, Theorem 1.195 **(d)** (applied to $\frac{f}{g}$ instead of f) yields $\left(\frac{f}{g} \cdot g\right)' = \left(\frac{f}{g}\right)' \cdot g + \frac{f}{g} \cdot g'$. In view of $\frac{f}{g} \cdot g = f$, this rewrites as $f' = \left(\frac{f}{g}\right)' \cdot g + \frac{f}{g} \cdot g'$. Solving this for $\left(\frac{f}{g}\right)'$, we find $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$. ☐

Lemma 1.206 (Theorem 1.195 **(f)**)**.** *If* $g \in K[[x]]$ *is an FPS, then* $(g^n)' = ng'g^{n-1}$ *for any* $n \in \mathbb{N}$ *(where the expression* $ng'g^{n-1}$ *is to be understood as* 0 *if* $n = 0$*).*

Proof. This follows by induction on n, using part **(d)** (in the induction step) and $1' = 0$ (in the induction base). ☐

Lemma 1.207 (Theorem 1.195 **(g)**)**.** *Given two FPSs* $f, g \in K[[x]]$, *we have*

$$(f \circ g)' = (f' \circ g) \cdot g'$$

if f *is a polynomial or if* $[x^0]\, g = 0$. *(This is known as the* chain rule.*)*

Proof. Let $f, g \in K[[x]]$ be two FPSs such that f is a polynomial or $[x^0]\, g = 0$. Write the FPS f in the form $f = \sum_{n \in \mathbb{N}} f_n x^n$ with $f_0, f_1, f_2, \ldots \in K$. Then, $f[g] = \sum_{n \in \mathbb{N}} f_n g^n$. Hence,

$$\begin{aligned}
(f[g])' &= \left(\sum_{n \in \mathbb{N}} f_n g^n\right)' = \sum_{n \in \mathbb{N}} \underbrace{(f_n g^n)'}_{\substack{= f_n (g^n)' \\ \text{(by part (c))}}} \qquad \text{(by part (b))} \\
&= \sum_{n \in \mathbb{N}} f_n (g^n)' = f_0 \underbrace{(g^0)'}_{= 1' = 0} + \sum_{n > 0} f_n \underbrace{(g^n)'}_{\substack{= ng'g^{n-1} \\ \text{(by part (f))}}} = \sum_{n > 0} f_n n g' g^{n-1}. \qquad (1.5)
\end{aligned}$$

On the other hand, from $f = \sum_{n \in \mathbb{N}} f_n x^n$, we obtain $f' = \sum_{n > 0} n f_n x^{n-1} = \sum_{m \in \mathbb{N}} (m+1) f_{m+1} x^m$ (here, we have substituted $m + 1$ for n in the sum). Hence,

$$f'[g] = \sum_{m \in \mathbb{N}} (m+1) f_{m+1} g^m = \sum_{n > 0} n f_n g^{n-1}$$

(here, we have substituted $n - 1$ for m in the sum). Multiplying both sides of this equality by g', we find

$$f'[g] \cdot g' = \left(\sum_{n > 0} n f_n g^{n-1}\right) \cdot g' = \sum_{n > 0} n f_n g^{n-1} g'.$$

Comparing this with (1.5), we find $(f[g])' = f'[g] \cdot g'$. In other words, $(f \circ g)' = (f' \circ g) \cdot g'$ (since $f \circ g$ is a synonym for $f[g]$). ☐

Lemma 1.208 (Theorem 1.195 **(h)**)**.** *If K is a $\mathbb{Q}$-algebra, and if two FPSs $f, g \in K[[x]]$ satisfy $f' = g'$, then $f - g$ is constant.*

Proof. The proof can be found in [Grinbe17, Lemma 0.3]. Let $n \geq 1$. From $f' = g'$, comparing coefficients at index $n-1$ yields $n \cdot [x^n]f = n \cdot [x^n]g$. Since K is a $\mathbb{Q}$-algebra (hence torsion-free), we can divide by n to conclude $[x^n]f = [x^n]g$, i.e., $[x^n](f-g) = 0$.

Note that the condition that K be a $\mathbb{Q}$-algebra is crucial, since it allows dividing by positive integers. (For example, if K could be $\mathbb{Z}/2$, then we would easily get a counterexample, e.g., by taking $f = x^2$ and $g = 0$.) $\square$

Lemma 1.209. *If K is a $\mathbb{Q}$-algebra, and if two FPSs $f, g \in K[[x]]$ satisfy $f' = g'$ and $[x^0]f = [x^0]g$, then $f = g$.*

Proof. By Lemma 1.208, $f - g$ is constant. Since $[x^0](f-g) = [x^0]f - [x^0]g = 0$, we have $f - g = 0$, so $f = g$. $\square$

1.8 Exponentials and logarithms

Convention 1.210. Throughout this section (unless otherwise stated), we assume that K is not just a commutative ring, but actually a commutative $\mathbb{Q}$-algebra.

1.8.1 Definitions

Definition 1.211. Define three FPS exp, $\overline{\log}$ and $\overline{\exp}$ in $K[[x]]$ by

$$\exp := \sum_{n\in\mathbb{N}} \frac{1}{n!}x^n,$$

$$\overline{\log} := \sum_{n\geq 1} \frac{(-1)^{n-1}}{n}x^n,$$

$$\overline{\exp} := \exp - 1 = \sum_{n\geq 1} \frac{1}{n!}x^n.$$

(The last equality sign here follows from $\exp = \sum_{n\in\mathbb{N}} \frac{1}{n!}x^n = \underbrace{\frac{1}{0!}}_{=1} \underbrace{x^0}_{=1} + \sum_{n\geq 1} \frac{1}{n!}x^n = 1 + \sum_{n\geq 1} \frac{1}{n!}x^n$.)

1.8.2 Helpers for derivative computations

Lemma 1.212 (The series $(1+x)^{-1}$)**.** *Define the FPS*

$$\iota := \sum_{n\in\mathbb{N}} (-1)^n x^n.$$

This is the geometric series for $(1+x)^{-1}$; indeed, $\iota \cdot (1+x) = 1$.

Proof. Direct verification: the product is computed coefficient-wise and shown to equal 1. $\square$

Lemma 1.213 (Derivative of $\overline{\log}$)**.** *We have $\overline{\log}' = \iota = (1+x)^{-1}$.*

Proof. By computing the derivative of $\overline{\log} = \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n$ term-by-term: $\overline{\log}' = \sum_{n\geq 1} (-1)^{n-1} x^{n-1} = \sum_{n\geq 0} (-1)^n x^n = \iota$. $\square$

Lemma 1.214 (Derivative of $\overline{\exp}$)**.** *We have* $\overline{\exp}' = \exp$.

Proof. Since $\overline{\exp} = \exp - 1$, we have $\overline{\exp}' = \exp' - 0 = \exp' = \exp$, where the last equality is the standard fact $\exp' = \exp$. $\square$

Lemma 1.215 (Substitution of $(1+x)^{-1}$)**.** *Let* $g \in K[\![x]\!]$ *with* $[x^0]g = 0$. *Then* $\iota \circ g = (1+g)^{-1}$.

Proof. We have $(\iota \circ g) \cdot (1+g) = (\iota \cdot (1+x)) \circ g = 1 \circ g = 1$. Since $[x^0](1+g) \neq 0$, this gives $\iota \circ g = (1+g)^{-1}$. $\square$

1.8.3 The exponential and the logarithm are inverse

Proposition 1.216. *Let* $g \in K[[x]]$ *with* $\left[x^0\right] g = 0$. *Then:*

- ***(a)*** *We have*
$$(\overline{\exp} \circ g)' = (\exp \circ g)' = (\exp \circ g) \cdot g'.$$
- ***(b)*** *We have*
$$\left(\overline{\log} \circ g\right)' = (1+g)^{-1} \cdot g'.$$

Proof. **(a)** Let us first show that $\overline{\exp}' = \exp' = \exp$. Indeed, $\overline{\exp} = \exp - 1$, so that $\exp = \overline{\exp} + 1$ and therefore

$$\exp' = \left(\overline{\exp} + 1\right)' = \overline{\exp}' + \underbrace{1'}_{=0} = \overline{\exp}'. \tag{1.6}$$

Next, we recall that $\exp = \sum_{n\in\mathbb{N}} \dfrac{1}{n!} x^n$. Hence, the definition of a derivative yields

$$\begin{aligned}\exp' &= \sum_{n\geq 1} \underbrace{n \cdot \frac{1}{n!}}_{\substack{=\dfrac{1}{(n-1)!} \\ (\text{since } n! = n\cdot(n-1)!)}} x^{n-1} = \sum_{n\geq 1} \frac{1}{(n-1)!} x^{n-1} = \sum_{n\in\mathbb{N}} \frac{1}{n!} x^n \\ &\qquad\qquad (\text{here, we have substituted } n \text{ for } n-1 \text{ in the sum}) \\ &= \exp. \end{aligned} \tag{1.7}$$

Comparing this with (1.6), we find
$$\overline{\exp}' = \exp. \tag{1.8}$$

Now, we can apply the chain rule to $f = \overline{\exp}$ (since $\left[x^0\right] g = 0$), and thus obtain

$$(\overline{\exp} \circ g)' = \left(\underbrace{\overline{\exp}'}_{=\exp} \circ g\right) \cdot g' = (\exp \circ g) \cdot g'.$$

The same computation (but with $\overline{\exp}$ replaced by $\exp$) yields $(\exp \circ g)' = (\exp \circ g) \cdot g'$. Combining these two formulas, we obtain $(\overline{\exp} \circ g)' = (\exp \circ g)' = (\exp \circ g) \cdot g'$. This proves part **(a)**.

(b) We have $\overline{\log} = \sum_{n\geq 1} \dfrac{(-1)^{n-1}}{n} x^n$. Thus,

$$\overline{\log}' = \left(\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n \right)' = \sum_{n\geq 1} \underbrace{\frac{(-1)^{n-1}}{n} n}_{=(-1)^{n-1}} \underbrace{x'}_{=1} x^{n-1} = \sum_{n\geq 1} (-1)^{n-1} x^{n-1} = \sum_{n\in\mathbb{N}} (-1)^n x^n$$

(here, we have substituted n for $n-1$ in the sum). On the other hand, $(1+x)^{-1} = \sum_{n\in\mathbb{N}} (-1)^n x^n$. Comparing these two equalities, we find

$$\overline{\log}' = (1+x)^{-1}. \tag{1.9}$$

Now, we can apply the chain rule to $f = \overline{\log}$ (since $[x^0]\, g = 0$), and thus obtain

$$\left(\overline{\log} \circ g\right)' = \left(\underbrace{\overline{\log}'}_{=(1+x)^{-1}} \circ g \right) \cdot g' = \left((1+x)^{-1} \circ g \right) \cdot g'. \tag{1.10}$$

However, we claim that $(1+x)^{-1} \circ g = (1+g)^{-1}$. Indeed, $\dfrac{1}{1+x} \circ g = \dfrac{1 \circ g}{(1+x) \circ g}$ (since the FPS $1+x$ is invertible), and $1 \circ g = \underline{1}$ and $(1+x) \circ g = 1+g$, so $(1+x)^{-1} \circ g = (1+g)^{-1}$. Hence, (1.10) becomes

$$\left(\overline{\log} \circ g\right)' = (1+g)^{-1} \cdot g'.$$

This proves part **(b)**. ☐

1.8.4 Helpers for Theorem 1.221

Lemma 1.217 (ODE uniqueness: $h' = (1+h) \cdot g$)**.** *Let $h_1, h_2, g \in K[\![x]\!]$. If $h_1' = (1+h_1) \cdot g$ and $h_2' = (1+h_2) \cdot g$ and $[x^0]h_1 = [x^0]h_2$, then $h_1 = h_2$.*

Proof. By strong induction on the coefficient index. The base case uses the matching constant terms. For the inductive step, the ODE relates $[x^{n+1}]h \cdot (n+1)$ to a convolution involving only lower coefficients; by the induction hypothesis, these coincide for h_1 and h_2, and one cancels $n+1$ using the $\mathbb{Q}$-algebra structure. ☐

Lemma 1.218 (ODE uniqueness: $h' = 1$)**.** *Let $h_1, h_2 \in K[\![x]\!]$. If $h_1' = 1$ and $h_2' = 1$ and $[x^0]h_1 = [x^0]h_2$, then $h_1 = h_2$.*

Proof. As in Lemma 1.217, by strong induction on the coefficient index, cancelling $n+1$ at each step. ☐

Lemma 1.219 ($\iota \circ \overline{\exp}$ times exp)**.** *We have $(\iota \circ \overline{\exp}) \cdot \exp = 1$.*

Proof. We substitute $\overline{\exp}$ into the identity $\iota \cdot (1+x) = 1$, noting that $(1+x) \circ \overline{\exp} = 1 + \overline{\exp} = \exp$. ☐

Lemma 1.220. *Let $f, g \in K[[x]]$ be two FPSs with $[x^0]\, g = 0$. Then, $[x^0]\,(f \circ g) = [x^0]\, f$.*

Proof. Write f in the form $f = \sum_{n\in\mathbb{N}} f_n x^n$ with $f_0, f_1, f_2, \ldots \in K$. Thus, $f_0 = [x^0]\, f$. Now, $f[g] = \sum_{n\in\mathbb{N}} f_n g^n$. However, $[x^0] \left(\sum_{n\in\mathbb{N}} f_n g^n\right) = f_0 = [x^0]\, f$. In view of $f \circ g = f[g] = \sum_{n\in\mathbb{N}} f_n g^n$, we can rewrite this as $[x^0]\,(f \circ g) = [x^0]\, f$. ☐

Theorem 1.221. *We have*

$$\overline{\exp} \circ \overline{\log} = x \qquad \text{and} \qquad \overline{\log} \circ \overline{\exp} = x.$$

Proof. Let us first prove that $\overline{\log} \circ \overline{\exp} = x$.

The idea of this proof is to show that $\overline{\log} \circ \overline{\exp}$ and x are two FPSs with the same constant term (namely, 0) and with the same derivative. Once this is proved, they must be equal (since a FPS is determined by its constant term and its derivative over a $\mathbb{Q}$-algebra).

We have $[x^0] \overline{\exp} = 0$ (since $\overline{\exp} = \sum_{n \geq 1} \frac{1}{n!} x^n$). Hence, Lemma 1.220 (applied to $f = \overline{\log}$ and $g = \overline{\exp}$) yields $[x^0] (\overline{\log} \circ \overline{\exp}) = [x^0] \overline{\log} = 0$ (since $\overline{\log} = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} x^n$). Now,

$$[x^0] (\overline{\log} \circ \overline{\exp} - x) = \underbrace{[x^0] (\overline{\log} \circ \overline{\exp})}_{=0} - \underbrace{[x^0]\, x}_{=0} = 0. \tag{1.11}$$

However, $\overline{\exp} = \exp - 1$ and thus $1 + \overline{\exp} = \exp$. Now, Proposition 1.216 **(b)** (applied to $g = \overline{\exp}$) yields

$$(\overline{\log} \circ \overline{\exp})' = \Big(\underbrace{1 + \overline{\exp}}_{=\exp}\Big)^{-1} \cdot \underbrace{\overline{\exp}'}_{=\exp} = \exp^{-1} \cdot \exp = 1 = x'$$

(where the underbrace uses (1.8), and $x' = 1$). Hence, $\overline{\log} \circ \overline{\exp} - x$ is constant (since its derivative is 0). In view of (1.11), this constant must be 0. Thus, $\overline{\log} \circ \overline{\exp} = x$.

Now it remains to prove that $\overline{\exp} \circ \overline{\log} = x$.

We shall first show that $\exp \circ \overline{\log} = 1 + x$.

To wit: The FPS $1+x$ is invertible. Applying the quotient rule to $f = \exp \circ \overline{\log}$ and $g = 1+x$, we obtain

$$\left(\frac{\exp \circ \overline{\log}}{1+x}\right)' = \frac{(\exp \circ \overline{\log})' \cdot (1+x) - (\exp \circ \overline{\log}) \cdot (1+x)'}{(1+x)^2}.$$

In view of

$$(\exp \circ \overline{\log})' = (\exp \circ \overline{\log}) \cdot \underbrace{\overline{\log}'}_{=(1+x)^{-1}} = (\exp \circ \overline{\log}) \cdot (1+x)^{-1}$$

(where we used (1.9)) and $(1+x)' = 1$, we can rewrite this as

$$\left(\frac{\exp \circ \overline{\log}}{1+x}\right)' = \frac{(\exp \circ \overline{\log}) - (\exp \circ \overline{\log})}{(1+x)^2} = 0 = 0'.$$

Thus, $\frac{\exp \circ \overline{\log}}{1+x}$ is constant, say equal to $\underline{a}$ for some $a \in K$. Then $\exp \circ \overline{\log} = a(1+x)$, so $[x^0] (\exp \circ \overline{\log}) = a$. However, it is easy to see that $[x^0] (\exp \circ \overline{\log}) = 1$ (using Lemma 1.220). Comparing, we find $a = 1$. Thus, $\exp \circ \overline{\log} = 1 + x$.

Now, $\overline{\exp} = \exp - 1 = \exp + \underline{-1}$. Hence,

$$\overline{\exp} \circ \overline{\log} = \underbrace{\exp \circ \overline{\log}}_{=1+x} + \underbrace{\underline{-1} \circ \overline{\log}}_{=\underline{-1}} = 1 + x + \underline{-1} = x.$$

This completes the proof. $\square$

1.8.5 The exponential and the logarithm of an FPS

Definition 1.222. **(a)** We let $K[[x]]_0$ denote the set of all FPSs $f \in K[[x]]$ with $[x^0] f = 0$.

(b) We let $K[[x]]_1$ denote the set of all FPSs $f \in K[[x]]$ with $[x^0] f = 1$.

(c) We define two maps

$$\begin{aligned} \operatorname{Exp} : K[[x]]_0 &\to K[[x]]_1, \\ g &\mapsto \exp \circ g \end{aligned}$$

and

$$\begin{aligned} \operatorname{Log} : K[[x]]_1 &\to K[[x]]_0, \\ f &\mapsto \overline{\log} \circ (f-1). \end{aligned}$$

(These two maps are well-defined according to parts **(c)** and **(d)** of Lemma 1.223 below.)

Lemma 1.223. *(a) For any $f, g \in K[[x]]_0$, we have $f \circ g \in K[[x]]_0$.*

(b) For any $f \in K[[x]]_1$ and $g \in K[[x]]_0$, we have $f \circ g \in K[[x]]_1$.

(c) For any $g \in K[[x]]_0$, we have $\exp \circ g \in K[[x]]_1$.

(d) For any $f \in K[[x]]_1$, we have $f - 1 \in K[[x]]_0$ and $\overline{\log} \circ (f-1) \in K[[x]]_0$.

Proof. **(a)** Let $f, g \in K[[x]]_0$. Then $[x^0] f = 0$ and $[x^0] g = 0$. Hence, Lemma 1.220 yields $[x^0](f \circ g) = [x^0] f = 0$. In other words, $f \circ g \in K[[x]]_0$.

(b) This is analogous to the proof of part **(a)**.

(c) Let $g \in K[[x]]_0$. From $\exp = \sum_{n \in \mathbb{N}} \frac{1}{n!} x^n$, we obtain $[x^0] \exp = 1$, so that $\exp \in K[[x]]_1$. Hence, part **(b)** (applied to $f = \exp$) yields $\exp \circ g \in K[[x]]_1$.

(d) Let $f \in K[[x]]_1$. Thus, $[x^0] f = 1$. Now, $[x^0](f-1) = [x^0] f - [x^0] 1 = 1 - 1 = 0$, so that $f - 1 \in K[[x]]_0$. Furthermore, $[x^0] \overline{\log} = 0$ and thus $\overline{\log} \in K[[x]]_0$. Hence, part **(a)** (applied to $\overline{\log}$ and $f-1$ instead of f and g) yields $\overline{\log} \circ (f-1) \in K[[x]]_0$. $\square$

Lemma 1.224. *The maps* Exp *and* Log *are mutually inverse bijections between $K[[x]]_0$ and $K[[x]]_1$.*

Proof. First, we make a simple auxiliary observation: Each $g \in K[[x]]_0$ satisfies

$$\exp \circ g = \overline{\exp} \circ g + 1. \tag{1.12}$$

[*Proof of (1.12):* Recall that $\overline{\exp} = \exp - 1$, so that $\exp = \overline{\exp} + \underline{1}$. Hence, $\exp \circ g = (\overline{\exp} + \underline{1}) \circ g = \overline{\exp} \circ g + \underline{1} \circ g$. But $\underline{1} \circ g = \underline{1} = 1$. Hence, $\exp \circ g = \overline{\exp} \circ g + 1$.]

Now, let us show that $\operatorname{Exp} \circ \operatorname{Log} = \operatorname{id}$. Indeed, we fix some $f \in K[[x]]_1$. Then, $f - 1 \in K[[x]]_0$ (by Lemma 1.223 **(d)**). By associativity of composition and Theorem 1.221, we get

$$\overline{\exp} \circ \left(\overline{\log} \circ (f-1)\right) = \underbrace{\left(\overline{\exp} \circ \overline{\log}\right)}_{=x} \circ (f-1) = x \circ (f-1) = f - 1.$$

Hence

$$\begin{aligned} (\operatorname{Exp} \circ \operatorname{Log})(f) &= \exp \circ (\operatorname{Log} f) \\ &= \overline{\exp} \circ \underbrace{(\operatorname{Log} f)}_{= \overline{\log} \circ (f-1)} + 1 \end{aligned}$$

(by (1.12)), so

$$(\operatorname{Exp} \circ \operatorname{Log})(f) = \underbrace{\overline{\exp} \circ \left(\overline{\log} \circ (f-1)\right)}_{=f-1} + 1 = (f-1) + 1 = f.$$

Using a similar argument (with $\overline{\log} \circ \overline{\exp} = x$ from Theorem 1.221), we can show that $\operatorname{Log} \circ \operatorname{Exp} = \operatorname{id}$. Combining, Exp and Log are mutually inverse bijections. □

1.8.6 Helpers for the group structure

Lemma 1.225 ($f - 1 \in K[\![x]\!]_0$ when $f \in K[\![x]\!]_1$)**.** *If $f \in K[\![x]\!]_1$ (i.e., $[x^0]f = 1$), then $f - 1 \in K[\![x]\!]_0$ (i.e., $[x^0](f-1) = 0$).*

Proof. $[x^0](f-1) = [x^0]f - 1 = 1 - 1 = 0$. □

Lemma 1.226 (ODE uniqueness: $h' = h \cdot g$)**.** *Let $h_1, h_2, g \in K[\![x]\!]$. If $h_1' = h_1 \cdot g$ and $h_2' = h_2 \cdot g$ and $[x^0]h_1 = [x^0]h_2$, then $h_1 = h_2$.*

Proof. By strong induction on the coefficient index. The ODE gives $[x^{n+1}]h \cdot (n+1) = [x^n](h \cdot g)$; the right side is a convolution involving only coefficients of h of index $\leq n$, which agree by induction hypothesis. Cancelling $n+1$ uses the $\mathbb{Q}$-algebra structure. □

Lemma 1.227 ($\operatorname{Exp}(0) = 1$ and $\operatorname{Log}(1) = 0$)**.** *The map* Exp *sends* $0 \in K[\![x]\!]_0$ *to* $1 \in K[\![x]\!]_1$*, and* Log *sends* 1 *to* 0*.*

Proof. $\operatorname{Exp}(0) = \overline{\exp} \circ 0 = 1$ (since $[x^0]\overline{\exp} = 1$ and substituting 0 picks out the constant term). $\operatorname{Log}(1) = \overline{\log} \circ (1-1) = \overline{\log} \circ 0 = 0$ (since $[x^0]\overline{\log} = 0$). □

1.8.7 Addition to multiplication

Lemma 1.228. ***(a)*** *For any $f, g \in K[[x]]_0$, we have*

$$\operatorname{Exp}(f+g) = \operatorname{Exp} f \cdot \operatorname{Exp} g.$$

(b) *For any $f, g \in K[[x]]_1$, we have*

$$\operatorname{Log}(fg) = \operatorname{Log} f + \operatorname{Log} g.$$

Proof. **(a)** Let $f, g \in K[[x]]_0$. Thus, $\left[x^0\right] f = 0$ and $\left[x^0\right] g = 0$. By the definition of Exp, we have

$$\operatorname{Exp} f = \overline{\exp} \circ f = \sum_{n \in \mathbb{N}} \frac{1}{n!} f^n.$$

Similarly, $\operatorname{Exp} g = \sum_{n\in\mathbb{N}} \frac{1}{n!} g^n$ and $\operatorname{Exp}(f+g) = \sum_{n\in\mathbb{N}} \frac{1}{n!}(f+g)^n$.

Now, the latter equality becomes

$$\operatorname{Exp}(f+g) = \sum_{n\in\mathbb{N}} \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} f^k g^{n-k} = \sum_{k\in\mathbb{N}} \sum_{\ell\in\mathbb{N}} \frac{1}{k!\ell!} f^k g^\ell.$$

Comparing this with $\operatorname{Exp} f \cdot \operatorname{Exp} g = \sum_{k\in\mathbb{N}} \sum_{\ell\in\mathbb{N}} \frac{1}{k!\ell!} f^k g^\ell$, we obtain $\operatorname{Exp}(f+g) = \operatorname{Exp} f \cdot \operatorname{Exp} g$.

(In the Lean formalization, this is proved via ODE uniqueness: both sides satisfy the ODE $h' = h \cdot (f' + g')$ with the same initial condition.)

(b) This follows from part **(a)**, since we know that Log is inverse to Exp. Setting $u = \operatorname{Log} f$ and $v = \operatorname{Log} g$, part **(a)** gives $\operatorname{Exp}(u+v) = \operatorname{Exp} u \cdot \operatorname{Exp} v$, and applying Log (using Lemma 1.224) yields $\operatorname{Log}(fg) = \operatorname{Log} f + \operatorname{Log} g$. ☐

Proposition 1.229. ***(a)*** *The subset $K[[x]]_0$ of $K[[x]]$ is closed under addition and subtraction and contains 0, and thus forms a group $(K[[x]]_0, +, 0)$.*

(b) *The subset $K[[x]]_1$ of $K[[x]]$ is closed under multiplication and division and contains 1, and thus forms a group $(K[[x]]_1, \cdot, 1)$.*

Proof. **(a)** It is clear that the set $K[[x]]_0$ contains the FPS 0 (since $[x^0]\, 0 = 0$). If $f, g \in K[[x]]_0$, then $[x^0]\, f = 0$ and $[x^0]\, g = 0$, and therefore $f + g \in K[[x]]_0$ (since $[x^0]\,(f+g) = 0 + 0 = 0$) and $f - g \in K[[x]]_0$ (by a similar argument).

(b) Any $a \in K[[x]]_1$ is invertible in $K[[x]]$ (since $[x^0]\, a = 1$ is invertible in K). Next, $K[[x]]_1$ is closed under multiplication: if $f, g \in K[[x]]_1$, then $[x^0]\, f = 1$ and $[x^0]\, g = 1$, and therefore $[x^0]\,(fg) = 1 \cdot 1 = 1$. Similarly, $K[[x]]_1$ is closed under division. ☐

Theorem 1.230. *The maps*

$$\operatorname{Exp} : (K[[x]]_0, +, 0) \to (K[[x]]_1, \cdot, 1)$$

and

$$\operatorname{Log} : (K[[x]]_1, \cdot, 1) \to (K[[x]]_0, +, 0)$$

are mutually inverse group isomorphisms.

Proof. Lemma 1.228 yields that these two maps are group homomorphisms. Lemma 1.224 shows that they are mutually inverse. Combining these results, we conclude that these two maps are mutually inverse group isomorphisms. ☐

1.8.8 Helpers for the logarithmic derivative

Lemma 1.231 ($\iota = (1+x)^{-1}$)**.** *Over a field (or $\mathbb{Q}$-algebra), ι equals the formal inverse $(1+x)^{-1}$ in $K[\![x]\!]$.*

Proof. Both sides multiply with $(1+x)$ to give 1; since the constant coefficient of $1+x$ is nonzero, the inverse is unique. ☐

Lemma 1.232 (Substitution of inverses)**.** *If $a \cdot b = 1$ in $K[\![x]\!]$ and g has constant term 0, then $(a \circ g)^{-1} = b \circ g$.*

Proof. Substitution is a ring homomorphism, so $(a \circ g) \cdot (b \circ g) = (a \cdot b) \circ g = 1 \circ g = 1$. Since $[x^0](a \circ g) \neq 0$, the inverse is unique. ☐

1.8.9 The logarithmic derivative

Definition 1.233. In this definition, we do **not** use Convention 1.210; thus, K can be an arbitrary commutative ring. However, we set $K[[x]]_1 = \{f \in K[[x]] \mid [x^0]\, f = 1\}$.

For any FPS $f \in K[[x]]_1$, we define the *logarithmic derivative* $\operatorname{loder} f \in K[[x]]$ to be the FPS $\dfrac{f'}{f}$. (This is well-defined, since f is easily seen to be invertible.)

Lemma 1.234 (Invertibility of FPS with constant term 1)**.** *If $[x^0]f = 1$, then f is a unit in $K[\![x]\!]$, i.e., $f \cdot f^{-1} = 1$ and $f^{-1} \cdot f = 1$. Moreover, $[x^0](f^{-1}) = 1$. More generally, f is a unit iff $[x^0]f$ is a unit in K.*

Proof. Since $[x^0]f$ is a unit, the invertibility criterion for power series gives the result (a power series is invertible if and only if its constant coefficient is invertible). The constant coefficient of f^{-1} is the inverse of $[x^0]f = 1$. $\square$

Lemma 1.235 ($\operatorname{loder}(1) = 0$)**.** $\operatorname{loder}(1) = 0$.

Proof. $1' = 0$, so $\operatorname{loder}(1) = 0 \cdot 1^{-1} = 0$. $\square$

Proposition 1.236. *Let K be a commutative $\mathbb{Q}$-algebra. Let $f \in K[[x]]_1$ be an FPS. Then,* $\operatorname{loder} f = (\operatorname{Log} f)'$.

Proof. The definition of Log yields $\operatorname{Log} f = \overline{\log} \circ (f-1)$.

From $f \in K[[x]]_1$, we obtain $\left[x^0\right] f = 1$. Now, $\left[x^0\right](f-1) = \left[x^0\right] f - \left[x^0\right] 1 = 0$. Hence, Proposition 1.216 **(b)** (applied to $g = f - 1$) yields

$$\left(\overline{\log} \circ (f-1)\right)' = \Bigg(\underbrace{1 + (f-1)}_{=f}\Bigg)^{-1} \cdot (f-1)' = f^{-1} \cdot (f-1)' = \frac{(f-1)'}{f}.$$

In view of $\operatorname{Log} f = \overline{\log} \circ (f-1)$, we can rewrite this as $(\operatorname{Log} f)' = \dfrac{(f-1)'}{f}$.

Since $(f-1)' = f'$, we can rewrite this further as $(\operatorname{Log} f)' = \dfrac{f'}{f} = \operatorname{loder} f$. $\square$

Proposition 1.237. *Let $f, g \in K[[x]]_1$ be two FPSs. Then,* $\operatorname{loder}(fg) = \operatorname{loder} f + \operatorname{loder} g$.

*(Here, we do **not** use Convention 1.210; thus, K can be an arbitrary commutative ring.)*

Proof. The definition of a logarithmic derivative yields $\operatorname{loder} f = \dfrac{f'}{f}$ and $\operatorname{loder} g = \dfrac{g'}{g}$, but it also yields

$$\operatorname{loder}(fg) = \frac{(fg)'}{fg} = \frac{f'g + fg'}{fg} \qquad \text{(by the product rule)} = \underbrace{\frac{f'}{f}}_{=\operatorname{loder} f} + \underbrace{\frac{g'}{g}}_{=\operatorname{loder} g} = \operatorname{loder} f + \operatorname{loder} g.$$

$\square$

Corollary 1.238. *Let $f_1, f_2, \ldots, f_k$ be any k FPSs in $K[[x]]_1$. Then,*

$$\operatorname{loder}(f_1 f_2 \cdots f_k) = \operatorname{loder}(f_1) + \operatorname{loder}(f_2) + \cdots + \operatorname{loder}(f_k).$$

*(Here, we do **not** use Convention 1.210; thus, K can be an arbitrary commutative ring.)*

Proof. Induct on k. The *base case* ($k = 0$) requires showing that $\operatorname{loder} 1 = 0$, which is easy to see from the definition of a logarithmic derivative (since $1' = 0$). The *induction step* follows easily from Proposition 1.237. $\square$

Corollary 1.239. *Let f be any FPS in $K[[x]]_1$. Then,* $\operatorname{loder}\left(f^{-1}\right) = -\operatorname{loder} f$.

*(Here, we do **not** use Convention 1.210; thus, K can be an arbitrary commutative ring.)*

Proof. First of all, f is invertible. This shows that f^{-1} is well-defined.

Proposition 1.237 (applied to $g = f^{-1}$) yields $\operatorname{loder}\left(ff^{-1}\right) = \operatorname{loder} f + \operatorname{loder}\left(f^{-1}\right)$. However, we also have $\operatorname{loder}\left(\underbrace{ff^{-1}}_{=1}\right) = \operatorname{loder} 1 = 0$ (since $1' = 0$). Comparing these two equalities, we obtain $\operatorname{loder} f + \operatorname{loder}\left(f^{-1}\right) = 0$. In other words, $\operatorname{loder}\left(f^{-1}\right) = -\operatorname{loder} f$. $\square$

1.8.10 Summable and multipliable families

Definition 1.240 (Summable and multipliable families)**.** A family $(f_i)_{i\in I}$ of FPSs in $K[\![x]\!]_0$ is *summable* if for each coefficient index n, only finitely many $[x^n](f_i)$ are nonzero.

A family $(f_i)_{i\in I}$ of FPSs in $K[\![x]\!]_1$ is *multipliable* if for each coefficient index n, all but finitely many f_i satisfy $[x^k](f_i - 1) = 0$ for all $1 \le k \le n$.

For summable families, the coefficient-wise sum $\sum_{i\in I} f_i$ belongs to $K[\![x]\!]_0$. For multipliable families, the product $\prod_{i\in I} f_i$ belongs to $K[\![x]\!]_1$.

Proof. For the sum: the constant coefficient of each f_i is 0, so the sum of constant coefficients is 0. For the product: the constant coefficient of $\prod f_i$ is $\prod [x^0](f_i) = \prod 1 = 1$. $\square$

1.8.11 Exp and Log for infinite products and sums

Lemma 1.241 (Summability of Log-images)**.** *If $(f_i)_{i\in I}$ is a multipliable family in $K[\![x]\!]_1$, then $(\operatorname{Log} f_i)_{i\in I}$ is a summable family in $K[\![x]\!]_0$ (meaning that for each coefficient index n, only finitely many $[\operatorname{Log} f_i]_n$ are nonzero).*

Proof. For each n, let M be an x^n-approximator for the product family. If $i \notin M$, then $[x^k](f_i) = 0$ for $1 \le k \le n$ (by the approximator property). This forces $[(f_i - 1)]_k = 0$ for $0 \le k \le n$, and a substitution argument shows $[\operatorname{Log} f_i]_n = 0$. Hence the set of i with nonzero n-th coefficient is contained in M. $\square$

Proposition 1.242 (Log of infinite product)**.** *If $(f_i)_{i\in I}$ is a multipliable family in $K[\![x]\!]_1$, then*

$$\operatorname{Log}\Big(\prod_{i\in I} f_i\Big) = \sum_{i\in I} \operatorname{Log}(f_i),$$

where the right side is the coefficient-wise sum (which is well-defined by Lemma 1.241).

Proof. For each n, choose an x^n-approximator M. By the finite version (Log of a finite product equals the sum of Logs, which follows from Lemma 1.228(b) by induction), we have $\operatorname{Log}(\prod_{i\in M} f_i) = \sum_{i\in M} \operatorname{Log}(f_i)$. The n-th coefficient of the infinite product agrees with that of the finite product over M, and the n-th coefficient of the sum is the finite sum over M (since $[\operatorname{Log} f_i]_n = 0$ for $i \notin M$). A coefficient comparison argument completes the proof. $\square$

Lemma 1.243 (Multipliability of Exp-images)**.** *If $(g_i)_{i\in I}$ is a summable family in $K[\![x]\!]_0$, then $(\operatorname{Exp} g_i)_{i\in I}$ is a multipliable family in $K[\![x]\!]_1$.*

Proof. Since $\operatorname{Exp}(g_i) = 1 + \overline{\exp} \circ g_i$, the family is of the form $(1 + h_i)$ where $h_i = \overline{\exp} \circ g_i$. The key observation is that if $[x^k](g_i) = 0$ for all $k \le n$, then $[x^n](\overline{\exp} \circ g_i) = 0$ as well (since $\overline{\exp}$ has constant term 0, so the substitution preserves vanishing of low-order terms). The summability of (g_i) then implies the multipliability of $(1 + h_i)$. $\square$

Proposition 1.244 (Exp of infinite sum)**.** *If $(g_i)_{i\in I}$ is a summable family in $K[\![x]\!]_0$, then*

$$\mathrm{Exp}\Big(\sum_{i\in I} g_i\Big) = \prod_{i\in I} \mathrm{Exp}(g_i).$$

Proof. Let $f_i = \mathrm{Exp}(g_i)$. By Proposition 1.401, $\mathrm{Log}(\prod f_i) = \sum \mathrm{Log}(f_i) = \sum g_i$ (using $\mathrm{Log} \circ \mathrm{Exp} = \mathrm{id}$). Applying Exp to both sides and using $\mathrm{Exp} \circ \mathrm{Log} = \mathrm{id}$ yields the result. ☐

1.9 Non-integer powers

1.9.1 Definition

Definition 1.245. An FPS $f \in K[[x]]$ has *constant term one* if $[x^0]f = 1$. We write $f \in K[[x]]_1$ for this condition.

Lemma 1.246. *If $f, g \in K[[x]]_1$, then $fg \in K[[x]]_1$.*

Proof. $[x^0](fg) = ([x^0]f)([x^0]g) = 1 \cdot 1 = 1$. ☐

Lemma 1.247. $1 + x \in K[[x]]_1$.

Proof. $[x^0](1 + x) = 1$. ☐

The Log map

Definition 1.248. The *logarithm series* (or $\overline{\log}$) is the FPS

$$\overline{\log} := \sum_{n\geq 1} \frac{(-1)^{n-1}}{n} x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots \in K[[x]].$$

Its constant term is 0.

Lemma 1.249. $[x^0]\overline{\log} = 0$.

Proof. Immediate from the definition. ☐

Definition 1.250. For $f \in K[[x]]_1$, we define

$$\mathrm{Log}(f) := \overline{\log} \circ (f - 1).$$

Since f has constant term 1, the FPS $f-1$ has constant term 0, so the substitution is well-defined.

Lemma 1.251. $\mathrm{Log}(1) = 0$.

Proof. $\mathrm{Log}(1) = \overline{\log} \circ (1 - 1) = \overline{\log} \circ 0 = 0$, since every term in the substitution sum is zero. ☐

Lemma 1.252. *For $f \in K[[x]]_1$, we have $[x^0]\,\mathrm{Log}(f) = 0$.*

Proof. The constant term of a substitution $g \circ h$ where both $[x^0]g = 0$ and $[x^0]h = 0$ is 0. ☐

Theorem 1.253. *For $f, g \in K[[x]]_1$,*

$$\mathrm{Log}(fg) = \mathrm{Log}(f) + \mathrm{Log}(g).$$

Proof. Let $h = \mathrm{Log}(fg) - \mathrm{Log}(f) - \mathrm{Log}(g)$. We show $h' = 0$ and $[x^0]h = 0$ using the derivative characterization: $(d/dx)\,\mathrm{Log}(f) = (d/dx\overline{\log}) \circ (f - 1) \cdot f'$, and the key identity $(d/dx\,\overline{\log}) \circ (f - 1) \cdot f = 1$ (since $d/dx\,\overline{\log} = 1/(1 + x)$ and substituting $f - 1$ gives $1/f$). The Leibniz rule for $(fg)'$ then shows $h' = 0$. Since $[x^0]h = 0 - 0 - 0 = 0$ and $h' = 0$, uniqueness gives $h = 0$. ☐

The Exp map

Definition 1.254. For $g \in K[[x]]_0$ (FPS with constant term 0), we define

$$\operatorname{Exp}(g) := \exp \circ g = \left(\sum_{n \in \mathbb{N}} \frac{x^n}{n!} \right) \circ g.$$

Since g has constant term 0, the substitution is well-defined.

Lemma 1.255. $\operatorname{Exp}(0) = 1$.

Proof. Only the $m = 0$ term in the substitution sum is nonzero, giving 1. □

Lemma 1.256. *For $g \in K[[x]]_0$, we have $\operatorname{Exp}(g) \in K[[x]]_1$.*

Proof. The constant term of $\operatorname{Exp}(g) = \exp \circ g$ is $[x^0] \exp = 1/0! = 1$, since all higher powers of g contribute 0 to the constant term. □

Theorem 1.257. *For $u, v \in K[[x]]_0$,*

$$\operatorname{Exp}(u + v) = \operatorname{Exp}(u) \cdot \operatorname{Exp}(v).$$

Proof. Follows from the multiplicativity of the exponential map, established in Mathlib's Exp/Log theory for formal power series. □

The power definition

Definition 1.258. Assume that K is a commutative $\mathbb{Q}$-algebra. Let $f \in K[[x]]_1$ and $c \in K$. Then, we define an FPS

$$f^c := \operatorname{Exp}(c \operatorname{Log} f) \in K[[x]]_1 .$$

Lemma 1.259. *For any $f \in K[[x]]$, we have $f^0 = 1$.*

Proof. $f^0 = \operatorname{Exp}(0 \cdot \operatorname{Log} f) = \operatorname{Exp}(0) = 1$. □

Lemma 1.260. *For $f \in K[[x]]_1$ and $c \in K$, we have $f^c \in K[[x]]_1$.*

Proof. Since $[x^0] \operatorname{Log} f = 0$, we have $[x^0](c \operatorname{Log} f) = 0$, so $\operatorname{Exp}(c \operatorname{Log} f) \in K[[x]]_1$. □

Lemma 1.261. *For $f \in K[[x]]_1$, we have $f^1 = f$.*

Proof. $f^1 = \operatorname{Exp}(1 \cdot \operatorname{Log} f) = \operatorname{Exp}(\operatorname{Log} f) = f$, using the inverse property $\operatorname{Exp} \circ \operatorname{Log} = \mathrm{id}$ on $K[[x]]_1$. The formal proof bridges the local definitions of Log and Exp to Mathlib's Exp/Log framework (via Lemmas 1.273 and 1.274) and then applies the inverse property. □

1.9.2 Rules of exponents

Theorem 1.262. *Assume that K is a commutative $\mathbb{Q}$-algebra. For any $a, b \in K$ and $f, g \in K[[x]]_1$, we have*

$$f^{a+b} = f^a f^b, \qquad (fg)^a = f^a g^a, \qquad (f^a)^b = f^{ab}.$$

Proof. **First rule ($f^{a+b} = f^a f^b$):**

$$\begin{aligned} f^{a+b} &= \operatorname{Exp}((a+b)\operatorname{Log} f) \\ &= \operatorname{Exp}(a \operatorname{Log} f + b \operatorname{Log} f) \\ &= \operatorname{Exp}(a \operatorname{Log} f) \cdot \operatorname{Exp}(b \operatorname{Log} f) \quad (\text{by } \operatorname{Exp}(x+y) = \operatorname{Exp}(x)\operatorname{Exp}(y)) \\ &= f^a \cdot f^b. \end{aligned}$$

Second rule ($(fg)^a = f^a g^a$):

$$\begin{aligned} (fg)^a &= \operatorname{Exp}(a \operatorname{Log}(fg)) \\ &= \operatorname{Exp}(a(\operatorname{Log} f + \operatorname{Log} g)) \quad (\text{by } \operatorname{Log}(fg) = \operatorname{Log} f + \operatorname{Log} g) \\ &= \operatorname{Exp}(a \operatorname{Log} f + a \operatorname{Log} g) \\ &= \operatorname{Exp}(a \operatorname{Log} f) \cdot \operatorname{Exp}(a \operatorname{Log} g) = f^a \cdot g^a. \end{aligned}$$

Third rule ($(f^a)^b = f^{ab}$):

$$\begin{aligned} (f^a)^b &= \operatorname{Exp}(b \operatorname{Log}(f^a)) \\ &= \operatorname{Exp}(b \operatorname{Log}(\operatorname{Exp}(a \operatorname{Log} f))) \\ &= \operatorname{Exp}(b \cdot a \operatorname{Log} f) \quad (\text{by } \operatorname{Log} \circ \operatorname{Exp} = \operatorname{id}) \\ &= \operatorname{Exp}((ab)\operatorname{Log} f) = f^{ab}. \end{aligned}$$

□

Lemma 1.263. *For $f \in K[[x]]_1$ and $n \in \mathbb{N}$, we have $f^n = \underbrace{f \cdot f \cdots f}_{n \text{ times}}$. That is, our definition of f^c is consistent with the standard power when $c \in \mathbb{N}$.*

Proof. By induction on n: the base case $n = 0$ gives $f^0 = 1$ (by Lemma 1.259), and the inductive step uses $f^{n+1} = f^n \cdot f^1 = f^n \cdot f$ (by the first rule of exponents and Lemma 1.261). □

1.9.3 The Newton binomial formula for arbitrary exponents

Theorem 1.264 (Generalized Newton binomial formula)**.** *Assume that K is a commutative $\mathbb{Q}$-algebra. Let $c \in K$. Then,*

$$(1+x)^c = \sum_{k \in \mathbb{N}} \binom{c}{k} x^k.$$

Proof. The definition of Log yields $\operatorname{Log}(1+x) = \overline{\log} \circ x = \overline{\log}$. By Definition 1.258, we have

$$\begin{aligned} (1+x)^c &= \operatorname{Exp}(c \operatorname{Log}(1+x)) = \operatorname{Exp}(c\,\overline{\log}) \\ &= \exp \circ (c\,\overline{\log}) = \sum_{m \in \mathbb{N}} \frac{1}{m!} \left(\sum_{n \geq 1} \frac{(-1)^{n-1}}{n} c x^n \right)^m . \end{aligned}$$

Expanding and collecting by powers of x, we obtain

$$(1+x)^c = \sum_{k \in \mathbb{N}} \left(\sum_{(n_1, \ldots, n_m) \in \operatorname{Comp}(k)} \frac{1}{m!} \cdot \frac{(-1)^{n_1 + \cdots + n_m - m}}{n_1 n_2 \cdots n_m} c^m \right) x^k,$$

where $\mathrm{Comp}(k)$ denotes the set of all compositions of k.

It remains to show that the coefficient of x^k equals $\binom{c}{k}$. We use the *polynomial identity trick*: for each $k \in \mathbb{N}$, both sides of the coefficient equality

$$\sum_{(n_1,\ldots,n_m)\in\mathrm{Comp}(k)} \frac{1}{m!} \cdot \frac{(-1)^{n_1+\cdots+n_m-m}}{n_1\cdots n_m} c^m = \binom{c}{k}$$

are polynomials in c with rational coefficients. (The left-hand side is a finite sum of "rational number times a power of c" expressions. The right-hand side is $\frac{c(c-1)(c-2)\cdots(c-k+1)}{k!}$.)

For $c = n \in \mathbb{N}$, the equality holds by the standard Newton binomial formula (Theorem 1.119), since both sides equal $\binom{n}{k}$. Since two polynomials with rational coefficients that agree on all natural numbers must be equal (having infinitely many common values), the equality holds for all $c \in K$.

In the formal proof:

- The base case $k = 0$: both sides equal 1.
- The inductive case $k = k' + 1$: both the LHS and the RHS are expressed as evaluations of explicit polynomials over $\mathbb{Q}$, mapped to K. The LHS polynomial is built by expanding $\exp \circ (c\,\overline{\log})$ as $\sum_{d\geq 0} \frac{1}{d!}(c\,\overline{\log})^d$; the scalar factoring $(c\,\overline{\log})^d = c^d\,\overline{\log}^d$ uses Lemma 1.275, and the truncation of the sum at $d = k$ uses the vanishing $[x^k](\overline{\log}^d) = 0$ for $d > k$ (Lemma 1.276). The polynomial identity principle (Lemma 1.278) then shows these polynomials are equal, since they agree on all $n \in \mathbb{N}$.

$\square$

Corollary 1.265. *For any $c \in K$ and $k \in \mathbb{N}$,*

$$[x^k](1+x)^c = \binom{c}{k}.$$

Proof. Immediate from Theorem 1.264 by reading off coefficients. $\square$

Theorem 1.266. *For any $n \in K$,*

$$(1+x)^{-n} = \sum_{i\in\mathbb{N}} (-1)^i \binom{n+i-1}{i} x^i.$$

Proof. Apply Theorem 1.264 with $c = -n$ and use the upper negation identity $\binom{-n}{i} = (-1)^i\binom{n+i-1}{i}$. $\square$

1.9.4 Further rules of exponents

Lemma 1.267. *For any $c \in K$, we have $1^c = 1$.*

Proof. $1^c = \mathrm{Exp}(c \cdot \mathrm{Log}\, 1) = \mathrm{Exp}(c \cdot 0) = \mathrm{Exp}(0) = 1$. $\square$

Lemma 1.268. *For $f \in K[[x]]_1$ and $c \in K$, we have $[x^0](f^c) = 1$.*

Proof. Immediate from $f^c \in K[[x]]_1$. $\square$

Lemma 1.269. *For $f \in K'[[x]]_1$ (over a field K') and $c \in K'$,*

$$f^{-c} = (f^c)^{-1}.$$

Proof. We have $f^{-c} \cdot f^c = f^{-c+c} = f^0 = 1$, so f^{-c} is the inverse of f^c. □

Lemma 1.270. *For integer exponents, f^c agrees with the standard power: for $n \geq 0$, $f^n = f^n$ (standard), and for $n < 0$, $f^n = (f^{|n|})^{-1}$.*

Proof. For $n \geq 0$, use Lemma 1.263. For $n < 0$, combine $f^{-c} \cdot f^c = 1$ with $f^{|n|}$ computed via the natural-number case. □

1.9.5 Another application

Proposition 1.271. *Let $n \in \mathbb{C}$ and $k \in \mathbb{N}$. Then,*

$$\sum_{i=0}^{k} \binom{n+i-1}{i} \binom{n}{k-2i} = \binom{n+k-1}{k}.$$

Proof. Define two FPSs $f, g \in \mathbb{C}[[x]]$ by

$$f = \sum_{i \in \mathbb{N}} \binom{n+i-1}{i} x^{2i}$$

$$\text{and} \qquad g = \sum_{j \in \mathbb{N}} \binom{n}{j} x^j.$$

Multiplying and collecting by powers of x, we find that

$$[x^k](fg) = \sum_{i=0}^{k} \binom{n+i-1}{i} \binom{n}{k-2i},$$

which is the left-hand side of our identity.

For g, the generalized Newton formula (Theorem 1.264) gives $g = (1+x)^n$. For f, the anti-Newton binomial formula (Theorem 1.266) gives $(1+x)^{-n} = \sum_{i \in \mathbb{N}} (-1)^i \binom{n+i-1}{i} x^i$. Substituting $-x^2$ for x, we get $f = (1-x^2)^{-n}$.

Thus

$$\begin{aligned} fg &= (1-x^2)^{-n}(1+x)^n \\ &= \left(\frac{1+x}{1-x^2}\right)^n = \left(\frac{1-x^2}{1+x}\right)^{-n} \\ &= (1-x)^{-n}. \end{aligned}$$

Substituting $-x$ for x in the anti-Newton formula gives

$$(1-x)^{-n} = \sum_{i \in \mathbb{N}} \binom{n+i-1}{i} x^i.$$

Hence $[x^k](fg) = \binom{n+k-1}{k}$.

The formal proof establishes that $fg = (1-x)^{-n}$ by first showing $f = (1-x^2)^{-n}$. The identity $f = (1-x^2)^{-n}$ is proved by comparing coefficients using the polynomial identity principle: for natural N, both sides of the coefficient identity follow from the inverse of $(1-x^2)^N$ expressed via expansion of $(1-x)^{-N}$, and uniqueness of inverses gives agreement; the polynomial identity principle then extends the result to all $n \in K$. The odd-coefficient vanishing is established by a pairing argument: for odd k, each pair $(i, k-i)$ contributes terms with opposite signs (since i and $k-i$ have opposite parities), giving a sum of 0. $\square$

1.9.6 Helper lemmas from the formalization

Lemma 1.272. *Our $\overline{\log}$ series equals the logarithm series from Mathlib's Exp/Log framework.*

Proof. Coefficient comparison. $\square$

Lemma 1.273. *The definition of* Log *from Definition 1.250 is equivalent to the definition from Mathlib's Exp/Log framework for FPS with constant term one.*

Proof. Unfold definitions and use Lemma 1.272. $\square$

Lemma 1.274. *For g with $[x^0]g = 0$, the definition of* $\operatorname{Exp}(g)$ *from Definition 1.254 is equivalent to the definition from Mathlib's Exp/Log framework for FPS with constant term zero.*

Proof. Unfold definitions. $\square$

Lemma 1.275. $(c \cdot g)^m = c^m \cdot g^m$ *for $c \in K$, $g \in K[[x]]$, $m \in \mathbb{N}$.*

Proof. Induction on m. $\square$

Lemma 1.276. *For $k < m$, we have $[x^k](\overline{\log}^m) = 0$ (since $\overline{\log}$ has order ≥ 1, so $\overline{\log}^m$ has order $\geq m$).*

Proof. The order of $\overline{\log}^m$ is at least m since $[x^0]\overline{\log} = 0$. $\square$

Theorem 1.277. *Let R be a commutative domain of characteristic 0, and let $p \in R[x]$ be a polynomial such that $p(n) = 0$ for all $n \in \mathbb{N}$. Then $p = 0$.*

Proof. The set of roots of p contains all natural numbers, which is an infinite subset of R (since R has characteristic 0). A nonzero polynomial of degree d has at most d roots, so p must be zero. $\square$

Lemma 1.278. *If two polynomials over $\mathbb{Q}$ agree when evaluated at all natural numbers (after mapping to K via the algebra map), then they are equal: if $p(n) = q(n)$ for all $n \in \mathbb{N}$, then $p = q$.*

Proof. Apply the polynomial identity trick to $p - q$. $\square$

Lemma 1.279. *Pascal's rule for generalized binomial coefficients:*

$$\binom{c}{k+1} = \binom{c-1}{k} + \binom{c-1}{k+1}.$$

Proof. Follows from the Pascal recurrence for generalized binomial coefficients. $\square$

Lemma 1.280. *The coefficients of $(1+x)^c$ satisfy Pascal's recurrence:*

$$[x^{k+1}](1+x)^c = [x^k](1+x)^{c-1} + [x^{k+1}](1+x)^{c-1}.$$

Proof. Since $(1+x)^c = (1+x)^{c-1} \cdot (1+x)$, the coefficient $[x^{k+1}]$ of the product is $[x^k](1+x)^{c-1} + [x^{k+1}](1+x)^{c-1}$. $\square$

1.9.7 Helpers for Theorem 1.264

The generalized Newton binomial formula requires showing that both the LHS (coefficients of $(1+x)^c$) and the RHS ($\binom{c}{k}$) are polynomial functions of c that agree on all natural numbers.

Lemma 1.281. *The generalized binomial coefficient $\binom{c}{k}$ equals the evaluation of a polynomial in c: there exists a polynomial $p_k \in \mathbb{Q}[x]$ such that $\binom{c}{k} = p_k(c)$ for all $c \in K$.*

Proof. The polynomial is $p_k(x) = x(x-1)\cdots(x-k+1)/k!$, evaluated at c via the algebra map $\mathbb{Q} \to K$. $\square$

Lemma 1.282. *The coefficient $[x^k](1+x)^c$ equals the evaluation of a polynomial in c:*

$$[x^k](1+x)^c = q_k(c)$$

for some $q_k \in \mathbb{Q}[x]$.

Proof. Expand $(1+x)^c = \mathrm{Exp}(c \cdot \overline{\log}) = \sum_{d \geq 0} \frac{1}{d!}(c\,\overline{\log})^d$. Using $(c\,\overline{\log})^d = c^d\,\overline{\log}^d$ and the vanishing $[x^k](\overline{\log}^d) = 0$ for $d > k$, the coefficient $[x^k]$ is a polynomial in c of degree $\leq k$. $\square$

Lemma 1.283. *For $n \in K$ and $k \in \mathbb{N}$,*

$$[x^k](1-x)^{-n} = \binom{n+k-1}{k}.$$

Proof. Use the upper negation identity $\binom{-n}{k} = (-1)^k\binom{n+k-1}{k}$ and the sign cancellation $(-1)^k \cdot (-1)^k = 1$. $\square$

1.9.8 Helpers for Proposition 1.271

The proof of the binomial identity requires establishing that the product $f \cdot g$ (where $f = (1-x^2)^{-n}$ and $g = (1+x)^n$) equals $(1-x)^{-n}$. This is done in several steps:

1. Define $f = \sum_{i \in \mathbb{N}} \binom{n+i-1}{i} x^{2i}$ and $g = \sum_{j \in \mathbb{N}} \binom{n}{j} x^j$.
2. Show $fg = (1-x)^{-n}$ (Lemma 1.290).
3. Read off coefficients.

Definition 1.284. Define two FPSs for the binomial identity:

$$f(n) := \sum_{i \in \mathbb{N}} \binom{n+i-1}{i} x^{2i} \quad \text{(i.e., the even-indexed coefficients of } (1-x^2)^{-n}\text{)},$$
$$g(n) := \sum_{j \in \mathbb{N}} \binom{n}{j} x^j = (1+x)^n.$$

Lemma 1.285. *The coefficient of x^k in $f(n) \cdot g(n)$ equals*

$$[x^k](f \cdot g) = \sum_{i=0}^{\lfloor k/2 \rfloor} \binom{n+i-1}{i}\binom{n}{k-2i}.$$

Proof. By expanding the Cauchy product and using the fact that f has nonzero coefficients only at even positions. $\square$

Lemma 1.286. *For natural N,*

$$(1-x)^{-N} \cdot (1+x)^{-N} \cdot (1-x^2)^N = 1.$$

That is, $(1-x)^{-N} \cdot (1+x)^{-N}$ is the inverse of $(1-x^2)^N$.

Proof. Factor $(1-x^2)^N = (1-x)^N(1+x)^N$ and use the inverse relations $(1-x)^{-N} \cdot (1-x)^N = 1$ and $(1+x)^{-N} \cdot (1+x)^N = 1$. $\square$

Lemma 1.287. *For natural N, $(1-x)^{-N}\big|_{x\mapsto x^2} \cdot (1-x^2)^N = 1$. That is, $(1-x)^{-N}\big|_{x\mapsto x^2}$ is also the inverse of $(1-x^2)^N$.*

Proof. Since the substitution $x \mapsto x^2$ is a ring homomorphism and $(1-x)\big|_{x\mapsto x^2} = 1-x^2$, we have $(1-x)^{-N}\big|_{x\mapsto x^2} \cdot (1-x)^N\big|_{x\mapsto x^2} = 1\big|_{x\mapsto x^2} = 1$. $\square$

Lemma 1.288. *For natural N,*

$$(1-x)^{-N} \cdot (1+x)^{-N} = (1-x)^{-N}\big|_{x\mapsto x^2}.$$

Both sides are inverses of $(1-x^2)^N$, so they are equal by uniqueness of inverses.

Proof. Both $P := (1-x)^{-N} \cdot (1+x)^{-N}$ and $E := (1-x)^{-N}\big|_{x\mapsto x^2}$ satisfy $(\cdot) \cdot (1-x^2)^N = 1$. Since $(1-x^2)^N$ is a unit, $P = E$. $\square$

Lemma 1.289. *For natural $N \geq 1$ and $m \in \mathbb{N}$,*

$$[x^{2m}]\big((1-x)^{-N} \cdot (1+x)^{-N}\big) = \binom{N+m-1}{m}.$$

Proof. By Lemma 1.288, the product equals $(1-x)^{-N}\big|_{x\mapsto x^2}$, so the coefficient at $2m$ is $[x^m](1-x)^{-N} = \binom{N+m-1}{m}$. $\square$

Lemma 1.290 (Key product identity)**.** *For all $n \in K$,*

$$f(n) \cdot g(n) = (1-x)^{-n}.$$

In other words, $(1-x^2)^{-n} \cdot (1+x)^n = (1-x)^{-n}$.

Proof. First show $f(n) = (1-x^2)^{-n}$ by establishing that $f(n)$ equals $(1+x)^{-n} \cdot (1-x)^{-n}$. The even-coefficient identity uses Lemma 1.289 for natural N and then extends to all n by the polynomial identity principle (Theorem 1.277). The odd-coefficient vanishing follows from a pairing argument: for odd k, each pair $(i, k-i)$ contributes terms with opposite signs. Then $f(n) \cdot g(n) = (1-x^2)^{-n} \cdot (1+x)^n$, and the cancellation $(1+x)^{-n} \cdot (1+x)^n = 1$ gives the result. $\square$

1.10 Integer compositions

1.10.1 Compositions

Definition 1.291. **(a)** An *(integer) composition* means a (finite) tuple of positive integers.

(b) The *size* of an integer composition $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ is defined to be the integer $\alpha_1 + \alpha_2 + \cdots + \alpha_m$. It is written $|\alpha|$.

(c) The *length* of an integer composition $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ is defined to be the integer m. It is written $\ell(\alpha)$.

(d) Let $n \in \mathbb{N}$. A *composition of* n means a composition whose size is n.

(e) Let $n \in \mathbb{N}$ and $k \in \mathbb{N}$. A *composition of* n *into* k *parts* is a composition whose size is n and whose length is k.

Lemma 1.292. *The size of the empty composition is* 0.

Proof. By definition. ☐

Lemma 1.293. *The length of the empty composition is* 0.

Proof. By definition. ☐

Lemma 1.294. *The size of a composition* $(a) \cdot \alpha$ *(prepending* a *to* α*) is* $a + |\alpha|$.

Proof. By definition. ☐

Lemma 1.295. *The length of a composition* $(a) \cdot \alpha$ *(prepending* a *to* α*) is* $\ell(\alpha) + 1$.

Proof. By definition of length. ☐

Lemma 1.296. *The size of a composition is at least its length:* $\ell(\alpha) \leq |\alpha|$*, since each part is a positive integer (at least* 1*).*

Proof. By induction on the composition. The base case (empty composition) is trivial. For the inductive step, $|\alpha| = a + |\beta| \geq 1 + \ell(\beta) = \ell(\alpha)$ since $a \geq 1$. ☐

Lemma 1.297. *The empty list is the only composition of size* 0.

Proof. If $\alpha = (a) \cdot \beta$, then $|\alpha| = a + |\beta| \geq 1 > 0$ since a is a positive integer. Hence a composition of size 0 must be empty. ☐

Lemma 1.298. *The definition of compositions of* n *from Definition 1.291 (as lists of positive integers with given sum) is equivalent to the definition from Mathlib's composition type.*

Proof. Convert between lists of positive integers and lists of natural numbers with positivity proofs. ☐

Lemma 1.299. *There is an equivalence between compositions of* n *into* k *parts and the subtype of Mathlib compositions of* n *having length* k.

Proof. Follows from the equivalence in Lemma 1.298, restricting to compositions of the appropriate length. ☐

1.10.2 Helpers for Theorem 1.304

Lemma 1.300. *There is an equivalence between Mathlib's compositions of n and subsets of $\{1, 2, \ldots, n-1\}$, obtained by composing two Mathlib equivalences.*

Proof. This is the composition of the Mathlib equivalences between compositions of n, composition-as-set representations, and finite subsets of $\{1, 2, \ldots, n-1\}$. $\square$

Lemma 1.301. *For $n > 0$ and a composition-as-set c of n, the cardinality of the associated finset of interior boundary points $S(c)$ satisfies $|S(c)| + 1 = \ell(c)$. This holds because the equivalence extracts the interior boundary points (those other than 0 and n), and a composition of length k has exactly $k-1$ such points.*

Proof. The boundary set of c has $\ell(c)+1$ elements (including 0 and n). The equivalence maps to the interior points, obtained by removing 0 and n. The resulting finset has $\ell(c)+1-2 = \ell(c)-1$ elements. $\square$

Lemma 1.302. *For $n > 0$ and a composition c of n, the finset $S(c)$ corresponding to c under the composition-to-finset equivalence satisfies $|S(c)| + 1 = \ell(c)$.*

Proof. Follows directly from Lemma 1.301 applied to the composition-as-set representation of c. $\square$

Lemma 1.303. *For $n > 0$ and $k > 0$, the number of compositions of n into k parts (using Mathlib's composition type) equals $\binom{n-1}{k-1}$. The proof establishes a bijection between compositions of length k and $(k-1)$-element subsets of $\{1, 2, \ldots, n-1\}$ via Lemma 1.302, then uses the formula for counting subsets of a given size.*

Proof. Compositions of length k correspond (via the composition-to-finset equivalence) to $(k-1)$-element subsets of $\{1, 2, \ldots, n-1\}$. These are counted by $\binom{n-1}{k-1}$. $\square$

Theorem 1.304. *Let $n, k \in \mathbb{N}$. Then, the # of compositions of n into k parts is*

$$\binom{n-1}{n-k} = \begin{cases} \dbinom{n-1}{k-1}, & \textit{if } n > 0; \\ \delta_{k,0}, & \textit{if } n = 0. \end{cases}$$

Proof. Fix $k \in \mathbb{N}$, but don't fix n. Let

$$a_{n,k} = (\text{\# of compositions of } n \text{ into } k \text{ parts}) . \tag{1.13}$$

We want to find $a_{n,k}$. We define the generating function

$$A_k := \sum_{n \in \mathbb{N}} a_{n,k} x^n = (a_{0,k}, a_{1,k}, a_{2,k}, \ldots) \in \mathbb{Q}[[x]] . \tag{1.14}$$

Let us write $\mathbb{P}$ for the set $\{1, 2, 3, \ldots\}$. Then, a composition of n into k parts is nothing but a k-tuple $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{P}^k$ satisfying $\alpha_1 + \alpha_2 + \cdots + \alpha_k = n$. Hence, (1.13) can be rewritten as

$$\begin{aligned} a_{n,k} &= \left(\text{\# of all } k\text{-tuples } (\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{P}^k \text{ satisfying } \alpha_1 + \alpha_2 + \cdots + \alpha_k = n\right) \\ &= \sum_{\substack{(\alpha_1, \alpha_2, \ldots, \alpha_k) \in \mathbb{P}^k; \\ \alpha_1 + \alpha_2 + \cdots + \alpha_k = n}} 1. \end{aligned} \tag{1.15}$$

Thus, we can rewrite the equality $A_k = \sum_{n\in\mathbb{N}} a_{n,k} x^n$ as

$$\begin{aligned}
A_k &= \sum_{n\in\mathbb{N}} \left(\sum_{\substack{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\mathbb{P}^k;\\ \alpha_1+\alpha_2+\cdots+\alpha_k=n}} 1 \right) x^n = \sum_{n\in\mathbb{N}} \sum_{\substack{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\mathbb{P}^k;\\ \alpha_1+\alpha_2+\cdots+\alpha_k=n}} \underbrace{x^n}_{\substack{=x^{\alpha_1+\alpha_2+\cdots+\alpha_k}\\ (\text{since } \alpha_1+\alpha_2+\cdots+\alpha_k=n)}} \\
&= \underbrace{\sum_{n\in\mathbb{N}} \sum_{\substack{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\mathbb{P}^k;\\ \alpha_1+\alpha_2+\cdots+\alpha_k=n}}}_{=\sum_{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\mathbb{P}^k}} \underbrace{x^{\alpha_1+\alpha_2+\cdots+\alpha_k}}_{=x^{\alpha_1}x^{\alpha_2}\cdots x^{\alpha_k}} = \sum_{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\mathbb{P}^k} x^{\alpha_1}x^{\alpha_2}\cdots x^{\alpha_k} \\
&= \left(\sum_{\alpha_1\in\mathbb{P}} x^{\alpha_1}\right)\left(\sum_{\alpha_2\in\mathbb{P}} x^{\alpha_2}\right)\cdots\left(\sum_{\alpha_k\in\mathbb{P}} x^{\alpha_k}\right) \\
&= \left(\sum_{n\in\mathbb{P}} x^n\right)^k
\end{aligned}$$

(here, we have renamed all the k summation indices as n, and realized that all k sums are identical). Since

$$\sum_{n\in\mathbb{P}} x^n = x^1 + x^2 + x^3 + \cdots = x \underbrace{\left(1 + x + x^2 + \cdots\right)}_{=\dfrac{1}{1-x}} = x \cdot \frac{1}{1-x} = \frac{x}{1-x},$$

this can be rewritten further as

$$A_k = \left(\frac{x}{1-x}\right)^k = x^k (1-x)^{-k}. \tag{1.16}$$

In order to simplify this, we need to expand $(1-x)^{-k}$. This is routine by now: Theorem 1.119 (applied to $-k$ instead of n) yields

$$(1+x)^{-k} = \sum_{j\in\mathbb{N}} \binom{-k}{j} x^j.$$

Substituting $-x$ for x on both sides of this equality (i.e., applying the map $f \mapsto f \circ (-x)$), we obtain

$$\begin{aligned}
(1-x)^{-k} &= \sum_{j\in\mathbb{N}} \underbrace{\binom{-k}{j}}_{=(-1)^j \dbinom{j+k-1}{j}} \underbrace{(-x)^j}_{=(-1)^j x^j} = \sum_{j\in\mathbb{N}} (-1)^j \binom{j+k-1}{j} \cdot (-1)^j x^j \\
&= \sum_{j\in\mathbb{N}} \underbrace{\left((-1)^j\right)^2}_{=1} \binom{j+k-1}{j} x^j = \sum_{j\in\mathbb{N}} \binom{j+k-1}{j} x^j \qquad (1.17) \\
&= \sum_{n\geq k} \binom{n-1}{n-k} x^{n-k}
\end{aligned}$$

(where we used Theorem 1.120, applied to k and j instead of n and k, to simplify $\binom{-k}{j}$; and we have substituted $n-k$ for j in the sum). Hence, our above computation of A_k can be completed as follows:

$$\begin{aligned} A_k &= x^k \underbrace{(1-x)^{-k}}_{=\sum_{n\geq k}\binom{n-1}{n-k}x^{n-k}} = x^k \sum_{n\geq k}\binom{n-1}{n-k}x^{n-k} \\ &= \sum_{n\geq k}\binom{n-1}{n-k}\underbrace{x^k x^{n-k}}_{=x^n} = \sum_{n\geq k}\binom{n-1}{n-k}x^n = \sum_{n\in\mathbb{N}}\binom{n-1}{n-k}x^n \end{aligned}$$

(here, we have extended the range of the summation from all $n \geq k$ to all $n \in \mathbb{N}$; this did not change the sum, since all the newly introduced addends with $n < k$ are 0). Comparing coefficients, we thus obtain

$$a_{n,k} = \binom{n-1}{n-k} \qquad \text{for each } n \in \mathbb{N}. \tag{1.18}$$

If $n > 0$, then we can rewrite the right hand side of this equality as $\binom{n-1}{k-1}$ (using Theorem 1.9). However, if $n = 0$, then this right hand side equals $\delta_{k,0}$ instead (where we are using Definition 1.167). Thus, we can rewrite (1.18) as

$$a_{n,k} = \begin{cases} \binom{n-1}{k-1}, & \text{if } n > 0; \\ \delta_{k,0}, & \text{if } n = 0 \end{cases} \qquad \text{for each } n \in \mathbb{N}. \tag{1.19}$$

In the Lean formalization, this proceeds by:

1. Using the equivalence between our compositions and Mathlib's composition type (Lemma 1.299).
2. Using Mathlib's equivalence between compositions of n and subsets of $\{1, 2, \ldots, n-1\}$.
3. Showing that compositions of length k correspond to subsets of size $k-1$.
4. Computing $|\{S \subseteq \{1, \ldots, n-1\} \mid |S| = k-1\}| = \binom{n-1}{k-1}$ using the formula for counting subsets of a given size.

The edge cases are:

- For $n > 0$ and $k = 0$: There are no compositions (since parts are positive, and a tuple of zero positive integers sums to 0, not n). This gives $0 = \binom{n-1}{-1} = 0$.
- For $n = 0$: The empty tuple is the only composition of 0 (by Lemma 1.297), and it has length 0. So the count is $\delta_{k,0}$.

□

Theorem 1.305. *Let $n \in \mathbb{N}$. Then, the # of compositions of n is*

$$\begin{cases} 2^{n-1}, & \text{if } n > 0; \\ 1, & \text{if } n = 0. \end{cases}$$

Proof. If $n = 0$, then the # of compositions of n is 1 (since the empty tuple () is the only composition of 0). Thus, for the rest of this proof, we WLOG assume that $n \neq 0$. Hence, we must prove that the # of compositions of n is 2^{n-1}.

If $(n_1, n_2, \ldots, n_m)$ is a composition of n, then $m \in \{1, 2, \ldots, n\}$. In other words, any composition of n is a composition of n into k parts for some $k \in \{1, 2, \ldots, n\}$. Hence,

$$\begin{aligned}
&(\#\text{ of compositions of } n) \\
&= \sum_{k\in\{1,2,\ldots,n\}} \underbrace{(\#\text{ of compositions of } n \text{ into } k \text{ parts})}_{=\binom{n-1}{n-k}} \\
&= \sum_{k\in\{1,2,\ldots,n\}} \binom{n-1}{n-k} = \sum_{k=1}^{n} \binom{n-1}{n-k} = \sum_{k=0}^{n-1} \binom{n-1}{k}
\end{aligned}$$

(where the second equality uses Theorem 1.304; and we have substituted k for $n-k$ in the sum). Comparing this with

$$\begin{aligned}
2^{n-1} = (1+1)^{n-1} &= \sum_{k=0}^{n-1} \binom{n-1}{k} \underbrace{1^k 1^{n-1-k}}_{=1} \qquad \text{(by the binomial theorem)} \\
&= \sum_{k=0}^{n-1} \binom{n-1}{k},
\end{aligned}$$

we obtain $(\#\text{ of compositions of } n) = 2^{n-1}$.

In the Lean formalization, this is proved by establishing an equivalence with Mathlib's composition type (Lemma 1.298) and using Mathlib's theorem on the cardinality of compositions. □

1.10.3 Weak compositions

Definition 1.306. **(a)** An *(integer) weak composition* means a (finite) tuple of nonnegative integers.

(b) The *size* of a weak composition $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ is defined to be the integer $\alpha_1 + \alpha_2 + \cdots + \alpha_m$. It is written $|\alpha|$.

(c) The *length* of a weak composition $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_m)$ is defined to be the integer m. It is written $\ell(\alpha)$.

(d) Let $n \in \mathbb{N}$. A *weak composition of n* means a weak composition whose size is n.

(e) Let $n \in \mathbb{N}$ and $k \in \mathbb{N}$. A *weak composition of n into k parts* is a weak composition whose size is n and whose length is k.

Lemma 1.307. *The list-based representation of weak compositions of n into k parts is equivalent to the function-based representation $\{f : \{0, 1, \ldots, k-1\} \to \mathbb{N} \mid \sum_i f(i) = n\}$.*

Proof. The forward direction converts a list $[\alpha_0, \alpha_1, \ldots, \alpha_{k-1}]$ to the function $i \mapsto \alpha_i$. The backward direction converts a function f to the list $[f(0), f(1), \ldots, f(k-1)]$. □

Lemma 1.308. *The function-based weak compositions of n into k parts are equivalent to multisets of size n from a k-element alphabet.*

Proof. The bijection sends $f : \{0, \ldots, k-1\} \to \mathbb{N}$ with $\sum_i f(i) = n$ to the multiset containing $f(i)$ copies of i for each i, and the inverse sends a multiset m to the function $i \mapsto \operatorname{count}(i, m)$. $\square$

Lemma 1.309. *There is a bijection between weak compositions of n into k parts and compositions of $n+k$ into k parts, defined by adding 1 to each entry.*

Proof. If $(\alpha_1, \alpha_2, \ldots, \alpha_k)$ is a weak composition of n into k parts, then $(\alpha_1 + 1, \alpha_2 + 1, \ldots, \alpha_k + 1)$ is a composition of $n+k$ into k parts (since adding 1 to each nonnegative entry yields a positive integer, and the sum increases by k). The inverse map subtracts 1 from each entry. $\square$

Lemma 1.310. *The number of function-based weak compositions of n into k parts equals $\binom{n+k-1}{n}$. The proof uses the equivalence with multisets of size n from a k-element alphabet (Lemma 1.308) and the formula for counting such multisets.*

Proof. By the equivalence with multisets of size n from a k-element alphabet and the formula $\binom{n+k-1}{n}$ for counting such multisets. $\square$

Theorem 1.311. *Let $n, k \in \mathbb{N}$. Then, the # of weak compositions of n into k parts is*

$$\binom{n+k-1}{n} = \begin{cases} \dbinom{n+k-1}{k-1}, & \text{if } k > 0; \\ \delta_{n,0}, & \text{if } k = 0. \end{cases}$$

Proof. Adding 1 to each entry of a weak composition of n into k parts gives a composition of $n+k$ into k parts (Lemma 1.309). This map is a bijection (its inverse subtracts 1 from each entry). Thus, by the bijection principle, we have

$$\begin{aligned} &|\{\text{weak compositions of } n \text{ into } k \text{ parts}\}| \\ &= |\{\text{compositions of } n+k \text{ into } k \text{ parts}\}| \\ &= (\# \text{ of compositions of } n+k \text{ into } k \text{ parts}) \\ &= \binom{n+k-1}{n+k-k} = \binom{n+k-1}{n}. \end{aligned}$$

(Here, the last step follows by Theorem 1.304, applied to $n+k$ instead of n.) It remains to prove that this equals $\begin{cases} \dbinom{n+k-1}{k-1}, & \text{if } k > 0; \\ \delta_{n,0}, & \text{if } k = 0 \end{cases}$ as well. If $k = 0$, then it is clear by inspection; otherwise it follows from Theorem 1.9.

In the Lean formalization, the primary proof uses the "stars and bars" technique: weak compositions of n into k parts are equivalent to multisets of size n from a k-element alphabet, via Lemmas 1.307 and 1.308. These are counted by $\binom{n+k-1}{n}$ using the formula for counting multisets. $\square$

1.10.4 Weak compositions with entries from $\{0, 1, \ldots, p-1\}$

Lemma 1.312. *The geometric series identity: $\left(\sum_{i=0}^{p-1} x^i\right)(1-x) = 1 - x^p$.*

Proof. By induction on p and ring arithmetic. $\square$

Lemma 1.313. *The geometric series equals* $(1-x^p)\cdot(1-x)^{-1}$*:*

$$\sum_{i=0}^{p-1} x^i = (1-x^p)\cdot(1-x)^{-1}.$$

Proof. Multiply both sides by $(1-x)$ and use Lemma 1.312. $\square$

Lemma 1.314. *For all* $k \in \mathbb{N}$, $\left((1-x)^{-1}\right)^k = (1-x)^{-k}$*. That is, the* k*-th power of the formal inverse of* $(1-x)$ *equals the formal inverse of* $(1-x)^k$*.*

Proof. By induction on k, using the identity $(1-x)^{-(a+b)} = (1-x)^{-a}\cdot(1-x)^{-b}$ from Mathlib. $\square$

Lemma 1.315. *For* $k > 0$*, the coefficient of* x^n *in* $(1-x)^{-k}$ *is* $\binom{k-1+n}{k-1}$*.*

Proof. This is a standard result from Mathlib's power series library. $\square$

Lemma 1.316. *The generating function for bounded weak compositions satisfies* $W_{k,p} = (1-x^p)^k\cdot(1-x)^{-k}$*.*

Proof. Use Lemma 1.313 and raise both sides to the k-th power, using Lemma 1.314 to simplify $\left((1-x)^{-1}\right)^k = (1-x)^{-k}$. $\square$

Lemma 1.317. *The coefficient of* x^n *in the generating function* $W_{k,p}$ *equals the number of bounded weak compositions of* n *into* k *parts with entries in* $\{0,1,\ldots,p-1\}$*.*

Proof. Use the product rule for power series coefficients and show that the product of indicator functions picks out exactly the bounded weak compositions summing to n. $\square$

Lemma 1.318. *The coefficient of* x^n *in* $W_{k,p}$ *equals the inclusion-exclusion formula:*

$$[x^n]W_{k,p} = \sum_{\substack{j\in\{0,\ldots,k\}\\ pj\le n}} (-1)^j \binom{k}{j}\binom{n-pj+k-1}{n-pj}.$$

Proof. Expand the product $(1-x^p)^k\cdot(1-x)^{-k}$ using the binomial theorem for $(1-x^p)^k$ and the expansion of $(1-x)^{-k}$ via Lemma 1.315, then extract the coefficient of x^n via the convolution formula. $\square$

Theorem 1.319. *Let* $n,k,p \in \mathbb{N}$*. Then, the # of* k*-tuples* $(\alpha_1,\alpha_2,\ldots,\alpha_k) \in \{0,1,\ldots,p-1\}^k$ *satisfying* $\alpha_1+\alpha_2+\cdots+\alpha_k = n$ *is*

$$\sum_{j=0}^{k} (-1)^j \binom{k}{j}\binom{n-pj+k-1}{n-pj}.$$

Proof. We fix three nonnegative integers n, k and p. We are looking for the # of k-tuples $(\alpha_1,\alpha_2,\ldots,\alpha_k) \in \{0,1,\ldots,p-1\}^k$ satisfying $\alpha_1+\alpha_2+\cdots+\alpha_k = n$. In other words, we are looking for the # of all weak compositions of n into k parts with the property that each entry is $< p$. Let us denote this # by $w_{n,k,p}$.

We invoke a generating function. Define the FPS

$$W_{k,p} := \sum_{n\in\mathbb{N}} w_{n,k,p}x^n \in \mathbb{Q}[[x]].$$

For each $n \in \mathbb{N}$, we have

$$w_{n,k,p} = \sum_{\substack{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\{0,1,\ldots,p-1\}^k;\\ \alpha_1+\alpha_2+\cdots+\alpha_k=n}} 1.$$

Thus, we can rewrite $W_{k,p} = \sum_{n\in\mathbb{N}} w_{n,k,p} x^n$ as

$$\begin{aligned} W_{k,p} &= \sum_{(\alpha_1,\alpha_2,\ldots,\alpha_k)\in\{0,1,\ldots,p-1\}^k} x^{\alpha_1} x^{\alpha_2} \cdots x^{\alpha_k} \\ &= \left(\sum_{n\in\{0,1,\ldots,p-1\}} x^n \right)^k \end{aligned}$$

(by the product rule). Since

$$\sum_{n\in\{0,1,\ldots,p-1\}} x^n = x^0 + x^1 + \cdots + x^{p-1} = \frac{1-x^p}{1-x},$$

this can be rewritten further as

$$W_{k,p} = \left(\frac{1-x^p}{1-x} \right)^k = (1-x^p)^k (1-x)^{-k}. \tag{1.20}$$

In order to expand the right hand side, let us expand $(1-x^p)^k$ and $(1-x)^{-k}$ separately.

The binomial theorem yields

$$(1-x^p)^k = \sum_{j\in\mathbb{N}} (-1)^j \binom{k}{j} x^{pj}.$$

Multiplying this by (1.17), we obtain

$$\begin{aligned} &(1-x^p)^k (1-x)^{-k} \\ &= \sum_{n\in\mathbb{N}} \left(\sum_{\substack{(i,j)\in\mathbb{N}\times\mathbb{N};\\ pj+i=n}} (-1)^j \binom{k}{j} \binom{i+k-1}{i} \right) x^n. \end{aligned}$$

Thus, (1.20) becomes

$$W_{k,p} = \sum_{n\in\mathbb{N}} \left(\sum_{\substack{(i,j)\in\mathbb{N}\times\mathbb{N};\\ pj+i=n}} (-1)^j \binom{k}{j} \binom{i+k-1}{i} \right) x^n.$$

Comparing coefficients, we find that each $n \in \mathbb{N}$ satisfies

$$w_{n,k,p} = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \binom{n-pj+k-1}{n-pj}.$$

$\square$

Lemma 1.320. *Let $n, k \in \mathbb{N}$. The number of binary k-strings with exactly n ones is $\binom{k}{n}$.*

Proof. Establish a bijection between binary k-strings with n ones and n-element subsets of $\{1, 2, \ldots, k\}$ (a binary string corresponds to the set of positions where a 1 appears). Then use the formula for counting subsets of a given size. $\square$

Proposition 1.321. *Let $n, k \in \mathbb{N}$. Then,*

$$\binom{k}{n} = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \binom{n-2j+k-1}{n-2j}.$$

Proof. The k-tuples $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in \{0,1\}^k$ are just the *binary k-strings*, i.e., the k-tuples formed of 0s and 1s. Imposing the condition $\alpha_1 + \alpha_2 + \cdots + \alpha_k = n$ on such a k-tuple is tantamount to requiring that it contain exactly n many 1s. Therefore, the # of all k-tuples $(\alpha_1, \alpha_2, \ldots, \alpha_k) \in \{0,1\}^k$ satisfying $\alpha_1 + \alpha_2 + \cdots + \alpha_k = n$ is $\dbinom{k}{n}$, since we have to choose which n of its k positions will be occupied by 1s (Lemma 1.320). However, Theorem 1.319 (applied to $p = 2$) yields that this # equals $\sum_{j=0}^{k} (-1)^j \dbinom{k}{j} \dbinom{n-2j+k-1}{n-2j}$. Comparing these two results, we obtain the identity. $\square$

1.11 x^n-equivalence

Definition 1.322. Let $n \in \mathbb{N}$. Let $f, g \in K[[x]]$ be two FPSs. We write $f \overset{x^n}{\equiv} g$ if and only if

$$\text{each } m \in \{0, 1, \ldots, n\} \text{ satisfies } [x^m] f = [x^m] g.$$

Thus, we have defined a binary relation $\overset{x^n}{\equiv}$ on the set $K[[x]]$. We say that an FPS f is *x^n-equivalent* to an FPS g if and only if $f \overset{x^n}{\equiv} g$.

Lemma 1.323. *Let $m, n \in \mathbb{N}$ with $m \leq n$. If $f \overset{x^n}{\equiv} g$, then $f \overset{x^m}{\equiv} g$.*

Proof. If each $k \in \{0, 1, \ldots, n\}$ satisfies $[x^k]f = [x^k]g$, then in particular each $k \in \{0, 1, \ldots, m\} \subseteq \{0, 1, \ldots, n\}$ satisfies $[x^k]f = [x^k]g$. $\square$

Lemma 1.324. *Two FPSs $f, g \in K[[x]]$ satisfy $f \overset{x^0}{\equiv} g$ if and only if $[x^0]f = [x^0]g$.*

Proof. This is immediate from the definition, since $\{0, 1, \ldots, 0\} = \{0\}$. $\square$

1.11.1 Helpers for the equivalence relation

Lemma 1.325. *Let $n \in \mathbb{N}$. For any FPS $f \in K[[x]]$, we have $f \overset{x^n}{\equiv} f$.*

Proof. Immediate from the definition: $[x^m]f = [x^m]f$ for all m. $\square$

Lemma 1.326. *Let $n \in \mathbb{N}$. If $f, g \in K[[x]]$ satisfy $f \overset{x^n}{\equiv} g$, then $g \overset{x^n}{\equiv} f$.*

Proof. If $[x^m]f = [x^m]g$ for all $m \leq n$, then $[x^m]g = [x^m]f$ by symmetry of equality. $\square$

Lemma 1.327. *Let $n \in \mathbb{N}$. If $f, g, h \in K[[x]]$ satisfy $f \overset{x^n}{\equiv} g$ and $g \overset{x^n}{\equiv} h$, then $f \overset{x^n}{\equiv} h$.*

Proof. If $[x^m]f = [x^m]g$ and $[x^m]g = [x^m]h$ for all $m \leq n$, then $[x^m]f = [x^m]h$ by transitivity of equality. $\square$

Theorem 1.328. *Let $n \in \mathbb{N}$.*

(a) *The relation $\overset{x^n}{\equiv}$ on $K[[x]]$ is an equivalence relation. In other words:*

- *This relation is reflexive (i.e., we have $f \overset{x^n}{\equiv} f$ for each $f \in K[[x]]$).*
- *This relation is transitive (i.e., if three FPSs $f, g, h \in K[[x]]$ satisfy $f \overset{x^n}{\equiv} g$ and $g \overset{x^n}{\equiv} h$, then $f \overset{x^n}{\equiv} h$).*
- *This relation is symmetric (i.e., if two FPSs $f, g \in K[[x]]$ satisfy $f \overset{x^n}{\equiv} g$, then $g \overset{x^n}{\equiv} f$).*

(b) *If $a, b, c, d \in K[[x]]$ are four FPSs satisfying $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, then we also have*

$$a + c \overset{x^n}{\equiv} b + d; \tag{1.21}$$

$$a - c \overset{x^n}{\equiv} b - d; \tag{1.22}$$

$$ac \overset{x^n}{\equiv} bd. \tag{1.23}$$

(c) *If $a, b \in K[[x]]$ are two FPSs satisfying $a \overset{x^n}{\equiv} b$, then $\lambda a \overset{x^n}{\equiv} \lambda b$ for each $\lambda \in K$.*

(d) *If $a, b \in K[[x]]$ are two invertible FPSs satisfying $a \overset{x^n}{\equiv} b$, then $a^{-1} \overset{x^n}{\equiv} b^{-1}$.*

(e) *If $a, b, c, d \in K[[x]]$ are four FPSs satisfying $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, and if the FPSs c and d are invertible, then we also have*

$$\frac{a}{c} \overset{x^n}{\equiv} \frac{b}{d}. \tag{1.24}$$

(f) *Let S be a finite set. Let $(a_s)_{s\in S} \in K[[x]]^S$ and $(b_s)_{s\in S} \in K[[x]]^S$ be two families of FPSs such that*

$$\text{each } s \in S \text{ satisfies } a_s \overset{x^n}{\equiv} b_s. \tag{1.25}$$

Then, we have

$$\sum_{s\in S} a_s \overset{x^n}{\equiv} \sum_{s\in S} b_s; \tag{1.26}$$

$$\prod_{s\in S} a_s \overset{x^n}{\equiv} \prod_{s\in S} b_s. \tag{1.27}$$

Proof. All of these properties are analogous to familiar properties of integer congruences, except for Theorem 1.328 **(d)**, which is moot for integers (since there are not many integers that are invertible in $\mathbb{Z}$).

(a) Reflexivity, symmetry and transitivity follow directly from the corresponding properties of equality of coefficients.

(b) For addition: If $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, then for each $m \in \{0, 1, \ldots, n\}$ we have $[x^m](a + c) = [x^m]a + [x^m]c = [x^m]b + [x^m]d = [x^m](b + d)$. Subtraction is analogous.

For multiplication: We have $[x^m](ac) = \sum_{i=0}^{m} [x^i]a \cdot [x^{m-i}]c$. Since $m \leq n$, each $i \leq m$ satisfies $i \leq n$ and $m - i \leq n$, so $[x^i]a = [x^i]b$ and $[x^{m-i}]c = [x^{m-i}]d$. Thus $[x^m](ac) = [x^m](bd)$.

(c) For each $m \leq n$, we have $[x^m](\lambda a) = \lambda \cdot [x^m] a = \lambda \cdot [x^m] b = [x^m](\lambda b)$.

(d) Let $a, b \in K[[x]]$ be two invertible FPSs satisfying $a \overset{x^n}{\equiv} b$. We want to show that $a^{-1} \overset{x^n}{\equiv} b^{-1}$.

The FPS a is invertible; thus, its constant term $[x^0] a$ is invertible in K (by Proposition 1.111).

We prove that each $m \in \{0, 1, \ldots, n\}$ satisfies $[x^m](a^{-1}) = [x^m](b^{-1})$ by strong induction on m. Fix some $k \in \{0, 1, \ldots, n\}$, and assume (as induction hypothesis) that $[x^m](a^{-1}) = [x^m](b^{-1})$ for each $m \in \{0, 1, \ldots, k-1\}$.

We know that

$$\begin{aligned}
[x^k](aa^{-1}) &= \sum_{i=0}^{k} [x^i] a \cdot [x^{k-i}](a^{-1}) \\
&= [x^0] a \cdot [x^k](a^{-1}) + \sum_{i=1}^{k} [x^i] a \cdot [x^{k-i}](a^{-1})
\end{aligned}$$

(here, we have split off the addend for $i = 0$ from the sum). Thus,

$$[x^0] a \cdot [x^k](a^{-1}) + \sum_{i=1}^{k} [x^i] a \cdot [x^{k-i}](a^{-1}) = [x^k] \underbrace{(aa^{-1})}_{=1} = [x^k] 1.$$

We can solve this equation for $[x^k](a^{-1})$ (since $[x^0] a$ is invertible), and thus obtain

$$[x^k](a^{-1}) = \frac{1}{[x^0] a} \cdot \left([x^k] 1 - \sum_{i=1}^{k} [x^i] a \cdot [x^{k-i}](a^{-1}) \right).$$

The same argument (applied to b instead of a) yields

$$[x^k](b^{-1}) = \frac{1}{[x^0] b} \cdot \left([x^k] 1 - \sum_{i=1}^{k} [x^i] b \cdot [x^{k-i}](b^{-1}) \right).$$

The right hand sides of the latter two equalities are equal (since each $i \in \{1, 2, \ldots, k\}$ satisfies $[x^i] a = [x^i] b$ as a consequence of $a \overset{x^n}{\equiv} b$, and satisfies $[x^{k-i}](a^{-1}) = [x^{k-i}](b^{-1})$ as a consequence of the induction hypothesis, and since we have $[x^0] a = [x^0] b$ as a consequence of $a \overset{x^n}{\equiv} b$). Hence, the left hand sides must also be equal. In other words, $[x^k](a^{-1}) = [x^k](b^{-1})$, which is precisely what we wanted to prove. Thus, the induction step is complete, so that $a^{-1} \overset{x^n}{\equiv} b^{-1}$ is proved.

(e) Combine parts **(b)** and **(d)**: we have $a \overset{x^n}{\equiv} b$ and $c^{-1} \overset{x^n}{\equiv} d^{-1}$ (by **(d)**), so $ac^{-1} \overset{x^n}{\equiv} bd^{-1}$ (by **(b)**).

(f) The sum case follows by induction on $|S|$ using part **(b)** for the induction step (adding one element at a time). The product case is analogous, using the multiplication part of **(b)**. $\square$

Lemma 1.329. *Let $n \in \mathbb{N}$. If $a, b \in K[[x]]$ satisfy $a \overset{x^n}{\equiv} b$, then $-a \overset{x^n}{\equiv} -b$.*

Proof. For each $m \leq n$, we have $[x^m](-a) = -[x^m] a = -[x^m] b = [x^m](-b)$. $\square$

1.11.2 Helpers for inversion and division

Lemma 1.330. *Let $n \in \mathbb{N}$. Let $f, g \in K[[x]]$ be two FPSs satisfying $f \overset{x^n}{\equiv} g$, and let u, v be units of K satisfying $u = v$. Then the FPS obtained by inverting f using the unit u is x^n-equivalent to the FPS obtained by inverting g using the unit v.*

Proof. The proof proceeds by strong induction on the coefficient index, analogous to the proof of Theorem 1.328 **(d)**, using the equality of unit values $u = v$ and the x^n-equivalence of f and g. □

Lemma 1.331. *Let $n \in \mathbb{N}$. Let $a, b, c, d \in K[[x]]$ be four FPSs satisfying $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, and let c and d have invertible constant terms. Then $a \cdot c^{-1} \overset{x^n}{\equiv} b \cdot d^{-1}$.*

Proof. The constant terms of c and d agree (since $c \overset{x^n}{\equiv} d$), so Lemma 1.330 gives $c^{-1} \overset{x^n}{\equiv} d^{-1}$. Then $a \cdot c^{-1} \overset{x^n}{\equiv} b \cdot d^{-1}$ by the multiplication rule of Theorem 1.328 **(b)**. □

Lemma 1.332. *Let $n \in \mathbb{N}$. If $a, b \in K[[x]]$ satisfy $a \overset{x^n}{\equiv} b$, then $a^k \overset{x^n}{\equiv} b^k$ for every $k \in \mathbb{N}$.*

Proof. By induction on k. The base case $k = 0$ gives $a^0 = 1 = b^0$, which is trivially x^n-equivalent to itself. For the induction step, $a^{k+1} = a^k \cdot a \overset{x^n}{\equiv} b^k \cdot b = b^{k+1}$ by the induction hypothesis and (1.23). □

Lemma 1.333. *Let $n \in \mathbb{N}$. Two FPSs $f, g \in K[[x]]$ satisfy $f \overset{x^n}{\equiv} g$ if and only if* $\operatorname{trunc}_{n+1}(f) = \operatorname{trunc}_{n+1}(g)$.

Proof. The truncation $\operatorname{trunc}_{n+1}(f)$ is the polynomial $\sum_{k=0}^{n}([x^k]f) \cdot x^k$. Two such truncations are equal if and only if all their coefficients at degrees $0, 1, \ldots, n$ agree, which is exactly the condition $f \overset{x^n}{\equiv} g$. □

Lemma 1.334. *Let $n \in \mathbb{N}$ and $f \in K[[x]]$. There exists a polynomial $p \in K[x]$ such that $f \overset{x^n}{\equiv} p$.*

Proof. Take $p = \sum_{k=0}^{n}([x^k]f) \cdot x^k$. Then for each $m \le n$, $[x^m]p = [x^m]f$. □

1.11.3 Divisibility characterization

Proposition 1.335. *Let $n \in \mathbb{N}$. Let $f, g \in K[[x]]$ be two FPSs. Then, we have $f \overset{x^n}{\equiv} g$ if and only if the FPS $f - g$ is a multiple of x^{n+1}.*

Proof. A simple consequence of Lemma 1.130.

($\Longrightarrow$): Assume $f \overset{x^n}{\equiv} g$. Then for each $m < n + 1$ (i.e., $m \le n$), we have $[x^m](f - g) = [x^m]f - [x^m]g = 0$. Thus the first $n + 1$ coefficients of $f - g$ are 0, so $f - g$ is a multiple of x^{n+1} by Lemma 1.130.

($\Longleftarrow$): Assume $x^{n+1} \mid (f - g)$. Then for each $m < n + 1$, we have $[x^m](f - g) = 0$ (by the power series analogue of Lemma 1.130), so $[x^m]f = [x^m]g$. Since this holds for each $m \le n$, we obtain $f \overset{x^n}{\equiv} g$. □

Lemma 1.336. *Let $n \in \mathbb{N}$. Let $f, g \in K[[x]]$ be two FPSs. Then, $f \overset{x^n}{\equiv} g$ if and only if there exists a $q \in K[[x]]$ such that $f - g = x^{n+1}q$.*

Proof. This is simply a restatement of Proposition 1.335, since "$f - g$ is a multiple of x^{n+1}" means exactly that there exists such a q. □

1.11.4 Composition

Lemma 1.337. *Let $f \in K[[x]]$ be an FPS with $[x^0]f = 0$. Let $m, k \in \mathbb{N}$ with $m < k$. Then $[x^m](f^k) = 0$.*

Proof. Since $[x^0]f = 0$, the order of f is at least 1, and thus the order of f^k is at least $k > m$. Hence $[x^m](f^k) = 0$. $\square$

Proposition 1.338. *Let $n \in \mathbb{N}$. Let $a, b, c, d \in K[[x]]$ be four FPSs satisfying $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$ and $[x^0]c = 0$ and $[x^0]d = 0$. Then,*

$$a \circ c \overset{x^n}{\equiv} b \circ d.$$

Proof. Write a and b as $a = \sum_{i\in\mathbb{N}} a_i x^i$ and $b = \sum_{i\in\mathbb{N}} b_i x^i$ (with $a_i, b_i \in K$). Then, $a \overset{x^n}{\equiv} b$ means that $a_i = b_i$ for all $i \leq n$. Combine this with $c^i \overset{x^n}{\equiv} d^i$ (which holds for all $i \in \mathbb{N}$ as a consequence of $c \overset{x^n}{\equiv} d$ and of (1.27)) to obtain the relation $a_i c^i \overset{x^n}{\equiv} b_i d^i$ for all $i \leq n$. But this relation also holds for all $i > n$, since all such i satisfy $[x^m](c^i) = [x^m](d^i) = 0$ for all $m \in \{0, 1, \ldots, n\}$ (a consequence of Lemma 1.337 using $[x^0]c = 0$ and $[x^0]d = 0$). Thus, the relation $a_i c^i \overset{x^n}{\equiv} b_i d^i$ holds for all $i \in \mathbb{N}$. Summing over all i, we find $a \circ c \overset{x^n}{\equiv} b \circ d$. $\square$

1.11.5 Equality via x^n-equivalence

Lemma 1.339. *Two FPSs $f, g \in K[[x]]$ are equal if and only if $f \overset{x^n}{\equiv} g$ for all $n \in \mathbb{N}$.*

Proof. ($\Longrightarrow$): If $f = g$, then $f \overset{x^n}{\equiv} g$ for every n by reflexivity.

($\Longleftarrow$): If $f \overset{x^n}{\equiv} g$ for all n, then in particular for each $m \in \mathbb{N}$, taking $n = m$ gives $[x^m]f = [x^m]g$ (since $m \leq m$). Thus $f = g$ by extensionality (Lemma 1.86). $\square$

1.12 Infinite products of FPSs

1.12.1 Determining coefficients in products

Definition 1.340. Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a (possibly infinite) family of FPSs. Let $n \in \mathbb{N}$. Let M be a finite subset of I.

(a) We say that M *determines the x^n-coefficient in the sum of* $(\mathbf{a}_i)_{i\in I}$ if every finite subset J of I satisfying $M \subseteq J \subseteq I$ satisfies

$$[x^n]\left(\sum_{i\in J} \mathbf{a}_i\right) = [x^n]\left(\sum_{i\in M} \mathbf{a}_i\right).$$

(b) We say that M *determines the x^n-coefficient in the product of* $(\mathbf{a}_i)_{i\in I}$ if every finite subset J of I satisfying $M \subseteq J \subseteq I$ satisfies

$$[x^n]\left(\prod_{i\in J} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M} \mathbf{a}_i\right).$$

Definition 1.341. Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a (possibly infinite) family of FPSs. Let $n \in \mathbb{N}$.

(a) We say that *the x^n-coefficient in the sum of $(\mathbf{a}_i)_{i\in I}$ is finitely determined* if there is a finite subset M of I that determines the x^n-coefficient in the sum of $(\mathbf{a}_i)_{i\in I}$.

(b) We say that *the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$ is finitely determined* if there is a finite subset M of I that determines the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$.

Proposition 1.342. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a (possibly infinite) family of FPSs. Then:*

(a) *The family $(\mathbf{a}_i)_{i\in I}$ is summable if and only if each coefficient in its sum is finitely determined (i.e., for each $n \in \mathbb{N}$, the x^n-coefficient in the sum of $(\mathbf{a}_i)_{i\in I}$ is finitely determined).*

(b) *If the family $(\mathbf{a}_i)_{i\in I}$ is summable, then its sum $\sum_{i\in I} \mathbf{a}_i$ is the FPS whose x^n-coefficient (for any $n \in \mathbb{N}$) can be computed as follows: If $n \in \mathbb{N}$, and if M is a finite subset of I that determines the x^n-coefficient in the sum of $(\mathbf{a}_i)_{i\in I}$, then*

$$[x^n]\left(\sum_{i\in I} \mathbf{a}_i\right) = [x^n]\left(\sum_{i\in M} \mathbf{a}_i\right).$$

Proof. Easy and LTTR. ∎

1.12.2 Multipliable families and infinite products

Definition 1.343. Let $(\mathbf{a}_i)_{i\in I}$ be a (possibly infinite) family of FPSs. Then:

(a) The family $(\mathbf{a}_i)_{i\in I}$ is said to be *multipliable* if and only if each coefficient in its product is finitely determined.

(b) If the family $(\mathbf{a}_i)_{i\in I}$ is multipliable, then its *product* $\prod_{i\in I} \mathbf{a}_i$ is defined to be the FPS whose x^n-coefficient (for any $n \in \mathbb{N}$) can be computed as follows: If $n \in \mathbb{N}$, and if M is a finite subset of I that determines the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$, then

$$[x^n]\left(\prod_{i\in I} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M} \mathbf{a}_i\right).$$

Proposition 1.344. *This definition of $\prod_{i\in I} \mathbf{a}_i$ is well-defined – i.e., the coefficient $[x^n]\left(\prod_{i\in M} \mathbf{a}_i\right)$ does not depend on M (as long as M is a finite subset of I that determines the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$).*

Proof. Let $n \in \mathbb{N}$. We need to check that the coefficient $[x^n]\left(\prod_{i\in M} \mathbf{a}_i\right)$ does not depend on M. In other words, we need to check that if M_1 and M_2 are two finite subsets of I that each determine the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$, then

$$[x^n]\left(\prod_{i\in M_1} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M_2} \mathbf{a}_i\right).$$

So let us prove this. Since M_1 determines the x^n-coefficient, every finite subset J of I with $M_1 \subseteq J$ satisfies $[x^n]\left(\prod_{i\in J} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M_1} \mathbf{a}_i\right)$. Applying this to $J = M_1 \cup M_2$, we obtain

$$[x^n]\left(\prod_{i\in M_1\cup M_2} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M_1} \mathbf{a}_i\right).$$

The same argument (with the roles of M_1 and M_2 swapped) yields

$$[x^n]\left(\prod_{i\in M_2\cup M_1} \mathbf{a}_i\right) = [x^n]\left(\prod_{i\in M_2} \mathbf{a}_i\right).$$

Since $M_1 \cup M_2 = M_2 \cup M_1$, the left hand sides are equal, whence $[x^n] \left(\prod_{i \in M_1} \mathbf{a}_i\right) = [x^n] \left(\prod_{i \in M_2} \mathbf{a}_i\right)$. ☐

Proposition 1.345. *Let $(\mathbf{a}_i)_{i \in I}$ be a finite family of FPSs. Then, the product $\prod_{i \in I} \mathbf{a}_i$ defined according to Definition 1.343 **(b)** equals the finite product $\prod_{i \in I} \mathbf{a}_i$ defined in the usual way (i.e., defined as in any commutative ring).*

Proof. Argue that I itself is a subset of I that determines all coefficients in the product of $(\mathbf{a}_i)_{i \in I}$. See Section A.1 for a detailed proof. ☐

1.12.3 Irrelevance lemmas

Lemma 1.346. *If $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, then $ac \overset{x^n}{\equiv} bd$. In other words, x^n-equivalence is preserved under multiplication.*

Proof. This follows from the Cauchy product formula: the x^m-coefficient of a product ac depends only on the coefficients of a and c up to degree m. ☐

Lemma 1.347. *If $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$ with c and d invertible (i.e., $[x^0]c$ and $[x^0]d$ are units), then $a/c \overset{x^n}{\equiv} b/d$.*

Proof. First show that $c^{-1} \overset{x^n}{\equiv} d^{-1}$ using x^n-equivalence of inverses, then apply Lemma 1.346 to $a \cdot c^{-1}$ and $b \cdot d^{-1}$. ☐

Lemma 1.348. *Let $a, f \in K[[x]]$ be two FPSs. Let $n \in \mathbb{N}$. Assume that*

$$[x^m] f = 0 \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Then,

$$[x^m] (a (1 + f)) = [x^m] a \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Proof. The FPS af is a multiple of f (since $af = fa$). Hence, Lemma 1.132 yields that $[x^m] (af) = 0$ for each $m \in \{0, 1, \ldots, n\}$. Now, for each $m \in \{0, 1, \ldots, n\}$, we have

$$[x^m] (a (1 + f)) = [x^m] (a + af) = [x^m] a + \underbrace{[x^m] (af)}_{=0} = [x^m] a.$$

☐

Lemma 1.349. *Let $a \in K[[x]]$ be an FPS. Let $(f_i)_{i \in J} \in K[[x]]^J$ be a finite family of FPSs. Let $n \in \mathbb{N}$. Assume that each $i \in J$ satisfies*

$$[x^m] (f_i) = 0 \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Then,

$$[x^m] \left(a \prod_{i \in J} (1 + f_i)\right) = [x^m] a \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Proof. This is just Lemma 1.348, applied $|J|$ many times. See Section A.1 for a detailed proof. ☐

1.12.4 Multiplicability criterion

Theorem 1.350. *Let $(f_i)_{i\in I} \in K[[x]]^I$ be a (possibly infinite) summable family of FPSs. Then, the family $(1+f_i)_{i\in I}$ is multipliable.*

Proof. This is an easy consequence of Lemma 1.349. See Section A.1 for a detailed proof. $\square$

Proposition 1.351. *If all but finitely many entries of a family $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ equal 1 (that is, if all but finitely many $i \in I$ satisfy $\mathbf{a}_i = 1$), then this family is multipliable.*

Proof. LTTR. (See Section A.1 for a detailed proof.) $\square$

1.12.5 Further multipliability results

Lemma 1.352. *If every entry of a family $(\mathbf{a}_i)_{i\in I}$ equals 1, then the family is multipliable.*

Proof. The empty set determines every coefficient (the product over $\emptyset$ is 1). $\square$

Lemma 1.353. *Any family indexed by the empty type is multipliable.*

Proof. The empty set determines all coefficients. $\square$

Lemma 1.354. *Any family indexed by a unique (singleton) type is multipliable.*

Proof. The single-element set determines all coefficients. $\square$

Lemma 1.355. *If some $\mathbf{a}_j = 0$ in the family, then the family is multipliable (the product is 0).*

Proof. Any finite set containing j determines all coefficients (the product is 0). $\square$

Lemma 1.356. *If I is a finite set, then any family $(\mathbf{a}_i)_{i\in I}$ is multipliable.*

Proof. I itself determines all coefficients. $\square$

Lemma 1.357. *If I is a finite set and $(\mathbf{a}_i)_{i\in I}$ is a family of FPSs, then the infinite product $\prod_{i\in I} \mathbf{a}_i$ equals the ordinary finite product.*

Proof. This is a special case of Proposition 1.345. $\square$

Lemma 1.358. *If every entry of a family equals 1, then the infinite product equals 1.*

Proof. The empty set is an x^n-approximator for every n, and $\prod_{i\in\emptyset} \mathbf{a}_i = 1$. $\square$

1.12.6 Binary product example

Lemma 1.359. *For each $m \in \mathbb{N}$,*

$$\prod_{i=0}^{m} \left(1 + x^{2^i}\right) = \left(1 - x^{2^{m+1}}\right) \cdot \frac{1}{1-x}.$$

Proof. By induction on m, using the identity $(1-a)(1+a) = 1 - a^2$. $\square$

Lemma 1.360. *The family $\left(1 + x^{2^i}\right)_{i\in\mathbb{N}}$ is multipliable.*

Proof. Apply Theorem 1.350: the family $\left(x^{2^i}\right)_{i\in\mathbb{N}}$ is summable (for each n, at most one i satisfies $2^i = n$ since $i \mapsto 2^i$ is injective). $\square$

Proposition 1.361. *We have*

$$\prod_{i=0}^{\infty}\left(1+x^{2^i}\right) = \frac{1}{1-x}.$$

This relates to the unique binary representation of natural numbers.

Proof. For each n, the set $\{0,1,\ldots,n\}$ determines the x^n-coefficient: any factor $\left(1+x^{2^i}\right)$ with $i > n$ has $2^i > n$, so it does not affect coefficients $\leq n$. The finite product for $i \in \{0,\ldots,n\}$ equals $\left(1-x^{2^{n+1}}\right)/(1-x)$ by Lemma 1.359, and its x^n-coefficient equals that of $1/(1-x)$ since $2^{n+1} > n$. $\square$

1.12.7 x^n-approximators

Definition 1.362. Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a family of FPSs. Let $n \in \mathbb{N}$. An *x^n-approximator* for $(\mathbf{a}_i)_{i\in I}$ means a finite subset M of I that determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i\in I}$. (In other words, M has to determine the x^m-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$ for each $m \in \{0,1,\ldots,n\}$.)

Lemma 1.363. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a multipliable family of FPSs. Let $n \in \mathbb{N}$. Then, there exists an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$.*

Proof. This is an easy consequence of the fact that a union of finitely many finite sets is finite. A detailed proof can be found in Section A.1. $\square$

Proposition 1.364. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a family of FPSs. Let $n \in \mathbb{N}$. Let M be an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$. Then:*

(a) *Every finite subset J of I satisfying $M \subseteq J \subseteq I$ satisfies*

$$\prod_{i\in J}\mathbf{a}_i \overset{x^n}{\equiv} \prod_{i\in M}\mathbf{a}_i.$$

(b) *If the family $(\mathbf{a}_i)_{i\in I}$ is multipliable, then*

$$\prod_{i\in I}\mathbf{a}_i \overset{x^n}{\equiv} \prod_{i\in M}\mathbf{a}_i.$$

Proof. This follows easily from Definition 1.362 and Definition 1.343 **(b)**. See Section A.1 for a detailed proof. $\square$

1.12.8 Properties of infinite products

Proposition 1.365. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a family of FPSs. Let J be a subset of I. Assume that the subfamilies $(\mathbf{a}_i)_{i\in J}$ and $(\mathbf{a}_i)_{i\in I\setminus J}$ are multipliable. Then:*

(a) *The entire family $(\mathbf{a}_i)_{i\in I}$ is multipliable.*

(b) *We have*

$$\prod_{i\in I}\mathbf{a}_i = \left(\prod_{i\in J}\mathbf{a}_i\right)\cdot\left(\prod_{i\in I\setminus J}\mathbf{a}_i\right).$$

Proof. Fix $n \in \mathbb{N}$. Lemma 1.363 (applied to J instead of I) shows that there exists an x^n-approximator U for $(\mathbf{a}_i)_{i\in J}$. Lemma 1.363 shows that there exists an x^n-approximator V for $(\mathbf{a}_i)_{i\in I\setminus J}$. Note that $U \cup V$ is finite. Now, $U \cup V$ is an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$. Hence, the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in I}$ is finitely determined. This proves part **(a)**. Part **(b)** follows using Proposition 1.364 **(b)**.

See Section A.1 for the details. $\square$

Proposition 1.366. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ and $(\mathbf{b}_i)_{i\in I} \in K[[x]]^I$ be two multipliable families of FPSs. Then:*

(a) *The family $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ is multipliable.*

(b) *We have*

$$\prod_{i\in I} (\mathbf{a}_i\mathbf{b}_i) = \left(\prod_{i\in I} \mathbf{a}_i\right) \cdot \left(\prod_{i\in I} \mathbf{b}_i\right).$$

Proof. Fix $n \in \mathbb{N}$. Lemma 1.363 shows that there exists an x^n-approximator U for $(\mathbf{a}_i)_{i\in I}$. Lemma 1.363 (applied to $\mathbf{b}_i$ instead of $\mathbf{a}_i$) shows that there exists an x^n-approximator V for $(\mathbf{b}_i)_{i\in I}$. Note that $U \cup V$ is finite. Now, $U \cup V$ is an x^n-approximator for $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$. From here, proceed as in the proof of Proposition 1.365.

See Section A.1 for the details. $\square$

Proposition 1.367. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ and $(\mathbf{b}_i)_{i\in I} \in K[[x]]^I$ be two multipliable families of FPSs. Assume that the FPS $\mathbf{b}_i$ is invertible for each $i \in I$. Then:*

(a) *The family $\left(\dfrac{\mathbf{a}_i}{\mathbf{b}_i}\right)_{i\in I}$ is multipliable.*

(b) *We have*

$$\prod_{i\in I} \frac{\mathbf{a}_i}{\mathbf{b}_i} = \frac{\prod_{i\in I} \mathbf{a}_i}{\prod_{i\in I} \mathbf{b}_i}.$$

Proof. This is similar to the proof of Proposition 1.366, but using Lemma A.27 (x^n-equivalence of quotients) instead of Lemma 1.346. See Section A.1 for the details. $\square$

1.12.9 Subfamilies of multipliable families

Lemma 1.368. *Let $(\mathbf{a}_i)_{i\in I}$ be a family of invertible FPSs. Let U be an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$. Let $J \subseteq I$. Then $U \cap J$ (viewed as a subset of J) is an x^n-approximator for the subfamily $(\mathbf{a}_i)_{i\in J}$.*

Proof. Let $T \supseteq U \cap J$ be a finite subset of J. Then $T \cup (U \setminus J)$ and $(U \cap J) \cup (U \setminus J) = U$ are both finite supersets of U, so their products are x^n-equivalent to $\prod_{i\in U} \mathbf{a}_i$ by the approximator property. Writing each product as a product over the J-part times a product over $U \setminus J$, and canceling the invertible common factor $\prod_{i\in U\setminus J} \mathbf{a}_i$, we conclude $\prod_{i\in T} \mathbf{a}_i \overset{x^n}{\equiv} \prod_{i\in U\cap J} \mathbf{a}_i$. $\square$

Proposition 1.369. *Let $(\mathbf{a}_i)_{i\in I} \in K[[x]]^I$ be a multipliable family of invertible FPSs. Then, any subfamily of $(\mathbf{a}_i)_{i\in I}$ is multipliable.*

Proof. We must show that $(\mathbf{a}_i)_{i\in J}$ is multipliable whenever $J \subseteq I$. The idea is to show that if U is an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$, then $U \cap J$ determines the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ (and, in fact, is an x^n-approximator for $(\mathbf{a}_i)_{i\in J}$). This uses Lemma A.28 and relies on the invertibility of $\prod_{i\in U\setminus J} \mathbf{a}_i$; this is why the FPS $\mathbf{a}_i$ are required to be invertible in the proposition.

See Section A.1 for the details. $\square$

1.12.10 Reindexing

Proposition 1.370. *Let S and T be two sets. Let $f : S \to T$ be a bijection. Let $(\mathbf{a}_t)_{t\in T} \in K[[x]]^T$ be a multipliable family of FPSs. Then,*

$$\prod_{t\in T} \mathbf{a}_t = \prod_{s\in S} \mathbf{a}_{f(s)}$$

(and, in particular, the product on the right hand side is well-defined, i.e., the family $\left(\mathbf{a}_{f(s)}\right)_{s\in S}$ is multipliable).

Proof. This is a trivial consequence of the definitions of multipliability and infinite products. $\square$

1.12.11 Breaking products into subproducts

Lemma 1.371. *Let $n \in \mathbb{N}$ and let V be a finite set. Let $c, d : V \to K[[x]]$ be two families of FPSs such that $c_v \overset{x^n}{\equiv} d_v$ for each $v \in V$. Then*

$$\prod_{v\in V} c_v \overset{x^n}{\equiv} \prod_{v\in V} d_v.$$

Proof. By induction on $|V|$. The base case $V = \emptyset$ is trivial. In the inductive step, use x^n-equivalence of products: $\prod_{v\in V\cup\{w\}} c_v = c_w \cdot \prod_{v\in V} c_v$ and similarly for d, then apply the fact that x^n-equivalence is preserved under multiplication. $\square$

Proposition 1.372. *Let $(\mathbf{a}_s)_{s\in S} \in K[[x]]^S$ be a multipliable family of FPSs. Let W be a set. Let $f : S \to W$ be a map. Assume that for each $w \in W$, the family $(\mathbf{a}_s)_{s\in S,\ f(s)=w}$ is multipliable. Then,*

$$\prod_{s\in S} \mathbf{a}_s = \prod_{w\in W} \prod_{\substack{s\in S;\\ f(s)=w}} \mathbf{a}_s.$$

(In particular, the right hand side is well-defined – i.e., the family $\left(\prod_{\substack{s\in S;\\ f(s)=w}} \mathbf{a}_s\right)_{w\in W}$ is multipliable.)

Note: The Lean formalization of this proposition (without sorry) requires the additional assumption that all FPSs are invertible (i.e., $[x^0]\mathbf{a}_s$ is a unit for each s), which is used through Proposition 1.369 to guarantee multipliability of subfamilies. The versions without this invertibility hypothesis are formalized with sorry.

Proof. This can be derived from the analogous property of finite products, since all coefficients in a multipliable product are finitely determined (conveniently using x^n-approximators, which determine several coefficients at the same time). See Section A.1 for the details. $\square$

1.12.12 Fubini rule for infinite products

Proposition 1.373 (Fubini rule for infinite products of FPSs)**.** *Let I and J be two sets. Let $\left(\mathbf{a}_{(i,j)}\right)_{(i,j)\in I\times J} \in K[[x]]^{I\times J}$ be a multipliable family of FPSs. Assume that for each $i \in I$, the family $\left(\mathbf{a}_{(i,j)}\right)_{j\in J}$ is multipliable. Assume that for each $j \in J$, the family $\left(\mathbf{a}_{(i,j)}\right)_{i\in I}$ is multipliable. Then,*

$$\prod_{i\in I}\prod_{j\in J} \mathbf{a}_{(i,j)} = \prod_{(i,j)\in I\times J} \mathbf{a}_{(i,j)} = \prod_{j\in J}\prod_{i\in I} \mathbf{a}_{(i,j)}.$$

(In particular, all the products appearing in this equality are well-defined.)

Proof. The first equality follows by applying Proposition 1.372 to $S = I \times J$ and $W = I$ and $f(i,j) = i$ (and appropriately reindexing the products). The second equality is analogous. See Section A.1 for the details. $\square$

Proposition 1.374 (Fubini rule for infinite products, invertible case)**.** *Let I and J be two sets. Let $\left(\mathbf{a}_{(i,j)}\right)_{(i,j)\in I\times J} \in K[[x]]^{I\times J}$ be a multipliable family of invertible FPSs. Then,*

$$\prod_{i\in I}\ \prod_{j\in J} \mathbf{a}_{(i,j)} = \prod_{(i,j)\in I\times J} \mathbf{a}_{(i,j)} = \prod_{j\in J}\ \prod_{i\in I} \mathbf{a}_{(i,j)}.$$

(In particular, all the products appearing in this equality are well-defined.)

Proof. See Section A.1. $\square$

1.13 Product rules (generalized distributive laws)

Proposition 1.375. *Let L be a commutative ring. For every $n \in \mathbb{N}$, let $[n]$ denote the set $\{1, 2, \ldots, n\}$.*

Let $n \in \mathbb{N}$. For every $i \in [n]$, let $p_{i,1}, p_{i,2}, \ldots, p_{i,m_i}$ be finitely many elements of L. Then,

$$\prod_{i=1}^{n} \sum_{k=1}^{m_i} p_{i,k} = \sum_{(k_1,k_2,\ldots,k_n)\in[m_1]\times[m_2]\times\cdots\times[m_n]} \prod_{i=1}^{n} p_{i,k_i}. \tag{1.28}$$

Proof. The idea is to reduce it to the case $n = 2$ by induction, then to use the discrete Fubini rule. $\square$

Proposition 1.376. *For every $n \in \mathbb{N}$, let $[n]$ denote the set $\{1, 2, \ldots, n\}$.*

Let $n \in \mathbb{N}$. For every $i \in [n]$, let $(p_{i,k})_{k\in S_i}$ be a summable family of elements of $K[[x]]$. Then,

$$\prod_{i=1}^{n} \sum_{k\in S_i} p_{i,k} = \sum_{(k_1,k_2,\ldots,k_n)\in S_1\times S_2\times\cdots\times S_n} \prod_{i=1}^{n} p_{i,k_i}. \tag{1.29}$$

In particular, the family $\left(\prod_{i=1}^{n} p_{i,k_i}\right)_{(k_1,k_2,\ldots,k_n)\in S_1\times S_2\times\cdots\times S_n}$ is summable.

Proof. Same method as for Proposition 1.375, but now using the discrete Fubini rule for infinite sums. The proof is by induction on n:

- Base case ($n = 0$): Both sides equal 1.
- Inductive step: Split the product into the first n factors and the last factor, apply the induction hypothesis on the first n, then apply the Cauchy product formula for infinite sums.

Summability of the product family is established by showing that for each coefficient d, only finitely many functions f give a nonzero contribution (using finite support of each summable family in the discrete topology). $\square$

Helpers for the finite product rule

Lemma 1.377. *Let $n \in \mathbb{N}$. Let $(p_{i,k})$ be a family indexed by $\{1, 2, \ldots, n\} \times S_i$, and let f be a choice function selecting $f(i) \in S_i$ for each i. If for some $j \in \{1, 2, \ldots, n\}$, the FPS $p_{j,f(j)}$ satisfies $[x^m](p_{j,f(j)}) = 0$ for all $m \leq d$, then $[x^d]\left(\prod_i p_{i,f(i)}\right) = 0$.*

Proof. The product $\prod_i p_{i,f(i)}$ is divisible by $p_{j,f(j)}$, which has order $> d$. Since the order of a product is at least the sum of the orders of the factors, the product also has order $> d$. □

Lemma 1.378. ***(Summability of product families.)*** *Assume K has the discrete topology. Let $n \in \mathbb{N}$, and for each $i \in \{1, 2, \ldots, n\}$, let $(p_{i,k})_{k \in S_i}$ be a summable family in $K[[x]]$. Then the family $\left(\prod_i p_{i,f(i)}\right)_{f \in \prod_i S_i}$ is summable.*

Proof. For each coefficient d, define $T_i = \bigcup_{m \leq d} \operatorname{supp}([x^m] \circ p_i)$. Each T_i is finite (by discrete summability). If $f(i) \notin T_i$ for some i, then $[x^d](\prod_j p_{j,f(j)}) = 0$ by Lemma 1.377. The set $\{f \mid \forall i,\ f(i) \in T_i\}$ is finite (product of finite sets), so the support at coefficient d is finite. □

Lemma 1.379. ***(Product of two summable families.)*** *Assume K has the discrete topology. If $(f_\alpha)_{\alpha \in A}$ and $(g_\beta)_{\beta \in B}$ are summable families in $K[[x]]$, then the family $(f_\alpha \cdot g_\beta)_{(\alpha,\beta) \in A \times B}$ is summable.*

Proof. For each coefficient d, the support of $(\alpha, \beta) \mapsto [x^d](f_\alpha g_\beta)$ is contained in the finite product of coefficient supports, using the Cauchy product formula and the finiteness of supports. □

Definition 1.380. **(a)** A sequence $(k_1, k_2, k_3, \ldots)$ is said to be *essentially finite* if all but finitely many $i \in \{1, 2, 3, \ldots\}$ satisfy $k_i = 0$.

(b) A family $(k_i)_{i \in I}$ is said to be *essentially finite* if all but finitely many $i \in I$ satisfy $k_i = 0$.

Proposition 1.381. *Let $S_1, S_2, S_3, \ldots$ be infinitely many sets that all contain the number 0. Set*

$$\overline{S} = \{(i, k) \mid i \in \{1, 2, 3, \ldots\} \text{ and } k \in S_i \text{ and } k \neq 0\}.$$

For any $i \in \{1, 2, 3, \ldots\}$ and any $k \in S_i$, let $p_{i,k}$ be an element of $K[[x]]$. Assume that

$$p_{i,0} = 1 \qquad \text{for any } i \in \{1, 2, 3, \ldots\}. \tag{1.30}$$

Assume further that the family $(p_{i,k})_{(i,k) \in \overline{S}}$ is summable. Then, the product $\prod_{i=1}^{\infty} \sum_{k \in S_i} p_{i,k}$ is well-defined (i.e., the family $(p_{i,k})_{k \in S_i}$ is summable for each $i \in \{1, 2, 3, \ldots\}$, and the family $\left(\sum_{k \in S_i} p_{i,k}\right)_{i \in \{1,2,3,\ldots\}}$ is multipliable), and we have

$$\prod_{i=1}^{\infty} \sum_{k \in S_i} p_{i,k} = \sum_{\substack{(k_1, k_2, k_3, \ldots) \in S_1 \times S_2 \times S_3 \times \cdots \\ \text{is essentially finite}}} \prod_{i=1}^{\infty} p_{i,k_i}. \tag{1.31}$$

In particular, the family $(\prod_{i=1}^{\infty} p_{i,k_i})_{(k_1,k_2,k_3,\ldots) \in S_1 \times S_2 \times S_3 \times \cdots \text{ is essentially finite}}$ is summable.

Proof. This is the particular case of Proposition 1.382 for $I = \{1, 2, 3, \ldots\}$. □

Proposition 1.382. *Let I be a set. For any $i \in I$, let S_i be a set that contains the number 0. Set*

$$\overline{S} = \{(i, k) \mid i \in I \text{ and } k \in S_i \text{ and } k \neq 0\}.$$

For any $i \in I$ and any $k \in S_i$, let $p_{i,k}$ be an element of $K[[x]]$. Assume that

$$p_{i,0} = 1 \qquad \text{for any } i \in I. \tag{1.32}$$

Assume further that the family $(p_{i,k})_{(i,k)\in\overline{S}}$ is summable. Then, the product $\prod_{i\in I} \sum_{k\in S_i} p_{i,k}$ is well-defined (i.e., the family $(p_{i,k})_{k\in S_i}$ is summable for each $i \in I$, and the family $\left(\sum_{k\in S_i} p_{i,k}\right)_{i\in I}$ is multipliable), and we have

$$\prod_{i\in I} \sum_{k\in S_i} p_{i,k} = \sum_{\substack{(k_i)_{i\in I}\in\prod_{i\in I} S_i \\ \text{is essentially finite}}} \prod_{i\in I} p_{i,k_i}. \tag{1.33}$$

In particular, the family $\left(\prod_{i\in I} p_{i,k_i}\right)_{(k_i)_{i\in I}\in\prod_{i\in I} S_i \text{ is essentially finite}}$ is summable.

Proof. The proof is a long-winded reduction to the finite case. It proceeds by showing both sides have the same n-th coefficient for each n.

Setup. For each $m \in \mathbb{N}$, define $T'_n = \{(i,k) \in \overline{S} \mid \exists m \le n,\ [x^m](p_{i,k}) \neq 0\}$. This set is finite by coefficient-wise summability. Let $I_n = \{i \in I \mid \exists k : (i,k) \in T'_n\}$. Then I_n is finite.

RHS reduction (Claim 8). For the RHS, only essentially finite functions f with "graph" contained in T'_n can contribute to the n-th coefficient. If some $(j, f(j)) \notin T'_n$ with $f(j) \neq 0$, then $[x^m](p_{j,f(j)}) = 0$ for all $m \le n$, so $[x^n](\prod_i p_{i,f(i)}) = 0$ (by the finite product coefficient lemma). The set of such functions is finite (since their graphs inject into the finite powerset of T'_n).

LHS reduction (Claim 10). For the LHS, write $\sum_k p_{i,k} = 1 + \sum_{k\neq 0} p_{i,k}$. For $i \notin I_n$, the sum $\sum_{k\neq 0} p_{i,k}$ has all coefficients up to n equal to 0. By the irrelevance lemma (Lemma 1.384), the factors for $i \notin I_n$ do not affect the n-th coefficient. Thus $[x^n](\prod_i \sum_k p_{i,k}) = [x^n](\prod_{i\in I_n} \sum_k p_{i,k})$.

Finite product rule (Claim 11). Since I_n is finite, we can apply Proposition 1.383: $\prod_{i\in I_n} \sum_k p_{i,k} = \sum_{(k_i)_{i\in I_n}\in\prod_{i\in I_n} S_i} \prod_{i\in I_n} p_{i,k_i}$.

Combining Claims 8, 10, and 11 shows both sides have equal n-th coefficients. $\square$

Proposition 1.383. *Let N be a finite set. For any $i \in N$, let $(p_{i,k})_{k\in S_i}$ be a summable family of elements of $K[[x]]$. Then,*

$$\prod_{i\in N} \sum_{k\in S_i} p_{i,k} = \sum_{(k_i)_{i\in N}\in\prod_{i\in N} S_i} \prod_{i\in N} p_{i,k_i}. \tag{1.34}$$

In particular, the family $\left(\prod_{i\in N} p_{i,k_i}\right)_{(k_i)_{i\in N}\in\prod_{i\in N} S_i}$ is summable.

Proof. This is just Proposition 1.376, with the indexing set $\{1, 2, \ldots, n\}$ replaced by N. It can be proved by reindexing the products (using a bijection between N and $\{1, 2, \ldots, |N|\}$) or directly by induction on $|N|$. $\square$

Lemma 1.384. *Let $a \in K[[x]]$ be an FPS. Let $(f_i)_{i\in J} \in K[[x]]^J$ be a summable family of FPSs. Let $n \in \mathbb{N}$. Assume that each $i \in J$ satisfies*

$$[x^m](f_i) = 0 \qquad \text{for each } m \in \{0, 1, \ldots, n\}. \tag{1.35}$$

Then,

$$[x^m]\left(a \prod_{i\in J} (1 + f_i)\right) = [x^m]\, a \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Proof. The idea is to argue that the first $n+1$ coefficients of $a\prod_{i\in J}(1+f_i)$ agree with those of a.

First, one shows that for any finite subset s of J, if each f_i has $[x^m](f_i)=0$ for all $m\le n$, then $[x^m](\prod_{i\in s}(1+f_i))=[x^m](1)$ for all $m\le n$. This uses the expansion $\prod_{i\in s}(1+f_i)=\sum_{t\subseteq s}\prod_{i\in t}f_i$; for $t\ne\emptyset$, each $\prod_{i\in t}f_i$ has order $>n$, so its m-th coefficient is zero.

Then, for the infinite product, the result follows by taking limits: $\prod_{i\in J}(1+f_i)$ is the limit of the partial products $\prod_{i\in s}(1+f_i)$, and since $[x^m]$ is continuous, each partial product gives $[x^m](\prod_{i\in s}(1+f_i))=[x^m](1)$.

Finally, the coefficient of $a\cdot\prod_{i\in J}(1+f_i)$ uses the Cauchy product formula, which only involves coefficients of $\prod_{i\in J}(1+f_i)$ up to degree $m\le n$, all of which equal the corresponding coefficients of 1. Hence $[x^m](a\cdot\prod_{i\in J}(1+f_i))=[x^m](a\cdot 1)=[x^m](a)$.

If the family $(1+f_i)_{i\in J}$ is not multipliable, the infinite product defaults to 1 by convention, and the result holds trivially. $\square$

1.13.1 Helper lemmas for the infinite product rule

These lemmas implement the key technical steps in the proof of Proposition 1.382. They require K to have the discrete topology.

Lemma 1.385. ***(Finiteness of T'_n.)*** *Under the assumptions of Proposition 1.382 (with discrete K), define $T'_n=\{(i,k)\in\overline{S}\mid\exists m\le n,\ [x^m](p_{i,k})\ne 0\}$. Then T'_n is a finite set.*

Proof. We have $T'_n=\bigcup_{m\le n}\operatorname{supp}((i,k)\mapsto[x^m](p_{i,k}))$. Each support set is finite by coefficient-wise summability (summable families over a discrete topological space have finite support). A finite union of finite sets is finite. $\square$

Lemma 1.386. *If $j\in S$ and $[x^m](f_j)=0$ for all $m\le n$, then $[x^m]\big(\prod_{i\in S}f_i\big)=0$ for all $m\le n$.*

Proof. Write $\prod_{i\in S}f_i=f_j\cdot\prod_{i\in S\setminus\{j\}}f_i$. For each $m\le n$, use the Cauchy product formula: each term involves $[x^a](f_j)$ with $a\le m\le n$, which is 0 by hypothesis. $\square$

Lemma 1.387. *Let T be a finite set of pairs (i,k) with $k\ne 0$. Then the set of essentially finite functions $f:I\to\prod_i S_i$ whose "graph" $\{(i,f(i)):f(i)\ne 0\}$ is contained in T is finite.*

Proof. Each such function is determined by its graph, which is a subset of T. The map sending f to its graph is injective (since the graph determines f on nonzero values, and the rest are 0). Since the powerset of T is finite, the image is finite, so the set of such functions is finite. $\square$

Lemma 1.388. ***(Finite irrelevance for extra high-order factors.)*** *Let $s\subseteq t$ be finite sets of indices, and let $g:I\to K[[x]]$. Assume that for each $i\in t\setminus s$, we have $[x^m](g_i)=0$ for all $m\le n$. Then $[x^n]\big(\prod_{i\in t}(1+g_i)\big)=[x^n]\big(\prod_{i\in s}(1+g_i)\big)$.*

Proof. Write $\prod_{i\in t}(1+g_i)=\prod_{i\in s}(1+g_i)\cdot\prod_{i\in t\setminus s}(1+g_i)$. For the second factor, each g_i $(i\in t\setminus s)$ has order $>n$, so by expanding $\prod_{i\in t\setminus s}(1+g_i)$ via the powerset formula, all terms except the empty product have order $>n$. Hence $[x^m]\Big(\prod_{i\in t\setminus s}(1+g_i)\Big)=[x^m](1)$ for all $m\le n$, and the Cauchy product formula gives the result. $\square$

Lemma 1.389. ***(Infinite irrelevance: reduction to finite product.)*** *Assume K has the discrete topology. Let $(g_i)_{i\in I}$ be a family such that $(1+g_i)_{i\in I}$ is multipliable, and let $I_n\subseteq I$ be a finite set such that $[x^m](g_i)=0$ for all $i\notin I_n$ and all $m\le n$. Then*

$$[x^n]\left(\prod_{i\in I}(1+g_i)\right)=[x^n]\left(\prod_{i\in I_n}(1+g_i)\right).$$

Proof. The infinite product $\prod_{i\in I}(1+g_i)$ is the limit of partial products $\prod_{i\in s}(1+g_i)$ over finite sets s. By Lemma 1.388, for any $s \supseteq I_n$, the n-th coefficient of $\prod_{i\in s}(1+g_i)$ equals that of $\prod_{i\in I_n}(1+g_i)$. By continuity of $[x^n]$ and the eventual constancy, the limit also has this n-th coefficient. $\square$

1.13.2 Helper lemmas from the Lean formalization

Lemma 1.390. ***(Summability of fibers.)*** *Let I be a set. For any $i \in I$, let S_i be a set containing 0. For any $i \in I$ and $k \in S_i$, let $p_{i,k} \in K[[x]]$. Assume that the family $(p_{i,k})_{(i,k)\in\overline{S}}$ is summable (where $\overline{S} = \{(i,k) : k \neq 0\}$). Then, for each $i \in I$, the family $(p_{i,k})_{k\in S_i}$ is summable.*

Proof. The subfamily $(p_{i,k})_{k\in S_i, k\neq 0}$ is summable because it injects into the summable family $(p_{i,k})_{(i,k)\in\overline{S}}$ via $(i,\cdot)$. Adding the single term $p_{i,0}$ preserves summability. Formally, we reduce to coefficient-wise summability and split each coefficient sum at $k = 0$. $\square$

Lemma 1.391. ***(Multipliability of the sum family.)*** *Under the same assumptions as Lemma 1.390, and assuming $p_{i,0} = 1$ for all $i \in I$, the family $\left(\sum_{k\in S_i} p_{i,k}\right)_{i\in I}$ is multipliable.*

Proof. Write $\sum_k p_{i,k} = 1 + \sum_{k\neq 0} p_{i,k}$. We need to show $(1+g_i)_{i\in I}$ is multipliable, where $g_i = \sum_{k\neq 0} p_{i,k}$. By the multipliability criterion (Theorem 1.350), it suffices to show that $\sum_{s\in\mathcal{F}(I)} \prod_{i\in s} g_i$ is summable, where $\mathcal{F}(I)$ denotes the set of all finite subsets of I. This is proved coefficient-wise: for each n, the support of $s \mapsto [x^n](\prod_{i\in s} g_i)$ is contained in the set of finite subsets s with $s \subseteq I_n$ (the finite set of indices contributing to coefficients up to n), which is finite. $\square$

Lemma 1.392. ***(Summability of the RHS.)*** *Under the assumptions of Proposition 1.382 (with discrete K), the family $\left(\prod_{i\in I} p_{i,k_i}\right)_{(k_i)\text{ ess. fin.}}$ is summable.*

Proof. For each coefficient n, define $T'_n = \{(i,k) \in \overline{S} \mid \exists m \leq n,\ [x^m](p_{i,k}) \neq 0\}$. This is finite by Lemma 1.385. The support of $f \mapsto [x^n](\prod_{i\in I} p_{i,f(i)})$ is contained in the set of essentially finite functions whose "graph" $\{(i, f(i)) : f(i) \neq 0\}$ is a subset of T'_n (if $(j, f(j)) \notin T'_n$ for some j with $f(j) \neq 0$, then $[x^n](\prod_{i\in I} p_{i,f(i)}) = 0$ by Lemma 1.386). This set of functions is finite by Lemma 1.387. $\square$

1.13.3 Binary product rule

Proposition 1.393. *Assume K has the discrete topology. Let $(a_i)_{i\in\mathbb{N}}$ be a summable family in $K[[x]]$. Then*

$$\prod_{i\in\mathbb{N}}(1+a_i) = \sum_{S\subseteq\mathbb{N},\ S\text{ finite}} \prod_{i\in S} a_i.$$

In particular, the right-hand side is summable.

Proof. The identity $\prod_{i\in\mathbb{N}}(1+f_i) = \sum_{s\subseteq\mathbb{N},\ s\text{ finite}} \prod_{i\in s} f_i$ holds provided the right-hand side is summable. For the summability: for each coefficient d, define $T = \bigcup_{m\leq d} \operatorname{supp}([x^m] \circ a)$, which is finite by discrete summability. Any finset s not contained in T contributes 0 to coefficient d (since some factor a_i with $i \notin T$ has all low coefficients zero). The set of finsets contained in T is finite. $\square$

Proposition 1.394. *Alternative formulation: under the same assumptions,*

$$\prod_{i\in\mathbb{N}}(1+a_i)=\sum_{J\subseteq\mathbb{N},\ J\ \textit{finite}}\ \prod_{i\in J}^{\text{fin}}a_i,$$

where the RHS sums over all finite subsets J of $\mathbb{N}$ and uses the finitary product $\prod^{\text{fin}}$.

Proof. This follows from Proposition 1.393 by observing that both formulations sum over the same collection of finite subsets of $\mathbb{N}$. ☐

1.13.4 Another example

Proposition 1.395 (Euler)**.** *We have*

$$\prod_{i>0}\left(1-x^{2i-1}\right)^{-1}=\prod_{k>0}\left(1+x^k\right).$$

Proof. For each $k>0$, we have $1+x^k=\dfrac{1-x^{2k}}{1-x^k}$ (since $1-x^{2k}=\left(1-x^k\right)\left(1+x^k\right)$). Thus,

$$\begin{aligned}\prod_{k>0}\left(1+x^k\right)&=\prod_{k>0}\frac{1-x^{2k}}{1-x^k}=\frac{\prod_{k>0}\left(1-x^{2k}\right)}{\prod_{k>0}\left(1-x^k\right)}\\&=\frac{1}{(1-x^1)(1-x^3)(1-x^5)(1-x^7)\cdots}\\&=\prod_{i>0}\left(1-x^{2i-1}\right)^{-1},\end{aligned}$$

where the even factors cancel. The cancellation is justified by the fact that the family $\left(1-x^k\right)_{k>0}$ factors as a product over even and odd indices, by the partition of positive integers into evens and odds (by Proposition 1.365).

Lean proof strategy. The Lean proof proceeds differently: both sides are shown to equal the same partition-theoretic generating function. The RHS equals $\sum_n d_n\cdot x^n$ (generating function for partitions into distinct parts, i.e., parts of multiplicity <2). The LHS equals $\sum_n o_n\cdot x^n$ (generating function for partitions into odd parts). These are equal by Glaisher's theorem. ☐

Theorem 1.396 (Euler)**.** *Let $n\in\mathbb{N}$. Then, $d_n=o_n$, where*

- *d_n is the number of ways to write n as a sum of distinct positive integers, without regard to the order;*
- *o_n is the number of ways to write n as a sum of (not necessarily distinct) odd positive integers, without regard to the order.*

Proof. Proposition 1.395 shows that the generating functions for d_n and o_n are equal, hence $d_n=o_n$ for all n. ☐

Proposition 1.397. *Alternative formulation of Euler's identity: indexing the left-hand side over odd positive integers rather than all positive integers,*

$$\prod_{\substack{i\in\mathbb{N}^+\\ i\ \textit{odd}}}\left(1-x^i\right)^{-1}=\prod_{k>0}\left(1+x^k\right).$$

Proof. This follows from Proposition 1.395 by reindexing the product using the bijection $i\mapsto 2i-1$ between $\mathbb{N}^+$ and the odd positive integers. ☐

1.13.5 Infinite products and substitution

Lemma 1.398. *For any $a \in K$, the rescaling map* rescale$(a) : K[[x]] \to K[[x]]$ *defined by $f(x) \mapsto f(ax)$ is continuous (with respect to the coefficient-wise topology on $K[[x]]$).*

Proof. The n-th coefficient of rescale$(a)(f)$ is $a^n \cdot [x^n]f$. This is continuous because $[x^n]$ is continuous (it is a projection) and scalar multiplication by a^n is continuous. $\square$

Proposition 1.399. *If $(f_i)_{i\in I} \in K[[x]]^I$ is a multipliable family of FPSs, and if $g \in K[[x]]$ is an FPS satisfying $[x^0]\, g = 0$, then the family $(f_i \circ g)_{i\in I} \in K[[x]]^I$ is multipliable as well and we have $\left(\prod_{i\in I} f_i\right) \circ g = \prod_{i\in I} (f_i \circ g)$.*

Proof. The Lean formalization proves the special case of rescaling $g = ax$ (i.e., substituting ax for x). The proof uses Lemma 1.398 (the rescale map is continuous) together with the fact that it is a ring homomorphism, so it commutes with infinite products. $\square$

1.13.6 Exponentials, logarithms and infinite products

Proposition 1.400. *Assume that K is a commutative $\mathbb{Q}$-algebra. Let $(f_i)_{i\in I} \in K[[x]]^I$ be a summable family of FPSs in $K[[x]]_0$. Then, $(\operatorname{Exp} f_i)_{i\in I}$ is a multipliable family of FPS in $K[[x]]_1$, and we have $\sum_{i\in I} f_i \in K[[x]]_0$ and*

$$\operatorname{Exp}\left(\sum_{i\in I} f_i\right) = \prod_{i\in I} (\operatorname{Exp} f_i).$$

Proof. The Exp map converts infinite sums in $K[[x]]_0$ to infinite products in $K[[x]]_1$. This is because Exp is a continuous ring homomorphism from $K[[x]]_0$ (with addition) to $K[[x]]_1$ (with multiplication), and therefore commutes with infinite sums and products. $\square$

Proposition 1.401. *Assume that K is a commutative $\mathbb{Q}$-algebra. Let $(f_i)_{i\in I} \in K[[x]]^I$ be a multipliable family of FPSs in $K[[x]]_1$. Then, $(\operatorname{Log} f_i)_{i\in I}$ is a summable family of FPS in $K[[x]]_0$, and we have $\prod\limits_{i\in I} f_i \in K[[x]]_1$ and*

$$\operatorname{Log}\left(\prod_{i\in I} f_i\right) = \sum_{i\in I} (\operatorname{Log} f_i).$$

Proof. The Log map converts infinite products in $K[[x]]_1$ to infinite sums in $K[[x]]_0$. This follows from the fact that Log is a continuous ring homomorphism from $K[[x]]_1$ (with multiplication) to $K[[x]]_0$ (with addition), and therefore commutes with infinite products and sums. $\square$

1.14 The generating function of a weighted set

1.14.1 The theory

Definition 1.402. (a) A *weighted set* is a set A equipped with a function $w : A \to \mathbb{N}$, which is called the *weight function* of this weighted set. For each $a \in A$, the value $w(a)$ is denoted $|a|$ and is called the *weight* of a (in our weighted set).

Usually, instead of explicitly specifying the weight function w as a function, we will simply specify the weight $|a|$ for each $a \in A$. The weighted set consisting of the set A and the weight function w will be called (A, w) or simply A when the weight function w is clear from the context.

(b) A weighted set A is said to be *finite-type* if for each $n \in \mathbb{N}$, there are only finitely many $a \in A$ having weight $|a| = n$.

(c) If A is a finite-type weighted set, then the *weight generating function* of A is defined to be the FPS

$$\sum_{a \in A} x^{|a|} = \sum_{n \in \mathbb{N}} (\# \text{ of } a \in A \text{ having weight } n) \cdot x^n \in \mathbb{Z}[[x]].$$

This FPS is denoted by $\overline{A}$.

(d) An *isomorphism* between two weighted sets A and B means a bijection $\rho : A \to B$ that preserves the weight (i.e., each $a \in A$ satisfies $|\rho(a)| = |a|$).

(e) We say that two weighted sets A and B are *isomorphic* (this is written $A \cong B$) if there exists an isomorphism between A and B.

Proposition 1.403. *Let A and B be two isomorphic finite-type weighted sets. Then, $\overline{A} = \overline{B}$.*

Proof. The weighted sets A and B are isomorphic; thus, there exists an isomorphism $\rho : A \to B$. Consider this ρ. Then, ρ is a bijection and preserves the weight (since ρ is an isomorphism of weighted sets). The latter property says that we have

$$|\rho(a)| = |a| \qquad \text{for each } a \in A. \tag{1.36}$$

Now, the definition of $\overline{B}$ yields

$$\overline{B} = \sum_{b \in B} x^{|b|} = \sum_{a \in A} \underbrace{x^{|\rho(a)|}}_{\substack{= x^{|a|} \\ \text{(by the weight-preservation} \\ \text{property above)}}} \qquad \left(\begin{array}{c}\text{here, we have substituted } \rho(a) \text{ for } b \\ \text{in the sum, since } \rho \text{ is a bijection}\end{array}\right)$$
$$= \sum_{a \in A} x^{|a|}.$$

Comparing this with

$$\overline{A} = \sum_{a \in A} x^{|a|} \qquad \left(\text{by the definition of } \overline{A}\right),$$

we obtain $\overline{A} = \overline{B}$. $\square$

Definition 1.404. Let A and B be two weighted sets. Then, the weighted set $A + B$ is defined to be the disjoint union of A and B, with the weight function inherited from A and B (meaning that each element of A has the same weight that it had in A, and each element of B has the same weight that it had in B). Formally speaking, this means that $A + B$ is the set $(\{0\} \times A) \cup (\{1\} \times B)$, with the weight function given by

$$|(0, a)| = |a| \qquad \text{for each } a \in A \tag{1.37}$$

and

$$|(1, b)| = |b| \qquad \text{for each } b \in B. \tag{1.38}$$

Proposition 1.405. *Let A and B be two finite-type weighted sets. Then, $A + B$ is finite-type, too, and satisfies $\overline{A + B} = \overline{A} + \overline{B}$.*

Proof. The formal definition of $A+B$ says that $A+B = (\{0\}\times A) \cup (\{1\}\times B)$. Thus, the set $A+B$ is the union of the two disjoint sets $\{0\}\times A$ and $\{1\}\times B$ (indeed, these two sets are clearly disjoint, since $0 \neq 1$).

Let us first check that $A+B$ is finite-type.

For each $n \in \mathbb{N}$, there are only finitely many $a \in A$ having weight $|a| = n$ (since A is finite-type), and there are only finitely many $b \in B$ having weight $|b| = n$ (since B is finite-type). Hence, there are only finitely many $c \in A+B$ having weight $|c| = n$ (because any such c either has the form $(0,a)$ for some $a \in A$ having weight $|a| = n$, or has the form $(1,b)$ for some $b \in B$ having weight $|b| = n$). In other words, the weighted set $A+B$ is finite-type.

Let us now check that $\overline{A+B} = \overline{A} + \overline{B}$. Indeed, the definition of $\overline{A+B}$ yields

$$\begin{aligned}
\overline{A+B} &= \sum_{c\in A+B} x^{|c|} \\
&= \underbrace{\sum_{c\in\{0\}\times A} x^{|c|}}_{=\sum_{a\in A} x^{|(0,a)|}} + \underbrace{\sum_{c\in\{1\}\times B} x^{|c|}}_{=\sum_{b\in B} x^{|(1,b)|}} \\
&= \sum_{a\in A} \underbrace{x^{|(0,a)|}}_{\substack{=x^{|a|} \\ \text{(by the definition} \\ \text{of the weight on } A+B)}} + \sum_{b\in B} \underbrace{x^{|(1,b)|}}_{\substack{=x^{|b|} \\ \text{(by the definition} \\ \text{of the weight on } A+B)}} = \underbrace{\sum_{a\in A} x^{|a|}}_{=\overline{A}} + \underbrace{\sum_{b\in B} x^{|b|}}_{=\overline{B}} = \overline{A} + \overline{B}.
\end{aligned}$$

$\square$

Definition 1.406. Let A and B be two weighted sets. Then, the weighted set $A\times B$ is defined to be the Cartesian product of the sets A and B (that is, the set $\{(a,b) \mid a \in A \text{ and } b \in B\}$), with the weight function defined as follows: For any $(a,b) \in A\times B$, we set

$$|(a,b)| = |a| + |b|. \tag{1.39}$$

Proposition 1.407. *Let A and B be two finite-type weighted sets. Then, $A\times B$ is finite-type, too, and satisfies $\overline{A\times B} = \overline{A}\cdot\overline{B}$.*

Proof. The proof that $A\times B$ is finite-type is easy: Let $n \in \mathbb{N}$. We must prove that there are only finitely many pairs $(a,b) \in A\times B$ having weight $|(a,b)| = n$. Since $|(a,b)| = |a| + |b|$, these pairs have the property that $a \in A$ has weight k for some $k \in \{0,1,\ldots,n\}$, and that $b \in B$ has weight $n-k$ for the same k. This leaves only finitely many options for k, only finitely many options for a (since A is finite-type and thus has only finitely many elements of weight k), and only finitely many options for b (since B is finite-type and thus has only finitely many elements of $n-k$). Altogether, we thus obtain only finitely many options for (a,b).

Now, the definition of $\overline{A\times B}$ yields

$$\begin{aligned}
\overline{A\times B} &= \sum_{(a,b)\in A\times B} \underbrace{x^{|(a,b)|}}_{\substack{=x^{|a|+|b|} \\ \text{(by the definition} \\ \text{of the weight on } A\times B)}} \\
&= \sum_{(a,b)\in A\times B} \underbrace{x^{|a|+|b|}}_{=x^{|a|}\cdot x^{|b|}} \\
&= \sum_{(a,b)\in A\times B} x^{|a|}\cdot x^{|b|} = \underbrace{\left(\sum_{a\in A} x^{|a|}\right)}_{=\overline{A}} \underbrace{\left(\sum_{b\in B} x^{|b|}\right)}_{=\overline{B}} = \overline{A}\cdot\overline{B}.
\end{aligned}$$

□

The weight function on the Cartesian product $A_1 \times A_2 \times \cdots \times A_k$ of k weighted sets is defined by

$$|(a_1, a_2, \ldots, a_k)| = |a_1| + |a_2| + \cdots + |a_k|. \tag{1.40}$$

As a particular case of such Cartesian products, we obtain Cartesian powers when we multiply k copies of the same weighted set:

Definition 1.408. Let A be a weighted set. Then, A^k (for $k \in \mathbb{N}$) means the weighted set $\underbrace{A \times A \times \cdots \times A}_{k \text{ times}}$.

Proposition 1.409. *Let A be a finite-type weighted set. Let $k \in \mathbb{N}$. Then, A^k is finite-type, too, and satisfies $\overline{A^k} = \overline{A}^k$.*

Proof. By induction on k. The base case $k = 0$ is straightforward: A^0 consists of a single element (the empty tuple) with weight 0, so $\overline{A^0} = 1 = \overline{A}^0$.

For the inductive step, we observe that $A^{k+1} \cong A \times A^k$ as weighted sets (via the isomorphism that splits a $(k+1)$-tuple into its last entry and the remaining k-tuple). By Proposition 1.403, their generating functions are equal. By Proposition 1.407, $\overline{A \times A^k} = \overline{A} \cdot \overline{A^k}$. By the induction hypothesis, $\overline{A^k} = \overline{A}^k$. Hence $\overline{A^{k+1}} = \overline{A} \cdot \overline{A}^k = \overline{A}^{k+1}$. □

Lemma 1.410. *For any weighted set A and any $n \in \mathbb{N}$, there is a canonical isomorphism of weighted sets $A^{n+1} \cong A \times A^n$.*

Proof. The isomorphism sends a $(n+1)$-tuple $(a_0, a_1, \ldots, a_n)$ to the pair $(a_n, (a_0, \ldots, a_{n-1}))$. It preserves weight because the sum of entries is unchanged. □

Lemma 1.411. *Let A be a finite-type weighted set such that every element has weight ≥ 1 (i.e., A has positive weights). Then the infinite disjoint union $A^0 + A^1 + A^2 + \cdots$ (consisting of all finite tuples of elements of A) is also a finite-type weighted set.*

Proof. If every element has weight ≥ 1, then a tuple of length n has total weight $\geq n$. Therefore, for any fixed weight m, we only need to consider tuples of length $\leq m$. For each such length, there are only finitely many tuples of weight m (since A^k is finite-type by Proposition 1.409). Hence the disjoint union is finite-type. □

1.14.2 Domino tilings

Definition 1.412. (a) A *shape* means a subset of $\mathbb{Z}^2$.

We draw each $(i, j) \in \mathbb{Z}^2$ as a unit square with center at the point (i, j) (in Cartesian coordinates); thus, a shape can be drawn as a cluster of squares.

(b) For any $n, m \in \mathbb{N}$, the shape $R_{n,m}$ (called the *$n \times m$-rectangle*) is defined to be

$$\{1, 2, \ldots, n\} \times \{1, 2, \ldots, m\} = \left\{(i, j) \in \mathbb{Z}^2 \mid 1 \leq i \leq n \text{ and } 1 \leq j \leq m\right\}.$$

(c) A *domino* means a size-2 shape of the form

$$\begin{array}{ll} \{(i,j)\,,\ (i+1,j)\} \ \text{(a “}\textit{horizontal domino}\text{”)} & \text{or} \\ \{(i,j)\,,\ (i,j+1)\} \ \text{(a “}\textit{vertical domino}\text{”)} & \end{array}$$

for some $(i, j) \in \mathbb{Z}^2$.

(d) A *domino tiling* of a shape S is a set partition of S into dominos (i.e., a set of disjoint dominos whose union is S).

(e) For any $n, m \in \mathbb{N}$, let $d_{n,m}$ be the # of domino tilings of $R_{n,m}$.

We define the weighted set

$$D := \{\text{domino tilings of height-2 rectangles}\} = \{\text{domino tilings of } R_{n,2} \text{ with } n \in \mathbb{N}\},$$

with the weight of a tiling T of $R_{n,2}$ defined to be $|T| := n$. Thus, D is a finite-type weighted set, with weight generating function $\overline{D} = \sum_{n \in \mathbb{N}} d_{n,2} x^n$.

A *fault* of a domino tiling T is a vertical line ℓ such that each domino of T lies either left of ℓ or right of ℓ (but does not straddle ℓ), and there is at least one domino of T that lies left of ℓ, and at least one domino of T that lies right of ℓ.

A domino tiling is called *faultfree* if it is nonempty and has no fault. We define the weighted set

$$F := \{\textbf{faultfree} \text{ domino tilings of } R_{n,2} \text{ with } n \in \mathbb{N}\}$$

(with the same weights as in D).

Lemma 1.413. *Any domino tiling of a height-2 rectangle can be decomposed uniquely into a tuple of faultfree tilings of (usually smaller) height-2 rectangles, by cutting it along its faults. (If the original tiling was faultfree, then it decomposes into a 1-tuple. If the original tiling was empty, then it decomposes into a 0-tuple.)*

Moreover, the sum of the weights of the faultfree tilings in the tuple is the weight of the original tiling. (In other words, if a tiling T decomposes into the tuple $(T_1, T_2, \ldots, T_k)$, then $|T| = |T_1| + |T_2| + \cdots + |T_k|$.)

In the language of weighted sets, this yields an isomorphism

$$D \cong F^0 + F^1 + F^2 + F^3 + \cdots,$$

and therefore, by the infinite analogue of Proposition 1.405,

$$\overline{D} = \overline{F^0 + F^1 + F^2 + F^3 + \cdots} = \overline{F^0} + \overline{F^1} + \overline{F^2} + \overline{F^3} + \cdots.$$

By Proposition 1.409, this equals

$$\overline{F}^0 + \overline{F}^1 + \overline{F}^2 + \overline{F}^3 + \cdots = \frac{1}{1 - \overline{F}}.$$

Proof. The decomposition is defined by cutting the tiling along its faults. Given a tiling T of $R_{n,2}$:

- If $n = 0$ (empty tiling), the decomposition is the empty 0-tuple.
- If T has no faults, the decomposition is the 1-tuple (T).
- If T has faults, let k be the smallest fault position. Cut T at position k to obtain a faultfree left part (a tiling of $R_{k,2}$) and a right part (a tiling of $R_{n-k,2}$). Recursively decompose the right part, and prepend the left part.

The composition (inverse map) concatenates a tuple of faultfree tilings horizontally by shifting each component to the appropriate position. The weight-preservation property follows because the total width of the concatenated rectangle equals the sum of the widths of the components.

The proof that these two maps are inverse to each other requires showing:

1. Composing and then decomposing recovers the original tuple (by showing that faults in the composed tiling occur exactly at the partial width sums).
2. Decomposing and then composing recovers the original tiling (by showing that cutting at faults and reassembling gives back the original domino set).

□

Helpers for the decomposition isomorphism

Lemma 1.414. *At a fault position k of a tiling T of $R_{n,2}$, each domino of T lies entirely in the left part (all x-coordinates $\leq k$) or entirely in the right part (all x-coordinates $\geq k+1$).*

Proof. For horizontal dominos, the fault condition directly gives that both cells lie on the same side. For vertical dominos, both cells share the same x-coordinate, so they are trivially on one side. □

Lemma 1.415. *Given a tiling T of $R_{n,2}$ with a fault at position k, the set of dominos lying in the left part forms a valid tiling of $R_{k,2}$.*

Proof. Pairwise disjointness is inherited from T. For coverage: every cell (x, y) with $1 \leq x \leq k$ is covered by some domino $d \in T$, and by Lemma 1.414, d lies entirely in the left part. □

Lemma 1.416. *Given a tiling T of $R_{n,2}$ with a fault at position k, the set of dominos lying in the right part, after shifting left by k, forms a valid tiling of $R_{n-k,2}$.*

Proof. Each right-part domino is shifted by $-k$ in the x-coordinate. Disjointness is preserved since shifting is injective. Coverage follows because every cell (x, y) with $k+1 \leq x \leq n$ is covered by a right-part domino. □

Lemma 1.417. *The left restriction at the* minimum *fault position is faultfree. That is, if k is the smallest fault of T and $j < k$, then j is not a fault of the restricted tiling.*

Proof. A fault at $j < k$ in the left restriction would induce a fault at j in the original tiling T (since all horizontal dominos in the left part are also in T). But k is the minimum fault, so no fault at $j < k$ exists in T. □

Lemma 1.418. *Given a k-tuple of faultfree tilings $(T_1, \ldots, T_k)$ of rectangles $R_{m_1,2}, \ldots, R_{m_k,2}$, their horizontal concatenation (shifting each T_i by the cumulative width $m_1+\cdots+m_{i-1}$) produces a valid tiling of $R_{m_1+\cdots+m_k,2}$.*

Proof. Dominos from different components have disjoint x-coordinate ranges (Lemma 1.419), so they are pairwise disjoint. Coverage follows because each point in the total rectangle lies in some component's range and is covered by that component's shifted dominos. □

Lemma 1.419. *Dominos from different components $i_1 \neq i_2$ in the composition have disjoint shapes. This follows from the fact that their x-coordinate ranges are separated by the partial width sums.*

Proof. For $i_1 < i_2$, the partial width sum satisfies $\sum_{j<i_1} m_j + m_{i_1} \leq \sum_{j<i_2} m_j$, so the x-coordinate ranges $[1 + \text{offset}_{i_1}, m_{i_1} + \text{offset}_{i_1}]$ and $[1 + \text{offset}_{i_2}, m_{i_2} + \text{offset}_{i_2}]$ do not overlap. □

Lemma 1.420. *Given a tiling T of $R_{n,2}$, there exists a k-tuple of faultfree tilings whose horizontal concatenation reproduces T. The decomposition is constructed recursively: for $n = 0$ it returns the empty tuple; if T is faultfree it returns the singleton (T); otherwise it cuts at the minimum fault and recurses on the right part.*

Proof. Well-founded recursion on n: at each fault k, the right part has width $n - k < n$. $\square$

Lemma 1.421. *Decomposing a tiling T and then composing the resulting tuple recovers T:* $\mathrm{compose}(\mathrm{decompose}(T)) = T$.

Proof. By strong induction on n. The key step shows that the set of dominos of T equals the union of the left restriction's dominos and the shifted right restriction's dominos (Lemma 1.422), and by the induction hypothesis the right part reconstructs correctly. $\square$

Lemma 1.422. *At a fault position k, the original tiling's dominos equal the union of the left restriction's dominos and the right restriction's dominos shifted back by k.*

Proof. Every domino is in the left or right part. Left-part dominos are unchanged. Right-part dominos are shifted by $-k$ and then back by $+k$, recovering the original. $\square$

Lemma 1.423. *Composing a k-tuple of faultfree tilings and then decomposing recovers the original tuple:* $\mathrm{decompose}(\mathrm{compose}(t_1, \ldots, t_k)) = (t_1, \ldots, t_k)$.

Proof. By strong induction on k. For $k = 0$ and $k = 1$ the result is immediate. For $k \geq 2$: the composed tiling has a fault at position $m_1 = |t_1|$ (Lemma 1.424); no earlier fault exists (Lemma 1.425); the minimum fault is m_1 (Lemma 1.426); the left restriction equals t_1 (Lemma 1.427); and the right restriction equals the composition of the tail $(t_2, \ldots, t_k)$ (Lemma 1.428). The induction hypothesis on the tail completes the proof. $\square$

Lemma 1.424. *For $k \geq 2$, the composed tiling of a k-tuple has a fault at position m_1 (the width of the first component). No horizontal domino crosses this boundary because dominos from the first component have $x \leq m_1$ and dominos from later components have $x \geq m_1 + 1$.*

Proof. The width $m_1 > 0$ (since faultfree tilings are nonempty) and $m_1 < m_1 + m_2 + \cdots$ (since $k \geq 2$ and $m_2 > 0$). Each horizontal domino from component 0 has both x-coordinates $\leq m_1$; each horizontal domino from component $i \geq 1$ has both x-coordinates $\geq m_1 + 1$. $\square$

Lemma 1.425. *For $k \geq 1$, if $p < m_1$ (the width of the first component), then p is not a fault in the composed tiling. This is because a fault at p would induce a fault at p in the first (faultfree) component.*

Proof. Every domino from component 0 that is also in the composed tiling at position p witnesses the same crossing in the first component. But the first component is faultfree, so no such fault exists. $\square$

Lemma 1.426. *For $k \geq 2$, the minimum fault position of the composed tiling equals m_1 (the width of the first component).*

Proof. There is a fault at m_1 (Lemma 1.424) and no fault at any $p < m_1$ (Lemma 1.425). $\square$

Lemma 1.427. *For $k \geq 2$, the left restriction of the composed tiling at the first boundary m_1 equals the first component's tiling.*

Proof. Dominos in the left part at position m_1 come exactly from component 0 (since components $i \geq 1$ have $x \geq m_1 + 1$), and the partial width sum for component 0 is 0, so no shifting occurs. $\square$

Lemma 1.428. *For $k \geq 2$, the right restriction of the composed tiling at the first boundary m_1, after shifting, equals the composition of the tail $(t_2, \ldots, t_k)$.*

Proof. The dominos in the right part come from components $i \geq 1$. After unshifting by m_1, each component i's partial width sum decreases by m_1, matching the partial width sums of the tail composition. □

Lemma 1.429. *The sum of the widths of the faultfree tilings returned by the decomposition equals the original width n. That is, $\sum_i m_i = n$.*

Proof. Follows from compose(decompose(T)) $= T$ (Lemma 1.421) and the fact that compose produces a tiling whose width is the sum of the component widths. □

The only faultfree tilings of height-2 rectangles have width 1 (one vertical domino) or width 2 (two horizontal dominos stacked vertically). Thus, the weighted set F consists of just two tilings: one of weight 1 and one of weight 2. Hence $\overline{F} = x + x^2$, and so

$$\overline{D} = \frac{1}{1 - \overline{F}} = \frac{1}{1 - (x + x^2)} = \frac{1}{1 - x - x^2} = f_1 + f_2 x + f_3 x^2 + f_4 x^3 + \cdots ,$$

where $(f_0, f_1, f_2, \ldots)$ is the Fibonacci sequence. Thus $d_{n,2} = f_{n+1}$ for each $n \in \mathbb{N}$.

Lemma 1.430. *The only faultfree tilings of height-2 rectangles have width 1 or 2.*

Proof. Consider any faultfree tiling of a height-2 rectangle $R_{n,2}$ with $n \geq 3$. Look at the domino that covers the cell $(1,1)$. If it is a vertical domino $\{(1,1),(1,2)\}$, then this vertical domino constitutes the entire first column, so there is a fault at position 1 – contradiction. If it is a horizontal domino $\{(1,1),(2,1)\}$, then there must be a horizontal domino $\{(1,2),(2,2)\}$ stacked above it, and these two dominos cover the first two columns entirely, so there is a fault at position 2 – contradiction. □

Lemma 1.431. *The weight generating function of faultfree height-2 tilings is $x + x^2$.*

Proof. By the classification lemma, there is exactly one faultfree tiling of width 1 (the single vertical domino) and exactly one faultfree tiling of width 2 (the two horizontal dominos), and no faultfree tilings of any other width. Hence $\overline{F} = x + x^2$. □

Lemma 1.432. *For any $n \in \mathbb{N}$, $d_{n+2,2} = d_{n,2} + d_{n+1,2}$.*

Proof. We construct a bijection $\mathrm{Tiling}(R_{n+2,2}) \simeq \mathrm{Tiling}(R_{n,2}) \sqcup \mathrm{Tiling}(R_{n+1,2})$ by classifying tilings according to what covers the cell $(1,1)$:

- If a vertical domino $\{(1,1),(1,2)\}$ covers $(1,1)$, then removing it and shifting gives a tiling of $R_{n+1,2}$.
- If a horizontal domino $\{(1,1),(2,1)\}$ covers $(1,1)$, then $\{(1,2),(2,2)\}$ must also be present; removing both and shifting gives a tiling of $R_{n,2}$.

The inverse maps are given by prepending a vertical domino (resp. a pair of horizontal dominos) and shifting the remaining dominos. □

Helpers for the Fibonacci recurrence

Lemma 1.433. *For $n \geq 1$, every tiling T of $R_{n,2}$ contains a domino covering the cell $(1,1)$.*

Proof. The cell $(1,1) \in R_{n,2}$ is covered by the tiling. By the coverage condition, some domino in T contains $(1,1)$. □

Lemma 1.434. *If a domino d in a tiling T of $R_{n,2}$ covers $(1,1)$, then d is either the vertical domino $\{(1,1),(1,2)\}$ or the horizontal domino $\{(1,1),(2,1)\}$.*

Proof. A horizontal domino covering $(1,1)$ must be $\{(1,1),(2,1)\}$ since extending left to $(0,1)$ would leave the rectangle. A vertical domino covering $(1,1)$ must be $\{(1,1),(1,2)\}$ since extending down to $(1,0)$ would leave the rectangle. □

Lemma 1.435. *If a tiling T of $R_{n,2}$ (with $n \geq 2$) contains the horizontal domino $\{(1,1),(2,1)\}$, then T also contains $\{(1,2),(2,2)\}$.*

Proof. The cell $(1,2)$ must be covered. The vertical domino $\{(1,1),(1,2)\}$ would overlap with $\{(1,1),(2,1)\}$. So $(1,2)$ must be covered by a horizontal domino $\{(1,2),(2,2)\}$. □

Lemma 1.436. *If $n \geq 2$ and the tiling T of $R_{n,2}$ contains the vertical domino $\{(1,1),(1,2)\}$, then T has a fault at position 1. (The vertical domino covers the entire first column, so no horizontal domino crosses from column 1 to column 2.)*

Proof. The vertical domino covers both cells of column 1. Any other domino covering a cell in column 1 would overlap with the vertical domino. Hence all other dominos have x-coordinates ≥ 2, giving a fault at 1. □

Lemma 1.437. *If $n \geq 3$ and the tiling T of $R_{n,2}$ contains the horizontal domino $\{(1,1),(2,1)\}$, then T has a fault at position 2. The two horizontal dominos $\{(1,1),(2,1)\}$ and $\{(1,2),(2,2)\}$ cover columns 1–2 entirely.*

Proof. By Lemma 1.435, $\{(1,2),(2,2)\}$ is also present. These two dominos cover all four cells of columns 1–2. Any other domino must avoid all these cells, forcing all x-coordinates ≥ 3. □

Lemma 1.438. *Given a tiling T of $R_{m,2}$, prepending a vertical domino $\{(1,1),(1,2)\}$ and shifting all of T's dominos right by 1 gives a valid tiling of $R_{m+1,2}$.*

Proof. Disjointness holds because the vertical domino covers column 1 and the shifted dominos have $x \geq 2$. Coverage: column 1 is covered by the vertical domino; columns 2 through $m+1$ are covered by the shifted dominos. □

Lemma 1.439. *Given a tiling T of $R_{m,2}$, prepending horizontal dominos $\{(1,1),(2,1)\}$ and $\{(1,2),(2,2)\}$ and shifting all of T's dominos right by 2 gives a valid tiling of $R_{m+2,2}$.*

Proof. Disjointness holds because the two horizontal dominos cover columns 1–2 (and are disjoint from each other since they cover different rows) and the shifted dominos have $x \geq 3$. Coverage is verified similarly to Lemma 1.438. □

Lemma 1.440. *For any $n, m \in \mathbb{N}$, the set of all domino tilings of $R_{n,m}$ is finite.*

Proof. A tiling is determined by its set of dominos, which is a subset of the (finite) set of all dominos that fit inside $R_{n,m}$. □

The Fibonacci formula

Lemma 1.441. *For any $n \in \mathbb{N}$, $d_{n,2} = f_{n+1}$, where $(f_k)_{k\geq 0}$ is the Fibonacci sequence.*

Proof. By strong induction on n. Base cases: $d_{0,2} = 1 = f_1$ and $d_{1,2} = 1 = f_2$. Inductive step: $d_{n+2,2} = d_{n,2} + d_{n+1,2}$ (Lemma 1.432) $= f_{n+1} + f_{n+2}$ (induction hypothesis) $= f_{n+3}$ (Fibonacci recurrence). □

Lemma 1.442. *The count of elements of weight n in the weighted set D of height-2 tilings equals $d_{n,2}$.*

Proof. The elements of weight n in D are exactly the tilings of $R_{n,2}$. □

Lemma 1.443. *The weight generating function of D equals $\sum_n f_{n+1}x^n$, the Fibonacci generating function.*

Proof. The n-th coefficient is the number of elements of weight n in D, which is $d_{n,2} = f_{n+1}$. □

1.15 Limits of FPSs

1.15.1 Stabilization of scalars

Definition 1.444. Let $(a_i)_{i\in\mathbb{N}} = (a_0, a_1, a_2, \ldots) \in K^{\mathbb{N}}$ be a sequence of elements of K. Let $a \in K$.

We say that the sequence $(a_i)_{i\in\mathbb{N}}$ *stabilizes to a* if there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } a_i = a.$$

If a_i stabilizes to a as $i \to \infty$, then we write $\lim\limits_{i\to\infty} a_i = a$ and say that a is the *limit* (or *eventual value*) of $(a_i)_{i\in\mathbb{N}}$. This is legitimate, since a is uniquely determined by the sequence $(a_i)_{i\in\mathbb{N}}$.

Lemma 1.445. *If (a_i) stabilizes to both ℓ_1 and ℓ_2, then $\ell_1 = \ell_2$.*

Proof. Let N_1 and N_2 be as in the definition for ℓ_1 and ℓ_2 respectively. Then $a_{\max(N_1,N_2)} = \ell_1$ and $a_{\max(N_1,N_2)} = \ell_2$, so $\ell_1 = \ell_2$. □

Lemma 1.446. *Any constant sequence $(c, c, c, \ldots) \in K^{\mathbb{N}}$ stabilizes to c.*

Proof. Take $N = 0$. Then all $i \geq 0$ satisfy $a_i = c$. □

Lemma 1.447. *If a sequence (a_i) stabilizes to ℓ, and a sequence (b_i) is eventually equal to (a_i) (i.e., there exists N such that $a_i = b_i$ for all $i \geq N$), then (b_i) also stabilizes to ℓ.*

Proof. Let N_1 witness stabilization of (a_i) to ℓ and N_2 witness the eventual equality. Then for all $i \geq \max(N_1, N_2)$, we have $b_i = a_i = \ell$. □

Lemma 1.448. *If (a_i) stabilizes to ℓ_a and (b_i) stabilizes to ℓ_b, then $(a_i + b_i)$ stabilizes to $\ell_a + \ell_b$.*

Proof. Let N_a and N_b witness the two stabilizations. For $i \geq \max(N_a, N_b)$, we have $a_i + b_i = \ell_a + \ell_b$. □

Lemma 1.449. *If (a_i) stabilizes to ℓ_a and (b_i) stabilizes to ℓ_b, then $(a_i \cdot b_i)$ stabilizes to $\ell_a \cdot \ell_b$.*

Proof. Let N_a and N_b witness the two stabilizations. For $i \geq \max(N_a, N_b)$, we have $a_i \cdot b_i = \ell_a \cdot \ell_b$. □

Lemma 1.450. *If (a_i) stabilizes to ℓ, then $(-a_i)$ stabilizes to $-\ell$.*

Proof. Let N witness the stabilization. For $i \geq N$, we have $-a_i = -\ell$. □

Lemma 1.451. *Let S be a finite set, and for each $s \in S$, let $(a_{s,i})_{i\in\mathbb{N}}$ be a sequence stabilizing to ℓ_s. Then the sequence $\left(\sum_{s\in S} a_{s,i}\right)_{i\in\mathbb{N}}$ stabilizes to $\sum_{s\in S} \ell_s$.*

Proof. By induction on $|S|$, using Lemma 1.448 for the induction step. □

Lemma 1.452. *If $s \in K$ satisfies $s^k = 0$ for some $k \in \mathbb{N}$ (i.e., s is nilpotent), then the sequence $(s^i)_{i\in\mathbb{N}}$ stabilizes to 0.*

Proof. For $i \geq k$, we have $s^i = s^k \cdot s^{i-k} = 0 \cdot s^{i-k} = 0$. □

1.15.2 Coefficientwise stabilization of FPSs

Definition 1.453. Let $(f_i)_{i\in\mathbb{N}} \in K[[x]]^{\mathbb{N}}$ be a sequence of FPSs over K. Let $f \in K[[x]]$ be an FPS.

We say that $(f_i)_{i\in\mathbb{N}}$ *coefficientwise stabilizes to* f if for each $n \in \mathbb{N}$,

$$\text{the sequence } \left([x^n] f_i\right)_{i\in\mathbb{N}} \text{ stabilizes to } [x^n] f.$$

If f_i coefficientwise stabilizes to f as $i \to \infty$, then we write $\lim_{i\to\infty} f_i = f$ and say that f is the *limit* of $(f_i)_{i\in\mathbb{N}}$. This is legitimate, because f is uniquely determined by the sequence $(f_i)_{i\in\mathbb{N}}$.

Lemma 1.454. *If a sequence (f_i) of FPSs coefficientwise stabilizes to both ℓ_1 and ℓ_2, then $\ell_1 = \ell_2$.*

Proof. By extensionality (Lemma 1.86): for each n, the n-th coefficient of ℓ_1 equals that of ℓ_2 by Lemma 1.445. □

Lemma 1.455. *A constant sequence $(g, g, g, \ldots)$ of FPSs coefficientwise stabilizes to g.*

Proof. For each n, the sequence $([x^n]g, [x^n]g, \ldots)$ is constant and stabilizes to $[x^n]g$ by Lemma 1.446. □

Lemma 1.456. *The sequence $(x^i)_{i\in\mathbb{N}}$ of FPSs coefficientwise stabilizes to 0.*

Proof. For each $n \in \mathbb{N}$, the sequence $([x^n](x^i))_{i\in\mathbb{N}}$ consists of a single 1 (at $i = n$) and 0s elsewhere. For $i \geq n+1$, $[x^n](x^i) = 0$, so the sequence stabilizes to $0 = [x^n]0$. □

1.15.3 Some properties of limits

Helpers for x^n-equivalence and composition

Lemma 1.457. *Let $g \in K[[x]]$ with $[x^0]g = 0$. Then for any $n, d \in \mathbb{N}$ with $d > n$, we have $[x^n](g^d) = 0$. This follows from $\operatorname{ord}(g) \geq 1$ implying $\operatorname{ord}(g^d) \geq d > n$.*

Proof. Since $[x^0]g = 0$, we have $\operatorname{ord}(g) \geq 1$, so $\operatorname{ord}(g^d) \geq d > n$, and therefore $[x^n](g^d) = 0$. □

Lemma 1.458. *Let $f, g \in K[[x]]$ with $[x^0]g = 0$. Then*

$$[x^n](f \circ g) = \sum_{d=0}^{n} [x^d]f \cdot [x^n](g^d).$$

In other words, only the first $n+1$ coefficients of f contribute to $[x^n](f \circ g)$.

Proof. For $d > n$, we have $[x^n](g^d) = 0$ by Lemma 1.457, so those terms vanish from the infinite sum. □

Lemma 1.459. *The relation $\overset{x^n}{\equiv}$ is compatible with taking inverses. If $f \overset{x^n}{\equiv} g$ and u is a unit in K such that $[x^0]f = u = [x^0]g$, then $f^{-1} \overset{x^n}{\equiv} g^{-1}$ (where f^{-1} denotes the inverse of f as an invertible FPS with constant term u).*

Proof. By strong induction on the coefficient index k. For $k = 0$, both inverse coefficients equal u^{-1}. For $k > 0$, the recursive formula for inverse coefficients involves only lower-degree coefficients of the inverse (handled by the IH) and coefficients of f, g that agree by x^n-equivalence. □

Lemma 1.460. *The relation $\overset{x^n}{\equiv}$ is compatible with composition. If $f_1 \overset{x^n}{\equiv} f_2$ and $g_1 \overset{x^n}{\equiv} g_2$, with $[x^0]g_1 = [x^0]g_2 = 0$, then $f_1 \circ g_1 \overset{x^n}{\equiv} f_2 \circ g_2$.*

Proof. By Lemma 1.458, $[x^m](f \circ g)$ depends only on the first $m+1$ coefficients of f and on powers g^d for $d \leq m$. Since $f_1 \overset{x^n}{\equiv} f_2$ gives $[x^d]f_1 = [x^d]f_2$ for $d \leq m \leq n$, and $g_1 \overset{x^n}{\equiv} g_2$ implies $g_1^d \overset{x^n}{\equiv} g_2^d$ (by compatibility of $\overset{x^n}{\equiv}$ with powers), the result follows. □

Theorem 1.461. *Let $(f_i)_{i \in \mathbb{N}} \in K[[x]]^{\mathbb{N}}$ be a sequence of FPSs. Assume that for each $n \in \mathbb{N}$, there exists some $g_n \in K$ such that the sequence $([x^n] f_i)_{i \in \mathbb{N}}$ stabilizes to g_n. Then, the sequence $(f_i)_{i \in \mathbb{N}}$ coefficientwise stabilizes to $\sum_{n \in \mathbb{N}} g_n x^n$. (That is, $\lim\limits_{i \to \infty} f_i = \sum_{n \in \mathbb{N}} g_n x^n$.)*

Proof. We need to show that for each n, the sequence $([x^n]f_i)$ stabilizes to $[x^n](\sum_{m \in \mathbb{N}} g_m x^m) = g_n$. But this is exactly the hypothesis applied to n. □

Lemma 1.462. *Assume that $(f_i)_{i \in \mathbb{N}}$ is a sequence of FPSs, and that f is an FPS such that $\lim\limits_{i \to \infty} f_i = f$. Then, for each $n \in \mathbb{N}$, there exists some integer $N \in \mathbb{N}$ such that*

$$\text{all integers } i \geq N \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Proof. Let $n \in \mathbb{N}$. For each $k \in \{0, 1, \ldots, n\}$, we pick an $N_k \in \mathbb{N}$ such that all $i \geq N_k$ satisfy $\left[x^k\right] f_i = \left[x^k\right] f$ (such an N_k exists, since $\lim\limits_{i \to \infty} f_i = f$). Then, all integers $i \geq \max\{N_0, N_1, \ldots, N_n\}$ satisfy $f_i \overset{x^n}{\equiv} f$. □

Proposition 1.463. *Assume that $(f_i)_{i \in \mathbb{N}}$ and $(g_i)_{i \in \mathbb{N}}$ are two sequences of FPSs, and that f and g are two FPSs such that*

$$\lim_{i \to \infty} f_i = f \qquad \text{and} \qquad \lim_{i \to \infty} g_i = g.$$

Then,

$$\lim_{i \to \infty} (f_i + g_i) = f + g \qquad \text{and} \qquad \lim_{i \to \infty} (f_i g_i) = fg.$$

Proof. Let $n \in \mathbb{N}$.

Addition. The n-th coefficient of $f_i + g_i$ is $[x^n]f_i + [x^n]g_i$. Since $([x^n]f_i)$ stabilizes to $[x^n]f$ and $([x^n]g_i)$ stabilizes to $[x^n]g$, Lemma 1.448 shows that $([x^n]f_i + [x^n]g_i)$ stabilizes to $[x^n]f + [x^n]g = [x^n](f+g)$.

Multiplication. The n-th coefficient of $f_i g_i$ is $[x^n](f_i g_i) = \sum_{(a,b)\in\operatorname{antidiag}(n)} [x^a]f_i \cdot [x^b]g_i$. Each term $[x^a]f_i \cdot [x^b]g_i$ is a product of two stabilizing sequences (by Lemma 1.449), and a finite sum of stabilizing sequences stabilizes (by Lemma 1.451). Thus the sequence stabilizes to $\sum_{(a,b)} [x^a]f \cdot [x^b]g = [x^n](fg)$. □

Lemma 1.464. *If (f_i) coefficientwise stabilizes to f, then $(-f_i)$ coefficientwise stabilizes to $-f$.*

Proof. For each n, $([x^n](-f_i)) = (-[x^n]f_i)$ stabilizes to $-[x^n]f = [x^n](-f)$ by Lemma 1.450. □

Lemma 1.465. *If (f_i) coefficientwise stabilizes to f and (g_i) coefficientwise stabilizes to g, then $(f_i - g_i)$ coefficientwise stabilizes to $f - g$.*

Proof. Write $f_i - g_i = f_i + (-g_i)$ and apply Proposition 1.463 (addition) and Lemma 1.464. □

Corollary 1.466. *Let $k \in \mathbb{N}$. For each $i \in \{1, 2, \ldots, k\}$, let f_i be an FPS, and let $(f_{i,n})_{n\in\mathbb{N}}$ be a sequence of FPSs such that*

$$\lim_{n\to\infty} (f_{i,n}) = f_i$$

(note that it is n, not i, that goes to ∞ here!). Then,

$$\lim_{n\to\infty} \sum_{i=1}^{k} f_{i,n} = \sum_{i=1}^{k} f_i \qquad \text{and} \qquad \lim_{n\to\infty} \prod_{i=1}^{k} f_{i,n} = \prod_{i=1}^{k} f_i.$$

Proof. Follows by induction on k, using Proposition 1.463. □

Lemma 1.467. *Assume that (g_i) coefficientwise stabilizes to g, and that each g_i is invertible (i.e., its constant coefficient is a unit). Then g is also invertible (i.e., the constant coefficient of g is a unit).*

Proof. The 0-th coefficient of g_i stabilizes to the 0-th coefficient of g. For sufficiently large i, $[x^0]g_i = [x^0]g$. Since $[x^0]g_i$ is a unit, so is $[x^0]g$. □

Proposition 1.468. *Assume that $(f_i)_{i\in\mathbb{N}}$ and $(g_i)_{i\in\mathbb{N}}$ are two sequences of FPSs, and that f and g are two FPSs such that*

$$\lim_{i\to\infty} f_i = f \qquad \text{and} \qquad \lim_{i\to\infty} g_i = g.$$

Assume that each FPS g_i is invertible. Then, g is also invertible, and we have

$$\lim_{i\to\infty} \frac{f_i}{g_i} = \frac{f}{g}.$$

Proof. The invertibility of g follows from Lemma 1.467. For the limit equality, we write $f_i/g_i = f_i \cdot g_i^{-1}$ and show that the coefficients of g_i^{-1} stabilize to those of g^{-1}.

The key step is showing that the inverse coefficients stabilize. This is done by strong induction on the coefficient index n.

Base case ($n = 0$): The 0-th coefficient of g_i^{-1} is $([x^0]g_i)^{-1}$. Since $[x^0]g_i$ stabilizes to $[x^0]g$, the inverse stabilizes as well.

Induction step ($n > 0$): The n-th coefficient of g_i^{-1} is determined by the recursive formula

$$[x^n](g_i^{-1}) = -([x^0]g_i)^{-1} \sum_{\substack{(a,b)\in\text{antidiag}(n)\\ b<n}} [x^a]g_i \cdot [x^b](g_i^{-1}).$$

By the induction hypothesis, each $[x^b](g_i^{-1})$ with $b < n$ stabilizes. Combined with stabilization of $[x^a]g_i$ and $([x^0]g_i)^{-1}$, the products and sums stabilize by Lemmas 1.449 and 1.451.

Once we know g_i^{-1} coefficientwise stabilizes to g^{-1}, the result $\lim f_i g_i^{-1} = fg^{-1}$ follows from Proposition 1.463 (multiplication). □

Proposition 1.469. *Assume that $(f_i)_{i\in\mathbb{N}}$ and $(g_i)_{i\in\mathbb{N}}$ are two sequences of FPSs, and that f and g are two FPSs such that*

$$\lim_{i\to\infty} f_i = f \qquad \text{and} \qquad \lim_{i\to\infty} g_i = g.$$

Assume that $\left[x^0\right] g_i = 0$ for each $i \in \mathbb{N}$. Then, $\left[x^0\right] g = 0$ and

$$\lim_{i\to\infty} (f_i \circ g_i) = f \circ g.$$

Proof. First, $[x^0]g = 0$ since $[x^0]g_i = 0$ for all i and the 0-th coefficient stabilizes.

For each $n \in \mathbb{N}$, by Lemma 1.462 we can find N_f and N_g such that for $i \geq \max(N_f, N_g)$, $f_i \overset{x^n}{\equiv} f$ and $g_i \overset{x^n}{\equiv} g$. By Proposition 1.338 (compatibility of x^n-equivalence with composition), $f_i \circ g_i \overset{x^n}{\equiv} f \circ g$, so in particular $[x^n](f_i \circ g_i) = [x^n](f \circ g)$. □

Proposition 1.470. *Let $(f_n)_{n\in\mathbb{N}}$ be a sequence of FPSs, and let f be an FPS such that*

$$\lim_{i\to\infty} f_i = f.$$

Then,

$$\lim_{i\to\infty} f_i' = f'.$$

Proof. For each n, $[x^n](f_i') = (n+1)\cdot[x^{n+1}]f_i$. Since $([x^{n+1}]f_i)$ stabilizes to $[x^{n+1}]f$, multiplying by the constant $(n+1)$ preserves stabilization (by Lemma 1.449 with the constant sequence $(n+1)$). Thus the sequence stabilizes to $(n+1)\cdot[x^{n+1}]f = [x^n](f')$. □

1.15.4 Infinite sums and products

Definition 1.471. A family $(f_n)_{n\in\mathbb{N}}$ of FPSs is *summable* if for each $n \in \mathbb{N}$, the set $\{i \in \mathbb{N} : [x^n]f_i \neq 0\}$ is finite.

Definition 1.472. The *infinite sum* of a summable family $(f_n)_{n\in\mathbb{N}}$ is the FPS $\sum_{n\in\mathbb{N}} f_n$ whose n-th coefficient is $\sum_{i\in S_n}[x^n]f_i$, where $S_n = \{i : [x^n]f_i \neq 0\}$.

Definition 1.473. A family $(f_n)_{n\in\mathbb{N}}$ of FPSs is *multipliable* if (1) $[x^0]f_i = 1$ for all i, and (2) for each $n \in \mathbb{N}$, there exists N such that for all $i \geq N$ and all $k \leq n$, $[x^k]f_i = \delta_{k,0}$ (i.e., eventually all f_i are $\equiv 1 \pmod{x^{n+1}}$).

Definition 1.474. The *infinite product* of a multipliable family $(f_n)_{n\in\mathbb{N}}$ is the FPS $\prod_{n\in\mathbb{N}} f_n$ whose n-th coefficient equals $[x^n](\prod_{j=0}^{N} f_j)$ for any sufficiently large N.

Theorem 1.475. *Let $(f_n)_{n\in\mathbb{N}}$ be a summable sequence of FPSs. Then,*

$$\lim_{i\to\infty}\sum_{n=0}^{i} f_n = \sum_{n\in\mathbb{N}} f_n.$$

In other words, the infinite sum $\sum_{n\in\mathbb{N}} f_n$ is the limit of the finite partial sums $\sum_{n=0}^{i} f_n$.

Proof. For each coefficient index n, only finitely many f_j have nonzero n-th coefficient (by summability). Let S be the finite set of such indices, and let N be the maximum element of S (or 0 if S is empty). For $i \geq N+1$, the partial sum $\sum_{j=0}^{i}[x^n]f_j$ includes all nonzero contributors, so it equals the infinite sum $\sum_{j\in\mathbb{N}}[x^n]f_j$. Hence the n-th coefficient stabilizes. □

Theorem 1.476. *Let $(f_n)_{n\in\mathbb{N}}$ be a multipliable sequence of FPSs. Then,*

$$\lim_{i\to\infty}\prod_{n=0}^{i} f_n = \prod_{n\in\mathbb{N}} f_n.$$

In other words, the infinite product $\prod_{n\in\mathbb{N}} f_n$ is the limit of the finite partial products $\prod_{n=0}^{i} f_n$.

Proof. By the multipliability condition, for each n there exists N such that for all $j \geq N$, $f_j \equiv 1 \pmod{x^{n+1}}$ (i.e., $[x^k]f_j = \delta_{k,0}$ for $k \leq n$). Once i exceeds N, multiplying the partial product by f_{i+1} does not change the first $n+1$ coefficients, since f_{i+1} acts as 1 on those coefficients. Hence the n-th coefficient of the partial product stabilizes to the n-th coefficient of the infinite product. □

Corollary 1.477. *Each FPS is a limit of a sequence of polynomials. Indeed, if $a = \sum_{n\in\mathbb{N}} a_n x^n$ (with $a_n \in K$), then*

$$a = \lim_{i\to\infty}\sum_{n=0}^{i} a_n x^n.$$

More precisely, the sequence of truncations $(\mathrm{trunc}_{i+1}(a))_{i\in\mathbb{N}}$ coefficientwise stabilizes to a.

Proof. For each n, once $i \geq n$, the truncation $\mathrm{trunc}_{i+1}(a)$ has $[x^n](\mathrm{trunc}_{i+1}(a)) = [x^n]a$ (since $n < i+1$). □

Theorem 1.478. *Let $(f_0, f_1, f_2, \ldots)$ be a sequence of FPSs such that $\lim\limits_{i\to\infty}\sum_{n=0}^{i} f_n$ exists. Then, the family $(f_n)_{n\in\mathbb{N}}$ is summable, and satisfies*

$$\sum_{n\in\mathbb{N}} f_n = \lim_{i\to\infty}\sum_{n=0}^{i} f_n.$$

Proof. **Summability.** Fix a coefficient index n. Let N be such that $[x^n](\sum_{j=0}^{i} f_j) = [x^n]\ell$ for all $i \geq N$ (where ℓ is the limit). Suppose for contradiction that $i \geq N+1$ and $[x^n]f_i \neq 0$. Then

$$[x^n]\ell = [x^n]\sum_{j=0}^{i} f_j = [x^n]\sum_{j=0}^{i-1} f_j + [x^n]f_i = [x^n]\ell + [x^n]f_i,$$

which gives $[x^n]f_i = 0$, a contradiction. So the set $\{i : [x^n]f_i \neq 0\} \subseteq \{0, 1, \ldots, N\}$ is finite.

Equality. Once summability is established, Theorem 1.475 gives $\lim_{i\to\infty}\sum_{n=0}^{i} f_n = \sum_{n\in\mathbb{N}} f_n$. By uniqueness of limits, $\sum_{n\in\mathbb{N}} f_n = \ell$. □

Theorem 1.479. *Let $(f_0, f_1, f_2, \ldots)$ be a sequence of FPSs such that $\lim\limits_{i\to\infty} \prod_{n=0}^{i} f_n$ exists. Then, the family $(f_n)_{n\in\mathbb{N}}$ is multipliable, and satisfies*

$$\prod_{n\in\mathbb{N}} f_n = \lim_{i\to\infty} \prod_{n=0}^{i} f_n.$$

Proof. **Multipliability.** We first show that $[x^0]f_i = 1$ for all i. Since the 0-th coefficient of $\prod_{j=0}^{i} f_j$ stabilizes to $[x^0]\ell$, and $[x^0](\prod_{j=0}^{i} f_j) = \prod_{j=0}^{i} [x^0]f_j$, the product of the constant coefficients stabilizes; this forces $[x^0]f_i = 1$ for all sufficiently large i, and since the constant coefficients are units whose product stabilizes, we get $[x^0]f_i = 1$ for all i.

Next, we need to show that for each n, eventually $f_i \equiv 1 \pmod{x^{n+1}}$ (i.e., $[x^k]f_i = \delta_{k,0}$ for all $k \le n$).

We prove this by strong induction on k: for $0 < k \le n$ and sufficiently large i, $[x^k]f_i = 0$.

Let $P_i = \prod_{j=0}^{i-1} f_j$ denote the partial product (so that the partial product up to i is $P_i \cdot f_i$). By hypothesis, the k-th coefficient of both $P_{i+1} = P_i \cdot f_i$ and P_i eventually stabilizes to $[x^k]\ell$. Hence eventually $[x^k](P_i \cdot f_i) = [x^k]P_i$.

Expanding $[x^k](P_i \cdot f_i)$ via the convolution formula and using $[x^0]f_i = 1$, we get

$$[x^k]P_i + \sum_{\substack{(a,b)\in\text{antidiag}(k)\\ b\neq 0, b\neq k}} [x^a]P_i \cdot [x^b]f_i + [x^0]P_i \cdot [x^k]f_i = [x^k]P_i.$$

Since $[x^0]P_i = 1$ (as a product of FPSs with constant term 1), the last term equals $[x^k]f_i$. By the induction hypothesis, the middle sum vanishes (since $0 < b < k$ and $[x^b]f_i = 0$ for large i). Thus $[x^k]f_i = 0$.

Note: the formalization adds $[x^0]f_i = 1$ as an explicit hypothesis rather than deriving it from the existence of the limit.

Equality. Once multipliability is established, Theorem 1.476 gives $\lim \prod_{n=0}^{i} f_n = \prod_{n\in\mathbb{N}} f_n$, and uniqueness of limits gives the result. □

1.16 Laurent power series

1.16.1 Motivation

Fix a commutative ring K.

Definition 1.480. A *binary representation* of an integer n means an essentially finite sequence $(b_i)_{i\in\mathbb{N}} = (b_0, b_1, b_2, \ldots) \in \{0,1\}^{\mathbb{N}}$ such that

$$n = \sum_{i\in\mathbb{N}} b_i 2^i.$$

(Recall that "essentially finite" means "all but finitely many $i \in \mathbb{N}$ satisfy $b_i = 0$".)

Theorem 1.481. *Each $n \in \mathbb{N}$ has a unique binary representation.*

Proof. In the first formalization, the proof uses the canonical bit-index representation from Mathlib. In the second formalization, the proof proceeds by strong induction on n: the least significant bit is determined by n mod 2, and uniqueness of the remaining bits follows from the induction hypothesis applied to $n/2$. □

Definition 1.482. A *balanced ternary representation* of an integer n means an essentially finite sequence $(b_i)_{i\in\mathbb{N}} = (b_0, b_1, b_2, \ldots) \in \{0, 1, -1\}^{\mathbb{N}}$ such that

$$n = \sum_{i\in\mathbb{N}} b_i 3^i.$$

Theorem 1.483. *Each integer n has a unique balanced ternary representation.*

Proof. **Existence.** In the first formalization, existence is proved by finding k large enough that $|n| \leq M_k = 3^0 + 3^1 + \cdots + 3^k$, then using a cardinality/pigeonhole argument: the map from k-bounded representations (of which there are 3^{k+1}) to the interval $\{m \in \mathbb{Z} : |m| \leq M_k\}$ (which also has size 3^{k+1}) is injective, hence surjective. In the second formalization, existence is proved by strong induction on $|n|$: the least significant digit is determined by $n \bmod 3$ (choosing from $\{-1, 0, 1\}$), and the remaining digits represent $(n - d)/3$.

Uniqueness. Both formalizations prove uniqueness by showing that if the difference $c_i = b_i - b'_i$ of two representations satisfies $\sum c_i \cdot 3^i = 0$ with $c_i \in \{-2, \ldots, 2\}$ and all but finitely many $c_i = 0$, then all $c_i = 0$. This is proved by induction on the highest nonzero index: $c_0 \equiv 0 \pmod 3$ forces $c_0 = 0$, then dividing by 3 reduces to the induction hypothesis. □

Lemma 1.484 (Key uniqueness lemma for base-3 digit sums)**.** *Let $c : \mathbb{N} \to \mathbb{Z}$ be an essentially finite sequence with $c_i \in \{-2, \ldots, 2\}$ for all i. If $\sum_i c_i \cdot 3^i = 0$, then $c_i = 0$ for all i.*

Proof. By induction on the highest nonzero index k. Since $c_0 \equiv 0 \pmod 3$ and $|c_0| \leq 2$, we get $c_0 = 0$. Dividing the remaining sum by 3 gives a shifted sequence satisfying the same hypotheses with a smaller bound, and the induction hypothesis applies. □

Lemma 1.485 (k-bounded balanced ternary uniqueness)**.** *Each integer n with $|n| \leq 3^0 + 3^1 + \cdots + 3^k$ has a unique k-bounded balanced ternary representation.*

Proof. In the first formalization, this uses a cardinality argument: the number of k-bounded representations is 3^{k+1}, the target interval $\{m : |m| \leq M_k\}$ also has size 3^{k+1}, and the evaluation map is injective (by Lemma 1.484), hence bijective. In the second formalization, uniqueness follows directly from Lemma 1.484 applied to the difference of two representations. □

Lemma 1.486 (Balanced ternary digit type)**.** *A balanced ternary digit is one of -1, 0, or 1. There is an injection from the set of balanced ternary digits to $\mathbb{Z}$ sending each digit to its integer value. This injection is injective, and two digits are equal if and only if their integer values are congruent modulo 3.*

Proof. Exhaustive case analysis on the three values of a balanced ternary digit. □

Lemma 1.487 (Existence of balanced ternary representations by strong induction)**.** *Every integer n has a balanced ternary representation. The proof proceeds by strong induction on $|n|$: the least significant digit d is chosen by $n \bmod 3$ (from $\{-1, 0, 1\}$), and the remaining digits represent $(n - d)/3$, which satisfies $|(n - d)/3| < |n|$ for $n \neq 0$.*

Proof. Base case: $n = 0$ uses the zero representation (all digits zero). Inductive step: pick the digit d such that $n \equiv d \pmod 3$ with $d \in \{-1, 0, 1\}$, form $m = (n - d)/3$, and prepend d to the representation of m (obtained by the induction hypothesis since $|m| < |n|$). □

Lemma 1.488 (Partial product formula for balanced ternary)**.** *Define the partial product $P_k = \prod_{i=0}^{k}(1 + T(3^i) + T(-3^i))$ in $K[x^{\pm}]$. The geometric sum identity $2\sum_{i=0}^{k} 3^i = 3^{k+1} - 1$ is used to establish that P_k enumerates all integers in $[-M_k, M_k]$ where $M_k = (3^{k+1} - 1)/2$.*

Proof. The geometric sum identity is proved by induction on k. $\square$

Lemma 1.489 (Cardinality identity: $2M_k + 1 = 3^{k+1}$)**.** *For $M_k = \sum_{i=0}^{k} 3^i$, we have $2M_k + 1 = 3^{k+1}$.*

Proof. By induction on k. $\square$

Lemma 1.490 ($2\sum_{i<k} 3^i < 3^k$)**.** *For all $k \geq 0$, we have $2\sum_{i=0}^{k-1} 3^i < 3^k$.*

Proof. By induction on k. $\square$

Lemma 1.491 (Bounded representation values lie in $[-M_k, M_k]$)**.** *The value $\sum_{i=0}^{k} f(i) \cdot 3^i$ of any k-bounded balanced ternary representation f (with $f(i) \in \{-1, 0, 1\}$) lies in the interval $[-M_k, M_k]$.*

Proof. Each term $f(i) \cdot 3^i$ satisfies $-3^i \leq f(i) \cdot 3^i \leq 3^i$, so the total sum is bounded by $\pm \sum_{i=0}^{k} 3^i = \pm M_k$. $\square$

Lemma 1.492 (Injectivity of k-bounded evaluation)**.** *If two k-bounded balanced ternary representations have the same value, they must be equal. The proof uses a mod-3 argument: the lowest digit is determined by the value modulo 3, and dividing by 3 reduces to the induction hypothesis.*

Proof. By strong induction on k. Base case ($k = 0$): the value equals $f(0)$, so equality of values gives equality of digits. Inductive step: the value modulo 3 determines $f(0)$ (since $f(0) \in \{-1, 0, 1\}$), then dividing by 3 gives a $(k-1)$-bounded problem. $\square$

Lemma 1.493 (Conversion between bounded and finitely supported representations)**.** *A k-bounded representation $f : \{0, 1, \ldots, k\} \to \mathbb{Z}$ can be converted to a finitely supported function $\mathbb{N} \to \mathbb{Z}$ by extending with zeros. The weighted sum of the resulting function equals the k-bounded value.*

Proof. The finitely supported function is constructed by extending with zeros. The sum identity follows by reindexing. $\square$

1.16.2 The space $K[[x^{\pm}]]$

Definition 1.494. Let $K[[x^{\pm}]]$ be the K-module $K^{\mathbb{Z}}$ of all families $(a_n)_{n\in\mathbb{Z}} = (\ldots, a_{-2}, a_{-1}, a_0, a_1, a_2, \ldots)$ of elements of K. Its addition and its scaling are defined entrywise:

$$(a_n)_{n\in\mathbb{Z}} + (b_n)_{n\in\mathbb{Z}} = (a_n + b_n)_{n\in\mathbb{Z}};$$
$$\lambda (a_n)_{n\in\mathbb{Z}} = (\lambda a_n)_{n\in\mathbb{Z}} \qquad \text{for each } \lambda \in K.$$

An element of $K[[x^{\pm}]]$ will be called a *doubly infinite power series*. We use the notation $\sum_{n\in\mathbb{Z}} a_n x^n$ for a family $(a_n)_{n\in\mathbb{Z}} \in K[[x^{\pm}]]$.

Lemma 1.495 (Single monomials in $K[\![x^{\pm}]\!]$)**.** *For $n \in \mathbb{Z}$, the monomial $x^n \in K[\![x^{\pm}]\!]$ is the family $(\delta_{m,n})_{m\in\mathbb{Z}}$. Its coefficient at m is $\delta_{m,n}$. Scalar multiplication satisfies $\lambda \cdot x^n = (\lambda\delta_{m,n})_{m\in\mathbb{Z}}$.*

Proof. Immediate from the definitions. $\square$

Lemma 1.496 (Coefficient decomposition of doubly infinite power series)**.** *For any $f = (a_n) \in K[\![x^{\pm}]\!]$ and $m \in \mathbb{Z}$, f evaluated at position m gives a_m. Equivalently, the coefficient of f at m equals the scalar by which f scales the m-th basis vector $(\delta_{n,m})_{n\in\mathbb{Z}}$.*

Proof. Immediate from the definitions. $\square$

1.16.3 Laurent polynomials

Definition 1.497. Let $K[x^{\pm}]$ be the K-submodule of $K[[x^{\pm}]]$ consisting of all **essentially finite** families $(a_n)_{n\in\mathbb{Z}}$. The elements of $K[x^{\pm}]$ are called *Laurent polynomials* in the indeterminate x over K.

We define a multiplication on $K[x^{\pm}]$ by setting

$$(a_n)_{n\in\mathbb{Z}} \cdot (b_n)_{n\in\mathbb{Z}} = (c_n)_{n\in\mathbb{Z}}, \qquad \text{where} \qquad c_n = \sum_{i\in\mathbb{Z}} a_i b_{n-i}.$$

The sum $\sum_{i\in\mathbb{Z}} a_i b_{n-i}$ is well-defined because it is essentially finite.

We define an element $x \in K[x^{\pm}]$ by

$$x = (\delta_{i,1})_{i\in\mathbb{Z}}.$$

In Mathlib, Laurent polynomials are represented as finitely supported functions $\mathbb{Z} \to K$ (the group algebra of $\mathbb{Z}$ over K).

Theorem 1.498. *The K-module $K[x^{\pm}]$, equipped with the multiplication defined above, is a commutative K-algebra. Its unity is $(\delta_{i,0})_{i\in\mathbb{Z}}$. The element x is invertible in this K-algebra.*

Proof. The commutative ring and algebra structures are provided by Mathlib's group algebra structure. Invertibility of $x = T(1)$ is witnessed by $T(-1)$: we have $T(1)\cdot T(-1) = T(1+(-1)) = T(0) = 1$ and similarly $T(-1)\cdot T(1) = 1$. □

Lemma 1.499 (Unit structure of $T(n)$)**.** *For every $n \in \mathbb{Z}$, $T(n)$ is a unit in $K[x^{\pm}]$ with inverse $T(-n)$: we have $T(n)\cdot T(-n) = 1$ and $T(-n)\cdot T(n) = 1$.*

Proof. By $T(a)\cdot T(b) = T(a+b)$ and $T(0) = 1$. □

Lemma 1.500 (Laurent polynomial decomposition)**.** *Every Laurent polynomial $f \in K[x^{\pm}]$ can be written as $f = \sum_{n\in\operatorname{supp}(f)} f(n)\cdot T(n)$. Moreover, $K[x^{\pm}]$ is isomorphic to the group algebra $K[\mathbb{Z}]$.*

Proof. The decomposition follows from the fact that any finitely supported function equals the sum of its single-term components. The group algebra isomorphism is the identity on the underlying finitely supported functions $\mathbb{Z} \to K$. □

Lemma 1.501 (Multiplication by T shifts coefficients)**.** *If $a = (a_n)_{n\in\mathbb{Z}}$ is a Laurent polynomial, then $T(1)\cdot a = (a_{n-1})_{n\in\mathbb{Z}}$ and $T(-1)\cdot a = (a_{n+1})_{n\in\mathbb{Z}}$.*

Proof. Follows from the definition of multiplication: the convolution of a single-term function with a shifts the coefficients. □

Lemma 1.502 (Powers of T)**.** *For each $k \in \mathbb{Z}$, the Laurent polynomial $T(k)$ has coefficient 1 at position k and 0 elsewhere: $T(k)_i = \delta_{i,k}$. Moreover, $T(1)^k = T(k)$ for all $k \in \mathbb{Z}$ (using integer powers via the unit structure).*

Proof. The coefficient characterization follows from the fact that $T(k)$ is the family with 1 at position k and 0 elsewhere. The power identity $T(1)^k = T(k)$ is proved by induction on k (for $k \geq 0$, using $T(a)\cdot T(b) = T(a+b)$; for $k < 0$, using the inverse $T(-1)$). □

Lemma 1.503 (Coefficient of $T(k)$ at position n)**.** *For any $k \in \mathbb{Z}$ and $n \in \mathbb{Z}$, the coefficient $[x^n]T(k)$ equals $\delta_{n,k}$, i.e., 1 if $n = k$ and 0 otherwise. For power series embedded into Laurent series, the coefficient at non-negative index $n \in \mathbb{N}$ is preserved.*

Proof. Follows from the fact that $T(k)$ is the family $(\delta_{n,k})_{n\in\mathbb{Z}}$ and the definition of coefficients. ☐

Lemma 1.504 (Right-multiplication by T on Laurent series)**.** *A Laurent series $f \in K((x))$ equals the summable-family sum of the monomials $f_n \cdot x^n$ (each being the family $(\delta_{m,n} \cdot f_n)_{m\in\mathbb{Z}}$). This constructs an explicit summable family whose sum recovers f.*

Proof. Construct the summable family by sending $n \in \mathbb{Z}$ to the monomial $f_n \cdot x^n$. The support at any given position is a singleton (or empty), hence finite. Checking coefficients: at position m, only the m-th monomial contributes. ☐

Proposition 1.505. *Any doubly infinite power series $a = (a_i)_{i\in\mathbb{Z}} \in K[[x^{\pm}]]$ satisfies*

$$a = \sum_{i\in\mathbb{Z}} a_i x^i.$$

Here, the powers x^i are taken in the Laurent polynomial ring $K[x^{\pm}]$, but the infinite sum $\sum_{i\in\mathbb{Z}} a_i x^i$ is taken in the K-module $K[[x^{\pm}]]$.

For Laurent polynomials, the sum is finite and the identity holds in $K[x^{\pm}]$ itself. For Laurent series, the identity is formalized using a summable family construction.

Proof. For Laurent polynomials, this follows from the decomposition of a finitely supported function as a sum of its single-term components. For Laurent series, the proof constructs a summable family of monomials and shows that its sum equals the original series by checking coefficients: at each position n, only the n-th monomial contributes (using the coefficient characterization of x^k from Lemma 1.502 and Lemma A.114). ☐

1.16.4 Laurent series

Definition 1.506. We let $K((x))$ be the subset of $K[[x^{\pm}]]$ consisting of all families $(a_i)_{i\in\mathbb{Z}} \in K[[x^{\pm}]]$ such that the sequence $(a_{-1}, a_{-2}, a_{-3}, \ldots)$ is essentially finite – i.e., such that all sufficiently low $i \in \mathbb{Z}$ satisfy $a_i = 0$.

The elements of $K((x))$ are called *Laurent series* in one indeterminate x over K.

In Mathlib, Laurent series are implemented as Hahn series over $\mathbb{Z}$ with coefficients in K, which are functions $\mathbb{Z} \to K$ whose support is well-founded (equivalently, bounded below). The formalization also provides a predicate on doubly infinite power series that captures this textbook definition.

Lemma 1.507 (Mathlib's Laurent series satisfy the textbook definition)**.** *Mathlib's Laurent series satisfy the textbook definition (the sequence of negative-indexed coefficients is essentially finite). Moreover, the conversions between doubly infinite power series satisfying this condition and Mathlib's Laurent series are inverse to each other.*

Proof. The forward direction uses the well-foundedness of the support: a Hahn series has a minimum support element (if nonempty), and all indices below it have zero coefficient. The round-trip identities hold by direct computation on coefficients. ☐

Lemma 1.508 (Bounded-below subsets of $\mathbb{Z}$ are partially well-ordered)**.** *If $s \subseteq \mathbb{Z}$ is bounded below, then s is partially well-ordered.*

Proof. Given a sequence in s, shift it to $\mathbb{N}$ by subtracting the lower bound. By Ramsey's theorem (monotone subsequence in $\mathbb{N}$), extract a monotone subsequence. A strictly decreasing sequence in $\mathbb{N}$ is impossible, so the monotone case gives the result. ☐

Theorem 1.509. *The subset $K((x))$ is a K-submodule of $K[[x^{\pm}]]$. It has a multiplication given by the same convolution rule as $K[x^{\pm}]$: namely,*

$$(a_n)_{n\in\mathbb{Z}} \cdot (b_n)_{n\in\mathbb{Z}} = (c_n)_{n\in\mathbb{Z}}, \qquad \text{where} \qquad c_n = \sum_{i\in\mathbb{Z}} a_i b_{n-i}.$$

The sum $\sum_{i\in\mathbb{Z}} a_i b_{n-i}$ is well-defined because for sufficiently low i we have $a_i = 0$, and for sufficiently high i we have $b_{n-i} = 0$.

Equipped with this multiplication, $K((x))$ is a commutative K-algebra with unity $(\delta_{i,0})_{i\in\mathbb{Z}}$. The element x is invertible in this algebra.

Proof. The commutative ring and algebra structures follow from Mathlib's Laurent series ring structure, using the agreement between our definition and Mathlib's (Lemma 1.507). Well-definedness of the product follows from the partial well-orderedness of the support and the order multiplicativity (Lemma 1.510).

The algebra structure extends the FPS algebra via the power series embedding (Lemma 1.511) and includes Laurent polynomials via the Laurent polynomial embedding (Lemma 1.512).

Invertibility of x: the element $x = x^1$ has inverse x^{-1}, since $x^1 \cdot x^{-1} = x^{1+(-1)} = x^0 = 1$. □

Lemma 1.510 (Order of product)**.** *If f and g are Laurent series, then* $\operatorname{ord}(f) + \operatorname{ord}(g) \leq \operatorname{ord}(f \cdot g)$*. In particular, the coefficients of $f \cdot g$ vanish below* $\operatorname{ord}(f) + \operatorname{ord}(g)$*.*

Proof. This is a standard property of Hahn series from Mathlib. □

1.16.5 Embeddings

Lemma 1.511 (Power series embed into Laurent series)**.** *There is an injective ring homomorphism $K[[x]] \hookrightarrow K((x))$ that preserves coefficients.*

Proof. This is provided by Mathlib's embedding of power series into Hahn series, which is injective. □

Lemma 1.512 (Laurent polynomials embed into Laurent series)**.** *There is an injective ring homomorphism $K[x^{\pm}] \hookrightarrow K((x))$ that preserves coefficients.*

Proof. In the first formalization, the map sends a finitely supported function $p : \mathbb{Z} \to K$ to the Laurent series with the same coefficient function (whose support, being finite, is automatically bounded below). Additivity and multiplicativity are checked by comparing coefficients.

In the second formalization, the map is constructed as a ring homomorphism using the universal property of the group algebra, sending $n \in \mathbb{Z}$ to the monomial x^n in $K((x))$. □

Lemma 1.513 (Embedding preserves addition)**.** *The embedding $K[x^{\pm}] \hookrightarrow K((x))$ preserves addition: for Laurent polynomials p, q, the image of $p + q$ equals the sum of their images.*

Proof. By checking coefficients: both sides agree at every position $n \in \mathbb{Z}$. □

Lemma 1.514 (Embedding preserves 0 and 1)**.** *The embedding sends $0 \in K[x^{\pm}]$ to $0 \in K((x))$ and $1 \in K[x^{\pm}]$ to $1 \in K((x))$.*

Proof. By checking coefficients at each position. □

Lemma 1.515 (Embedding preserves coefficients)**.** *The embedding preserves coefficients: for any Laurent polynomial p and $n \in \mathbb{Z}$, the n-th coefficient of the image equals $p(n)$. Moreover, $T(n)$ maps to the monomial x^n in $K((x))$.*

Proof. Immediate from the definition of the embedding. □

1.16.6 The partial product formula

Lemma 1.516 (Geometric identity for balanced ternary factors)**.** *In the Laurent polynomial ring,*

$$(1 + T(1) + T(-1)) \cdot (T(1) \cdot (1 - T(1))) = 1 - T(3).$$

Proof. By direct algebraic computation using $T(a) \cdot T(b) = T(a + b)$ and the ring axioms. ☐

Lemma 1.517 (Invertibility of $1 - x^i$ for $i > 0$)**.** *For any positive integer i, the power series $1 - x^i$ is invertible in $K\,[[x]]$ and hence also in $K\,((x))$.*

Proof. The constant coefficient of $1 - x^i$ is 1, which is a unit. By the invertibility criterion for power series (a power series is invertible if and only if its constant coefficient is a unit), $1 - x^i$ is invertible. ☐

1.16.7 Laurent polynomial action and torsion

Lemma 1.518 (Laurent polynomial action on doubly infinite power series)**.** *A Laurent polynomial $p \in K[x^{\pm}]$ acts on a doubly infinite power series $f \in K[\![x^{\pm}]\!]$ by the convolution formula:*

$$(p \cdot f)_n = \sum_{m \in \operatorname{supp}(p)} p(m) \cdot f(n - m).$$

Proof. Well-definedness follows from the finite support of p, which makes the sum finite. ☐

Lemma 1.519 (Torsion example)**.** *The constant series $\mathbf{1} = (1, 1, 1, \ldots) \in K[\![x^{\pm}]\!]$ satisfies $(1 - x) \cdot \mathbf{1} = 0$. This shows that the $K[x^{\pm}]$-module $K[\![x^{\pm}]\!]$ has torsion: a nonzero module element is annihilated by a nonzero ring element.*

Proof. The coefficient at position n of $(1 - x) \cdot \mathbf{1}$ is $1 \cdot 1 + (-1) \cdot 1 = 0$ (the contribution from position n minus the contribution from position $n - 1$, both equal to 1). ☐

1.16.8 Equivalence of balanced ternary formulations

Lemma 1.520 (Equivalence between the two balanced ternary representations)**.** *The two formulations of balanced ternary representations (the finitely-supported-function formulation and the digit-sequence formulation) are equivalent: for each $n \in \mathbb{Z}$, the two types of balanced ternary representations of n are in bijection. The conversions are mutual inverses (up to the digit representation).*

Proof. The forward direction converts a finitely supported function with values in $\{-1, 0, 1\}$ to a digit sequence $\mathbb{N} \to \{-1, 0, 1\}$ via the natural identification. The backward direction converts a digit sequence to a finitely supported function via the integer embedding. Round-trip identities follow from the fact that the conversions between digits and integers are mutually inverse. ☐

Lemma 1.521 (Bounded representation predicate and sum)**.** *A balanced ternary representation is k-*bounded *if all digits at positions $> k$ are zero. For a k-bounded representation of n, the value equals the finite sum $\sum_{i=0}^{k} d_i \cdot 3^i$.*

Proof. The predicate is $\forall i > k,\ d_i = 0$. The sum identity follows from rewriting the infinite sum as a finite sum over $\{0, \ldots, k\}$ using the vanishing of higher digits. ☐

1.17 Multivariate formal power series

Definition 1.522. Let $k \in \mathbb{N}$. The K-algebra of all FPSs in k variables $x_1, x_2, \ldots, x_k$ over K will be denoted by $K[[x_1, x_2, \ldots, x_k]]$.

Sometimes we will use different names for our variables. For example, if we work with 2 variables, we will commonly call them x and y instead of x_1 and x_2. Correspondingly, we will use the notation $K[[x, y]]$ for the K-algebra of FPSs in these two variables.

1.17.1 Embedding univariate power series into bivariate power series

Definition 1.523. Let $f_0, f_1, f_2, \ldots$ be FPSs in a single variable x. Define the bivariate power series $\operatorname{embed}(f) := \sum_{k \in \mathbb{N}} f_k\, y^k \in K[[x, y]]$, whose coefficient at $x^n y^k$ is the coefficient of x^n in f_k.

Lemma 1.524. *Let $f_0, f_1, f_2, \ldots$ be FPSs in a single variable x. Then for all $n, k \in \mathbb{N}$,*

$$[x^n y^k] \operatorname{embed}(f) = [x^n] f_k.$$

Proof. Immediate from the definition of embed. □

1.17.2 Comparing coefficients of y^k

Proposition 1.525. *Let $f_0, f_1, f_2, \ldots$ and $g_0, g_1, g_2, \ldots$ be FPSs in a single variable x such that*

$$\sum_{k \in \mathbb{N}} f_k y^k = \sum_{k \in \mathbb{N}} g_k y^k \qquad \text{in } K[[x, y]]. \tag{1.41}$$

Then, $f_k = g_k$ for each $k \in \mathbb{N}$.

Proof. For each $k \in \mathbb{N}$, let us write the two FPSs f_k and g_k as $f_k = \sum_{n \in \mathbb{N}} f_{k,n} x^n$ and $g_k = \sum_{n \in \mathbb{N}} g_{k,n} x^n$ with $f_{k,n}, g_{k,n} \in K$. Then, the equality (1.41) can be rewritten as

$$\sum_{k \in \mathbb{N}} \sum_{n \in \mathbb{N}} f_{k,n} x^n y^k = \sum_{k \in \mathbb{N}} \sum_{n \in \mathbb{N}} g_{k,n} x^n y^k.$$

Now, comparing coefficients in front of $x^n y^k$ in this equality, we obtain $f_{k,n} = g_{k,n}$ for each $k, n \in \mathbb{N}$. Therefore, $f_k = g_k$ for each $k \in \mathbb{N}$. □

1.17.3 Application: binomial generating function identity

Definition 1.526. The *binomial generating function* is the bivariate power series

$$\operatorname{BinGen} := \sum_{n,k \in \mathbb{N}} \binom{n}{k} x^n y^k \in \mathbb{Q}[[x, y]],$$

whose coefficient at $x^n y^k$ is $\binom{n}{k}$.

Definition 1.527. The *closed form* of the binomial generating function is

$$\frac{1}{1 - x(1 + y)} \in \mathbb{Q}[[x, y]].$$

Theorem 1.528. *The binomial generating function equals its closed form:*

$$\sum_{n,k\in\mathbb{N}} \binom{n}{k} x^n y^k = \frac{1}{1-x(1+y)}.$$

Proof. It suffices to show that $(1-x(1+y))\cdot \mathrm{BinGen} = 1$, since the constant term of $1-x(1+y)$ is $1 \neq 0$ and inverses in power series rings are unique.

Since $1-x(1+y) = 1-x-xy$, the coefficient of $x^n y^k$ in $(1-x-xy)\cdot \mathrm{BinGen}$ is

$$\binom{n}{k} - \binom{n-1}{k} - \binom{n-1}{k-1}$$

(where we set $\binom{n-1}{k} = 0$ when $n = 0$ and $\binom{n-1}{k-1} = 0$ when $n = 0$ or $k = 0$).

For $n = 0$, this equals $\binom{0}{k} = \delta_{k,0}$, which is the coefficient of $x^0 y^k$ in 1.

For $n \geq 1$ and $k = 0$, this equals $\binom{n}{0} - \binom{n-1}{0} = 1 - 1 = 0$.

For $n \geq 1$ and $k \geq 1$, Pascal's identity gives $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$, so the expression is 0.

Thus $(1-x(1+y))\cdot \mathrm{BinGen} = 1$, and uniqueness of inverses yields $\mathrm{BinGen} = (1-x(1+y))^{-1}$. ☐

From the computation in the source text, comparing the two expressions for $\sum_{n,k}\binom{n}{k}x^n y^k$, we find

$$\sum_{k\in\mathbb{N}} \frac{x^k}{(1-x)^{k+1}} y^k = \sum_{k\in\mathbb{N}} \left(\sum_{n\in\mathbb{N}} \binom{n}{k} x^n\right) y^k. \tag{1.42}$$

1.17.4 Helper lemmas from the formalization

Lemma 1.529. *The inverse of $1-x$ in $\mathbb{Q}[[x]]$ equals the power series with all coefficients equal to 1:*

$$(1-x)^{-1} = \sum_{n\in\mathbb{N}} x^n.$$

Proof. We verify $(1-x)\cdot \sum_{n\in\mathbb{N}} x^n = 1$ by direct computation, and the constant term of $1-x$ is $1 \neq 0$. ☐

Lemma 1.530. *For each $k \in \mathbb{N}$,*

$$\frac{x^k}{(1-x)^{k+1}} = \sum_{n\in\mathbb{N}} \binom{n}{k} x^n \quad \textit{in } \mathbb{Q}[[x]].$$

Proof. We compare coefficients. By the formula for the inverse of $(1-x)^{k+1}$, $(1-x)^{-(k+1)} = \sum_n \binom{n+k}{k} x^n$. Multiplying by x^k shifts the index: the coefficient of x^n in $x^k \cdot (1-x)^{-(k+1)}$ is 0 for $n < k$ and $\binom{n-k+k}{k} = \binom{n}{k}$ for $n \geq k$. For $n < k$, $\binom{n}{k} = 0$ as well, so both sides agree. ☐

Theorem 1.531. *Equation (1.42) holds: as embedded sequences of univariate power series,*

$$\mathrm{embed}\left(k \mapsto \frac{x^k}{(1-x)^{k+1}}\right) = \mathrm{embed}\left(k \mapsto \sum_{n\in\mathbb{N}} \binom{n}{k} x^n\right).$$

Proof. Follows immediately from Lemma 1.530, which shows that the two sequences are equal term by term. ☐

Theorem 1.532. *For each $k \in \mathbb{N}$,*

$$\frac{x^k}{(1-x)^{k+1}} = \sum_{n \in \mathbb{N}} \binom{n}{k} x^n$$

*as an identity in $\mathbb{Q}[[x]]$. This is an equality between **univariate** FPSs, even though we have obtained it by manipulating **bivariate** FPSs.*

Proof. By Theorem 1.531, the two embedded sequences are equal. Apply Proposition 1.525 to extract the k-th component of this equality. $\square$

1.17.5 Partial derivatives

Definition 1.533. Let σ be a type of variable indices and let $i \in \sigma$. The *partial derivative* of a multivariate power series $f = \sum_{\mathbf{m}} a_{\mathbf{m}} \mathbf{x}^{\mathbf{m}}$ with respect to x_i is the power series

$$\frac{\partial f}{\partial x_i} = \sum_{\mathbf{m}} (m_i + 1) \cdot a_{\mathbf{m}+\mathbf{e}_i} \cdot \mathbf{x}^{\mathbf{m}},$$

where $\mathbf{e}_i$ is the i-th standard basis vector (i.e., $\mathbf{e}_i = (\delta_{j,i})_{j \in \sigma}$).

This defines an R-linear map $\partial/\partial x_i \colon R[[\mathbf{x}]] \to R[[\mathbf{x}]]$.

Lemma 1.534. *For any multivariate power series f and any multi-index $\mathbf{m}$,*

$$[\mathbf{x}^{\mathbf{m}}] \frac{\partial f}{\partial x_i} = (m_i + 1) \cdot [\mathbf{x}^{\mathbf{m}+\mathbf{e}_i}] f.$$

Proof. Immediate from the definition. $\square$

Lemma 1.535. *The map $(\mathbf{a}, \mathbf{b}) \mapsto (\mathbf{a} + \mathbf{e}_i, \mathbf{b})$ gives a bijection from* antidiagonal$(\mathbf{m})$ *to the subset of* antidiagonal$(\mathbf{m} + \mathbf{e}_i)$ *consisting of pairs $(\mathbf{a}, \mathbf{b})$ with $\mathbf{e}_i \leq \mathbf{a}$.*

Proof. Verified by checking the membership conditions in both directions. $\square$

Lemma 1.536. *The map $(\mathbf{a}, \mathbf{b}) \mapsto (\mathbf{a}, \mathbf{b} + \mathbf{e}_i)$ gives a bijection from* antidiagonal$(\mathbf{m})$ *to the subset of* antidiagonal$(\mathbf{m} + \mathbf{e}_i)$ *consisting of pairs $(\mathbf{a}, \mathbf{b})$ with $\mathbf{e}_i \leq \mathbf{b}$.*

Proof. Symmetric to Lemma 1.535. $\square$

Lemma 1.537. *In the sum over* antidiagonal$(\mathbf{m} + \mathbf{e}_i)$*, the terms with $a_i = 0$ (i.e., $\mathbf{e}_i \not\leq \mathbf{a}$) contribute zero to $\sum_{(\mathbf{a},\mathbf{b})} a_i \cdot [\mathbf{x}^{\mathbf{a}}] f \cdot [\mathbf{x}^{\mathbf{b}}] g$.*

Proof. Each such term has $a_i = 0$, so the factor a_i makes the whole summand vanish. $\square$

Lemma 1.538. *In the sum over* antidiagonal$(\mathbf{m} + \mathbf{e}_i)$*, the terms with $b_i = 0$ (i.e., $\mathbf{e}_i \not\leq \mathbf{b}$) contribute zero to $\sum_{(\mathbf{a},\mathbf{b})} [\mathbf{x}^{\mathbf{a}}] f \cdot b_i \cdot [\mathbf{x}^{\mathbf{b}}] g$.*

Proof. Each such term has $b_i = 0$, so the factor b_i makes the whole summand vanish. $\square$

Theorem 1.539 (Product rule for partial derivatives)**.** *The partial derivative satisfies the product rule:*

$$\frac{\partial (fg)}{\partial x_i} = \frac{\partial f}{\partial x_i} \cdot g + f \cdot \frac{\partial g}{\partial x_i}.$$

Proof. We compare coefficients at an arbitrary multi-index $\mathbf{m}$. Set $\mathbf{m}' = \mathbf{m} + \mathbf{e}_i$. On the left side, by Lemma 1.534 and the product formula,

$$[\mathbf{x}^{\mathbf{m}}]\, \frac{\partial (fg)}{\partial x_i} = m_i' \cdot \sum_{\mathbf{a}+\mathbf{b}=\mathbf{m}'} [\mathbf{x}^{\mathbf{a}}]\, f \cdot [\mathbf{x}^{\mathbf{b}}]\, g.$$

We distribute the scalar m_i' into each summand. For each pair $(\mathbf{a}, \mathbf{b})$ in antidiagonal$(\mathbf{m}')$, we have $m_i' = a_i + b_i$, so

$$m_i' \cdot [\mathbf{x}^{\mathbf{a}}]\, f \cdot [\mathbf{x}^{\mathbf{b}}]\, g = a_i \cdot [\mathbf{x}^{\mathbf{a}}]\, f \cdot [\mathbf{x}^{\mathbf{b}}]\, g + [\mathbf{x}^{\mathbf{a}}]\, f \cdot b_i \cdot [\mathbf{x}^{\mathbf{b}}]\, g.$$

Summing over the antidiagonal, the first sum equals $[\mathbf{x}^{\mathbf{m}}]\, (\frac{\partial f}{\partial x_i} \cdot g)$ and the second equals $[\mathbf{x}^{\mathbf{m}}]\, (f \cdot \frac{\partial g}{\partial x_i})$, after reindexing via Lemmas 1.535 and 1.536 and discarding the zero terms via Lemmas 1.537 and 1.538. □

Chapter 2

Integer Partitions

2.1 Partition basics

2.1.1 Definitions

Definition 2.1. **(a)** An *(integer) partition* means a (finite) weakly decreasing tuple of positive integers – i.e., a finite tuple $(\lambda_1, \lambda_2, \ldots, \lambda_m)$ of positive integers such that $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_m$.

Thus, partitions are the same as weakly decreasing compositions. Hence, the notions of *size* and *length* of a partition are automatically defined, since we have defined them for compositions (in Definition 1.291).

(b) The *parts* of a partition $(\lambda_1, \lambda_2, \ldots, \lambda_m)$ are simply its entries $\lambda_1, \lambda_2, \ldots, \lambda_m$.

(c) Let $n \in \mathbb{Z}$. A *partition of n* means a partition whose size is n.

(d) Let $n \in \mathbb{Z}$ and $k \in \mathbb{N}$. A *partition of n into k parts* is a partition whose size is n and whose length is k.

Lemma 2.2. *A partition of n has all parts $\leq n$.*

Proof. Since parts are positive and sum to n, each individual part is bounded by n. □

Lemma 2.3. *The partition of 0 has no parts (i.e., it is the empty tuple).*

Proof. By definition. □

Lemma 2.4. *The length of the partition of 0 is 0.*

Proof. Follows from the fact that the partition of 0 has no parts. □

Lemma 2.5. *The cardinality of a multiset of positive naturals is at most its sum.*

Proof. By induction on the multiset. □

Lemma 2.6. *A partition of n has at most n parts.*

Proof. Since each part is ≥ 1, the number of parts is at most the sum of parts, which is n. □

Lemma 2.7. *For $n > 0$, every partition of n has at least one part.*

Proof. If a partition of a positive n had no parts, its sum would be 0, a contradiction. □

Lemma 2.8. *For $n > 0$, every partition of n has a positive largest part.*

Proof. Since the partition has at least one part and all parts are positive, the maximum is positive. $\square$

Lemma 2.9. *For every $n \in \mathbb{N}$, we have $p(n) > 0$.*

Proof. The empty partition (for $n = 0$) or the indiscrete partition (n) (for $n > 0$) provides at least one partition. $\square$

Definition 2.10. **(a)** Let $n \in \mathbb{Z}$ and $k \in \mathbb{N}$. Then, we set

$$p_k(n) := (\# \text{ of partitions of } n \text{ into } k \text{ parts}).$$

(b) Let $n \in \mathbb{Z}$. Then, we set

$$p(n) := (\# \text{ of partitions of } n).$$

This is called the *n-th partition number*.

Definition 2.11. We will use the *Iverson bracket notation*: If $\mathcal{A}$ is a logical statement, then $[\mathcal{A}]$ means the *truth value* of $\mathcal{A}$; this is the integer $\begin{cases} 1, & \text{if } \mathcal{A} \text{ is true;} \\ 0, & \text{if } \mathcal{A} \text{ is false.} \end{cases}$

Note that the Kronecker delta notation is a particular case of the Iverson bracket: We have

$$\delta_{i,j} = [i = j] \qquad \text{for any objects } i \text{ and } j.$$

Lemma 2.12. *The Iverson bracket equals 1 iff the proposition is true.*

Proof. By case analysis on whether the proposition is true or false. $\square$

Lemma 2.13. *The Iverson bracket equals 0 iff the proposition is false (assuming $1 \neq 0$).*

Proof. By case analysis on whether the proposition is true or false. $\square$

Lemma 2.14. *The sum of Iverson brackets equals the cardinality of the filtered set: $\sum_{x \in s} [p(x)] = |s \cap \{x : p(x)\}|$.*

Proof. Follows from the definition of the Iverson bracket and the identity $\sum_{x \in s} [p(x)] = |\{x \in s : p(x)\}|$. $\square$

Lemma 2.15. *The Iverson bracket is multiplicative for conjunctions: $[P \wedge Q] = [P] \cdot [Q]$.*

Proof. By case split on both propositions. $\square$

Definition 2.16. Let a be a real number.

Then, $\lfloor a \rfloor$ (called the *floor* of a) means the largest integer that is $\leq a$.

Likewise, $\lceil a \rceil$ (called the *ceiling* of a) means the smallest integer that is $\geq a$.

2.1.2 Simple properties of partition numbers

Convention 2.17. We agree to say that the largest part of the empty partition () is 0 (even though this partition has no parts).

Lemma 2.18. *The largest part of the partition of* 0 *is* 0.

Proof. The partition of 0 has no parts, so the maximum (with default 0) is 0. □

Lemma 2.19. *For a partition of* n*, the largest part is at most* n.

Proof. Each part is bounded by n (by Lemma 2.2). □

Lemma 2.20. *A partition has largest part* $\leq m$ *if and only if all its parts are* $\leq m$.

Proof. The largest part is the maximum of all parts, so it is $\leq m$ iff every part is $\leq m$. □

Lemma 2.21. *The set of partitions of* n *whose largest part is* $\leq m$ *equals the set of partitions of* n *with all parts* $\leq m$.

Proof. Follows directly from Lemma 2.20. □

Proposition 2.22. *Let* $n \in \mathbb{Z}$ *and* $k \in \mathbb{N}$.

- ***(a)*** *We have* $p_k(n) = 0$ *whenever* $n < 0$ *and* $k \in \mathbb{N}$.
- ***(b)*** *We have* $p_k(n) = 0$ *whenever* $k > n$.
- ***(c)*** *We have* $p_0(n) = [n = 0]$.
- ***(d)*** *We have* $p_1(n) = [n > 0]$.
- ***(e)*** *We have* $p_k(n) = p_k(n-k) + p_{k-1}(n-1)$ *whenever* $k > 0$.
- ***(f)*** *We have* $p_2(n) = \lfloor n/2 \rfloor$ *whenever* $n \in \mathbb{N}$.
- ***(g)*** *We have* $p(n) = p_0(n) + p_1(n) + \cdots + p_n(n)$ *whenever* $n \in \mathbb{N}$.
- ***(h)*** *We have* $p(n) = 0$ *whenever* $n < 0$.

Proof. **(a)** The size of a partition is always nonnegative (being a sum of positive integers). Thus, a negative number n has no partitions whatsoever. Thus, $p_k(n) = 0$ whenever $n < 0$ and $k \in \mathbb{N}$. (In the formalization, n is a natural number, so this case is vacuous.)

(b) If $(\lambda_1, \lambda_2, \ldots, \lambda_k)$ is a partition, then $\lambda_i \geq 1$ for each $i \in \{1, 2, \ldots, k\}$ (because a partition is a tuple of positive integers, i.e., of integers ≥ 1). Hence, if $(\lambda_1, \lambda_2, \ldots, \lambda_k)$ is a partition of n into k parts, then

$$n = \lambda_1 + \lambda_2 + \cdots + \lambda_k \geq \underbrace{1 + 1 + \cdots + 1}_{k \text{ times}} = k.$$

Thus, a partition of n into k parts cannot satisfy $k > n$. Thus, no such partitions exist if $k > n$. In other words, $p_k(n) = 0$ if $k > n$.

(c) The integer 0 has a unique partition into 0 parts, namely the empty tuple (). A nonzero integer n cannot have any partitions into 0 parts, since the empty tuple has size $0 \neq n$. Thus, $p_0(n)$ equals 1 for $n = 0$ and equals 0 for $n \neq 0$. In other words, $p_0(n) = [n = 0]$.

(d) Any positive integer n has a unique partition into 1 part – namely, the 1-tuple (n). On the other hand, if n is not positive, then this 1-tuple is not a partition, so in this case n has no partition into 1 part. Thus, $p_1(n)$ equals 1 if n is positive and equals 0 otherwise. In other words, $p_1(n) = [n > 0]$.

(e) Assume that $k > 0$. We must prove that $p_k(n) = p_k(n-k) + p_{k-1}(n-1)$.
We consider all partitions of n into k parts. We classify these partitions into two types:

- *Type 1* consists of all partitions that have 1 as a part.
- *Type 2* consists of all partitions that don't.

Any type-1 partition has 1 as a part, therefore as its last part (because it is weakly decreasing). Hence, any type-1 partition has the form $(\lambda_1, \lambda_2, \ldots, \lambda_{k-1}, 1)$. If $(\lambda_1, \lambda_2, \ldots, \lambda_{k-1}, 1)$ is a type-1 partition (of n into k parts), then $(\lambda_1, \lambda_2, \ldots, \lambda_{k-1})$ is a partition of $n-1$ into $k-1$ parts. Thus, we have a bijection

$$\{\text{type-1 partitions of } n \text{ into } k \text{ parts}\} \to \{\text{partitions of } n-1 \text{ into } k-1 \text{ parts}\}$$

given by removing the trailing 1.

A type-2 partition does not have 1 as a part; hence, all its parts are larger than 1, and therefore we can subtract 1 from each part and still have a partition. If $(\lambda_1, \lambda_2, \ldots, \lambda_k)$ is a type-2 partition of n into k parts, then $(\lambda_1 - 1, \lambda_2 - 1, \ldots, \lambda_k - 1)$ is a partition of $n-k$ into k parts. This gives a bijection

$$\{\text{type-2 partitions of } n \text{ into } k \text{ parts}\} \to \{\text{partitions of } n-k \text{ into } k \text{ parts}\}.$$

Since any partition of n into k parts is either type-1 or type-2, $p_k(n) = p_k(n-k) + p_{k-1}(n-1)$.

(f) Let $n \in \mathbb{N}$. The partitions of n into 2 parts are

$$(n-1, 1), \quad (n-2, 2), \quad \ldots, \quad (\lceil n/2 \rceil, \lfloor n/2 \rfloor).$$

Thus there are $\lfloor n/2 \rfloor$ of them. In other words, $p_2(n) = \lfloor n/2 \rfloor$.

(g) Let $n \in \mathbb{N}$. Any partition of n must have k parts for some $k \in \mathbb{N}$. Thus,

$$p(n) = \sum_{k \in \mathbb{N}} p_k(n) = \sum_{k=0}^{n} p_k(n) = p_0(n) + p_1(n) + \cdots + p_n(n).$$

(h) Same argument as for **(a)**. □

Helpers for the recurrence (Proposition 2.22 (e))

The recurrence $p_k(n) = p_k(n-k) + p_{k-1}(n-1)$ requires two bijections. The following definitions and lemmas provide the infrastructure.

Definition 2.23. Given a partition of n that contains 1 as a part, removing one copy of 1 yields a partition of $n-1$. Conversely, adding one copy of 1 to a partition of n yields a partition of $n+1$.

Definition 2.24. Given a partition of n into k parts with all parts > 1, subtracting 1 from each part yields a partition of $n-k$ into k parts. Conversely, adding 1 to each part of a partition of n into k parts produces a partition of $n+k$ into k parts.

Lemma 2.25. *The set of partitions of n into k parts splits into those containing* 1 *and those not containing* 1*:*

$$p_k(n) = |\{\textit{partitions with 1}\}| + |\{\textit{partitions without 1}\}|.$$

Proof. By the disjoint union of the two filtered sets. □

Lemma 2.26. *For $k > 0$ and $n > 0$:*

$$|\{\textit{partitions of } n \textit{ into } k \textit{ parts containing 1}\}| = p_{k-1}(n-1).$$

The bijection removes/adds a trailing 1*.*

Proof. The forward map removes one copy of 1; the backward map adds one. Injectivity and surjectivity follow from the fact that removing and adding a trailing 1 are mutually inverse operations. □

Lemma 2.27. *For $k > 0$:*

$$|\{\textit{partitions of } n \textit{ into } k \textit{ parts not containing 1}\}| = p_k(n-k).$$

The bijection subtracts/adds 1 *from/to each part.*

Proof. The forward map subtracts 1 from each part (valid since all parts > 1). The backward map adds 1 to each part. Injectivity uses the fact that subtracting 1 from each entry is injective on multisets whose all elements are > 1. □

Helpers for $p_2(n) = \lfloor n/2 \rfloor$ (Proposition 2.22 (f))

Lemma 2.28. *Every partition of n into 2 parts has the form $\{n-b, b\}$ for some $1 \leq b \leq \lfloor n/2 \rfloor$.*

Proof. If the partition has parts $\{a, b\}$ with $b \leq a$, then $a + b = n$ and $b \geq 1$ (since parts are positive), so $a = n - b$ and $2b \leq n$. □

Lemma 2.29. *We have $p_k(n) = 0$ if and only if $k > n$ or ($k = 0$ and $n > 0$).*

Proof. The forward direction follows by constructing an explicit partition with k parts when $0 < k \leq n$ (namely $(1, 1, \ldots, 1, n-k+1)$). The backward direction uses parts (b) and (c) of Proposition 2.22. □

2.1.3 The generating function

Theorem 2.30. *In the FPS ring $\mathbb{Z}[[x]]$, we have*

$$\sum_{n \in \mathbb{N}} p(n) x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}.$$

(The product on the right hand side is well-defined, since multiplying a FPS by $\dfrac{1}{1-x^k}$ does not affect its first k coefficients.)

Proof. We have

$$\prod_{k=1}^{\infty}\frac{1}{1-x^k}=\prod_{k=1}^{\infty}\sum_{u\in\mathbb{N}}x^{ku}=\sum_{\substack{(u_1,u_2,u_3,\ldots)\in\mathbb{N}^\infty \text{ is}\\ \text{essentially finite}}}x^{1u_1+2u_2+3u_3+\cdots}=\sum_{n\in\mathbb{N}}|Q_n|\,x^n,$$

where

$$Q_n=\{(u_1,u_2,u_3,\ldots)\in\mathbb{N}^\infty \text{ essentially finite } \mid 1u_1+2u_2+3u_3+\cdots=n\}.$$

It suffices to show that $|Q_n|=p(n)$ for each $n\in\mathbb{N}$.

We construct a bijection $\pi:Q_n\to\{\text{partitions of } n\}$: for any $(u_1,u_2,u_3,\ldots)\in Q_n$, define

$$\begin{aligned}\pi(u_1,u_2,u_3,\ldots)&:=(\text{the partition that contains each } i \text{ exactly } u_i \text{ times})\\&=\left(\ldots,\underbrace{3,3,\ldots,3}_{u_3\text{ times}},\underbrace{2,2,\ldots,2}_{u_2\text{ times}},\underbrace{1,1,\ldots,1}_{u_1\text{ times}}\right).\end{aligned}$$

This is a partition of n. Its inverse map ρ sends each partition λ of n to $(\#\text{ of 1's in }\lambda,\ \#\text{ of 2's in }\lambda,\ldots)\in Q_n$. $\square$

Theorem 2.31. *Let $m\in\mathbb{N}$. For each $n\in\mathbb{N}$, let $p_{\text{parts}\le m}(n)$ be the # of partitions λ of n such that all parts of λ are $\le m$. Then,*

$$\sum_{n\in\mathbb{N}}p_{\text{parts}\le m}(n)\,x^n=\prod_{k=1}^{m}\frac{1}{1-x^k}.$$

Proof. This proof is analogous to the proof of Theorem 2.30, using m-tuples instead of infinite sequences. We have

$$\prod_{k=1}^{m}\frac{1}{1-x^k}=\prod_{k=1}^{m}\sum_{u\in\mathbb{N}}x^{ku}=\sum_{(u_1,u_2,\ldots,u_m)\in\mathbb{N}^m}x^{1u_1+2u_2+\cdots+mu_m}=\sum_{n\in\mathbb{N}}|Q_n|\,x^n,$$

where $Q_n=\{(u_1,u_2,\ldots,u_m)\in\mathbb{N}^m \mid 1u_1+2u_2+\cdots+mu_m=n\}$. It suffices to show $|Q_n|=p_{\text{parts}\le m}(n)$. The bijection $\pi:Q_n\to\{\text{partitions }\lambda\text{ of }n\text{ with all parts }\le m\}$ is analogous to that in Theorem 2.30. $\square$

Theorem 2.32. *Let I be a subset of $\{1,2,3,\ldots\}$. For each $n\in\mathbb{N}$, let $p_I(n)$ be the # of partitions λ of n such that all parts of λ belong to I. Then,*

$$\sum_{n\in\mathbb{N}}p_I(n)\,x^n=\prod_{k\in I}\frac{1}{1-x^k}.$$

Proof. Analogous to the proof of Theorem 2.31, with $\prod_{k=1}^m$ replaced by $\prod_{k\in I}$ and m-tuples replaced by essentially finite families $(u_i)_{i\in I}\in\mathbb{N}^I$. $\square$

2.1.4 Odd parts and distinct parts

Definition 2.33. Let $n\in\mathbb{Z}$.

(a) A *partition of n into odd parts* means a partition of n whose all parts are odd.

(b) A *partition of n into distinct parts* means a partition of n whose parts are distinct.

(c) Let

$$p_{\text{odd}}(n) := (\# \text{ of partitions of } n \text{ into odd parts}) \qquad \text{and}$$
$$p_{\text{dist}}(n) := (\# \text{ of partitions of } n \text{ into distinct parts}).$$

Lemma 2.34. *A partition is into odd parts iff all its parts are odd.*

Proof. By definition. □

Lemma 2.35. *A partition is into distinct parts iff its parts multiset has no duplicates.*

Proof. By definition. □

Lemma 2.36. *A partition is into distinct parts iff each part appears at most once.*

Proof. This is the characterization of "no duplicates" in terms of counts. □

Theorem 2.37 (Euler's odd-distinct identity)**.** *We have $p_{\text{odd}}(n) = p_{\text{dist}}(n)$ for each $n \in \mathbb{N}$.*

Proof. Let $n \in \mathbb{N}$. We construct a bijection

$$A : \{\text{partitions of } n \text{ into odd parts}\} \to \{\text{partitions of } n \text{ into distinct parts}\}.$$

Given a partition λ of n into odd parts, we repeatedly merge pairs of equal parts in λ until no more equal parts appear. The final result is $A(\lambda)$.

More precisely: if the original partition λ contains an odd part k exactly m times, and if the binary representation of m is $m = \sum_{j=0}^{i} m_j 2^j$ with $m_j \in \{0, 1\}$, then the partition $A(\lambda)$ contains the number $2^j k$ exactly m_j times for each j. Since the binary digits m_j are all ≤ 1, the partition $A(\lambda)$ has distinct parts.

The inverse map B starts with a partition λ into distinct parts. Each part $k \cdot 2^i$ (with k odd and $i \geq 0$) is replaced by 2^i copies of k. Equivalently, one repeatedly splits even parts into halves until only odd parts remain. The result $B(\lambda)$ is a partition into odd parts.

The maps A and B are mutually inverse, establishing the bijection. Thus $p_{\text{odd}}(n) = p_{\text{dist}}(n)$.

In the formalization, this is proved using Glaisher's theorem from Mathlib, which generalizes this result: for any positive integer d, the number of partitions with parts not divisible by d equals the number of partitions where no part is repeated d or more times. Euler's identity is the special case $d = 2$. □

2.1.5 Partitions with a given largest part

Lemma 2.38. *For any partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$, the* conjugate *(or* transpose*) λ^t is the partition whose Young diagram is obtained by transposing the Young diagram of λ. Explicitly, $\lambda^t = (\mu_1, \mu_2, \ldots, \mu_p)$ where p is the largest part of λ and*

$$\mu_i = (\# \text{ of parts of } \lambda \text{ that are } \geq i) \quad \text{for each } i \in \{1, 2, \ldots, p\}.$$

This satisfies $|\lambda^t| = |\lambda|$.

Proof. The sum of the new parts equals the original size by a double-counting argument: $\sum_i \mu_i = \sum_i \#\{j : \lambda_j \geq i\} = \sum_j \lambda_j = n$. □

Lemma 2.39. *The transpose preserves size: $|\lambda^t| = |\lambda|$, i.e., if λ is a partition of n, then λ^t is also a partition of n.*

Proof. This is a restatement of the defining property of the transpose. □

Lemma 2.40. *The transpose is an involution: $(\lambda^t)^t = \lambda$ for every partition λ.*

Proof. The proof uses the sorted list representation and the key combinatorial fact that for a sorted decreasing list, the transpose operation is controlled by the count of parts exceeding each threshold (Lemma 2.41). This establishes that applying the Young diagram transpose twice recovers the original partition. □

Lemma 2.41. *For a sorted decreasing list $\ell = (\ell_0, \ell_1, \ldots)$, indices $j < |\ell|$, and any $i \in \mathbb{N}$:*

$$|\{k : \ell_k > i\}| > j \iff \ell_j > i.$$

This is the key combinatorial lemma underlying the Young diagram transpose involution: it says the number of parts exceeding i is $> j$ precisely when the j-th part itself exceeds i.

Proof. **Forward:** If $|\{k : \ell_k > i\}| \leq j$, then $\ell_j \leq i$ (since the sorted property implies all elements from position j onward are $\leq \ell_j$, so at most j elements exceed i). Contrapositive gives the result.

Backward: If $\ell_j > i$, then all $\ell_0, \ldots, \ell_j$ are $> i$ (by the sorted property), so $|\{k : \ell_k > i\}| \geq j + 1 > j$. □

Lemma 2.42. *The length of the transpose equals the largest part of the original partition: $\ell(\lambda^t) =$ largest part of λ.*

Proof. The transpose has as many parts as there are columns in the Young diagram, which equals the number of boxes in the first row, i.e., the largest part. □

Lemma 2.43. *For $i <$ largest part of p, the set of parts of p that are $> i$ is nonempty.*

Proof. If all parts were $\leq i$, then the largest part would be $\leq i$, contradicting $i <$ largest part. □

Lemma 2.44. *For a partition p, filtering to keep only the positive parts preserves the cardinality (since all parts are already positive).*

Proof. Every part of a partition is positive, so restricting to positive parts does not change the count. □

Lemma 2.45. *The largest part of the transpose equals the length of the original partition: largest part of $\lambda^t = \ell(\lambda)$.*

Proof. The largest part of the transpose is the number of rows in the transposed diagram, which equals the number of rows of the original, i.e., $\ell(\lambda)$. □

Proposition 2.46. *Let $n \in \mathbb{N}$ and $k \in \mathbb{N}$. Then,*

$$p_k(n) = (\# \text{ of partitions of } n \text{ whose largest part is } k).$$

Proof. We use the transpose as a bijection between:

- partitions of n with k parts, and
- partitions of n with largest part k.

For any partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$, we define the *conjugate* λ^t by $\lambda^t = (\mu_1, \mu_2, \ldots, \mu_p)$, where p is the largest part of λ and $\mu_i = (\text{\# of parts of } \lambda \text{ that are } \geq i)$ for each $i \in \{1, 2, \ldots, p\}$. We have $|\lambda^t| = |\lambda|$ and $(\lambda^t)^t = \lambda$ for each partition λ, and the largest part of λ^t equals the length of λ. By Lemma 2.45, the transpose sends a partition with k parts to one whose largest part is k. By Lemma 2.40, the transpose is an involution, hence a bijection. This establishes the result. $\square$

Corollary 2.47. *Let $n \in \mathbb{N}$ and $k \in \mathbb{N}$. Then,*

$$\begin{aligned}&p_0(n) + p_1(n) + \cdots + p_k(n) \\ &= (\text{\# of partitions of } n \text{ whose largest part is } \leq k).\end{aligned}$$

Proof. We have

$$p_0(n) + p_1(n) + \cdots + p_k(n) = \sum_{i=0}^{k} p_i(n) = \sum_{i=0}^{k} (\text{\# of partitions of } n \text{ whose largest part is } i)$$

(by Proposition 2.46 applied to each i). This sum counts partitions of n whose largest part is $\leq k$. $\square$

Lemma 2.48. *The sum $p_0(n) + p_1(n) + \cdots + p_m(n)$ equals the count of partitions with all parts $\leq m$:*

$$\sum_{k=0}^{m} p_k(n) = p_{\text{parts}\leq m}(n).$$

This combines Corollary 2.47 with the equivalence "largest part $\leq m$" $\iff$ "all parts $\leq m$" (Lemma 2.20).

Proof. Direct from the two referenced results. $\square$

Theorem 2.49. *Let $m \in \mathbb{N}$. Then,*

$$\sum_{n \in \mathbb{N}} (p_0(n) + p_1(n) + \cdots + p_m(n)) x^n = \prod_{k=1}^{m} \frac{1}{1 - x^k}.$$

Proof. For each $n \in \mathbb{N}$, Corollary 2.47 shows that

$$p_0(n) + p_1(n) + \cdots + p_m(n) = (\text{\# of partitions of } n \text{ whose largest part is } \leq m) = p_{\text{parts}\leq m}(n),$$

where $p_{\text{parts}\leq m}(n)$ is defined as in Theorem 2.31. Hence,

$$\sum_{n \in \mathbb{N}} (p_0(n) + p_1(n) + \cdots + p_m(n)) x^n = \sum_{n \in \mathbb{N}} p_{\text{parts}\leq m}(n) x^n = \prod_{k=1}^{m} \frac{1}{1 - x^k}$$

(by Theorem 2.31). $\square$

2.1.6 Counting partitions by parts and largest part

Definition 2.50. Let $k, \ell \in \mathbb{N}$.

1. $q_{k,\ell}(n) := |\{p \vdash n : \ell(p) = k, \text{ largest part of } p = \ell\}|$.

2. $Q(k, \ell) := \sum_{n=k}^{k\ell} q_{k,\ell}(n)$, the total number of such partitions over all valid sizes.

Proposition 2.51. *For positive integers k and ℓ,*

$$Q(k, \ell) = \binom{k + \ell - 2}{k - 1}.$$

Proof. The proof uses a bijection between the disjoint union $\bigsqcup_{n \in [k,k\ell]} \{p \vdash n : \ell(p) = k, \text{ largest} = \ell\}$ and the set of $(k-1)$-element multisubsets of $\{0, 1, \ldots, \ell-1\}$, whose cardinality is $\binom{\ell+(k-1)-1}{k-1} = \binom{k+\ell-2}{k-1}$ by the stars-and-bars formula.

Forward map: A $(k-1)$-element multisubset $\{a_1, \ldots, a_{k-1}\} \subseteq \{0, 1, \ldots, \ell-1\}$ maps to the partition $(\ell, a_1 + 1, \ldots, a_{k-1} + 1)$ sorted in decreasing order. This has k parts, largest part ℓ, and size $\ell + \sum(a_i + 1) \in [k, k\ell]$.

Backward map: A partition $(\ell, \lambda_2, \ldots, \lambda_k)$ maps to the multiset $\{\lambda_2 - 1, \ldots, \lambda_k - 1\} \subseteq \{0, 1, \ldots, \ell - 1\}$, since each $\lambda_i \in \{1, \ldots, \ell\}$.

The maps are mutual inverses: incrementing each element of the multiset by 1 after decrementing undoes the decrement on parts ≥ 1, and prepending ℓ to the result of erasing ℓ recovers the original multiset. $\square$

Lemma 2.52. *Properties of the multiset $(\ell, a_1 + 1, a_2 + 1, \ldots, a_{k-1} + 1)$ for a $(k-1)$-element multisubset $(a_1, \ldots, a_{k-1})$ of $\{0, 1, \ldots, \ell - 1\}$:*

1. *All elements are positive (since $\ell \geq 1$ and each $a_i + 1 \geq 1$).*

2. *Cardinality is k (one ℓ plus $k - 1$ from the multisubset).*

3. *Largest part is ℓ.*

4. *Removing ℓ recovers the remaining parts $(a_1 + 1, \ldots, a_{k-1} + 1)$.*

5. *The sum lies in $[k, k\ell]$.*

Proof. Each property follows from elementary multiset operations. In particular, (3) uses $a_i + 1 \leq \ell$ for $a_i \in \{0, 1, \ldots, \ell - 1\}$, and (5) uses the bounds $1 \leq a_i + 1 \leq \ell$. $\square$

2.1.7 Partition number vs. sums of divisors

Lemma 2.53. *For any set $I \subseteq \{1, 2, 3, \ldots\}$ and $n \in \mathbb{N}$:*

$$n \cdot |\operatorname{restricted}(n, I)| = \sum_{p \in \operatorname{restricted}(n,I)} |\, p \,|.$$

Proof. Each partition p of n has size $|p| = n$, so the sum of a constant equals the constant times the count. $\square$

Lemma 2.54. *For any finite set s of partitions of n and any $d \in \mathbb{N}$:*

$$\sum_{p \in s} \operatorname{count}(d, p) = \sum_{j=1}^{n} |\{p \in s : \operatorname{count}(d, p) \geq j\}|.$$

Proof. This is the standard identity $\sum_i f(i) = \sum_{j\geq 1} |\{i : f(i) \geq j\}|$ applied with $f(p) = \text{count}(d,p)$, bounded by n since at most n/d copies of d fit in a partition of n. □

Lemma 2.55. *For any partition p of n,*

$$\textit{(sum of parts of } p\textit{)} = \sum_{d \text{ distinct in } p} d \cdot \text{count}(d,p).$$

Proof. By induction on the multiset of parts. □

Lemma 2.56. *For any partition p of n and $d > 0$, $\text{count}(d,p) \leq n/d$.*

Proof. At most n/d copies of d can fit in a partition of n. □

Lemma 2.57. *If a partition p of n contains j or more copies of a part d, then removing j copies of d yields a valid partition of $n - d \cdot j$. Conversely, adding j copies of a positive integer d to a partition of m yields a partition of $m + d \cdot j$.*

Proof. Removing copies preserves positivity of remaining parts and adjusts the sum. Adding copies preserves positivity since $d > 0$. □

Lemma 2.58. *Let $I \subseteq \{1, 2, 3, \ldots\}$, let $d \in I$, and let $d \cdot j \leq n$. Then*

$$\#\{p \in \text{restricted}(n, I) : \text{count}(d,p) \geq j\} = \#\,\text{restricted}(n - d \cdot j, I).$$

The bijection sends a partition p to the partition obtained by removing j copies of d, and its inverse adds j copies of d.

Proof. The forward map removes j copies of d from the partition. The backward map adds j copies. Both preserve the property that all parts are in I. Injectivity and surjectivity are verified by showing the maps are mutual inverses. □

Lemma 2.59. *For any set I of positive integers, any $n \in \mathbb{N}$, and any function f:*

$$\sum_{k=0}^{n-1} \sum_{\substack{d \mid k+1 \\ d \in I}} d \cdot f(n-k-1) = \sum_{\substack{d \in I \\ d \leq n}} d \cdot \sum_{j=1}^{\lfloor n/d \rfloor} f(n - d \cdot j).$$

The bijection is $(k,d) \leftrightarrow (d,j)$ where $k = d \cdot j - 1$.

Proof. The reindexing bijection is $(k,d) \leftrightarrow (d,j)$ where $k = d \cdot j - 1$. The forward map sends (k,d) to $(d, (k+1)/d)$ and the backward map sends (d,j) to $(d \cdot j - 1, d)$. □

Theorem 2.60. *Let I be a subset of $\{1, 2, 3, \ldots\}$. For each $n \in \mathbb{N}$, let $p_I(n)$ be the # of partitions λ of n such that all parts of λ belong to I.*

For any positive integer n, let $\sigma_I(n)$ denote the sum of all positive divisors of n that belong to I.

For any $n \in \mathbb{N}$, we have

$$np_I(n) = \sum_{k=1}^{n} \sigma_I(k)\, p_I(n-k).$$

Proof. The proof uses a double counting argument. Both sides count weighted pairs.

LHS Analysis:

$$n \cdot |\mathrm{restricted}(n, I)| = \sum_{p} |p| = \sum_{p} \sum_{\substack{d \text{ distinct} \\ \text{in } p}} d \cdot \mathrm{count}(d, p)$$

Swapping the order of summation and using the identity $\sum_p \mathrm{count}(d, p) = \sum_{j \geq 1} |\{p : \mathrm{count}(d, p) \geq j\}|$, together with the bijection from Lemma 2.58, gives:

$$= \sum_{\substack{d \in I \\ d \leq n}} d \cdot \sum_{j=1}^{\lfloor n/d \rfloor} |\,\mathrm{restricted}(n - d \cdot j, I)|.$$

RHS Analysis:

$$\sum_{k=0}^{n-1} \sigma_I(k+1) \cdot |\,\mathrm{restricted}(n-k-1, I)| = \sum_{k=0}^{n-1} \sum_{\substack{d \mid k+1 \\ d \in I}} d \cdot |\,\mathrm{restricted}(n-k-1, I)|.$$

By the reindexing lemma (Lemma 2.59), this equals the same expression. □

Theorem 2.61. *For any positive integer n, let $\sigma(n)$ denote the sum of all positive divisors of n. (For example, $\sigma(6) = 1 + 2 + 3 + 6 = 12$ and $\sigma(7) = 1 + 7 = 8$.)*

For any $n \in \mathbb{N}$, we have

$$np(n) = \sum_{k=1}^{n} \sigma(k)\, p(n-k).$$

Proof. This is the particular case of Theorem 2.60 for $I = \{1, 2, 3, \ldots\}$. We specialize the restricted version with $I = \{n \in \mathbb{N} : n > 0\}$. Since all parts of a partition are positive, the restricted set equals the set of all partitions, so $p_I(n) = p(n)$. Similarly, $\sigma_I(n) = \sigma(n)$ since all divisors of a positive integer are positive. □

Generating function proof of the recurrence (alternative approach)

The divisor sum recurrence can also be proved via generating functions. Define $P := \sum_{n \geq 0} p(n)x^n$ and $S := \sum_{k \geq 1} \sigma(k)x^k$. Then $X \cdot P' = S \cdot P$, and comparing coefficients gives the result.

Definition 2.62. The partition generating function $P := \sum_{n \geq 0} p(n)x^n \in \mathbb{Z}[[x]]$ and the divisor sum generating function $S := \sum_{k \geq 1} \sigma(k)x^k \in \mathbb{Z}[[x]]$.

Lemma 2.63. *The coefficient of x^n in P equals $p(n)$: $[x^n]P = p(n)$.*

Proof. Immediate from the definition of P. □

Lemma 2.64. *The coefficient of x^n in $X \cdot P'$ equals $n \cdot p(n)$: $[x^n](X \cdot P') = n \cdot [x^n]P$.*

Proof. Since $[x^n](X \cdot f') = n \cdot [x^n]f$ for any FPS f. □

Lemma 2.65. *The coefficient of x^n in $S \cdot P$ equals the RHS of the recurrence:*

$$[x^n](S \cdot P) = \sum_{k=1}^{n} \sigma(k) \cdot p(n-k).$$

Proof. Expand the Cauchy product $[x^n](S\cdot P)=\sum_{i+j=n}[x^i]S\cdot[x^j]P$, noting that $[x^0]S=0$ so the $i=0$ term vanishes. $\square$

Theorem 2.66. *The generating function identity $X\cdot P'=S\cdot P$ holds as an equality of formal power series in $\mathbb{Z}[[x]]$.*

Proof. Comparing coefficients at each x^n: $[x^n](X\cdot P')=n\cdot p(n)=\sum_{k=1}^{n}\sigma(k)\cdot p(n-k)=[x^n](S\cdot P)$, where the middle equality is Theorem 2.61. $\square$

2.2 Euler's pentagonal number theorem and Jacobi's triple product identity

2.2.1 Euler's pentagonal number theorem

Definition 2.67. For any $k\in\mathbb{Z}$, define a nonnegative integer $w_k\in\mathbb{N}$ by

$$w_k=\frac{(3k-1)\,k}{2}.$$

This is called the *k-th pentagonal number.*

Theorem 2.68 (Euler's pentagonal number theorem)**.** *We have*

$$\prod_{k=1}^{\infty}\left(1-x^k\right)=\sum_{k\in\mathbb{Z}}(-1)^k\,x^{w_k}.$$

Proof. Set $q=x^3$ and $z=-x$ in Theorem 2.81. (This means that we apply Theorem 2.81 to $a=3$ and $b=1$ and $u=1$ and $v=-1$.) We get

$$\begin{aligned}&\prod_{n>0}\left(\left(1+\left(x^3\right)^{2n-1}(-x)\right)\left(1+\left(x^3\right)^{2n-1}(-x)^{-1}\right)\left(1-\left(x^3\right)^{2n}\right)\right)\\&=\sum_{\ell\in\mathbb{Z}}\left(x^3\right)^{\ell^2}(-x)^{\ell}.\end{aligned}\tag{2.1}$$

The left hand side of this equality simplifies as follows:

$$\begin{aligned}&\prod_{n>0}\left(\left(1-x^{6n-2}\right)\left(1-x^{6n-4}\right)\left(1-x^{6n}\right)\right)\\&=\prod_{n>0}\left(\left(1-\left(x^2\right)^{3n-1}\right)\left(1-\left(x^2\right)^{3n-2}\right)\left(1-\left(x^2\right)^{3n}\right)\right)\\&=\prod_{k>0}\left(1-\left(x^2\right)^{k}\right),\end{aligned}$$

since each positive integer k can be uniquely represented as $3n-1$ or $3n-2$ or $3n$ for some positive integer n.

Comparing this with (2.1), we obtain

$$\begin{aligned}&\prod_{k>0}\left(1-\left(x^2\right)^{k}\right)\\&=\sum_{\ell\in\mathbb{Z}}(-1)^{\ell}\,x^{3\ell^2+\ell}=\sum_{\ell\in\mathbb{Z}}(-1)^{\ell}\,x^{2w_{-\ell}}\\&=\sum_{k\in\mathbb{Z}}(-1)^{k}\left(x^2\right)^{w_k}\end{aligned}\tag{2.2}$$

(here, we have substituted k for $-\ell$ in the sum, and used the identity $3\ell^2 + \ell = 2w_{-\ell}$).

Now, we "substitute x for x^2" in this equality using Lemma 2.119. As a result, we obtain

$$\prod_{k>0} \left(1 - x^k\right) = \sum_{k \in \mathbb{Z}} (-1)^k \, x^{w_k}.$$

This is Euler's pentagonal number theorem (Theorem 2.68).

The Lean proof proceeds by first proving the identity over $\mathbb{Q}$ (using Lemma 2.122, Lemma 2.120, Lemma 2.121, and Lemma 2.119), and then transferring to $\mathbb{Z}$ by comparing coefficients via the ring homomorphism $\mathbb{Z} \to \mathbb{Q}$ (using Lemma 2.123). □

Pentagonal number properties

Lemma 2.69. *The function $k \mapsto w_k$ is injective: if $w_j = w_k$ then $j = k$.*

Proof. By monotonicity on the positive integers (strictly increasing) and on the negative integers (also strictly increasing in absolute value), and the fact that $w_k = w_j$ forces k, j to have the same sign. □

Lemma 2.70. *For any $n \in \mathbb{N}$, the set $\{k \in \mathbb{Z} : w_k < n\}$ is finite.*

Proof. Since w_k grows quadratically (specifically $w_k \geq |k|$ for $k \neq 0$), only finitely many k satisfy $w_k < n$. □

Definition 2.71. The *pentagonal coefficient* at $n \in \mathbb{N}$ is

$$\mathrm{pentagonalCoeff}(n) = \begin{cases} (-1)^{|k|} & \text{if } n = w_k \text{ for some } k \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases}$$

This is well-defined by the injectivity of pentagonal numbers (Lemma 2.69).

Lemma 2.72. *For any $k \in \mathbb{Z}$, $\mathrm{pentagonalCoeff}(w_k) = (-1)^{|k|}$.*

Proof. Immediate from the definitions, since the inverse lookup recovers k from w_k by injectivity (Lemma 2.69). □

Lemma 2.73. *If $m \in \mathbb{N}$ is not a pentagonal number (i.e., $w_k \neq m$ for all $k \in \mathbb{Z}$), then* $\mathrm{pentagonalCoeff}(m) = 0$.

Proof. By definition of pentagonal coefficients. □

Generating function definitions

Definition 2.74. The *pentagonal series* is the formal power series

$$Q = \sum_{n \geq 0} \mathrm{pentagonalCoeff}(n) \, x^n = \sum_{k \in \mathbb{Z}} (-1)^k x^{w_k} \in R[[x]].$$

Definition 2.75. The *Euler product* is the infinite product

$$\prod_{k=1}^{\infty} (1 - x^k) \in R[[x]],$$

defined as a multipliable product in the Pi topology on $R[[x]]$.

Definition 2.76. The *partition generating function* is

$$P = \sum_{n \geq 0} p(n)\, x^n \in \mathbb{Z}[[x]],$$

where $p(n)$ denotes the number of partitions of n.

Lemma 2.77. *The partition generating function satisfies*

$$P = \sum_{n \geq 0} p(n)\, x^n = \prod_{k=1}^{\infty} \frac{1}{1 - x^k}.$$

Proof. By Mathlib's partition generating function infrastructure. $\square$

Lemma 2.78. *The Euler product times the partition generating function equals* 1*:*

$$\prod_{k=1}^{\infty} (1 - x^k) \cdot \prod_{k=1}^{\infty} \frac{1}{1 - x^k} = 1.$$

Proof. Each factor $(1 - x^k)$ cancels with its inverse, using the per-term identity $(1 + x^k + x^{2k} + \cdots)(1 - x^k) = 1$ and multipliability of the products. $\square$

Corollary 2.79. *For each positive integer n, we have*

$$\begin{aligned} p\,(n) &= \sum_{\substack{k \in \mathbb{Z};\\ k \neq 0}} (-1)^{k-1}\, p\,(n - w_k) \\ &= p\,(n-1) + p\,(n-2) - p\,(n-5) - p\,(n-7) \\ &\qquad + p\,(n-12) + p\,(n-15) - p\,(n-22) - p\,(n-26) \pm \cdots . \end{aligned}$$

Proof. We have

$$\sum_{m \in \mathbb{N}} p\,(m)\, x^m = \prod_{k=1}^{\infty} \frac{1}{1 - x^k}$$

(by Theorem 2.30) and

$$\sum_{k \in \mathbb{Z}} (-1)^k\, x^{w_k} = \prod_{k=1}^{\infty} \left(1 - x^k\right)$$

(by Theorem 2.68). Multiplying these two equalities, we obtain

$$\left(\sum_{m \in \mathbb{N}} p\,(m)\, x^m \right) \cdot \left(\sum_{k \in \mathbb{Z}} (-1)^k\, x^{w_k} \right) = \left(\prod_{k=1}^{\infty} \frac{1}{1 - x^k} \right) \cdot \left(\prod_{k=1}^{\infty} \left(1 - x^k\right) \right) = 1. \tag{2.3}$$

Now, let us fix a positive integer n. We shall compare the x^n-coefficients on both sides of (2.3).

The x^n-coefficient on the left hand side of (2.3) is

$$\sum_{k \in \mathbb{Z}} (-1)^k\, p\,(n - w_k) = p\,(n) + \sum_{\substack{k \in \mathbb{Z};\\ k \neq 0}} (-1)^k\, p\,(n - w_k)\,.$$

But the x^n-coefficient on the right hand side of (2.3) is 0 (since n is positive). Hence, comparing the coefficients yields

$$p(n) + \sum_{\substack{k \in \mathbb{Z}; \\ k \neq 0}} (-1)^k p(n - w_k) = 0.$$

Solving this for $p(n)$, we find

$$p(n) = \sum_{\substack{k \in \mathbb{Z}; \\ k \neq 0}} (-1)^{k-1} p(n - w_k).$$

Corollary 2.79 is thus proved. $\square$

2.2.2 Jacobi's triple product identity

The identity

Theorem 2.80 (Jacobi's triple product identity, take 1). *In the ring $(\mathbb{Z}[z^{\pm}])[[q]]$, we have*

$$\prod_{n>0} \left(\left(1 + q^{2n-1}z\right) \left(1 + q^{2n-1}z^{-1}\right) \left(1 - q^{2n}\right) \right) = \sum_{\ell \in \mathbb{Z}} q^{\ell^2} z^{\ell}.$$

Proof. Both sides are equal after multiplying by the partition generating function $\prod_{k>0}(1 - q^{2k})^{-1}$. This is shown in Lemma 2.112 (for the RHS) and Lemma 2.113 (for the LHS). Since the partition generating function is a unit (Lemma 2.111), the result follows by cancellation. $\square$

Theorem 2.81 (Jacobi's triple product identity, take 2). *Let a and b be two integers such that $a > 0$ and $a \geq |b|$. Let $u, v \in \mathbb{Q}$ be rational numbers with $v \neq 0$. In the ring $\mathbb{Q}[[x]]$, set $q = ux^a$ and $z = vx^b$. Then,*

$$\prod_{n>0} \left(\left(1 + q^{2n-1}z\right) \left(1 + q^{2n-1}z^{-1}\right) \left(1 - q^{2n}\right) \right) = \sum_{\ell \in \mathbb{Z}} q^{\ell^2} z^{\ell}.$$

Proof. Both sides are equal after multiplying by the partition generating function $\prod_{k>0}(1 - q^{2k})^{-1}$. This is shown in Lemma 2.116 (for the RHS) and Lemma 2.117 (for the LHS). Since the partition generating function is a unit (Lemma 2.114), the result follows by cancellation. $\square$

Borcherds' state infrastructure

Definition 2.82. A *level* is a number of the form $p + \frac{1}{2}$ with $p \in \mathbb{Z}$. A *state* is a set of levels that contains all but finitely many negative levels, and only finitely many positive levels.

For any state S, we define:

- the *energy* of S to be
$$\operatorname{energy} S := \sum_{\substack{p>0; \\ p \in S}} 2p - \sum_{\substack{p<0; \\ p \notin S}} 2p \in \mathbb{N};$$

- the *particle number* of S to be
$$\operatorname{parnum} S := (\# \text{ of positive levels in } S) - (\# \text{ of negative levels not in } S) \in \mathbb{Z}.$$

Definition 2.83. For $\ell \in \mathbb{Z}$, the *ℓ-ground state* is

$$G_\ell := \{\text{all levels } < \ell\}.$$

Theorem 2.84. *For any $\ell \in \mathbb{Z}$, we have* energy $G_\ell = \ell^2$ *and* parnum $G_\ell = \ell$.

Proof. If $\ell \geq 0$, the energy is $1+3+5+\cdots+(2\ell-1) = \ell^2$ and the particle number is $\ell - 0 = \ell$. If $\ell < 0$, the energy is $-(-1)-(-3)-\cdots-(2\ell+1) = \ell^2$ and the particle number is $0-(-\ell) = \ell$. □

Definition 2.85. If S is a state, $p \in S$, and q is a positive integer such that $p+q \notin S$, then the state

$$\operatorname{jump}_{p,q} S := (S \setminus \{p\}) \cup \{p+q\}$$

is said to be obtained from S by letting the electron at level p *jump* q steps to the right.

Theorem 2.86. *A jump by q steps keeps the particle number unchanged and raises the energy by $2q$.*

Proof. Three cases are checked: the particle jumps from a positive to a positive level, from a negative to a positive level, or from a negative to a negative level. □

Definition 2.87. For any partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_k)$ and any $\ell \in \mathbb{Z}$, the *excited state $E_{\ell,\lambda}$* is defined by starting with the ℓ-ground state G_ℓ and then successively letting the k electrons at the highest levels jump $\lambda_1, \lambda_2, \ldots, \lambda_k$ steps to the right, respectively:

$$E_{\ell,\lambda} = \{\text{all levels } < \ell - k\} \cup \left\{\ell - i + \tfrac{1}{2} + \lambda_i \;\middle|\; i \in \{1, \ldots, k\}\right\}.$$

Lemma 2.88. *The excited state $E_{\ell,\lambda}$ is reachable from the ground state G_ℓ by a sequence of jumps (in the sense of Definition 2.85).*

Proof. By induction on the number of parts: starting from G_ℓ, one performs k successive jumps of sizes $\lambda_1, \ldots, \lambda_k$. The intermediate state construction ensures each jump is valid (source is occupied, target is vacant). □

Theorem 2.89. *The excited state $E_{\ell,\lambda}$ has energy $\ell^2 + 2|\lambda|$ and particle number ℓ.*

Proof. The state $E_{\ell,\lambda}$ is obtained from G_ℓ by k jumps of $\lambda_1, \ldots, \lambda_k$ steps. Each jump by q steps raises the energy by $2q$ and keeps the particle number unchanged. Since the ground state has energy ℓ^2 and particle number ℓ, the result follows. □

The partition–state bijection

Lemma 2.90. *For fixed $\ell \in \mathbb{Z}$, the map $\lambda \mapsto E_{\ell,\lambda}$ is injective: if $E_{\ell,\mu_1} = E_{\ell,\mu_2}$, then $\mu_1 = \mu_2$.*

Proof. Two excited states are equal if and only if their level sets agree. The partition λ is recovered from the excited levels via $\lambda_i = (i\text{-th largest excited level}) - (\ell - i + \frac{1}{2})$, so equal level sets force equal partitions. The key step is that $i \mapsto \lambda_i - i$ is strictly decreasing for sorted parts, making the level encoding injective. □

Lemma 2.91. *For any $\ell \in \mathbb{Z}$ and any state S with* $\operatorname{parnum}(S) = \ell$, *there exists a partition μ such that $S = E_{\ell,\mu}$.*

Proof. Given S with particle number ℓ, we construct μ by listing the excited levels of S (those above the base level) in decreasing order and computing $\mu_i = (i\text{-th excited level}) - (\ell - i + \frac{1}{2})$. The proof shows these differences are positive integers forming a valid partition, and that the resulting excited state equals S. □

Theorem 2.92. *For each $\ell \in \mathbb{Z}$, the map*

$$\Phi_\ell : \{\textit{partitions}\} \to \{\textit{states with particle number } \ell\}, \quad \lambda \mapsto E_{\ell,\lambda}$$

is a bijection.

Proof. Injectivity is Lemma 2.90. Surjectivity is Lemma 2.91. ☐

Theorem 2.93. *The global map*

$$\Phi : \mathbb{Z} \times \{\textit{partitions}\} \to \{\textit{states}\}, \quad (\ell, \mu) \mapsto E_{\ell,\mu}$$

is a bijection. The inverse sends a state S to $(\mathrm{parnum}(S), \mu)$ where μ is the unique partition with $E_{\mathrm{parnum}(S),\mu} = S$.

Proof. Injectivity: if $E_{\ell_1,\mu_1} = E_{\ell_2,\mu_2}$, comparing particle numbers gives $\ell_1 = \ell_2$, and then injectivity of Φ_ℓ gives $\mu_1 = \mu_2$. Surjectivity: any state S has particle number $\ell = \mathrm{parnum}(S)$, and by the surjectivity of Φ_ℓ, there exists μ with $E_{\ell,\mu} = S$. ☐

Lemma 2.94. *For any state S, there exists $n \geq 0$ such that*

$$\mathrm{energy}(S) = |\,\mathrm{parnum}(S)|^2 + 2n.$$

In particular, the energy is always at least $|\,\mathrm{parnum}(S)|^2$, and the excess is always even.

Proof. By surjectivity (Lemma 2.91), $S = E_{\ell,\mu}$ for some $\ell = \mathrm{parnum}(S)$ and partition μ of some n. Then $\mathrm{energy}(S) = \ell^2 + 2n$ by Theorem 2.89. ☐

Finset pair bijection with states

Definition 2.95. Given a pair (P, N) of finite subsets of $\mathbb{N}$, define the state $\mathrm{fromFinsetPair}(P, N)$ whose levels are

$$\{p \geq 0 : p \in P\} \cup \{p < 0 : (-p - 1) \notin N\}.$$

Here P encodes the nonneg levels present, and N encodes the negative levels missing (with $n \in N$ meaning level $-(n + 1)$ is missing).

Lemma 2.96. *For any pair (P, N) of finite subsets of $\mathbb{N}$,*

$$\mathrm{energy}(\mathrm{fromFinsetPair}(P, N)) = \sum_{n \in P} (2n + 1) + \sum_{n \in N} (2n + 1).$$

Proof. For nonneg level $n \in P$: contribution is $2n + 1$. For missing negative level $-(n + 1)$ with $n \in N$: $|-(n+1)| = n + 1$, so contribution is $2(n + 1) - 1 = 2n + 1$. Both give $2n + 1$. ☐

Lemma 2.97. *For any pair (P, N) of finite subsets of $\mathbb{N}$,*

$$\mathrm{parnum}(\mathrm{fromFinsetPair}(P, N)) = |P| - |N|.$$

Proof. The number of nonneg levels present is $|P|$, and the number of negative levels missing is $|N|$. ☐

Theorem 2.98. *The map* $(P, N) \mapsto \text{fromFinsetPair}(P, N)$ *is a bijection from pairs of finite subsets of* $\mathbb{N}$ *to states.*

Proof. Injectivity: P and N are recovered from any state S by taking $P = \{p \geq 0 : p \in S\}$ and $N = \{n \in \mathbb{N} : -(n+1) \notin S\}$. Surjectivity: any state S equals $\text{fromFinsetPair}(P, N)$ where P and N are constructed as above. □

Lemma 2.99. *For any* $d \in \mathbb{N}$ *and function* $f : \mathbb{Z} \to \mathbb{Z}[z^{\pm}]$*,*

$$\sum_{\substack{(P,N):\\ \text{energy}=d}} f(|P| - |N|) = \sum_{\substack{(\ell,\mu):\\ \ell^2+2|\mu|=d}} f(\ell).$$

Both sums are over finite sets, and equality holds because both sides can be rewritten as $\sum_{S:\text{energy}(S)=d} f(\text{parnum}(S))$ *using the two bijections (finset pairs and integer–partition pairs with states).*

Proof. Both sides equal $\sum_{S:\text{energy}(S)=d} f(\text{parnum}(S))$ via the respective bijections. The LHS uses the finset pair bijection (Theorem 2.98), and the RHS uses the integer–partition bijection (Theorem 2.93). □

Jacobi ring definitions and state generating function

Definition 2.100. The *state monomial* with energy e and particle number p is the element $q^e z^p \in (\mathbb{Z}[z^{\pm}])[[q]]$.

Definition 2.101. The *state generating function* is the formal sum over all integer–partition pairs (ℓ, μ):

$$\mathcal{S} = \sum_{\ell \in \mathbb{Z}} \sum_{\mu \text{ partition}} q^{\ell^2+2|\mu|} z^{\ell} \in (\mathbb{Z}[z^{\pm}])[[q]].$$

It is defined as the formal sum $\sum'_{(\ell,\mu)} q^{\ell^2+2|\mu|} z^{\ell}$ over all integer–partition pairs.

Definition 2.102. The *partition generating function in JacobiRing* is

$$\sum_{\mu \text{ partition}} q^{2|\mu|} = \prod_{k>0} (1 - q^{2k})^{-1} \in (\mathbb{Z}[z^{\pm}])[[q]].$$

Definition 2.103. The *z-dependent product* is

$$\prod_{n>0} \left((1 + q^{2n-1} z)(1 + q^{2n-1} z^{-1})\right) \in (\mathbb{Z}[z^{\pm}])[[q]].$$

Definition 2.104. The *q-only product* is

$$\prod_{n>0} (1 - q^{2n}) \in (\mathbb{Z}[z^{\pm}])[[q]].$$

LHS factorization and product–state connection

Lemma 2.105. *The LHS of Jacobi's identity factors as*

$$\prod_{n>0} \left((1 + q^{2n-1} z)(1 + q^{2n-1} z^{-1})(1 - q^{2n})\right) = \left(\prod_{n>0} (1 + q^{2n-1} z)(1 + q^{2n-1} z^{-1})\right) \cdot \left(\prod_{n>0} (1 - q^{2n})\right).$$

Proof. Each factor in the LHS product is $(1+q^{2n-1}z)(1+q^{2n-1}z^{-1})(1-q^{2n})$, which splits into the product of the first two factors (which form the z-dependent product) and the third factor (which forms the q-only product). The infinite product distributes because both sub-products are multipliable. $\square$

Lemma 2.106. *The partition generating function times the q-only product equals 1:*

$$\prod_{k>0}(1-q^{2k})^{-1}\cdot\prod_{k>0}(1-q^{2k})=1.$$

Proof. Per-term cancellation: $(1+q^{2k}+q^{4k}+\cdots)(1-q^{2k})=1$ for each k. The per-term identity and multipliability of both products give the result. $\square$

Lemma 2.107. *The z-dependent product admits a binary expansion as a double sum:*

$$\prod_{n>0}((1+q^{2n-1}z)(1+q^{2n-1}z^{-1}))=\sum_{P,N\ \mathit{finite}}\left(\prod_{n\in P}(q^{2n+1}z)\right)\left(\prod_{n\in N}(q^{2n+1}z^{-1})\right).$$

Proof. The product of $(1+a_n)(1+b_n)$ over n expands by choosing a subset P of indices contributing a_n and a subset N contributing b_n. This uses the identities $\prod(1+x_n)=\sum_{S\text{ finite}}\prod_{n\in S}x_n$ for each factor, combined via the product rule for infinite products. $\square$

Lemma 2.108. *Each term of the binary expansion has the explicit form*

$$\prod_{n\in P}(q^{2n+1}z)\cdot\prod_{n\in N}(q^{2n+1}z^{-1})=q^{\sum_{n\in P}(2n+1)+\sum_{n\in N}(2n+1)}\cdot z^{|P|-|N|}.$$

Proof. The q-exponents add up, and the z-powers contribute $+1$ for each element of P and -1 for each element of N, giving $z^{|P|-|N|}$. $\square$

Lemma 2.109. *For each $d\in\mathbb{N}$, the q^d-coefficient of the double sum equals the q^d-coefficient of the state generating function:*

$$[q^d]\sum_{P,N}\prod_{n\in P}(q^{2n+1}z)\prod_{n\in N}(q^{2n+1}z^{-1})=[q^d]\sum_{(\ell,\mu)}q^{\ell^2+2|\mu|}z^{\ell}.$$

Proof. By Lemma 2.108, the LHS coefficient is a sum of $z^{|P|-|N|}$ over pairs with $\sum(2n+1)=d$. By Lemma 2.99, this equals the sum of z^ℓ over (ℓ,μ) with $\ell^2+2|\mu|=d$. $\square$

Lemma 2.110. *The z-dependent product equals the state generating function:*

$$\prod_{n>0}((1+q^{2n-1}z)(1+q^{2n-1}z^{-1}))=\mathcal{S}.$$

Proof. By Lemma 2.107, the LHS equals the double sum. By power series extensionality and Lemma 2.109, coefficients match those of the state generating function. $\square$

Proof infrastructure for Jacobi's triple product identity

Lemma 2.111. *The partition generating function $\prod_{k>0}(1-q^{2k})^{-1} \in (\mathbb{Z}[z^{\pm}])[[q]]$ is a unit (its constant term is 1).*

Proof. The constant coefficient is 1, which is a unit. □

Lemma 2.112. *The state generating function equals the RHS of Jacobi's identity times the partition generating function:*

$$\sum_{S\ \mathit{state}} q^{\text{energy } S} z^{\text{parnum } S} = \left(\sum_{\ell\in\mathbb{Z}} q^{\ell^2} z^{\ell}\right) \prod_{n>0} (1-q^{2n})^{-1}.$$

Proof. Using the bijection Φ_ℓ from Theorem 2.92, we rewrite the sum over states as

$$\sum_{\ell\in\mathbb{Z}} \sum_{\lambda \text{ partition}} q^{\ell^2+2|\lambda|} z^{\ell} = \left(\sum_{\ell\in\mathbb{Z}} q^{\ell^2} z^{\ell}\right) \left(\sum_{\lambda \text{ partition}} q^{2|\lambda|}\right).$$

The second factor is $\prod_{n>0}(1-q^{2n})^{-1}$ by the generating function for partitions. □

Lemma 2.113. *The state generating function equals the LHS of Jacobi's identity times the partition generating function:*

$$\sum_{S\ \mathit{state}} q^{\text{energy } S} z^{\text{parnum } S} = \prod_{n>0} \left((1+q^{2n-1}z)(1+q^{2n-1}z^{-1})(1-q^{2n})\right) \cdot \prod_{k>0} (1-q^{2k})^{-1}.$$

Proof. The proof chains the identities. By Lemma 2.110,

$$\mathcal{S} = \prod_{n>0}((1+q^{2n-1}z)(1+q^{2n-1}z^{-1})).$$

By Lemma 2.106, this equals

$$\prod_{n>0}((1+q^{2n-1}z)(1+q^{2n-1}z^{-1})) \cdot \left(\prod_{n>0}(1-q^{2n}) \cdot \prod_{n>0}(1-q^{2n})^{-1}\right).$$

By Lemma 2.105, this equals

$$\prod_{n>0} \left((1+q^{2n-1}z)(1+q^{2n-1}z^{-1})(1-q^{2n})\right) \cdot \prod_{n>0}(1-q^{2n})^{-1}.$$

□

Parameterized Jacobi identity infrastructure

Lemma 2.114. *For integers $a > 0$ and rational u, the evaluated partition generating function $\prod_{k>0}(1-u^{2k}x^{2ak})^{-1} \in \mathbb{Q}[[x]]$ is a unit (its constant term is 1).*

Proof. The constant coefficient is 1, which is a unit in $\mathbb{Q}$. □

Lemma 2.115. *For $a > 0$, the partition generating function times the Euler product (both evaluated at parameter a, u) equals 1:*

$$\prod_{k>0} \frac{1}{1-u^{2k}x^{2ak}} \cdot \prod_{k>0}(1-u^{2k}x^{2ak}) = 1.$$

Proof. Per-factor cancellation: each geometric series factor $(1+u^{2k}x^{2ak}+u^{4k}x^{4ak}+\cdots)$ multiplied by $(1-u^{2k}x^{2ak})$ gives 1. The proof stabilizes the partial products and shows the coefficients eventually agree. □

Lemma 2.116. *The product of the RHS of Jacobi's identity (evaluated at $q = ux^a$, $z = vx^b$) with the partition generating function equals the state generating function:*

$$\left(\sum_{\ell\in\mathbb{Z}} q^{\ell^2} z^\ell\right) \prod_{n>0}(1-q^{2n})^{-1} = \sum_{(\ell,\mu)} u^{\ell^2+2|\mu|} v^\ell \, x^{a(\ell^2+2|\mu|)+b\ell}.$$

Proof. Follows from the Cauchy product formula for infinite sums. □

Lemma 2.117. *The product of the LHS of Jacobi's identity (evaluated at $q = ux^a$, $z = vx^b$) with the partition generating function equals the state generating function.*

Proof. The proof requires: (1) showing that the $(1-q^{2n})$ factors cancel with the partition generating function (via Lemma 2.115), (2) applying binary expansion to the remaining product, (3) reindexing using the state bijection (Theorem 2.92). □

Jacobi implies Euler

Lemma 2.118. *For any $\ell \in \mathbb{Z}$,*

$$3\ell^2 + \ell = 2\,w_{-\ell}.$$

Proof. Direct computation: $3\ell^2+\ell = (3\ell+1)\ell = (3(-\ell)-1)(-\ell) = 2w_{-\ell}$. □

Lemma 2.119. *Let K be a commutative ring. Let f and g be two FPSs in $K[[x]]$. Assume that $f[x^2] = g[x^2]$. Then, $f = g$.*

Proof. Write $f = \sum_{n\in\mathbb{N}} f_n x^n$ and $g = \sum_{n\in\mathbb{N}} g_n x^n$. Then $f[x^2] = \sum_{n\in\mathbb{N}} f_n x^{2n}$ and $g[x^2] = \sum_{n\in\mathbb{N}} g_n x^{2n}$. Comparing x^{2n}-coefficients in $f[x^2] = g[x^2]$ yields $f_n = g_n$ for each $n \in \mathbb{N}$. Hence $f = g$. □

Lemma 2.120. *The LHS of Jacobi's identity evaluated at $a = 3$, $b = 1$, $u = 1$, $v = -1$ equals $\prod_{k>0}(1-x^k)[x^2]$, i.e. the Euler product with x replaced by x^2:*

$$\prod_{n>0}\left((1+q^{2n-1}z)(1+q^{2n-1}z^{-1})(1-q^{2n})\right)\Big|_{q=x^3,\, z=-x} = \left(\prod_{k>0}(1-x^k)\right)[x^2].$$

Proof. The LHS factors simplify: setting $q = x^3$, $z = -x$ gives factors $(1-x^{6n-2})(1-x^{6n-4})(1-x^{6n})$ for each $n > 0$, which is $\prod_{k>0}(1-(x^2)^k)$. □

Lemma 2.121. *The RHS of Jacobi's identity evaluated at $a = 3$, $b = 1$, $u = 1$, $v = -1$ equals the pentagonal series with x replaced by x^2:*

$$\sum_{\ell\in\mathbb{Z}} q^{\ell^2} z^\ell \Big|_{q=x^3,\, z=-x} = \left(\sum_{k\in\mathbb{Z}} (-1)^k x^{w_k}\right)[x^2].$$

Proof. The RHS sum is $\sum_{\ell\in\mathbb{Z}}(-1)^\ell x^{3\ell^2+\ell}$. By Lemma 2.118, $3\ell^2+\ell = 2w_{-\ell}$, so the exponents are all even and the sum equals $\sum_{k\in\mathbb{Z}}(-1)^k (x^2)^{w_k}$, which is the pentagonal series evaluated at x^2. □

Lemma 2.122. *Euler's pentagonal number theorem holds over $\mathbb{Q}$:*

$$\prod_{k=1}^{\infty}(1-x^k) = \sum_{k\in\mathbb{Z}}(-1)^k x^{w_k} \quad \text{in } \mathbb{Q}[[x]].$$

Proof. Apply Lemma 2.119 to reduce to showing $f[x^2] = g[x^2]$. By Lemma 2.120 and Lemma 2.121, both sides squared equal the LHS and RHS of Jacobi's identity evaluated at $a = 3$, $b = 1$, $u = 1$, $v = -1$, which are equal by Theorem 2.81. □

Lemma 2.123. *The coefficients of the Euler product are preserved by the ring homomorphism $\mathbb{Z} \to \mathbb{Q}$:*

$$\iota\left([x^n]\prod_{k>0}(1-x^k)_{\mathbb{Z}}\right) = [x^n]\prod_{k>0}(1-x^k)_{\mathbb{Q}},$$

where $\iota : \mathbb{Z} \hookrightarrow \mathbb{Q}$ is the inclusion.

Proof. The ring homomorphism $\mathbb{Z} \to \mathbb{Q}$ sends the $\mathbb{Z}$-product to the $\mathbb{Q}$-product by continuity of the induced map on power series, and coefficients commute with the map. □

Application: A recursion for the sum of divisors

Lemma 2.124. *We have $P \cdot Q = 1$, where $P = \sum_{n\geq 0} p(n)x^n$ is the partition generating function and $Q = \sum_{k\in\mathbb{Z}}(-1)^k x^{w_k}$ is the pentagonal series.*

Proof. This follows from the partition generating function identity $P = \prod_{k\geq 1}(1-x^k)^{-1}$ (Lemma 2.77) and Euler's pentagonal number theorem $Q = \prod_{k\geq 1}(1-x^k)$ (Theorem 2.68). Their product $PQ = 1$ is established by Lemma 2.78. □

Definition 2.125. The *sum-of-divisors generating function* is

$$S = \sum_{k>0}\sigma(k)\, x^k \in \mathbb{Z}[[x]],$$

where $\sigma(k) = \sigma_1(k) = \sum_{d\mid k} d$ is the sum of positive divisors of k.

Lemma 2.126. *For each $n \in \mathbb{N}$,*

$$n \cdot p(n) = \sum_{k=1}^{n}\sigma(k)\cdot p(n-k).$$

Proof. The proof uses a combinatorial double-counting argument: a marked partition (p, i) (a partition p of n together with a marked cell $i \in \{1, \ldots, n\}$) can equivalently be described by (d, m, j, μ) where d is the row length of the marked cell, m is the number of rows of length d, j is the column of the cell, and μ is the partition obtained by removing all rows of length d. The identity follows by summing $\sum_{k=1}^{n}\sigma(k)p(n-k) = \sum_{d\mid k} d\cdot p(n-k)$ and reindexing. □

Lemma 2.127. *We have $xP' = SP$, where P is the partition generating function and $S = \sum_{k>0}\sigma(k)x^k$ is the sum-of-divisors generating function.*

Proof. Both sides have the same coefficients: the coefficient of x^n on the left is $n\cdot p(n)$, and on the right is $\sum_{k=1}^{n}\sigma(k)\cdot p(n-k)$. These are equal by the combinatorial identity (Lemma 2.126). □

Lemma 2.128. *We have $xQ' = -QS$, where Q is the pentagonal series and S is the sum-of-divisors generating function.*

Proof. From $PQ = 1$, differentiate to get $P'Q + PQ' = 0$, so $PQ' = -P'Q$. Multiply by x: $xPQ' = -xP'Q = -SPQ = -S$ (using $xP' = SP$ and $PQ = 1$). Since $PQ = 1$ is a unit, multiply both sides by Q to get $xQ' = -QS$. $\square$

Lemma 2.129. *The n-th coefficient of xQ' is determined by pentagonal numbers:*

$$[x^n](xQ') = \begin{cases} (-1)^{|k|} \cdot n & \textit{if } n = w_k \textit{ for some } k \in \mathbb{Z}, \\ 0 & \textit{otherwise.} \end{cases}$$

This follows from $xQ' = \sum_{k\in\mathbb{Z}}(-1)^k w_k x^{w_k}$.

Proof. Taking the derivative of $Q = \sum_n \text{pentagonalCoeff}(n)x^n$ and multiplying by x gives $[x^n](xQ') = n \cdot \text{pentagonalCoeff}(n)$. The result follows from Lemma 2.72 and Lemma 2.73. $\square$

Theorem 2.130. *For any positive integer n, let $\sigma(n)$ denote the sum of all positive divisors of n.*

Then, for each positive integer n, we have

$$\sum_{\substack{k\in\mathbb{Z};\\ w_k<n}} (-1)^k \sigma(n - w_k) = \begin{cases} (-1)^{k-1} n, & \textit{if } n = w_k \textit{ for some } k \in \mathbb{Z}; \\ 0, & \textit{if not.} \end{cases}$$

(Here, w_k is the k-th pentagonal number, defined in Definition 2.67.)

Proof. From Lemma 2.128, we have $xQ' = -QS$, where $Q = \sum_{k\in\mathbb{Z}}(-1)^k x^{w_k}$ and $S = \sum_{k>0}\sigma(k)x^k$.

Taking the derivative:

$$xQ' = \sum_{k\in\mathbb{Z}}(-1)^k w_k x^{w_k}.$$

On the other hand:

$$-QS = -\left(\sum_{k\in\mathbb{Z}}(-1)^k x^{w_k}\right)\left(\sum_{i>0}\sigma(i)x^i\right) = \sum_{m>0}\sum_{\substack{k\in\mathbb{Z};\\ m>w_k}}(-1)^{k-1}\sigma(m-w_k)x^m.$$

Comparing the coefficient of x^n on both sides:

- On the left, the coefficient is $(-1)^k w_k$ if $n = w_k$ for some k, and 0 otherwise (Lemma 2.129);
- On the right, the coefficient is $\sum_{\substack{k\in\mathbb{Z};\\ w_k<n}}(-1)^{k-1}\sigma(n-w_k)$.

Rearranging gives the result. $\square$

Alternative proof via evaluation homomorphism

Definition 2.131. The *evaluation map* from $(\mathbb{Z}[z^{\pm}])[[q]]$ to $\mathbb{Q}[[x]]$ is defined by sending a power series $f = \sum_e c_e q^e$ (where $c_e \in \mathbb{Z}[z^{\pm}]$) to

$$\text{eval}_{a,b,u,v}(f) = \sum_{e\geq 0} u^e \cdot \widehat{c_e}(a,b,v,e),$$

where $\widehat{c_e}(a,b,v,e)$ denotes the shifted evaluation that replaces each z^ℓ in c_e by $v^\ell x^{ae+b\ell}$.

Lemma 2.132. *Evaluating the RHS of Jacobi's identity gives the parameterized RHS:*

$$\operatorname{eval}_{a,b,u,v}\left(\sum_{\ell\in\mathbb{Z}} q^{\ell^2} z^{\ell}\right) = \sum_{\ell\in\mathbb{Z}} u^{\ell^2} v^{\ell}\, x^{a\ell^2+b\ell}.$$

Proof. Each sum term $q^{\ell^2} z^{\ell}$ is evaluated to $u^{\ell^2} v^{\ell} x^{a\ell^2+b\ell}$. The evaluation distributes over the infinite sum by summability and continuity. □

Theorem 2.133. *The parameterized Jacobi triple product identity (Theorem 2.81) can also be derived from Theorem 2.80 by applying the evaluation map* $\operatorname{eval}_{a,b,u,v}$.

Proof. Apply $\operatorname{eval}_{a,b,u,v}$ to both sides of the identity from Theorem 2.80 (i.e., the LHS = RHS in $(\mathbb{Z}[z^{\pm}])[[q]]$). By Lemma 2.132, the RHS evaluates to the parameterized RHS. The LHS evaluates to the parameterized LHS by a corresponding evaluation lemma. □

2.3 q-Binomial Coefficients

Convention 2.134. In this section, we will mostly be using FPSs in the indeterminate q. That is, we call the indeterminate q rather than x. The ring of FPSs in the indeterminate q over a commutative ring K will be denoted by $K[[q]]$. The ring of polynomials in the indeterminate q over K will be denoted by $K[q]$.

2.3.1 Motivation

Proposition 2.135. *For any positive integers k and ℓ, we have*

$$(\#\text{ of partitions with } k \text{ parts and largest part } \ell) = \binom{k+\ell-2}{k-1}.$$

Proof. Consider the Young diagram of a partition with k parts and largest part ℓ. Its lower boundary is a lattice path from $(0,0)$ to (ℓ,k) consisting of east-steps and north-steps. This path begins with an east-step (since otherwise, the partition would have fewer than k parts) and ends with a north-step (since otherwise, the largest part would be $<\ell$). There are precisely $k+\ell$ steps total, and the first and last are predetermined, so it remains to choose which $k-1$ of the remaining $k+\ell-2$ steps are north-steps. The number of such choices is $\binom{k+\ell-2}{k-1}$. □

2.3.2 Definition

Definition 2.136. Let $n\in\mathbb{N}$ and $k\in\mathbb{Z}$.

(a) The q-binomial coefficient (or Gaussian binomial coefficient) $\binom{n}{k}_q$ is defined to be the polynomial

$$\sum_{\substack{\lambda \text{ is a partition}\\ \text{with largest part } \leq n-k\\ \text{and length } \leq k}} q^{|\lambda|} \in \mathbb{Z}[q].$$

(b) If a is any element of a ring A, then we set

$$\binom{n}{k}_a := \binom{n}{k}_q [a] = \sum_{\substack{\lambda \text{ is a partition}\\ \text{with largest part } \leq n-k\\ \text{and length } \leq k}} a^{|\lambda|} \in A.$$

2.3.3 Basic properties

Lemma 2.137. *For each $s \leq k \cdot m$, the number of weakly increasing tuples $(f_0, f_1, \ldots, f_{k-1}) \in \{0, 1, \ldots, m\}^k$ with value sum s equals the number of partitions of s fitting in a $k \times m$ box.*

Proof. The forward map sends f to the partition whose parts are the nonzero values of f; the inverse pads the partition's parts with zeros to length k and sorts. This is a sum-preserving bijection. ☐

Lemma 2.138. *Let $n, k \in \mathbb{N}$ with $k \leq n$. Then*

$$\binom{n}{k}_q = \sum_{m=0}^{k(n-k)} \left|\{\text{partitions of } m \text{ fitting in a } k \times (n-k) \text{ box}\}\right| \cdot q^m.$$

Proof. This is a restatement of the definition, grouping the sum by partition size $m = |\lambda|$, and applying Lemma 2.137 to count the monotone functions with each given value sum. ☐

Lemma 2.139. *The number of weakly increasing k-tuples from $\{0, 1, \ldots, \ell\}$ equals $\binom{k+\ell}{k}$, the number of multisets of size k from $\ell + 1$ elements.*

Proof. The set of weakly increasing tuples is in bijection with the set of multisets of size k from $\ell + 1$ elements, whose cardinality is $\binom{k+\ell}{k}$ by the stars-and-bars formula. ☐

Proposition 2.140. *Let $n \in \mathbb{N}$ and $k \in \mathbb{Z}$.*

(a) *We have*

$$\binom{n}{k}_q = \sum_{0 \leq i_1 \leq i_2 \leq \cdots \leq i_k \leq n-k} q^{i_1+i_2+\cdots+i_k}.$$

Here, the sum ranges over all weakly increasing k-tuples $(i_1, i_2, \ldots, i_k) \in \{0, 1, \ldots, n-k\}^k$. If $k > n$, then this is an empty sum. If $k < 0$, then this is also an empty sum.

(b) *Set* sum $S = \sum_{s \in S} s$ *for any finite set S of integers. Then, we have*

$$\binom{n}{k}_q = \sum_{\substack{S \subseteq \{1,2,\ldots,n\};\\ |S|=k}} q^{\operatorname{sum} S - (1+2+\cdots+k)}.$$

(c) *We have*

$$\binom{n}{k}_1 = \binom{n}{k}.$$

Proof. **(a)** The definition of $\binom{n}{k}_q$ yields

$$\binom{n}{k}_q = \sum_{\substack{\lambda \text{ is a partition}\\ \text{with largest part } \leq n-k\\ \text{and length } \leq k}} q^{|\lambda|} = \sum_{\ell=0}^{k} \sum_{\substack{\lambda \text{ is a partition}\\ \text{with largest part } \leq n-k\\ \text{and length } \ell}} q^{|\lambda|}.$$

For each fixed $\ell \in \{0, 1, \ldots, k\}$, the partitions λ with largest part $\leq n-k$ and length ℓ correspond bijectively to k-tuples $(\lambda_1, \lambda_2, \ldots, \lambda_k) \in \mathbb{N}^k$ satisfying $n-k \geq \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 0$ and having exactly ℓ positive entries (by padding with zeros). Summing over all ℓ and then reversing the k-tuple (substituting $(i_k, i_{k-1}, \ldots, i_1)$ for $(\lambda_1, \ldots, \lambda_k)$) gives part **(a)**.

(b) There is a bijection from weakly increasing k-tuples $(i_1, \ldots, i_k) \in \{0, 1, \ldots, n-k\}^k$ to strictly increasing k-tuples $(s_1, \ldots, s_k) \in \{1, 2, \ldots, n\}^k$, given by $(i_1, \ldots, i_k) \mapsto (i_1 + 1,\ i_2 + 2,\ \ldots,\ i_k + k)$. Moreover, strictly increasing k-tuples from $\{1, \ldots, n\}$ correspond bijectively to k-element subsets $S \subseteq \{1, \ldots, n\}$. Combining with **(a)** gives **(b)**.

(c) Substituting 1 for q in **(b)** gives $\dbinom{n}{k}_1 = |\{S \subseteq \{1, \ldots, n\} : |S| = k\}| = \dbinom{n}{k}$. ∎

Proposition 2.141. *Let $n, k \in \mathbb{N}$ satisfy $k > n$. Then, $\dbinom{n}{k}_q = 0$.*

Proof. From $k > n$, we obtain $n - k < 0$. The sum in the definition is empty, since there exists no partition with largest part $\leq n - k < 0$. Thus $\dbinom{n}{k}_q = 0$. ∎

Proposition 2.142. *We have $\dbinom{n}{0}_q = \dbinom{n}{n}_q = 1$ for each $n \in \mathbb{N}$.*

Proof. For $\dbinom{n}{0}_q$: the only weakly increasing 0-tuple is the empty one, whose value sum is 0, giving $q^0 = 1$.

For $\dbinom{n}{n}_q$: when $k = n$, we have $n - k = 0$, so the values must lie in $\{0\}$, and the only weakly increasing tuple sends everything to 0; the value sum is 0, giving $q^0 = 1$. ∎

Convention 2.143. Let $n \in \mathbb{N}$. For any $k \notin \mathbb{Z}$, we set $\dbinom{n}{k}_q := 0$.

2.3.4 Recurrence relations

Theorem 2.144. *Let n be a positive integer. Let $k \in \mathbb{N}$. Then:*

(a) *We have*

$$\binom{n}{k}_q = q^{n-k} \binom{n-1}{k-1}_q + \binom{n-1}{k}_q.$$

(b) *We have*

$$\binom{n}{k}_q = \binom{n-1}{k-1}_q + q^k \binom{n-1}{k}_q.$$

Proof. **(a)** If $k = 0$, the claim reduces to $1 = q^{n-k} \cdot 0 + 1$, which holds by Proposition 2.142 and Convention 2.143. So assume $k > 0$.

Using Proposition 2.140 **(b)**, we write $\dbinom{n}{k}_q$ as a sum over k-element subsets $S \subseteq \{1, \ldots, n\}$. We split these subsets into type-1 (containing n) and type-2 (not containing n).

The type-2 subsets are exactly the k-element subsets of $\{1, \ldots, n-1\}$, contributing $\dbinom{n-1}{k}_q$.

The type-1 subsets biject to $(k-1)$-element subsets of $\{1, \ldots, n-1\}$ via $S \mapsto S \setminus \{n\}$. Since $\operatorname{sum}(S \cup \{n\}) = \operatorname{sum} S + n$, the exponent shifts by $n - k$, giving $q^{n-k} \dbinom{n-1}{k-1}_q$.

(b) This is proved similarly, or by combining part **(a)** with symmetry (applied locally before the main symmetry theorem is established). ∎

2.3.5 Product and quotient formulas

Lemma 2.145. *Over a field, if $q^n \neq 1$ then $[n]_q \neq 0$.*

Proof. By Lemma 2.149, $(1-q)[n]_q = 1-q^n$. If $[n]_q = 0$, then $1-q^n = 0$, contradicting $q^n \neq 1$. $\square$

Lemma 2.146. *Over a field, if $q^i \neq 1$ for all $i = 1, \ldots, n$, then $[n]_q! \neq 0$.*

Proof. Since $[n]_q! = \prod_{i=1}^n [i]_q$ and each factor is nonzero by Lemma 2.145, the product is nonzero. $\square$

Theorem 2.147. *Let $n, k \in \mathbb{N}$ satisfy $n \geq k$. Then:*

(a) *We have*

$$(1-q^k)(1-q^{k-1})\cdots(1-q^1)\cdot\binom{n}{k}_q = (1-q^n)(1-q^{n-1})\cdots(1-q^{n-k+1}).$$

(b) *We have*

$$\binom{n}{k}_q = \frac{(1-q^n)(1-q^{n-1})\cdots(1-q^{n-k+1})}{(1-q^k)(1-q^{k-1})\cdots(1-q^1)}$$

(in the ring $\mathbb{Z}[[q]]$ or in the field of rational functions over $\mathbb{Q}$).

Proof. **(a)** By induction on n (strong induction). For $k = 0$ both sides equal 1. For $k > 0$, we use the recurrence from Theorem 2.144 **(a)** to split $\binom{n}{k}_q$, multiply both sides by the product $(1-q^k)\cdots(1-q)$, apply the inductive hypothesis to the two resulting terms (at $n-1$), and combine using the identity $(1-q^{k+1})q^{n-k-1} + (1-q^{n-k-1}) = 1-q^n$ (after appropriate index shifts).

(b) Divide both sides of **(a)** by $(1-q^k)(1-q^{k-1})\cdots(1-q)$, which is nonzero when q is not a root of unity of order $\leq k$. $\square$

2.3.6 q-integers and q-factorials

Definition 2.148. **(a)** For any $n \in \mathbb{N}$, define the q-integer $[n]_q$ to be

$$[n]_q := q^0 + q^1 + \cdots + q^{n-1} \in \mathbb{Z}[q].$$

(b) For any $n \in \mathbb{N}$, define the q-factorial $[n]_q!$ to be

$$[n]_q! := [1]_q[2]_q\cdots[n]_q \in \mathbb{Z}[q].$$

(c) As usual, if a is an element of a ring A, then $[n]_a$ and $[n]_a!$ will mean the results of substituting a for q in $[n]_q$ and $[n]_q!$, respectively. Thus, explicitly, $[n]_a = a^0 + a^1 + \cdots + a^{n-1}$ and $[n]_a! = [1]_a[2]_a\cdots[n]_a$.

Lemma 2.149. *For any $n \in \mathbb{N}$, we have*

$$(1-q)\cdot[n]_q = 1-q^n$$

and hence $[n]_q = \dfrac{1-q^n}{1-q}$ (in $\mathbb{Z}[[q]]$ or in the field of rational functions over $\mathbb{Q}$).

Proof. Telescoping: $(1-q)(q^0+q^1+\cdots+q^{n-1}) = (q^0+\cdots+q^{n-1})-(q^1+\cdots+q^n) = 1-q^n$. $\square$

Lemma 2.150. *For any $n \in \mathbb{N}$, we have $[n]_1 = n$ and $[n]_1! = n!$.*

Proof. $[n]_1 = 1^0+1^1+\cdots+1^{n-1} = \underbrace{1+1+\cdots+1}_{n \text{ times}} = n$. Then $[n]_1! = [1]_1[2]_1\cdots[n]_1 = 1{\cdot}2\cdots n = n!$. $\square$

Lemma 2.151. *For $k \leq n$, we have*

$$[n]_q! = \big([n]_q \cdot [n{-}1]_q \cdots [n{-}k{+}1]_q\big) \cdot [n{-}k]_q!\,.$$

Proof. By induction on k. The base case $k=0$ is trivial. For $k+1$, split off the last factor $[n-k]_q$ from the falling product and use $[n-k]_q! = [n-k-1]_q! \cdot [n-k]_q$. $\square$

Lemma 2.152. *Over a field, when q is not a root of unity of order $\leq k$,*

$$\binom{n}{k}_q = \prod_{i=0}^{k-1} \frac{[n-i]_q}{[i+1]_q}.$$

Proof. Divide the product formula from Theorem 2.147 **(b)** by $(1-q)^k$ to replace each factor $(1-q^m)/(1-q)$ with $[m]_q$. $\square$

Theorem 2.153. *Let $n, k \in \mathbb{N}$ with $n \geq k$. Then,*

$$\binom{n}{k}_q = \frac{[n]_q[n-1]_q\cdots[n-k+1]_q}{[k]_q!} = \frac{[n]_q!}{[k]_q! \cdot [n-k]_q!}$$

(in the ring $\mathbb{Z}[[q]]$ or in the ring of rational functions over $\mathbb{Q}$).

Proof. The first equality follows from Theorem 2.147 **(b)** by rewriting each factor $(1-q^m)/(1-q)$ as $[m]_q$ (using Lemma 2.149). The second equality follows from the splitting $[n]_q! = [n]_q[n{-}1]_q\cdots[n{-}k{+}1]_q \cdot [n{-}k]_q!$ (Lemma 2.151). $\square$

2.3.7 Symmetry

Helpers for Proposition 2.156

Lemma 2.154. *Let $(f_0, f_1, \ldots, f_{k-1})$ be a weakly increasing tuple in $\{0,1,\ldots,m\}^k$. Define the transpose $(g_0, g_1, \ldots, g_{m-1})$ with $g_j = \#\{i : f_i > m{-}1{-}j\}$. Then applying the transpose operation twice recovers the original tuple.*

Proof. For each index i, the number of $j \in \{0,1,\ldots,m{-}1\}$ with $g_j > k{-}1{-}i$ equals the number of j with $j \geq m - f_i$, which equals f_i. The proof uses the complement counting identity and the characterization of the profile of f via the predicate $f_{i'} \leq m{-}1{-}j$. $\square$

Lemma 2.155. *The transpose operation preserves the sum of values: $\sum_j g(j) = \sum_i f(i)$.*

Proof. By swapping the order of summation in the double sum $\sum_j \#\{i : f(i) > m{-}1{-}j\}$ and using the identity $\#\{j : j \geq m-v\} = v$ for $0 \leq v \leq m$. $\square$

Proposition 2.156. *Let $n, k \in \mathbb{N}$. Then,*

$$\binom{n}{k}_q = \binom{n}{n-k}_q.$$

Proof. We construct an explicit involution on weakly increasing tuples. Given a weakly increasing tuple $(f_0, f_1, \ldots, f_{k-1}) \in \{0, 1, \ldots, m\}^k$ (where $m = n - k$), define the *transpose* $(g_0, g_1, \ldots, g_{m-1}) \in \{0, 1, \ldots, k\}^m$ by

$$g_j = \#\{i : f_i > m - 1 - j\}.$$

This tuple is again weakly increasing. The transpose operation is an involution (applying it twice recovers the original tuple) and preserves the sum of values ($\sum_j g_j = \sum_i f_i$). It therefore gives a sum-preserving bijection between weakly increasing k-tuples in $\{0, \ldots, n-k\}$ and weakly increasing $(n-k)$-tuples in $\{0, \ldots, k\}$, proving $\binom{n}{k}_q = \binom{n}{n-k}_q$. □

2.4 q-Binomial formulas

2.4.1 Combinatorial definition of q-binomial coefficients

This section formalizes Definition 2.136 from the combinatorial side: the q-binomial as a sum over partitions, and proves that this agrees with the recurrence-based definition from the previous chapter.

Definition 2.157. The q-binomial coefficient as a polynomial in $\mathbb{Z}[q]$ via the partition sum:

$$\binom{n}{k}_q = \sum_{s=0}^{k(n-k)} c(s, k, n-k) \cdot q^s$$

where $c(s, k, m)$ is the number of partitions of s fitting in a $k \times m$ box. The evaluation $\binom{n}{k}_a$ is obtained by substituting a for q.

Lemma 2.158. *The combinatorial q-binomial satisfies the q-Pascal identity as polynomials:*

$$\binom{n+1}{k+1}_q = \binom{n}{k}_q + q^{k+1} \cdot \binom{n}{k+1}_q.$$

This is the polynomial version; the evaluated version follows by evaluation.

Proof. Partitions in a $(k+1) \times (n-k)$ box split into those with $\leq k$ parts and those with exactly $k+1$ parts. The latter biject with partitions in a $(k+1) \times (n-k-1)$ box via "removing a column" (subtracting 1 from each of the $k+1$ parts), with size decreased by $k+1$ (giving the factor q^{k+1}). □

Lemma 2.159. *For $m \geq 1$ and $s \geq k+1$, the number of partitions of s with exactly $k+1$ parts, each $\leq m$, equals the number of partitions of $s-(k+1)$ fitting in a $(k+1) \times (m-1)$ box:*

$$|\text{partitionsInBoxExact}(s, k+1, m)| = c(s-(k+1), k+1, m-1).$$

Proof. The forward bijection subtracts 1 from each of the $k+1$ parts (and filters zeros); the inverse adds 1 to each part and pads with 1s to reach exactly $k+1$ parts. Injectivity follows from the recovery lemma for filtered multiset subtraction; surjectivity is verified directly. □

Theorem 2.160. *The combinatorial definition (sum over partitions) agrees with the recurrence-based definition from Definition 2.136: for all $n, k \in \mathbb{N}$ and ring element q,*

$$\text{qBinomialEval}(n, k, q) = \binom{n}{k}_q.$$

Proof. Both definitions satisfy the same boundary conditions ($\binom{n}{0}_q = 1$, $\binom{n}{k}_q = 0$ for $k > n$) and the same q-Pascal recurrence. By strong induction on n with case analysis on k, the two definitions agree. $\square$

2.4.2 Alternative characterizations from the recurrence

Lemma 2.161. *For $k \leq n$, the recurrence-based q-binomial equals the sum over monotone functions:*

$$\binom{n}{k}_q = \sum_{f \in \text{monotoneFunctions}(k, n-k)} q^{\text{sumValues}(f)}.$$

Proof. By induction on n, splitting the monotone functions into those starting at 0 and those starting at a positive value, using the bijections dropFirst/prependZero and shiftDown/shiftUp. $\square$

Lemma 2.162. *For all n, k:*

$$\binom{n}{k}_q = \sum_{\substack{S \subseteq \{1,\ldots,n\} \\ |S|=k}} q^{\text{sum}(S)-(1+2+\cdots+k)}.$$

Proof. Via the bijection from monotone k-tuples in $\{0, \ldots, n-k\}$ to k-element subsets of $\{1, \ldots, n\}$: $(i_1, \ldots, i_k) \mapsto \{i_1+1, i_2+2, \ldots, i_k+k\}$. For $k > n$, both sides are 0. $\square$

Lemma 2.163. *Over a field, when $[i+1]_q \neq 0$ for $i = 0, \ldots, k-1$:*

$$\binom{n}{k}_q = \prod_{i=0}^{k-1} \frac{[n-i]_q}{[i+1]_q}.$$

Proof. By strong induction on n, using the q-Pascal recurrence and the splitting identity $[n+1]_q = [k+1]_q + q^{k+1}[n-k]_q$. $\square$

2.4.3 Product expansion rule

Lemma 2.164. *Let L be a commutative ring. Let $n \in \mathbb{N}$. Let $[n]$ denote the set $\{1, 2, \ldots, n\}$. Let $a_1, a_2, \ldots, a_n$ be n elements of L. Let $b_1, b_2, \ldots, b_n$ be n further elements of L. Then,*

$$\prod_{i=1}^{n} (a_i + b_i) = \sum_{S \subseteq [n]} \left(\prod_{i \in S} a_i \right) \left(\prod_{i \in [n] \setminus S} b_i \right). \tag{2.4}$$

Proof. When expanding the product $\prod_{i=1}^{n} (a_i + b_i) = (a_1 + b_1)(a_2 + b_2) \cdots (a_n + b_n)$, you obtain a sum of 2^n terms, each of which is a product of one addend chosen from each of the n sums $a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n$. This is precisely what the right hand side of (2.4) is. A rigorous proof can be found in [Grinbe15, Exercise 6.1 **(a)**]. The formal proof follows by induction on n, using the distributive law at each step. $\square$

2.4.4 First q-binomial theorem

Helpers for Theorem 2.167

Lemma 2.165. *For all $n, k \in \mathbb{N}$:*

$$\sum_{\substack{S \subseteq \{0,1,\ldots,n-1\} \\ |S|=k}} q^{\sum_{i \in S} i} = q^{k(k-1)/2} \cdot \binom{n}{k}_q .$$

This relates the subset sum over $\{0, 1, \ldots, n-1\}$ to the q-binomial coefficient.

Proof. By induction on n, using the q-Pascal recurrence. The subsets of $\{0, \ldots, n\}$ of size $k+1$ split into those containing n and those not containing n. The key algebraic identity is $q^k \binom{n}{k+1}_q + q^n \binom{n}{k}_q = q^k \binom{n}{k}_q + q^{k+1} \cdot q^k \binom{n}{k+1}_q$, which is proved by induction using algebraic manipulation. $\square$

Lemma 2.166. *For all $n, k \in \mathbb{N}$:*

$$q^k \binom{n}{k+1}_q + q^n \binom{n}{k}_q = q^k \binom{n}{k}_q + q^{k+1} \cdot q^k \binom{n}{k+1}_q .$$

Proof. By induction on n, using the q-Pascal recurrence and algebraic manipulation. $\square$

Theorem 2.167 (1st q-binomial theorem)**.** *Let K be a commutative ring. Let $a, b \in K$ and $n \in \mathbb{N}$. In the polynomial ring $K[q]$, we have*

$$\left(aq^0 + b\right)\left(aq^1 + b\right) \cdots \left(aq^{n-1} + b\right) = \sum_{k=0}^{n} q^{k(k-1)/2} \binom{n}{k}_q a^k b^{n-k}.$$

Proof. Let $[n]$ denote the set $\{1, 2, \ldots, n\}$. We have

$$\left(aq^0 + b\right)\left(aq^1 + b\right) \cdots \left(aq^{n-1} + b\right) = \prod_{i=1}^{n} \left(aq^{i-1} + b\right)$$

$$= \sum_{S \subseteq [n]} \left(\prod_{i \in S} \left(aq^{i-1}\right)\right) \left(\prod_{i \in [n] \setminus S} b\right)$$

by Lemma 2.164, applied to $L = K[q]$, $a_i = aq^{i-1}$ and $b_i = b$. Now,

$$= \sum_{S \subseteq [n]} a^{|S|} \left(\prod_{i \in S} q^{i-1}\right) b^{|[n] \setminus S|} = \sum_{k=0}^{n} \sum_{\substack{S \subseteq [n]; \\ |S|=k}} a^k q^{\operatorname{sum} S - k} b^{n-k}$$

(since $\prod_{i \in S} q^{i-1} = q^{\operatorname{sum} S - |S|}$ and $|[n] \setminus S| = n-k$). Using $q^{\operatorname{sum} S - k} = q^{\operatorname{sum} S - (1+2+\cdots+k)} \cdot q^{k(k-1)/2}$ (since $1 + 2 + \cdots + (k-1) = k(k-1)/2$), we obtain

$$= \sum_{k=0}^{n} q^{k(k-1)/2} \underbrace{\left(\sum_{\substack{S \subseteq [n]; \\ |S|=k}} q^{\operatorname{sum} S - (1+2+\cdots+k)}\right)}_{= \binom{n}{k}_q} a^k b^{n-k} = \sum_{k=0}^{n} q^{k(k-1)/2} \binom{n}{k}_q a^k b^{n-k}.$$

(Here, the underbrace equality follows from Proposition 2.140 **(b)**.) This proves Theorem 2.167. $\square$

Lemma 2.168. *The first q-binomial theorem at $q = 1$ recovers the ordinary binomial formula:*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}.$$

Proof. Specializing Theorem 2.167 at $q = 1$: the LHS becomes $(a+b)^n$ and each $\binom{n}{k}_1 = \binom{n}{k}$. $\square$

2.4.5 Second q-binomial theorem

Helpers for Theorem 2.170

Lemma 2.169. *Let A be an L-algebra, let $\omega \in L$, and let $a, b \in A$ satisfy $ba = \omega \cdot ab$. Then for all $k \in \mathbb{N}$:*

$$b \cdot a^k = \omega^k \cdot (a^k \cdot b).$$

Proof. By induction on k. The base case is trivial. For the inductive step: $b \cdot a^{k+1} = b \cdot a^k \cdot a = \omega^k(a^k b) \cdot a = \omega^k \cdot a^k \cdot (ba) = \omega^k \cdot a^k \cdot \omega(ab) = \omega^{k+1}(a^{k+1}b)$. $\square$

Theorem 2.170 (2nd q-binomial theorem, aka Potter's binomial theorem)**.** *Let L be a commutative ring. Let $\omega \in L$. Let A be a noncommutative L-algebra. Let $a, b \in A$ be such that $ba = \omega ab$. Then,*

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k}_{\omega} a^k b^{n-k}.$$

Proof. By induction on n. The base case $n = 0$ is trivial. For the inductive step, write $(a + b)^{n+1} = (a+b) \cdot (a+b)^n$ and apply the induction hypothesis to $(a+b)^n$. Distribute $(a+b)$ over the sum. The key step is the relation $b \cdot a^k = \omega^k \cdot a^k \cdot b$, which follows from $ba = \omega ab$ by induction on k (proved in Lemma 2.169). After reindexing and combining terms, the coefficients match via the q-Pascal recurrence $\binom{n+1}{k+1}_{\omega} = \binom{n}{k}_{\omega} + \omega^{k+1}\binom{n}{k+1}_{\omega}$. $\square$

2.4.6 Counting subspaces of vector spaces

Helpers for Theorem 2.176

Definition 2.171. The *span map* sends a linearly independent k-tuple $(v_1, \ldots, v_k)$ in V to its span $\operatorname{span}(v_1, \ldots, v_k)$, viewed as a k-dimensional subspace.

Lemma 2.172. *Each k-dimensional subspace W has exactly $\prod_{i=0}^{k-1}(|F|^k - |F|^i)$ preimages under the span map. These are precisely the ordered bases of W: the linearly independent k-tuples in W that automatically span W by dimension count.*

Proof. The fiber over W is in bijection with the set of linearly independent k-tuples in W. Since W is k-dimensional, applying Lemma 2.174 (with W in place of V, and k in place of n) gives the count $\prod_{i=0}^{k-1}(|F|^k - |F|^i)$. $\square$

Lemma 2.173. *Let F be a field. Let V be an F-vector space. Let $(v_1, v_2, \ldots, v_k)$ be a k-tuple of vectors in V. Then, $(v_1, v_2, \ldots, v_k)$ is linearly independent if and only if each $i \in \{1, 2, \ldots, k\}$ satisfies $v_i \notin \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$ (where the span $\operatorname{span}()$ of an empty list is understood to* *be the set $\{\mathbf{0}\}$ consisting only of the zero vector $\mathbf{0}$). In other words, $(v_1, v_2, \ldots, v_k)$ is linearly independent if and only if we have*

$$\begin{aligned} &v_1 \notin \underbrace{\operatorname{span}()}_{=\{\mathbf{0}\}} \qquad \textit{and} \\ &v_2 \notin \operatorname{span}(v_1) \qquad \textit{and} \\ &v_3 \notin \operatorname{span}(v_1, v_2) \qquad \textit{and} \\ &\quad \cdots \qquad \textit{and} \\ &v_k \notin \operatorname{span}(v_1, v_2, \ldots, v_{k-1}). \end{aligned}$$

Proof. We prove both directions:

$\Longrightarrow$: Assume that the k-tuple $(v_1, v_2, \ldots, v_k)$ is linearly independent. Let $i \in \{1, 2, \ldots, k\}$. If we had $v_i \in \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$, then we could write v_i in the form $v_i = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_{i-1} v_{i-1}$ for some coefficients $\alpha_1, \alpha_2, \ldots, \alpha_{i-1} \in F$, and therefore $\alpha_1 v_1 + \cdots + \alpha_{i-1} v_{i-1} + (-1) v_i + 0 v_{i+1} + \cdots + 0 v_k = \mathbf{0}$, which would be a nontrivial linear dependence relation (since v_i appears with the nonzero coefficient -1); this would contradict the linear independence of $(v_1, v_2, \ldots, v_k)$. Hence, $v_i \notin \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$.

$\Longleftarrow$: Assume that each $i \in \{1, 2, \ldots, k\}$ satisfies $v_i \notin \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$. Assume for contradiction that the k-tuple $(v_1, v_2, \ldots, v_k)$ is linearly dependent. Thus, there exist coefficients $\beta_1, \beta_2, \ldots, \beta_k \in F$ that satisfy $\beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_k v_k = \mathbf{0}$ and that are not all zero. Pick the **largest** i with $\beta_i \neq 0$. Then $\beta_{i+1} = \beta_{i+2} = \cdots = \beta_k = 0$, so $\beta_i v_i = -(\beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_{i-1} v_{i-1}) \in \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$. Since $\beta_i \neq 0$, we obtain $v_i \in \operatorname{span}(v_1, v_2, \ldots, v_{i-1})$, contradicting our assumption. $\square$

Lemma 2.174. *Let F be a finite field. Let $n, k \in \mathbb{N}$. Let V be an n-dimensional F-vector space. Then,*

$$\begin{aligned} &(\#\textit{ of linearly independent } k\textit{-tuples of vectors in } V) \\ &= \left(|F|^n - |F|^0\right)\left(|F|^n - |F|^1\right) \cdots \left(|F|^n - |F|^{k-1}\right) = \prod_{i=0}^{k-1} \left(|F|^n - |F|^i\right). \end{aligned}$$

Proof. We have $|V| = |F|^n$ (since V is an n-dimensional F-vector space).

By Lemma 2.173, a k-tuple $(v_1, v_2, \ldots, v_k)$ of vectors in V is linearly independent if and only if each $v_i \notin \operatorname{span}(v_1, \ldots, v_{i-1})$.

We construct such a tuple entry by entry:

- First, we choose $v_1 \in V \setminus \{\mathbf{0}\}$; thus, there are $|V| - 1$ options. Once v_1 has been chosen, $\operatorname{span}(v_1)$ has dimension 1 and thus size $|F|^1$.
- Next, we choose $v_2 \in V \setminus \operatorname{span}(v_1)$; thus, there are $|V| - |F|^1$ options. Once v_2 has been chosen, $\operatorname{span}(v_1, v_2)$ has dimension 2 and thus size $|F|^2$.
- And so on, until the last vector v_k has been chosen.

The total count is $\prod_{i=0}^{k-1} \left(|V| - |F|^i\right) = \prod_{i=0}^{k-1} \left(|F|^n - |F|^i\right)$ (since $|V| = |F|^n$). When $k > n$, both sides are 0 since there are no linearly independent k-tuples in an n-dimensional space. $\square$

Lemma 2.175 (Multijection principle)**.** *Let A and B be two finite sets. Let $m \in \mathbb{N}$. Let $f : A \to B$ be any map. Assume that each $b \in B$ has exactly m preimages under f (that is, for each $b \in B$, there are exactly m many elements $a \in A$ such that $f(a) = b$). Then,*

$$|A| = m \cdot |B|.$$

Proof. By the fiber decomposition: $|A| = \sum_{b \in B} |f^{-1}(b)|$, where each fiber has cardinality m by assumption. Hence $|A| = \sum_{b \in B} m = m \cdot |B|$. $\square$

Theorem 2.176. *Let F be a finite field. Let $n, k \in \mathbb{N}$. Let V be an n-dimensional F-vector space. Then,*

$$\binom{n}{k}_{|F|} = (\text{\# of } k\text{-dimensional vector subspaces of } V).$$

Proof. If $k > n$, then $\binom{n}{k}_{|F|} = 0$ (by Proposition 2.141) and $(\text{\# of } k\text{-dimensional vector subspaces of } V) = 0$ (since the dimension of a subspace of V is never larger than the dimension of V). Hence, for the rest of this proof, we WLOG assume that $k \leq n$.

Define a map

$$f : \{\text{linind } k\text{-tuples of vectors in } V\} \to \{k\text{-dimensional vector subspaces of } V\},$$
$$(v_1, v_2, \ldots, v_k) \mapsto \operatorname{span}(v_1, v_2, \ldots, v_k).$$

Observation 1: Each k-dimensional vector subspace of V has exactly $\left(|F|^k - |F|^0\right)\left(|F|^k - |F|^1\right)\cdots\left(|F|^k - |F|^k\right.$ preimages under f.

Indeed, let W be a k-dimensional vector subspace of V. A preimage of W under f is a linind k-tuple of vectors in W that spans W. Since W is k-dimensional, a k-tuple of vectors in W spans W if and only if it is linind. Hence, the number of preimages is the number of linind k-tuples in W, which equals $\prod_{i=0}^{k-1}(|F|^k - |F|^i)$ by Lemma 2.174 (applied to W and k instead of V and n).

By the multijection principle (Lemma 2.175):

$$\begin{aligned}&\left(|F|^k - |F|^0\right)\left(|F|^k - |F|^1\right)\cdots\left(|F|^k - |F|^{k-1}\right)\\ &\qquad \cdot (\text{\# of } k\text{-dimensional vector subspaces of } V)\\ &= \left(|F|^n - |F|^0\right)\left(|F|^n - |F|^1\right)\cdots\left(|F|^n - |F|^{k-1}\right)\end{aligned}$$

(where the right hand side comes from Lemma 2.174). Therefore,

$$\begin{aligned}&(\text{\# of } k\text{-dimensional vector subspaces of } V)\\ &= \frac{\prod_{i=0}^{k-1}\left(|F|^n - |F|^i\right)}{\prod_{i=0}^{k-1}\left(|F|^k - |F|^i\right)} = \prod_{i=0}^{k-1}\frac{|F|^{n-i} - 1}{|F|^{k-i} - 1} = \frac{\left(1 - |F|^n\right)\left(1 - |F|^{n-1}\right)\cdots\left(1 - |F|^{n-k+1}\right)}{\left(1 - |F|^k\right)\left(1 - |F|^{k-1}\right)\cdots\left(1 - |F|^1\right)}\\ &= \binom{n}{k}_{|F|}\end{aligned}$$

(by Theorem 2.147 **(b)**). $\square$

2.4.7 Limits of q-binomial coefficients

Helpers for Proposition 2.182

Lemma 2.177. *For all $d, k, n, m \in \mathbb{N}$ with $n \geq d + k$ and $m \geq d + k$, the d-th coefficient of $\binom{n}{k}_q$ equals the d-th coefficient of $\binom{m}{k}_q$ (as power series). In other words, the coefficient $[q^d]\binom{n}{k}_q$ stabilizes for $n \geq d + k$.*

Proof. By strong induction on $d + k$, using the q-Pascal recurrence to relate coefficients of $\binom{n}{k}_q$ and $\binom{m}{k}_q$ through $\binom{n-1}{\cdot}_q$ and $\binom{m-1}{\cdot}_q$ at lower coefficients. $\square$

Lemma 2.178. *The stable value of the d-th coefficient equals the d-th coefficient of the product formula:*

$$[q^d]\binom{d+k}{k}_q = [q^d]\prod_{i=1}^{k}\frac{1}{1-q^i}.$$

Proof. By induction on $d+k$. The product formula satisfies a recurrence matching the q-Pascal identity: extracting the last factor $(1-q^{k+1})^{-1}$ gives $\prod_{i=1}^{k+1}(1-q^i)^{-1} = \prod_{i=1}^{k}(1-q^i)^{-1} + q^{k+1}\cdot\prod_{i=1}^{k+1}(1-q^i)^{-1}$. Comparing coefficients and applying the inductive hypothesis completes the proof. □

Lemma 2.179. *Young diagram conjugation (transposition) is an involution on partitions of n that interchanges "all parts $\leq k$" with "at most k parts". Specifically, if p is a partition of n with all parts $\leq k$, then its conjugate has at most k parts, and vice versa. Moreover, conjugating twice recovers the original partition.*

Proof. The conjugate is constructed using Young diagram transposition. The involution property follows from the fact that transposing twice recovers the original diagram. The interchange of "max part" and "number of parts" follows from the relationship between row lengths and column lengths under transposition. □

Lemma 2.180. *The number of partitions of n with all parts $\leq k$ equals the number of partitions of n with at most k parts:*

$$|\{p\vdash n : \textit{all parts} \leq k\}| = |\{p\vdash n : \textit{at most } k \textit{ parts}\}|.$$

Proof. Young diagram conjugation provides a bijection between these two sets (Lemma 2.179). □

Theorem 2.181. *The product formula equals the generating function for partitions with at most k parts:*

$$\prod_{i=1}^{k}\frac{1}{1-q^i} = \sum_{n=0}^{\infty}|\{p\vdash n : \textit{at most } k \textit{ parts}\}|\cdot q^n.$$

Proof. The product $\prod_{i=1}^{k}(1-q^i)^{-1}$ equals the generating function $\sum_n |\{\text{partitions of } n \text{ with all parts } \leq k\}|\cdot q^n$ by the theorem on the product form of restricted partition generating functions. Applying Lemma 2.180 converts the restricted partition count to the count of partitions with at most k parts. □

Proposition 2.182. *Let $k\in\mathbb{N}$ be fixed. Then,*

$$\lim_{n\to\infty}\binom{n}{k}_q = \sum_{n\in\mathbb{N}}\left(p_0(n)+p_1(n)+\cdots+p_k(n)\right)q^n = \prod_{i=1}^{k}\frac{1}{1-q^i}.$$

*(See Definition 2.10 **(a)** for the meaning of $p_i(n)$.)*

First proof (sketched). For each integer $n\geq k$, we have

$$\binom{n}{k}_q = \frac{(1-q^n)\left(1-q^{n-1}\right)\cdots\left(1-q^{n-k+1}\right)}{(1-q^k)\left(1-q^{k-1}\right)\cdots\left(1-q^1\right)}$$
$$= \frac{1}{\prod_{i=1}^{k}\left(1-q^i\right)}\cdot\prod_{i=1}^{k}\left(1-q^{n-k+i}\right).$$

(Here, the first equality holds by Theorem 2.147 **(b)**.) Since $\lim_{n\to\infty} q^n = 0$ (coefficientwise), for each $i \in \{1, \ldots, k\}$ we have $\lim_{n\to\infty}(1 - q^{n-k+i}) = 1$. Hence, by Corollary 1.466,

$$\lim_{n\to\infty} \binom{n}{k}_q = \frac{1}{\prod_{i=1}^{k}(1-q^i)} \cdot 1 = \prod_{i=1}^{k} \frac{1}{1-q^i}.$$

The equality $\sum_{n\in\mathbb{N}} (p_0(n) + p_1(n) + \cdots + p_k(n))\, q^n = \prod_{i=1}^{k} \frac{1}{1-q^i}$ follows from Theorem 2.49 (with the letters x, m and k renamed as q, k and i).

The proof establishes coefficientwise stabilization via Lemma 2.177 (showing that for $n \geq d + k$, the d-th coefficient of $\binom{n}{k}_q$ is constant) and Lemma 2.178 (identifying the stable value with the product formula). The equality with the partition generating function is proved in Theorem 2.181 using Young diagram conjugation to biject partitions with parts $\leq k$ and partitions with $\leq k$ parts. $\square$

Chapter 3

Permutations

3.1 Permutations: basic definitions

3.1.1 Basic definitions

Convention 3.1. In the Lean formalization, we use $\operatorname{Fin} n = \{0, 1, \ldots, n-1\}$ instead of the textbook's $[n] = \{1, 2, \ldots, n\}$. Both are n-element sets, so their symmetric groups are isomorphic. Permutations are represented by the type of bijections $X \to X$ (denoted $\operatorname{Equiv.Perm} X$ in Lean), which is a group under composition. Composition $\alpha\beta$ is the group multiplication $\alpha * \beta$, sending x to $\alpha(\beta(x))$.

Definition 3.2. Let X be a set.

(a) A *permutation* of X means a bijection from X to X.

(b) It is known that the set of all permutations of X is a group under composition. This group is called the *symmetric group* of X, and is denoted by S_X. Its neutral element is the identity map $\operatorname{id}_X : X \to X$. Its size is $|X|!$ when X is finite.

(c) As usual in group theory, we will write $\alpha\beta$ for the composition $\alpha \circ \beta$ when $\alpha, \beta \in S_X$. This is the map that sends each $x \in X$ to $\alpha(\beta(x))$.

(d) If $\alpha \in S_X$ and $i \in \mathbb{Z}$, then α^i shall denote the i-th power of α in the group S_X. Note that $\alpha^i = \underbrace{\alpha \circ \alpha \circ \cdots \circ \alpha}_{i \text{ times}}$ if $i \geq 0$. Also, $\alpha^0 = \operatorname{id}_X$. Also, α^{-1} is the inverse of α in the group S_X, that is, the inverse of the map α.

Definition 3.3. Let $n \in \mathbb{Z}$. Then, $[n]$ shall mean the set $\{1, 2, \ldots, n\}$. This is an n-element set if $n \geq 0$, and is an empty set if $n \leq 0$.

The symmetric group $S_{[n]}$ (consisting of all permutations of $[n]$) will be denoted S_n and called the *n-th symmetric group*. Its size is $n!$ (when $n \geq 0$).

Proposition 3.4. *Let X and Y be two sets, and let $f : X \to Y$ be a bijection. Then, for each permutation σ of X, the map $f \circ \sigma \circ f^{-1} : Y \to Y$ is a permutation of Y. Furthermore, the map*

$$S_f : S_X \to S_Y,$$
$$\sigma \mapsto f \circ \sigma \circ f^{-1}$$

is a group isomorphism; thus, we obtain $S_X \cong S_Y$.

Proof. Easy and LTTR. $\square$

Definition 3.5. Let $n \in \mathbb{N}$ and $\sigma \in S_n$. We introduce three notations for σ:

(a) A *two-line notation* of σ means a $2 \times n$-array

$$\begin{pmatrix} p_1 & p_2 & \cdots & p_n \\ \sigma(p_1) & \sigma(p_2) & \cdots & \sigma(p_n) \end{pmatrix},$$

where the entries $p_1, p_2, \ldots, p_n$ of the top row are the n elements of $[n]$ in some order. Commonly, we pick $p_i = i$.

(b) The *one-line notation* (short, *OLN*) of σ means the n-tuple $(\sigma(1), \sigma(2), \ldots, \sigma(n))$.

(c) The *cycle digraph* of σ is the directed graph with vertices $1, 2, \ldots, n$ and arcs $i \to \sigma(i)$ for all $i \in [n]$.

Lemma 3.6. *Two vertices i and j are in the same connected component of the cycle digraph of σ if and only if there exists $k \in \mathbb{Z}$ with $\sigma^k(i) = j$.*

Proof. By induction on reachability paths: each step corresponds to applying σ or σ^{-1}, so the integer exponent k accumulates additively. Conversely, for any integer k, the sequence $i, \sigma(i), \sigma^2(i), \ldots$ (or the reverse via σ^{-1}) gives a path in the cycle digraph. □

3.1.2 Transpositions, cycles and involutions

Transpositions

Definition 3.7. Let i and j be two distinct elements of a set X.

Then, the *transposition* $t_{i,j}$ is the permutation of X that sends i to j, sends j to i, and leaves all other elements of X unchanged.

Lemma 3.8. *For any two distinct elements i, j of a set X, we have $t_{i,j} = t_{j,i}$.*

Proof. Immediate from the definitions: both $t_{i,j}$ and $t_{j,i}$ swap i and j and fix all other elements. □

Lemma 3.9. *Every transposition is an involution: $t_{i,j}^2 = \mathrm{id}$.*

Proof. Case-split on each element x: if $x = i$, then $t_{i,j}(t_{i,j}(i)) = t_{i,j}(j) = i$; similarly for $x = j$; otherwise $t_{i,j}$ fixes x. □

Lemma 3.10. *A transposition is its own inverse: $t_{i,j}^{-1} = t_{i,j}$.*

Proof. Immediate from $t_{i,j}^2 = \mathrm{id}$. □

Lemma 3.11. *The number of transpositions in S_X (for a finite set X) is $\binom{|X|}{2}$.*

Proof. Each 2-element subset $\{i, j\} \subseteq X$ gives rise to exactly one transposition $t_{i,j}$. The bijection between swaps and non-diagonal elements of $\mathrm{Sym}^2(X)$ establishes the count. □

Definition 3.12. Let $n \in \mathbb{N}$ and $i \in [n-1]$. Then, the *simple transposition* s_i is defined by

$$s_i := t_{i,i+1} \in S_n.$$

Lemma 3.13. *Every simple transposition s_i is a swap (i.e., a 2-cycle).*

Proof. Witness the pair $(i, i+1)$ with $i \neq i+1$. □

Lemma 3.14. *Every simple transposition s_i is a cycle (specifically, a 2-cycle).*

Proof. A swap is always a cycle: any permutation that exchanges exactly two distinct elements is a 2-cycle. □

Lemma 3.15. *The support of s_i is exactly $\{i, i+1\}$, and has cardinality* 2*.*

Proof. The support of a swap of two distinct elements is exactly $\{i, i+1\}$, since $i \neq i+1$. □

Lemma 3.16. *For $n \geq 2$ and $i \in [n-1]$, we have $s_i \neq \mathrm{id}$.*

Proof. The support of s_i has cardinality 2, but the identity has empty support. □

Proposition 3.17. *Let $n \in \mathbb{N}$.*

(a) *We have $s_i^2 = \mathrm{id}$ for all $i \in [n-1]$. In other words, we have $s_i = s_i^{-1}$ for all $i \in [n-1]$.*

(b) *We have $s_i s_j = s_j s_i$ for any $i, j \in [n-1]$ with $|i-j| > 1$.*

(c) *We have $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ for any $i \in [n-2]$.*

Proof. To prove that two permutations α and β of $[n]$ are identical, it suffices to show that $\alpha(k) = \beta(k)$ for each $k \in [n]$. Using this strategy, we can prove all three parts straightforwardly (distinguishing cases corresponding to the relative positions of k, i, $i+1$, j and $j+1$). This is done in detail for part **(c)** in [Grinbe15, solution to Exercise 5.1 **(a)**]; the proofs of parts **(a)** and **(b)** are easier and LTTR. □

Cycles

Definition 3.18. Let X be a set. Let $i_1, i_2, \ldots, i_k$ be k distinct elements of X. Then,

$$\mathrm{cyc}_{i_1, i_2, \ldots, i_k}$$

means the permutation of X that sends

$$\begin{aligned} & i_1 \text{ to } i_2, \\ & i_2 \text{ to } i_3, \\ & \ldots, \\ & i_{k-1} \text{ to } i_k, \\ & i_k \text{ to } i_1 \end{aligned}$$

and leaves all other elements of X unchanged. In other words, $\mathrm{cyc}_{i_1, i_2, \ldots, i_k}$ means the permutation of X that satisfies

$$\mathrm{cyc}_{i_1, \ldots, i_k}(p) = \begin{cases} i_{j+1}, & \text{if } p = i_j \text{ for some } j \in \{1, 2, \ldots, k\}; \\ p, & \text{otherwise} \end{cases}$$

for every $p \in X$, where i_{k+1} means i_1.

This permutation is called a *k-cycle*.

Lemma 3.19. *A* 2*-cycle is exactly a transposition:* $\mathrm{cyc}_{i,j} = t_{i,j}$*.*

Proof. Immediate from the definitions: both sides send i to j, send j to i, and fix all other elements. □

Lemma 3.20. *If the list $[i_1, \ldots, i_k]$ has no duplicates and $k \geq 2$, then $\mathrm{cyc}_{i_1, \ldots, i_k}$ is a cycle (i.e., it has exactly one nontrivial orbit).*

Proof. The cycle permutation of a list with no duplicates and length ≥ 2 acts with exactly one nontrivial orbit, hence is a cycle. $\square$

Lemma 3.21. *Cyclic rotation of the list does not change the cycle:* $\mathrm{cyc}_{i_1,i_2,\ldots,i_k} = \mathrm{cyc}_{i_2,i_3,\ldots,i_k,i_1} = \cdots = \mathrm{cyc}_{i_k,i_1,\ldots,i_{k-1}}$.

Proof. A k-cycle constructed from a list is invariant under cyclic rotation of that list. $\square$

Lemma 3.22. *For a nodup list $l = [i_1, \ldots, i_k]$ and $x \in l$, we have $\mathrm{cyc}(l)(x)$ equal to the next element after x in l (wrapping around at the end).*

Proof. This is a standard property of cycle permutations constructed from lists. $\square$

Lemma 3.23. *For a nodup list of length ≥ 2, the support of $\mathrm{cyc}_{i_1,\ldots,i_k}$ equals $\{i_1, \ldots, i_k\}$.*

Proof. The support of a cycle is exactly the set of elements in the list. $\square$

Lemma 3.24. *For nodup lists l, l' of length ≥ 2, $\mathrm{cyc}(l) = \mathrm{cyc}(l')$ if and only if l and l' are cyclic rotations of each other.*

Proof. Two cycle permutations constructed from nodup lists of length ≥ 2 are equal if and only if the lists are cyclic rotations of each other. $\square$

Lemma 3.25. *If σ is a cycle with $|\mathrm{support}(\sigma)| = 2$, then σ is a transposition: there exist distinct i, j with $\sigma = t_{i,j}$.*

Proof. Since σ is a cycle with support of size 2, its order is 2, so $\sigma^2 = 1$. Then $\sigma(\sigma(x)) = x$ for all x. A cycle of order 2 must be a swap of two elements, giving the result. $\square$

Exercise 3.26. Let $n \in \mathbb{N}$ and let $k \in [n]$. Let X be an n-element set. How many k-cycles exist in S_X?

Proof. First, we note that there is exactly one 1-cycle in S_X (for $n > 0$), since a 1-cycle is just the identity map. This should be viewed as a degenerate case; thus, we WLOG assume that $k > 1$.

For any k distinct elements $i_1, i_2, \ldots, i_k$ of X, we have $\mathrm{cyc}_{i_1,i_2,\ldots,i_k} = \mathrm{cyc}_{i_2,i_3,\ldots,i_k,i_1} = \cdots = \mathrm{cyc}_{i_k,i_1,i_2,\ldots,i_{k-1}}$. That is, $\mathrm{cyc}_{i_1,i_2,\ldots,i_k}$ does not change if we cyclically rotate the list $(i_1, i_2, \ldots, i_k)$.

Any k-cycle $\mathrm{cyc}_{i_1,i_2,\ldots,i_k}$ uniquely determines the elements $i_1, i_2, \ldots, i_k$ up to cyclic rotation (since $k > 1$). Therefore, if $\sigma \in S_X$ is a k-cycle, then there are precisely k lists $(i_1, i_2, \ldots, i_k)$ for which $\sigma = \mathrm{cyc}_{i_1,i_2,\ldots,i_k}$.

Hence, the map

$$f : \{k\text{-tuples of distinct elements of } X\} \to \{k\text{-cycles in } S_X\},$$
$$(i_1, i_2, \ldots, i_k) \mapsto \mathrm{cyc}_{i_1,i_2,\ldots,i_k}$$

is a k-to-1 map. Therefore,

$$\begin{aligned}(\#\text{ of } k\text{-cycles in } S_X) &= \frac{1}{k} \cdot n(n-1)(n-2)\cdots(n-k+1) \\ &= \binom{n}{k} \cdot (k-1)!.\end{aligned}$$

$\square$

Involutions

Definition 3.27. Let X be a set. An *involution* of X means a map $f : X \to X$ that satisfies $f \circ f = \mathrm{id}$. Clearly, an involution is always a permutation, and equals its own inverse.

Lemma 3.28. *Any transposition $t_{i,j}$ is an involution.*

Proof. This is exactly $t_{i,j}^2 = \mathrm{id}$. ☐

Lemma 3.29. *The cycle digraph of an involution of a finite set consists of 1-cycles and 2-cycles only. That is, every cycle factor of an involution has support of size at most 2.*

Proof. Let c be a cycle factor of the involution σ. Since c agrees with σ on its support and $\sigma^2 = 1$, we get $c^2 = 1$ (by checking on both the support and its complement). For a cycle, $\mathrm{orderOf}(c) = |\mathrm{supp}(c)|$, so $|\mathrm{supp}(c)| \mid 2$, giving $|\mathrm{supp}(c)| \leq 2$. ☐

3.1.3 Further properties of involutions

Lemma 3.30. *The order of an involution divides 2.*

Proof. From $\sigma^2 = 1$, we get $\mathrm{orderOf}(\sigma) \mid 2$. ☐

Lemma 3.31. *A nontrivial involution (i.e., $\sigma \neq \mathrm{id}$) has order exactly 2.*

Proof. Since $\mathrm{orderOf}(\sigma) \mid 2$ and 2 is prime, $\mathrm{orderOf}(\sigma) \in \{1, 2\}$. The case $\mathrm{orderOf}(\sigma) = 1$ gives $\sigma = 1$, contradicting $\sigma \neq 1$. ☐

Lemma 3.32. *A k-cycle with $k > 2$ is never an involution.*

Proof. If σ is a k-cycle, then $\mathrm{orderOf}(\sigma) = k$. If σ were an involution, then $k \mid 2$, so $k \leq 2$, contradicting $k > 2$. ☐

Lemma 3.33. *A product of pairwise disjoint transpositions is an involution.*

Proof. Let $\sigma = t_1 t_2 \cdots t_m$ where the t_i are pairwise disjoint transpositions. Since disjoint permutations commute, $\sigma^2 = t_1^2 t_2^2 \cdots t_m^2 = 1$. ☐

Lemma 3.34. *Every involution of a finite set is a product of pairwise disjoint transpositions.*

Proof. By Lemma 3.29, every cycle factor of σ has support of size at most 2. The cycle factorization of σ only contains cycles of length ≥ 2, so each cycle factor is a transposition. The cycle factors are pairwise disjoint, and their product equals σ. ☐

3.2 Inversions, length and Lehmer codes

Throughout this section, let $n \in \mathbb{N}$. We write S_n for the symmetric group on $[n] = \{1, 2, \ldots, n\}$.

3.2.1 Inversions and lengths

Definition 3.35. Let $n \in \mathbb{N}$ and $\sigma \in S_n$.

(a) An *inversion* of σ means a pair (i, j) of elements of $[n]$ such that $i < j$ and $\sigma(i) > \sigma(j)$.

(b) The *length* (also known as the *Coxeter length*) of σ is the # of inversions of σ. It is called $\ell(\sigma)$.

Lemma 3.36. *The identity permutation* id $\in S_n$ *has no inversions:* $\operatorname{inv}(\mathrm{id}) = \emptyset$ *and* $\ell(\mathrm{id}) = 0$.

Proof. The identity preserves the natural order: if $i < j$ then $\mathrm{id}(i) = i < j = \mathrm{id}(j)$, so no pair (i, j) with $i < j$ satisfies $\mathrm{id}(i) > \mathrm{id}(j)$. $\square$

Proposition 3.37. *Let $n \in \mathbb{N}$.*

(a) *For any $\sigma \in S_n$, we have $\ell(\sigma) \in \left\{0, 1, \ldots, \binom{n}{2}\right\}$.*

(b) *We have*

$$(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = 0) = 1.$$

Indeed, the only permutation $\sigma \in S_n$ with $\ell(\sigma) = 0$ is the identity map id.

(c) *We have*

$$\left(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = \binom{n}{2}\right) = 1.$$

Indeed, the only permutation $\sigma \in S_n$ with $\ell(\sigma) = \binom{n}{2}$ is the permutation with OLN $n(n-1)(n-2)\cdots 21$, often called w_0.

(d) *If $n \geq 1$, then*

$$(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = 1) = n - 1.$$

Indeed, the only permutations $\sigma \in S_n$ with $\ell(\sigma) = 1$ are the simple transpositions s_i with $i \in [n-1]$.

(e) *If $n \geq 2$, then*

$$(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = 2) = \frac{(n-2)(n+1)}{2}.$$

Indeed, the only permutations $\sigma \in S_n$ with $\ell(\sigma) = 2$ are the products $s_i s_j$ with $1 \leq i < j < n$ as well as the products $s_i s_{i-1}$ with $i \in \{2, 3, \ldots, n-1\}$. If $n \geq 2$, then there are $\dfrac{(n-2)(n+1)}{2}$ such products (and they are all distinct).

(f) *For any $k \in \mathbb{Z}$, we have*

$$(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = k) = \left(\# \text{ of } \sigma \in S_n \text{ with } \ell(\sigma) = \binom{n}{2} - k\right).$$

Proof. **(a)** The set of inversions of σ is a subset of the set of all pairs (i, j) with $i < j$, and the latter has cardinality $\binom{n}{2}$.

(b) If σ has no inversions, then σ is strictly monotone on $[n]$; a strictly monotone permutation on a finite set must be the identity (since if any $\sigma(i) > i$, the sum $\sum_j \sigma(j)$ would strictly exceed $\sum_j j$, contradicting bijectivity). Conversely, id clearly has no inversions.

(c) The longest element w_0 reverses the ordering, so $w_0(i) = n + 1 - i$, making every pair (i, j) with $i < j$ an inversion. Thus $\ell(w_0) = \binom{n}{2}$. Conversely, if $\ell(\sigma) = \binom{n}{2}$ then $\operatorname{inv}(\sigma)$ equals the set of all ordered pairs, so σ is strictly antitone. A strictly antitone permutation on $[n]$ must send i to $n+1-i$, i.e., $\sigma = w_0$.

(d) Each simple transposition $s_i = \mathrm{swap}(i, i+1)$ has exactly one inversion. Conversely, if σ has exactly one inversion $\{(a,b)\}$, then $b = a+1$ (otherwise an element between a and b would create an additional inversion), and $\sigma = \mathrm{swap}(a,b) = s_a$.

(e) The permutations with exactly 2 inversions are exactly: products $s_i s_j$ with $1 \le i < j < n$ (non-adjacent transpositions, contributing $\binom{n-2}{2}$ permutations) and products $s_i s_{i-1}$ with $i \in \{2, \ldots, n-1\}$ (adjacent 3-cycles, contributing $n-2$ permutations). These give $\binom{n-2}{2} + (n-2) = \frac{(n-2)(n+1)}{2}$ in total.

(f) The bijection $\sigma \mapsto w_0\sigma$ satisfies $\ell(w_0\sigma) = \binom{n}{2} - \ell(\sigma)$, since w_0 swaps inversions and non-inversions. ☐

3.2.2 Helpers for Proposition 3.37

Lemma 3.38. *The longest element $w_0 \in S_n$ is the reversal permutation $w_0(i) = n+1-i$. It is an involution: $w_0^2 = \mathrm{id}$ and $w_0^{-1} = w_0$.*

Proof. Direct computation. ☐

Lemma 3.39. *The longest element $w_0 \in S_n$ equals the standard reversal permutation from Mathlib. That is, $w_0(i) = n-1-i$ for all $i \in [n]$.*

Proof. Both send i to $n-1-i$; the equality follows by extensionality. ☐

Lemma 3.40. *The set of inversions of the longest element w_0 is the set of all pairs (i,j) with $i < j$. That is, every ordered pair is an inversion of w_0.*

Proof. Since $w_0(i) = n+1-i$, if $i < j$ then $w_0(i) = n+1-i > n+1-j = w_0(j)$. ☐

Definition 3.41. The *non-inversions* of a permutation $\sigma \in S_n$ are the pairs (i,j) with $i < j$ and $\sigma(i) < \sigma(j)$. These are exactly the ordered pairs that are not inversions. The inversions and non-inversions partition the set of all $\binom{n}{2}$ ordered pairs.

Lemma 3.42. *For any $\sigma \in S_n$, we have $\ell(\sigma^{-1}) = \ell(\sigma)$.*

Proof. The map $(a,b) \mapsto (\sigma^{-1}(b), \sigma^{-1}(a))$ is a bijection between the inversions of σ^{-1} and the inversions of σ. ☐

Lemma 3.43. *For any $\sigma \in S_n$, the inversions of $w_0\sigma$ are exactly the non-inversions of σ, and*

$$\ell(w_0\sigma) = \binom{n}{2} - \ell(\sigma).$$

Proof. A pair (i,j) with $i < j$ is an inversion of $w_0\sigma$ iff $(w_0\sigma)(i) > (w_0\sigma)(j)$ iff $\sigma(i) < \sigma(j)$ (since w_0 reverses the order), which is the condition for (i,j) to be a non-inversion of σ. The cardinality identity follows since inversions and non-inversions partition the $\binom{n}{2}$ ordered pairs. ☐

Lemma 3.44. *For any $\sigma, \tau \in S_n$,*

$$\ell(\sigma\tau) \le \ell(\sigma) + \ell(\tau).$$

This is the triangle inequality for the inversion count.

Proof. An inversion (i,j) of $\sigma\tau$ (i.e., $i < j$ and $(\sigma\tau)(i) > (\sigma\tau)(j)$) is either an inversion of τ (if $\tau(i) > \tau(j)$) or corresponds to an inversion of σ at $(\tau(i), \tau(j))$ (if $\tau(i) < \tau(j)$). Thus $\mathrm{inv}(\sigma\tau) \subseteq \mathrm{inv}(\tau) \cup \tau^{-1}(\mathrm{inv}(\sigma))$, and the result follows by cardinality. ☐

3.2.3 Lehmer codes

Definition 3.45. Let $n \in \mathbb{N}$.

(a) For each $\sigma \in S_n$ and $i \in [n]$, we set

$$\begin{aligned} \ell_i(\sigma) &:= (\text{\# of all } j \in [n] \text{ satisfying } i < j \text{ and } \sigma(i) > \sigma(j)) \\ &= (\text{\# of all } j \in \{i+1, i+2, \ldots, n\} \text{ such that } \sigma(i) > \sigma(j)). \end{aligned}$$

(b) For each $m \in \mathbb{Z}$, we let $[m]_0$ denote the set $\{0, 1, \ldots, m\}$. (This is an empty set when $m < 0$.)

(c) We let H_n denote the set

$$\begin{aligned} &[n-1]_0 \times [n-2]_0 \times \cdots \times [n-n]_0 \\ &= \{(j_1, j_2, \ldots, j_n) \in \mathbb{N}^n \mid j_i \leq n - i \text{ for each } i \in [n]\}. \end{aligned}$$

This set H_n has size $n!$.

(d) Each $\sigma \in S_n$ and each $i \in [n]$ satisfy $\ell_i(\sigma) \in \{0, 1, \ldots, n-i\} = [n-i]_0$.

(e) We define the map

$$\begin{aligned} L : S_n &\to H_n, \\ \sigma &\mapsto (\ell_1(\sigma), \ell_2(\sigma), \ldots, \ell_n(\sigma)). \end{aligned}$$

If $\sigma \in S_n$ is a permutation, then the n-tuple $L(\sigma)$ is called the *Lehmer code* (or just the *code*) of σ.

Proposition 3.46. *Let $n \in \mathbb{N}$ and $\sigma \in S_n$. Then, $\ell(\sigma) = \ell_1(\sigma) + \ell_2(\sigma) + \cdots + \ell_n(\sigma)$.*

Proof. The inversions of σ are pairs (i, j) with $i < j$ and $\sigma(i) > \sigma(j)$. The Lehmer entry $\ell_i(\sigma)$ counts the $j > i$ with $\sigma(i) > \sigma(j)$. Hence the sum $\sum_i \ell_i(\sigma)$ partitions the set of inversions by first coordinate, so equals $\ell(\sigma)$. □

Theorem 3.47. *Let $n \in \mathbb{N}$. Then, the map $L : S_n \to H_n$ is a bijection.*

First proof of Theorem 3.47 (sketched). Let $\sigma \in S_n$, $i \in [n]$. We can rewrite

$$\ell_i(\sigma) = (\text{\# of elements of } [n] \setminus \{\sigma(1), \ldots, \sigma(i-1)\} \text{ that are smaller than } \sigma(i)). \tag{3.1}$$

Therefore,

$$\sigma(i) = (\text{the } (\ell_i(\sigma) + 1)\text{-st smallest element of } [n] \setminus \{\sigma(1), \ldots, \sigma(i-1)\}). \tag{3.2}$$

This allows recovering $\sigma(i)$ from $\ell_i(\sigma)$ and $\sigma(1), \ldots, \sigma(i-1)$ by induction on i, proving that L is injective.

Conversely, given any $\mathbf{j} = (j_1, \ldots, j_n) \in H_n$, define $M(\mathbf{j})$ to be the map $\sigma : [n] \to [n]$ whose values are determined by

$$\sigma(i) = (\text{the } (j_i + 1)\text{-st smallest element of } [n] \setminus \{\sigma(1), \ldots, \sigma(i-1)\}).$$

This is well-defined (since $j_i \leq n - i$) and gives a permutation (since $\sigma(i) \notin \{\sigma(1), \ldots, \sigma(i-1)\}$). The maps L and M are mutually inverse, so L is bijective. □

Lemma 3.48. *For $\sigma \in S_n$ and $i \in [n]$,*

$$\ell_i(\sigma) = |\{v < \sigma(i) : v \notin \sigma(\{1, \ldots, i-1\})\}|.$$

Proof. The elements $j > i$ with $\sigma(j) < \sigma(i)$ are in bijection (via $j \mapsto \sigma(j)$) with elements $v < \sigma(i)$ not in the image $\sigma(\{1, \ldots, i-1\})$, since these are precisely $[n] \setminus \{\sigma(1), \ldots, \sigma(i)\}$ intersected with $\{v : v < \sigma(i)\}$. □

Lemma 3.49. *For $\sigma \in S_n$ and $i \in [n]$,*

$$\ell_i(\sigma) < n - i.$$

This strict bound (compared to the weak bound $\ell_i(\sigma) \leq n - i$ from Definition 3.45 part (d)) holds because $\ell_i(\sigma)$ counts elements among $\{i+1, \ldots, n-1\}$ where $\sigma(j) < \sigma(i)$, and there are only $n - 1 - i$ such elements.

Proof. The set $\{j : j > i \text{ and } \sigma(j) < \sigma(i)\}$ is a subset of $\{j : j > i\}$, which has cardinality $n - 1 - i < n - i$. □

3.2.4 Lexicographic order

Definition 3.50. Let $(a_1, a_2, \ldots, a_n)$ and $(b_1, b_2, \ldots, b_n)$ be two n-tuples of integers. We say that

$$(a_1, a_2, \ldots, a_n) <_{\text{lex}} (b_1, b_2, \ldots, b_n)$$

if and only if

- there exists some $k \in [n]$ such that $a_k \neq b_k$, and
- the **smallest** such k satisfies $a_k < b_k$.

The relation $<_{\text{lex}}$ is a strict total order on $\mathbb{Z}^n$ (the lexicographic order).

Proposition 3.51. *If $\mathbf{a}$ and $\mathbf{b}$ are two distinct n-tuples of integers, then we have either $\mathbf{a} <_{\text{lex}} \mathbf{b}$ or $\mathbf{b} <_{\text{lex}} \mathbf{a}$.*

Proof. Since $\mathbf{a} \neq \mathbf{b}$, there exists some index where they differ. By well-founded induction on $\mathrm{Fin}\, n$, let k be the minimal such index. Then $a_i = b_i$ for all $i < k$, and $a_k \neq b_k$. If $a_k < b_k$ then $\mathbf{a} <_{\text{lex}} \mathbf{b}$; otherwise $b_k < a_k$ and $\mathbf{b} <_{\text{lex}} \mathbf{a}$. □

Proposition 3.52. *Let $\sigma \in S_n$ and $\tau \in S_n$ be such that*

$$(\sigma(1), \sigma(2), \ldots, \sigma(n)) <_{\text{lex}} (\tau(1), \tau(2), \ldots, \tau(n)).$$

Then,

$$(\ell_1(\sigma), \ell_2(\sigma), \ldots, \ell_n(\sigma)) <_{\text{lex}} (\ell_1(\tau), \ell_2(\tau), \ldots, \ell_n(\tau)).$$

(In other words, $L(\sigma) <_{\text{lex}} L(\tau)$.)

Proof of Proposition 3.52 (sketched). The assumption says there exists a smallest $k \in [n]$ with $\sigma(k) \neq \tau(k)$ and $\sigma(k) < \tau(k)$. For $i < k$: $\sigma(i) = \tau(i)$, so the images $\sigma(\{1, \ldots, i-1\})$ and $\tau(\{1, \ldots, i-1\})$ agree. Since $\sigma(i) = \tau(i)$ as well, we get $\ell_i(\sigma) = \ell_i(\tau)$.

Let $Z = [n] \setminus \{\sigma(1), \ldots, \sigma(k-1)\} = [n] \setminus \{\tau(1), \ldots, \tau(k-1)\}$. Then $\ell_k(\sigma)$ counts elements of Z smaller than $\sigma(k)$, and $\ell_k(\tau)$ counts elements of Z smaller than $\tau(k)$. Since $\sigma(k) < \tau(k)$ and $\sigma(k) \in Z$, every element of Z smaller than $\sigma(k)$ is also smaller than $\tau(k)$, and $\sigma(k)$ itself is smaller than $\tau(k)$ but not smaller than $\sigma(k)$. Thus $\ell_k(\sigma) < \ell_k(\tau)$.

Combining, we get $L(\sigma) <_{\text{lex}} L(\tau)$. □

Second proof of Theorem 3.47 (sketched). We first show that L is injective. Let σ and τ be two distinct permutations in S_n. Their OLNs are distinct, so by Proposition 3.51 we have either $\operatorname{OLN}(\sigma) <_{\text{lex}} \operatorname{OLN}(\tau)$ or vice versa. In either case, Proposition 3.52 gives $L(\sigma) <_{\text{lex}} L(\tau)$ or $L(\tau) <_{\text{lex}} L(\sigma)$. In particular $L(\sigma) \neq L(\tau)$, so L is injective. Since $|S_n| = n! = |H_n|$ and L is injective between finite sets of equal size, the Pigeonhole Principle shows L is bijective. $\square$

3.2.5 Generating function for lengths

Proposition 3.53. *Let $n \in \mathbb{N}$. Then,*

$$\begin{aligned}\sum_{\sigma \in S_n} x^{\ell(\sigma)} &= \prod_{i=1}^{n-1} \left(1 + x + x^2 + \cdots + x^i\right) \\ &= (1+x)\left(1+x+x^2\right)\left(1+x+x^2+x^3\right)\cdots\left(1+x+x^2+\cdots+x^{n-1}\right) \\ &= [n]_x!.\end{aligned}$$

Proof of Proposition 3.53. Each $\sigma \in S_n$ satisfies

$$\ell(\sigma) = \ell_1(\sigma) + \ell_2(\sigma) + \cdots + \ell_n(\sigma)$$

by Proposition 3.46, whence $x^{\ell(\sigma)} = \prod_{i=1}^{n} x^{\ell_i(\sigma)}$. Therefore

$$\sum_{\sigma \in S_n} x^{\ell(\sigma)} = \sum_{\sigma \in S_n} \prod_{i=1}^{n} x^{\ell_i(\sigma)} = \sum_{(j_1,\ldots,j_n) \in H_n} x^{j_1} x^{j_2} \cdots x^{j_n}$$

where we substituted $(j_1, \ldots, j_n)$ for $L(\sigma)$, which is valid since $L : S_n \to H_n$ is a bijection (Theorem 3.47). Since $H_n = [n-1]_0 \times [n-2]_0 \times \cdots \times [0]_0$, the product rule gives

$$\begin{aligned}\sum_{(j_1,\ldots,j_n) \in H_n} x^{j_1} \cdots x^{j_n} &= \left(\sum_{j_1=0}^{n-1} x^{j_1}\right)\left(\sum_{j_2=0}^{n-2} x^{j_2}\right)\cdots\left(\sum_{j_n=0}^{0} x^{j_n}\right) \\ &= \prod_{i=1}^{n-1}\left(1+x+x^2+\cdots+x^i\right) = [n]_x!.\end{aligned}$$

$\square$

3.3 More about lengths and simples

Let us continue studying lengths of permutations.

Proposition 3.54. *Let $n \in \mathbb{N}$ and $\sigma \in S_n$. Then, $\ell(\sigma^{-1}) = \ell(\sigma)$.*

Proof. Recall that $\ell(\sigma)$ is the number of inversions of σ, while $\ell(\sigma^{-1})$ is the number of inversions of σ^{-1}.

An inversion of σ is a pair $(i,j) \in [n] \times [n]$ such that $i < j$ and $\sigma(i) > \sigma(j)$. Likewise, an inversion of σ^{-1} is a pair $(u,v) \in [n] \times [n]$ such that $u < v$ and $\sigma^{-1}(u) > \sigma^{-1}(v)$.

Thus, if (i,j) is an inversion of σ, then $(\sigma(j), \sigma(i))$ is an inversion of σ^{-1}. Hence, we obtain a map

$$\begin{aligned}\{\text{inversions of } \sigma\} &\to \{\text{inversions of } \sigma^{-1}\}, \\ (i,j) &\mapsto (\sigma(j), \sigma(i)).\end{aligned}$$

This map is bijective (indeed, it has an inverse map, which sends each $(u, v) \in \{\text{inversions of } \sigma^{-1}\}$ to $(\sigma^{-1}(v), \sigma^{-1}(u))$). Thus, the bijection principle yields

$$(\text{number of inversions of } \sigma) = (\text{number of inversions of } \sigma^{-1}).$$

In other words, $\ell(\sigma) = \ell(\sigma^{-1})$. (See [Grinbe15, Exercise 5.2 **(f)**] for details.) ☐

Definition 3.55. The explicit bijection between inversions of σ and inversions of σ^{-1}, sending (i, j) to $(\sigma(j), \sigma(i))$.

Lemma 3.56 (Single swap lemma)**.** *Let $n \in \mathbb{N}$, $\sigma \in S_n$ and $k \in [n-1]$. Then:*

- ***(a)*** *We have*
$$\ell(\sigma s_k) = \begin{cases} \ell(\sigma) + 1, & \text{if } \sigma(k) < \sigma(k+1); \\ \ell(\sigma) - 1, & \text{if } \sigma(k) > \sigma(k+1). \end{cases}$$

- ***(b)*** *We have*
$$\ell(s_k \sigma) = \begin{cases} \ell(\sigma) + 1, & \text{if } \sigma^{-1}(k) < \sigma^{-1}(k+1); \\ \ell(\sigma) - 1, & \text{if } \sigma^{-1}(k) > \sigma^{-1}(k+1). \end{cases}$$

Proof. This combines parts **(a)** and **(b)**. ☐

Theorem 3.57. *Part **(a)** of Lemma 3.56: When multiplying a permutation σ on the right by the simple transposition s_k, we have $\ell(\sigma s_k) = \ell(\sigma) + 1$ if $\sigma(k) < \sigma(k+1)$, and $\ell(\sigma s_k) = \ell(\sigma) - 1$ if $\sigma(k) > \sigma(k+1)$.*

Proof. The proof partitions the inversions of σ and σs_k into two kinds. Call an inversion (i, j) of a permutation τ the *key pair* if $(i, j) = (k, k+1)$; all other inversions are *non-key*.

Key observation: There is a bijection between non-key inversions of σ and non-key inversions of σs_k, given by applying the swap $k \leftrightarrow k+1$ to both coordinates. This bijection preserves the strict ordering of the pair coordinates because there is no element strictly between k and $k+1$.

Thus
$$(\text{non-key inversions of } \sigma s_k) = (\text{non-key inversions of } \sigma).$$

For the key pair $(k, k+1)$: it is an inversion of σ if and only if $\sigma(k+1) < \sigma(k)$, and it is an inversion of σs_k if and only if $\sigma(k) < \sigma(k+1)$. So the key pair's inversion status *flips* when we pass from σ to σs_k.

Combining these two observations: the number of inversions changes by exactly ± 1, depending on whether the key pair is added or removed. ☐

Theorem 3.58. *Part **(b)** of Lemma 3.56: When multiplying a permutation σ on the left by the simple transposition s_k, we have $\ell(s_k \sigma) = \ell(\sigma) + 1$ if $\sigma^{-1}(k) < \sigma^{-1}(k+1)$, and $\ell(s_k \sigma) = \ell(\sigma) - 1$ if $\sigma^{-1}(k) > \sigma^{-1}(k+1)$.*

Proof. We use the identity $s_k \sigma = (\sigma^{-1} s_k)^{-1}$ (since s_k is its own inverse) and then apply Proposition 3.54 and part **(a)**. ☐

Proposition 3.59. *Let $n \in \mathbb{N}$ and $\sigma \in S_n$. Let i and j be two elements of $[n]$ such that $i < j$. Then:*

- ***(a)*** *We have $\ell(\sigma t_{i,j}) < \ell(\sigma)$ if $\sigma(i) > \sigma(j)$. We have $\ell(\sigma t_{i,j}) > \ell(\sigma)$ if $\sigma(i) < \sigma(j)$.*

(b) *We have*

$$\ell(\sigma t_{i,j}) = \begin{cases} \ell(\sigma) - 2|Q| - 1, & \text{if } \sigma(i) > \sigma(j); \\ \ell(\sigma) + 2|R| + 1, & \text{if } \sigma(i) < \sigma(j), \end{cases}$$

where

$$Q = \{k \in \{i+1, i+2, \ldots, j-1\} \mid \sigma(i) > \sigma(k) > \sigma(j)\} \quad \text{and}$$
$$R = \{k \in \{i+1, i+2, \ldots, j-1\} \mid \sigma(i) < \sigma(k) < \sigma(j)\}.$$

Proof. This combines parts **(a)** and **(b)**. □

Definition 3.60. The set Q from Proposition 3.59: $Q = \{k \in \{i+1, \ldots, j-1\} \mid \sigma(j) < \sigma(k) < \sigma(i)\}$.

Definition 3.61. The set R from Proposition 3.59: $R = \{k \in \{i+1, \ldots, j-1\} \mid \sigma(i) < \sigma(k) < \sigma(j)\}$.

3.3.1 Helpers for Proposition 3.59

Definition 3.62. The set A from the transposition length proof: $A = \{a < i \mid \sigma(j) < \sigma(a) < \sigma(i)\}$. These elements produce paired lost and gained inversions: (a, j) is lost and (a, i) is gained.

Definition 3.63. The set B from the transposition length proof: $B = \{b > j \mid \sigma(j) < \sigma(b) < \sigma(i)\}$. These elements produce paired lost and gained inversions: (i, b) is lost and (j, b) is gained.

Lemma 3.64. *The set Q for $\sigma \cdot t_{i,j}$ equals the set R for σ. This key symmetry allows reducing the case $\sigma(i) < \sigma(j)$ to the case $\sigma(i) > \sigma(j)$ in the transposition length formula.*

Proof. For k between i and j with $k \neq i, j$, the swap $t_{i,j}$ fixes k, so $(\sigma \cdot t_{i,j})(k) = \sigma(k)$. The conditions $(\sigma t_{i,j})(j) < \sigma(k) < (\sigma t_{i,j})(i)$ become $\sigma(i) < \sigma(k) < \sigma(j)$, which defines R. □

Lemma 3.65. *When $\sigma(j) < \sigma(i)$, the pair (i, j) itself is a lost inversion: $(i, j) \in \operatorname{inv}(\sigma) \setminus \operatorname{inv}(\sigma \cdot t_{i,j})$.*

Proof. Since $(\sigma t_{i,j})(i) = \sigma(j) < \sigma(i) = (\sigma t_{i,j})(j)$, the pair (i, j) is no longer an inversion after applying $t_{i,j}$. □

Lemma 3.66. *For $k \in Q$: the pair (i, k) is a lost inversion. Before the swap, $\sigma(i) > \sigma(k)$ so (i, k) is an inversion. After the swap, $(\sigma t_{i,j})(i) = \sigma(j) < \sigma(k)$, so it is not.*

Proof. Direct verification using the definition of Q and the swap. □

Lemma 3.67. *For $k \in Q$: the pair (k, j) is a lost inversion. Before the swap, $\sigma(k) > \sigma(j)$ so (k, j) is an inversion. After the swap, $(\sigma t_{i,j})(j) = \sigma(i) > \sigma(k)$, so it is not.*

Proof. Direct verification using the definition of Q and the swap. □

Lemma 3.68. *For $a \in A$: the pair (a, j) is a lost inversion. Before the swap, $\sigma(a) > \sigma(j)$. After the swap, $(\sigma t_{i,j})(j) = \sigma(i) > \sigma(a)$, so (a, j) is no longer an inversion.*

Proof. Direct verification. □

Lemma 3.69. *For $b \in B$: the pair (i, b) is a lost inversion. Before the swap, $\sigma(i) > \sigma(b)$. After the swap, $(\sigma t_{i,j})(i) = \sigma(j) < \sigma(b)$, so (i, b) is no longer an inversion.*

Proof. Direct verification. $\square$

Lemma 3.70. *For $a \in A$: the pair (a, i) is a gained inversion. Before the swap, $\sigma(a) < \sigma(i)$ so (a, i) is not an inversion. After the swap, $(\sigma t_{i,j})(i) = \sigma(j) < \sigma(a)$, so it becomes one.*

Proof. Direct verification. $\square$

Lemma 3.71. *For $b \in B$: the pair (j, b) is a gained inversion. Before the swap, $\sigma(j) < \sigma(b)$ so (j, b) is not an inversion. After the swap, $(\sigma t_{i,j})(j) = \sigma(i) > \sigma(b)$, so it becomes one.*

Proof. Direct verification. $\square$

Lemma 3.72. *Completeness of the lost inversion classification: every lost inversion $(a, b) \in \operatorname{inv}(\sigma) \setminus \operatorname{inv}(\sigma \cdot t_{i,j})$ (with $\sigma(j) < \sigma(i)$) is one of five types:*

1. *$(a, b) = (i, j)$,*
2. *$a \in A$ and $b = j$,*
3. *$a = i$ and $b \in B$,*
4. *$a = i$ and $b \in Q$,*
5. *$a \in Q$ and $b = j$.*

Proof. By case analysis on whether a or b equals i or j, and using the fact that the swap only changes the values at positions i and j. $\square$

Lemma 3.73. *Completeness of the gained inversion classification: every gained inversion $(a, b) \in \operatorname{inv}(\sigma \cdot t_{i,j}) \setminus \operatorname{inv}(\sigma)$ (with $\sigma(j) < \sigma(i)$) is one of two types:*

1. *$a \in A$ and $b = i$,*
2. *$a = j$ and $b \in B$.*

Proof. By case analysis, similar to the lost inversion completeness. $\square$

Lemma 3.74. *Let $\sigma \in S_n$ and $i < j$ with $\sigma(j) < \sigma(i)$. Then the number of "lost" inversions (inversions of σ that are not inversions of $\sigma \cdot t_{i,j}$) exceeds the number of "gained" inversions (inversions of $\sigma \cdot t_{i,j}$ that are not inversions of σ) by exactly $2|Q| + 1$.*

The proof partitions lost inversions into five disjoint types: the pair (i, j) itself; pairs (a, j) and (i, b) for $a \in A$, $b \in B$; and pairs (i, k) and (k, j) for $k \in Q$. Gained inversions are pairs (a, i) for $a \in A$ and (j, b) for $b \in B$. The sets A and B cancel, leaving $2|Q| + 1$.

Proof. By explicit enumeration and counting of the five types of lost inversions and two types of gained inversions. $\square$

Theorem 3.75. *Part **(b)** of Proposition 3.59: The exact formula for length change when multiplying by an arbitrary transposition $t_{i,j}$.*

Proof. **Case $\sigma(i) > \sigma(j)$:** This is [Grinbe15, Exercise 5.20]. Apply Lemma 3.74 directly. The symmetric difference formula $|\operatorname{inv}(\sigma t_{i,j})| + |\text{lost}| = |\operatorname{inv}(\sigma)| + |\text{gained}|$ together with $|\text{lost}| = |\text{gained}| + 2|Q| + 1$ gives $\ell(\sigma t_{i,j}) = \ell(\sigma) - 2|Q| - 1$.

Case $\sigma(i) < \sigma(j)$: Apply the previous case to $\sigma t_{i,j}$ instead of σ. Since $(\sigma t_{i,j})(i) = \sigma(j) > \sigma(i) = (\sigma t_{i,j})(j)$, and $\sigma t_{i,j} t_{i,j} = \sigma$, and the sets Q and R trade places when we replace σ by $\sigma t_{i,j}$, we obtain $\ell(\sigma) = \ell(\sigma t_{i,j}) - 2|R| - 1$, hence $\ell(\sigma t_{i,j}) = \ell(\sigma) + 2|R| + 1$. $\square$

Theorem 3.76. *Part **(a)** of Proposition 3.59: The length decreases if (i, j) was an inversion, and increases if it was not.*

Proof. This follows immediately from part **(b)**: when $\sigma(i) > \sigma(j)$, the formula gives $\ell(\sigma t_{i,j}) = \ell(\sigma) - 2|Q| - 1 < \ell(\sigma)$, and when $\sigma(i) < \sigma(j)$, the formula gives $\ell(\sigma t_{i,j}) = \ell(\sigma) + 2|R| + 1 > \ell(\sigma)$. □

Convention 3.77. A *simple transposition* in S_n means one of the $n-1$ transpositions $s_0, s_1, \ldots, s_{n-2}$. We shall occasionally abbreviate "simple transposition" as "simple".

Definition 3.78. A *word* (of length q) in S_n is a finite sequence $(k_1, k_2, \ldots, k_q)$ where each $k_i \in \{0, 1, \ldots, n-2\}$ is an index of a simple transposition.

Definition 3.79. The *product* of a word $w = (k_1, \ldots, k_q)$ is the permutation $s_{k_1} s_{k_2} \cdots s_{k_q}$.

Definition 3.80. A word $w = (k_1, \ldots, k_q)$ is *reduced* if its length q equals $\ell(s_{k_1} \cdots s_{k_q})$.

Theorem 3.81 (First reduced word theorem for the symmetric group)**.** *Let $n \in \mathbb{N}$ and $\sigma \in S_n$. Then:*

***(a)** We can write σ as a composition (i.e., product) of $\ell(\sigma)$ simples.*

***(b)** The number $\ell(\sigma)$ is the smallest $p \in \mathbb{N}$ such that we can write σ as a composition of p simples.*

***Keep in mind:** The composition of* 0 *simples is* id*, since* id *is the neutral element of the group S_n.*

Proof. This combines parts **(a)** and **(b)**. □

Theorem 3.82. *Part **(a)** of Theorem 3.81: Every permutation $\sigma \in S_n$ can be written as a composition of $\ell(\sigma)$ simple transpositions. Equivalently, there exists a reduced word w whose product $s_{k_1} \cdots s_{k_q} = \sigma$.*

Proof. (See [Grinbe15, Exercise 5.2 **(e)**] for details.)

We proceed by strong induction on $\ell(\sigma)$.

Base case: If $\ell(\sigma) = 0$, then $\sigma = \mathrm{id}$ (a permutation with no inversions is the identity), so the empty word works.

Induction step: If $\ell(\sigma) = h+1 > 0$, then σ has at least one inversion, so there exists $k \in [n-1]$ with $\sigma(k) > \sigma(k+1)$. By Lemma 3.56 **(a)**, $\ell(\sigma s_k) = \ell(\sigma) - 1 = h$. By the induction hypothesis, σs_k has a reduced word w of length h. Then $w +\!\!+ [k]$ is a reduced word of length $h+1$ for $\sigma s_k \cdot s_k = \sigma$. □

Theorem 3.83. *Part **(b)** of Theorem 3.81: For any word $w = (k_1, \ldots, k_q)$, we have $\ell(s_{k_1} \cdots s_{k_q}) \leq q$. In other words, $\ell(\sigma)$ is the minimum word length.*

Proof. (See [Grinbe15, Exercise 5.2 **(g)**] for details.)

We prove $\ell(s_{k_1} s_{k_2} \cdots s_{k_g}) \leq g$ by induction on g. The base case $g = 0$ gives $\ell(\mathrm{id}) = 0$. For the induction step, Lemma 3.56 **(b)** gives $\ell(s_k \cdot \tau) \leq \ell(\tau) + 1$ for any τ and k, so $\ell(s_{k_1} \cdots s_{k_g}) \leq \ell(s_{k_2} \cdots s_{k_g}) + 1 \leq (g-1) + 1 = g$. □

Theorem 3.84. *Combining parts **(a)** and **(b)**: $\ell(\sigma)$ is exactly the minimum word length for σ.*

Proof. Immediate from parts **(a)** and **(b)**. □

Corollary 3.85. *Let $n \in \mathbb{N}$.*

(a) *We have $\ell(\sigma\tau) \equiv \ell(\sigma) + \ell(\tau) \pmod 2$ for all $\sigma, \tau \in S_n$.*

(b) *We have $\ell(\sigma\tau) \leq \ell(\sigma) + \ell(\tau)$ for all $\sigma, \tau \in S_n$.*

(c) *Let $k_1, k_2, \ldots, k_q \in [n-1]$, and let $\sigma = s_{k_1} s_{k_2} \cdots s_{k_q}$. Then $q \geq \ell(\sigma)$ and $q \equiv \ell(\sigma) \pmod 2$.*

Proof. This combines parts **(a)**, **(b)**, and **(c)**. □

Theorem 3.86. *Part **(a)** of Corollary 3.85: The length of a product has the same parity as the sum of lengths.*

Proof. (See [Grinbe15, Exercise 5.2 **(b)**] for details.)

Write $\tau = s_{k_1} s_{k_2} \cdots s_{k_q}$ with $q = \ell(\tau)$ (using Theorem 3.81 **(a)**). Then

$$\begin{aligned}
\ell(\sigma\tau) &= \ell(\sigma s_{k_1} s_{k_2} \cdots s_{k_q}) \\
&\equiv \ell(\sigma s_{k_1} \cdots s_{k_{q-1}}) + 1 \\
&\equiv \cdots \\
&\equiv \ell(\sigma) + q = \ell(\sigma) + \ell(\tau) \pmod 2.
\end{aligned}$$

□

Theorem 3.87. *Part **(b)** of Corollary 3.85: The triangle inequality $\ell(\sigma\tau) \leq \ell(\sigma) + \ell(\tau)$.*

Proof. (See [Grinbe15, Exercise 5.2 **(c)**] for details.)

Get reduced words w_1, w_2 for σ, τ. Then $w_1 \mathbin{+\!\!+} w_2$ represents $\sigma\tau$, so $\ell(\sigma\tau) \leq |w_1 \mathbin{+\!\!+} w_2| = |w_1| + |w_2| = \ell(\sigma) + \ell(\tau)$ by Theorem 3.81 **(b)**. □

Theorem 3.88. *Part **(c)** of Corollary 3.85: For any word representing σ, the word length is at least $\ell(\sigma)$ and has the same parity as $\ell(\sigma)$.*

Proof. The inequality $q \geq \ell(\sigma)$ is Theorem 3.81 **(b)**. The parity claim $q \equiv \ell(\sigma) \pmod 2$ is proved by the same telescoping argument as part **(a)**, starting from $\ell(\sigma) = \ell(s_{k_1} \cdots s_{k_q})$ and peeling off one simple at a time. □

Corollary 3.89. *Let $n \in \mathbb{N}$. Then, the group S_n is generated by the simples $s_0, s_1, \ldots, s_{n-2}$.*

Proof. This follows directly from Theorem 3.81 **(a)**: every $\sigma \in S_n$ can be written as a product of simple transpositions, so every σ lies in the subgroup generated by the simples. □

3.3.2 Rothe diagram and Lehmer entry properties

Definition 3.90. A cell (i, j) is in the *Rothe diagram* of σ if $j < \sigma(i)$ and $i < \sigma^{-1}(j)$. These are the cells that are not "hit" by either the row or column of a permutation matrix entry.

Lemma 3.91. *The number of cells in the Rothe diagram of σ equals $\ell(\sigma)$. The bijection sends (i, j) in the Rothe diagram to the inversion $(i, \sigma^{-1}(j))$.*

Proof. The map $(i, j) \mapsto (i, \sigma^{-1}(j))$ is a bijection from Rothe diagram cells to inversions: the condition $j < \sigma(i)$ becomes $\sigma(\sigma^{-1}(j)) < \sigma(i)$, which is $\sigma(k) < \sigma(i)$ with $k = \sigma^{-1}(j) > i$. □

Lemma 3.92. *For $\sigma \in S_n$ and $i \in [n]$, we have $\ell_i(\sigma) < n - i$. This strict bound holds because $\ell_i(\sigma)$ counts elements among $\{i + 1, \ldots, n - 1\}$ where $\sigma(j) < \sigma(i)$, and there are only $n - 1 - i$ such elements.*

Proof. The set counted by $\ell_i(\sigma)$ is a subset of $\{j : j > i\}$, which has cardinality $n-1-i < n-i$. $\square$

Lemma 3.93. *For $\sigma \in S_n$ with $n > 0$, we have $\sigma(0) = \ell_0(\sigma)$. This is because $\ell_0(\sigma)$ counts how many elements among $\sigma(1), \ldots, \sigma(n-1)$ are less than $\sigma(0)$, and since σ is a bijection onto $\{0, \ldots, n-1\}$, this count equals $\sigma(0)$.*

Proof. The elements $j > 0$ with $\sigma(j) < \sigma(0)$ are in bijection (via $j \mapsto \sigma(j)$) with elements $v < \sigma(0)$. Since σ is a bijection, there are exactly $\sigma(0)$ such elements. $\square$

Lemma 3.94. *For $\sigma \in S_n$ and i with $i+1 < n$, let $i' = i+1$. Then*

$$\ell_i(\sigma) \geq \ell_{i'}(\sigma) + 1 \iff \sigma(i) > \sigma(i').$$

This characterizes when consecutive Lehmer entries differ by at least 1, which is crucial for determining when a Lehmer block shifts a position.

Proof. The Lehmer entry $\ell_i(\sigma)$ counts $j > i$ with $\sigma(j) < \sigma(i)$. We split this count by whether $j = i'$ or $j > i'$: $\ell_i(\sigma) = [[\sigma(i') < \sigma(i)]] + |\{j > i' : \sigma(j) < \sigma(i)\}|$. The set $\{j > i' : \sigma(j) < \sigma(i)\}$ contains $\{j > i' : \sigma(j) < \sigma(i')\}$ (counted by $\ell_{i'}(\sigma)$) when $\sigma(i') < \sigma(i)$, from which the equivalence follows. $\square$

Lemma 3.95. *Two permutations $\sigma, \tau \in S_n$ with the same inversions are equal: if for all $i < j$ we have $\sigma(j) < \sigma(i) \iff \tau(j) < \tau(i)$, then $\sigma = \tau$.*

Proof. For each i, the value $\sigma(i)$ equals $|\{j : \sigma(j) < \sigma(i)\}|$ (since σ is a bijection onto $\{0, \ldots, n-1\}$). The hypothesis implies $\{j : \sigma(j) < \sigma(i)\} = \{j : \tau(j) < \tau(i)\}$ for each i, so $\sigma(i) = \tau(i)$. $\square$

3.3.3 The Lehmer code representation

Definition 3.96. For each $i \in [n]$, we define the *Lehmer block* a_i as

$$a_i := s_{i'-1} s_{i'-2} \cdots s_i$$

where $i' = i + \ell_i(\sigma)$ and $\ell_i(\sigma)$ is the Lehmer entry at position i. If $\ell_i(\sigma) = 0$, then $a_i = \mathrm{id}$ (the empty product). The product a_i equals the cycle $(i', i'-1, \ldots, i)$.

Lemma 3.97. *The length of the Lehmer block a_i (as a word) equals the Lehmer entry $\ell_i(\sigma)$.*

Proof. The block consists of $\ell_i(\sigma)$ simple transpositions by construction. $\square$

Definition 3.98. The *canonical reduced word* for σ is the concatenation of Lehmer blocks:

$$w(\sigma) = a_0 \mathbin{+\!\!+} a_1 \mathbin{+\!\!+} \cdots \mathbin{+\!\!+} a_{n-1}.$$

Lemma 3.99. *The length of the canonical reduced word equals $\ell(\sigma)$. This follows from $\sum_i \ell_i(\sigma) = \ell(\sigma)$.*

Proof. Each block a_i has length $\ell_i(\sigma)$, and the total length is $\sum_i \ell_i(\sigma) = \ell(\sigma)$ (since each inversion contributes to exactly one Lehmer entry). $\square$

Lemma 3.100. *The sum of Lehmer entries equals the length: $\sum_{i \in [n]} \ell_i(\sigma) = \ell(\sigma)$.*

Proof. By a bijection between the set of pairs $\{(i,j) : i \in [n],\ j \text{ such that } (i,j) \text{ is an inversion}\}$ and the set of all inversions. $\square$

3.3.4 Helpers for Proposition 3.107

Definition 3.101. For $\sigma \in S_n$ and $i \in [n]$, define $s(\sigma, i) = |\{j < i : \sigma(j) < \sigma(i)\}|$. This counts positions before i with smaller values.

Definition 3.102. For $\sigma \in S_n$ and $i \in [n]$, define $g(\sigma, i) = |\{j < i : \sigma(j) > \sigma(i)\}|$. This counts inversions with i as the second element.

Lemma 3.103. *For $\sigma \in S_n$ and $i \in [n]$:*

$$s(\sigma, i) + g(\sigma, i) = i.$$

This is because the elements $j < i$ are partitioned into those with $\sigma(j) < \sigma(i)$ and those with $\sigma(j) > \sigma(i)$ (equality is impossible since σ is injective).

Proof. The set $\{j : j < i\}$ has cardinality i and splits into $\{j < i : \sigma(j) < \sigma(i)\}$ and $\{j < i : \sigma(j) > \sigma(i)\}$, which are disjoint (since σ is injective, $\sigma(j) \neq \sigma(i)$ for $j \neq i$). □

Lemma 3.104. *For each $i \in [n]$, we have $\sigma(i) = s(\sigma, i) + \ell_i(\sigma)$.*

Proof. The elements j with $\sigma(j) < \sigma(i)$ partition into those with $j < i$ (counted by $s(\sigma, i)$) and those with $j > i$ (counted by ℓ_i). Their total count is $\sigma(i)$. □

Lemma 3.105. *For $\sigma \in S_n$ and $p \in [n]$:*

$$p + \ell_p(\sigma) - g(\sigma, p) = \sigma(p).$$

This key formula shows that the final position after applying all Lehmer blocks to position p equals $\sigma(p)$.

Proof. From $\sigma(p) = s(\sigma, p) + \ell_p(\sigma)$ and $p = s(\sigma, p) + g(\sigma, p)$, we get $p + \ell_p(\sigma) - g(\sigma, p) = \sigma(p)$. □

Lemma 3.106. *For $\sigma \in S_n$ and $i \in [n]$:*

$$i + \ell_i(\sigma) = \sigma(i) + g(\sigma, i).$$

This gives the upper bound of block i's range in terms of the permutation value and the count of larger elements before position i.

Proof. Follows from the formulas $\sigma(i) = s(\sigma, i) + \ell_i(\sigma)$ and $i = s(\sigma, i) + g(\sigma, i)$. □

Proposition 3.107. *Let $n \in \mathbb{N}$ and $\sigma \in S_n$. For each $i \in [n]$, set*

$$a_i := s_{i'-1} s_{i'-2} \cdots s_i,$$

where $i' = i + \ell_i(\sigma)$. Then $\sigma = a_0 a_1 \cdots a_{n-1}$.

In other words, the product of the canonical reduced word equals σ.

Proof. We refer to [Grinbe15, Exercise 5.21 parts **(b)** and **(c)**] for a detailed proof.

We prove by extensionality: for each position p, we track where it ends up after applying all blocks.

By tracking position p through all blocks, the final position is

$$p + \ell_p(\sigma) - g(\sigma, p).$$

Since $p = s(\sigma, p) + g(\sigma, p)$ and $\sigma(p) = s(\sigma, p) + \ell_p(\sigma)$ (Lemma 3.104), we get

$$p + \ell_p(\sigma) - g(\sigma, p) = \sigma(p).$$

□

Theorem 3.108. *The canonical reduced word is indeed reduced: it is a word of length $\ell(\sigma)$ whose product is σ.*

Proof. Combine Proposition 3.107 (the product of the canonical reduced word equals σ) with the length equality from Lemma 3.99 ($|w(\sigma)| = \ell(\sigma)$). $\square$

3.4 Signs of permutations

The notion of the *sign* (aka *signature*) of a permutation is a simple consequence of that of its length; moreover, it is rather well-known, due to its role in the definition of a determinant. Thus we will survey its properties quickly and without proofs.

3.4.1 Definition and basic properties

Definition 3.109. Let $n \in \mathbb{N}$. The *sign* of a permutation $\sigma \in S_n$ is defined to be the integer $(-1)^{\ell(\sigma)}$.

It is denoted by $(-1)^{\sigma}$ or $\operatorname{sgn}(\sigma)$ or $\operatorname{sign}(\sigma)$ or $\varepsilon(\sigma)$. It is also known as the *signature* of σ.

Proposition 3.110. *Let $n \in \mathbb{N}$.*

(a) *The sign of the identity permutation* $\operatorname{id} \in S_n$ *is* $(-1)^{\operatorname{id}} = 1$.

(b) *For any two distinct elements i and j of $[n]$, the transposition $t_{i,j} \in S_n$ has sign $(-1)^{t_{i,j}} = -1$.*

(c) *For any positive integer k and any distinct elements $i_1, i_2, \ldots, i_k \in [n]$, the k-cycle $\operatorname{cyc}_{i_1,i_2,\ldots,i_k}$ has sign $(-1)^{\operatorname{cyc}_{i_1,i_2,\ldots,i_k}} = (-1)^{k-1}$.*

(d) *We have $(-1)^{\sigma\tau} = (-1)^{\sigma} \cdot (-1)^{\tau}$ for any $\sigma \in S_n$ and $\tau \in S_n$.*

(e) *We have $(-1)^{\sigma_1\sigma_2\cdots\sigma_p} = (-1)^{\sigma_1}(-1)^{\sigma_2}\cdots(-1)^{\sigma_p}$ for any $\sigma_1, \sigma_2, \ldots, \sigma_p \in S_n$.*

(f) *We have $(-1)^{\sigma^{-1}} = (-1)^{\sigma}$ for any $\sigma \in S_n$. (The left hand side here has to be understood as $(-1)^{(\sigma^{-1})}$.)*

(g) *We have*

$$(-1)^{\sigma} = \prod_{1 \le i < j \le n} \frac{\sigma(i) - \sigma(j)}{i - j} \qquad \text{for each } \sigma \in S_n.$$

(The product sign "$\prod_{1 \le i < j \le n}$" means a product over all pairs (i,j) of integers satisfying $1 \le i < j \le n$. There are $\binom{n}{2}$ such pairs.)

(h) *If $x_1, x_2, \ldots, x_n$ are any elements of some commutative ring, and if $\sigma \in S_n$, then*

$$\prod_{1 \le i < j \le n} \left(x_{\sigma(i)} - x_{\sigma(j)}\right) = (-1)^{\sigma} \prod_{1 \le i < j \le n} \left(x_i - x_j\right).$$

Proof. Most of this follows easily from what we have proved above, but here are references to complete proofs:

(a) This is [Grinbe15, Proposition 5.15 **(a)**], and follows easily from $\ell(\operatorname{id}) = 0$.

(d) This is [Grinbe15, Proposition 5.15 **(c)**], and follows easily from Corollary 3.85 **(a)**.

(b) This is [Grinbe15, Exercise 5.10 **(b)**], and follows easily from the fact that a permutation has even length if and only if it has an even number of even-length cycles.

(c) This is [Grinbe15, Exercise 5.17 **(d)**], and follows easily from the fact that a cycle of length ℓ has exactly $\ell - 1$ inversions, combined with Proposition 3.110 **(d)**.

(e) This is [Grinbe15, Proposition 5.28], and follows by induction from Proposition 3.110 **(d)**.

(f) This is [Grinbe15, Proposition 5.15 **(d)**], and follows easily from Proposition 3.110 **(d)** or from Proposition 3.54.

(h) This is [Grinbe15, Exercise 5.13 **(a)**] (or, rather, the straightforward generalization of [Grinbe15, Exercise 5.13 **(a)**] to arbitrary commutative rings). Each factor $x_{\sigma(i)} - x_{\sigma(j)}$ on the left hand side appears also on the right hand side, albeit with a different sign if (i, j) is an inversion of σ. Thus, the products on both sides agree up to a sign, which is precisely $(-1)^{\ell(\sigma)} = (-1)^{\sigma}$.

(g) This is [Grinbe15, Exercise 5.13 **(c)**], and is a particular case of Proposition 3.110 **(h)**. □

3.4.2 The sign homomorphism

Corollary 3.111. *Let $n \in \mathbb{N}$. The map*

$$S_n \to \{1, -1\},$$
$$\sigma \mapsto (-1)^{\sigma}$$

is a group homomorphism from the symmetric group S_n to the order-2 group $\{1, -1\}$. (Of course, $\{1, -1\}$ is a group with respect to multiplication.)

This map is known as the *sign homomorphism.*

Proof. Proposition 3.110 **(d)** shows that this map respects multiplication (i.e., sends products to products). However, if a map between two groups respects multiplication, then it is automatically a group homomorphism. Thus, Corollary 3.111 follows. □

3.4.3 Even and odd permutations

Definition 3.112. Let $n \in \mathbb{N}$. A permutation $\sigma \in S_n$ is said to be

- *even* if $(-1)^{\sigma} = 1$ (that is, if $\ell(\sigma)$ is even);
- *odd* if $(-1)^{\sigma} = -1$ (that is, if $\ell(\sigma)$ is odd).

Lemma 3.113. *Every permutation σ is either even or odd.*

Proof. $(-1)^{\sigma} \in \{1, -1\}$. □

Lemma 3.114. *A permutation cannot be both even and odd.*

Proof. $1 \neq -1$. □

Lemma 3.115. *The identity permutation is even.*

Proof. By Proposition 3.110 **(a)**, $(-1)^{\mathrm{id}} = 1$. □

Lemma 3.116. *A transposition of two distinct elements is an odd permutation.*

Proof. By Proposition 3.110 **(b)**, $(-1)^{t_{i,j}} = -1$. □

Lemma 3.117. *A permutation σ is even if and only if $\sigma \in A_n$.*

Proof. This follows directly from the definition of A_n as the kernel of the sign homomorphism: $\sigma \in A_n$ iff $\mathrm{sign}(\sigma) = 1$ iff σ is even. □

Lemma 3.118. *A permutation σ is odd if and only if $\sigma \notin A_n$.*

Proof. Since every permutation is either even or odd (Lemma 3.113), and σ is even iff $\sigma \in A_n$ (Lemma 3.117), it follows that σ is odd iff $\sigma \notin A_n$. □

3.4.4 The alternating group

Corollary 3.119. *Let $n \in \mathbb{N}$. The set of all even permutations in S_n is a normal subgroup of S_n.*

This subgroup is known as the n-th *alternating group* (commonly called A_n).

Proof. The set of all even permutations in S_n is the kernel of the group homomorphism $S_n \to \{1, -1\}$ from Corollary 3.111. Thus, it is a normal subgroup of S_n (since any kernel is a normal subgroup). $\square$

3.4.5 Counting even and odd permutations

Corollary 3.120. *Let $n \geq 2$. Then,*

$$(\# \text{ of even permutations } \sigma \in S_n) = (\# \text{ of odd permutations } \sigma \in S_n) = n!/2.$$

Proof. The symmetric group S_n contains the simple transposition s_1 (since $n \geq 2$). If $\sigma \in S_n$, then by Prop. 3.110 **(d)**, we get

$$(-1)^{\sigma s_1} = (-1)^{\sigma} \cdot \underbrace{(-1)^{s_1}}_{\substack{=-1 \\ \text{(by Prop. } \textbf{(b)}\text{,} \\ \text{since } s_1 = t_{1,2})}} = -(-1)^{\sigma},$$

where the underbrace uses Prop. 3.110 **(b)**. Hence, a permutation $\sigma \in S_n$ is even if and only if σs_1 is odd. Hence, the map

$$\{\text{even permutations } \sigma \in S_n\} \to \{\text{odd permutations } \sigma \in S_n\},$$
$$\sigma \mapsto \sigma s_1$$

is well-defined. This map is furthermore a bijection (since S_n is a group). Thus, the bijection principle yields

$$(\# \text{ of even permutations } \sigma \in S_n) = (\# \text{ of odd permutations } \sigma \in S_n).$$

Both sides of this equality must furthermore equal $n!/2$, since they add up to $|S_n| = n!$. This proves Corollary 3.120. (See [Grinbe15, Exercise 5.4] for details.) $\square$

As a consequence of Corollary 3.120, we see that

$$\sum_{\sigma \in S_n} (-1)^{\sigma} = 0 \qquad \text{for each } n \geq 2. \tag{3.3}$$

(Indeed, the sum $\sum_{\sigma \in S_n} (-1)^{\sigma}$ can be rewritten as

$$(\# \text{ of even permutations } \sigma \in S_n) - (\# \text{ of odd permutations } \sigma \in S_n),$$

since the addends corresponding to the even permutations $\sigma \in S_n$ are equal to 1 whereas the addends corresponding to the odd permutations $\sigma \in S_n$ are equal to -1.)

Lemma 3.121. *For $n \geq 2$, $\sum\limits_{\sigma \in S_n} (-1)^{\sigma} = 0$.*

Proof. This is a restatement of (3.3). $\square$

3.4.6 Sign for permutations of arbitrary finite sets

We note that the sign can be defined not only for a permutation $\sigma \in S_n$, but also for any permutation of any finite set X (even if the set X has no chosen total order on it, as the set $[n]$ has). Here is one way to do so:

Proposition 3.122. *Let X be a finite set. We want to define the sign of any permutation of X.*

Fix a bijection $\phi : X \to [n]$ for some $n \in \mathbb{N}$. (Such a bijection always exists, since X is finite.) For every permutation σ of X, set

$$(-1)^\sigma_\phi := (-1)^{\phi \circ \sigma \circ \phi^{-1}}.$$

Here, the right hand side is well-defined, since $\phi \circ \sigma \circ \phi^{-1}$ is a permutation of $[n]$. Now:

(a) *This number $(-1)^\sigma_\phi$ depends only on the permutation σ, but not on the bijection ϕ. (In other words, if ϕ_1 and ϕ_2 are two bijections from X to $[n]$, then $(-1)^\sigma_{\phi_1} = (-1)^\sigma_{\phi_2}$.)*

Thus, we shall denote $(-1)^\sigma_\phi$ by $(-1)^\sigma$ from now on. We refer to this number $(-1)^\sigma$ as the sign *of the permutation $\sigma \in S_X$. (When $X = [n]$, this notation does not clash with Definition 3.109, since we can pick the bijection $\phi = \text{id}$ and obtain $(-1)^\sigma_\phi = (-1)^{\text{id} \circ \sigma \circ \text{id}^{-1}} = (-1)^\sigma$.)*

(b) *The identity permutation $\text{id} : X \to X$ satisfies $(-1)^{\text{id}} = 1$.*

(c) *We have $(-1)^{\sigma\tau} = (-1)^\sigma \cdot (-1)^\tau$ for any two permutations σ and τ of X.*

Proof. This all follows quite easily from Proposition 3.110 **(d)**. See [Grinbe15, Exercise 5.12] for a detailed proof. ∎

3.5 The cycle decomposition

Theorem 3.123 (disjoint cycle decomposition of permutations)**.** *Let X be a finite set. Let σ be a permutation of X. Then:*

(a) *There is a list*

$$\Big((a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}), \\ (a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}), \\ \ldots, \\ (a_{k,1}, a_{k,2}, \ldots, a_{k,n_k}) \Big)$$

of nonempty lists of elements of X such that:

- *each element of X appears exactly once in the composite list*

$$(a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}, \\ a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}, \\ \ldots, \\ a_{k,1}, a_{k,2}, \ldots, a_{k,n_k}),$$

  and

- *we have*

$$\sigma = \text{cyc}_{a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}} \circ \text{cyc}_{a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}} \circ \cdots \circ \text{cyc}_{a_{k,1}, a_{k,2}, \ldots, a_{k,n_k}}.$$

Such a list is called a disjoint cycle decomposition *(or short* DCD*) of σ. Its entries (which themselves are lists of elements of X) are called the* cycles *of σ.*

(b) *Any two DCDs of σ can be obtained from each other by (repeatedly) swapping the cycles with each other, and rotating each cycle (i.e., replacing $(a_{i,1}, a_{i,2}, \ldots, a_{i,n_i})$ by $(a_{i,2}, a_{i,3}, \ldots, a_{i,n_i}, a_{i,1})$).*

(c) *Now assume that X is a set of integers (or, more generally, any totally ordered finite set). Then, there is a unique DCD*

$$\begin{aligned}\Big(& (a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}), \\ & (a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}), \\ & \ldots, \\ & (a_{k,1}, a_{k,2}, \ldots, a_{k,n_k}) \Big)\end{aligned}$$

of σ that satisfies the additional requirements that

- *we have $a_{i,1} \leq a_{i,p}$ for each $i \in [k]$ and each $p \in [n_i]$ (that is, each cycle in this DCD is written with its smallest entry first), and*
- *we have $a_{1,1} > a_{2,1} > \cdots > a_{k,1}$ (that is, the cycles appear in this DCD in the order of decreasing first entries).*

Proof. **(a)** Let $\mathcal{D}$ be the cycle digraph of σ. This cycle digraph $\mathcal{D}$ has the following two properties:

- *Outbound uniqueness:* For each node i, there is exactly one arc outgoing from i. (Indeed, this is the arc from i to $\sigma(i)$.)
- *Inbound uniqueness:* For each node i, there is exactly one arc incoming into i. (Indeed, this is the arc from $\sigma^{-1}(i)$ to i, since σ is a permutation and therefore invertible.)

Using these two properties, we will now show that the cycle digraph $\mathcal{D}$ consists of several node-disjoint cycles.

Indeed, if two cycles C and D of $\mathcal{D}$ have a node in common, then they are identical (because the outbound uniqueness property prevents these cycles from ever separating after meeting at the common node). In other words, any two cycles C and D of $\mathcal{D}$ are either identical or node-disjoint.

Now, let i be any node of $\mathcal{D}$. Then, if we start at i and follow the outgoing arcs, then we obtain an infinite walk

$$\sigma^0(i) \to \sigma^1(i) \to \sigma^2(i) \to \sigma^3(i) \to \cdots$$

along our digraph $\mathcal{D}$. Since X is finite, the Pigeonhole Principle guarantees that this walk will eventually revisit a node it has already been to; i.e., there exist two integers $u, v \in \mathbb{N}$ with $u < v$ and $\sigma^u(i) = \sigma^v(i)$. Let us pick two such integers u, v with the **smallest** possible v. Thus,

$$\text{the } v \text{ nodes } \sigma^0(i), \sigma^1(i), \ldots, \sigma^{v-1}(i) \text{ are distinct} \tag{3.4}$$

(since otherwise, v would not be smallest possible). Now, σ is a permutation and thus has an inverse σ^{-1}. Applying the map σ^{-1} to both sides of the equality $\sigma^u(i) = \sigma^v(i)$, we obtain $\sigma^{u-1}(i) = \sigma^{v-1}(i)$. However, if we had $u \geq 1$, then (3.4) would entail $\sigma^{u-1}(i) \neq \sigma^{v-1}(i)$, which would contradict $\sigma^{u-1}(i) = \sigma^{v-1}(i)$. Thus, we cannot have $u \geq 1$. Hence, $u < 1$, so that $u = 0$. Therefore, $\sigma^u(i) = \sigma^0(i)$, so that $\sigma^0(i) = \sigma^u(i) = \sigma^v(i)$. This shows that our walk is circular:

it comes back to its starting node $\sigma^0(i) = i$ after v steps. We have thus found a cycle in our digraph $\mathcal{D}$:

$$\sigma^0(i) \to \sigma^1(i) \to \sigma^2(i) \to \cdots \to \sigma^v(i) = \sigma^0(i).$$

This shows that the node i lies on a cycle C_i of $\mathcal{D}$.

We thus have shown that each node i of $\mathcal{D}$ lies on a cycle C_i. Any arc of our digraph $\mathcal{D}$ must belong to one of these chosen cycles.

Combining these facts, we conclude that $\mathcal{D}$ consists of several node-disjoint cycles. Let us label these cycles as

$$\begin{aligned}
&(a_{1,1} \to a_{1,2} \to \cdots \to a_{1,n_1} \to a_{1,1}), \\
&(a_{2,1} \to a_{2,2} \to \cdots \to a_{2,n_2} \to a_{2,1}), \\
&\ldots, \\
&(a_{k,1} \to a_{k,2} \to \cdots \to a_{k,n_k} \to a_{k,1})
\end{aligned}$$

(making sure to label each cycle only once). Then, each element of X appears exactly once in the composite list

$$\begin{aligned}
(&a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}, \\
&a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}, \\
&\ldots, \\
&a_{k,1}, a_{k,2}, \ldots, a_{k,n_k}),
\end{aligned}$$

and we have

$$\sigma = \operatorname{cyc}_{a_{1,1},a_{1,2},\ldots,a_{1,n_1}} \circ \operatorname{cyc}_{a_{2,1},a_{2,2},\ldots,a_{2,n_2}} \circ \cdots \circ \operatorname{cyc}_{a_{k,1},a_{k,2},\ldots,a_{k,n_k}}$$

(since σ moves any node $i \in X$ one step forward along its chosen cycle). This proves Theorem 3.123 **(a)**.

(b) See [Goodma15, Theorem 1.5.3] or [Bourba74, Chapter I, §5.7, Proposition 7]. Let

$$\begin{aligned}
\big(&(a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}), \\
&(a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}), \\
&\ldots, \\
&(a_{k,1}, a_{k,2}, \ldots, a_{k,n_k})\big)
\end{aligned}$$

be a DCD of σ. Then, for each $i \in X$, the cycle of this DCD that contains i is uniquely determined by σ and i up to cyclic rotation (indeed, it is a rotated version of the list $\left(i, \sigma(i), \sigma^2(i), \ldots, \sigma^{r-1}(i)\right)$, where r is the smallest positive integer satisfying $\sigma^r(i) = i$). Therefore, all cycles of this DCD are uniquely determined by σ up to cyclic rotation and up to the relative order in which these cycles appear in the DCD. This is the claim of Theorem 3.123 **(b)**.

(c) In order to obtain a DCD of σ that satisfies these two requirements, it suffices to

- start with an arbitrary DCD of σ,
- then rotate each cycle of this DCD so that it begins with its smallest entry, and
- then repeatedly swap these cycles so they appear in the order of decreasing first entries.

It is clear that the result of this procedure is uniquely determined (a consequence of Theorem 3.123 **(b)**). Thus, Theorem 3.123 **(c)** is proven. ☐

Definition 3.124. Let X be a finite set. Let σ be a permutation of X.

(a) The *cycles* of σ are defined to be the cycles in the DCD of σ (as defined in Theorem 3.123 **(a)**). (This includes 1-cycles, if there are any in the DCD of σ.)

We shall equate a cycle of σ with any of its cyclic rotations; thus, for example, $(3, 1, 4)$ and $(1, 4, 3)$ shall be regarded as being the same cycle (but $(3, 1, 4)$ and $(3, 4, 1)$ shall not).

(b) The *cycle lengths partition* of σ shall denote the partition of $|X|$ obtained by writing down the lengths of the cycles of σ in weakly decreasing order.

3.5.1 Properties of the cycle decomposition

Lemma 3.125. *The cycles of σ are pairwise disjoint, i.e., their supports are pairwise disjoint.*

Proof. This is the cycle factors pairwise disjointness lemma from Mathlib. ☐

Lemma 3.126. *The product of the cycles of σ (in any order, since they commute) equals σ.*

Proof. The cycles are pairwise disjoint, hence pairwise commuting. Their product equals σ by the cycle factorization theorem from Mathlib. ☐

Lemma 3.127. *If x is not a fixed point of σ (i.e., $\sigma(x) \neq x$), then the cycle of σ containing x is among the cycles of σ.*

Proof. This is the characterization of cycle factor membership from Mathlib. ☐

Lemma 3.128. *Two permutations σ, τ of a finite set X have the same cycle type (i.e., the same multiset of cycle lengths) if and only if they are conjugate, i.e., there exists a permutation g of X such that $\tau = g\sigma g^{-1}$.*

Proof. This is the conjugacy-iff-same-cycle-type characterization from Mathlib. ☐

Definition 3.129. The *cycle lengths* of σ is the multiset of lengths of the nontrivial cycles (length ≥ 2) in the cycle decomposition of σ.

Lemma 3.130. *The sum of the cycle lengths (of nontrivial cycles) equals the number of non-fixed-point elements of σ.*

Proof. Follows from the cycle type summation lemma in Mathlib. ☐

Lemma 3.131. *Every element of the cycle lengths multiset is at least 2.*

Proof. Follows from the fact that each cycle type entry is at least 2, which is proved in Mathlib. ☐

Lemma 3.132. *The parts of the cycle lengths partition sum to $|X|$.*

Proof. This is the defining property of a partition of $|X|$. ☐

3.5.2 Auxiliary lemmas on cycles

Lemma 3.133. *Let l be a list with no duplicate entries, and let $k \in \mathbb{N}$. Then the cycle permutation of l equals the cycle permutation of the k-fold rotation of l.*

Proof. This is the rotation invariance of cycle permutations from Mathlib. □

Lemma 3.134. *Let l_1, l_2 be lists such that l_1 is a cyclic rotation of l_2, and l_1 has no duplicate entries. Then l_1 and l_2 give rise to the same cycle permutation.*

Proof. Follows from Lemma 3.133. □

Proposition 3.135. *Let X be a finite set. Let $i, j \in X$. Let σ be a permutation of X. Then, we have the following chain of equivalences:*

$$\begin{aligned} & (i \text{ and } j \text{ belong to the same cycle of } \sigma) \\ \iff & (i = \sigma^p (j) \text{ for some } p \in \mathbb{N}) \\ \iff & (j = \sigma^p (i) \text{ for some } p \in \mathbb{N}) . \end{aligned}$$

Proof. The forward direction uses that for finite types, $\mathbb{Z}$-powers can be replaced by $\mathbb{N}$-powers (reducing k modulo the order of σ). The backward direction is immediate.

The symmetric version (the third equivalence) follows by applying the first equivalence to σ^{-1}. □

Lemma 3.136. *Let X be a finite set, σ a permutation of X, and $i, j \in X$ with i not a fixed point of σ. Then i and j belong to the support of a common cycle of σ if and only if i and j belong to the same cycle of σ.*

Proof. For the forward direction, if i and j are both in the support of a cycle c, then i and j are in the same cycle of c, and since c is the cycle of σ containing i, this transfers to i and j being in the same cycle of σ. For the backward direction, the cycle of σ containing i is among the cycles of σ, and both i and j belong to its support. □

3.5.3 Sign from cycle decomposition

Lemma 3.137. *The total number of cycles of a permutation σ of a finite set X (including 1-cycles / fixed points) equals the number of nontrivial cycles (those of length ≥ 2) plus the number of fixed points of σ, i.e., the number of elements $x \in X$ with $\sigma(x) = x$.*

Proof. Immediate from the definitions. □

Lemma 3.138. *Let $\sigma \in S_n$ be a k-cycle (i.e., σ is a cycle and moves exactly k elements). Then $(-1)^\sigma = (-1)^{k-1}$.*

Proof. This follows from the sign-of-a-cycle lemma in Mathlib, which gives $(-1)^\sigma = -(-1)^k$ for a k-cycle σ. Since $-(-1)^k = (-1)^{k-1}$ for $k \geq 1$, the result follows. □

Proposition 3.139. *Let $n \in \mathbb{N}$. Let $\sigma \in S_n$. Let $k \in \mathbb{N}$ be such that σ has exactly k cycles (including the 1-cycles). Then, $(-1)^\sigma = (-1)^{n-k}$.*

Proof. Let

$$\begin{aligned}&(a_{1,1}, a_{1,2}, \ldots, a_{1,n_1}),\\&(a_{2,1}, a_{2,2}, \ldots, a_{2,n_2}),\\&\ldots,\\&(a_{k,1}, a_{k,2}, \ldots, a_{k,n_k})\end{aligned}$$

be the k cycles of σ. Thus,

$$\sigma = \operatorname{cyc}_{a_{1,1},a_{1,2},\ldots,a_{1,n_1}} \circ \operatorname{cyc}_{a_{2,1},a_{2,2},\ldots,a_{2,n_2}} \circ \cdots \circ \operatorname{cyc}_{a_{k,1},a_{k,2},\ldots,a_{k,n_k}}$$

so that, by Proposition 3.110 **(e)** and **(c)**,

$$\begin{aligned}(-1)^{\sigma} &= (-1)^{n_1-1} \cdot (-1)^{n_2-1} \cdot \cdots \cdot (-1)^{n_k-1}\\&= (-1)^{(n_1-1)+(n_2-1)+\cdots+(n_k-1)}.\end{aligned}$$

Since each element of $[n]$ appears exactly once in the DCD, we have $n_1 + n_2 + \cdots + n_k = n$. Hence,

$$(n_1 - 1) + (n_2 - 1) + \cdots + (n_k - 1) = \underbrace{(n_1 + n_2 + \cdots + n_k)}_{=n} - k = n - k.$$

Thus, $(-1)^{\sigma} = (-1)^{n-k}$. This proves Proposition 3.139. □

Lemma 3.140. *Let $\sigma \in S_n$. Let $m_1, m_2, \ldots, m_j$ be the lengths of the nontrivial cycles (of length ≥ 2) of σ. Then $(-1)^{\sigma} = (-1)^{(m_1-1)+(m_2-1)+\cdots+(m_j-1)}$.*

Proof. This follows from the sign-from-cycle-type formula in Mathlib by observing that $(m_1 + \cdots + m_j) + j$ and $(m_1 + \cdots + m_j) - j$ have the same parity (they differ by $2j$). □

3.5.4 Canonical DCD construction helpers

Definition 3.141. Let c be a cycle permutation (i.e., c is a cycle in the sense of Definition 3.124). The *minimum element* of c is the smallest element in the support of c.

Definition 3.142. Let c be a cycle permutation. The *canonical list representation* of c is the list $(a, c(a), c^2(a), \ldots)$ where a is the minimum element of c (Definition 3.141).

Lemma 3.143. *The first element of the canonical list is minimal among all elements of the list.*

Proof. The first element is the minimum element of c, and every element in the canonical list belongs to the support of c, so the minimum is $\leq$ every element. □

Lemma 3.144. *The cycle permutation determined by the canonical list equals c.*

Proof. The canonical list is the orbit of the minimum element under c. The cycle permutation of this list recovers c because the minimum element is moved by c. □

Lemma 3.145. *The minimum element of a cycle c belongs to the support of c.*

Proof. By definition, the minimum element of c is the minimum of the support, which is an element of the support. □

Lemma 3.146. *The canonical list representation of a cycle has no duplicate entries.*

Proof. Follows from the no-duplicates property of permutation cycle lists in Mathlib. □

Lemma 3.147. *An element x is in the canonical list of c if and only if x is in the support of c.*

Proof. The canonical list enumerates the orbit of the minimum element under c, which is exactly the support of c (since c is a cycle). □

Lemma 3.148. *If two cycles c and d are disjoint (as permutations), then their canonical lists are disjoint (as lists).*

Proof. By Lemma 3.147, membership in the canonical list is equivalent to membership in the support. Since c and d are disjoint, their supports are disjoint. □

Definition 3.149. Given a list of lists (each representing a cycle), the corresponding permutation is obtained by composing the cycle permutations $\mathrm{cyc}_{a_1,a_2,\ldots,a_m}$ for each list $(a_1, a_2, \ldots, a_m)$ in the input.

Lemma 3.150. *The product of the cycle permutations of the canonical list representations of σ's cycles equals σ.*

Proof. Each canonical list determines a cycle permutation that recovers the original cycle (by Lemma 3.144), and the product of all cycles equals σ (by the cycle factorization theorem from Mathlib). □

Lemma 3.151. *The flattened list of all canonical cycle lists has no duplicate entries.*

Proof. Each individual canonical list is nodup (Lemma 3.146), and different cycles have disjoint canonical lists (Lemma 3.148). □

Lemma 3.152. *An element y appears in the flattened list of canonical cycle lists if and only if y is not a fixed point of σ.*

Proof. By Lemma 3.147, each canonical list covers exactly the support of its cycle, and the union of all cycle supports equals σ's support. □

Chapter 4

Signed Counting

4.1 Cancellations in alternating sums

4.1.1 Acceptable sets, partners, and signs

Definition 4.1. The *acceptable sets* for parameters $n, m \in \mathbb{N}$ are the subsets of $\{0, 1, \ldots, n-1\}$ with cardinality at most m:

$$\mathcal{A}(n,m) := \{I \subseteq \{0, \ldots, n-1\} : |I| \leq m\}.$$

Definition 4.2. The *partner* of a finite set I of natural numbers is the symmetric difference $I' := I \triangle \{0\}$. If $0 \in I$, then $I' = I \setminus \{0\}$; if $0 \notin I$, then $I' = I \cup \{0\}$.

Definition 4.3. The *sign* of a finite set I of natural numbers is $\operatorname{sign}(I) := (-1)^{|I|}$.

Lemma 4.4. *The partner operation is an involution:* $(I')' = I$ *for all finite sets* I.

Proof. By the self-cancellation property of symmetric difference: $(I \triangle \{0\}) \triangle \{0\} = I$. ☐

Lemma 4.5. *The partner reverses the sign:* $\operatorname{sign}(I') = -\operatorname{sign}(I)$ *for all finite sets* I.

Proof. If $0 \in I$, then $|I'| = |I| - 1$; if $0 \notin I$, then $|I'| = |I| + 1$. In either case, $(-1)^{|I'|} = -(-1)^{|I|}$. ☐

Proposition 4.6 (Negative hockey-stick identity)**.** *Let* $n \in \mathbb{C}$ *and* $m \in \mathbb{N}$*. Then,*

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m}.$$

Proof. The Lean proof delegates to `Int.alternating_sum_range_choose_eq_choose` from Mathlib.

The source gives a combinatorial proof using sign-reversing involutions. Both sides of the equality are polynomial functions in n, so we can WLOG assume that n is a positive integer (by the polynomial identity trick). Set $[n] := \{1, 2, \ldots, n\}$. Define an *acceptable set* to be a subset of $[n]$ that has size $\leq m$. Define the *sign* of a finite set I to be $(-1)^{|I|}$. Then

$$\big(\text{the sum of the signs of all acceptable sets}\big) = \sum_{k=0}^{m} (-1)^k \binom{n}{k}.$$

For each finite set I, define the *partner* of I to be the set $I' := I \triangle \{1\}$, where $\triangle$ denotes symmetric difference. Each finite set I satisfies $I'' = I$ and $|I'| = |I| \pm 1$, so that $(-1)^{|I'|} = -(-1)^{|I|}$. If both I and I' are acceptable sets, then their contributions to the sum cancel each other.

Claim 1: Let I be an acceptable set. Then, the partner I' of I is non-acceptable if and only if ($1 \notin I$ and $|I| = m$).

[*Proof of Claim 1:* If $1 \notin I$ and $|I| = m$, then $I' = I \cup \{1\}$ has size $|I| + 1 > m$, so it is non-acceptable. Conversely, if I' is non-acceptable, then $1 \notin I$ (otherwise $I' = I \setminus \{1\} \subseteq I$ would be acceptable) and $|I| = m$ (otherwise $|I| \leq m - 1$ would give $|I'| = |I| + 1 \leq m$, making I' acceptable).]

The acceptable sets with non-acceptable partners are precisely the m-element subsets of $[n]$ that do not contain 1, i.e., the m-element subsets of $[n] \setminus \{1\}$. There are $\binom{n-1}{m}$ such subsets, each with sign $(-1)^m$. Hence,

$$\sum_{k=0}^{m} (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m}.$$

This proves Proposition 4.6. $\square$

Theorem 4.7. *Let $n \geq 1$, $m \in \mathbb{N}$, and let I be an acceptable set (i.e., $I \subseteq \{0, \ldots, n-1\}$ with $|I| \leq m$). Then the partner I' of I is non-acceptable if and only if $0 \notin I$ and $|I| = m$.*

Proof. By case analysis on whether $0 \in I$ and the cardinality of I, as outlined in Claim 1 in the proof of Proposition 4.6. $\square$

Theorem 4.8. *For $n \geq 1$ and $m \in \mathbb{N}$,*

$$\sum_{I \in \mathcal{A}} \operatorname{sign}(I) = (-1)^m \binom{n-1}{m},$$

where $\mathcal{A} = \mathcal{A}(n, m)$ is the set of acceptable sets (subsets of $\{0, \ldots, n-1\}$ with size $\leq m$) and $\operatorname{sign}(I) = (-1)^{|I|}$.

Proof. Partition $\mathcal{A}$ into "moving" sets (those whose partner is also acceptable) and "fixed" sets (those whose partner is non-acceptable). The sum over moving sets is 0 by a sign-reversing involution argument: the partner pairs them up with opposite signs. The sum over fixed sets equals $(-1)^m \binom{n-1}{m}$, since by Theorem 4.7, the fixed sets are precisely the m-element subsets of $[n] \setminus \{0\}$, each contributing $(-1)^m$. $\square$

4.1.2 Cancellation principles

Lemma 4.9 (Cancellation principle, take 1)**.** *Let $\mathcal{A}$ be a finite set. Let $\mathcal{X}$ be a subset of $\mathcal{A}$.*

For each $I \in \mathcal{A}$, let $\operatorname{sign} I$ be an element of an additive abelian group R with no 2-torsion (i.e., $2a = 0$ implies $a = 0$). Let $f : \mathcal{X} \to \mathcal{X}$ be a bijection with the property that

$$\operatorname{sign}(f(I)) = -\operatorname{sign} I \qquad \textit{for all } I \in \mathcal{X}.$$

(Such a bijection f is called sign-reversing*.) Then,*

$$\sum_{I \in \mathcal{A}} \operatorname{sign} I = \sum_{I \in \mathcal{A} \setminus \mathcal{X}} \operatorname{sign} I.$$

Proof. We have

$$\begin{aligned}\sum_{I\in\mathcal{X}} \operatorname{sign} I &= \sum_{I\in\mathcal{X}} \underbrace{\operatorname{sign}(f(I))}_{=-\operatorname{sign} I} \qquad \text{(substituting } f(I) \text{ for } I\text{, since } f \text{ is a bijection)}\\ &= \sum_{I\in\mathcal{X}} (-\operatorname{sign} I) = -\sum_{I\in\mathcal{X}} \operatorname{sign} I.\end{aligned}$$

Adding $\sum_{I\in\mathcal{X}} \operatorname{sign} I$ to both sides, we obtain $2 \cdot \sum_{I\in\mathcal{X}} \operatorname{sign} I = 0$. Since R has no 2-torsion, $\sum_{I\in\mathcal{X}} \operatorname{sign} I = 0$.

Now, $\mathcal{X} \subseteq \mathcal{A}$; hence,

$$\sum_{I\in\mathcal{A}} \operatorname{sign} I = \underbrace{\sum_{I\in\mathcal{X}} \operatorname{sign} I}_{=0} + \sum_{I\in\mathcal{A}\setminus\mathcal{X}} \operatorname{sign} I = \sum_{I\in\mathcal{A}\setminus\mathcal{X}} \operatorname{sign} I.$$

□

Lemma 4.10 (Cancellation principle, take 2)**.** *Let $\mathcal{A}$ be a finite set. Let $\mathcal{X}$ be a subset of $\mathcal{A}$.*

For each $I \in \mathcal{A}$, let $\operatorname{sign} I$ be an element of some additive abelian group. Let $f : \mathcal{X} \to \mathcal{X}$ be an involution (i.e., a map satisfying $f \circ f = \mathrm{id}$) that has no fixed points. Assume that

$$\operatorname{sign}(f(I)) = -\operatorname{sign} I \qquad \textit{for all } I \in \mathcal{X}.$$

Then,

$$\sum_{I\in\mathcal{A}} \operatorname{sign} I = \sum_{I\in\mathcal{A}\setminus\mathcal{X}} \operatorname{sign} I.$$

Proof. All addends corresponding to the $I \in \mathcal{X}$ cancel out from the sum $\sum_{I\in\mathcal{A}} \operatorname{sign} I$ (because they come in pairs of addends with opposite signs). The Lean proof uses Mathlib's `Finset.sum_involution`. □

Lemma 4.11 (Cancellation principle, take 3)**.** *Let $\mathcal{A}$ be a finite set. Let $\mathcal{X}$ be a subset of $\mathcal{A}$.*

For each $I \in \mathcal{A}$, let $\operatorname{sign} I$ be an element of some additive abelian group. Let $f : \mathcal{X} \to \mathcal{X}$ be an involution (i.e., a map satisfying $f \circ f = \mathrm{id}$). Assume that

$$\operatorname{sign}(f(I)) = -\operatorname{sign} I \qquad \textit{for all } I \in \mathcal{X}.$$

Assume furthermore that

$$\operatorname{sign} I = 0 \qquad \textit{for all } I \in \mathcal{X} \textit{ satisfying } f(I) = I.$$

Then,

$$\sum_{I\in\mathcal{A}} \operatorname{sign} I = \sum_{I\in\mathcal{A}\setminus\mathcal{X}} \operatorname{sign} I.$$

Proof. This is similar to Lemma 4.10, except that the addends corresponding to the $I \in \mathcal{X}$ satisfying $f(I) = I$ don't cancel (but are already zero and thus can be removed right away). The Lean proof uses Mathlib's `Finset.sum_involution`, which handles both the pairing and the fixed-point condition. □

4.1.3 Switching operations and the q-binomial at $q = -1$

Definition 4.12. For $i \in \mathbb{N}$, define the *switch operation* switch_i on finite subsets S of $\mathbb{N}$ by

$$\mathrm{switch}_i(S) := \begin{cases} S \triangle \{i, i+1\}, & \text{if } |S \cap \{i, i+1\}| = 1; \\ S, & \text{otherwise.} \end{cases}$$

In other words, if exactly one of i and $i+1$ belongs to S, then we replace it by the other; otherwise we leave S unchanged.

Lemma 4.13. *The switch operation* switch_i *is an involution, i.e.,* $\mathrm{switch}_i(\mathrm{switch}_i(S)) = S$ *for all finite sets* S*.*

Proof. By case analysis: if exactly one of $\{i, i+1\}$ is in S, then the symmetric difference $S \triangle \{i, i+1\}$ restores S after two applications. Otherwise the operation is the identity. $\square$

Lemma 4.14. *The switch operation preserves cardinality:* $|\mathrm{switch}_i(S)| = |S|$ *for all finite sets* S*.*

Proof. When $|S \cap \{i, i+1\}| = 1$, the symmetric difference removes one element and adds another. Otherwise the set is unchanged. $\square$

4.1.4 Roots of unity

Definition 4.15. Let K be a field. Let d be a positive integer.

(a) A *d-th root of unity* in K means an element ω of K satisfying $\omega^d = 1$.

(b) A *primitive d-th root of unity* in K means an element ω of K satisfying $\omega^d = 1$ but $\omega^i \neq 1$ for each $i \in \{1, 2, \ldots, d-1\}$. In other words, a primitive d-th root of unity in K means an element of the multiplicative group $K^\times$ whose order is d.

Lemma 4.16. *In any integral domain where* $-1 \neq 1$*, the element* -1 *is a primitive* 2*nd root of unity.*

Proof. We have $(-1)^2 = 1$ and $(-1)^1 = -1 \neq 1$. $\square$

Lemma 4.17. *If* ω *is a primitive d-th root of unity with* $d > 1$*, then*

$$1 + \omega + \omega^2 + \cdots + \omega^{d-1} = 0.$$

Proof. This is `IsPrimitiveRoot.geom_sum_eq_zero` from Mathlib. $\square$

4.1.5 The q-Lucas theorem

Definition 4.18. The *q-binomial coefficient* $\binom{n}{k}_q$ is the polynomial in $\mathbb{Z}[q]$ defined by

$$\binom{n}{k}_q := \sum_{\substack{S \subseteq \{0, \ldots, n-1\} \\ |S| = k}} q^{\mathrm{sum}(S) - (0+1+\cdots+(k-1))},$$

where $\mathrm{sum}(S) = \sum_{x \in S} x$. This is the local definition used in the formalization; it agrees with Mathlib's q-binomial coefficient (see Lemma 4.19).

Lemma 4.19. *The local q-binomial coefficient* $\binom{n}{k}_q$ *(Definition 4.18) equals Mathlib's q-binomial coefficient* $\binom{n}{k}_q$ *(as a polynomial in* $\mathbb{Z}[q]$*).*

Proof. Both definitions agree: they are sums over k-element subsets of $\{0, \ldots, n-1\}$, with monomials $q^{(\text{sum}(S)-k(k-1)/2)}$. The proof establishes the identity by showing the exponents coincide using the order embedding of S into $\{0, \ldots, n-1\}$. $\square$

Lemma 4.20. *For $n, k \in \mathbb{N}$, $\binom{n}{k}_q\big|_{q=1} = \binom{n}{k}$.*

Proof. Each term evaluates to 1 at $q = 1$, so the sum counts the number of k-element subsets of $[n]$, which is $\binom{n}{k}$. $\square$

Lemma 4.21. *For $n < k$, $\binom{n}{k}_q = 0$.*

Proof. There are no k-element subsets of $[n]$ when $n < k$. $\square$

Lemma 4.22. *For all $n \in \mathbb{N}$, $\binom{n}{0}_q = 1$.*

Proof. The only 0-element subset is $\varnothing$, contributing $q^0 = 1$. $\square$

Lemma 4.23. *For all $n \in \mathbb{N}$, $\binom{n}{n}_q = 1$.*

Proof. The only n-element subset of $[n]$ is $[n]$ itself, contributing $q^0 = 1$. $\square$

Definition 4.24. A subset $S \subseteq \{0, \ldots, n-1\}$ is *d-blocky* if $S \subseteq \{0, \ldots, n-1\}$ and for each block index $i \in \{0, \ldots, \lfloor n/d \rfloor - 1\}$, either all elements of $\{id, id+1, \ldots, id+d-1\}$ are in S or none are.

Definition 4.25. The set $\binom{[n]}{k}$ of k-element subsets of $\{0, \ldots, n-1\}$:

$$\binom{[n]}{k} := \big\{S \subseteq \{0, \ldots, n-1\} : |S| = k\big\}.$$

Definition 4.26. The set of d-blocky k-element subsets of $\{0, \ldots, n-1\}$:

$$\mathcal{B}(d, n, k) := \left\{S \in \binom{[n]}{k} : S \text{ is } d\text{-blocky}\right\}.$$

Lemma 4.27. *For $d > 0$, the number of d-blocky k-element subsets of $\{0, \ldots, n-1\}$ is*

$$|\mathcal{B}(d, n, k)| = \begin{cases} \binom{n/d}{k/d} \cdot \binom{n \bmod d}{k \bmod d}, & \textit{if } k \bmod d \leq n \bmod d; \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. A d-blocky k-element subset is uniquely determined by the pair (B, P) where $B \subseteq \{0, \ldots, n/d - 1\}$ selects which complete d-blocks are fully included (giving $|B| \cdot d$ elements) and $P \subseteq \{0, \ldots, n \bmod d - 1\}$ selects which elements of the partial block are included (giving $|P|$ elements). The constraint $|S| = k$ forces $|B| = k/d$ and $|P| = k \bmod d$. If $k \bmod d > n \bmod d$, no valid P exists. $\square$

Theorem 4.28 (q-Lucas theorem)**.** *Let K be a field. Let d be a positive integer. Let ω be a primitive d-th root of unity in K. Let $n, k \in \mathbb{N}$. Let q and u be the quotients obtained when dividing n and k by d with remainder, and let r and v be the respective remainders. Then,*

$$\binom{n}{k}_\omega = \binom{q}{u} \cdot \binom{r}{v}_\omega.$$

Proof. The proof splits into cases:

Case $d = 1$: Then $\omega = 1$, and both sides reduce to $\binom{n}{k}$ by Lemma 4.20.

Case $k > n$: Both sides are 0.

Case $n < d$: Then $n/d = 0$, $n \bmod d = n$, and the formula reduces to $\binom{n}{k}_\omega = \binom{0}{k/d} \cdot \binom{n}{k \bmod d}_\omega$.

Main case ($d \geq 2$, $k \leq n$, $n \geq d$): The proof has three steps:

1. Rewrite $\binom{n}{k}_\omega$ as $\sum_{S \in \mathcal{A}} \omega^{(\mathrm{sum}(S) - \mathrm{baseline})}$ where $\mathcal{A}$ is the set of k-element subsets of $[n]$.

2. Split the sum into contributions from *d-blocky* subsets (those that consist of entire d-blocks) and non-d-blocky subsets.

3. Show that non-blocky contributions cancel (Lemma 4.29) and that the blocky contributions give the product formula (Lemma 4.31).

A subset S of $[n]$ is *d-blocky* if for each complete d-block $\{id, id+1, \ldots, id+d-1\}$ contained in $[n]$, either all elements of the block are in S or none are. ☐

Lemma 4.29. *Let K be a field, $d \geq 2$, and ω a primitive d-th root of unity. For $n, k \in \mathbb{N}$,*

$$\sum_{\substack{S \subseteq [n],\ |S| = k, \\ S \text{ not } d\text{-blocky}}} \omega^{\mathrm{sum}(S)} = 0.$$

Proof. For each non-d-blocky set S, there exists a "split block" i: a block index where S contains some but not all elements of $\{id, id+1, \ldots, id+d-1\}$. Define the rotation operation that cyclically permutes elements within block i (replacing each element x in the block by $id + (x - id + 1) \bmod d$).

The non-blocky sets are partitioned into orbits under rotation in the smallest split block. Each orbit has size $m = d/\gcd(d, j)$ where $j = |S \cap \mathrm{block}_i|$, and $m > 1$ since $0 < j < d$. The sum over each orbit is 0 by Lemma 4.30. Therefore the total sum is 0.

The Lean proof applies Lemma 4.41 to the full set of non-blocky subsets, verifying orbit-closure (rotation preserves membership via Lemmas 4.36 and 4.37). ☐

Lemma 4.30. *Let K be a field, $d \geq 2$, and ω a primitive d-th root of unity. For $0 < j < d$ and any $b \in \mathbb{N}$, let $m = d/\gcd(d, j)$. Then*

$$\sum_{k=0}^{m-1} \omega^{b + k \cdot j} = 0.$$

Proof. Factor out ω^b: $\sum_{k=0}^{m-1} \omega^{b+kj} = \omega^b \sum_{k=0}^{m-1} (\omega^j)^k$. Since $0 < j < d$, we have $\omega^j \neq 1$ and $(\omega^j)^m = 1$. The sum $\sum_{k=0}^{m-1} (\omega^j)^k$ can be evaluated using the geometric series formula $\frac{(\omega^j)^m - 1}{\omega^j - 1} = 0$. ☐

Lemma 4.31. *Let K be a field, d a positive integer, and ω a primitive d-th root of unity. For $n, k \in \mathbb{N}$,*

$$\sum_{\substack{S \subseteq [n],\ |S| = k, \\ S \text{ is } d\text{-blocky}}} \omega^{\mathrm{sum}(S) - (0 + 1 + \cdots + (k-1))} = \binom{n/d}{k/d} \cdot \binom{n \bmod d}{k \bmod d}_\omega.$$

Proof. A d-blocky k-element subset of $[n]$ decomposes into two independent parts: which complete d-blocks to include (choosing k/d blocks from n/d available) and which elements of the partial block $\{(n/d) \cdot d, \ldots, n-1\}$ to include (choosing $k \bmod d$ elements from $n \bmod d$ available).

The complete-block part contributes a factor of 1 (since $\omega^d = 1$) and the partial-block part contributes $\omega^{(\text{sum of partial part})}$. The number of complete-block configurations is $\binom{n/d}{k/d}$, and the partial-block sum gives $\binom{n \bmod d}{k \bmod d}_\omega$. $\square$

Lemma 4.32. *The rotation function within block i is injective (for $d > 0$).*

Proof. By case analysis on whether x and y are in block i, and using injectivity of the modular arithmetic operation. $\square$

Lemma 4.33. *Rotating d times within block i returns to the identity:* $(\text{rotate}_i)^d(x) = x$ *for all x.*

Proof. For x in block i, after k rotations the offset becomes $(x - id + k) \bmod d$. After d rotations, the offset returns to its original value. For x outside block i, the rotation is the identity. $\square$

Lemma 4.34. *For a primitive d-th root of unity ω,*

$$\omega^{\text{sum}(\text{rotate}_i^k(S))} = \omega^{\text{sum}(S) + k\cdot|\mathit{blockOffsets}(d,i,S)|}.$$

Proof. The rotation changes $\text{sum}(S)$ by $|\text{blockOffsets}(d, i, S)| \pmod d$ per iteration. Since $\omega^d = 1$, the congruence modulo d suffices. $\square$

Lemma 4.35. *For $0 < j < d$ where $j = |\mathit{blockOffsets}(d, i, S)|$ and $m = d/\gcd(d, j)$, the map $k \mapsto \text{rotate}_i^k(S)$ is injective on $\{0, 1, \ldots, m-1\}$.*

Proof. If $\text{rotate}_i^{k_1}(S) = \text{rotate}_i^{k_2}(S)$ with $k_1 < k_2 < m$, then $k_1 \cdot j \equiv k_2 \cdot j \pmod d$. By modular arithmetic cancellation, $m \mid (k_2 - k_1)$, contradicting $0 < k_2 - k_1 < m$. $\square$

Lemma 4.36. *Rotation preserves cardinality: for $d > 0$, $|\text{rotate}_i^k(S)| = |S|$.*

Proof. The rotation function rotate_i is injective (Lemma 4.32), so the image of S under rotate_i has the same cardinality. The result follows by induction on k. $\square$

Lemma 4.37. *Rotation preserves the ambient set: for $d > 0$ and $i < n/d$, if $S \subseteq \{0, \ldots, n-1\}$ then $\text{rotate}_i^k(S) \subseteq \{0, \ldots, n-1\}$.*

Proof. Elements outside block i are unchanged. Elements inside block i stay within $\{id, \ldots, id + d - 1\} \subseteq \{0, \ldots, n-1\}$ by the modular arithmetic of the rotation. $\square$

Lemma 4.38. *Rotating d times is the identity on sets: $\text{rotate}_i^d(S) = S$ for all sets S.*

Proof. Since $(\text{rotate}_i)^d = \text{id}$ on elements (Lemma 4.33), the image of S after d rotations equals S. $\square$

Lemma 4.39. *Let ω be a primitive d-th root of unity with $d \geq 2$. For $0 < j < d$ and any $b \in \mathbb{N}$,*

$$\sum_{k=0}^{d-1} \omega^{b+k\cdot j} = 0.$$

Proof. Since $m = d/\gcd(d, j)$ divides d, the sum over the full range $\{0, \ldots, d-1\}$ is a sum of d/m copies of the orbit sum $\sum_{k=0}^{m-1} \omega^{b+kj}$, which is 0 by Lemma 4.30. $\square$

Lemma 4.40. *For $d > 0$ and any set S, the fibers of the map $k \mapsto \text{rotate}_i^k(S)$ on $\{0, \ldots, d-1\}$ all have the same cardinality. That is, there exists c such that for every set T in the image, the number of $k \in \{0, \ldots, d-1\}$ with $\text{rotate}_i^k(S) = T$ is exactly c, and $c \mid d$.*

Proof. By the orbit-stabilizer theorem for the cyclic group $\mathbb{Z}/d$ acting on sets via rotation. The stabilizer of S consists of those k with $\text{rotate}_i^k(S) = S$; all fibers have the same size as the stabilizer. The result follows from $|\,\text{orbit}\,| \cdot |\,\text{stab}\,| = d$. $\square$

Lemma 4.41. *For any orbit-closed subset A of the non-blocky k-element subsets of $[n]$,*

$$\sum_{S \in A} \omega^{\text{sum}(S)} = 0.$$

Here "orbit-closed" means that A is closed under rotation in the smallest split block of each element: for each $S \in A$, the entire orbit of S under rotation in its smallest split block is contained in A.

Proof. By strong induction on $|A|$. If A is empty, the sum is 0. Otherwise, pick $S_0 \in A$, let i_0 be its smallest split block index, and let O be the orbit of S_0 under rotation in block i_0 (all d rotations). Since A is orbit-closed, $O \subseteq A$. The sum over O is 0 by Lemma 4.39 and Lemma 4.40 (the sum over the image equals a constant times the sum over the fibers, which is zero). Then $\sum_{S \in A} \omega^{\text{sum}(S)} = \sum_{S \in O}(\cdots) + \sum_{S \in A \setminus O}(\cdots) = 0 + 0$ by the inductive hypothesis on $A \setminus O$ (which is orbit-closed and has strictly smaller cardinality). $\square$

4.1.6 The q-binomial at $q = -1$

Theorem 4.42. *Let $n, k \in \mathbb{N}$. Then:*

$$\binom{n}{k}_{-1} = \begin{cases} 0, & \text{if } n \text{ is even and } k \text{ is odd;} \\ \binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}, & \text{otherwise.} \end{cases}$$

Proof. By Theorem 4.28 with $d = 2$ and $\omega = -1$ (which is a primitive 2nd root of unity by Lemma 4.16):

$$\binom{n}{k}_{-1} = \binom{n/2}{k/2} \cdot \binom{n \bmod 2}{k \bmod 2}_{-1}.$$

The only possible remainders upon division by 2 are 0 and 1, and the base-case (-1)-binomial coefficients are:

$$\binom{0}{0}_{-1} = 1, \quad \binom{0}{1}_{-1} = 0, \quad \binom{1}{0}_{-1} = 1, \quad \binom{1}{1}_{-1} = 1.$$

If n is even and k is odd, then $n \bmod 2 = 0$ and $k \bmod 2 = 1$, so $\binom{n \bmod 2}{k \bmod 2}_{-1} = 0$ and the result is 0. Otherwise, $\binom{n \bmod 2}{k \bmod 2}_{-1} = 1$ and the result is $\binom{\lfloor n/2 \rfloor}{\lfloor k/2 \rfloor}$.

The Lean proof applies the q-Lucas theorem (Theorem 4.28) with $d = 2$ over $\mathbb{Q}$, then uses injectivity of $\mathbb{Z} \hookrightarrow \mathbb{Q}$ to transfer the result back to $\mathbb{Z}$. $\square$

4.2 The principles of inclusion and exclusion

4.2.1 The size version

Theorem 4.43 (Size version of the PIE)**.** *Let $n \in \mathbb{N}$. Let U be a finite set. Let $A_1, A_2, \ldots, A_n$ be n subsets of U. Then,*

$$(\text{\# of } u \in U \text{ that satisfy } u \notin A_i \text{ for all } i \in [n]) = \sum_{I \subseteq [n]} (-1)^{|I|} (\text{\# of } u \in U \text{ that satisfy } u \in A_i \text{ for all } i \in I).$$

Equivalently,

$$|U \setminus (A_1 \cup A_2 \cup \cdots \cup A_n)| = \sum_{I \subseteq [n]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|,$$

where $\bigcap_{i \in I} A_i$ *denotes* $\{u \in U \mid u \in A_i$ *for all* $i \in I\}$ *(so that* $\bigcap_{i \in \varnothing} A_i = U$*).*

Proof. This is the inclusion-exclusion cardinality formula from Mathlib. □

4.2.2 Alternative formulations of the size version

Lemma 4.44 (Complement-of-union formulation)**.** *The complement of the union* $A_1 \cup A_2 \cup \cdots \cup A_n$ *satisfies*

$$|(A_1 \cup A_2 \cup \cdots \cup A_n)^c| = \sum_{I \subseteq [n]} (-1)^{|I|} \left| \bigcap_{i \in I} A_i \right|.$$

This is the more common textbook formulation of the PIE.

Proof. Rewrite $(A_1 \cup \cdots \cup A_n)^c = \bigcap_i A_i^c$, then apply Theorem 4.43. □

Lemma 4.45 (PIE specialized to $\{0, 1, \ldots, n-1\}$)**.** *Let* α *be a finite type and* $A_0, A_1, \ldots, A_{n-1}$ *be* n *subsets of* α *indexed by* $\{0, 1, \ldots, n-1\}$*. Then*

$$\#\{u \in \alpha \mid u \notin A_i \ \forall\, i \in \{0, \ldots, n-1\}\} = \sum_{I \subseteq \{0, \ldots, n-1\}} (-1)^{|I|} \, \# \left(\bigcap_{i \in I} A_i \right).$$

Proof. This is the specialization of Theorem 4.43 to the index set $\{0, 1, \ldots, n-1\}$. □

Lemma 4.46 (Rule-breaking interpretation of PIE)**.** *Given* n *"rules" encoded as subsets* $A_0, \ldots, A_{n-1}$ *of a finite type* α *(where* A_i *is the set of elements satisfying rule* i*), the number of elements violating all rules is*

$$\#\{u \in \alpha \mid u \notin A_i \ \forall\, i\} = \sum_{I \subseteq \{0, \ldots, n-1\}} (-1)^{|I|} \, \#\{u \in \alpha \mid u \in A_i \ \forall\, i \in I\}.$$

Proof. Rewrite both sides in terms of infima of complements and intersections, then apply Lemma 4.45. □

4.2.3 Counting surjections

Definition 4.47 (Number of surjections)**.** The *number of surjective maps* from $[m]$ to $[n]$ is defined as

$$\mathrm{numSurj}(m, n) := \#\{f \colon [m] \to [n] \mid f \text{ is surjective}\}.$$

Theorem 4.48. *Let* $n, m \in \mathbb{N}$*. Then,*

$$(\textit{\# of surjective maps } f \colon [m] \to [n]) = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m.$$

Proof. A map $f\colon [m] \to [n]$ is surjective if and only if it takes each $i \in [n]$ as a value. Thus, imposing n rules on a map $f\colon [m] \to [n]$, where rule i says "thou shalt not take i as a value," the surjective maps are precisely the maps that violate all n rules. By the PIE (Theorem 4.43):

$$\begin{aligned}&(\#\text{ of surjective maps } f\colon [m] \to [n])\\ &= \sum_{I \subseteq [n]} (-1)^{|I|} (\#\text{ of maps } f\colon [m] \to [n] \text{ that satisfy all rules in } I).\end{aligned}$$

For each $I \subseteq [n]$, the maps satisfying all rules in I are the maps from $[m]$ to $[n] \setminus I$, of which there are $(n - |I|)^m$. Substituting and splitting the sum by $|I| = k$:

$$\sum_{k=0}^{n} \binom{n}{k} (-1)^k (n-k)^m.$$

In Lean, we relate the surjection count to the PIE via the "avoid" sets $S_i = \{f \mid i \notin \operatorname{im} f\}$, apply the PIE (Theorem 4.43), substitute the cardinality formula for each subset, and rewrite the sum over the powerset as a sum over cardinalities. □

Lemma 4.49 (Cardinality of functions avoiding a set)**.** *Let $m, n \in \mathbb{N}$ and let I be a subset of $[n]$. The number of functions $f\colon [m] \to [n]$ such that $i \notin \operatorname{im} f$ for all $i \in I$ is $(n - |I|)^m$.*

Proof. Such functions are exactly the functions from $[m]$ to I^c, and $|I^c| = n - |I|$. □

Lemma 4.50 (Cardinality of intersection of avoid-sets)**.** *For $t \subseteq [n]$, the cardinality of the intersection $\bigcap_{i \in t} S_{\text{avoid}}(i)$ is $(n - |t|)^m$, where $S_{\text{avoid}}(i) = \{f\colon [m] \to [n] \mid i \notin \operatorname{im} f\}$.*

Proof. The intersection equals the set of functions whose range avoids all elements of t, which by Lemma 4.49 has cardinality $(n - |t|)^m$. □

4.2.4 Consequences of the surjection formula

Corollary 4.51. *Let $n \in \mathbb{N}$. Then:*

- ***(a)*** *We have $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m = 0$ for any $m \in \mathbb{N}$ satisfying $m < n$.*
- ***(b)*** *We have $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^n = n!$.*
- ***(c)*** *We have $\sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m \geq 0$ for each $m \in \mathbb{N}$.*
- ***(d)*** *We have $n! \mid \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)^m$ for each $m \in \mathbb{N}$.*

Proof. **(a)** By Theorem 4.48, the sum equals the number of surjective maps $f\colon [m] \to [n]$. Since $m < n$, no such surjection exists (by the Pigeonhole Principle), so the sum is 0.

(b) By Theorem 4.48 with $m = n$, the sum counts surjections $f\colon [n] \to [n]$, which are precisely the permutations of $[n]$, of which there are $n!$.

(c) The sum equals the number of surjections, which is a nonnegative integer.

(d) The symmetric group S_n acts on the set $U := \{\text{surjective maps } f\colon [m] \to [n]\}$ by post-composition ($\sigma \cdot f = \sigma \circ f$). This action is free: if $\sigma \circ f = f$ and f is surjective, then for each $j \in [n]$, pick i with $f(i) = j$; then $\sigma(j) = \sigma(f(i)) = f(i) = j$, so $\sigma = \mathrm{id}$ (Lemma 4.54). By the orbit–stabilizer theorem, each orbit has size $n!$, so the orbits partition U into blocks of size $n!$, whence $n! \mid |U|$ (Lemma 4.55). □

4.2.5 Helpers for the divisibility proof

Lemma 4.52 (Post-composition preserves surjectivity)**.** *If σ is a permutation of $[n]$ and $f\colon [m] \to [n]$ is surjective, then $\sigma \circ f$ is surjective.*

Proof. Given $y \in [n]$, since f is surjective there exists x with $f(x) = \sigma^{-1}(y)$, so $\sigma(f(x)) = y$. □

Lemma 4.53 (Action of S_n on surjections)**.** *The symmetric group S_n acts on the set of surjective functions $f\colon [m] \to [n]$ by post-composition: $\sigma \cdot f := \sigma \circ f$.*

Proof. Well-definedness follows from Lemma 4.52. The action laws $1 \cdot f = f$ and $(\sigma\tau) \cdot f = \sigma \cdot (\tau \cdot f)$ are immediate from composition. □

Lemma 4.54 (Free action on surjections)**.** *Let $f\colon [m] \to [n]$ be surjective. Then the stabilizer of f under the action of S_n by post-composition is trivial.*

Proof. If $\sigma \circ f = f$ and f is surjective, then for each $y \in [n]$, pick x with $f(x) = y$; then $\sigma(y) = \sigma(f(x)) = f(x) = y$. So $\sigma = \mathrm{id}$. □

Lemma 4.55 (Factorial divides number of surjections)**.** *We have $n! \mid \mathrm{numSurj}(m, n)$.*

Proof. By the orbit–stabilizer theorem, each orbit of the free action of S_n on surjections has size $|S_n|/|\{\mathrm{id}\}| = n!$, so the orbits partition the set of surjections into blocks of size $n!$. □

4.2.6 Derangements

Definition 4.56. A *derangement* of a set X means a permutation of X that has no fixed points.

We write D_n for the number of derangements of $[n]$.

Lemma 4.57 (Equivalence to Mathlib's derangements)**.** *The set of derangements of α (as defined in Definition 4.56) is in bijection with the set of fixed-point-free permutations from Mathlib.*

Proof. Both are defined by the same predicate (having no fixed points) on the set of permutations of α. □

Lemma 4.58 (Derangements of the empty type)**.** *The identity permutation is a derangement of the empty type (vacuously, it has no fixed points). Hence $D_0 = 1$.*

Proof. Since the type is empty, there are no elements to be fixed points. □

Lemma 4.59 (Identity is not a derangement of nonempty types)**.** *If α is nonempty, then no derangement of α equals the identity permutation. In particular $D_1 = 0$.*

Proof. If d is a derangement with $d = \mathrm{id}$, pick any $x \in \alpha$; then $d(x) = x$, contradicting the fixed-point-free property. □

Lemma 4.60 (Small derangement counts)**.** *The first few values of the derangement count are: $D_2 = 1$ (the only derangement of $\{0, 1\}$ is the swap) and $D_3 = 2$ (the two 3-cycles).*

Proof. By computation (verified by decision procedure in Lean). □

Lemma 4.61 (Derangement type matches Mathlib)**.** *The number of derangements of $[n]$ (as defined in Definition 4.56) agrees with Mathlib's derangement count.*

Proof. Follows from Lemma 4.57 and Mathlib's computation of the derangement count. □

Theorem 4.62. *Let $n \in \mathbb{N}$. Then, the number of derangements of $[n]$ is*

$$D_n = \sum_{k=0}^{n} (-1)^k \binom{n}{k} (n-k)! = n! \cdot \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

Proof. Set $U := S_n$ (the set of all permutations of $[n]$). Impose n rules $1, 2, \ldots, n$ on a permutation $\sigma \in U$, with rule i being "thou shalt leave the element i fixed" (i.e., rule i requires $\sigma(i) = i$). Derangements are precisely the permutations violating all n rules. By the PIE (Theorem 4.43):

$$D_n = \sum_{I \subseteq [n]} (-1)^{|I|} \left(\# \text{ of permutations } u \in U \text{ that satisfy all rules in } I\right).$$

For each $I \subseteq [n]$, the permutations satisfying all rules in I are the permutations of $[n]$ that leave each $i \in I$ fixed; there are $(n - |I|)!$ such permutations (they are essentially the permutations of the $(n - |I|)$-element set $[n] \setminus I$). Substituting and grouping by $|I| = k$:

$$D_n = \sum_{k=0}^{n} \binom{n}{k} (-1)^k (n-k)! = \sum_{k=0}^{n} (-1)^k \frac{n!}{k!} = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

In Lean, the first equality is derived from Mathlib's derangement sum formula using the identity $(n-k)! \cdot \binom{n}{k} = n!/k!$. $\square$

Theorem 4.63 (Derangement formula, rational form)**.** *Let $n \in \mathbb{N}$. Then,*

$$D_n = n! \cdot \sum_{k=0}^{n} \frac{(-1)^k}{k!}.$$

(This is the same formula as in Theorem 4.62, but stated over $\mathbb{Q}$ so that the division $1/k!$ is literal.)

Proof. Convert Theorem 4.62 from $\mathbb{Z}$ to $\mathbb{Q}$, then use $\binom{n}{k}(n-k)! = n!/k!$. $\square$

4.2.7 Euler's totient formula

Theorem 4.64. *Let c be a positive integer with prime factorization $p_1^{a_1} p_2^{a_2} \cdots p_n^{a_n}$, where $p_1, p_2, \ldots, p_n$ are distinct primes and $a_1, a_2, \ldots, a_n$ are positive integers. Then,*

$$(\# \textit{ of all } u \in [c] \textit{ that are coprime to } c) = c \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^{n} \left(p_i^{a_i} - p_i^{a_i - 1}\right).$$

Proof. A number $u \in [c]$ is coprime to c if and only if it is not divisible by any of the prime factors $p_1, \ldots, p_n$ of c. Thus u must break all n rules $1, 2, \ldots, n$, where rule i says "thou shalt be divisible by p_i." By the PIE (Theorem 4.43):

$$\begin{aligned}
\phi(c) &= \sum_{I \subseteq [n]} (-1)^{|I|} \underbrace{\left(\# \text{ of all } u \in [c] \text{ that are divisible by all } p_i \text{ with } i \in I\right)}_{= c / \prod_{i \in I} p_i} \\
&= c \cdot \underbrace{\sum_{I \subseteq [n]} (-1)^{|I|} \prod_{i \in I} \frac{1}{p_i}}_{= \prod_{i=1}^{n} (1 - 1/p_i)} \\
&= \underbrace{c}_{= \prod_{i=1}^{n} p_i^{a_i}} \cdot \prod_{i=1}^{n} \left(1 - \frac{1}{p_i}\right) = \prod_{i=1}^{n} \left(p_i^{a_i} - p_i^{a_i - 1}\right).
\end{aligned}$$

(The factorization of the sum over subsets into a product uses the identity $\sum_{I\subseteq[n]}\prod_{i\in I}x_i = \prod_{i=1}^{n}(1+x_i)$.)

In Lean, this is derived from Mathlib's totient formula. $\square$

4.2.8 The weighted version

Theorem 4.65 (Weighted version of the PIE)**.** *Let $n \in \mathbb{N}$, and let U be a finite set. Let $A_1, A_2, \ldots, A_n$ be n subsets of U. Let G be any additive abelian group. Let $w\colon U \to G$ be any map. Then,*

$$\sum_{\substack{u\in U\\ u\notin A_i \text{ for all } i\in[n]}} w(u) = \sum_{I\subseteq[n]} (-1)^{|I|} \sum_{\substack{u\in U\\ u\in A_i \text{ for all } i\in I}} w(u).$$

Proof. This is the weighted inclusion-exclusion formula from Mathlib. $\square$

Theorem 4.66 (Weighted PIE, explicit index set)**.** *Let U be a finite set, s a finite index set, A_i subsets of U for $i \in s$, G an additive abelian group, and $w\colon U \to G$ a weight function. Then,*

$$\sum_{\substack{u\in U\\ u\notin A_i\ \forall i\in s}} w(u) = \sum_{I\subseteq s}(-1)^{|I|} \sum_{\substack{u\in U\\ u\in A_i\ \forall i\in I}} w(u).$$

Proof. Specialization of Theorem 4.65 to an explicit index set. $\square$

Lemma 4.67 (Size PIE from the weighted PIE)**.** *The size version of the PIE (Theorem 4.43) follows from the weighted version (Theorem 4.65) by taking the weight function $w(u) = 1$ for all u. Explicitly, for a finite type U, finite index set s, and subsets A_i of U:*

$$\left|\bigcap_{i\in s} A_i^c\right| = \sum_{I\in s.\text{powerset}} (-1)^{|I|} \left|\bigcap_{i\in I} A_i\right|.$$

Proof. Specialize Theorem 4.66 to $G = \mathbb{Z}$ and $w \equiv 1$. Each weighted sum $\sum_{u\in S} w(u) = \sum_{u\in S} 1 = |S|$. $\square$

4.3 Boolean Möbius inversion

4.3.1 Alternating sums over supersets

Lemma 4.68 (Alternating sum over supersets)**.** *Let $P \subseteq Q$ be two finite sets.*

(a) *We have*

$$\sum_{\substack{I\subseteq Q;\\ P\subseteq I}} (-1)^{|I|} = (-1)^{|P|}\,[P = Q]\,. \tag{4.1}$$

(b) *We have*

$$\sum_{\substack{I\subseteq Q;\\ P\subseteq I}} (-1)^{|Q\setminus I|} = [P = Q]\,. \tag{4.2}$$

Proof. **(a)** We must prove the equality (4.1). If $P = Q$, then the only subset I of Q satisfying $P \subseteq I$ is Q itself, so the sum equals $(-1)^{|Q|} = (-1)^{|P|}\cdot[P = Q]$ (since $[Q = Q] = 1$).

For the rest of this proof, we assume $P \neq Q$. Thus, $[P = Q] = 0$.

Now, P is a proper subset of Q, so there exists some $q \in Q$ such that $q \notin P$. Fix such a q.

Let

$$\mathcal{A} := \{I \subseteq Q \ \mid \ P \subseteq I\},$$

and let $\operatorname{sign} I := (-1)^{|I|}$ for each $I \in \mathcal{A}$. Then

$$\sum_{I \in \mathcal{A}} \operatorname{sign} I = \sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|I|}. \tag{4.3}$$

We define a map $f : \mathcal{A} \to \mathcal{A}$ by

$$f(I) := I \triangle \{q\} = \begin{cases} I \setminus \{q\}, & \text{if } q \in I; \\ I \cup \{q\}, & \text{if } q \notin I \end{cases} \qquad \text{for each } I \in \mathcal{A}.$$

This map f is well-defined and it is an involution (the symmetric difference with $\{q\}$ undoes itself). This involution has no fixed points (since $f(I) = I \triangle \{q\} \neq I$). Furthermore, for each $I \in \mathcal{A}$, the set $f(I) = I \triangle \{q\}$ differs from I in exactly one element (namely q), so $|f(I)| = |I| \pm 1$. Therefore

$$(-1)^{|f(I)|} = -(-1)^{|I|},$$

i.e., $\operatorname{sign}(f(I)) = -\operatorname{sign} I$. Since f is a sign-reversing fixed-point-free involution on $\mathcal{A}$, the elements of $\mathcal{A}$ pair up into pairs with opposite signs, giving

$$\sum_{I \in \mathcal{A}} \operatorname{sign} I = 0.$$

Comparing with (4.3), we obtain

$$\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|I|} = 0 = (-1)^{|P|} [P = Q]$$

(since $(-1)^{|P|} \cdot 0 = 0$). This proves (4.1).

(b) If I is any subset of Q, then $|Q \setminus I| = |Q| - |I| \equiv |Q| + |I| \pmod 2$, so

$$(-1)^{|Q \setminus I|} = (-1)^{|Q| + |I|} = (-1)^{|Q|} (-1)^{|I|}$$

(since $|Q \setminus I| \equiv |Q| + |I| \pmod 2$). Hence

$$\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q \setminus I|} = \sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q|} (-1)^{|I|} = (-1)^{|Q|} \underbrace{\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|I|}}_{\substack{= (-1)^{|P|} [P = Q] \\ \text{(by part } \textbf{(a)}\text{)}}}$$

$$= (-1)^{|Q|} (-1)^{|P|} [P = Q].$$

It remains to check that

$$(-1)^{|Q|} (-1)^{|P|} [P = Q] = [P = Q]. \tag{4.4}$$

If $P \neq Q$, then $[P = Q] = 0$ and both sides are 0. If $P = Q$, then $(-1)^{|Q|} (-1)^{|P|} = (-1)^{2|Q|} = 1$, so both sides equal 1. Thus

$$\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q \setminus I|} = [P = Q].$$

This proves (4.2). $\square$

4.3.2 The Boolean Möbius inversion formula

Lemma 4.69. *Let Q be a finite set, A an additive abelian group, and f a function that assigns to each pair (J, P) of subsets of Q an element of A. Then*

$$\sum_{J \subseteq Q} \sum_{P \subseteq J} f(J, P) = \sum_{P \subseteq Q} \sum_{\substack{J \subseteq Q; \\ P \subseteq J}} f(J, P).$$

Proof. Both sides sum the same function f over all pairs (J, P) of subsets of Q satisfying $P \subseteq J \subseteq Q$; only the order of summation differs. $\square$

Theorem 4.70 (Boolean Möbius inversion)**.** *Let S be a finite set. Let A be any additive abelian group.*

For each subset I of S, let a_I and b_I be two elements of A.

Assume that

$$b_I = \sum_{J \subseteq I} a_J \qquad \text{for all } I \subseteq S. \tag{4.5}$$

Then, we also have

$$a_I = \sum_{J \subseteq I} (-1)^{|I \setminus J|} b_J \qquad \text{for all } I \subseteq S. \tag{4.6}$$

Proof. Fix a subset Q of S. We shall prove that

$$a_Q = \sum_{I \subseteq Q} (-1)^{|Q \setminus I|} b_I.$$

We begin by rewriting the right hand side. By (4.5), we have $b_I = \sum_{J \subseteq I} a_J$ for each I, so

$$\begin{aligned}
\sum_{I \subseteq Q} (-1)^{|Q \setminus I|} b_I &= \sum_{I \subseteq Q} (-1)^{|Q \setminus I|} \underbrace{\sum_{J \subseteq I} a_J}_{= \sum_{P \subseteq I} a_P} = \sum_{I \subseteq Q} (-1)^{|Q \setminus I|} \sum_{P \subseteq I} a_P \\
&= \sum_{I \subseteq Q} \sum_{P \subseteq I} (-1)^{|Q \setminus I|} a_P.
\end{aligned} \tag{4.7}$$

The two summation signs "$\sum_{I \subseteq Q} \sum_{P \subseteq I}$" on the right hand side of this equality result in a sum over all pairs (I, P) of subsets of Q satisfying $P \subseteq I \subseteq Q$. The same result can be obtained by the two summation signs "$\sum_{P \subseteq Q} \sum_{\substack{I \subseteq Q; \\ P \subseteq I}}$" (by Lemma 4.69). Hence, (4.7) rewrites as follows:

$$\begin{aligned}
\sum_{I \subseteq Q} (-1)^{|Q \setminus I|} b_I &= \sum_{P \subseteq Q} \sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q \setminus I|} a_P \\
&= \sum_{P \subseteq Q} \left(\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q \setminus I|} \right) a_P.
\end{aligned} \tag{4.8}$$

We want to prove that this equals a_Q. To this end, we show that the coefficient $\sum_{\substack{I \subseteq Q; \\ P \subseteq I}} (-1)^{|Q \setminus I|}$ in front of a_P is 0 whenever $P \neq Q$, and is 1 whenever $P = Q$. That is, every subset P of Q

satisfies

$$\sum_{\substack{I\subseteq Q;\\ P\subseteq I}} (-1)^{|Q\setminus I|} = [P=Q]. \tag{4.9}$$

This is precisely Lemma 4.68 **(b)**.

Now, (4.8) becomes (using (4.9))

$$\begin{aligned}
\sum_{I\subseteq Q} (-1)^{|Q\setminus I|} b_I &= \sum_{P\subseteq Q} \underbrace{\left(\sum_{\substack{I\subseteq Q;\\ P\subseteq I}} (-1)^{|Q\setminus I|}\right)}_{=[P=Q]} a_P = \sum_{P\subseteq Q} [P=Q]\, a_P \\
&= \underbrace{[Q=Q]}_{\substack{=1\\ (\text{since } Q=Q)}} a_Q + \sum_{\substack{P\subseteq Q;\\ P\neq Q}} \underbrace{[P=Q]}_{\substack{=0\\ (\text{since } P\neq Q)}} a_P \\
&= a_Q + \underbrace{\sum_{\substack{P\subseteq Q;\\ P\neq Q}} 0\cdot a_P}_{=0} = a_Q.
\end{aligned}$$

In other words, $a_Q = \sum_{I\subseteq Q} (-1)^{|Q\setminus I|} b_I$.

Since Q was an arbitrary subset of S, we have shown that

$$a_I = \sum_{J\subseteq I} (-1)^{|I\setminus J|} b_J \qquad \text{for all } I\subseteq S.$$

This proves Theorem 4.70. □

Lemma 4.71. *Alternative formulation of Boolean Möbius inversion (Theorem 4.70) using the $\mathbb{Z}$-module scalar multiplication: under the same hypotheses, for all $I\subseteq S$,*

$$a_I = \sum_{J\subseteq I} (-1)^{|I\setminus J|}\cdot b_J,$$

where the multiplication $(-1)^{|I\setminus J|}\cdot b_J$ denotes the $\mathbb{Z}$-module scalar action.

Proof. Immediate from Theorem 4.70, since the integer scalar multiplication $n\cdot a$ in any additive abelian group coincides with n times a when the group carries a $\mathbb{Z}$-module structure. □

4.4 More subtractive methods

4.4.1 Sums with varying signs

Definition 4.72. Let $n\in\mathbb{N}$ and $d\in\mathbb{N}$. An n-tuple $(x_1,x_2,\ldots,x_n)\in[d]^n$ is said to be *all-even* if each element of $[d]$ occurs an even number of times in this n-tuple (i.e., if for each $k\in[d]$, the number of all $i\in[n]$ satisfying $x_i=k$ is even).

In the Lean formalization, sign tuples $(e_1,e_2,\ldots,e_d)\in\{1,-1\}^d$ are represented as functions $e:\operatorname{Fin} d\to\mathbb{Z}/2\mathbb{Z}$, where 0 represents $+1$ and 1 represents -1. The conversion is performed by a sign conversion function. The product $e_{x_1}e_{x_2}\cdots e_{x_n}$ is represented by a sign product function, and the sum $e_1+e_2+\cdots+e_d$ (in $\mathbb{Z}$) by a sign sum function.

Definition 4.73. For an n-tuple $x : \operatorname{Fin} n \to \operatorname{Fin} d$ and $k \in \operatorname{Fin} d$, the *multiplicity* of k in x is the number of all $i \in \operatorname{Fin} n$ satisfying $x(i) = k$.

Definition 4.74 (Sign conversion)**.** The function toSign : $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}$ converts elements of $\mathbb{Z}/2\mathbb{Z}$ to signs: $\operatorname{toSign}(0) = 1$ and $\operatorname{toSign}(1) = -1$.

Definition 4.75 (Sign product)**.** For a sign tuple $e : \operatorname{Fin} d \to \mathbb{Z}/2\mathbb{Z}$ and an index tuple $x : \operatorname{Fin} n \to \operatorname{Fin} d$, the *sign product* is

$$\operatorname{signProduct}(e, x) := \prod_{i=1}^{n} \operatorname{toSign}(e_{x_i}).$$

Definition 4.76 (Sign sum)**.** For a sign tuple $e : \operatorname{Fin} d \to \mathbb{Z}/2\mathbb{Z}$, the *sign sum* is the integer

$$\operatorname{signSum}(e) := \sum_{k=1}^{d} \operatorname{toSign}(e_k).$$

Definition 4.77 (Set of all-even tuples)**.** The set of all-even n-tuples in $[d]^n$ is

$$\operatorname{allEvenTuples}(n, d) := \{x \in (\operatorname{Fin} n \to \operatorname{Fin} d) \mid x \text{ is all-even}\}.$$

Theorem 4.78. *Let $n \in \mathbb{N}$ and $d \in \mathbb{N}$. Then, the number of all all-even n-tuples $(x_1, x_2, \ldots, x_n) \in [d]^n$ is*

$$\frac{1}{2^d} \sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n .$$

More precisely (avoiding division), the number of all-even n-tuples multiplied by 2^d equals $\sum_{k=0}^{d} \binom{d}{k} (d-2k)^n$.

Proof. Lemma 4.80 yields

$$\begin{aligned} &\sum_{(e_1,e_2,\ldots,e_d)\in\{1,-1\}^d} (e_1 + e_2 + \cdots + e_d)^n \\ &= \sum_{(x_1,x_2,\ldots,x_n)\in[d]^n} \ \sum_{(e_1,e_2,\ldots,e_d)\in\{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n}. \end{aligned}$$

By Lemma 4.90 (which splits according to whether each tuple is all-even or not, using Lemma 4.88 and Lemma 4.89), the right-hand side equals (# of all-even tuples) $\cdot\, 2^d$.

On the other hand, a d-tuple $(e_1, e_2, \ldots, e_d) \in \{1, -1\}^d$ is uniquely determined by the set $S = \{i \in [d] \mid e_i = -1\}$, and for this tuple we have $e_1 + e_2 + \cdots + e_d = d - 2\,|S|$. By Lemma 4.96, we can rewrite the sum over sign tuples as a sum over subsets. By Lemma 4.97, grouping by cardinality $|S| = k$ yields

$$\sum_{(e_1,e_2,\ldots,e_d)\in\{1,-1\}^d} (e_1 + e_2 + \cdots + e_d)^n = \sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n .$$

Comparing the two expressions gives

$$(\#\text{ of all-even tuples}) \cdot 2^d = \sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n .$$

$\square$

4.4.2 Lemmas for the proof

Lemma 4.79. *Let $n, d \in \mathbb{N}$. Let $e = (e_1, \ldots, e_d) \in \{1, -1\}^d$ and $x = (x_1, \ldots, x_n) \in [d]^n$. For each $k \in [d]$, let m_k be the multiplicity of k in x. Then*

$$e_{x_1} e_{x_2} \cdots e_{x_n} = e_1^{m_1} e_2^{m_2} \cdots e_d^{m_d}.$$

Proof. Each factor of the product $e_{x_1} e_{x_2} \cdots e_{x_n}$ is one of the d entries $e_1, e_2, \ldots, e_d$. Moreover, each given entry e_k appears exactly m_k times in the product $e_{x_1} e_{x_2} \cdots e_{x_n}$, because k appears exactly m_k times among the subscripts $x_1, x_2, \ldots, x_n$. Thus the product equals $e_1^{m_1} e_2^{m_2} \cdots e_d^{m_d}$. □

Lemma 4.80. *Let $n, d \in \mathbb{N}$. Then,*

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} (e_1 + e_2 + \cdots + e_d)^n = \sum_{(x_1, x_2, \ldots, x_n) \in [d]^n} \; \sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n}.$$

Proof. For each $(e_1, e_2, \ldots, e_d) \in \{1, -1\}^d$, we have

$$(e_1 + e_2 + \cdots + e_d)^n = \left(\sum_{k=1}^{d} e_k \right)^n = \sum_{(x_1, x_2, \ldots, x_n) \in [d]^n} e_{x_1} e_{x_2} \cdots e_{x_n}$$

(by the generalized distributive law / product rule). Summing over all $(e_1, e_2, \ldots, e_d) \in \{1, -1\}^d$ and interchanging the order of summation yields the result. □

Lemma 4.81. *Let $n, d \in \mathbb{N}$. Let $(x_1, x_2, \ldots, x_n) \in [d]^n$.*

(a) *If the n-tuple $(x_1, x_2, \ldots, x_n)$ is not all-even, then $\sum_{(e_1, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 0$.*

(b) *If the n-tuple $(x_1, x_2, \ldots, x_n)$ is all-even, then $\sum_{(e_1, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 2^d$.*

Proof. This combines parts **(a)** and **(b)**. □

4.4.3 Involution argument for part (a)

Definition 4.82. For $k \in [d]$, the map $\mathrm{flipSign}_k : \{1, -1\}^d \to \{1, -1\}^d$ sends a d-tuple $(e_1, \ldots, e_d)$ to the tuple obtained by flipping the sign of the k-th entry: i.e., replacing e_k by $-e_k$ and leaving all other entries unchanged.

Lemma 4.83. *For any $k \in [d]$, the map $\mathrm{flipSign}_k$ is an involution.*

Proof. Flipping the sign of the k-th entry twice restores the original sign. □

Lemma 4.84. *For any $k \in [d]$ and $e \in \{1, -1\}^d$, we have $\mathrm{flipSign}_k(e) \neq e$.*

Proof. Since $e_k \in \{1, -1\}$, flipping the sign of e_k produces a different value ($-e_k \neq e_k$), so the resulting tuple differs from the original. □

Lemma 4.85 (flipSign at the flipped position)**.** *For any $k \in [d]$ and $e \in \{1, -1\}^d$, $(\mathrm{flipSign}_k(e))_k = e_k + 1$ (in $\mathbb{Z}/2\mathbb{Z}$).*

Proof. By definition of flipSign, which updates position k. □

Lemma 4.86 (flipSign leaves other positions unchanged)**.** *For any $k, j \in [d]$ with $j \neq k$ and $e \in \{1, -1\}^d$, $(\mathrm{flipSign}_k(e))_j = e_j$.*

Proof. The update only affects position k, so all other positions are unchanged. $\square$

Lemma 4.87. *Let $e \in \{1, -1\}^d$, $x \in [d]^n$, and $k \in [d]$ such that the multiplicity m_k of k in x is odd. Then*

$$\mathrm{signProduct}(\mathrm{flipSign}_k(e), x) = -\,\mathrm{signProduct}(e, x).$$

Proof. Using the product-of-powers representation (Lemma 4.79), flipping the k-th entry multiplies the product by $(-1)^{m_k} = -1$ (since m_k is odd), while all other factors remain unchanged. $\square$

Lemma 4.88. *Let $n, d \in \mathbb{N}$. Let $(x_1, x_2, \ldots, x_n) \in [d]^n$. If the n-tuple $(x_1, x_2, \ldots, x_n)$ is not all-even, then*

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 0.$$

Proof. Since the tuple is not all-even, there exists some $k \in [d]$ such that m_k is odd.

Define $f : \{1, -1\}^d \to \{1, -1\}^d$ as $\mathrm{flipSign}_k$ (flipping the sign of the k-th entry). By Lemma 4.83, f is an involution. By Lemma 4.84, f has no fixed points. By Lemma 4.87, $\mathrm{signProduct}(f(e), x) = -\,\mathrm{signProduct}(e, x)$.

Hence, the terms cancel in pairs, giving

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 0.$$

The Lean proof uses the involution summation lemma from Mathlib. $\square$

Lemma 4.89. *Let $n, d \in \mathbb{N}$. Let $(x_1, x_2, \ldots, x_n) \in [d]^n$. If the n-tuple $(x_1, x_2, \ldots, x_n)$ is all-even, then*

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 2^d.$$

Proof. Since the tuple is all-even, all the multiplicities $m_1, m_2, \ldots, m_d$ are even.

Let $(e_1, e_2, \ldots, e_d) \in \{1, -1\}^d$. For each $k \in [d]$, we have $e_k \in \{1, -1\}$ and m_k is even, so $e_k^{m_k} = 1$. By Lemma 4.79, $e_{x_1} e_{x_2} \cdots e_{x_n} = e_1^{m_1} e_2^{m_2} \cdots e_d^{m_d} = 1$.

Since every sign product equals 1 and there are 2^d sign tuples,

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = 2^d.$$

$\square$

Lemma 4.90. *Let $n, d \in \mathbb{N}$. Then*

$$\sum_{(x_1, x_2, \ldots, x_n) \in [d]^n} \; \sum_{(e_1, e_2, \ldots, e_d) \in \{1,-1\}^d} e_{x_1} e_{x_2} \cdots e_{x_n} = (\#\ \textit{of all-even tuples in}\ [d]^n) \cdot 2^d.$$

Proof. Split the outer sum according to whether each tuple is all-even or not. By Lemma 4.88, the non-all-even tuples contribute 0. By Lemma 4.89, each all-even tuple contributes 2^d. Hence the total is $(\#\ \text{of all-even tuples}) \cdot 2^d$. $\square$

4.4.4 Computing the left-hand side

Lemma 4.91. *For a sign tuple $e \in \{1, -1\}^d$, let $S = \{i \in [d] \mid e_i = -1\}$. Then*

$$e_1 + e_2 + \cdots + e_d = d - 2|S|.$$

Proof. Each entry e_i equals -1 if $i \in S$ and 1 if $i \notin S$. The sum has $|S|$ addends equal to -1 and $d - |S|$ addends equal to 1, giving $|S| \cdot (-1) + (d - |S|) \cdot 1 = d - 2|S|$. □

Definition 4.92 (Sign tuple to subset)**.** The map signTupleToSubset : $(\operatorname{Fin} d \to \mathbb{Z}/2\mathbb{Z}) \to \mathcal{P}(\operatorname{Fin} d)$ sends a sign tuple e to the set $S = \{k \in \operatorname{Fin} d \mid e_k = 1\}$ (i.e., the positions where $\mathrm{toSign}(e_k) = -1$).

Definition 4.93 (Subset to sign tuple)**.** The inverse map subsetToSignTuple : $\mathcal{P}(\operatorname{Fin} d) \to (\operatorname{Fin} d \to \mathbb{Z}/2\mathbb{Z})$ sends a subset $S \subseteq \operatorname{Fin} d$ to the sign tuple where position k has value 1 (representing -1) if $k \in S$, and 0 (representing $+1$) otherwise.

Lemma 4.94. *There is a bijection between sign tuples $\{1, -1\}^d$ and subsets of $[d]$, sending each tuple e to the set $S = \{i \in [d] \mid e_i = -1\}$.*

Proof. The inverse sends a subset $S \subseteq [d]$ to the sign tuple where position k has $e_k = -1$ if $k \in S$ and $e_k = 1$ otherwise. □

Lemma 4.95 (Sign sum under the subset correspondence)**.** *For a subset $S \subseteq [d]$, the sign sum of the corresponding sign tuple is*

$$\mathrm{signSum}(\mathrm{subsetToSignTuple}(S)) = d - 2|S|.$$

Proof. By Lemma 4.91, the sign sum equals $d - 2|T|$ where $T = \mathrm{signTupleToSubset}(\mathrm{subsetToSignTuple}(S)) = S$. □

Lemma 4.96. *Let $n, d \in \mathbb{N}$. Then*

$$\sum_{(e_1, e_2, \ldots, e_d) \in \{1, -1\}^d} (e_1 + e_2 + \cdots + e_d)^n = \sum_{S \subseteq [d]} (d - 2\,|S|)^n .$$

Proof. Substitute via the bijection of Lemma 4.94 and use Lemma 4.91. □

Lemma 4.97. *Let $n, d \in \mathbb{N}$. Then*

$$\sum_{S \subseteq [d]} (d - 2\,|S|)^n = \sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n .$$

Proof. Split the sum over subsets S according to the value of $|S| = k$. For each k, all subsets of size k contribute the same summand $(d - 2k)^n$, and the number of k-element subsets of $[d]$ is $\binom{d}{k}$. □

4.4.5 Helper lemmas for toSign

Lemma 4.98. *For any $b \in \mathbb{Z}/2\mathbb{Z}$, we have* $(\operatorname{toSign} b)^2 = 1$.

Proof. By case analysis: both $1^2 = 1$ and $(-1)^2 = 1$. □

Lemma 4.99. *For any $b \in \mathbb{Z}/2\mathbb{Z}$ and even $m \in \mathbb{N}$, we have* $(\operatorname{toSign} b)^m = 1$.

Proof. Write $m = 2k$. Then $(\operatorname{toSign} b)^m = ((\operatorname{toSign} b)^2)^k = 1^k = 1$. □

Lemma 4.100 (Odd powers of a sign)**.** *For any $b \in \mathbb{Z}/2\mathbb{Z}$ and odd $m \in \mathbb{N}$, we have* $(\operatorname{toSign} b)^m = \operatorname{toSign} b$.

Proof. Write $m = 2k + 1$. Then $(\operatorname{toSign} b)^m = ((\operatorname{toSign} b)^2)^k \cdot \operatorname{toSign} b = 1^k \cdot \operatorname{toSign} b = \operatorname{toSign} b$.
□

Lemma 4.101. *For any $b \in \mathbb{Z}/2\mathbb{Z}$, we have* $\operatorname{toSign}(b + 1) = -\operatorname{toSign}(b)$.

Proof. By case analysis on $b \in \{0, 1\}$: adding 1 in $\mathbb{Z}/2\mathbb{Z}$ toggles between 0 and 1, and $\operatorname{toSign}(0) = 1 = -(-1) = -\operatorname{toSign}(1)$ while $\operatorname{toSign}(1) = -1 = -(1) = -\operatorname{toSign}(0)$. □

4.4.6 Corollaries of the counting formula

Corollary 4.102 (Divisibility by 2^d)**.** *For any $n, d \in \mathbb{N}$, we have*

$$2^d \mid \sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n.$$

Proof. By Theorem 4.78, the sum equals $(\#\text{ of all-even tuples}) \cdot 2^d$, which is divisible by 2^d. □

Corollary 4.103 (Nonnegativity)**.** *For any $n, d \in \mathbb{N}$, we have*

$$\sum_{k=0}^{d} \binom{d}{k} (d - 2k)^n \geq 0.$$

This is not obvious from the formula when n is odd, since the summands can be negative.

Proof. By Theorem 4.78, the sum equals $(\#\text{ of all-even tuples}) \cdot 2^d$, which is a product of two nonnegative integers. □

Chapter 5

Determinants

5.1 Determinants: Basic Properties

5.1.1 Conventions

Convention 5.1. For the rest of this section, we fix a commutative ring K. In most examples, K will be $\mathbb{Z}$ or $\mathbb{Q}$ or a polynomial ring.

Convention 5.2. Let $n, m \in \mathbb{N}$.

(a) If A is an $n \times m$-matrix, then $A_{i,j}$ shall mean the (i, j)-th entry of A, that is, the entry of A in row i and column j.

(b) If $a_{i,j}$ is an element of K for each $i \in [n]$ and each $j \in [m]$, then

$$(a_{i,j})_{1 \leq i \leq n,\ 1 \leq j \leq m}$$

shall denote the $n \times m$-matrix whose (i, j)-th entry is $a_{i,j}$ for all $i \in [n]$ and $j \in [m]$.

(c) We let $K^{n \times m}$ denote the set of all $n \times m$-matrices with entries in K. This is a K-module. If $n = m$, this is also a K-algebra.

(d) Let $A \in K^{n \times m}$ be an $n \times m$-matrix. The *transpose* A^T of A is defined to be the $m \times n$-matrix whose entries are given by

$$\left(A^T\right)_{i,j} = A_{j,i} \quad \text{for all } i \in [m] \text{ and } j \in [n].$$

5.1.2 Definition

Definition 5.3. Let $n \in \mathbb{N}$. Let $A \in K^{n \times n}$ be an $n \times n$-matrix. The *determinant* $\det A$ of A is defined to be the element

$$\sum_{\sigma \in S_n} (-1)^\sigma \underbrace{A_{1,\sigma(1)} A_{2,\sigma(2)} \cdots A_{n,\sigma(n)}}_{= \prod_{i=1}^n A_{i,\sigma(i)}}$$

of K. Here:

- we let S_n denote the n-th symmetric group (i.e., the group of permutations of $[n] = \{1, 2, \ldots, n\}$);
- we let $(-1)^\sigma$ denote the sign of the permutation σ (as defined in Definition 3.109).

5.1.3 Small determinants and basic API

Lemma 5.4. *The determinant of a 0×0 matrix is 1 (empty product convention).*

Proof. The sum over $S_0 = \{\text{id}\}$ gives $(-1)^{\text{id}} \cdot 1 = 1$ (empty product is 1). $\square$

Lemma 5.5. *The determinant of a 1×1 matrix is its single entry:* $\det(a) = a$.

Proof. The sum over $S_1 = \{\text{id}\}$ gives $(-1)^{\text{id}} \cdot a = a$. $\square$

Lemma 5.6. *The determinant of a 2×2 matrix satisfies* $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$.

Proof. S_2 has two elements: the identity (sign $+1$) and the transposition (sign -1). The sum gives $a \cdot d - b \cdot c$. $\square$

Lemma 5.7. *The determinant of the identity matrix is 1:* $\det I_n = 1$.

Proof. By the fact that the identity matrix has determinant 1. $\square$

Lemma 5.8. *For $n \geq 1$, the determinant of the zero matrix is 0. (For $n = 0$, the zero matrix is the empty matrix, and $\det(\emptyset) = 1$.)*

Proof. The zero matrix has a zero row, so $\det = 0$ by Theorem 5.20 **(b)**. $\square$

Lemma 5.9. *The determinant of a diagonal matrix equals the product of its diagonal entries:* $\det(\text{diag}(d_1, \ldots, d_n)) = d_1 d_2 \cdots d_n$.

Proof. A diagonal matrix is both upper- and lower-triangular, so this follows from Theorem 5.19. $\square$

Lemma 5.10. *For $n \geq 2$, a matrix with all entries equal to c has determinant 0. This is a corollary of Proposition 5.12 with $x_i = 1$ and $y_j = c$.*

Proof. The constant matrix $(c)_{i,j}$ is the outer product of $\mathbf{1}$ and $(c, c, \ldots, c)$. By Proposition 5.12, its determinant is 0. $\square$

Lemma 5.11. *Let $n \geq 2$. Then,*

$$\sum_{\sigma \in S_n} (-1)^\sigma = 0.$$

Proof. The proof uses a sign-reversing involution: for $n \geq 2$, pick two distinct elements i, j and their transposition $t_{i,j}$. The map $\sigma \mapsto \sigma \cdot t_{i,j}$ pairs up permutations with opposite signs, so all summands cancel. $\square$

Proposition 5.12. *Let $n \in \mathbb{N}$ be such that $n \geq 2$. Let $x_1, x_2, \ldots, x_n \in K$ and $y_1, y_2, \ldots, y_n \in K$. Then,*

$$\det \left((x_i y_j)_{1 \leq i \leq n,\ 1 \leq j \leq n} \right) = 0.$$

Proof. The definition of the determinant yields (using Lemma 5.11 for the last step)

$$\begin{aligned}\det\left((x_i y_j)_{1\le i\le n,\ 1\le j\le n}\right) &= \sum_{\sigma\in S_n}(-1)^\sigma \underbrace{(x_1 y_{\sigma(1)})(x_2 y_{\sigma(2)})\cdots(x_n y_{\sigma(n)})}_{=(x_1x_2\cdots x_n)(y_{\sigma(1)}y_{\sigma(2)}\cdots y_{\sigma(n)})} \\ &= \sum_{\sigma\in S_n}(-1)^\sigma (x_1x_2\cdots x_n)\underbrace{(y_{\sigma(1)}y_{\sigma(2)}\cdots y_{\sigma(n)})}_{\substack{=y_1y_2\cdots y_n \\ \text{(since } \sigma \text{ is a bijection)}}} \\ &= (x_1x_2\cdots x_n)(y_1y_2\cdots y_n)\underbrace{\sum_{\sigma\in S_n}(-1)^\sigma}_{=0} = 0.\end{aligned}$$

□

Lemma 5.13. *Let A be a 5×5 matrix such that $A_{i,j} = 0$ whenever $i, j \in \{2,3,4\}$. Then $\det A = 0$. This formalizes the 5×5 hollow matrix example: a matrix with a 3×3 block of zeros in the middle has determinant 0.*

Proof. By the pigeonhole principle: any permutation σ of $[5]$ must map some element of $\{2,3,4\}$ to an element of $\{2,3,4\}$, because $|\{2,3,4\}| = 3 > 2 = |\{1,5\}|$. Hence, for every σ, the product $\prod_i A_{i,\sigma(i)}$ contains a zero factor, making every term in the determinant sum equal to 0. □

5.1.4 Helpers for Proposition 5.17

Lemma 5.14. *The factor matrix A in the factorization of the sum matrix $(x_i + y_j)$ has a zero column (column index 2) when $n \ge 3$. The matrix A has x_i in the first column, 1 in the second column, and 0 elsewhere.*

Proof. By direct computation: for column index 2, neither $j = 0$ nor $j = 1$, so the entry is 0. □

Lemma 5.15. *For $n \ge 3$, the factor matrix A has determinant 0.*

Proof. By Lemma 5.14, column 2 is zero. By Theorem 5.21 **(b)**, $\det A = 0$. □

Lemma 5.16. *For $n \ge 2$, the matrix $(x_i + y_j)_{i,j}$ factors as $A \cdot B$ where:*

$$A_{i,j} = \begin{cases} x_i & \text{if } j = 0, \\ 1 & \text{if } j = 1, \\ 0 & \text{otherwise,}\end{cases} \qquad B_{i,j} = \begin{cases} 1 & \text{if } i = 0, \\ y_j & \text{if } i = 1, \\ 0 & \text{otherwise.}\end{cases}$$

Proof. Direct matrix multiplication: $(AB)_{i,j} = \sum_k A_{i,k}B_{k,j} = x_i \cdot 1 + 1 \cdot y_j = x_i + y_j$. □

Proposition 5.17. *Let $n \in \mathbb{N}$ be such that $n \ge 3$. Let $x_1, x_2, \ldots, x_n \in K$ and $y_1, y_2, \ldots, y_n \in K$. Then,*

$$\det\left((x_i + y_j)_{1\le i\le n,\ 1\le j\le n}\right) = 0.$$

Proof. Define two $n\times n$-matrices

$$A := \begin{pmatrix} x_1 & 1 & 0 & \cdots & 0 \\ x_2 & 1 & 0 & \cdots & 0 \\ x_3 & 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_n & 1 & 0 & \cdots & 0 \end{pmatrix} \quad \text{and} \quad B := \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ y_1 & y_2 & y_3 & \cdots & y_n \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Then $(x_i + y_j)_{1 \leq i \leq n,\ 1 \leq j \leq n} = AB$, so

$$\det\left((x_i + y_j)_{1 \leq i \leq n,\ 1 \leq j \leq n}\right) = \det(AB) = \det A \cdot \det B$$

(by Theorem 5.23). However, the matrix A has a zero column (since $n \geq 3$), and thus $\det A = 0$ (by Theorem 5.21 **(b)**). Hence the product is 0. $\square$

5.1.5 Basic properties

Theorem 5.18 (Transposes preserve determinants)**.** *Let $n \in \mathbb{N}$. If $A \in K^{n \times n}$ is any $n \times n$-matrix, then $\det(A^T) = \det A$.*

Proof. See [Strick13, Corollary B.16] or [Grinbe15, Exercise 6.4] or [Laue15, §5.3.2]. $\square$

Theorem 5.19 (Determinants of triangular matrices)**.** *Let $n \in \mathbb{N}$. Let $A \in K^{n \times n}$ be a triangular (i.e., lower-triangular or upper-triangular) $n \times n$-matrix. Then, the determinant of the matrix A is the product of its diagonal entries. That is,*

$$\det A = A_{1,1} A_{2,2} \cdots A_{n,n}.$$

Proof. See [Strick13, Proposition B.11] or [Grinbe15, Exercise 6.3 and the paragraph after Exercise 6.4]. $\square$

Theorem 5.20 (Row operation properties)**.** *Let $n \in \mathbb{N}$. Let $A \in K^{n \times n}$ be an $n \times n$-matrix. Then:*

(a) *If we swap two rows of A, then $\det A$ gets multiplied by -1.*

(b) *If A has a zero row (i.e., a row that consists entirely of zeroes), then $\det A = 0$.*

(c) *If A has two equal rows, then $\det A = 0$.*

(d) *Let $\lambda \in K$. If we multiply a row of A by λ (that is, we multiply all entries of this one row by λ, while leaving all other entries of A unchanged), then $\det A$ gets multiplied by λ.*

(e) *If we add a row of A to another row of A (that is, we add each entry of the former row to the corresponding entry of the latter), then $\det A$ stays unchanged.*

(f) *Let $\lambda \in K$. If we add λ times a row of A to another row of A (that is, we add λ times each entry of the former row to the corresponding entry of the latter), then $\det A$ stays unchanged.*

(g) *Let $B, C \in K^{n \times n}$ be two further $n \times n$-matrices. Let $k \in [n]$. Assume that*

$$(\text{the } k\text{-th row of } C) = (\text{the } k\text{-th row of } A) + (\text{the } k\text{-th row of } B),$$

whereas each $i \neq k$ satisfies

$$(\text{the } i\text{-th row of } C) = (\text{the } i\text{-th row of } A) = (\text{the } i\text{-th row of } B).$$

Then,

$$\det C = \det A + \det B.$$

Proof. **(a)** See [Grinbe15, Exercise 6.7 **(a)**]. This is also a particular case of [Strick13, Corollary B.19].

(b) See [Grinbe15, Exercise 6.7 **(c)**]. This is also near-obvious from Definition 5.3.

(c) See [Grinbe15, Exercise 6.7 **(e)**] or [Laue15, §5.3.3, property (iii)].

(d) See [Grinbe15, Exercise 6.7 **(g)**] or [Laue15, §5.3.3, property (ii)].

(f) See [Grinbe15, Exercise 6.8 **(a)**].

(e) This is the particular case of part **(f)** for $\lambda = 1$.

(g) See [Grinbe15, Exercise 6.7 **(i)**] or [Laue15, §5.3.3, property (i)]. $\square$

Theorem 5.21 (Column operation properties). *Theorem 5.20 also holds if we replace "row" by "column" throughout it.*

Proof. Theorem 5.18 shows that the determinant of a matrix does not change when we replace it by its transpose; however, the rows of this transpose A^T are the transposes of the columns of A. Thus, Theorem 5.21 follows by applying Theorem 5.20 to the transposes of all the matrices involved. □

Corollary 5.22. *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ and $\tau \in S_n$. Then,*

$$\det\left(\left(A_{\tau(i),j}\right)_{1\le i\le n,\ 1\le j\le n}\right) = (-1)^{\tau} \cdot \det A \tag{5.1}$$

and

$$\det\left(\left(A_{i,\tau(j)}\right)_{1\le i\le n,\ 1\le j\le n}\right) = (-1)^{\tau} \cdot \det A. \tag{5.2}$$

Proof. Let us first prove (5.2).

The definition of $\det A$ yields

$$\det A = \sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} A_{i,\sigma(i)}.$$

The definition of $\det\left(\left(A_{i,\tau(j)}\right)_{1\le i\le n,\ 1\le j\le n}\right)$ yields

$$\det\left(\left(A_{i,\tau(j)}\right)_{1\le i\le n,\ 1\le j\le n}\right) = \sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} A_{i,\tau(\sigma(i))}.$$

However, S_n is a group. Thus, the map $\sigma \mapsto \tau^{-1}\sigma$ is a bijection on S_n. Substituting $\tau^{-1}\sigma$ for σ, we obtain (using Proposition 3.110 **(d)** and **(f)**)

$$\begin{aligned}
&\sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} A_{i,\tau(\sigma(i))} \\
&= \sum_{\sigma\in S_n} \underbrace{(-1)^{\tau^{-1}\sigma}}_{=(-1)^{\tau^{-1}}\cdot(-1)^{\sigma}} \prod_{i=1}^{n} \underbrace{A_{i,\tau((\tau^{-1}\sigma)(i))}}_{=A_{i,\sigma(i)}} \\
&= \underbrace{(-1)^{\tau^{-1}}}_{=(-1)^{\tau}} \cdot \underbrace{\sum_{\sigma\in S_n} (-1)^{\sigma} \prod_{i=1}^{n} A_{i,\sigma(i)}}_{=\det A} \\
&= (-1)^{\tau} \cdot \det A.
\end{aligned}$$

This proves (5.2).

Now, (5.1) follows by applying (5.2) to A^T instead of A and using Theorem 5.18. □

Theorem 5.23 (Multiplicativity of the determinant). *Let $n \in \mathbb{N}$. Let $A, B \in K^{n\times n}$ be two $n \times n$-matrices. Then,*

$$\det(AB) = \det A \cdot \det B.$$

Proof. See [Strick13, Theorem B.17] or [Grinbe15, Theorem 6.23] or [Zeilbe85, §5] or [Ford21, Lemma 4.7.5 (1)] or [Laue15, Theorem 5.7]. □

Corollary 5.24. *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ and $d_1, d_2, \dots, d_n \in K$. Then,*

$$\det\left((d_i A_{i,j})_{1\le i\le n,\ 1\le j\le n}\right) = d_1 d_2 \cdots d_n \cdot \det A \tag{5.3}$$

and

$$\det\left((d_j A_{i,j})_{1\le i\le n,\ 1\le j\le n}\right) = d_1 d_2 \cdots d_n \cdot \det A. \tag{5.4}$$

Proof. The definition of a determinant yields

$$\det A = \sum_{\sigma\in S_n} (-1)^\sigma A_{1,\sigma(1)} A_{2,\sigma(2)} \cdots A_{n,\sigma(n)}$$

and

$$\begin{aligned}
&\det\left((d_i A_{i,j})_{1\le i\le n,\ 1\le j\le n}\right)\\
&= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{(d_1 A_{1,\sigma(1)})(d_2 A_{2,\sigma(2)})\cdots(d_n A_{n,\sigma(n)})}_{=(d_1 d_2\cdots d_n)(A_{1,\sigma(1)}A_{2,\sigma(2)}\cdots A_{n,\sigma(n)})}\\
&= d_1 d_2\cdots d_n \cdot \underbrace{\sum_{\sigma\in S_n} (-1)^\sigma A_{1,\sigma(1)} A_{2,\sigma(2)} \cdots A_{n,\sigma(n)}}_{=\det A}\\
&= d_1 d_2 \cdots d_n \cdot \det A
\end{aligned}$$

and

$$\begin{aligned}
&\det\left((d_j A_{i,j})_{1\le i\le n,\ 1\le j\le n}\right)\\
&= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{(d_{\sigma(1)} A_{1,\sigma(1)})(d_{\sigma(2)} A_{2,\sigma(2)})\cdots(d_{\sigma(n)} A_{n,\sigma(n)})}_{=(d_{\sigma(1)} d_{\sigma(2)}\cdots d_{\sigma(n)})(A_{1,\sigma(1)}A_{2,\sigma(2)}\cdots A_{n,\sigma(n)})}\\
&= \sum_{\sigma\in S_n} (-1)^\sigma \underbrace{(d_{\sigma(1)} d_{\sigma(2)}\cdots d_{\sigma(n)})}_{\substack{=d_1 d_2\cdots d_n\\ \text{(since } \sigma \text{ is a permutation of } [n])}} (A_{1,\sigma(1)}A_{2,\sigma(2)}\cdots A_{n,\sigma(n)})\\
&= d_1 d_2 \cdots d_n \cdot \det A.
\end{aligned}$$

□

5.1.6 Permutation images and submatrix determinants

Definition 5.25. The *image* of a finite subset $P \subseteq [n]$ under a permutation $\sigma \in S_n$ is $\sigma(P) = \{\sigma(i) \mid i \in P\}$.

Lemma 5.26. *The image of a finset under a permutation has the same cardinality: $|\sigma(P)| = |P|$.*

Proof. Since σ is injective, the image has the same cardinality as the original set. □

Lemma 5.27. *The image of a complement equals the complement of the image: $\sigma(P^c) = \sigma(P)^c$.*

Proof. By the injectivity of σ: $x \in \sigma(P^c)$ iff $\sigma^{-1}(x) \notin P$ iff $x \notin \sigma(P)$. □

Lemma 5.28. *If $\sigma(P) = Q$, then $\sigma^{-1}(Q) = P$.*

Proof. Apply σ^{-1} to both sides of $\sigma(P) = Q$ and use $\sigma^{-1} \circ \sigma = \mathrm{id}$. $\square$

Definition 5.29. The set of permutations mapping P to Q: $\mathrm{permsMapping}(P, Q) = \{\sigma \in S_n \mid \sigma(P) = Q\}$.

Lemma 5.30. *If $|P| \neq |Q|$, then no permutation maps P to Q:* $\mathrm{permsMapping}(P, Q) = \emptyset$.

Proof. If $\sigma(P) = Q$, then $|Q| = |\sigma(P)| = |P|$ by Lemma 5.26, contradicting $|P| \neq |Q|$. $\square$

Definition 5.31. The *submatrix determinant* of an $n \times n$ matrix A with row set P and column set Q (both subsets of $[n]$) is

$$\mathrm{submatrixDet}(A, P, Q) = \begin{cases} \det(A[P, Q]) & \text{if } |P| = |Q|, \\ 0 & \text{otherwise,} \end{cases}$$

where $A[P, Q]$ is the submatrix of A with rows indexed by P and columns indexed by Q, in the order determined by their natural ordering.

Lemma 5.32. *When $|P| = |Q|$, the submatrix determinant equals the actual determinant of the submatrix $A[P, Q]$.*

Proof. Immediate from the definitions, using the hypothesis $|P| = |Q|$. $\square$

Lemma 5.33. *When $|P| \neq |Q|$, the submatrix determinant is* 0.

Proof. Immediate from the definitions, using the hypothesis $|P| \neq |Q|$. $\square$

Lemma 5.34. *The submatrix determinant of the empty row and column sets is* 1*:* $\mathrm{submatrixDet}(A, \emptyset, \emptyset) = 1$.

Proof. Both sets are empty (so $|P| = |Q| = 0$), and the determinant of the 0×0 submatrix is 1. $\square$

5.2 Cauchy–Binet and related formulas

We work over a commutative ring K. We formalize the Cauchy–Binet formula for determinants of products of non-square matrices, the expansion of $\det(A+B)$ as a sum over pairs of subsets, and several applications: formulas for $\det(A + D)$ when D is diagonal, the characteristic polynomial in terms of principal minors, and the determinant of the Pascal matrix via LU decomposition.

Throughout, $[n] = \{1, 2, \ldots, n\}$ denotes the set of the first n positive integers (in the Lean formalization, indexing is 0-based). For any finite set S of integers we write $\mathrm{sum}\, S := \sum_{s \in S} s$.

5.2.1 Submatrices and minors

Definition 5.35. Let $n, m \in \mathbb{N}$. Let A be an $n \times m$-matrix.

Let U be a subset of $[n]$. Let V be a subset of $[m]$.

Writing the two sets U and V as

$$U = \{u_1, u_2, \ldots, u_p\} \quad \text{and} \quad V = \{v_1, v_2, \ldots, v_q\}$$

with

$$u_1 < u_2 < \cdots < u_p \quad \text{and} \quad v_1 < v_2 < \cdots < v_q,$$

we set

$$\operatorname{sub}_U^V A := \left(A_{u_i,v_j}\right)_{1\le i\le p,\ 1\le j\le q}.$$

Roughly speaking, $\operatorname{sub}_U^V A$ is the matrix obtained from A by focusing only on the i-th rows for $i \in U$ (that is, removing all the other rows) and only on the j-th columns for $j \in V$ (that is, removing all the other columns).

This matrix $\operatorname{sub}_U^V A$ is called the *submatrix of A obtained by restricting to the U-rows and the V-columns.* If this matrix is square (i.e., if $|U| = |V|$), then its determinant $\det(\operatorname{sub}_U^V A)$ is called a *minor* of A.

Lemma 5.36. *The minor of the empty sets is* 1*:* $\det(\operatorname{sub}_\varnothing^\varnothing A) = 1$*.*

Proof. The 0×0 matrix has determinant 1. This is `Mathlib.Matrix.det_isEmpty` from Mathlib. ☐

5.2.2 The Cauchy–Binet formula

Helpers for the Cauchy–Binet formula

Lemma 5.37 (Non-injective cancellation)**.** *Let $A \in K^{n\times m}$, $B \in K^{m\times n}$, and let $f\colon [n] \to [m]$ be a non-injective function. Then*

$$\sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} A_{\sigma(i),f(i)}\, B_{f(i),i} = 0.$$

Proof. Since f is not injective, there exist $i \neq j$ with $f(i) = f(j)$. Pairing each σ with $\sigma\cdot\operatorname{swap}(i,j)$ gives a sign-reversing involution, so the sum cancels to 0. ☐

Lemma 5.38 (Subset factorization)**.** *For a fixed n-element subset $S \subseteq [m]$,*

$$\sum_{\tau\in S_n}\sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} A_{\sigma(i),S_{\tau(i)}}\, B_{S_{\tau(i)},i} = \det(\operatorname{cols}_S A)\cdot\det(\operatorname{rows}_S B),$$

where S_k denotes the k-th element of S in sorted order.

Proof. Split the product $\prod_i A_{\sigma(i),e(\tau(i))}\, B_{e(\tau(i)),i}$ into a product of A-factors and a product of B-factors. The B-factors depend only on τ and can be pulled out of the σ-sum. The remaining σ-sum is a Leibniz expansion for $\det(\operatorname{cols}_S A)$, and summing the τ-terms gives $\det(\operatorname{rows}_S B)$. ☐

Theorem 5.39 (Cauchy–Binet formula)**.** *Let $n, m \in \mathbb{N}$. Let $A \in K^{n\times m}$ be an $n \times m$-matrix, and let $B \in K^{m\times n}$ be an $m \times n$-matrix. Then,*

$$\det(AB) = \sum_{\substack{(g_1,g_2,\ldots,g_n)\in[m]^n;\\ g_1<g_2<\cdots<g_n}} \det(\operatorname{cols}_{g_1,g_2,\ldots,g_n} A)\cdot\det(\operatorname{rows}_{g_1,g_2,\ldots,g_n} B).$$

Here, $\operatorname{cols}_{g_1,\ldots,g_n} A$ *is the $n \times n$-matrix obtained from A by keeping only columns $g_1, g_2, \ldots, g_n$, and* $\operatorname{rows}_{g_1,\ldots,g_n} B$ *is the $n \times n$-matrix obtained from B by keeping only rows $g_1, g_2, \ldots, g_n$.*

Equivalently, the sum runs over all n-element subsets S of $[m]$:

$$\det(AB) = \sum_{\substack{S\subseteq[m]\\ |S|=n}} \det(\operatorname{cols}_S A)\cdot\det(\operatorname{rows}_S B).$$

Proof. The proof expands $\det(AB)$ via the Leibniz formula:

$$\det(AB) = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} (AB)_{\sigma(i),i} = \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n}\sum_{k=1}^{m} A_{\sigma(i),k}B_{k,i}.$$

Expanding the product over i as a sum over functions $f\colon [n]\to[m]$ and swapping sums gives

$$\det(AB) = \sum_{f\colon [n]\to[m]} \sum_{\sigma\in S_n} (-1)^\sigma \prod_{i=1}^{n} A_{\sigma(i),f(i)}B_{f(i),i}.$$

When f is not injective, the inner sum vanishes by Lemma 5.37.

Therefore only injective f contribute. Each injective f has image $S = \operatorname{im}(f)$ with $|S| = n$. The injective functions with a given image S are in bijection with permutations of $[n]$. Partitioning by S and reindexing via this bijection, the sum transforms into a sum over n-element subsets S, and for each S the inner sum factors as $\det(\mathrm{cols}_S\, A)\cdot\det(\mathrm{rows}_S\, B)$ by Lemma 5.38. □

Theorem 5.40 (Cauchy–Binet for square matrices)**.** *When $m = n$, the Cauchy–Binet formula specializes to* $\det(AB) = \det A\cdot\det B$.

Proof. This is Mathlib's `Matrix.det_mul`. □

5.2.3 The formula for $\det(A + B)$

Basic definitions

Definition 5.41 (Subset sum and permutation sets)**.** For a subset $P\subseteq[n]$, define $\mathrm{sum}(P) = \sum_{i\in P} i$.

For subsets $P, Q\subseteq[n]$, define $\mathrm{Perm}(P,Q) = \{\sigma\in S_n : \sigma(P) = Q\}$ where $\sigma(P) = \{\sigma(i) \mid i\in P\}$ denotes the image of P under σ.

Lemma 5.42. *If $|P|\neq|Q|$, then* $\mathrm{Perm}(P,Q) = \varnothing$.

Proof. A permutation is a bijection, so $|\sigma(P)| = |P|$. If $|P|\neq|Q|$ then $\sigma(P)\neq Q$ for all σ. □

Expansion steps

Lemma 5.43. *For a permutation $\sigma\in S_n$ and a subset $P\subseteq[n]$, $\sigma(\overline{P}) = \overline{\sigma(P)}$.*

Proof. Since σ is a bijection on $[n]$, the complement is preserved. □

Permutation decomposition infrastructure

The proof of Theorem 5.61 requires decomposing permutations $\sigma\in S_n$ with $\sigma(P) = Q$ into independent parts acting on P and on $\overline{P}$.

Definition 5.44 (Extraction of component permutations)**.** Let $P, Q\subseteq[n]$ with $|P| = |Q| = k$ and let $\sigma\in S_n$ satisfy $\sigma(P) = Q$. Write the elements of P in increasing order as $p_1 < \cdots < p_k$, of Q as $q_1 < \cdots < q_k$, of $\overline{P}$ as $p'_1 < \cdots < p'_\ell$, and of $\overline{Q}$ as $q'_1 < \cdots < q'_\ell$.

Define $\alpha_\sigma\in S_k$ by $\sigma(p_i) = q_{\alpha_\sigma(i)}$ and $\beta_\sigma\in S_\ell$ by $\sigma(p'_i) = q'_{\beta_\sigma(i)}$.

Definition 5.45 (Construction of permutation from components)**.** Given $P, Q \subseteq [n]$ with $|P| = |Q|$ and permutations $\alpha \in S_k$, $\beta \in S_\ell$ (with $k = |P|$, $\ell = |\overline{P}|$), define $\sigma \in S_n$ by:

$$\sigma(p_i) = q_{\alpha(i)} \quad \text{and} \quad \sigma(p'_i) = q'_{\beta(i)}.$$

This is the inverse of the map $\sigma \mapsto (\alpha_\sigma, \beta_\sigma)$.

Lemma 5.46 (Round-trip properties)**.** *The maps $\sigma \mapsto (\alpha_\sigma, \beta_\sigma)$ and $(\alpha, \beta) \mapsto \sigma_{(\alpha,\beta)}$ are mutual inverses:*

1. *Extracting α from the permutation constructed from (α, β) returns α.*
2. *Extracting β from the permutation constructed from (α, β) returns β.*
3. *Constructing a permutation from the components $(\alpha_\sigma, \beta_\sigma)$ extracted from σ returns σ.*

Proof. By unfolding the definitions and showing that the sorted-order embeddings and their inverses compose to the identity. □

Lemma 5.47 (Extract specification)**.** *For σ with $\sigma(P) = Q$:*

1. *$\sigma(p_j) = q_{\alpha_\sigma(j)}$ for all j,*
2. *$\sigma(p'_j) = q'_{\beta_\sigma(j)}$ for all j.*

Proof. By definition of α_σ and β_σ, using the order-embedding properties of sorted finsets. □

Lemma 5.48 (Product factorization via extract)**.** *For σ with $\sigma(P) = Q$:*

1. $\displaystyle\prod_{i \in P} A_{\sigma(i),i} = \prod_{j=1}^{k} A_{q_{\alpha(j)},\, p_j}$,
2. $\displaystyle\prod_{i \in \overline{P}} B_{\sigma(i),i} = \prod_{j=1}^{\ell} B_{q'_{\beta(j)},\, p'_j}$.

Proof. Reindex the product over sorted elements of P (resp. $\overline{P}$) using Lemma 5.47. □

Sign decomposition infrastructure

The sign decomposition (Lemma 5.59) is proved by reducing any (P, Q) to the base case $P = Q = [k]$ via "left shifts" and "left co-shifts."

Lemma 5.49 (Shift existence)**.** *Let $0 < k < n$ and let $P \subseteq [n]$ with $|P| = k$ and $P \neq [k]$. Then there exists i such that $i \notin P$ and $i + 1 \in P$ (with $i + 1 < n$).*

Proof. If $P = [k]$ (the first k elements), then P is already the prefix. Otherwise, some element of P is "too large," meaning there exists j with $p_j > j$ (the j-th smallest element is above position j). A downward-closure argument produces the required adjacent pair. □

Lemma 5.50 (Sum decrement under shift)**.** *If $i \notin P$ and $i + 1 \in P$, let $P' = s_i(P)$ where $s_i = \mathrm{swap}(i, i+1)$. Then $\mathrm{sum}(P') = \mathrm{sum}(P) - 1$.*

Proof. P' is obtained from P by replacing $i+1$ with i, so the sum decreases by $(i+1)-i = 1$. □

Lemma 5.51 (Combined sign preserved under left shift)**.** *If $i \notin P$ and $i+1 \in P$, let $P' = s_i(P)$ and $\sigma' = \sigma \cdot s_i$. Then*

$$(-1)^{\operatorname{sum} P' + \operatorname{sum} Q} \cdot (-1)^{\sigma'} = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \cdot (-1)^{\sigma}.$$

Proof. $(-1)^{\sigma'} = (-1)^{\sigma} \cdot (-1)^{s_i} = -(-1)^{\sigma}$ (since adjacent transpositions are odd), and $(-1)^{\operatorname{sum} P'} = -(-1)^{\operatorname{sum} P}$ (since $\operatorname{sum} P' = \operatorname{sum} P - 1$ by Lemma 5.50). The two sign flips cancel. □

Lemma 5.52 (Extract preserved under left shift)**.** *Under the left shift $P' = s_i(P)$, $\sigma' = \sigma \cdot s_i$:*

1. $(-1)^{\alpha_{\sigma'}} = (-1)^{\alpha_\sigma}$,
2. $(-1)^{\beta_{\sigma'}} = (-1)^{\beta_\sigma}$.

Proof. The sorted order of P' is s_i applied to the sorted order of P (since replacing $i+1$ by i preserves relative order among all other elements). Therefore $\sigma'(P'_j) = \sigma(s_i(s_i(P_j))) = \sigma(P_j)$, so $\alpha_{\sigma'} = \alpha_\sigma$ (and similarly for β). □

Lemma 5.53 (Left shift invariance)**.** *If the sign decomposition formula holds for (P', Q, σ') (where $P' = s_i(P)$, $\sigma' = \sigma \cdot s_i$), then it holds for (P, Q, σ).*

Proof. Combine Lemma 5.51 (combined sign preserved) with Lemma 5.52 (α and β signs preserved). □

Lemma 5.54 (Left co-shift invariance)**.** *If the sign decomposition formula holds for (P, Q', σ') (where $Q' = s_i(Q)$, $\sigma' = s_i \cdot \sigma$ for $i \notin Q$, $i+1 \in Q$), then it holds for (P, Q, σ).*

Proof. Analogous to Lemma 5.53: the combined sign is preserved under left co-shifts, and the extracted component permutations α and β are unchanged. □

Lemma 5.55 (Sign decomposition base case)**.** *When $P = Q = [k]$ (the prefix finset), the sign decomposition formula*

$$(-1)^{\sigma} = (-1)^{\operatorname{sum}[k] + \operatorname{sum}[k]} \cdot (-1)^{\alpha_\sigma} \cdot (-1)^{\beta_\sigma} = (-1)^{\alpha_\sigma} \cdot (-1)^{\beta_\sigma}$$

holds, where the sign factor vanishes because $\operatorname{sum}[k] + \operatorname{sum}[k] = k(k-1)$ *is even.*

Proof. Since σ preserves $[k]$, it decomposes as $\sigma = \alpha \oplus \beta$ (a direct sum of the restrictions to $[k]$ and its complement). The sign of a direct sum satisfies $(-1)^{\sigma} = (-1)^{\alpha} \cdot (-1)^{\beta}$ (this is `Equiv.Perm.sign_subtypeCongr` from Mathlib). The factor $(-1)^{k(k-1)} = 1$ since $k(k-1)$ is even. □

Lemma 5.56. *For $n \times n$-matrices A and B,*

$$\det(A+B) = \sum_{\sigma \in S_n} (-1)^{\sigma} \sum_{P \subseteq [n]} \Big(\prod_{i \in P} A_{\sigma(i),i} \Big) \Big(\prod_{i \in \overline{P}} B_{\sigma(i),i} \Big).$$

Proof. Expand $\det(A+B) = \sum_\sigma (-1)^\sigma \prod_i (A_{\sigma(i),i} + B_{\sigma(i),i})$ and distribute each factor using the product rule for sums (this is `Fintype.prod_add` from Mathlib). □

Lemma 5.57. *Swapping summation order:*

$$\det(A+B) = \sum_{P \subseteq [n]} \sum_{\sigma \in S_n} (-1)^{\sigma} \Big(\prod_{i \in P} A_{\sigma(i),i} \Big) \Big(\prod_{i \in \overline{P}} B_{\sigma(i),i} \Big).$$

Proof. By swapping the order of summation (this is `Finset.sum_comm` from Mathlib). ☐

Lemma 5.58. *Partitioning by $Q = \sigma(P)$:*

$$\det(A+B) = \sum_{P\subseteq[n]} \sum_{Q\subseteq[n]} \sum_{\substack{\sigma\in S_n;\\ \sigma(P)=Q}} (-1)^\sigma \Big(\prod_{i\in P} A_{\sigma(i),i}\Big)\Big(\prod_{i\in \overline{P}} B_{\sigma(i),i}\Big).$$

Proof. Split $\sum_{\sigma\in S_n}$ according to the value of $Q = \sigma(P)$, using pairwise disjointness of the fibers. ☐

Lemma 5.59 (Sign decomposition). *Let $P, Q \subseteq [n]$ with $|P| = |Q|$ and let $\sigma \in S_n$ satisfy $\sigma(P) = Q$. Write the elements of P in increasing order as $p_1 < p_2 < \cdots < p_k$, the elements of Q as $q_1 < q_2 < \cdots < q_k$, the elements of $\overline{P}$ as $p'_1 < \cdots < p'_\ell$, and those of $\overline{Q}$ as $q'_1 < \cdots < q'_\ell$.*

Define $\alpha_\sigma \in S_k$ by $\sigma(p_i) = q_{\alpha_\sigma(i)}$ and $\beta_\sigma \in S_\ell$ by $\sigma(p'_i) = q'_{\beta_\sigma(i)}$.

Then

$$(-1)^\sigma = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \cdot (-1)^{\alpha_\sigma} \cdot (-1)^{\beta_\sigma}.$$

Proof. The proof uses strong induction on $\operatorname{sum} P + \operatorname{sum} Q$.

Base case. When $P = Q = [k]$, the formula holds by Lemma 5.55.

Inductive case. If $P \neq [k]$, Lemma 5.49 provides an index i with $i \notin P$ and $i+1 \in P$. The left shift $P' = s_i(P)$, $\sigma' = \sigma \cdot s_i$ reduces $\operatorname{sum} P$ by 1, so the induction hypothesis applies to (P', Q, σ'). Lemma 5.53 lifts the result back to (P, Q, σ).

Similarly, if $P = [k]$ but $Q \neq [k]$, Lemma 5.49 applied to Q gives an index for a left co-shift, and Lemma 5.54 lifts the result. ☐

Lemma 5.60 (Key factorization). *For fixed subsets $P, Q \subseteq [n]$ with $|P| = |Q|$:*

$$\sum_{\substack{\sigma\in S_n;\\ \sigma(P)=Q}} (-1)^\sigma \Big(\prod_{i\in P} A_{\sigma(i),i}\Big)\Big(\prod_{i\in \overline{P}} B_{\sigma(i),i}\Big) = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \det(\operatorname{sub}_P^Q A) \cdot \det(\operatorname{sub}_{\overline{P}}^{\overline{Q}} B).$$

Proof. The proof uses the bijection $\sigma \mapsto (\alpha_\sigma, \beta_\sigma)$ (Definition 5.44) from $\{\sigma \in S_n : \sigma(P) = Q\}$ to $S_k \times S_\ell$, with inverse given by the construction in Definition 5.45. The round-trip properties (Lemma 5.46) establish that this is a bijection.

After reindexing by (α, β):

1. The sign identity (Lemma 5.59) gives $(-1)^\sigma = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \cdot (-1)^\alpha \cdot (-1)^\beta$.
2. The products over P and $\overline{P}$ factor (Lemma 5.48).
3. The double sum $\sum_\alpha \sum_\beta$ factors into a product of two Leibniz expansions, giving $\det(\operatorname{sub}_P^Q A) \cdot \det(\operatorname{sub}_{\overline{P}}^{\overline{Q}} B)$.

☐

Theorem 5.61. *Let $n \in \mathbb{N}$. For any subset I of $[n]$, let $\widetilde{I} = [n] \setminus I$ denote its complement.*

Let A and B be two $n \times n$-matrices in $K^{n\times n}$. Then,

$$\det(A+B) = \sum_{P\subseteq[n]} \sum_{\substack{Q\subseteq[n];\\ |P|=|Q|}} (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \det(\operatorname{sub}_P^Q A) \cdot \det(\operatorname{sub}_{\widetilde{P}}^{\widetilde{Q}} B).$$

Proof. The expansion from Lemma 5.58 gives a triple sum $\sum_P \sum_Q \sum_{\sigma:\sigma(P)=Q}$. Applying Lemma 5.60 to each inner sum collapses the σ-sum. When $|P| \neq |Q|$, the set of permutations mapping P to Q is empty (Lemma 5.42), so the contribution is 0. ☐

5.2.4 Minors of diagonal matrices

Lemma 5.62. *Let $n \in \mathbb{N}$. Let $d_1, d_2, \ldots, d_n \in K$. Let*

$$D := \operatorname{diag}(d_1, d_2, \ldots, d_n) = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix} \in K^{n \times n}.$$

(a) *We have $\det(\operatorname{sub}_P^P D) = \prod_{i \in P} d_i$ for any subset P of $[n]$.*

(b) *Let P and Q be two distinct subsets of $[n]$ satisfying $|P| = |Q|$. Then, $\det(\operatorname{sub}_P^Q D) = 0$.*

Proof. **(a)** The submatrix $\operatorname{sub}_P^P D$ is itself diagonal with entries d_i for $i \in P$. Thus its determinant is $\prod_{i \in P} d_i$.

The submatrix of a diagonal matrix via an order-preserving embedding is again diagonal; then the determinant of a diagonal matrix is the product of its diagonal entries (this is `Matrix.det_diagonal` from Mathlib).

(b) Since $P \neq Q$ but $|P| = |Q|$, there exists $i \in P \setminus Q$. The row of $\operatorname{sub}_P^Q D$ corresponding to i is all zeros (since $D_{i,j} = 0$ for $j \neq i$ and $i \notin Q$), so the determinant is 0 (by `Matrix.det_eq_zero_of_row_eq_zero` from Mathlib). □

5.2.5 Determinant of $A + D$

Theorem 5.63. *Let $n \in \mathbb{N}$. Let A and D be two $n \times n$-matrices in $K^{n \times n}$ such that the matrix D is diagonal. Let $d_1, d_2, \ldots, d_n$ be the diagonal entries of the diagonal matrix D. Then,*

$$\det(A + D) = \sum_{P \subseteq [n]} \det(\operatorname{sub}_P^P A) \cdot \prod_{i \in [n] \setminus P} d_i.$$

The minors $\det(\operatorname{sub}_P^P A)$ *are called the* principal minors *of A.*

Proof. Theorem 5.61 (applied to $B = D$) yields a double sum. By Lemma 5.62**(b)**, the terms with $P \neq Q$ vanish (since $\det(\operatorname{sub}_{\widetilde{P}}^{\widetilde{Q}} D) = 0$). In the surviving terms ($P = Q$), the sign factor $(-1)^{\operatorname{sum} P + \operatorname{sum} P} = 1$, and Lemma 5.62**(a)** gives $\det(\operatorname{sub}_{\widetilde{P}}^{\widetilde{P}} D) = \prod_{i \in \widetilde{P}} d_i$. □

5.2.6 Determinant of $x + D$

Lemma 5.64. *The matrix F with entries $F_{i,j} = x + d_i\,[i{=}j]$ decomposes as $F = A + D$ where A is the $n \times n$ constant matrix with all entries equal to x and $D = \operatorname{diag}(d_1, \ldots, d_n)$.*

Proof. Entry-wise verification. □

Lemma 5.65. *For a singleton $P = \{p\}$, the 1×1 submatrix of the constant-x matrix has determinant x.*

Proof. The 1×1 matrix is (x), which has determinant x (this is `Matrix.det_unique` from Mathlib). □

Lemma 5.66. *Let A be the $n \times n$ constant matrix with all entries equal to x. If $P \subseteq [n]$ has $|P| \geq 2$, then $\det(\operatorname{sub}_P^P A) = 0$.*

Proof. When $|P| \geq 2$, the submatrix $\operatorname{sub}_P^P A$ is a constant matrix of size ≥ 2. All its rows are identical, so the determinant is 0. $\square$

Proposition 5.67. *Let $n \in \mathbb{N}$. Let $d_1, d_2, \ldots, d_n \in K$ and $x \in K$. Let F be the $n \times n$-matrix*

$$\begin{pmatrix} x+d_1 & x & \cdots & x \\ x & x+d_2 & \cdots & x \\ \vdots & \vdots & \ddots & \vdots \\ x & x & \cdots & x+d_n \end{pmatrix} \in K^{n\times n}.$$

Then,

$$\det F = d_1 d_2 \cdots d_n + x \sum_{i=1}^{n} d_1 d_2 \cdots \widehat{d_i} \cdots d_n,$$

where the hat over "d_i" means "omit the d_i factor."

Proof. Write $F = A + D$ (Lemma 5.64) where A is the $n \times n$ constant matrix with all entries x and $D = \operatorname{diag}(d_1, \ldots, d_n)$. By Theorem 5.63,

$$\det(A + D) = \sum_{P \subseteq [n]} \det(\operatorname{sub}_P^P A) \cdot \prod_{i \in [n] \setminus P} d_i.$$

For $|P| \geq 2$, $\det(\operatorname{sub}_P^P A) = 0$ (Lemma 5.66), so only $P = \varnothing$ and singleton $P = \{p\}$ contribute:

$$\begin{aligned} \det F &= \underbrace{\det(\operatorname{sub}_\varnothing^\varnothing A)}_{=1} \cdot \underbrace{\prod_{i \in [n]} d_i}_{=d_1 d_2 \cdots d_n} + \sum_{p=1}^{n} \underbrace{\det(\operatorname{sub}_{\{p\}}^{\{p\}} A)}_{=x} \cdot \prod_{i \in [n] \setminus \{p\}} d_i \\ &= d_1 d_2 \cdots d_n + x \sum_{i=1}^{n} d_1 d_2 \cdots \widehat{d_i} \cdots d_n. \end{aligned}$$

$\square$

5.2.7 Characteristic polynomial coefficients

Proposition 5.68. *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix. Let $x \in K$. Let I_n denote the $n \times n$ identity matrix. Then,*

$$\det(A + xI_n) = \sum_{P \subseteq [n]} \det(\operatorname{sub}_P^P A) \cdot x^{n-|P|} = \sum_{k=0}^{n} \Bigg(\sum_{\substack{P \subseteq [n]; \\ |P| = n-k}} \det(\operatorname{sub}_P^P A) \Bigg) x^k.$$

Proof. The matrix xI_n is diagonal with all entries equal to x. Theorem 5.63 (applied to $D = xI_n$ and $d_i = x$) yields

$$\det(A + xI_n) = \sum_{P \subseteq [n]} \det(\operatorname{sub}_P^P A) \cdot \underbrace{\prod_{i \in [n] \setminus P} x}_{=x^{n-|P|}}.$$

This is the first equality. The second follows by grouping the sum according to the value of $|P|$ and substituting $n - k$ for k.

In the Lean formalization, xI_n is written as $x \cdot I_n$ and the diagonal form is obtained from the identity $x \cdot I_n = \operatorname{diag}(x, x, \ldots, x)$. The second form regroups by partitioning $\mathcal{P}([n])$ into fibers by cardinality. $\square$

5.2.8 The Pascal matrix

Definition 5.69 (Pascal matrix and its LU factors)**.** Define the $n \times n$ Pascal matrix, the lower triangular factor L, and the upper triangular factor U by:

$$A_{i,j} = \binom{i+j}{i}, \qquad L_{i,k} = \binom{i}{k}, \qquad U_{k,j} = \binom{j}{k},$$

for $0 \le i, j, k \le n-1$.

Lemma 5.70 (Triangularity and unit diagonal)**.** *L is lower triangular ($L_{i,k} = 0$ for $i < k$) with $L_{i,i} = 1$. U is upper triangular ($U_{k,j} = 0$ for $k > j$) with $U_{k,k} = 1$.*

Proof. $\binom{i}{k} = 0$ when $i < k$, and $\binom{i}{i} = 1$. Similarly for U. □

Lemma 5.71. *Then $A = L \cdot U$ (over $\mathbb{Z}$).*

Proof. For all i, j, we have (by Vandermonde's identity)

$$A_{i,j} = \binom{i+j}{i} = \sum_{k=0}^{n-1} \binom{i}{k}\binom{j}{k} = \sum_{k=0}^{n-1} L_{i,k} U_{k,j} = (LU)_{i,j}.$$

The key combinatorial identity $\sum_k \binom{m}{k}\binom{n}{k} = \binom{m+n}{n}$ (Vandermonde's identity) is used to establish the LU decomposition. □

Lemma 5.72. *The lower triangular factor L has* $\det L = 1$.

Proof. L is lower triangular ($\binom{i}{k} = 0$ when $i < k$) and its diagonal entries are $L_{i,i} = \binom{i}{i} = 1$ (Lemma 5.70). The determinant of a triangular matrix is the product of its diagonal entries, which is 1. □

Lemma 5.73. *The upper triangular factor U has* $\det U = 1$.

Proof. U is upper triangular ($\binom{j}{k} = 0$ when $k > j$) and its diagonal entries are $U_{k,k} = \binom{k}{k} = 1$ (Lemma 5.70). The determinant of a triangular matrix is the product of its diagonal entries, which is 1. □

Proposition 5.74. *Let $n \in \mathbb{N}$. Let A be the $n \times n$-matrix*

$$\left(\binom{i+j-2}{i-1}\right)_{1 \le i \le n,\ 1 \le j \le n} = \begin{pmatrix} \binom{0}{0} & \binom{1}{0} & \cdots & \binom{n-1}{0} \\ \binom{1}{1} & \binom{2}{1} & \cdots & \binom{n}{1} \\ \vdots & \vdots & \ddots & \vdots \\ \binom{n-1}{n-1} & \binom{n}{n-1} & \cdots & \binom{2n-2}{n-1} \end{pmatrix}.$$

Then $\det A = 1$.

Proof. By Lemma 5.71, $A = LU$. Therefore

$$\det A = \det(LU) = \det L \cdot \det U = 1 \cdot 1 = 1,$$

using the multiplicativity of the determinant (Theorem 5.40) and Lemmas 5.72 and 5.73. □

5.3 Determinants: Factor Hunting, Laplace Expansion, and Desnanot–Jacobi

5.3.1 Factor hunting and the Vandermonde determinant

Theorem 5.75 (Vandermonde determinant). *Let $n \in \mathbb{N}$. Let $a_1, a_2, \ldots, a_n$ be n elements of K. Then:*

(a) *We have*

$$\det\left(\left(a_i^{n-j}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq i<j\leq n}(a_i - a_j).$$

(b) *We have*

$$\det\left(\left(a_j^{n-i}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq i<j\leq n}(a_i - a_j).$$

(c) *We have*

$$\det\left(\left(a_i^{j-1}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq j<i\leq n}(a_i - a_j).$$

(d) *We have*

$$\det\left(\left(a_j^{i-1}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq j<i\leq n}(a_i - a_j).$$

Proof. **(a)** Follows from Lemma 5.76 by substituting $a_1, a_2, \ldots, a_n$ for the indeterminates $x_1, x_2, \ldots, x_n$. More precisely, the equality

$$\det\left(\left(a_i^{n-j}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq i<j\leq n}(a_i - a_j)$$

follows from the polynomial identity (5.5) by applying the evaluation homomorphism $x_i \mapsto a_i$.

(b) The matrix $\left(a_j^{n-i}\right)_{1\leq i\leq n,\ 1\leq j\leq n}$ is the transpose of the matrix $\left(a_i^{n-j}\right)_{1\leq i\leq n,\ 1\leq j\leq n}$. Thus part (b) follows from part (a) using $\det(A^T) = \det A$.

(c) The matrix $\left(a_i^{j-1}\right)$ is obtained from $\left(a_i^{n-j}\right)$ by reversing the column order. Let $\tau \in S_n$ send each j to $n+1-j$. Then

$$\det\left(\left(a_i^{j-1}\right)\right) = (-1)^\tau \cdot \det\left(\left(a_i^{n-j}\right)\right) = (-1)^\tau \cdot \prod_{1\leq i<j\leq n}(a_i - a_j).$$

Since each factor $(a_i - a_j)$ gets negated by $(-1)^\tau$, the result becomes $\prod_{1\leq j<i\leq n}(a_i - a_j)$.

(d) Follows from part (c) using $\det(A^T) = \det A$. $\square$

Lemma 5.76. *Let $n \in \mathbb{N}$. Consider the polynomial ring $\mathbb{Z}[x_1, x_2, \ldots, x_n]$ in n indeterminates $x_1, x_2, \ldots, x_n$ with integer coefficients. In this ring, we have*

$$\det\left(\left(x_i^{n-j}\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq i<j\leq n}(x_i - x_j). \tag{5.5}$$

Proof. Set

$$f := \det\left(\left(x_i^{n-j}\right)_{1\le i\le n,\ 1\le j\le n}\right) \quad \text{and} \quad g := \prod_{1\le i<j\le n}(x_i - x_j).$$

We must prove that $f = g$.

By the definition of the determinant,

$$f = \sum_{\sigma\in S_n}(-1)^\sigma x_1^{n-\sigma(1)}x_2^{n-\sigma(2)}\cdots x_n^{n-\sigma(n)},$$

which is a homogeneous polynomial of degree $\frac{n(n-1)}{2}$. The monomial $x_1^{n-1}x_2^{n-2}\cdots x_n^0$ appears with coefficient 1 (from the identity permutation $\sigma = \mathrm{id}$).

For any $u < v$ in $[n]$, setting $x_u = x_v$ makes two rows of the matrix identical, so f vanishes. By the root factoring-off theorem (viewing f as a polynomial in x_u over $\mathbb{Z}[x_1,\ldots,\widehat{x_u},\ldots,x_n]$), we conclude that $x_u - x_v$ divides f. Since the ring $\mathbb{Z}[x_1,\ldots,x_n]$ is a UFD and the linear factors $x_u - x_v$ are irreducible and mutually non-associate, the product $g = \prod_{1\le i<j\le n}(x_i - x_j)$ divides f. Thus $f = gq$ for some $q \in \mathbb{Z}[x_1,\ldots,x_n]$.

Since $\deg f \le \frac{n(n-1)}{2} = \deg g$, we get $\deg q \le 0$, so $q \in \mathbb{Z}$. Comparing the coefficient of $x_1^{n-1}x_2^{n-2}\cdots x_n^0$ on both sides gives $1 = q\cdot 1$, hence $q = 1$ and $f = g$.

The formalization relates our matrix convention to Mathlib's via column reversal, and uses the fact that swapping the order of subtraction in each factor introduces a sign that cancels with the sign from the column permutation. $\square$

Proposition 5.77. *Let $n \in \mathbb{N}$. Let $x_1, x_2, \ldots, x_n \in K$ and $y_1, y_2, \ldots, y_n \in K$. Then,*

$$\det\left(\left((x_i + y_j)^{n-1}\right)_{1\le i\le n,\ 1\le j\le n}\right) = \left(\prod_{k=0}^{n-1}\binom{n-1}{k}\right)\left(\prod_{1\le i<j\le n}(x_i - x_j)\right)\left(\prod_{1\le i<j\le n}(y_j - y_i)\right).$$

Proof. Let $C = \left((x_i + y_j)^{n-1}\right)_{1\le i\le n,\ 1\le j\le n}$. By the binomial formula, for any $i, j \in [n]$,

$$C_{i,j} = (x_i + y_j)^{n-1} = \sum_{k=1}^{n}\binom{n-1}{k-1}x_i^{k-1}y_j^{n-k}.$$

Define two $n\times n$-matrices

$$P := \left(\binom{n-1}{k-1}x_i^{k-1}\right)_{1\le i\le n,\ 1\le k\le n} \quad \text{and} \quad Q := \left(y_j^{n-k}\right)_{1\le k\le n,\ 1\le j\le n}.$$

Then $C = PQ$, so $\det C = \det P \cdot \det Q$.

From $Q = \left(y_j^{n-i}\right)$, Theorem 5.75 (b) gives $\det Q = \prod_{1\le i<j\le n}(y_i - y_j)$.

From $P = \left(\binom{n-1}{j-1}x_i^{j-1}\right)$, factoring out the binomial coefficients from each column and applying Theorem 5.75 (c) gives

$$\det P = \left(\prod_{k=0}^{n-1}\binom{n-1}{k}\right)\cdot\prod_{1\le j<i\le n}(x_i - x_j) = \left(\prod_{k=0}^{n-1}\binom{n-1}{k}\right)\cdot\prod_{1\le i<j\le n}(x_j - x_i).$$

Hence,

$$\begin{aligned}\det C &= \det P \cdot \det Q \\ &= \left(\prod_{k=0}^{n-1} \binom{n-1}{k}\right) \cdot \prod_{1\le i<j\le n} \underbrace{(x_j - x_i)(y_i - y_j)}_{=(x_i-x_j)(y_j-y_i)} \\ &= \left(\prod_{k=0}^{n-1} \binom{n-1}{k}\right) \left(\prod_{1\le i<j\le n} (x_i - x_j)\right) \left(\prod_{1\le i<j\le n} (y_j - y_i)\right).\end{aligned}$$

□

5.3.2 Laplace expansion

Convention 5.78. Let $n \in \mathbb{N}$. Let A be an $n \times n$-matrix. Let $i, j \in [n]$. Then, we set

$$A_{\sim i,\sim j} := \operatorname{sub}_{[n]\setminus\{i\}}^{[n]\setminus\{j\}} A.$$

This is the $(n-1)\times(n-1)$-matrix obtained from A by removing its i-th row and its j-th column.

Theorem 5.79 (Laplace expansion)**.** *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix.*

(a) *For every $p \in [n]$, we have*

$$\det A = \sum_{q=1}^{n} (-1)^{p+q} A_{p,q} \det(A_{\sim p,\sim q}).$$

(b) *For every $q \in [n]$, we have*

$$\det A = \sum_{p=1}^{n} (-1)^{p+q} A_{p,q} \det(A_{\sim p,\sim q}).$$

Proof. See [Grinbe15, Theorem 6.82] or [Ford21, Lemma 4.7.7] or [Laue15, 5.8 and 5.8'] or [Strick13, Proposition B.24 and Proposition B.25] or [Loehr11, Theorem 9.48].

Part (a) is Mathlib's `Matrix.det_succ_row`; part (b) is `Matrix.det_succ_column`. □

Proposition 5.80. *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix. Let $r \in [n]$.*

(a) *For every $p \in [n]$ satisfying $p \neq r$, we have*

$$0 = \sum_{q=1}^{n} (-1)^{p+q} A_{r,q} \det(A_{\sim p,\sim q}).$$

(b) *For every $q \in [n]$ satisfying $q \neq r$, we have*

$$0 = \sum_{p=1}^{n} (-1)^{p+q} A_{p,r} \det(A_{\sim p,\sim q}).$$

Proof. **(a)** Let $p \in [n]$ satisfy $p \neq r$. Let C be the matrix A with its p-th row replaced by its r-th row. Then C has two equal rows, so $\det C = 0$. On the other hand, expanding $\det C$ along the p-th row (using Theorem 5.79 (a) applied to C) yields

$$\det C = \sum_{q=1}^{n} (-1)^{p+q} \underbrace{C_{p,q}}_{=A_{r,q}} \det \underbrace{(C_{\sim p,\sim q})}_{=A_{\sim p,\sim q}} = \sum_{q=1}^{n} (-1)^{p+q} A_{r,q} \det(A_{\sim p,\sim q}).$$

Comparing gives the claim. Part (b) follows similarly.

The formalization uses the adjugate: $(-1)^{p+q}\det(A_{\sim p,\sim q}) = (\operatorname{adj} A)_{q,p}$, so the sum becomes $\sum_q A_{r,q}(\operatorname{adj} A)_{q,p} = (A \cdot \operatorname{adj} A)_{r,p} = (\det A) \cdot I_{r,p} = 0$ when $r \neq p$. □

Definition 5.81 (Adjugate matrix)**.** Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix. We define a new $n \times n$-matrix $\operatorname{adj} A \in K^{n\times n}$ by

$$\operatorname{adj} A = \left((-1)^{i+j} \det(A_{\sim j,\sim i})\right)_{1\leq i\leq n,\ 1\leq j\leq n}.$$

This matrix $\operatorname{adj} A$ is called the *adjugate* (or *classical adjoint*) of the matrix A. Note the index swap: the (i,j) entry involves removing row j and column i.

The formalization shows that this definition equals Mathlib's `Matrix.adjugate`, which is defined via $(\operatorname{adj} A)_{i,j} = \det(A')$ where A' is the matrix A with row j replaced by the i-th standard basis vector.

Theorem 5.82. *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix. Let I_n denote the $n \times n$ identity matrix. Then,*

$$A \cdot (\operatorname{adj} A) = (\operatorname{adj} A) \cdot A = (\det A) \cdot I_n.$$

Proof. See [Grinbe15, Theorem 6.100] or [Ford21, Lemma 4.7.9] or [Loehr11, Theorem 9.50] or [Strick13, Proposition B.28].

To show $A \cdot (\operatorname{adj} A) = (\det A) \cdot I_n$, it suffices to check that the (i,j)-th entry of $A \cdot (\operatorname{adj} A)$ equals $\det A$ when $i = j$ and equals 0 otherwise.

Case $i = j$: The (i,i) entry is $\sum_k A_{i,k}(\operatorname{adj} A)_{k,i} = \sum_k (-1)^{i+k} A_{i,k} \det(A_{\sim i,\sim k}) = \det A$ by Laplace expansion along row i (Theorem 5.79 (a)).

Case $i \neq j$: The (i,j) entry is $\sum_k A_{i,k}(\operatorname{adj} A)_{k,j} = \sum_k (-1)^{j+k} A_{i,k} \det(A_{\sim j,\sim k}) = 0$ by Proposition 5.80 (a) (using row i entries with row j cofactors).

Similarly, $(\operatorname{adj} A) \cdot A = (\det A) \cdot I_n$ follows using column versions. □

5.3.3 Laplace expansion along multiple rows

Convention 5.83. For any subset I of $[n]$, we write $\widetilde{I} = [n] \setminus I$ for its complement, and $\operatorname{sum} S = \sum_{s\in S} s$ for any finite set S of integers.

Theorem 5.84 (Laplace expansion along multiple rows/columns)**.** *Let $n \in \mathbb{N}$. Let $A \in K^{n\times n}$ be an $n \times n$-matrix.*

(a) *For every subset P of $[n]$, we have*

$$\det A = \sum_{\substack{Q\subseteq[n] \\ |Q|=|P|}} (-1)^{\operatorname{sum} P+\operatorname{sum} Q} \det\left(\operatorname{sub}_P^Q A\right) \det\left(\operatorname{sub}_{\widetilde{P}}^{\widetilde{Q}} A\right).$$

(b) *For every subset Q of $[n]$, we have*

$$\det A = \sum_{\substack{P\subseteq[n] \\ |P|=|Q|}} (-1)^{\operatorname{sum} P+\operatorname{sum} Q} \det\left(\operatorname{sub}_P^Q A\right) \det\left(\operatorname{sub}_{\widetilde{P}}^{\widetilde{Q}} A\right).$$

Proof. See [Grinbe15, Theorem 6.156].

The proof partitions the symmetric group based on how permutations map P to subsets Q of the same size. For each such Q, the permutations σ with $\sigma(P) = Q$ decompose as a bijection $\tau : P \to Q$ (contributing to $\det(\operatorname{sub}_P^Q A)$) and a bijection $\rho : \widetilde{P} \to \widetilde{Q}$ (contributing to $\det(\operatorname{sub}_{\widetilde{P}}^{\widetilde{Q}} A)$). The sign decomposes as $(-1)^\sigma = (-1)^{\operatorname{sum} P+\operatorname{sum} Q}(-1)^\tau(-1)^\rho$.

Part (b) follows from part (a) applied to A^T using $\det(A^T) = \det A$. □

5.3.4 Desnanot–Jacobi and Dodgson condensation

Theorem 5.85 (Desnanot–Jacobi formula, take 1)**.** *Let $n \in \mathbb{N}$ be such that $n \geq 2$. Let $A \in K^{n \times n}$ be an $n \times n$-matrix.*

Let A' be the $(n-2) \times (n-2)$-matrix

$$\operatorname{sub}_{\{2,3,\ldots,n-1\}}^{\{2,3,\ldots,n-1\}} A = (A_{i+1,j+1})_{1 \leq i \leq n-2,\ 1 \leq j \leq n-2}.$$

(This is what remains of A when we remove the first row, the last row, the first column and the last column.) Then,

$$\begin{aligned}\det A \cdot \det(A') &= \det(A_{\sim 1,\sim 1}) \cdot \det(A_{\sim n,\sim n}) - \det(A_{\sim 1,\sim n}) \cdot \det(A_{\sim n,\sim 1}) \\ &= \det \begin{pmatrix} \det(A_{\sim 1,\sim 1}) & \det(A_{\sim 1,\sim n}) \\ \det(A_{\sim n,\sim 1}) & \det(A_{\sim n,\sim n}) \end{pmatrix}.\end{aligned}$$

Proof. See [Grinbe15, Proposition 6.122] or [Bresso99, §3.5, proof of Theorem 3.12] or [Zeilbe98].

The proof uses the adjugate matrix.

The right-hand side equals $(\operatorname{adj} A)_{0,0} \cdot (\operatorname{adj} A)_{n-1,n-1} - (\operatorname{adj} A)_{0,n-1} \cdot (\operatorname{adj} A)_{n-1,0}$, which is the 2×2 determinant of the corner submatrix of $\operatorname{adj} A$.

By Jacobi's complementary minor theorem for the 2×2 case (a special case proved independently), this 2×2 determinant of the adjugate equals $\det A \cdot \det(A')$, giving the left-hand side.

The proof of the special case uses the polynomial ring approach: reduce to the generic matrix over the multivariate polynomial ring, then work in its field of fractions where the matrix is invertible. $\square$

Theorem 5.86 (Cauchy determinant)**.** *Let $n \in \mathbb{N}$. Let $x_1, x_2, \ldots, x_n$ be n elements of K. Let $y_1, y_2, \ldots, y_n$ be n elements of K. Assume that $x_i + y_j$ is invertible in K for each $(i,j) \in [n]^2$. Then,*

$$\det\left(\left(\frac{1}{x_i + y_j}\right)_{1 \leq i \leq n,\ 1 \leq j \leq n}\right) = \frac{\prod_{1 \leq i < j \leq n} \left((x_i - x_j)(y_i - y_j)\right)}{\prod_{(i,j) \in [n]^2} (x_i + y_j)}.$$

Proof. The proof uses the polynomial version of the identity (factor hunting).

Step 1: Clear denominators. Multiply both sides by $\prod_{i,j}(x_i + y_j)$ to obtain the "cleared" polynomial identity:

$$\det\left(\left(\prod_{k \neq j} (x_i + y_k)\right)_{i,j}\right) = \prod_{1 \leq i < j \leq n} \left((x_i - x_j)(y_i - y_j)\right).$$

Step 2: Base cases. For $n = 0, 1, 2, 3$: verified by direct computation.

Step 3: Factor hunting ($n \geq 4$): The determinant on the left is a polynomial in $x_1, \ldots, x_n, y_1, \ldots, y_n$. Setting $x_i = x_j$ (for $i \neq j$) makes two rows identical, so $(x_i - x_j)$ divides the determinant. Similarly, $(y_i - y_j)$ divides the determinant for $i \neq j$. Since these linear factors are pairwise coprime irreducibles in the UFD $\mathbb{Z}[x_1, \ldots, x_n, y_1, \ldots, y_n]$, their product divides the determinant.

Step 4: Degree comparison. Both the determinant and the product have total degree $n(n-1)$, so the quotient is a constant. Evaluating at a specific point shows the constant is 1.

Step 5: Recover the Cauchy formula. Dividing the cleared identity by $\prod_{i,j}(x_i + y_j)$ yields the Cauchy determinant formula. $\square$

Lemma 5.87. *Let $n \in \mathbb{N}$. Let $x_1, \ldots, x_n \in R$ and $y_1, \ldots, y_n \in R$ be elements of a commutative ring R. Then*

$$\det\left(\left(\prod_{k\neq j}(x_i + y_k)\right)_{1\leq i\leq n,\ 1\leq j\leq n}\right) = \prod_{1\leq i<j\leq n}(x_i - x_j)(y_i - y_j).$$

This is equivalent to the Cauchy determinant formula after dividing both sides by $\prod_{(i,j)\in[n]^2}(x_i + y_j)$.

Proof. The proof uses factor hunting in the polynomial ring. Both sides are polynomials of degree $n(n-1)$ in the x_i's and y_j's. The left-hand side vanishes when $x_u = x_v$ (two rows become identical) and when $y_u = y_v$ (two columns become proportional). Thus the product $\prod_{i<j}(x_i - x_j)(y_i - y_j)$ divides the left-hand side. Degree comparison shows the quotient is a constant, and coefficient comparison shows the constant is 1. $\square$

5.3.5 Generalized Desnanot–Jacobi

Theorem 5.88. *Let $n \in \mathbb{N}$ be such that $n \geq 2$. Let p, q, u, v be four elements of $[n]$ such that $p < q$ and $u < v$. Let A be an $n \times n$-matrix. Then,*

$$\begin{aligned}&\det A \cdot \det\left(\operatorname{sub}_{[n]\setminus\{p,q\}}^{[n]\setminus\{u,v\}} A\right)\\ &= \det(A_{\sim p,\sim u}) \cdot \det(A_{\sim q,\sim v}) - \det(A_{\sim p,\sim v}) \cdot \det(A_{\sim q,\sim u}).\end{aligned}$$

Proof. See [Grinbe15, Theorem 6.126].

The proof combines two identities:

1. **Adjugate identity** (Lemma 5.94, generalized): The 2×2 determinant $(\operatorname{adj} A)_{u,p}(\operatorname{adj} A)_{v,q} - (\operatorname{adj} A)_{u,q}(\operatorname{adj} A)_{v,p}$ equals $(-1)^{p+u+q+v}$ times the product-minus-product expression.

2. **Jacobi 2×2 identity** (Theorem 5.89 for $|P| = |Q| = 2$): The same 2×2 determinant of the adjugate equals $(-1)^{p+u+q+v} \det A \cdot \det(\operatorname{sub}_{[n]\setminus\{p,q\}}^{[n]\setminus\{u,v\}} A)$.

Equating the two expressions and canceling $(-1)^{p+u+q+v}$ (which squares to 1) yields the result.

Theorem 5.85 is the particular case $p = 1$, $q = n$, $u = 1$, $v = n$. $\square$

5.3.6 Jacobi's complementary minor theorem

Theorem 5.89 (Jacobi's complementary minor theorem for adjugates)**.** *Let $n \in \mathbb{N}$. For any subset I of $[n]$, let $\widetilde{I}$ denote the complement $[n] \setminus I$. Set $\operatorname{sum} S = \sum_{s\in S} s$ for any finite set S of integers.*

Let $A \in K^{n\times n}$ be an $n \times n$-matrix. Let P and Q be two subsets of $[n]$ such that $|P| = |Q| \geq 1$. Then,

$$\det\left(\operatorname{sub}_P^Q(\operatorname{adj} A)\right) = (-1)^{\operatorname{sum} P + \operatorname{sum} Q}(\det A)^{|Q|-1} \det\left(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A\right).$$

Proof. The proof uses the polynomial ring approach (following Prasolov's *Problems and Theorems in Linear Algebra*).

Step 1: Reduce to generic matrix. Let A' be the generic $n \times n$ matrix with polynomial entries in $\mathbb{Z}[(x_{i,j})_{1\leq i,j\leq n}]$. The identity for arbitrary A over any commutative ring R follows by applying the evaluation homomorphism $x_{i,j} \mapsto A_{i,j}$.

Step 2: Work in the field of fractions. Since $\det(A')$ is a nonzero polynomial, we can embed A' into the field of fractions K of the polynomial ring, where A' becomes invertible.

Step 3: Use $\operatorname{adj}(A) = \det(A) \cdot A^{-1}$. For invertible A with $|P| = |Q| = k$:

$$\det(\operatorname{sub}_P^Q(\operatorname{adj} A)) = (\det A)^k \cdot \det(\operatorname{sub}_P^Q(A^{-1})).$$

Step 4: Complementary minor theorem for inverse. The classical result for invertible A states:

$$\det(\operatorname{sub}_P^Q(A^{-1})) = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \frac{\det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A)}{\det A}.$$

This is proved using block matrix decomposition.

Step 5: Combine. $\det(\operatorname{sub}_P^Q(\operatorname{adj} A)) = (\det A)^k \cdot (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \cdot \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A)/\det A = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} (\det A)^{k-1} \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A)$.

Step 6: Pull back. Since the identity holds in K and the canonical map from the polynomial ring to K is injective, the identity holds in the polynomial ring, and hence for all matrices A over any commutative ring.

Theorem 5.88 is the particular case of Theorem 5.89 for $P = \{u, v\}$ and $Q = \{p, q\}$. Theorem 5.85 is, of course, the particular case of Theorem 5.88 for $p = 1$, $q = n$, $u = 1$, $v = n$. ☐

5.3.7 Helpers for the Vandermonde determinant proof

Lemma 5.90. *Let $v_1, \ldots, v_m$ be elements of a commutative ring S. Then*

$$\prod_{1 \le i < j \le m} (v_i - v_j) = (-1)^{m(m-1)/2} \prod_{1 \le i < j \le m} (v_j - v_i).$$

Proof. Each factor $(v_i - v_j) = -(v_j - v_i)$, and there are $\binom{m}{2} = m(m-1)/2$ such factors. ☐

Lemma 5.91. *The sign of the column reversal permutation τ on $\{1, 2, \ldots, m\}$ (sending j to $m + 1 - j$) satisfies*

$$\operatorname{sign}(\tau) = (-1)^{m(m-1)/2}.$$

Proof. For $i < j$, we have $\tau(i) > \tau(j)$, so every pair is an inversion. The total number of inversions is $\binom{m}{2}$. ☐

Lemma 5.92. *Let A be an $m \times m$ matrix and σ a permutation of $[m]$. Then $\det(A \circ \sigma) = \operatorname{sign}(\sigma) \cdot \det A$, where $A \circ \sigma$ denotes the column permutation $A_{i,\sigma(j)}$.*

Proof. Transpose to a row permutation, apply the row permutation determinant formula, then transpose back. ☐

Lemma 5.93. *The determinant of the polynomial Vandermonde matrix (with entries x_i^{m-1-j}) equals*

$$\det\left(\left(x_i^{m-1-j}\right)_{i,j}\right) = \operatorname{sign}(\tau) \cdot \det\left(\left(x_i^j\right)_{i,j}\right),$$

where τ is the column reversal permutation (sending j to $m - 1 - j$). This relates the textbook matrix convention to Mathlib's via column reversal.

Proof. The polynomial Vandermonde matrix is Mathlib's Vandermonde matrix with columns permuted by the reversal permutation, so the determinant picks up the sign factor. ☐

5.3.8 Helpers for the Desnanot–Jacobi proof

Lemma 5.94. *Let $A \in K^{n\times n}$ with $n \geq 2$. Then*

$$\det(A_{\sim 1,\sim 1}) \cdot \det(A_{\sim n,\sim n}) - \det(A_{\sim 1,\sim n}) \cdot \det(A_{\sim n,\sim 1})$$
$$= (\operatorname{adj} A)_{0,0} \cdot (\operatorname{adj} A)_{n-1,n-1} - (\operatorname{adj} A)_{0,n-1} \cdot (\operatorname{adj} A)_{n-1,0}.$$

Proof. Direct computation using $(\operatorname{adj} A)_{i,j} = (-1)^{i+j} \det(A_{\sim j,\sim i})$. □

Lemma 5.95. *The Desnanot–Jacobi identity holds for 2×2 matrices by direct computation.*

Proof. The identity $\det A \cdot \det(A') = \det(A_{\sim 1,\sim 1}) \cdot \det(A_{\sim 2,\sim 2}) - \det(A_{\sim 1,\sim 2}) \cdot \det(A_{\sim 2,\sim 1})$ reduces to $\det A \cdot 1 = A_{2,2} \cdot A_{1,1} - A_{1,2} \cdot A_{2,1}$ for a 2×2 matrix, which is the definition of $\det A$. □

Lemma 5.96. *For a 2×2 matrix A, the 2×2 determinant of the corner entries of $\operatorname{adj} A$ satisfies*

$$(\operatorname{adj} A)_{0,0} \cdot (\operatorname{adj} A)_{1,1} - (\operatorname{adj} A)_{0,1} \cdot (\operatorname{adj} A)_{1,0} = \det A.$$

This is the base case of the Jacobi complementary minor theorem (since $\det(A') = 1$ for the 0×0 inner submatrix).

Proof. Direct computation using the explicit 2×2 adjugate formula. □

Lemma 5.97. *For a 3×3 matrix A, the 2×2 determinant of the corner entries of $\operatorname{adj} A$ at positions $\{0, 2\} \times \{0, 2\}$ satisfies*

$$(\operatorname{adj} A)_{0,0} \cdot (\operatorname{adj} A)_{2,2} - (\operatorname{adj} A)_{0,2} \cdot (\operatorname{adj} A)_{2,0} = \det A \cdot A_{1,1}.$$

This verifies the Jacobi complementary minor theorem for 3×3 matrices with $P = Q = \{0, 2\}$.

Proof. Direct expansion of the adjugate entries and the 3×3 determinant, followed by algebraic simplification. □

5.3.9 Helpers for the generalized Desnanot–Jacobi proof

Convention 5.98. Let $n \in \mathbb{N}$ with $n \geq 2$. Let A be an $n \times n$-matrix. For $p < q$ in $[n]$ and $u < v$ in $[n]$, define

$$\operatorname{sub}_{[n]\setminus\{p,q\}}^{[n]\setminus\{u,v\}} A$$

as the $(n - 2) \times (n - 2)$-submatrix of A obtained by removing rows p, q and columns u, v. The row and column indexing uses the order-preserving injection $[n - 2] \hookrightarrow [n]$ that skips the two removed indices in order.

Lemma 5.99. *Let A be an $(m + 2) \times (m + 2)$ matrix, and let $p < q$ and $u < v$ be indices. Then*

$$\det\left(\operatorname{sub}_{[n]\setminus\{p,q\}}^{[n]\setminus\{u,v\}} A\right) = \det\left(\operatorname{sub}_{\{p,q\}^c}^{\{u,v\}^c} A\right),$$

where the right-hand side uses the complement-based submatrix construction. This connects the explicit order-preserving injection used in the submatrix construction to the general complement-based definition, enabling the use of Jacobi's complementary minor theorem.

Proof. The proof shows that the order-preserving injection that skips p and q, being strictly monotone and having range equal to $\{p, q\}^c$, coincides with the canonical order-preserving enumeration of the complement set, so the two submatrix constructions yield the same matrix. □

5.3.10 Helpers for the adjugate and inverse relationship

Lemma 5.100. *Let K be a field and $A \in K^{n\times n}$ with $\det A \neq 0$. Then for all i, j,*

$$A^{-1}_{i,j} = \frac{(\operatorname{adj} A)_{i,j}}{\det A}.$$

Proof. Follows from $A^{-1} = (\det A)^{-1} \cdot \operatorname{adj} A$. □

Lemma 5.101. *Let A be an $n \times n$ matrix with $\det A$ a unit. Then for any index functions f and g,*

$$(\operatorname{adj} A)_{f(i),g(j)} = \det A \cdot (A^{-1})_{f(i),g(j)}.$$

That is, the adjugate submatrix equals $\det A$ times the corresponding inverse submatrix.

Proof. Entry-wise, $(\operatorname{adj} A)_{i,j} = \det A \cdot A^{-1}_{i,j}$ for invertible A. □

Lemma 5.102. *Let A be an $m\times m$ matrix with $\det A$ a unit. Let $P, Q \subseteq [m]$ with $|P| = |Q| = k$. Then*

$$\det\Big(\operatorname{sub}^Q_P(\operatorname{adj} A)\Big) = (\det A)^k \cdot \det\Big(\operatorname{sub}^Q_P(A^{-1})\Big).$$

This reduces Jacobi's complementary minor theorem to the complementary minor theorem for inverse matrices.

Proof. By Lemma 5.101, the adjugate submatrix is $\det A$ times the inverse submatrix. The determinant of a scalar multiple is $(\det A)^k$ times the original determinant. □

5.3.11 Helpers for block matrix decomposition

Lemma 5.103. *For a block matrix $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with D invertible and the Schur complement $A - BD^{-1}C$ invertible, we have*

$$\det(M) \cdot \det\big((M^{-1})_{\textit{top-left}}\big) = \det D.$$

Proof. By the Schur complement formula, $\det(M) = \det(D) \cdot \det(A - BD^{-1}C)$. The top-left block of M^{-1} is $(A - BD^{-1}C)^{-1}$, whose determinant is $\det(A - BD^{-1}C)^{-1}$. Multiplying gives $\det(D)$. □

Lemma 5.104. *Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ be an invertible block matrix with D invertible and Schur complement $A - BD^{-1}C$ invertible. Let $k = |A|$ (the size of the top-left block) with $k \geq 1$. Then*

$$\det((\operatorname{adj} M)_{\textit{top-left}}) = (\det M)^{k-1} \cdot \det D.$$

Proof. Since $\operatorname{adj}(M) = \det(M) \cdot M^{-1}$ for invertible M, the top-left block of $\operatorname{adj}(M)$ is $\det(M) \cdot (M^{-1})_{\text{top-left}}$. Taking the $k\times k$ determinant yields $(\det M)^k \cdot \det((M^{-1})_{\text{top-left}})$. By Lemma 5.103, $\det(M) \cdot \det((M^{-1})_{\text{top-left}}) = \det D$, so $\det((M^{-1})_{\text{top-left}}) = \det D/ \det M$, giving the result. □

Definition 5.105. Given a subset P of $[m]$, the *sorting equivalence* is a bijection

$$\{1, \ldots, |P|\} \sqcup \{1, \ldots, m - |P|\} \xrightarrow{\sim} [m]$$

that sends the i-th element of the left component to the i-th smallest element of P and the j-th element of the right component to the j-th smallest element of P^c. This equivalence is used to decompose matrices into block form with respect to the partition $[m] = P \sqcup P^c$.

Lemma 5.106. *Let K be a field, $A \in K^{m\times m}$ with $\det A \neq 0$, and σ, τ bijections from $[n]$ to $[m]$. Then*

$$(A_{\sigma,\tau})^{-1} = (A^{-1})_{\tau,\sigma},$$

where subscripts denote the submatrix obtained by reindexing rows and columns via the given bijections.

Proof. Verify directly that $A_{\sigma,\tau} \cdot (A^{-1})_{\tau,\sigma} = I$ using $(A \cdot A^{-1})_{\sigma(i),\sigma(j)} = \delta_{ij}$. □

Lemma 5.107. *Let $A \in K^{m\times m}$ and $f, g : [n] \xrightarrow{\sim} [m]$ be bijections. Then*

$$\det(A_{f,g}) = \operatorname{sign}(g^{-1} \circ f) \cdot \det A.$$

Proof. Rewrite the submatrix as a reindexed matrix and apply the standard determinant reindexing formula. □

5.3.12 Helpers for the complementary minor theorem proof

Lemma 5.108. *Let K be a field. Let $A \in K^{m\times m}$ be an invertible matrix. Let $P, Q \subseteq [m]$ with $|P| = |Q|$. Then*

$$\det A \cdot \det(\operatorname{sub}_P^Q(A^{-1})) = (-1)^{\operatorname{sum} P + \operatorname{sum} Q} \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A).$$

Proof. The proof uses block matrix decomposition. Define sorting equivalences σ and τ for Q and P respectively (Definition 5.105), which reorder indices so that P-elements come first. The permuted matrix $M = A_{\sigma,\tau}$ decomposes into blocks: $M_{\text{top-left}} = \operatorname{sub}_Q^P(A)$ and $M_{\text{bottom-right}} = \operatorname{sub}_{Q^c}^{P^c}(A)$. By Lemma 5.106, $(M^{-1})_{\text{top-left}} = \operatorname{sub}_P^Q(A^{-1})$. By Lemma 5.103, $\det(M)\cdot\det((M^{-1})_{\text{top-left}}) = \det(M_{\text{bottom-right}})$. The sign factor $(-1)^{\operatorname{sum} P+\operatorname{sum} Q}$ arises from the determinant of the sorting permutation (Lemma 5.107). □

Lemma 5.109. *Let K be a field. Let $A \in K^{m\times m}$ with $\det A \neq 0$. Let $P, Q \subseteq [m]$ with $|P| = |Q| \geq 1$. Then*

$$\det(\operatorname{sub}_P^Q(\operatorname{adj} A)) = (-1)^{\operatorname{sum} P+\operatorname{sum} Q}(\det A)^{|Q|-1} \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A).$$

Proof. By Lemma 5.102,

$$\det(\operatorname{sub}_P^Q(\operatorname{adj} A)) = (\det A)^{|P|} \cdot \det(\operatorname{sub}_P^Q(A^{-1})).$$

By Lemma 5.108,

$$\det A \cdot \det(\operatorname{sub}_P^Q(A^{-1})) = (-1)^{\operatorname{sum} P+\operatorname{sum} Q} \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A),$$

so $\det(\operatorname{sub}_P^Q(A^{-1})) = (-1)^{\operatorname{sum} P+\operatorname{sum} Q} \det(\operatorname{sub}_{\widetilde{Q}}^{\widetilde{P}} A)/\det A$. Since $|P| = |Q|$, the exponent becomes $|Q| - 1$. □

5.3.13 Helpers for the Cauchy determinant proof (factor hunting)

Definition 5.110. The *variable substitution* $\varphi_{i\to j}$ is the algebra homomorphism $R[x_1,\dots,x_n] \to R[x_1,\dots,x_n]$ that sends $x_i \mapsto x_j$ and fixes all other variables.

Lemma 5.111. *Let $p \in R[x_1,\dots,x_n]$ be a multivariate polynomial and $i \neq j$. Then $(x_i - x_j) \mid p - p|_{x_i=x_j}$, where $p|_{x_i=x_j}$ denotes the substitution of x_j for x_i.*

Proof. By induction on the structure of p (using the polynomial recursor on constants, sums, and products by a variable). The base case (constants) is trivial. The additive case follows from divisibility of sums. The multiplicative case $p \cdot x_k$ splits: if $k = i$, factor as $(p - p|_{x_i=x_j}) \cdot x_i + p|_{x_i=x_j} \cdot (x_i - x_j)$; otherwise, factor as $(p - p|_{x_i=x_j}) \cdot x_k$. □

Lemma 5.112. *Let R be an integral domain, and let $P \in R[x_0, x_1,\dots,x_{m+1}]$. If P vanishes when x_0 is replaced by x_1 (i.e., $P|_{x_0=x_1} = 0$), then $(x_0 - x_1) \mid P$.*

Proof. View P as a univariate polynomial in x_0 over $R[x_1,\dots,x_{m+1}]$ (using the canonical isomorphism between $R[x_0, x_1,\dots,x_{m+1}]$ and $R[x_1,\dots,x_{m+1}][x_0]$). The hypothesis says that x_1 is a root of this univariate polynomial, so $(X - x_1)$ divides it by the polynomial root theorem. Pulling back through the isomorphism yields $(x_0 - x_1) \mid P$. □

Lemma 5.113. *For distinct variables x_i, x_j in $\mathbb{Z}[x_1,\dots,x_n]$, the polynomial $x_i - x_j$ is irreducible.*

Proof. Since $x_i - x_j$ has total degree 1 and is primitive (its content is 1), it is irreducible. The degree-1 argument: any nontrivial factorization would require one factor of degree 0 (a constant), but a primitive polynomial of positive degree cannot have a non-unit constant factor. □

Lemma 5.114. *For distinct i, j, the total degree of $x_i - x_j$ in $\mathbb{Z}[x_1,\dots,x_n]$ is 1.*

Proof. The upper bound follows from $\deg(f - g) \leq \max(\deg f, \deg g)$ and $\deg(X_i) = 1$. The lower bound follows from the monomial X_i^1 having nonzero coefficient in $x_i - x_j$. □

Lemma 5.115. *For distinct i, j, the polynomial $x_i - x_j \in \mathbb{Z}[x_1,\dots,x_n]$ is primitive (its content is 1).*

Proof. The coefficient of the monomial x_i^1 is 1, so the GCD of the coefficients is 1. □

Lemma 5.116. *Let $i \neq j$ and $k \neq l$ be pairs of distinct indices such that $\{i, j\}$ and $\{k, l\}$ are not equal as unordered pairs. Then $(x_i - x_j)$ and $(x_k - x_l)$ are coprime in $\mathbb{Z}[x_1,\dots,x_n]$.*

Proof. Both are irreducible by Lemma 5.113, and they are non-associate (not equal up to unit multiple) since they involve different variable pairs. Distinct irreducibles in a UFD are coprime. □

Lemma 5.117. *Let $C(m)$ be the $m \times m$ matrix with entry $(i, j) = \prod_{k\neq j}(X_i + Y_k)$ over $\mathbb{Z}[X_1,\dots,X_m, Y_1,\dots,Y_m]$. For distinct i, j, we have $(X_i - X_j) \mid \det(C(m))$.*

Proof. Substituting X_j for X_i makes rows i and j identical, so the determinant vanishes. By Lemma 5.112, $(X_i - X_j)$ divides the determinant. □

Lemma 5.118. *For distinct i, j, we have $(Y_i - Y_j) \mid \det(C(m))$, where $C(m)$ is the polynomial Cauchy matrix from Lemma 5.117.*

Proof. Substituting Y_j for Y_i makes columns i and j proportional (both become $\prod_{k\neq i,k\neq j}(X_r + Y_k) \cdot (X_r + Y_j)$), so the determinant vanishes. The divisibility follows by Lemma 5.112. □

Lemma 5.119. *Let K be a field. Let $x_1, \dots, x_m, y_1, \dots, y_m \in K$ with $x_i + y_j \neq 0$ for all i, j. Then*

$$\det\left(\frac{1}{x_i + y_j}\right)_{i,j} = \frac{\prod_{i<j}(x_i - x_j)(y_i - y_j)}{\prod_{i,j}(x_i + y_j)}.$$

Proof. The polynomial Cauchy identity (Lemma 5.87) gives $\det(\prod_{k \neq j}(x_i + y_k)) = \prod_{i<j}(x_i - x_j)(y_i - y_j)$. The left-hand side equals $\prod_{i,j}(x_i + y_j) \cdot \det(1/(x_i + y_j))$ since each row of the Cauchy matrix can be multiplied by $\prod_j (x_i + y_j)$ to clear denominators. Dividing gives the result. $\square$

5.4 The Lindström–Gessel–Viennot lemma

5.4.1 Definitions

Definition 5.120. We consider the infinite simple digraph with vertex set $\mathbb{Z}^2$ (so the vertices are pairs of integers) and arcs

$$(i, j) \to (i + 1, j) \qquad \text{for all } (i, j) \in \mathbb{Z}^2$$

and

$$(i, j) \to (i, j + 1) \qquad \text{for all } (i, j) \in \mathbb{Z}^2.$$

The arcs of the first form are called *east-steps* or *right-steps*; the arcs of the second form are called *north-steps* or *up-steps*.

The vertices of this digraph will be called *lattice points* or *grid points* or simply *points*.

The entire digraph will be denoted by $\mathbb{Z}^2$ and called the *integer lattice* or *integer grid*.

Any path is uniquely determined by its starting point and its step sequence.

5.4.2 Helpers for lattice path infrastructure

Lemma 5.121. *Let p_1, p_2 be two lattice paths and A a starting point. Then the endpoint of the concatenation $p_1 \cdot p_2$ starting from A equals the endpoint of p_2 starting from the endpoint of p_1:*

$$\text{endpoint}(p_1 \cdot p_2, A) = \text{endpoint}(p_2, \text{endpoint}(p_1, A)).$$

Proof. By induction on p_1: applying the steps of $p_1 \cdot p_2$ starting from A is the same as first applying the steps of p_1 (reaching the endpoint of p_1) and then applying the steps of p_2. $\square$

Lemma 5.122. *If a path p goes from A to B, then for any n, the first n steps of p form a path from A to the intermediate vertex M, and the remaining steps form a path from M to B.*

Proof. Follows from the fact that p equals its first n steps concatenated with its remaining steps, together with Lemma 5.121. $\square$

Lemma 5.123. *If path h goes from A to v and path t goes from v to B, then $h \cdot t$ goes from A to B.*

Proof. By Lemma 5.121: $\text{endpoint}(h \cdot t, A) = \text{endpoint}(t, \text{endpoint}(h, A)) = \text{endpoint}(t, v) = B$. $\square$

Lemma 5.124. *For any lattice path p starting at A, the coordinate sum of the endpoint satisfies*

$$\text{cs}(\text{endpoint}(p, A)) = \text{cs}(A) + \ell(p),$$

where $\ell(p)$ is the number of steps in p.

Proof. By induction on p: each step increases the coordinate sum by exactly 1. $\square$

Lemma 5.125. *The x-coordinate of the endpoint of a path p starting at A satisfies*

$$\mathrm{x}(\mathrm{endpoint}(p, A)) = \mathrm{x}(A) + (\#\ \textit{of east-steps in } p).$$

Proof. By induction: each east-step increments the x-coordinate by 1, and each north-step leaves it unchanged. $\square$

Lemma 5.126. *For any two lattice points A, B, the set of lattice paths from A to B is finite.*

Proof. If B is not componentwise $\geq A$, the set is empty. Otherwise, every path from A to B has a bounded number of steps (exactly $\mathrm{cs}(B) - \mathrm{cs}(A)$), and there are finitely many lists of bounded length over the two-element set $\{\mathrm{east}, \mathrm{north}\}$. $\square$

Lemma 5.127. *If a path p goes from A to B, then* $\mathrm{x}(A) \leq \mathrm{x}(B)$ *and* $\mathrm{y}(A) \leq \mathrm{y}(B)$.

Proof. By induction on the path: each east-step increments the x-coordinate and each north-step increments the y-coordinate. $\square$

5.4.3 Counting paths from (a, b) to (c, d)

Lemma 5.128. *Let $(a, b), (c, d) \in \mathbb{Z}^2$.*
Observation 1: *Each path from (a, b) to (c, d) has exactly $c + d - a - b$ steps.*
Observation 2: *Each path from (a, b) to (c, d) has exactly $c - a$ east-steps.*

Proof. **Observation 1:** We define the *coordinate sum* of a point $(x, y) \in \mathbb{Z}^2$ to be $x + y$, denoted $\mathrm{cs}(x, y)$. The coordinate sum increases by exactly 1 along each arc. If the path visits vertices $v_0, v_1, \ldots, v_n$ with $v_0 = (a, b)$ and $v_n = (c, d)$, then by the telescope principle,

$$n = \sum_{i=1}^{n} \big(\mathrm{cs}(v_i) - \mathrm{cs}(v_{i-1})\big) = \mathrm{cs}(c, d) - \mathrm{cs}(a, b) = c + d - a - b.$$

Observation 2: The x-coordinate of a point increases by exactly 1 along each east-step and stays unchanged along each north-step. By telescoping on the x-coordinate:

$$(\#\text{ of east-steps}) = \mathrm{x}(c, d) - \mathrm{x}(a, b) = c - a.$$

$\square$

Proposition 5.129. *Let $(a, b) \in \mathbb{Z}^2$ and $(c, d) \in \mathbb{Z}^2$ be two points. Then,*

$$(\#\ \textit{of paths from } (a, b) \textit{ to } (c, d)) = \begin{cases} \dbinom{c + d - a - b}{c - a}, & \textit{if } c + d \geq a + b; \\ 0, & \textit{if } c + d < a + b. \end{cases}$$

Proof. Observation 1 immediately shows that no path from (a, b) to (c, d) exists when $c + d < a + b$ (it would need a negative number of steps), so the count is 0.

Otherwise, $c + d \geq a + b$, so $c + d - a - b \in \mathbb{N}$. Observations 1 and 2 show that every path from (a, b) to (c, d) has exactly $c + d - a - b$ steps, of which exactly $c - a$ are east-steps.

Observation 3: Let p be a path that starts at the point (a, b) and has exactly $c + d - a - b$ steps. Assume that exactly $c - a$ of these steps are east-steps. Then, this path p ends at (c, d).

(This follows by applying Observations 1 and 2 to the endpoint of p and comparing.)

Combining Observations 1, 2 and 3, the paths from (a, b) to (c, d) are precisely the paths that start at (a, b) and have exactly $c+d-a-b$ steps and exactly $c-a$ east-steps. Such a path is uniquely determined by which $c-a$ of its $c+d-a-b$ steps are east-steps. By the bijection principle,

$$(\# \text{ of paths from } (a, b) \text{ to } (c, d)) = \binom{c+d-a-b}{c-a}.$$

□

5.4.4 Path tuples, nipats and ipats

Definition 5.130. Let $k \in \mathbb{N}$.

(a) A *k-vertex* means a k-tuple of lattice points.

(b) If $\mathbf{A} = (A_1, A_2, \ldots, A_k)$ is a k-vertex, and if $\sigma \in S_k$ is a permutation, then $\sigma(\mathbf{A})$ shall denote the k-vertex $\left(A_{\sigma(1)}, A_{\sigma(2)}, \ldots, A_{\sigma(k)}\right)$.

(c) If $\mathbf{A} = (A_1, A_2, \ldots, A_k)$ and $\mathbf{B} = (B_1, B_2, \ldots, B_k)$ are two k-vertices, then a *path tuple* from $\mathbf{A}$ to $\mathbf{B}$ means a k-tuple $(p_1, p_2, \ldots, p_k)$, where each p_i is a path from A_i to B_i.

(d) A path tuple $(p_1, p_2, \ldots, p_k)$ is said to be *non-intersecting* if no two of the paths $p_1, p_2, \ldots, p_k$ have any vertex in common. We abbreviate "non-intersecting path tuple" as *nipat*.

(e) A path tuple $(p_1, p_2, \ldots, p_k)$ is said to be *intersecting* if it is not non-intersecting (i.e., if two of its paths have a vertex in common). We abbreviate "intersecting path tuple" as *ipat*.

5.4.5 Helpers for path tuple finiteness

Lemma 5.131. *For any two k-vertices $\mathbf{A}$, $\mathbf{B}$, the set of path tuples from $\mathbf{A}$ to $\mathbf{B}$ is finite.*

Proof. The set of path tuples embeds injectively into the product $\prod_{i=1}^{k}\{\text{paths from } A_i \text{ to } B_i\}$, which is finite by Lemma 5.126. □

Lemma 5.132. *For any two k-vertices $\mathbf{A}$, $\mathbf{B}$, the set of nipats from $\mathbf{A}$ to $\mathbf{B}$ is finite.*

Proof. Subset of the finite set of all path tuples (Lemma 5.131). □

Lemma 5.133. *For any two k-vertices $\mathbf{A}$, $\mathbf{B}$, the set of ipats from $\mathbf{A}$ to $\mathbf{B}$ is finite.*

Proof. Subset of the finite set of all path tuples (Lemma 5.131). □

5.4.6 Crowded vertices and the minimum crowded path index

Definition 5.134. Let $\mathbf{p} = (p_1, \ldots, p_k)$ be a path tuple.

(a) A vertex v is *crowded* in $\mathbf{p}$ if v lies on at least two distinct paths p_i, p_j with $i \neq j$.

(b) The *crowded path indices* of $\mathbf{p}$ is the set of indices i such that p_i contains a vertex shared with some other path p_j.

(c) For an ipat $\mathbf{p}$, the *minimum crowded path index* is the smallest i in the crowded path indices.

(d) The *crowded vertices on path i* is the set of vertices on p_i that also appear on some other path.

Lemma 5.135. *A path tuple is intersecting if and only if it has at least one crowded vertex. Equivalently, a path tuple is intersecting if and only if its set of crowded path indices is nonempty.*

Proof. Immediate from the definitions: the negation of "all pairs of paths are disjoint" is equivalent to the existence of a shared vertex. □

5.4.7 The LGV lemma for two paths

Signed path tuple infrastructure

Definition 5.136. Given four lattice points A, A', B, B':

(a) A *signed path tuple* is a pair of paths (p_0, p_1) together with a sign flag indicating whether the destinations are (B, B') (sign $+1$) or (B', B) (sign -1).

(b) The *path count matrix* is the 2×2 integer matrix with entries $\#\text{paths}(A_i \to B_j)$.

(c) The number of nipats from (A, A') to (B, B').

Lemma 5.137. *The sets of signed path tuples (all, non-intersecting, and intersecting) are finite.*

Proof. The signed path tuples inject into the product of two finite path sets times a two-element set. All three subsets are finite as subsets of this finite set. □

First intersection point

Lemma 5.138. *Given an intersecting signed path tuple (p_0, p_1), the* first intersection index *is the smallest index i such that the i-th vertex of p_0 belongs to the vertices of p_1. This index is strictly less than the number of vertices of p_0.*

Proof. The existence of a shared vertex guarantees that a valid index is found before the end of the vertex list. □

Lemma 5.139. *The* first intersection point *v of an intersecting signed path tuple (p_0, p_1) satisfies:*

1. *v lies on p_0 (at the first intersection index).*
2. *v lies on p_1.*
3. *v is* first*: no vertex of p_0 at an earlier index belongs to p_1.*

Proof. Property (1) is by definition. Property (2) follows from the search semantics: the predicate tested is membership in p_1's vertices. Property (3) follows from the minimality of the search: all earlier indices fail the predicate. □

Lemma 5.140. *Splitting a path at a vertex v (by taking the first steps up to the index of v and the remaining steps after) produces a head ending at v and a tail starting at v.*

Proof. By Lemma 5.122, taking n steps gives a path from A to the intermediate point, and dropping n steps gives a path from that point to B. The key fact is that the vertex at the split position equals v, so the intermediate point is v. □

Lemma 5.141. *After applying the ipat involution (exchanging tails at the first intersection point v), the first intersection point of the resulting path tuple is still v.*

This is the key technical lemma for proving the involution is involutive. It relies on:

- *The prefix of p_0 up to the split index is unchanged.*
- *v is at the same index in the new path.*
- *No earlier vertex of p_0 lies in p_1's vertices (by minimality).*
- *Vertices at distinct positions on a lattice path are distinct (because the coordinate sum strictly increases).*

Proof. The new path $q_0 = \mathrm{head}(p_0) \cdot \mathrm{tail}(p_1)$ shares the same prefix as p_0 up to the split index. The value v at this index is preserved. For j less than the split index, the j-th vertex of q_0 equals the j-th vertex of p_0, which is not among the new p_1's vertices (the head of old p_1 is a subset of old p_1's vertices, and the tail of old p_0 cannot contain an earlier vertex because coordinate sums strictly increase). Thus the search on q_0 returns the same index. □

Lemma 5.142. *The sign-reversing involution on intersecting path tuples for $k = 2$.*

Given an intersecting path tuple (p, p') in the set $\mathcal{X}$ of ipats from (A, A') to (B, B') or to (B', B):

1. *Since $(p, p') \in \mathcal{X}$, the paths p and p' have a vertex in common. Let v be the first one. (The first one on p or the first one on p'? These are the same, since our digraph is acyclic.) We call this vertex v the* first intersection *of (p, p').*
2. *Split each path at v into a head (before v) and a tail (after v).*
3. *Exchange the tails: $q = (\text{head of } p) \cup (\text{tail of } p')$ and $q' = (\text{head of } p') \cup (\text{tail of } p)$.*
4. *Define $f(p, p') = (q, q')$.*

Then f is a sign-reversing involution on $\mathcal{X}$ with no fixed points.

Proof. If (p, p') is an ipat from (A, A') to (B, B'), then $f(p, p') = (q, q')$ is an ipat from (A, A') to (B', B) (exchanging tails swaps the endpoints), and vice versa. Thus f is well-defined and sign-reversing.

To show $f \circ f = \mathrm{id}$: after exchanging tails, the heads are unchanged, so v is still the first intersection point (Lemma 5.141). Exchanging tails again undoes the swap.

Since f is sign-reversing, it has no fixed points. □

Lemma 5.143. *For any four lattice points A, A', B, B',*

$$\sum_{(p,p') \in \mathcal{X}} \mathrm{sign}(p, p') = 0,$$

where $\mathcal{X}$ is the set of intersecting path tuples and the sign is $+1$ for path tuples to (B, B') and -1 for path tuples to (B', B).

Proof. By Lemma 4.10 applied to the sign-reversing involution of Lemma 5.142. □

Lemma 5.144. *The sum over all signed path tuples decomposes as:*

$$\big(\#\ \textit{of path tuples from } (A, A') \textit{ to } (B, B')\big) - \big(\#\ \textit{of path tuples from } (A, A') \textit{ to } (B', B)\big)$$
$$= \sum_{(p,p') \in \mathcal{A}} \mathrm{sign}(p, p'),$$

and the sum over nipats gives:

$$\sum_{(p,p') \in \mathcal{A} \setminus \mathcal{X}} \mathrm{sign}(p, p') = \big(\#\ \textit{of nipats from } (A, A') \textit{ to } (B, B')\big)$$
$$- \big(\#\ \textit{of nipats from } (A, A') \textit{ to } (B', B)\big).$$

Proof. The first identity follows from the product rule: the product of path counts equals the count of path tuples.

The second identity is immediate from the definitions: among the nipats, those going to (B, B') contribute $+1$ each and those going to (B', B) contribute -1 each. $\square$

Proposition 5.145 (LGV lemma for two paths)**.** *Let (A, A') and (B, B') be two 2-vertices (i.e., let A, A', B, B' be four lattice points). Then,*

$$\begin{aligned} &\det \begin{pmatrix} (\#\text{ of paths from } A \text{ to } B) & (\#\text{ of paths from } A \text{ to } B') \\ (\#\text{ of paths from } A' \text{ to } B) & (\#\text{ of paths from } A' \text{ to } B') \end{pmatrix} \\ &= (\#\text{ of nipats from } (A, A') \text{ to } (B, B')) \\ &\qquad - (\#\text{ of nipats from } (A, A') \text{ to } (B', B)). \end{aligned}$$

Proof. We have

$$\begin{aligned} &\det \begin{pmatrix} (\#\text{ of paths from } A \text{ to } B) & (\#\text{ of paths from } A \text{ to } B') \\ (\#\text{ of paths from } A' \text{ to } B) & (\#\text{ of paths from } A' \text{ to } B') \end{pmatrix} \\ &= (\#\text{ of path tuples from } (A, A') \text{ to } (B, B')) \\ &\qquad - (\#\text{ of path tuples from } (A, A') \text{ to } (B', B)) \end{aligned}$$

(by the product rule). By Lemma 5.144, the right hand side equals $\sum_{(p,p') \in \mathcal{A}} \operatorname{sign}(p, p')$.

By Lemma 4.10, using the sign-reversing involution of Lemma 5.142 (which uses Lemma 5.143), we obtain

$$\sum_{(p,p') \in \mathcal{A}} \operatorname{sign}(p, p') = \sum_{(p,p') \in \mathcal{A} \setminus \mathcal{X}} \operatorname{sign}(p, p').$$

The right hand side equals

$$\begin{aligned} &(\#\text{ of nipats from } (A, A') \text{ to } (B, B')) \\ &\qquad - (\#\text{ of nipats from } (A, A') \text{ to } (B', B)) \end{aligned}$$

by Lemma 5.144. $\square$

5.4.8 The baby Jordan curve theorem

Proposition 5.146 (baby Jordan curve theorem)**.** *Let A, B, A' and B' be four lattice points satisfying*

$$\begin{aligned} \mathrm{x}(A') &\leq \mathrm{x}(A), & \mathrm{y}(A') &\geq \mathrm{y}(A), \\ \mathrm{x}(B') &\leq \mathrm{x}(B), & \mathrm{y}(B') &\geq \mathrm{y}(B). \end{aligned}$$

(That is, A' is weakly northwest of A, and B' is weakly northwest of B.)

Let p be any path from A to B'. Let p' be any path from A' to B. Then, p and p' have a vertex in common.

In particular, there are no nipats from (A, A') to (B', B).

Proof. The proof is by strong induction on $\ell(p) + \ell(p')$, the sum of path lengths.

Case 3 ($A' = A$): Then A is a common vertex of both paths and we are done.

Case 1 ($\mathrm{y}(A') > \mathrm{y}(A)$): Let P be the second vertex of p. Every vertex of the subpath r from P to B' is a vertex of p, so r and p' have no vertex in common. Also $\ell(r) + \ell(p') < \ell(p) + \ell(p')$. If $\mathrm{y}(A') \geq \mathrm{y}(P)$, then we could apply the induction hypothesis to r and p', contradicting $r \cap p' = \emptyset$.

Hence $\mathrm{y}(A') < \mathrm{y}(P)$, so $\mathrm{y}(A') \leq \mathrm{y}(P) - 1$. If the arc from A to P were an east-step, we would have $\mathrm{y}(P) = \mathrm{y}(A)$, contradicting $\mathrm{y}(A) \leq \mathrm{y}(A') < \mathrm{y}(P)$. So the arc is a north-step, giving $\mathrm{y}(P) = \mathrm{y}(A) + 1$, hence $\mathrm{y}(A') \leq \mathrm{y}(A)$, contradicting $\mathrm{y}(A') > \mathrm{y}(A)$.

Case 2 ($\mathrm{x}(A') < \mathrm{x}(A)$): Symmetric to Case 1, looking at the first step of p' and using reflection across the line $x = y$.

The consequence that there are no nipats from (A, A') to (B', B) follows immediately: if any pair of paths (one from A to B', one from A' to B) must intersect, then there can be no nipats from (A, A') to (B', B). ☐

5.4.9 Log-concavity of binomial coefficients

Corollary 5.147. *Let $n, k \in \mathbb{N}$. Then,* $\dbinom{n}{k}^2 \geq \dbinom{n}{k-1} \cdot \dbinom{n}{k+1}$.

Proof. Combinatorial proof (from the source text, using LGV): Define four lattice points $A = (1, 0)$ and $A' = (0, 1)$ and $B = (k+1, n-k)$ and $B' = (k, n-k+1)$. Then, Proposition 5.145 yields

$$\begin{aligned}
&\det \begin{pmatrix} (\text{\# of paths from } A \text{ to } B) & (\text{\# of paths from } A \text{ to } B') \\ (\text{\# of paths from } A' \text{ to } B) & (\text{\# of paths from } A' \text{ to } B') \end{pmatrix} \\
&= (\text{\# of nipats from } (A, A') \text{ to } (B, B')) - \underbrace{(\text{\# of nipats from } (A, A') \text{ to } (B', B))}_{=0} \\
&= (\text{\# of nipats from } (A, A') \text{ to } (B, B')) \geq 0.
\end{aligned}$$

(Here, the second term is 0 by Proposition 5.146.) However, Proposition 5.129 yields

$$\begin{pmatrix} (\text{\# of paths from } A \text{ to } B) & (\text{\# of paths from } A \text{ to } B') \\ (\text{\# of paths from } A' \text{ to } B) & (\text{\# of paths from } A' \text{ to } B') \end{pmatrix} = \begin{pmatrix} \dbinom{n}{k} & \dbinom{n}{k-1} \\ \dbinom{n}{k+1} & \dbinom{n}{k} \end{pmatrix},$$

so the determinant is $\dbinom{n}{k}^2 - \dbinom{n}{k-1} \cdot \dbinom{n}{k+1} \geq 0$.

Lean formalization (algebraic proof): The Lean proof avoids the LGV machinery and proceeds purely algebraically, using the recurrences

$$\begin{aligned}
\binom{n}{k+1}(k+1) &= \binom{n}{k}(n-k), \\
\binom{n}{k} \cdot k &= \binom{n}{k-1}(n-k+1).
\end{aligned}$$

These allow rewriting both sides after multiplying by $k(k+1) > 0$. The inequality reduces to $(n-k+1)(k+1) \geq (n-k) \cdot k$, which holds because the difference is $n + 1 \geq 0$. This proof is complete. ☐

5.4.10 The LGV lemma for k paths

Definitions for general k

Definition 5.148. Let $k \in \mathbb{N}$ and let $\mathbf{A}, \mathbf{B}$ be two k-vertices.

(a) The *path count matrix* M is the $k \times k$ integer matrix with entries $M_{i,j} = \#\text{paths}(A_i \to B_j)$.

(b) For $\sigma \in S_k$, the number of nipats from $\mathbf{A}$ to $\sigma(\mathbf{B})$.

(c) For $\sigma \in S_k$, the number of ipats from $\mathbf{A}$ to $\sigma(\mathbf{B})$.

Bridge between LGV1 and LGV2

Lemma 5.149. *The lattice step types in LGV1 and LGV2 are equivalent: the two representations of lattice steps are isomorphic, and this equivalence preserves the step application function. Lattice paths can be converted between representations, preserving endpoints and vertices.*

Proof. Both types are two-element enumerations with the same constructors. The equivalence and endpoint preservation follow by case analysis. □

Lemma 5.150. *The ipat count (defined via one representation of path tuples) equals the cardinality of the ipat set (defined via the other representation). This is the key bridge that allows the proof to invoke the sign-reversing involution from the weighted LGV formalization.*

Proof. The proof constructs an equivalence between the two representations of intersecting path tuples, converting between list-of-steps and list-of-vertices representations, then verifies that the intersection property is preserved at each step. □

Product rule and partition lemmas

Lemma 5.151. *Let $k \in \mathbb{N}$ and let $\mathbf{A}, \mathbf{B}$ be two k-vertices. Then:*

(a) *(Product rule) For any $\sigma \in S_k$,*

$$\prod_{i=1}^{k} \big(\#\text{ of paths from } A_i \text{ to } B_{\sigma(i)}\big) = \big(\#\text{ of path tuples from } \mathbf{A} \text{ to } \sigma(\mathbf{B})\big).$$

(b) *(Partition) Each path tuple is either a nipat or an ipat, so*

$$\big(\#\text{ of path tuples}\big) = \big(\#\text{ of nipats}\big) + \big(\#\text{ of ipats}\big).$$

(c) *(Determinant expansion)*

$$\det\Big(\big(\#\text{ of paths from } A_i \text{ to } B_j\big)_{1 \le i,j \le k}\Big) = \sum_{\sigma \in S_k} (-1)^{\sigma} \prod_{i=1}^{k} \big(\#\text{ of paths from } A_i \text{ to } B_{\sigma(i)}\big).$$

Proof. **(a)** A path tuple from $\mathbf{A}$ to $\sigma(\mathbf{B})$ is just a tuple of independent choices (one path per index), so by the product rule the count is the product.

(b) Every path tuple is intersecting or not (classical logic); the sets are disjoint and their union is the set of all path tuples.

(c) This is the Leibniz formula $\det(M) = \sum_{\sigma} (-1)^{\sigma} \prod_i M_{i,\sigma(i)}$. □

Lemma 5.152. *The ipat cancellation lemma for general k:*

$$\sum_{\sigma \in S_k} (-1)^{\sigma} \cdot \big(\#\text{ of ipats from } \mathbf{A} \text{ to } \sigma(\mathbf{B})\big) = 0.$$

This is proved by a sign-reversing involution on pairs $(\sigma, \mathbf{p})$ where $\mathbf{p}$ is an ipat from $\mathbf{A}$ to $\sigma(\mathbf{B})$. The involution:

1. *A point u is* crowded *if it is contained in at least two of the paths $p_1, p_2, \ldots, p_k$.*

2. *Pick the smallest $i \in [k]$ such that p_i contains a crowded point.*

3. *Let v be the first crowded point on p_i.*

4. *Let $j > i$ be the largest index such that $v \in p_j$.*

5. *Exchange the tails of p_i and p_j at v.*

6. *Replace σ by $\sigma \cdot t_{i,j}$ (composing with the transposition swapping i and j), which flips the sign (by Proposition 3.110 **(b)** and **(d)**).*

The map $f : (\sigma, \mathbf{p}) \mapsto (\sigma', \mathbf{q})$ is an involution because the heads of p_i and p_j are unchanged, so the same v, i, j are selected again.

Proof. The proof uses:

- The partition of path tuples (Lemma 5.151).

- The sign-reversing involution described above.

- Lemma 4.10 to cancel the signed sum of ipats.

- The bridge Lemma 5.150 to invoke the sign-reversing involution from the weighted LGV formalization.

□

Proposition 5.153 (LGV lemma, lattice counting version)**.** *Let $k \in \mathbb{N}$. Let $\mathbf{A} = (A_1, A_2, \ldots, A_k)$ and $\mathbf{B} = (B_1, B_2, \ldots, B_k)$ be two k-vertices. Then,*

$$\begin{aligned}&\det\left((\text{\# of paths from } A_i \text{ to } B_j)_{1\leq i\leq k,\ 1\leq j\leq k}\right)\\&= \sum_{\sigma\in S_k} (-1)^{\sigma} (\text{\# of nipats from } \mathbf{A} \text{ to } \sigma(\mathbf{B})).\end{aligned}$$

Proof. Define $\mathcal{A} := \{(\sigma, \mathbf{p}) \mid \sigma \in S_k, \text{ and } \mathbf{p} \text{ is a path tuple from } \mathbf{A} \text{ to } \sigma(\mathbf{B})\}$ and $\mathcal{X} := \{(\sigma, \mathbf{p}) \in \mathcal{A} \mid \mathbf{p}$ is intersecti
Set $\operatorname{sign}(\sigma, \mathbf{p}) := (-1)^{\sigma}$.

By Lemma 4.10 applied to the sign-reversing involution of Lemma 5.152,

$$\sum_{(\sigma,\mathbf{p})\in\mathcal{A}} (-1)^{\sigma} = \sum_{(\sigma,\mathbf{p})\in\mathcal{A}\setminus\mathcal{X}} (-1)^{\sigma}.$$

The left hand side equals

$$\sum_{\sigma\in S_k} (-1)^{\sigma} \prod_{i=1}^{k} (\text{\# of paths from } A_i \text{ to } B_{\sigma(i)}) = \det\left((\text{\# of paths from } A_i \text{ to } B_j)_{1\leq i\leq k,\ 1\leq j\leq k}\right)$$

(by Lemma 5.151(a) and the definition of the determinant), and the right hand side equals

$$\sum_{\sigma\in S_k} (-1)^{\sigma} (\text{\# of nipats from } \mathbf{A} \text{ to } \sigma(\mathbf{B})).$$

□

5.5 The LGV lemma: weighted and generalized versions

5.5.1 The weighted version

Theorem 5.154 (LGV lemma, lattice weight version)**.** *Let K be a commutative ring.*

For each arc a of the digraph $\mathbb{Z}^2$, let $w(a)$ be an element of K. We call $w(a)$ the weight *of a. For each path p of $\mathbb{Z}^2$, define the* weight *$w(p)$ of p by*

$$w(p) := \prod_{a \text{ is an arc of } p} w(a).$$

For each path tuple $\mathbf{p} = (p_1, p_2, \ldots, p_k)$, define the weight *$w(\mathbf{p})$ of $\mathbf{p}$ by*

$$w(\mathbf{p}) := w(p_1)\, w(p_2) \cdots w(p_k).$$

Let $k \in \mathbb{N}$. Let $\mathbf{A} = (A_1, A_2, \ldots, A_k)$ and $\mathbf{B} = (B_1, B_2, \ldots, B_k)$ be two k-vertices. Then,

$$\det\left(\left(\sum_{p:A_i \to B_j} w(p)\right)_{1\le i\le k,\ 1\le j\le k}\right) = \sum_{\sigma \in S_k} (-1)^\sigma \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p}).$$

Here, "$p : A_i \to B_j$" means "p is a path from A_i to B_j".

Proof. This is the special case of Theorem 5.155 for the integer lattice $\mathbb{Z}^2$, which is a path-finite acyclic digraph. $\square$

5.5.2 Generalization to acyclic digraphs

Theorem 5.155 (LGV lemma, digraph weight version)**.** *Let K be a commutative ring.*

Let D be a path-finite (but possibly infinite) acyclic digraph.

For each arc a of D, let $w(a) \in K$. For each path p of D, define $w(p) := \prod_{a \text{ is an arc of } p} w(a)$. For each path tuple $\mathbf{p} = (p_1, \ldots, p_k)$, define $w(\mathbf{p}) := w(p_1) \cdots w(p_k)$.

Let $k \in \mathbb{N}$. Let $\mathbf{A} = (A_1, \ldots, A_k)$ and $\mathbf{B} = (B_1, \ldots, B_k)$ be two k-tuples of vertices of D. Then,

$$\det\left(\left(\sum_{p:A_i \to B_j} w(p)\right)_{1\le i\le k,\ 1\le j\le k}\right) = \sum_{\sigma \in S_k} (-1)^\sigma \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p}).$$

Proof. The same argument as for Proposition 5.153 can be used here; just replace $\operatorname{sign}(\sigma, \mathbf{p}) := (-1)^\sigma$ by $\operatorname{sign}(\sigma, \mathbf{p}) = (-1)^\sigma \cdot w(\mathbf{p})$. The only new observation required is that when we exchange the tails of two paths in our path tuple, the weight of the path tuple does not change. This is clear: the weight of a path tuple is the product of the weights of all arcs in all paths of the tuple; when we exchange the tails of two paths, some arcs get moved from one path to the other, but the total product stays unchanged.

More precisely, the proof expands the determinant via the Leibniz formula. Each summand $(-1)^\sigma \prod_i M_{i,\sigma(i)}$ expands (by distributivity) into a sum over all path tuples from $\mathbf{A}$ to $\sigma(\mathbf{B})$, weighted by $(-1)^\sigma\, w(\mathbf{p})$. Split the sum into non-intersecting and intersecting path tuples.

For the intersecting path tuples, a sign-reversing involution φ (Lemma 5.161) is defined as follows: given an intersecting path tuple paired with a permutation $(\sigma, \mathbf{p})$, find the canonical first intersection point, exchange the tails of the two relevant paths at that point, and compose σ with the corresponding transposition. This involution reverses the sign factor $(-1)^\sigma$ (Lemma 5.162) while preserving the weight $w(\mathbf{p})$ (Lemma 5.163), and has no fixed points among intersecting path tuples (Lemma 5.164). Therefore the contributions of intersecting path tuples cancel in pairs, leaving only the non-intersecting ones. ☐

5.5.3 Helper lemmas for the sign-reversing involution

Lemma 5.156 (Vertices are distinct in acyclic digraphs)**.** *In an acyclic digraph D, all vertices on a path are distinct.*

Proof. If a vertex appeared twice at positions $i < j$, we could extract the subpath from i to j, which forms a cycle. By acyclicity, this subpath has length 1, so $i = j$. ☐

Lemma 5.157 (Exchange of tails preserves weight)**.** *Let D be an acyclic digraph with arc weight function w. Let p and q be two paths sharing a vertex v. Let (p', q') be the result of exchanging the tails of p and q at v (i.e., p' takes the head of p up to v followed by the tail of q from v, and symmetrically for q'). Then $w(p) \cdot w(q) = w(p') \cdot w(q')$.*

Proof. The weight of a path is the product of the weights of its arcs. When we exchange tails at v, arcs are redistributed between the two paths, but the total multiset of arcs is preserved. Hence the product is unchanged. ☐

Lemma 5.158 (Exchange of tails is an involution)**.** *Let D be an acyclic digraph, and let p, q be paths sharing a vertex v. Let (p', q') be the result of exchanging the tails of p and q at v. Then exchanging the tails of p' and q' at v returns (p, q).*

Proof. Exchanging tails at v twice returns each path to its original form, since the head and tail of each path at v are restored. ☐

Lemma 5.159 (Sign-reversing map preserves intersection)**.** *Let D be a path-finite acyclic digraph. If $(\sigma, \mathbf{p})$ is an intersecting path tuple with permutation, then $\varphi(\sigma, \mathbf{p})$ is also intersecting.*

Proof. By Lemma 5.158, exchanging tails is an involution. If the result were non-intersecting, applying φ again would give a non-intersecting tuple equal to the original (contradicting the hypothesis that it is intersecting). ☐

Lemma 5.160 (Canonical intersection data is invariant)**.** *Let D be an acyclic digraph and $(\sigma, \mathbf{p})$ an intersecting path tuple with canonical intersection data (i, j, v). After applying the sign-reversing map, the canonical intersection data of the resulting tuple $\varphi(\sigma, \mathbf{p})$ is still (i, j, v).*

Proof. The head of path i (vertices before v, inclusive) is unchanged by the tail exchange, so v remains the first crowded vertex on path i. Furthermore, i is still the smallest crowded path index (indices below i have unchanged paths), and j is still the largest index sharing vertex v with i (the set of path indices containing v is preserved). ☐

Lemma 5.161 (Sign-reversing involution is an involution)**.** *Let D be a path-finite acyclic digraph. Let $(\sigma, \mathbf{p})$ be an intersecting path tuple with permutation. Then the sign-reversing map φ satisfies: if $\varphi(\sigma, \mathbf{p})$ is intersecting, then $\varphi(\varphi(\sigma, \mathbf{p})) = (\sigma, \mathbf{p})$.*

Proof. After applying φ, the canonical intersection data (i, j, v) is preserved (Lemma 5.160). Applying φ again thus uses the same (i, j, v), and the paths and permutation are restored by Lemma 5.158 and the identity $(\sigma \cdot (i\ j)) \cdot (i\ j) = \sigma$. ☐

Lemma 5.162 (Sign-reversing involution reverses sign)**.** *The sign-reversing map φ satisfies: if $\varphi(\sigma, \mathbf{p}) = (\sigma', \mathbf{p}')$, then $(-1)^{\sigma'} = -(-1)^{\sigma}$.*

Proof. The permutation component of $\varphi(\sigma, \mathbf{p})$ is $\sigma \cdot (i\ j)$ for distinct $i \neq j$ (Lemma 5.165). Since transpositions have sign -1, the sign of the new permutation is $-(-1)^\sigma$. □

Lemma 5.163 (Sign-reversing involution preserves weight)**.** *The sign-reversing map φ satisfies: if $\varphi(\sigma, \mathbf{p}) = (\sigma', \mathbf{p}')$, then $w(\mathbf{p}') = w(\mathbf{p})$.*

Proof. Only paths i and j are modified (their tails are exchanged), and the remaining paths are unchanged (Lemma 5.166). By Lemma 5.157, the product of the weights of paths i and j is preserved. Hence the total weight $w(\mathbf{p}) = \prod_l w(p_l)$ is unchanged. □

Lemma 5.164 (Sign-reversing involution has no fixed points)**.** *The sign-reversing map φ has no fixed points among intersecting path tuples: if $\mathbf{p}$ is intersecting, then $\varphi(\sigma, \mathbf{p}) \neq (\sigma, \mathbf{p})$.*

Proof. The permutation component changes by a non-trivial transposition, so it cannot equal σ. □

Lemma 5.165 (Permutation of sign-reversing map)**.** *The sign-reversing map φ applied to $(\sigma, \mathbf{p})$ produces a permutation $\sigma' = \sigma \cdot (i\ j)$ for some distinct indices $i \neq j$.*

Proof. By definition: the canonical intersection data selects indices i and j, and φ composes σ with the transposition $(i\ j)$. □

Lemma 5.166 (Paths of sign-reversing map)**.** *The sign-reversing map φ applied to $(\sigma, \mathbf{p})$ produces paths $\mathbf{p}'$ such that: paths at indices i and j have their tails exchanged at the crowded vertex v, while all other paths remain unchanged.*

Proof. By definition: φ selects the canonical intersection data (i, j, v) and exchanges the tails of paths i and j at v, leaving all other paths untouched. □

5.5.4 Infrastructure for the LGV proof

Lemma 5.167 (Signed sum over intersecting path tuples is zero)**.** *For any path-finite acyclic digraph D with arc weights w,*

$$\sum_{\substack{(\sigma,\mathbf{p}) \\ \mathbf{p} \text{ intersecting}}} (-1)^\sigma\, w(\mathbf{p}) = 0.$$

Proof. Apply the sign-reversing involution φ as a fixed-point-free involution. Each pair $\{(\sigma, \mathbf{p}),\ \varphi(\sigma, \mathbf{p})\}$ contributes $(-1)^\sigma w(\mathbf{p}) + (-(-1)^\sigma) w(\mathbf{p}) = 0$. □

Lemma 5.168 (Determinant equals sum over all path tuples with permutation)**.** *For any path-finite digraph D with arc weights w,*

$$\det(M) = \sum_{(\sigma,\mathbf{p})} (-1)^\sigma\, w(\mathbf{p}),$$

where the sum ranges over all pairs of a permutation σ and a path tuple $\mathbf{p}$ from $\mathbf{A}$ to $\sigma(\mathbf{B})$.

Proof. Expand $\det(M)$ by the Leibniz formula. Each summand $(-1)^\sigma \prod_i M_{i,\sigma(i)}$ expands (using the definition $M_{i,j} = \sum_{p:A_i \to B_j} w(p)$ and distributivity) into a sum over all path tuples from $\mathbf{A}$ to $\sigma(\mathbf{B})$. Combine over all σ to obtain the sum over all pairs $(\sigma, \mathbf{p})$. □

Lemma 5.169 (Nipat sum over permutations equals weighted nipat sum)**.** *The sum over non-intersecting path tuples with permutation equals*

$$\sum_{\sigma \in S_k} (-1)^\sigma \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p}).$$

Proof. Decompose the set of non-intersecting path tuples with permutation as a disjoint union over permutations σ and nipats from $\mathbf{A}$ to $\sigma(\mathbf{B})$, then factor out $(-1)^\sigma$ from each inner sum. $\square$

Lemma 5.170 (Product of path weight sums equals sum over path tuples)**.** *For a path-finite digraph D with arc weights w,*

$$\prod_{j=1}^{k} \left(\sum_{p: A_j \to B_j} w(p) \right) = \sum_{\substack{\mathbf{p} \text{ is a path tuple} \\ \text{from } \mathbf{A} \text{ to } \mathbf{B}}} w(\mathbf{p}).$$

Proof. Expand the finite product of finite sums using distributivity. The resulting index set bijects with all path tuples (choosing one path for each component). $\square$

5.5.5 The nonpermutable case

Lemma 5.171 (Discrete intermediate value theorem)**.** *Let $f : \mathbb{Z} \to \mathbb{Z}$ satisfy $|f(n+1) - f(n)| \le 1$ for all $n \in [a, b)$. If $f(a) \ge 0$ and $f(b) \le 0$, then there exists $c \in [a, b]$ with $f(c) = 0$.*

Proof. By induction on $b - a$. If $f(a) = 0$, take $c = a$. Otherwise, $f(a) > 0$ and the step bound gives $f(a+1) \ge 0$. If $f(a+1) \le 0$, take $c = a + 1$; otherwise apply the induction hypothesis to $[a + 1, b]$. $\square$

Lemma 5.172 (Baby Jordan curve theorem for lattice paths)**.** *Let $A, A', B, B' \in \mathbb{Z}^2$ with $\mathrm{x}(A') \le \mathrm{x}(A)$, $\mathrm{y}(A) \le \mathrm{y}(A')$, $\mathrm{x}(B') \le \mathrm{x}(B)$, and $\mathrm{y}(B) \le \mathrm{y}(B')$. If p is a path from A to B' and p' is a path from A' to B, then p and p' share a vertex.*

Proof. Consider the "coordinate sum" $s = x + y$ along each path. The range of s-values for p is $[A_x + A_y,\ B'_x + B'_y]$, and for p' it is $[A'_x + A'_y,\ B_x + B_y]$. These ranges overlap. At the low end of the overlap, the x-coordinate of p is $\ge$ that of p'; at the high end, it is $\le$. Since the x-coordinate changes by at most 1 per step, a discrete intermediate value argument (Lemma 5.171) yields a value of s where p and p' have the same x-coordinate. Since they also have the same coordinate sum s, the y-coordinates agree as well, so they share that vertex. $\square$

Lemma 5.173 (Monotone permutation is identity)**.** *If $\sigma \in S_k$ satisfies $\sigma(i) \le \sigma(j)$ whenever $i < j$, then $\sigma = \mathrm{id}$.*

Proof. By strong induction. If σ is monotone, then for the smallest element i we have $\sigma(i) \ge i$ (otherwise injectivity is violated) and $\sigma(i) \le i$ (otherwise the preimage of i would have to be smaller, violating monotonicity). Hence $\sigma(i) = i$, and the claim follows by induction. $\square$

Lemma 5.174 (No nipats for non-identity permutations under sorting)**.** *Under the sorting conditions of Corollary 5.175, if $\sigma \in S_k$ is not the identity permutation, then there are no non-intersecting path tuples from $\mathbf{A}$ to $\sigma(\mathbf{B})$.*

Proof. Since $\sigma \ne \mathrm{id}$, by Lemma 5.173 there exists $i \in [k-1]$ with $\sigma(i) > \sigma(i+1)$. The sorting conditions then satisfy the hypotheses of Lemma 5.172 for paths $p_i : A_i \to B_{\sigma(i+1)}$ and $p_{i+1} : A_{i+1} \to B_{\sigma(i)}$, so these two paths must intersect. Hence no nipat exists. $\square$

Corollary 5.175 (LGV lemma, nonpermutable lattice weight version)**.** *Consider the setting of Theorem 5.154, but additionally assume that*

$$\mathrm{x}(A_1) \geq \mathrm{x}(A_2) \geq \cdots \geq \mathrm{x}(A_k); \tag{5.6}$$

$$\mathrm{y}(A_1) \leq \mathrm{y}(A_2) \leq \cdots \leq \mathrm{y}(A_k); \tag{5.7}$$

$$\mathrm{x}(B_1) \geq \mathrm{x}(B_2) \geq \cdots \geq \mathrm{x}(B_k); \tag{5.8}$$

$$\mathrm{y}(B_1) \leq \mathrm{y}(B_2) \leq \cdots \leq \mathrm{y}(B_k). \tag{5.9}$$

Here, $\mathrm{x}(P)$ *and* $\mathrm{y}(P)$ *denote the two coordinates of any point* $P \in \mathbb{Z}^2$*.*

Then, there are no nipats from $\mathbf{A}$ *to* $\sigma(\mathbf{B})$ *when* $\sigma \in S_k$ *is not the identity permutation* $\mathrm{id} \in S_k$*. Therefore, the claim of Theorem 5.154 simplifies to*

$$\det\left(\left(\sum_{p:A_i \to B_j} w(p)\right)_{1\leq i\leq k,\ 1\leq j\leq k}\right) = \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \mathbf{B}}} w(\mathbf{p}). \tag{5.10}$$

Proof. Let $\sigma \in S_k$ be a permutation that is not the identity permutation $\mathrm{id} \in S_k$. By Lemma 5.174, there are no nipats from $\mathbf{A}$ to $\sigma(\mathbf{B})$, so

$$\sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p}) = 0.$$

Now, Theorem 5.154 yields

$$\begin{aligned}
&\det\left(\left(\sum_{p:A_i \to B_j} w(p)\right)_{1\leq i\leq k,\ 1\leq j\leq k}\right) \\
&= \sum_{\sigma\in S_k} (-1)^{\sigma} \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p}) \\
&= \underbrace{(-1)^{\mathrm{id}}}_{=1} \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \mathbf{B}}} w(\mathbf{p}) + \sum_{\substack{\sigma\in S_k \\ \sigma\neq\mathrm{id}}} (-1)^{\sigma} \underbrace{\sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \sigma(\mathbf{B})}} w(\mathbf{p})}_{=0} \\
&= \sum_{\substack{\mathbf{p} \text{ is a nipat} \\ \text{from } \mathbf{A} \text{ to } \mathbf{B}}} w(\mathbf{p}).
\end{aligned}$$

$\square$

5.5.6 Applications

Nonnegativity of binomial coefficient determinants

Lemma 5.176 (FKG inequality for binomial coefficients)**.** *Let* a, b, c, d *be nonnegative integers with* $a \geq b$ *and* $c \geq d$*. Then*

$$\binom{a}{d}\binom{b}{c} \leq \binom{a}{c}\binom{b}{d}.$$

Proof. By induction on $c - d$. The base case $c = d$ is trivial. For the inductive step, use the identity $\binom{n}{m+1}(m+1) = \binom{n}{m}(n-m)$ to reduce the $(c+1)$ case to the c case, together with the inequality $b - m \leq a - m$ (from $b \leq a$). $\square$

Lemma 5.177 (Lattice path count equals binomial coefficient)**.** *The number of lattice paths in* $\mathbb{Z}^2$ *from* $(0,-a)$ *to* $(b,-b)$ *equals* $\binom{a}{b}$*.*

Proof. A path from $(0,-a)$ to $(b,-b)$ has exactly a steps (the coordinate sum $x+y$ increases from $-a$ to 0), of which exactly b are east-steps (the x-coordinate increases from 0 to b). Choosing which b of the a steps are east-steps gives a bijection with b-element subsets of $[a]$, so the count is $\binom{a}{b}$. □

Lemma 5.178 (Lattice paths biject to subsets)**.** *There is a bijection between paths from* $(0,-a)$ *to* $(b,-b)$ *in the integer lattice and* b*-element subsets of* $\{1,\ldots,a\}$*.*

Proof. A path is uniquely determined by the positions of its east-steps among all a steps. This gives a bijection to b-element subsets of $\{1,\ldots,a\}$. □

Helpers for Corollary 5.181

Lemma 5.179 (Binomial matrix equals path weight matrix)**.** *With* $A_i = (0,-a_i)$*,* $B_j = (b_j,-b_j)$*, and unit arc weights, the path weight matrix equals the matrix* $\left(\binom{a_i}{b_j}\right)$*.*

Proof. The (i,j) entry of the path weight matrix (with unit weights) is the number of paths from A_i to B_j, which by Lemma 5.177 equals $\binom{a_i}{b_j}$. □

Lemma 5.180 (Signed ipat count is zero (unit weight case))**.** *With unit arc weights on the integer lattice,*

$$\sum_{\sigma\in S_k}(-1)^{\sigma}\left|\{\mathbf{p}\ \text{ipat from }\mathbf{A}\text{ to }\sigma(\mathbf{B})\}\right| = 0.$$

Proof. This is the specialization of the signed ipat cancellation (Lemma 5.167) to unit weights, where the weight of each path tuple is 1 and the sum reduces to a signed cardinality count. □

Corollary 5.181. *Let* $k\in\mathbb{N}$*. Let* $a_1,a_2,\ldots,a_k$ *and* $b_1,b_2,\ldots,b_k$ *be nonnegative integers such that*

$$a_1\geq a_2\geq\cdots\geq a_k \qquad \text{and} \qquad b_1\geq b_2\geq\cdots\geq b_k.$$

Then,

$$\det\left(\left(\binom{a_i}{b_j}\right)_{1\leq i\leq k,\ 1\leq j\leq k}\right)\geq 0.$$

Proof. Set $K=\mathbb{Z}$ and $w(a):=1$ for each arc a of $\mathbb{Z}^2$. Define the lattice points

$$A_i := (0,-a_i) \qquad \text{and} \qquad B_i := (b_i,-b_i)$$

for all $i\in[k]$. These lattice points satisfy the assumptions of Corollary 5.175. Hence, (5.10) entails

$$\det\Big((\#\text{ of paths from }A_i\text{ to }B_j)_{1\leq i\leq k,\ 1\leq j\leq k}\Big) = \#\text{ of nipats from }\mathbf{A}\text{ to }\mathbf{B}.$$

Using Proposition 5.129, the matrix on the left hand side is $\left(\binom{a_i}{b_j}\right)_{1\leq i\leq k,\ 1\leq j\leq k}$. The right hand side is ≥ 0, so the determinant is ≥ 0.

In Lean, the cases $k=0$ and $k=1$ are proved directly (empty and 1×1 determinants), $k=2$ uses the FKG inequality (Lemma 5.176), and $k\geq 3$ uses the full LGV machinery. □

Catalan Hankel determinant

Lemma 5.182 (Dyck digraph is acyclic)**.** *The Dyck digraph (with vertex set* $\mathbb{Z} \times \mathbb{N}$ *and arcs* $(i, j) \to (i + 1, j + 1)$ *and* $(i, j) \to (i + 1, j - 1)$ *for* $j > 0$*) is acyclic.*

Proof. Every arc increases the first coordinate by 1, so any path is strictly increasing in the first coordinate. Hence there can be no cycle. ☐

Lemma 5.183 (Dyck digraph is path-finite)**.** *The Dyck digraph is path-finite.*

Proof. Since the first coordinate increases by exactly 1 on each arc, a path of length n visits vertices whose first coordinates form a consecutive sequence. The second coordinate is bounded by the first coordinate difference, so there are only finitely many possible vertex sequences. ☐

Lemma 5.184 (Dyck paths count equals Catalan number)**.** *The number of paths from* $(0, 0)$ *to* $(2n, 0)$ *in the Dyck digraph equals the Catalan number* c_n*.*

Proof. A path from $(0, 0)$ to $(2n, 0)$ in the Dyck digraph consists of $2n$ steps, each going either up-right or down-right, and never going below the x-axis. Such a path is exactly a Dyck path of semilength n. The bijection with Dyck words (Lemma 5.185) establishes the correspondence with the Catalan number. ☐

Helpers for the Dyck path–DyckWord bijection

Lemma 5.185 (Dyck path to DyckWord bijection)**.** *There is a bijection between paths from* $(0, 0)$ *to* $(2n, 0)$ *in the Dyck digraph and Dyck words of semilength* n*. The forward direction converts each arc into an up or down step; the backward direction constructs a vertex list from a Dyck word.*

Proof. The forward map sends each arc $(x, y) \to (x + 1, y + 1)$ to an up-step and each arc $(x, y) \to (x+1, y-1)$ to a down-step, verifying that the resulting list of steps forms a well-formed Dyck word. The backward map constructs vertices by scanning the Dyck word and accumulating y-coordinates. Round-trip equalities (the two maps are mutual inverses) are verified by induction on the step list. ☐

Lemma 5.186 (Shifted Dyck paths count equals Catalan number)**.** *The number of paths from* $(-2i, 0)$ *to* $(2j, 0)$ *in the Dyck digraph equals* c_{i+j}*.*

Proof. By translation invariance (Lemma 5.187), the set of paths from $(-2i, 0)$ to $(2j, 0)$ bijects with paths from $(0, 0)$ to $(2(i + j), 0)$, which has cardinality c_{i+j} by Lemma 5.184. ☐

Lemma 5.187 (Translation is a bijection on Dyck paths)**.** *For any integer* d*, translation by* $(d, 0)$ *is a bijection on paths in the Dyck digraph: if* p *is a path from* A *to* B*, then translating each vertex by* $(d, 0)$ *gives a path from* $A+(d, 0)$ *to* $B+(d, 0)$*. The map is injective and surjective (the inverse translates by* $(-d, 0)$*).*

Proof. Each arc $(x, y) \to (x + 1, y \pm 1)$ is preserved by horizontal translation, so the translated vertex list forms a valid path. Injectivity follows from injectivity of the translation map on vertices. Surjectivity follows from the inverse translation. ☐

Lemma 5.188 (Catalan Hankel matrix equals path weight matrix)**.** *The Catalan Hankel matrix* $(c_{i+j})_{0\le i,j<k}$ *equals the path weight matrix of the Dyck digraph with unit arc weights and* k*-vertices* $A_i = (-2i, 0)$*,* $B_j = (2j, 0)$*.*

Proof. The (i, j) entry of the path weight matrix (with unit weights) counts paths from $(-2i, 0)$ to $(2j, 0)$, which equals c_{i+j} by Lemma 5.186. ☐

Helpers for the unique nipat (Catalan case)

Lemma 5.189 (Nested Dyck path construction)**.** *For each $i \in \mathbb{N}$, define the* nested Dyck path *γ_i as the path in the Dyck digraph from $(-2i, 0)$ to $(2i, 0)$ that goes straight up (northeast) from $(-2i, 0)$ to $(0, 2i)$ and then straight down (southeast) from $(0, 2i)$ to $(2i, 0)$. This path has $4i$ arcs and visits the vertices $(j - 2i,\ \min(j,\ 4i - j))$ for $j = 0, 1, \ldots, 4i$.*

Proof. The vertex list is constructed explicitly. The start and finish are verified by evaluating the vertex list at positions 0 and $4i$. Each consecutive pair of vertices forms a valid Dyck arc (either up-right or down-right, with $y \geq 0$). $\square$

Lemma 5.190 (Nested Dyck paths are disjoint)**.** *For $i \neq j$, the nested Dyck paths γ_i and γ_j share no vertices.*

Proof. A vertex on γ_i at position m has $y = \min(m, 4i - m)$. At any shared x-coordinate, the corresponding positions yield different y-coordinates when $i \neq j$, because the maximum height $2i$ differs. $\square$

Lemma 5.191 (Dyck path parity constraint)**.** *For any path p in the Dyck digraph, the y-coordinate at step m has the same parity as $y_{\text{start}} + m$, where y_{start} is the y-coordinate of the starting vertex. (Each arc changes y by ± 1, so the parity alternates.)*

Proof. By induction on m. Each arc changes y by ± 1, flipping the parity. $\square$

Lemma 5.192 (Dyck path height upper bound)**.** *For a path p in the Dyck digraph from $(-2i, 0)$ to $(2i, 0)$, the y-coordinate at step m satisfies $y_m \leq \min(m,\ 4i - m)$.*

Proof. Each step changes y by at most $+1$, so $y_m \leq m$. Symmetrically, y must return to 0 in $4i - m$ remaining steps, so $y_m \leq 4i - m$. $\square$

Lemma 5.193 (Nested Dyck paths form the unique nipat)**.** *Define k-vertices $A_i := (-2(i-1), 0)$ and $B_i := (2(i - 1), 0)$ for $i \in [k]$. There is exactly one non-intersecting path tuple from **A** to **B** in the Dyck digraph: the* nested *Dyck paths, where path i goes from $(-2i, 0)$ up to $(0, 2i)$ and back down to $(2i, 0)$.*

Proof. **Existence and non-intersection:** The nested Dyck path for index i visits only vertices (x, y) with $|x| \leq 2i$ and $y \leq 2i$, and at the midpoint $x = 0$ it reaches height exactly $2i$. Two nested paths for indices $i \neq j$ are disjoint because they reach different maximum heights (Lemma 5.190).

Uniqueness: Any nipat must equal the nested one. At the midpoint of path i (step $2i$), the y-coordinate must be exactly $2i$: it is at most $2i$ (by Lemma 5.192), has the correct parity (by Lemma 5.191), and cannot equal $2m$ for any $m < i$ because that would force a shared vertex with the (inductively determined) path m. Once the midpoint heights are fixed, the entire path is forced (it must go straight up, then straight down) by parity and the non-intersection constraint.

For $k \geq 2$, the full proof uses strong induction on the path index. $\square$

Corollary 5.194. *Let $k \in \mathbb{N}$. Recall the Catalan numbers $c_n = \frac{1}{n+1}\binom{2n}{n}$ for all $n \in \mathbb{N}$. Then,*

$$\det\Big(\left(c_{i+j-2}\right)_{1 \leq i \leq k,\ 1 \leq j \leq k}\Big) = \det \begin{pmatrix} c_0 & c_1 & \cdots & c_{k-1} \\ c_1 & c_2 & \cdots & c_k \\ \vdots & \vdots & \ddots & \vdots \\ c_{k-1} & c_k & \cdots & c_{2k-2} \end{pmatrix} = 1.$$

Proof. We use not the lattice $\mathbb{Z}^2$, but a different digraph: the simple digraph with vertex set $\mathbb{Z} \times \mathbb{N}$ and arcs $(i, j) \to (i + 1, j + 1)$ for all $(i, j) \in \mathbb{Z} \times \mathbb{N}$ and $(i, j) \to (i + 1, j - 1)$ for all $(i, j) \in \mathbb{Z} \times \mathbb{P}$. It is easy to see that this digraph is acyclic and path-finite.

Set $K = \mathbb{Z}$ and $w(a) = 1$ for each arc a. Define $A_i := (-2(i - 1), 0)$ and $B_i := (2(i - 1), 0)$ for $i \in [k]$.

The Catalan number c_n counts the paths from $(0, 0)$ to $(2n, 0)$ on this digraph (Dyck paths). Hence c_n also counts the paths from $(i, 0)$ to $(2n + i, 0)$ for any $i \in \mathbb{N}$. By Lemma 5.184, the number of paths from A_i to B_j is c_{i+j-2}. Hence the path weight matrix is the Catalan Hankel matrix (Lemma 5.188).

It is not hard to show (Lemma 5.193) that there is only one nipat from $\mathbf{A}$ to $\mathbf{B}$. Moreover, there are no nipats from $\mathbf{A}$ to $\sigma(\mathbf{B})$ for $\sigma \neq \mathrm{id}$ (by an argument analogous to Corollary 5.175). Hence, by Theorem 5.155, the determinant equals 1. □

5.5.7 Bridge between path representations

Lemma 5.195 (Lattice path conversion is a bijection)**.** *There is a bijection between step-based lattice paths (lists of east/north steps starting at a given point) and vertex-based digraph paths in $\mathbb{Z}^2$ starting at that point. The forward direction computes the list of visited vertices; the backward direction recovers the step sequence from consecutive vertex pairs.*

Proof. The forward map is injective (Lemma 5.196) and surjective (Lemma 5.197), establishing the equivalence. □

Lemma 5.196 (Step-to-vertex map is injective)**.** *Different step sequences starting from the same point produce different vertex-based paths.*

Proof. By induction on the path length. If two step sequences produce the same vertex list, the second vertices must agree. Since each step (east or north) produces a distinct target from the same source, the first steps must be equal. Apply the induction hypothesis to the tails. □

Lemma 5.197 (Step-to-vertex map is surjective)**.** *Every vertex-based path in the integer lattice starting at a given point is the image of some step sequence.*

Proof. Convert the path's vertex list to a step list by recovering each step from consecutive vertex pairs: if the x-coordinate increases by 1, the step is east; otherwise it is north. The round-trip identity follows by construction. □

Lemma 5.198 (Path tuple bijection preserves intersection)**.** *A step-based lattice path tuple is intersecting if and only if the corresponding vertex-based path tuple is intersecting.*

Proof. Both representations check whether any two paths share a vertex. Since the step-to-vertex map preserves the vertex list, a vertex v appears in two step-based paths if and only if it appears in the corresponding vertex-based paths. □

Chapter 6

Symmetric Functions

6.1 Definitions and examples of symmetric polynomials

6.1.1 Setup

Convention 6.1. Fix a commutative ring K. Fix an $N \in \mathbb{N}$. Throughout this chapter, we will keep K and N fixed. Let S_N denote the symmetric group, i.e., the group of all permutations of $[N] := \{1, 2, \ldots, N\}$.

6.1.2 The polynomial ring and the symmetric group action

Definition 6.2. **(a)** Let $\mathcal{P}$ be the polynomial ring $K[x_1, x_2, \ldots, x_N]$ in N variables over K. This is not just a ring; it is a commutative K-algebra.

(b) The symmetric group S_N acts on the set $\mathcal{P}$ according to the formula

$$\sigma \cdot f = f[x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(N)}] \quad \text{for any } \sigma \in S_N \text{ and any } f \in \mathcal{P}.$$

Here, $f[a_1, a_2, \ldots, a_N]$ means the result of substituting $a_1, a_2, \ldots, a_N$ for the indeterminates $x_1, x_2, \ldots, x_N$ in a polynomial $f \in \mathcal{P}$.

Roughly speaking, the group S_N is thus acting on $\mathcal{P}$ by permuting variables: A permutation $\sigma \in S_N$ transforms a polynomial f by substituting $x_{\sigma(i)}$ for each x_i.

Note that this action of S_N on $\mathcal{P}$ is a well-defined group action (as we will see in Proposition 6.3 below).

(c) A polynomial $f \in \mathcal{P}$ is said to be *symmetric* if it satisfies

$$\sigma \cdot f = f \quad \text{for all } \sigma \in S_N.$$

(d) We let $\mathcal{S}$ be the set of all symmetric polynomials $f \in \mathcal{P}$.

6.1.3 The action is well-defined

Proposition 6.3. *The action of S_N on $\mathcal{P}$ is a well-defined group action. In other words, the following holds:*

(a) *We have $\mathrm{id}_{[N]} \cdot f = f$ for every $f \in \mathcal{P}$.*

(b) *We have $(\sigma\tau) \cdot f = \sigma \cdot (\tau \cdot f)$ for every $\sigma, \tau \in S_N$ and $f \in \mathcal{P}$.*

Proof. **(a)** If $f \in \mathcal{P}$, then the definition of $\mathrm{id}_{[N]} \cdot f$ yields

$$\mathrm{id}_{[N]} \cdot f = f[x_{\mathrm{id}(1)}, x_{\mathrm{id}(2)}, \ldots, x_{\mathrm{id}(N)}] = f[x_1, x_2, \ldots, x_N] = f.$$

This proves part **(a)**.

(b) Let $\sigma, \tau \in S_N$ and $f \in \mathcal{P}$. The definition of $\sigma \cdot (\tau \cdot f)$ yields

$$\sigma \cdot (\tau \cdot f) = (\tau \cdot f)[x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(N)}]. \tag{6.1}$$

Write the polynomial f in the form

$$f = \sum_{(a_1, a_2, \ldots, a_N) \in \mathbb{N}^N} f_{(a_1, a_2, \ldots, a_N)} x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}. \tag{6.2}$$

The definition of $\tau \cdot f$ yields

$$\tau \cdot f = \sum_{(a_1, \ldots, a_N) \in \mathbb{N}^N} f_{(a_1, \ldots, a_N)} x_{\tau(1)}^{a_1} x_{\tau(2)}^{a_2} \cdots x_{\tau(N)}^{a_N}.$$

Substituting $x_{\sigma(1)}, x_{\sigma(2)}, \ldots, x_{\sigma(N)}$ for $x_1, x_2, \ldots, x_N$ on both sides, we obtain

$$\begin{aligned}(\tau \cdot f)[x_{\sigma(1)}, \ldots, x_{\sigma(N)}] &= \sum f_{(a_1, \ldots, a_N)} x_{\sigma(\tau(1))}^{a_1} \cdots x_{\sigma(\tau(N))}^{a_N} \\ &= \sum f_{(a_1, \ldots, a_N)} x_{(\sigma\tau)(1)}^{a_1} \cdots x_{(\sigma\tau)(N)}^{a_N} = (\sigma\tau) \cdot f.\end{aligned}$$

Comparing with (6.1), we obtain $(\sigma\tau) \cdot f = \sigma \cdot (\tau \cdot f)$. ☐

6.1.4 The action is by algebra automorphisms

Proposition 6.4. *The group S_N acts on $\mathcal{P}$ by K-algebra automorphisms. In other words, for each $\sigma \in S_N$, the map*

$$\begin{aligned}\mathcal{P} &\to \mathcal{P}, \\ f &\mapsto \sigma \cdot f\end{aligned}$$

is a K-algebra automorphism of $\mathcal{P}$.

Proof. Fix $\sigma \in S_N$. For any $f, g \in \mathcal{P}$, we have

$$\sigma \cdot (fg) = (\sigma \cdot f) \cdot (\sigma \cdot g) \tag{6.3}$$

(by expanding the substitution definition of the action). Thus, the map $f \mapsto \sigma \cdot f$ respects multiplication. Similarly, this map respects addition, respects scaling, respects the zero and respects the unity. Hence, this map is a K-algebra morphism from $\mathcal{P}$ to $\mathcal{P}$. Furthermore, this map is invertible, since its inverse is the map $f \mapsto \sigma^{-1} \cdot f$. Thus, this map is a K-algebra automorphism of $\mathcal{P}$. ☐

6.1.5 Symmetric polynomials form a subalgebra

Theorem 6.5. *The subset $\mathcal{S}$ is a K-subalgebra of $\mathcal{P}$.*

Proof. We need to show that $\mathcal{S}$ is closed under addition, multiplication and scaling, and that $\mathcal{S}$ contains the zero and the unity of $\mathcal{P}$. Let us show that $\mathcal{S}$ is closed under multiplication (since all the other claims are equally easy): Let $f, g \in \mathcal{S}$. We must show that $fg \in \mathcal{S}$.

The polynomial f is symmetric (since $f \in \mathcal{S}$); in other words, $\sigma \cdot f = f$ for each $\sigma \in S_N$. Similarly, $\sigma \cdot g = g$ for each $\sigma \in S_N$. Now, from (6.3), we see that for each $\sigma \in S_N$, we have $\sigma \cdot (fg) = (\sigma \cdot f) \cdot (\sigma \cdot g) = fg$. This shows that fg is symmetric, i.e., $fg \in \mathcal{S}$. The other claims are proved similarly. $\square$

Definition 6.6. The K-subalgebra $\mathcal{S}$ of $\mathcal{P}$ is called the *ring of symmetric polynomials in N variables over K.*

6.1.6 Monomials

Definition 6.7. **(a)** A *monomial* is an expression of the form $x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}$ with $a_1, a_2, \ldots, a_N \in \mathbb{N}$.

(b) The *degree* $\deg \mathfrak{m}$ of a monomial $\mathfrak{m} = x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}$ is defined to be $a_1 + a_2 + \cdots + a_N \in \mathbb{N}$.

(c) A monomial $\mathfrak{m} = x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}$ is said to be *squarefree* if $a_1, a_2, \ldots, a_N \in \{0, 1\}$. (This is saying that no square or higher power of an indeterminate appears in $\mathfrak{m}$; thus the name "squarefree".)

(d) A monomial $\mathfrak{m} = x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}$ is said to be *primal* if there is at most one $i \in [N]$ satisfying $a_i > 0$. (This is saying that the monomial $\mathfrak{m}$ contains no two distinct indeterminates. Thus, a primal monomial is just 1 or a power of an indeterminate.)

6.1.7 Elementary, complete homogeneous, and power sum symmetric polynomials

Definition 6.8. **(a)** For each $n \in \mathbb{Z}$, define a symmetric polynomial $e_n \in \mathcal{S}$ by

$$e_n = \sum_{\substack{(i_1, i_2, \ldots, i_n) \in [N]^n; \\ i_1 < i_2 < \cdots < i_n}} x_{i_1} x_{i_2} \cdots x_{i_n} = (\text{sum of all squarefree monomials of degree } n).$$

This e_n is called the n-th *elementary symmetric polynomial* in $x_1, x_2, \ldots, x_N$.

(b) For each $n \in \mathbb{Z}$, define a symmetric polynomial $h_n \in \mathcal{S}$ by

$$h_n = \sum_{\substack{(i_1, i_2, \ldots, i_n) \in [N]^n; \\ i_1 \le i_2 \le \cdots \le i_n}} x_{i_1} x_{i_2} \cdots x_{i_n} = (\text{sum of all monomials of degree } n).$$

This h_n is called the n-th *complete homogeneous symmetric polynomial* in $x_1, x_2, \ldots, x_N$.

(c) For each $n \in \mathbb{Z}$, define a symmetric polynomial $p_n \in \mathcal{S}$ by

$$p_n = \begin{cases} x_1^n + x_2^n + \cdots + x_N^n, & \text{if } n > 0; \\ 1, & \text{if } n = 0; \\ 0, & \text{if } n < 0 \end{cases}$$
$$= (\text{sum of all primal monomials of degree } n).$$

This p_n is called the n-th *power sum* in $x_1, x_2, \ldots, x_N$.

6.1.8 Elementary symmetric polynomials vanish for large degree

Proposition 6.9. *For each integer $n > N$, we have $e_n = 0$.*

Proof. Let $n > N$ be an integer. Then, the set $[N]$ has no n distinct elements. Thus, there exists no n-tuple $(i_1, i_2, \ldots, i_n) \in [N]^n$ satisfying $i_1 < i_2 < \cdots < i_n$. Now, the definition of e_n yields

$$e_n = \sum_{\substack{(i_1,\ldots,i_n)\in[N]^n; \\ i_1<\cdots<i_n}} x_{i_1} \cdots x_{i_n} = (\text{empty sum}) = 0.$$

□

6.1.9 Generating function identities

Proposition 6.10. ***(a)*** *In the polynomial ring $\mathcal{P}[t]$, we have*

$$\prod_{i=1}^{N} (1 - tx_i) = \sum_{n\in\mathbb{N}} (-1)^n t^n e_n.$$

(b) *In the polynomial ring $\mathcal{P}[u, v]$, we have*

$$\prod_{i=1}^{N} (u - vx_i) = \sum_{n=0}^{N} (-1)^n u^{N-n} v^n e_n.$$

(c) *In the FPS ring $\mathcal{P}[[t]]$, we have*

$$\prod_{i=1}^{N} \frac{1}{1 - tx_i} = \sum_{n\in\mathbb{N}} t^n h_n.$$

Proof. **(a)** For each $i \in \{1, 2, \ldots, N\}$, we have $1 + tx_i = \sum_{a\in\{0,1\}} (tx_i)^a$. Multiplying these equalities over all $i \in \{1, 2, \ldots, N\}$, we obtain

$$\begin{aligned}
\prod_{i=1}^{N} (1 + tx_i) &= \sum_{(a_1,\ldots,a_N)\in\{0,1\}^N} t^{a_1+\cdots+a_N} x_1^{a_1} \cdots x_N^{a_N} \\
&= \sum_{\mathfrak{m} \text{ squarefree}} t^{\deg \mathfrak{m}} \mathfrak{m} = \sum_{n\in\mathbb{N}} t^n e_n.
\end{aligned}$$

Substituting $-t$ for t gives the stated identity.

(b) This is similar to part **(a)**, using two variables u and v.

(c) For each $i \in \{1, 2, \ldots, N\}$, we have $\frac{1}{1-tx_i} = \sum_{a\in\mathbb{N}} (tx_i)^a$. Multiplying these equalities over all $i \in \{1, 2, \ldots, N\}$, we obtain

$$\begin{aligned}
\prod_{i=1}^{N} \frac{1}{1 - tx_i} &= \sum_{(a_1,\ldots,a_N)\in\mathbb{N}^N} t^{a_1+\cdots+a_N} x_1^{a_1} \cdots x_N^{a_N} \\
&= \sum_{\mathfrak{m} \text{ monomial}} t^{\deg \mathfrak{m}} \mathfrak{m} = \sum_{n\in\mathbb{N}} t^n h_n.
\end{aligned}$$

□

Helpers for the generating function proofs

Lemma 6.11. *For each i, the geometric series identity $(1 - tx_i) \cdot \sum_{k\geq 0}(tx_i)^k = 1$ holds in the power series ring $\mathcal{P}[[t]]$. Taking the product over all $i \in [N]$, we obtain the generating function identity $E(t) \cdot H(t) = 1$, where $E(t) = \prod_{i=1}^{N}(1 - tx_i)$ and $H(t) = \prod_{i=1}^{N} \sum_{k\geq 0}(tx_i)^k$.*

Proof. The identity $(1-tx_i)\cdot\sum_{k\geq 0}(tx_i)^k = 1$ is proved by extracting coefficients: the coefficient of t^n on the left is $x_i^n - x_i \cdot x_i^{n-1} = 0$ for $n > 0$ and 1 for $n = 0$. The product identity follows from $\prod_i(1 - tx_i) \cdot \prod_i \sum_k(tx_i)^k = \prod_i 1 = 1$. $\square$

6.1.10 Newton–Girard formulas

Theorem 6.12 (Newton–Girard formulas)**.** *For any positive integer n, we have*

$$\sum_{j=0}^{n}(-1)^j e_j h_{n-j} = 0; \tag{6.4}$$

$$\sum_{j=1}^{n}(-1)^{j-1} e_{n-j} p_j = ne_n; \tag{6.5}$$

$$\sum_{j=1}^{n} h_{n-j} p_j = nh_n. \tag{6.6}$$

Proof. We prove formulas (6.4) and (6.6); the formula (6.5) can be derived by differentiating the generating function $\prod_{i=1}^{N}(1 - tx_i) = \sum_n(-1)^n t^n e_n$ with respect to t, and comparing coefficients (see the source for details).

Proof of (6.4)**:** In the FPS ring $\mathcal{P}[[t]]$, by Proposition 6.10 (a) and (c) we have

$$\prod_{i=1}^{N}(1 - tx_i) = \sum_{n\in\mathbb{N}}(-1)^n t^n e_n$$

and

$$\prod_{i=1}^{N} \frac{1}{1 - tx_i} = \sum_{n\in\mathbb{N}} t^n h_n.$$

Multiplying these two equalities, the left-hand side becomes

$$\left(\prod_{i=1}^{N}(1 - tx_i)\right)\left(\prod_{i=1}^{N} \frac{1}{1 - tx_i}\right) = \prod_{i=1}^{N} 1 = 1.$$

The right-hand side product is $\sum_{n\in\mathbb{N}} \left(\sum_{j=0}^{n}(-1)^j e_j h_{n-j}\right) t^n$. Comparing coefficients of t^n for $n > 0$, we obtain (6.4).

Proof of (6.6)**:** The proof is by a counting argument on monomials. Each monomial in h_n corresponds to a multiset s of size n from $[N]$. On the left-hand side, each such monomial appears exactly n times: once for each way to pick an element i in the support of s and a positive integer $j \leq \operatorname{count}(i, s)$, contributing to the term $h_{n-j}p_j$. For each s, the total count is $\sum_{i=1}^{N} \operatorname{count}(i, s) = n$, matching the right-hand side. $\square$

6.1.11 Fundamental Theorem of Symmetric Polynomials

Theorem 6.13 (Fundamental Theorem of Symmetric Polynomials)**.** ***(a)*** *The elementary symmetric polynomials $e_1, e_2, \ldots, e_N$ are algebraically independent (over K) and generate the K-algebra $\mathcal{S}$.*

In other words, each $f \in \mathcal{S}$ can be uniquely written as a polynomial in $e_1, e_2, \ldots, e_N$.

In yet other words, the map

$$K[y_1, y_2, \ldots, y_N] \to \mathcal{S},$$
$$g \mapsto g[e_1, e_2, \ldots, e_N]$$

is a K-algebra isomorphism.

(b) *The complete homogeneous symmetric polynomials $h_1, h_2, \ldots, h_N$ are algebraically independent (over K) and generate the K-algebra $\mathcal{S}$.*

In other words, each $f \in \mathcal{S}$ can be uniquely written as a polynomial in $h_1, h_2, \ldots, h_N$.

In yet other words, the map

$$K[y_1, y_2, \ldots, y_N] \to \mathcal{S},$$
$$g \mapsto g[h_1, h_2, \ldots, h_N]$$

is a K-algebra isomorphism.

(c) *Now assume that K is a commutative $\mathbb{Q}$-algebra (e.g., a field of characteristic 0). Then, the power sums $p_1, p_2, \ldots, p_N$ are algebraically independent (over K) and generate the K-algebra $\mathcal{S}$.*

In other words, each $f \in \mathcal{S}$ can be uniquely written as a polynomial in $p_1, p_2, \ldots, p_N$.

In yet other words, the map

$$K[y_1, y_2, \ldots, y_N] \to \mathcal{S},$$
$$g \mapsto g[p_1, p_2, \ldots, p_N]$$

is a K-algebra isomorphism.

Proof. Various proofs can be found in [Benede25, Theorem 171], [BluCos16, proof of Theorem 1], [Dumas08, Theorem 1.2.1], [Goodma15, Theorem 9.6.6], [Smith95, §1.1].

Part (a): Part (a) is proved by constructing the evaluation map $K[y_1, \ldots, y_N] \to \mathcal{S}$ sending $y_k \mapsto e_k$ and showing it is a K-algebra isomorphism. Algebraic independence follows from injectivity; generation follows from surjectivity.

Part (b): The proof uses the Newton–Girard relations to show that each e_k (for $k \leq N$) can be expressed as a polynomial in $h_1, \ldots, h_N$, by strong induction on k. The key identity is: from (6.4), $(-1)^{k+1}e_{k+1} = -\sum_{j=0}^{k}(-1)^j e_j h_{k+1-j}$, where by the induction hypothesis $e_0, \ldots, e_k$ are already in the range of the evaluation map at $h_1, \ldots, h_N$. The algebraic independence of the h_i then follows by a transcendence degree argument: the composition $K[y_1, \ldots, y_N] \to \mathcal{S} \xrightarrow{\sim} K[y_1, \ldots, y_N]$ (via the isomorphism from part (a)) is a surjective endomorphism, hence bijective since polynomial rings over integral domains have finite transcendence degree.

Note: The Lean proof of part (b) requires the additional hypothesis that K is an integral domain.

Part (c): The argument is analogous to part (b). Newton's identity (6.5) gives $n \cdot e_n = \sum_{j=1}^{n}(-1)^{j-1}e_{n-j}p_j$. Since K is a $\mathbb{Q}$-algebra, n is invertible in K, so we can solve for e_n as a polynomial in $e_0, \ldots, e_{n-1}$ and $p_1, \ldots, p_n$. By strong induction, each e_k lies in the range of the evaluation map at $p_1, \ldots, p_N$. Algebraic independence again follows by the transcendence degree argument.

Note: The Lean formalization of part (c) contains sorry in some helper lemmas (weight constraint and triangular structure arguments). $\square$

6.1.12 Simple transpositions suffice

Lemma 6.14. *For each $i \in [N-1]$, let $s_i \in S_N$ denote the simple transposition that swaps i and $i+1$.*

Let $f \in \mathcal{P}$. Assume that

$$s_k \cdot f = f \quad \text{for each } k \in [N-1]. \tag{6.7}$$

Then, the polynomial f is symmetric.

Proof. The proof proceeds by showing that invariance under simple transpositions implies invariance under all transpositions of the form $\operatorname{swap}(i, j)$ for any $i, j \in [N]$.

Step 1: Invariance under adjacent transpositions implies invariance under arbitrary transpositions. We prove by induction on $|j - i|$ that $\operatorname{swap}(i, j) \cdot f = f$. The base case $|j - i| = 1$ is the hypothesis (6.7). For the inductive step, use the conjugation identity: $\operatorname{swap}(i, j) = \operatorname{swap}(i, i+1) \cdot \operatorname{swap}(i+1, j) \cdot \operatorname{swap}(i, i+1)$.

Step 2: Every permutation $\sigma \in S_N$ can be written as a product of transpositions (this is a standard fact about the symmetric group). Since each transposition fixes f, so does σ. $\square$

6.1.13 Helper lemmas from the Lean formalization

The following results appear in the Lean formalization but not in the TeX source. They provide API for the definitions above.

Lemma 6.15. *A squarefree monomial has degree at most N.*

Proof. If $\mathfrak{m} = x_1^{a_1} \cdots x_N^{a_N}$ is squarefree, then each $a_i \leq 1$, so $\deg \mathfrak{m} = a_1 + \cdots + a_N \leq N$. $\square$

Lemma 6.16. *For any multiset s of elements from $[N]$ of size m, the sum of counts $\sum_{i=1}^{N} \mathrm{count}(i, s)$ equals m.*

Proof. This is the standard identity that the sum of all multiplicities in a multiset equals its cardinality. The proof splits the sum over elements that appear in s and elements that do not, the latter contributing zero. $\square$

Lemma 6.17. *The elementary symmetric polynomial e_n, the complete homogeneous symmetric polynomial h_n, and the power sum p_n are all symmetric (i.e., belong to $\mathcal{S}$) for every n.*

Proof. For e_n: permuting the variables permutes the strictly increasing n-tuples $(i_1 < \cdots < i_n)$, so the sum is unchanged. For h_n and p_n: the argument is analogous, using the fact that permuting variables permutes the non-decreasing tuples (for h_n) or the individual terms x_i^n (for p_n). $\square$

Lemma 6.18. *A monomial $\mathfrak{m}$ is primal if and only if $\mathfrak{m} = 1$ or $\mathfrak{m} = x_i^k$ for some $i \in [N]$ and some $k > 0$.*

Proof. If $\mathfrak{m}$ is primal, its support has at most one element. If the support is empty, then $\mathfrak{m} = 1$. If the support is $\{i\}$, then $\mathfrak{m} = x_i^{m_i}$ for some $m_i > 0$. The converse is immediate. $\square$

6.1.14 Monomial API

The following lemmas develop the API for monomials, connecting the exponent vector representation with the polynomial representation.

Lemma 6.19. *A monomial* $\mathfrak{m}$ *with exponent vector* $(a_1, \ldots, a_N)$ *corresponds to the polynomial* $x_1^{a_1} \cdots x_N^{a_N}$*. The polynomial corresponding to the zero exponent vector is* 1*, the polynomial corresponding to the* i*-th standard basis vector is* x_i*, and the polynomial corresponding to a sum of exponent vectors* $m_1 + m_2$ *is the product of the corresponding polynomials.*

Proof. Direct from the definition of monomials in a multivariable polynomial ring. $\square$

Lemma 6.20. *For a subset* $S \subseteq [N]$*, the monomial* $\mathfrak{m}_S = \prod_{i \in S} x_i$ *is squarefree, has degree* $|S|$*, and has support equal to* S*.*

Proof. The monomial $\mathfrak{m}_S$ is defined as $\sum_{i \in S} e_i$ where e_i is the i-th standard basis vector. Each entry is 0 or 1 (squarefree), the sum of entries equals $|S|$ (degree), and the support is exactly S. All properties are proved by induction on $|S|$. $\square$

Lemma 6.21. *The degree of the product of two monomials is the sum of their degrees:* $\deg(\mathfrak{m}_1 \cdot \mathfrak{m}_2) = \deg \mathfrak{m}_1 + \deg \mathfrak{m}_2$*.*

Proof. The degree is defined as the sum of all exponents. Since the exponent vector of $\mathfrak{m}_1 \cdot \mathfrak{m}_2$ is the pointwise sum of the exponent vectors, the degree of the product equals the sum of the degrees. $\square$

6.1.15 Basic values of e_n, h_n, p_n

Lemma 6.22. *We have the following basic values:*

- $e_0 = h_0 = 1$*;*
- $e_1 = h_1 = p_1 = \sum_{i=1}^N x_i$*;*
- $p_0 = N$ *(the number of variables; note this differs from the textbook convention* $p_0 = 1$*).*

Proof. These follow directly from the definitions. For e_0 and h_0, the only 0-tuple is the empty tuple, giving an empty product equal to 1. For e_1 and h_1, the 1-tuples are the singletons (i) for $i \in [N]$, giving $\sum_{i=1}^N x_i$. For p_1, this is $\sum_{i=1}^N x_i^1 = \sum_{i=1}^N x_i$. For p_0, the Lean definition follows Mathlib's convention, which gives $p_0 = \sum_{i=1}^N x_i^0 = N$. $\square$

6.1.16 Alternative characterizations of e_n, h_n, p_n

Lemma 6.23. *The defining formulas for* e_n *and* p_n *are:*

$$e_n = \sum_{\substack{S \subseteq [N] \\ |S| = n}} \prod_{i \in S} x_i, \qquad p_n = \sum_{i=1}^N x_i^n.$$

Proof. Both follow immediately from the definitions: e_n is the sum over strictly increasing n-tuples, which bijects with n-element subsets $S \subseteq [N]$; and $p_n = \sum_{i=1}^N x_i^n$ is the definition for $n > 0$. $\square$

6.1.17 Integer-indexed versions

Lemma 6.24. *The integer-indexed versions e_n, h_n, p_n for $n \in \mathbb{Z}$ satisfy:*

- $e_n = 0$ *for* $n < 0$*, and* e_n *agrees with Definition 6.8* ***(a)*** *for* $n \geq 0$*;*
- $h_n = 0$ *for* $n < 0$*, and* h_n *agrees with Definition 6.8* ***(b)*** *for* $n \geq 0$*;*
- $p_n = 0$ *for* $n < 0$*,* $p_0 = 1$*, and* $p_n = x_1^n + x_2^n + \cdots + x_N^n$ *for* $n > 0$*.*

All are symmetric for every $n \in \mathbb{Z}$*.*

Proof. By case analysis on the sign of n. For non-negative n, symmetry follows from Lemma 6.17. For negative n, the polynomial is 0 which is trivially symmetric. □

6.1.18 Adding a variable recurrence

Lemma 6.25. *For each* $n \geq 0$*, the elementary symmetric polynomial satisfies the recurrence*

$$e_{n+1}(x_1, \ldots, x_N, y) = e_{n+1}(x_1, \ldots, x_N) + y \cdot e_n(x_1, \ldots, x_N).$$

This follows by partitioning the n*-element subsets of* $\{1, \ldots, N+1\}$ *into those containing* $N+1$ *and those not.*

Proof. Partition $\binom{[N+1]}{n+1}$ into subsets not containing $N+1$ and subsets containing $N+1$. The first group yields $e_{n+1}(x_1, \ldots, x_N)$ (via a bijection with $\binom{[N]}{n+1}$). The second group yields $y \cdot e_n(x_1, \ldots, x_N)$ (via a bijection with $\binom{[N]}{n}$, factoring out the variable $x_{N+1} = y$). The proof establishes these bijections explicitly. □

6.1.19 Antisymmetric polynomials

Definition 6.26. A polynomial $f \in \mathcal{P}$ is *antisymmetric* if

$$\sigma \cdot f = \operatorname{sign}(\sigma) \cdot f \quad \text{for all } \sigma \in S_N.$$

Lemma 6.27. *The antisymmetric polynomials form a* K*-submodule of* $\mathcal{P}$*: the zero polynomial is antisymmetric, and the sum, scalar multiple, and negation of antisymmetric polynomials are antisymmetric.*

Proof. For each property, apply a permutation σ and use the linearity of the action. For instance, $\sigma \cdot (f + g) = \sigma \cdot f + \sigma \cdot g = \operatorname{sign}(\sigma)(f + g)$. □

Lemma 6.28. *The square of an antisymmetric polynomial is symmetric.*

Proof. If f is antisymmetric, then $\sigma \cdot (f^2) = (\sigma \cdot f)^2 = (\operatorname{sign}(\sigma) \cdot f)^2 = \operatorname{sign}(\sigma)^2 \cdot f^2 = f^2$, using $\operatorname{sign}(\sigma)^2 = 1$. □

6.1.20 The sum of variables is symmetric

Lemma 6.29. *The polynomial* $\sum_{i=1}^N x_i$ *is symmetric.*

Proof. This equals e_1, which is symmetric by Lemma 6.17. □

6.2 N-partitions and monomial symmetric polynomials

Throughout this section, let K be a commutative semiring and N a nonnegative integer. We work with the polynomial ring $\mathcal{P} = K[x_1, x_2, \ldots, x_N]$ and the subring $\mathcal{S}$ of symmetric polynomials in $\mathcal{P}$.

6.2.1 N-partitions

Definition 6.30. An *N-partition* will mean a weakly decreasing N-tuple of nonnegative integers. In other words, an N-partition means an N-tuple $(\lambda_1, \lambda_2, \ldots, \lambda_N) \in \mathbb{N}^N$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$.

The *size* (or weight) of an N-partition $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_N)$ is defined to be $|\lambda| := \lambda_1 + \lambda_2 + \cdots + \lambda_N$.

Proposition 6.31. *There is a bijection*

$$\{\textit{partitions of length } \leq N\} \to \{N\textit{-partitions}\},$$
$$(\lambda_1, \lambda_2, \ldots, \lambda_\ell) \mapsto \left(\lambda_1, \lambda_2, \ldots, \lambda_\ell, \underbrace{0, 0, \ldots, 0}_{N-\ell \text{ zeroes}}\right).$$

Proof. Straightforward. The forward map pads a partition of length $\ell \leq N$ with $N - \ell$ trailing zeros, and the inverse map strips trailing zeros. The padded tuple is weakly decreasing because the original partition is weakly decreasing and its entries are positive (hence ≥ 0). The composition in both directions is the identity: padding then stripping recovers the original partition, and stripping then padding recovers the original N-partition (since an N-partition is antitone, all zeros are trailing). ☐

Basic N-partition API

Lemma 6.32. *Let $\lambda = (\lambda_1, \ldots, \lambda_N)$ be an N-partition.*

- *Each entry is bounded by the size: $\lambda_i \leq |\lambda|$ for all i.*
- *$|\lambda| = 0$ if and only if $\lambda = (0, \ldots, 0)$.*

Proof. The bound $\lambda_i \leq \sum_j \lambda_j$ holds because each summand is nonnegative. The second claim uses the bound to show all parts must be zero when the size is. ☐

Lemma 6.33. *The* length *$\ell(\lambda)$ of an N-partition λ is the number of nonzero entries. We have $\ell(\lambda) \leq N$ and $\ell(\lambda) = 0 \iff \lambda = (0, \ldots, 0)$.*

Proof. The length is the cardinality of $\{i : \lambda_i \neq 0\}$, which is a subset of $[N]$. ☐

Lemma 6.34. *The componentwise sum of two N-partitions is an N-partition, and $|\mu + \nu| = |\mu| + |\nu|$.*

Proof. Antitone is preserved under addition since each coordinate sum preserves the inequality. The size formula follows from rearranging a sum. ☐

Definition 6.35. We define a partial order on N-partitions by componentwise comparison: $\mu \leq \nu$ iff $\mu_i \leq \nu_i$ for all $i \in [N]$.

Lemma 6.36. *If $\mu \leq \nu$ (componentwise), then $|\mu| \leq |\nu|$.*

Proof. Sum the componentwise inequalities. ☐

Young diagrams of N-partitions

Definition 6.37. The *Young diagram* $Y(\lambda)$ of an N-partition λ is the finite set of cells

$$Y(\lambda) = \{(i, j) : i \in [N],\ 0 \le j < \lambda_i\}.$$

Lemma 6.38. *The cardinality of the Young diagram equals the size:* $|Y(\lambda)| = |\lambda|$.

Proof. The Young diagram is the disjoint union of rows, where row i has λ_i cells. Thus $|Y(\lambda)| = \sum_i \lambda_i = |\lambda|$. ☐

Lemma 6.39. *The number of cells in row i of the Young diagram equals λ_i:* $|\{(i, j) \in Y(\lambda)\}| = \lambda_i$.

Proof. By definition, row i consists of cells (i, j) with $0 \le j < \lambda_i$, which has exactly λ_i elements. ☐

Lemma 6.40. $\mu \le \nu$ *if and only if* $Y(\mu) \subseteq Y(\nu)$.

Proof. Forward: if $\mu_i \le \nu_i$ for all i, then $j < \mu_i$ implies $j < \nu_i$, so $(i, j) \in Y(\mu)$ implies $(i, j) \in Y(\nu)$. Backward: if $Y(\mu) \subseteq Y(\nu)$ and $\mu_i > \nu_i$ for some i, then $(i, \nu_i) \in Y(\mu) \setminus Y(\nu)$, a contradiction. ☐

Definition 6.41. The *skew Young diagram* $Y(\lambda/\mu)$ for N-partitions λ, μ is the set difference $Y(\lambda) \setminus Y(\mu)$, consisting of cells (i, j) with $\mu_i \le j < \lambda_i$.

Lemma 6.42. *The cardinality of the skew Young diagram satisfies* $|Y(\lambda/\mu)| = \sum_{i=1}^{N} (\lambda_i - \mu_i)$. *When $\mu \le \lambda$, this equals* $|\lambda| - |\mu|$.

Proof. The skew diagram is a subset of the Young diagram, and each row contributes $\max(\lambda_i - \mu_i, 0)$ cells. The identity with sizes uses telescoping. ☐

Row and column partitions

Lemma 6.43. *The* row partition $(n, 0, \dots, 0)$ *has size* n. *The* column partition $(1, 1, \dots, 1, 0, \dots, 0)$ *(with n ones, $n \le N$) has size n.*

Proof. Direct computation of sums. ☐

6.2.2 Monomials and sorting

Definition 6.44. Let $a = (a_1, a_2, \dots, a_N) \in \mathbb{N}^N$. Then:

(a) We let x^a denote the monomial $x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}$.

(b) We let $\operatorname{sort} a$ mean the N-partition obtained from a by sorting the entries of a in weakly decreasing order.

Definition 6.45. For $a = (a_1, a_2, \dots, a_N) \in \mathbb{N}^N$, $\operatorname{sort} a$ is the N-partition obtained by sorting the entries of a in weakly decreasing order.

Lemma 6.46. *Let $a \in \mathbb{N}^N$ and let $\sigma \in S_N$ be a permutation. Then* $\operatorname{sort}(a \circ \sigma) = \operatorname{sort}(a)$.

Proof. Sorting depends only on the multiset of values, which is invariant under permutation of the indices. ☐

Lemma 6.47. *For any $a \in \mathbb{N}^N$,* $|\operatorname{sort}(a)| = \sum_i a_i$.

Proof. The sum of entries is preserved by sorting. □

Lemma 6.48. *If $a \in \mathbb{N}^N$ is already weakly decreasing, then* $\mathrm{sort}(a) = a$.

Proof. A weakly decreasing list is already sorted. □

Lemma 6.49. *For any N-partition μ,* $\mathrm{sort}(\mu) = \mu$.

Proof. An N-partition is antitone by definition, so the result follows from Lemma 6.48. □

Monomial API

Lemma 6.50. *The monomial $x^a = \prod_i x_i^{a_i}$ equals the Mathlib monomial* $\mathrm{monomial}(a, 1)$.

Proof. Immediate from the definitions of monomial and product of powers. □

Lemma 6.51. *The coefficient of x^a in x^a is 1. The coefficient of x^b in x^a is 0 when $a \neq b$.*

Proof. By Lemma 6.50, $x^a = \mathrm{monomial}(a, 1)$, and the coefficient extraction from a single monomial is immediate. □

Lemma 6.52. *The monomial map is multiplicative: $x^{a+b} = x^a \cdot x^b$, and $x^0 = 1$.*

Proof. The identity $x^{a+b} = x^a \cdot x^b$ follows from $x_i^{a_i+b_i} = x_i^{a_i} \cdot x_i^{b_i}$ and distributivity of products. □

Lemma 6.53. *The total degree of x^a equals $\sum_i a_i$.*

Proof. By Lemma 6.50, this reduces to the standard result that the total degree of a monomial equals the sum of its exponents. □

Characterization of equal sorts

Lemma 6.54. *Two tuples $a, b \in \mathbb{N}^N$ have the same sort if and only if they have the same multiset of values (equivalently, b is a permutation of a):*

$$\mathrm{sort}(a) = \mathrm{sort}(b) \iff \{a_1, \ldots, a_N\} = \{b_1, \ldots, b_N\} \quad \textit{(as multisets)}.$$

Proof. Forward: if the sorts are equal as N-partitions, then the sorted lists agree pointwise, hence are equal as lists, hence as multisets — and the sorted list of a multiset equals the multiset itself. Backward: if the multisets of values are equal, sorting them gives the same list. □

Lemma 6.55. *Sorting is idempotent:* $\mathrm{sort}(\mathrm{sort}(a)) = \mathrm{sort}(a)$.

Proof. $\mathrm{sort}(a)$ is an N-partition, so by Lemma 6.49 it is its own sort. □

The sort preimage

Definition 6.56. For an N-partition μ, the *sort preimage* is the finite set

$$\{a \in \mathbb{N}^N : \mathrm{sort}(a) = \mu,\ a_i < |\mu| + 1 \text{ for all } i\}.$$

The bound $a_i < |\mu| + 1$ makes the set finite (it is contained in $\{0, \ldots, |\mu|\}^N$).

Lemma 6.57. *For any N-partition μ, its parts tuple μ itself belongs to the sort preimage of μ: $\mu \in \mathrm{sortPreimage}(\mu)$.*

Proof. $\mathrm{sort}(\mu) = \mu$ by Lemma 6.49, and each $\mu_i \leq |\mu| < |\mu| + 1$ by Lemma 6.32. □

6.2.3 Monomial symmetric polynomials

Definition 6.58. Let λ be any N-partition. Then, we define a symmetric polynomial $m_\lambda \in \mathcal{S}$ by

$$m_\lambda := \sum_{\substack{a \in \mathbb{N}^N; \\ \operatorname{sort} a = \lambda}} x^a.$$

This is called the *monomial symmetric polynomial corresponding to* λ.

Lemma 6.59. *For any N-partition μ, the polynomial m_μ is symmetric.*

Proof. For any permutation $\sigma \in S_N$, we have $\sigma \cdot m_\mu = \sum_{a:\operatorname{sort}(a)=\mu} x^{a\circ\sigma^{-1}}$. Composition with σ^{-1} is a bijection on $\{a \in \mathbb{N}^N : \operatorname{sort}(a) = \mu\}$ by Lemma 6.46, so the sum is unchanged. $\square$

Lemma 6.60. *For any N-partition μ, the coefficient of x^μ (i.e. $x_1^{\mu_1} x_2^{\mu_2} \cdots x_N^{\mu_N}$) in m_μ is 1.*

Proof. The tuple μ itself is in the sort preimage of μ (by Lemma 6.57), and no other element of the sort preimage contributes to the coefficient of x^μ. $\square$

Lemma 6.61. *For any two distinct N-partitions $\mu \neq \nu$, the coefficient of x^μ in m_ν is 0.*

Proof. Every a in the sort preimage of ν satisfies $\operatorname{sort}(a) = \nu$. If $a = \mu$ (as a tuple), then $\operatorname{sort}(\mu) = \mu \neq \nu$ (by Lemma 6.49), a contradiction. So no element of the sort preimage of ν equals μ, hence the coefficient is 0. $\square$

Lemma 6.62. *The monomial symmetric polynomial m_μ is homogeneous of degree $|\mu|$.*

Proof. Each monomial x^a in the sum defining m_μ has degree $\sum_i a_i = |\operatorname{sort}(a)| = |\mu|$ by Lemma 6.47. $\square$

6.2.4 Elementary, homogeneous, and power sum via monomial symmetric polynomials

Proposition 6.63. ***(a)*** *For each $n \in \{0, 1, \ldots, N\}$, we have*

$$e_n = m_{(1,1,\ldots,1,0,0,\ldots,0)},$$

where $(1, 1, \ldots, 1, 0, 0, \ldots, 0)$ is the N-tuple that begins with n many 1's and ends with $N - n$ many 0's.

(b) *For each $n \in \mathbb{N}$, we have*

$$h_n = \sum_{\substack{\lambda \text{ is an } N\text{-partition;} \\ |\lambda| = n}} m_\lambda,$$

where the size *$|\lambda|$ of an N-partition λ is defined to be the sum of its entries (i.e., if $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_N)$, then $|\lambda| := \lambda_1 + \lambda_2 + \cdots + \lambda_N$).*

(c) *Assume that $N > 0$. For each $n \in \mathbb{N}$, we have*

$$p_n = m_{(n,0,0,\ldots,0)},$$

where $(n, 0, 0, \ldots, 0)$ is the N-tuple that begins with an n and ends with $N - 1$ zeroes.

Proof. This combines parts **(a)**, **(b)**, and **(c)**. $\square$

Definition 6.64. The N-partition $(1, 1, \ldots, 1, 0, 0, \ldots, 0)$ with n ones and $N - n$ zeros.

Definition 6.65. The N-partition $(n, 0, 0, \ldots, 0)$ with n in the first position and zeros elsewhere.

Theorem 6.66. *For each $n \in \{0, 1, \ldots, N\}$, we have $e_n = m_{(1,1,\ldots,1,0,0,\ldots,0)}$ where the N-tuple has n ones followed by $N - n$ zeros.*

Proof. We have $e_n = \sum_{S \subseteq \{1,\ldots,N\},\, |S|=n} \prod_{i \in S} x_i$. Each product $\prod_{i \in S} x_i = x^{\chi_S}$ where χ_S is the indicator function of S. Every indicator function of an n-element subset sorts to $(1, \ldots, 1, 0, \ldots, 0)$ (with n ones). Conversely, every 0-1 tuple in the sort preimage of $(1, \ldots, 1, 0, \ldots, 0)$ is the indicator function of some n-element subset. This gives a bijection between n-element subsets and the sort preimage, proving the result. □

Definition 6.67. The finite set of N-partitions of size n.

Theorem 6.68. *For each $n \in \mathbb{N}$, we have $h_n = \sum_{\substack{\lambda \text{ is an } N\text{-partition;} \\ |\lambda|=n}} m_\lambda$.*

Proof. We have $h_n = \sum_{\substack{a \in \mathbb{N}^N \\ a_1+\cdots+a_N=n}} x^a$. We partition the sum by $\operatorname{sort}(a)$: the tuples a with $\sum a_i = n$ are partitioned into groups indexed by N-partitions μ of size n, and each group contributes m_μ. □

Theorem 6.69. *Assume $N > 0$. For each $n \in \mathbb{N}$ with $n > 0$, we have $p_n = m_{(n,0,0,\ldots,0)}$.*

Proof. We have $p_n = \sum_i x_i^n$. Each x_i^n equals $x^{e_i \cdot n}$ where e_i is the i-th standard basis vector scaled by n. All such tuples $(0, \ldots, 0, n, 0, \ldots, 0)$ sort to $(n, 0, \ldots, 0)$. When $n > 0$, these are the *only* elements of the sort preimage of $(n, 0, \ldots, 0)$, because if $\operatorname{sort}(a) = (n, 0, \ldots, 0)$ then the multiset of values of a contains exactly one n and $N - 1$ zeros, so a must be a standard basis vector times n. □

Edge cases for p_0 and m_0

Lemma 6.70. *$p_0 = \sum_{i=1}^N x_i^0 = N$ (the constant polynomial N). This differs from the convention $p_0 = 1$ used in some textbooks; when $N > 0$ the identity $p_n = m_{(n,0,\ldots,0)}$ requires $n > 0$.*

Proof. Direct from the definition $p_0 = \sum_{i=1}^N x_i^0 = N$. □

Lemma 6.71. *When $N > 0$, $m_{(0,0,\ldots,0)} = 1$. The sort preimage of the zero partition consists only of the zero tuple, and $x^{(0,\ldots,0)} = 1$.*

Proof. The zero partition has size 0, so each entry a_j of any a in sortPreimage(0) satisfies $a_j < 0 + 1 = 1$, forcing $a_j = 0$. Thus the sort preimage is $\{(0, \ldots, 0)\}$ and $x^0 = 1$. □

6.2.5 Permutation action on polynomial coefficients

Proposition 6.72. *Let $\sigma \in S_N$ and $f \in \mathcal{P}$. Then,*

$$\left[x_1^{a_1} x_2^{a_2} \cdots x_N^{a_N}\right] (\sigma \cdot f) = \left[x_1^{a_{\sigma(1)}} x_2^{a_{\sigma(2)}} \cdots x_N^{a_{\sigma(N)}}\right] f$$

for any $(a_1, a_2, \ldots, a_N) \in \mathbb{N}^N$.

Proof. The permutation action on f sends $f(x_1, \ldots, x_N)$ to $f(x_{\sigma(1)}, \ldots, x_{\sigma(N)})$. The coefficient of $x^a = x_1^{a_1} \cdots x_N^{a_N}$ in $\sigma \cdot f$ equals the coefficient of $x^{a \circ \sigma} = x_1^{a_{\sigma(1)}} \cdots x_N^{a_{\sigma(N)}}$ in f, which is exactly the stated identity. □

Lemma 6.73. *Let $f \in \mathcal{P}$ be symmetric and let $\sigma \in S_N$. For any monomial exponent d, we have $[x^{\sigma\cdot d}]f = [x^d]f$.*

Proof. Since f is symmetric, $\sigma \cdot f = f$. By Proposition 6.72, the coefficient of $x^{\sigma\cdot d}$ in $\sigma \cdot f$ equals the coefficient of x^d in f. $\square$

Lemma 6.74. *Let $f \in \mathcal{P}$ be symmetric. If two exponent tuples d_1 and d_2 satisfy* $\mathrm{sort}(d_1) = \mathrm{sort}(d_2)$*, then $[x^{d_1}]f = [x^{d_2}]f$.*

Proof. Equal sorts imply equal multisets of values (by Lemma 6.54), which means d_1 and d_2 differ by a permutation σ. The result then follows from Lemma 6.73. $\square$

Helpers for the spanning proof

Lemma 6.75. *If $f \in \mathcal{P}$ is symmetric and d is in the support of f, then $\sigma \cdot d$ is also in the support of f for any permutation σ.*

Proof. By Lemma 6.73, $[x^{\sigma\cdot d}]f = [x^d]f \neq 0$. $\square$

Definition 6.76. The finitely-supported version of sort: for a finitely-supported function d from $[N]$ to $\mathbb{N}$, $\mathrm{sortFinsupp}(d) = \mathrm{sort}(d)$, where d is viewed as a function $[N] \to \mathbb{N}$.

Lemma 6.77. *The finitely-supported sort is invariant under permutation:* $\mathrm{sortFinsupp}(\sigma \cdot d) = \mathrm{sortFinsupp}(d)$.

Proof. Unfold the definition and apply Lemma 6.46. $\square$

6.2.6 Basis theorem

Theorem 6.78. ***(a)*** *The family $(m_\lambda)_{\lambda \text{ is an } N\text{-partition}}$ is a basis of the K-module $\mathcal{S}$.*

(b) *Each symmetric polynomial $f \in \mathcal{S}$ satisfies*

$$f = \sum_{\substack{\lambda=(\lambda_1,\lambda_2,\ldots,\lambda_N) \\ \text{is an } N\text{-partition}}} \left(\left[x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_N^{\lambda_N} \right] f \right) m_\lambda.$$

(c) *Let $n \in \mathbb{N}$. Let*

$$\mathcal{S}_n := \{\text{homogeneous symmetric polynomials } f \in \mathcal{P} \text{ of degree } n\}$$

(where we understand the zero polynomial $0 \in \mathcal{P}$ to be homogeneous of every degree). Then, $\mathcal{S}_n$ is a K-submodule of $\mathcal{S}$.

(d) *The family $(m_\lambda)_{\lambda \text{ is an } N\text{-partition of size } n}$ is a basis of the K-module $\mathcal{S}_n$.*

Proof. This combines parts **(a)** through **(d)**. $\square$

Theorem 6.79. *The family $(m_\mu)_{\mu \text{ is an } N\text{-partition}}$ spans the K-module $\mathcal{S}$.*

Proof. Let $f \in \mathcal{S}$. Write $f = \sum_{d \in \mathrm{supp}(f)} [x^d]f \cdot x^d$. Partition the support by $\mathrm{sort}(d)$: for each N-partition μ, group together all d with $\mathrm{sort}(d) = \mu$. Since f is symmetric, all monomials in the same group have the same coefficient (by Lemma 6.74), which can be factored out. Since f is symmetric, the support is closed under permutation, so the full sort preimage of each μ is present. The remaining sum of monomials in each group equals m_μ, showing f is in the span. $\square$

Theorem 6.80. *Each symmetric polynomial $f \in \mathcal{S}$ satisfies*

$$f = \sum_{\lambda \text{ is an } N\text{-partition}} \left([x_1^{\lambda_1} x_2^{\lambda_2} \cdots x_N^{\lambda_N}] f \right) m_\lambda .$$

Proof. We verify that both sides have the same coefficient for each exponent d. On the right-hand side, extract the coefficient of x^d: using Lemmas 6.60 and 6.61, only the term with $\lambda = \operatorname{sort}(d)$ survives, contributing $[x^\lambda]f$. By Lemma 6.74, $[x^\lambda]f = [x^d]f$ since $\operatorname{sort}(d) = \operatorname{sort}(\lambda)$. □

Definition 6.81. Let $n \in \mathbb{N}$. The submodule of homogeneous symmetric polynomials of degree n is

$$\mathcal{S}_n := \{ f \in \mathcal{P} \mid f \text{ is symmetric and homogeneous of degree } n \}.$$

This is a K-submodule of $\mathcal{S}$.

Theorem 6.82. *The family $(m_\mu)_{\mu \text{ is an } N\text{-partition of size } n}$ is linearly independent over K.*

Proof. Extract the coefficient of x^ν for each N-partition ν of size n; only the $\mu = \nu$ term survives (by Lemmas 6.60 and 6.61), isolating each coefficient. □

Theorem 6.83. *The family $(m_\mu)_{\mu \text{ is an } N\text{-partition of size } n}$ spans the K-module $\mathcal{S}_n$.*

Proof. Let $f \in \mathcal{S}_n$. Since f is symmetric, it lies in the span of all m_μ (by Theorem 6.79). Since f is homogeneous of degree n, applying the degree-n homogeneous component extracts exactly the m_μ with $|\mu| = n$ (by Lemma 6.62, m_μ is homogeneous of degree $|\mu|$, so the homogeneous component is m_μ when $|\mu| = n$ and 0 otherwise). □

Theorem 6.84. *The family $(m_\mu)_{\mu \text{ is an } N\text{-partition of size } n}$ is a basis of the K-module $\mathcal{S}_n$.*

Proof. Combines linear independence (Theorem 6.82) and spanning (Theorem 6.83). □

6.3 Schur polynomials

6.3.1 Alternants

We work over a fixed positive integer N and the polynomial ring $\mathcal{P} = \mathbb{Z}[x_1, x_2, \ldots, x_N]$. An N-partition is a weakly decreasing N-tuple $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_N) \in \mathbb{N}^N$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N$.

Definition 6.85. **(a)** We let ρ be the N-tuple $(N-1, N-2, \ldots, N-N) \in \mathbb{N}^N$.

(b) For any N-tuple $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_N) \in \mathbb{N}^N$, we define

$$a_\alpha := \det \left(\underbrace{\left(x_i^{\alpha_j} \right)_{1 \leq i \leq N,\ 1 \leq j \leq N}}_{\in \mathcal{P}^{N \times N}} \right) \in \mathcal{P}.$$

This is called the α-*alternant* (of $x_1, x_2, \ldots, x_N$).

Lemma 6.86. *The ρ vector is strictly decreasing: if $i < j$ (as indices in $\{0, 1, \ldots, N-1\}$), then $\rho_i > \rho_j$.*

Proof. Immediate from $\rho_i = N - 1 - i$. □

Lemma 6.87. *The sum of the ρ vector equals the triangular number:*

$$\sum_{i=0}^{N-1} \rho_i = \frac{N(N-1)}{2}.$$

Proof. Follows from $\sum_{i=0}^{N-1}(N-1-i) = \sum_{k=0}^{N-1} k = N(N-1)/2$. $\square$

Definition 6.88. The ρ vector $(N-1, N-2, \ldots, 0)$ is itself an N-partition (since it is strictly decreasing, hence weakly decreasing).

Theorem 6.89. *We have $a_\rho = \prod_{1 \leq i < j \leq N}(x_i - x_j)$.*

Proof. This is the classical Vandermonde determinant identity. The matrix for a_ρ has entry (i, j) equal to x_i^{N-j}, and its determinant equals $\prod_{i<j}(x_i - x_j)$ by the Vandermonde formula. $\square$

Lemma 6.90. *Let $\alpha \in \mathbb{N}^N$.*

(a) *If the N-tuple α has two equal entries, then $a_\alpha = 0$.*

(b) *Let $\beta \in \mathbb{N}^N$ be an N-tuple obtained from α by swapping two entries. Then, $a_\beta = -a_\alpha$.*

Proof. **(a)** If $\alpha_i = \alpha_j$, then two columns of the matrix $(x_k^{\alpha_\ell})$ are equal, so its determinant is 0.

(b) Swapping entries α_i and α_j amounts to permuting two columns of the matrix by a transposition, which multiplies the determinant by -1. $\square$

6.3.2 Young diagrams and Schur polynomials

Definition 6.91. Let λ be an N-partition.

The *Young diagram* of λ is defined as the set

$$\{(i, j) \mid i \in [N] \text{ and } j \in [\lambda_i]\} \subseteq \{1, 2, 3, \ldots\}^2.$$

We visually represent each element (i, j) of this Young diagram as a box in row i and column j.

We denote the Young diagram of λ by $Y(\lambda)$.

Lemma 6.92. *A cell (i, j) belongs to $Y(\lambda)$ if and only if $j < \lambda_i$.*

Proof. Immediate from the definition. $\square$

Definition 6.93. Let λ be an N-partition.

A *Young tableau* of shape λ means a way of filling the boxes of $Y(\lambda)$ with elements of $[N]$ (one element per box). Formally speaking, it is defined as a map $T : Y(\lambda) \to [N]$.

Definition 6.94. Let λ be an N-partition.

A Young tableau T of shape λ is said to be *semistandard* if its entries

- increase weakly along each row (from left to right);
- increase strictly down each column (from top to bottom).

Formally speaking, this means that a Young tableau $T : Y(\lambda) \to [N]$ is semistandard if and only if

- we have $T(i, j) \leq T(i, j+1)$ for any $(i, j) \in Y(\lambda)$ satisfying $(i, j+1) \in Y(\lambda)$;

- we have $T(i,j) < T(i+1,j)$ for any $(i,j) \in Y(\lambda)$ satisfying $(i+1,j) \in Y(\lambda)$.

We let $\mathrm{SSYT}(\lambda)$ denote the set of all semistandard Young tableaux of shape λ.

Lemma 6.95. *Let T be a semistandard Young tableau of shape λ. If (i,j_1) and (i,j_2) are cells of $Y(\lambda)$ with $j_1 \leq j_2$, then $T(i,j_1) \leq T(i,j_2)$.*

Proof. If $j_1 = j_2$ the claim is trivial; otherwise $j_1 < j_2$ and we apply the row-weak condition. ☐

Lemma 6.96. *Let T be a semistandard Young tableau of shape λ. If (i_1,j) and (i_2,j) are cells of $Y(\lambda)$ with $i_1 \leq i_2$, then $T(i_1,j) \leq T(i_2,j)$.*

Proof. If $i_1 = i_2$ the claim is trivial; otherwise $i_1 < i_2$ and $T(i_1,j) < T(i_2,j)$ by column-strictness. ☐

Lemma 6.97. *In a semistandard tableau T of shape λ, we have $T(i,j) \geq i$ for every cell $(i,j) \in Y(\lambda)$.*

Proof. By induction on i. For $i = 0$, this holds since entries are nonneg. For the inductive step, column-strictness gives $T(i,j) > T(i-1,j) \geq i-1$, so $T(i,j) \geq i$. ☐

Lemma 6.98. *For any N-partition λ, the tableau that fills every cell (i,j) with the value i is semistandard. This is called the* highest weight tableau.

Proof. Row-weakness holds because each row is constant. Column-strictness holds because entry values equal row indices, which are strictly increasing down columns. ☐

Lemma 6.99. *The highest weight tableau achieves the minimum possible entry at each cell: for any semistandard tableau T of shape λ and any cell $c \in Y(\lambda)$, the entry of the highest weight tableau at c is $\leq T(c)$.*

Proof. The highest weight entry at (i,j) is i, and by Lemma 6.97, $T(i,j) \geq i$. ☐

Lemma 6.100. *The size of a Young tableau (the number of cells in $Y(\lambda)$) equals $\sum_{i=1}^{N} \lambda_i$.*

Proof. The Young diagram is the disjoint union of rows, and row i has λ_i cells. ☐

Definition 6.101. Let λ be an N-partition. If T is any Young tableau of shape λ, then we define the corresponding monomial

$$x_T := \prod_{c \text{ is a box of } Y(\lambda)} x_{T(c)} = \prod_{(i,j) \in Y(\lambda)} x_{T(i,j)} = \prod_{k=1}^{N} x_k^{(\# \text{ of times } k \text{ appears in } T)}.$$

Lemma 6.102. *For any Young tableau T of shape λ,*

$$x_T = \prod_{k=1}^{N} x_k^{(\textit{occurrences of } k \textit{ in } T)}.$$

Proof. Reindex the product over cells by partitioning according to the entry value. ☐

Definition 6.103. Let λ be an N-partition. We define the *Schur polynomial* $s_\lambda \in \mathcal{P}$ by

$$s_\lambda := \sum_{T \in \mathrm{SSYT}(\lambda)} x_T.$$

6.3.3 Examples of Schur polynomials

Theorem 6.104. *The Schur polynomial $s_{(n,0,\ldots,0)}$ equals the complete homogeneous symmetric polynomial h_n:*

$$s_{(n,0,\ldots,0)} = h_n = \sum_{i_1 \le i_2 \le \cdots \le i_n} x_{i_1} x_{i_2} \cdots x_{i_n}.$$

Proof. For a single-row partition $(n, 0, \ldots, 0)$, the Young diagram has n cells in row 0. An SSYT filling is a weakly increasing sequence of n elements from $[N]$, which corresponds bijectively to a multiset of size n. The monomials correspond under this bijection, so the sums are equal. □

Theorem 6.105. *The Schur polynomial $s_{(1^n,0^{N-n})}$ (with n ones followed by zeros) equals the elementary symmetric polynomial e_n:*

$$s_{(1,1,\ldots,1,0,\ldots,0)} = e_n = \sum_{i_1 < i_2 < \cdots < i_n} x_{i_1} x_{i_2} \cdots x_{i_n}.$$

Proof. For a single-column partition (1^n), the Young diagram has n cells in column 0. An SSYT filling assigns strictly increasing values (by column-strictness), which corresponds bijectively to an n-element subset of $[N]$. The monomials match under this bijection. □

Theorem 6.106. *For $N \ge 2$, the Schur polynomial $s_{(2,1,0,\ldots,0)}$ equals $e_2 \cdot e_1 - e_3$.*

Proof. The Young diagram of $(2, 1)$ has three cells: $(0, 0)$, $(0, 1)$, $(1, 0)$. An SSYT filling assigns values (i, j, k) with $i \le j$ and $i < k$. Summing $x_i x_j x_k$ over all such triples and comparing with the expansion $e_2 \cdot e_1 = 3e_3 + \sum_{a<b} x_a^2 x_b + \sum_{a<b} x_a x_b^2$ yields the result. □

6.3.4 Symmetry of Schur polynomials

Theorem 6.107. *Let λ be an N-partition. Then:*

- ***(a)*** *The polynomial s_λ is symmetric.*
- ***(b)*** *We have*

$$a_{\lambda+\rho} = a_\rho \cdot s_\lambda.$$

Here, the addition on $\mathbb{N}^N$ is defined entrywise.

Proof. **(a)** This follows from Theorem 6.127 applied to $\mu = \mathbf{0}$, since $s_{\lambda/\mathbf{0}} = s_\lambda$. The proof of Theorem 6.127 uses Bender–Knuth involutions.

(b) The proof uses Stembridge's Lemma applied to $\mu = 0$ and $\nu = 0$. The only 0-Yamanouchi semistandard tableau of shape λ is the highest weight tableau T_0 (where each row i contains only i's). The content of T_0 equals λ, so the result reduces to $a_\rho \cdot s_\lambda = a_{\lambda+\rho}$. □

6.3.5 Skew Young diagrams and skew Schur polynomials

Definition 6.108. Let λ and μ be two N-partitions.

We say that $\mu \subseteq \lambda$ if and only if $Y(\mu) \subseteq Y(\lambda)$. Equivalently, $\mu \subseteq \lambda$ if and only if

$$\text{each } i \in [N] \text{ satisfies } \mu_i \le \lambda_i.$$

Thus we have defined a partial order $\subseteq$ on the set of all N-partitions.

Lemma 6.109. *We have $\mu \le \lambda$ if and only if $Y(\mu) \subseteq Y(\lambda)$.*

Proof. (⇒): If $\mu_i \leq \lambda_i$ for all i and $(i,j) \in Y(\mu)$ (i.e. $j < \mu_i$), then $j < \lambda_i$, so $(i,j) \in Y(\lambda)$. (⇐): If $\mu_i > \lambda_i$ for some i, then $(i, \lambda_i) \in Y(\mu) \setminus Y(\lambda)$. ☐

Definition 6.110. Let λ and μ be two N-partitions such that $\mu \subseteq \lambda$. Then, we define the *skew Young diagram* $Y(\lambda/\mu)$ to be the set difference

$$\begin{aligned} Y(\lambda) \setminus Y(\mu) &= \{(i,j) \mid i \in [N] \text{ and } j \in [\lambda_i] \setminus [\mu_i]\} \\ &= \{(i,j) \mid i \in [N] \text{ and } j \in \mathbb{Z} \text{ and } \mu_i < j \leq \lambda_i\}. \end{aligned}$$

Lemma 6.111. *A cell (i,j) belongs to $Y(\lambda/\mu)$ if and only if $\mu_i \leq j < \lambda_i$ (in 0-indexed coordinates).*

Proof. Follows from the definition $Y(\lambda/\mu) = Y(\lambda) \setminus Y(\mu)$ and the membership characterization of Young diagrams. ☐

Lemma 6.112. *The skew Young diagram $Y(\lambda/\mathbf{0})$ equals the full Young diagram $Y(\lambda)$.*

Proof. Since $\mathbf{0}_i = 0$ for all i, the condition $\mu_i \leq j$ is automatic, and the skew diagram reduces to the full diagram. ☐

Lemma 6.113 (Convexity of skew Young diagrams)**.** *Let λ and μ be two N-partitions such that $\mu \subseteq \lambda$. Let (a,b) and (e,f) be two elements of $Y(\lambda/\mu)$. Let $(c,d) \in \mathbb{Z}^2$ satisfy $a \leq c \leq e$ and $b \leq d \leq f$. Then, $(c,d) \in Y(\lambda/\mu)$.*

Proof. We need $\mu_c \leq d$ and $d < \lambda_c$. Since μ is weakly decreasing and $a \leq c$, we have $\mu_c \leq \mu_a \leq b \leq d$. Since λ is weakly decreasing and $c \leq e$, we have $d \leq f < \lambda_e \leq \lambda_c$. ☐

Definition 6.114. Let λ and μ be two N-partitions such that $\mu \subseteq \lambda$. A *Young tableau* of shape λ/μ means a way of filling the boxes of $Y(\lambda/\mu)$ with elements of $[N]$ (one element per box). Formally speaking, it is defined as a map $T : Y(\lambda/\mu) \to [N]$.

Young tableaux of shape λ/μ are often called *skew Young tableaux.*

If we don't have $\mu \subseteq \lambda$, then we agree that there are no Young tableaux of shape λ/μ.

Definition 6.115. Let λ and μ be two N-partitions.

A Young tableau T of shape λ/μ is said to be *semistandard* if its entries

- increase weakly along each row (from left to right);
- increase strictly down each column (from top to bottom).

Formally speaking, this means that a Young tableau $T : Y(\lambda/\mu) \to [N]$ is semistandard if and only if

- we have $T(i,j) \leq T(i,j+1)$ for any $(i,j) \in Y(\lambda/\mu)$ satisfying $(i,j+1) \in Y(\lambda/\mu)$;
- we have $T(i,j) < T(i+1,j)$ for any $(i,j) \in Y(\lambda/\mu)$ satisfying $(i+1,j) \in Y(\lambda/\mu)$.

We let $\mathrm{SSYT}(\lambda/\mu)$ denote the set of all semistandard Young tableaux of shape λ/μ.

Lemma 6.116. *Let λ and μ be two N-partitions. Let T be a semistandard Young tableau of shape λ/μ. Then:*

(a) *If (i,j_1) and (i,j_2) are two elements of $Y(\lambda/\mu)$ satisfying $j_1 \leq j_2$, then $T(i,j_1) \leq T(i,j_2)$.*

(b) *If* (i_1, j) *and* (i_2, j) *are two elements of* $Y(\lambda/\mu)$ *satisfying* $i_1 \le i_2$*, then* $T(i_1, j) \le T(i_2, j)$*.*

(c) *If* (i_1, j) *and* (i_2, j) *are two elements of* $Y(\lambda/\mu)$ *satisfying* $i_1 < i_2$*, then* $T(i_1, j) < T(i_2, j)$*.*

(d) *If* (i_1, j_1) *and* (i_2, j_2) *are two elements of* $Y(\lambda/\mu)$ *satisfying* $i_1 \le i_2$ *and* $j_1 \le j_2$*, then* $T(i_1, j_1) \le T(i_2, j_2)$*.*

(e) *If* (i_1, j_1) *and* (i_2, j_2) *are two elements of* $Y(\lambda/\mu)$ *satisfying* $i_1 < i_2$ *and* $j_1 \le j_2$*, then* $T(i_1, j_1) < T(i_2, j_2)$*.*

Proof. **(a)** If $j_1 = j_2$ the claim is trivial; otherwise apply row-weakness.

(b) If $i_1 = i_2$ the claim is trivial; otherwise $T(i_1, j) < T(i_2, j)$ by column-strictness.

(c) This is just the column-strict condition.

(d) By convexity (Lemma 6.113), $(i_2, j_1) \in Y(\lambda/\mu)$. Then $T(i_1, j_1) \le T(i_2, j_1) \le T(i_2, j_2)$ using parts (b) and (a).

(e) Similarly, $(i_2, j_1) \in Y(\lambda/\mu)$ by convexity. Then $T(i_1, j_1) < T(i_2, j_1) \le T(i_2, j_2)$ using parts (c) and (a). ☐

Definition 6.117. Let λ and μ be two N-partitions. If T is any Young tableau of shape λ/μ, then we define the corresponding monomial

$$x_T := \prod_{c \text{ is a box of } Y(\lambda/\mu)} x_{T(c)} = \prod_{(i,j) \in Y(\lambda/\mu)} x_{T(i,j)} = \prod_{k=1}^{N} x_k^{(\# \text{ of times } k \text{ appears in } T)}.$$

Lemma 6.118. *For any skew Young tableau* T *of shape* λ/μ*,*

$$x_T = \prod_{k=1}^{N} x_k^{(\textit{occurrences of } k \textit{ in } T)}.$$

Proof. Reindex the product over cells by partitioning according to the entry value. ☐

Lemma 6.119. *The skew Young diagram* $Y(\lambda/\lambda)$ *is empty for any* N*-partition* λ*.*

Proof. By the membership characterization, $(i, j) \in Y(\lambda/\lambda)$ iff $\lambda_i \le j < \lambda_i$, which is impossible. ☐

Lemma 6.120. *For a skew Young tableau* T *of shape* λ/μ*, if* $j < \mu_i$ *then* $T(i, j) = 0$*. That is, entries to the left of the inner partition are zero.*

Proof. If $j < \mu_i$, then $(i, j) \notin Y(\lambda/\mu)$, so the support condition gives $T(i, j) = 0$. ☐

Lemma 6.121. *For a skew Young tableau* T *of shape* λ/μ*, if* $j \ge \lambda_i$ *then* $T(i, j) = 0$*. That is, entries to the right of the outer partition are zero.*

Proof. If $j \ge \lambda_i$, then $(i, j) \notin Y(\lambda/\mu)$, so the support condition gives $T(i, j) = 0$. ☐

6.3.6 Filling infrastructure for Schur polynomials

The formal definition of Schur polynomials represents SSYT as functions from the diagram to $[N]$ ("fillings"), filters for those satisfying the semistandard conditions, and sums the associated monomials.

Definition 6.122. A *filling* of a skew Young diagram $Y(\lambda/\mu)$ is a function $f : Y(\lambda/\mu) \to [N]$. A filling is *semistandard* if it satisfies the row-weak and column-strict conditions. The *filling monomial* is $x_f = \prod_{c \in Y(\lambda/\mu)} x_{f(c)}$.

Definition 6.123. The *content* of a filling f of $Y(\lambda/\mu)$ is the function $\mathrm{cont}(f) : [N] \to \mathbb{N}$ defined by

$$\mathrm{cont}(f)(i) = \#\{c \in Y(\lambda/\mu) : f(c) = i\}.$$

Lemma 6.124. *For any filling f of $Y(\lambda/\mu)$,*

$$x_f = \prod_{i=0}^{N-1} x_i^{\mathrm{cont}(f)(i)}.$$

Proof. Reindex the product over cells by partitioning according to the entry value (fiberwise product decomposition). □

Definition 6.125. Let λ and μ be two N-partitions. We define the *skew Schur polynomial* $s_{\lambda/\mu} \in \mathcal{P}$ by

$$s_{\lambda/\mu} := \sum_{T \in \mathrm{SSYT}(\lambda/\mu)} x_T.$$

Lemma 6.126. *For any N-partition λ, we have $s_{\lambda/\mathbf{0}} = s_\lambda$.*

Proof. The semistandard tableaux of shape $\lambda/\mathbf{0}$ are precisely the semistandard tableaux of shape λ, since $Y(\lambda/\mathbf{0}) = Y(\lambda)$ by Lemma 6.112. □

Theorem 6.127. *Let λ and μ be any two N-partitions. Then, the polynomial $s_{\lambda/\mu}$ is symmetric.*

Proof. The proof uses the Bender–Knuth involutions. For each $k \in \{0, 1, \ldots, N-2\}$, the k-th Bender–Knuth involution BK_k is a bijection on $\mathrm{SSYT}(\lambda/\mu)$ satisfying:

1. BK_k is an involution: $\mathrm{BK}_k \circ \mathrm{BK}_k = \mathrm{id}$;

2. BK_k preserves the shape of the tableau;

3. $x_{\mathrm{BK}_k(T)} = s_k \cdot x_T$ (where s_k swaps x_k and x_{k+1}).

Since the simple transpositions $s_0, s_1, \ldots, s_{N-2}$ generate S_N, and each BK_k establishes that $s_k \cdot s_{\lambda/\mu} = s_{\lambda/\mu}$ (Lemma 6.137), we conclude that $s_{\lambda/\mu}$ is symmetric.

The reduction from arbitrary permutations to adjacent transpositions uses Lemma 6.138 (strong induction on the distance $|j - i|$). □

6.3.7 Bender–Knuth involutions (helper lemmas)

The following helper lemmas formalize the key properties of the Bender–Knuth involution, used in the proof of Theorem 6.127.

Definition 6.128. The *cell bijection* between two representations of skew Young diagrams. The first uses 0-indexed columns ($(i, j) \in Y(\lambda/\mu)$ iff $\mu_i \leq j < \lambda_i$), and the second uses 1-indexed columns ($(i, j) \in Y(\lambda/\mu)$ iff $\mu_i < j \leq \lambda_i$). The bijection is $(i, j) \mapsto (i, j + 1)$.

Definition 6.129. The *filling bijection* converts fillings between the two skew diagram representations by composing with the cell bijection.

Lemma 6.130. *The filling bijection preserves the semistandard property: a filling f is semistandard if and only if its image under the filling bijection is semistandard in the 1-indexed representation.*

Proof. Both row-weak and column-strict conditions depend only on the relative ordering of indices, which is preserved by the column shift $j \mapsto j + 1$. ☐

Lemma 6.131. *The filling bijection preserves content: for any filling f and index i, the content of f at i equals the content of the corresponding filling in the 1-indexed representation at i.*

Proof. The cell bijection induces a bijection between $\{c : f(c) = i\}$ and the corresponding set in the 1-indexed representation, since the entry values are unchanged. ☐

Definition 6.132. The *Bender–Knuth involution* BK_k on fillings of $Y(\lambda/\mu)$. For a semistandard filling f, $\mathrm{BK}_k(f)$ swaps certain entries k and $k+1$ while preserving semistandardness. For non-semistandard fillings, it returns the input unchanged.

Lemma 6.133. *The Bender–Knuth involution BK_k preserves SSYT membership: if f is a semistandard filling, then so is $\mathrm{BK}_k(f)$.*

Proof. Transfers through the filling bijection: the image is semistandard by the Bender–Knuth semistandard preservation theorem, and the bijection preserves semistandardness. ☐

Lemma 6.134. *The Bender–Knuth map is an involution: applying it twice returns the original filling.*

Proof. Transfers through the filling bijection using the Bender–Knuth involutivity theorem. ☐

Lemma 6.135. *Let f be a semistandard filling and $f' = \mathrm{BK}_k(f)$. Then the content of f' at index k equals the content of f at $k+1$, the content of f' at $k+1$ equals the content of f at k, and the content at all other indices is unchanged.*

Proof. This is the key property of the Bender–Knuth involution. Proved via the Bender–Knuth content swap theorem, transferred through the filling bijection using Lemma 6.131. ☐

Lemma 6.136. *The monomial effect of the Bender–Knuth involution: $x_{\mathrm{BK}_k(f)} = (\mathit{swap}\ x_k\ x_{k+1}) \cdot x_f$.*

This says that BK_k effectively swaps the roles of variables x_k and x_{k+1} in the monomial.

Proof. By Lemma 6.135, the content of $\mathrm{BK}_k(f)$ is obtained from that of f by swapping entries k and $k+1$. Since the monomial is $\prod_i x_i^{\mathrm{cont}(f)(i)}$, this transposition of the content corresponds to renaming $x_k \leftrightarrow x_{k+1}$. ☐

Lemma 6.137. *The skew Schur polynomial is invariant under simple transpositions:*

$$s_k \cdot s_{\lambda/\mu} = s_{\lambda/\mu}$$

for each $k \in \{0, 1, \ldots, N-2\}$.

Proof. Apply s_k to the sum $s_{\lambda/\mu} = \sum_f x_f$. By Lemma 6.136, $s_k \cdot x_f = x_{\mathrm{BK}_k(f)}$. Since BK_k is an involution on the SSYT fillings (Lemmas 6.133 and 6.134), reindexing the sum by BK_k shows $\sum_f x_{\mathrm{BK}_k(f)} = \sum_f x_f$. □

Lemma 6.138. *If a polynomial P is invariant under every simple transposition s_k (swapping x_k and x_{k+1}), then P is invariant under every transposition (swapping any two variables x_i and x_j).*

Proof. By strong induction on the distance $|j - i|$. If $|j - i| = 1$, the swap is already a simple transposition. Otherwise, decompose $\mathrm{swap}(i, j) = \mathrm{swap}(k, j) \cdot \mathrm{swap}(i, k) \cdot \mathrm{swap}(k, j)$ where $k = j - 1$, and apply the inductive hypothesis to the smaller swap. □

Lemma 6.139. *Renaming variables in a filling monomial equals applying the permutation to each entry:*

$$\mathrm{rename}(\sigma)(x_f) = x_{\sigma \circ f}.$$

Proof. Follows from $\mathrm{rename}(\sigma)(X_i) = X_{\sigma(i)}$ applied to each factor of the product. □

6.3.8 Equivalences between Schur polynomial definitions

The project maintains multiple representations of SSYT and Schur polynomials for different proof contexts. This subsection records the key equivalence results.

Theorem 6.140. *The two definitions of Schur polynomials in this project (one using $\mathbb{Z}$ coefficients and N-partitions, the other using generic coefficients and raw N-tuples) are equal.*

Proof. Both definitions sum over semistandard Young tableaux of shape λ. The cell bijection $(i, j) \mapsto (i, j + 1)$ between the two indexing conventions preserves the SSYT conditions and the monomials. □

Definition 6.141. An equivalence between the two SSYT representations in this project:

- The first uses a total function $T : [N] \times \mathbb{N} \to [N]$ with a support condition (entries outside the diagram are 0).
- The second uses a dependent function where the column index is bounded by the partition part: $T(i, j)$ for $j < \lambda_i$.

The equivalence converts between the total function and the bounded representations.

Theorem 6.142. *The two Schur polynomial definitions agree over $\mathbb{Z}$. More precisely, the Schur polynomial defined via fillings of the Young diagram equals the one defined via the bounded representation.*

Proof. Both are sums over SSYT with the same monomials. The equivalence on tableaux preserves the monomial at each tableau, so the sums are equal. □

Theorem 6.143. *Corollary of Theorem 6.142: the filling-based Schur polynomial equals the bounded-representation Schur polynomial (with appropriate conversion of the partition).*

Proof. Direct application of Theorem 6.142 with the partition conversion being an involution. □

6.4 The Littlewood–Richardson rule

6.4.1 Tuple arithmetic

Definition 6.144. **(a)** We let $\mathbf{0}$ denote the N-tuple $(0, 0, \ldots, 0) \in \mathbb{N}^N$.

(b) Let $\alpha = (\alpha_1, \alpha_2, \ldots, \alpha_N)$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_N)$ be two N-tuples in $\mathbb{N}^N$. Then, we set

$$\alpha + \beta := (\alpha_1 + \beta_1, \alpha_2 + \beta_2, \ldots, \alpha_N + \beta_N) \qquad \text{and}$$
$$\alpha - \beta := (\alpha_1 - \beta_1, \alpha_2 - \beta_2, \ldots, \alpha_N - \beta_N).$$

Note that $\alpha + \beta \in \mathbb{N}^N$, whereas $\alpha - \beta \in \mathbb{Z}^N$.

6.4.2 Content of a tableau

Definition 6.145. Let λ and μ be two N-partitions. Let T be a tableau of shape λ/μ. We define the *content* of T to be the N-tuple $(a_1, a_2, \ldots, a_N)$, where

$$a_i := (\# \text{ of } i\text{'s in } T) = (\# \text{ of boxes } c \text{ of } T \text{ such that } T(c) = i).$$

We denote this N-tuple by $\operatorname{cont} T$.

6.4.3 Column restriction of tableaux

Definition 6.146. Let λ and μ be two N-partitions. Let T be a tableau of shape λ/μ. Let j be a positive integer. Then, $\operatorname{col}_{\geq j} T$ means the restriction of T to columns $j, j+1, j+2, \ldots$ (that is, the result of removing the first $j-1$ columns from T). Formally speaking, this means the restriction of the map T to the set $\{(u, v) \in Y(\lambda/\mu) \mid v \geq j\}$.

6.4.4 Yamanouchi tableaux

Definition 6.147. Let λ, μ, ν be three N-partitions. A semistandard tableau T of shape λ/μ is said to be *ν-Yamanouchi* (this is an adjective) if for each positive integer j, the N-tuple $\nu + \operatorname{cont}(\operatorname{col}_{\geq j} T) \in \mathbb{N}^N$ is an N-partition (i.e., weakly decreasing).

6.4.5 Content and column restriction helpers

Lemma 6.148 (Content of column-restricted tableau)**.** *Let T be a tableau of shape λ/μ. Then $\operatorname{cont}(\operatorname{col}_{\geq 1} T) = \operatorname{cont}(T)$. In other words, since every cell in $Y(\lambda/\mu)$ has column ≥ 1, the column restriction to $j = 1$ recovers the full content.*

Proof. Every cell $(i, v) \in Y(\lambda/\mu)$ satisfies $v > \mu_i \geq 0$, hence $v \geq 1$, so $\operatorname{col}_{\geq 1} T$ has the same domain as T. $\square$

Lemma 6.149 (Content monotonicity under column restriction)**.** *For any tableau T of shape λ/μ, row index i, and positive integer j:*

$$\operatorname{cont}(\operatorname{col}_{\geq j+1} T)_i \;\leq\; \operatorname{cont}(\operatorname{col}_{\geq j} T)_i.$$

That is, restricting to higher columns can only decrease the count of any entry.

Proof. The cells counted by $\operatorname{cont}(\operatorname{col}_{\geq j+1} T)_i$ form a subset of those counted by $\operatorname{cont}(\operatorname{col}_{\geq j} T)_i$. $\square$

Lemma 6.150 (Column restriction vanishes beyond the shape)**.** *If $j > \lambda_i$ for every row index i, then* $\operatorname{cont}(\operatorname{col}_{\geq j} T) = \mathbf{0}$.

Proof. When j exceeds all column lengths, the domain $\{(u, v) \in Y(\lambda/\mu) \mid v \geq j\}$ is empty. $\square$

Lemma 6.151 (Content equality from pointwise characterization)**.** *If two tableaux T, T' of the same shape satisfy $T'(c) = i \iff T(c) = i$ for all cells c, then* $\operatorname{cont}(T')_i = \operatorname{cont}(T)_i$.

Proof. The sets $\{c \mid T'(c) = i\}$ and $\{c \mid T(c) = i\}$ are equal by the hypothesis (each cell belongs to one set if and only if it belongs to the other), so they have the same cardinality. $\square$

6.4.6 Properties of alternants

Lemma 6.152. *Let $\alpha \in \mathbb{N}^N$.*

- ***(a)*** *If the N-tuple α has two equal entries, then $a_\alpha = 0$.*
- ***(b)*** *Let $\beta \in \mathbb{N}^N$ be an N-tuple obtained from α by swapping two entries. Then, $a_\beta = -a_\alpha$.*

Proof. Both parts follow from the definition of alternants as signed sums over permutations.

(a) If $\alpha_i = \alpha_j$ with $i \neq j$, then the map $\sigma \mapsto (i\ j) \circ \sigma$ is a sign-reversing involution on S_N that preserves $x^{\alpha \circ \sigma}$ (since α is invariant under swapping i and j). The terms cancel in pairs, giving $a_\alpha = 0$.

(b) If $\beta = \alpha \circ (i\ j)$, then reindexing the sum defining a_β via $\tau \mapsto (i\ j) \circ \tau$ introduces a factor of $\operatorname{sign}((i\ j)) = -1$, giving $a_\beta = -a_\alpha$. $\square$

6.4.7 Regular elements and cancellation

Definition 6.153. Let L be a commutative ring. Let $a \in L$. The element a of L is said to be *regular* if and only if every $x \in L$ satisfying $ax = 0$ satisfies $x = 0$.

Lemma 6.154. *Let L be a commutative ring. Let $a, u, v \in L$ be such that a is regular. Assume that $au = av$. Then, $u = v$.*

Proof. We have $a(u - v) = au - av = 0$ (since $au = av$). However, a is regular; in other words, every $x \in L$ satisfying $ax = 0$ satisfies $x = 0$ (by the definition of "regular"). Applying this to $x = u - v$, we obtain $u - v = 0$ (since $a(u - v) = 0$). Thus, $u = v$. $\square$

Lemma 6.155. *The element a_ρ of the polynomial ring $\mathcal{P}$ is regular.*

Proof. Each of the indeterminates $x_1, x_2, \ldots, x_N$ is regular (as an element of $\mathcal{P}$). Indeed, multiplying a polynomial f by an indeterminate x_i merely shifts the coefficients of f to different monomials; thus, if $x_i f = 0$, then $f = 0$. Next, the polynomial $x_i - x_j \in \mathcal{P}$ is regular whenever $1 \leq i < j \leq N$. Indeed, this polynomial $x_i - x_j$ is the image of the indeterminate x_i under a K-algebra automorphism of $\mathcal{P}$ (the automorphism that sends x_i to $x_i - x_j$ while leaving all other indeterminates unchanged), and therefore is regular because x_i is regular (and because any ring automorphism sends regular elements to regular elements). Any finite product of regular elements is again regular. Thus, the element $\prod_{1 \leq i < j \leq N} (x_i - x_j) \in \mathcal{P}$ is regular. Since $a_\rho = \prod_{1 \leq i < j \leq N} (x_i - x_j)$ (the Vandermonde determinant), the element $a_\rho \in \mathcal{P}$ is regular.

Lean note: The Lean formalization takes a shortcut by assuming R is an integral domain, which makes the polynomial ring an integral domain; regularity then follows from the Vandermonde polynomial being nonzero. $\square$

6.4.8 Bender–Knuth involution

Definition 6.156 (Forced and free cells)**.** Let T be a semistandard tableau and let $k < N - 1$. A cell $c = (i, j)$ with $T(c) = k$ is *forced* if there exists a cell $(i + 1, j)$ directly below with $T(i + 1, j) = k + 1$. Similarly for $T(c) = k + 1$ with a cell directly above having value k. Free cells are those that are not forced.

Among the free cells, the *parenthesis-matching algorithm* classifies each as *matched* or *unmatched*: reading free entries left-to-right in each row, each free $(k{+}1)$ matches with the nearest unmatched free k to its left. Unmatched entries are those remaining after this process.

Definition 6.157 (Bender–Knuth involution β_k)**.** Let T be a semistandard tableau of shape λ/μ and let $k < N - 1$. The *Bender–Knuth involution* $\beta_k(T)$ is obtained from T by swapping only the *unmatched* free k's and unmatched free $(k{+}1)$'s:

$$\beta_k(T)(c) = \begin{cases} k + 1 & \text{if } c \text{ is an unmatched free } k, \\ k & \text{if } c \text{ is an unmatched free } k{+}1, \\ T(c) & \text{otherwise.} \end{cases}$$

Forced cells and matched free cells are left unchanged.

Lemma 6.158 (Bender–Knuth preserves semistandardness)**.** *If T is semistandard and λ, μ are N-partitions, then $\beta_k(T)$ is semistandard.*

Proof. Row-weak ordering is preserved because in each row, unmatched $(k{+}1)$'s (which become k) lie to the left of unmatched k's (which become $k{+}1$), by the structure of the parenthesis matching (Lemma 6.159). Column-strict ordering is preserved because the dangerous case (an unmatched k above an unmatched $k{+}1$ in the same column) is ruled out by the partition property, which forces adjacent $k, k{+}1$ pairs to be forced (Lemma 6.160). □

Lemma 6.159 (Bender–Knuth row-weak preservation)**.** *For cells (i, j_1) and (i, j_2) in the same row with $j_1 < j_2$, $\beta_k(T)(i, j_1) \le \beta_k(T)(i, j_2)$.*

Proof. The proof proceeds by case analysis on the status of c_1 and c_2 (forced, matched free, unmatched free, or having value other than k or $k{+}1$). The key insight is that unmatched $(k{+}1)$'s are the *leftmost* free $(k{+}1)$'s and unmatched k's are the *rightmost* free k's in each row. □

Lemma 6.160 (Bender–Knuth column-strict preservation)**.** *For cells (i_1, j) and (i_2, j) in the same column with $i_1 < i_2$, $\beta_k(T)(i_1, j) < \beta_k(T)(i_2, j)$.*

Proof. The proof proceeds by case analysis on whether c_1 and c_2 are unmatched free k, unmatched free $(k{+}1)$, or unchanged. The dangerous case is when c_1 (above) is an unmatched free k (becoming $k{+}1$) and c_2 (below) is an unmatched free $(k{+}1)$ (becoming k), which would violate strict ordering. But this cannot happen: if $T(c_1) = k$ and $T(c_2) = k{+}1$ in the same column, the partition property forces them to be in adjacent rows, making them a forced pair (not free), so they are unchanged by β_k. All other case combinations preserve strict ordering. □

Lemma 6.161 (Bender–Knuth is an involution)**.** *For N-partitions λ, μ and a semistandard tableau T of shape λ/μ, the tableau $T' = \beta_k(T)$ is semistandard, and $\beta_k(T') = T$.*

Proof. The matching structure is symmetric under BK: an unmatched free k becomes an unmatched free $(k{+}1)$ in T' and vice versa; matched and forced cells are unchanged. Applying BK twice therefore returns every cell to its original value. □

Lemma 6.162 (Forced cells are preserved under BK)**.** *If c is a forced k cell in T, then c is also a forced k cell in $\beta_k(T)$. Similarly for forced $(k+1)$ cells.*

Proof. A forced k cell has a forced $(k+1)$ partner directly below in the same column. Since forced cells are never unmatched, both cells retain their values under β_k, preserving the forced structure. $\square$

Lemma 6.163 (Unmatched cells swap status under BK)**.** *An unmatched free k in T becomes an unmatched free $(k+1)$ in $\beta_k(T)$, and vice versa.*

Proof. After swapping the value from k to $k+1$, the cell is free (not forced) because there is no cell directly above with value k. The counting argument shows that the cumulative counts shift appropriately to make the cell unmatched in its new role. $\square$

Lemma 6.164 (Matched cells stay matched under BK)**.** *A matched free k in T remains a matched free k in $\beta_k(T)$, and similarly for matched free $(k+1)$.*

Proof. Matched cells retain their original values. The counting argument for the parenthesis matching is preserved because the symmetric swap of unmatched cells leaves the matching structure invariant. $\square$

Lemma 6.165 (BK content swap)**.** *Let $T' = \beta_k(T)$. Then*

$$\operatorname{cont}(T')_k = \operatorname{cont}(T)_{k+1}, \quad \operatorname{cont}(T')_{k+1} = \operatorname{cont}(T)_k, \quad \operatorname{cont}(T')_i = \operatorname{cont}(T)_i \textit{ for } i \notin \{k, k+1\}.$$

Proof. For $i \notin \{k, k+1\}$, cells with value i are unchanged by BK. For the swap of k and $k+1$ counts, decompose each set into forced cells, matched free cells, and unmatched free cells. The forced bijection gives equal forced counts; matched counts are equal by the matching definition; and unmatched cells swap their values, contributing to the other count. $\square$

Lemma 6.166 (BK and the monomial of a tableau)**.** *For the monomial associated to a tableau, $x^{\operatorname{cont}(\beta_k(T))} = \operatorname{rename}_{(k\ k+1)}(x^{\operatorname{cont}(T)})$. That is, the BK involution acts on monomials by the transposition $(k\ k+1)$.*

Proof. This follows directly from the content swap: renaming by $(k\ k+1)$ exchanges the exponents of x_k and x_{k+1}. $\square$

6.4.9 Symmetry of skew Schur polynomials

Lemma 6.167 (Skew Schur polynomial is symmetric)**.** *For N-partitions λ and μ, the skew Schur polynomial $s_{\lambda/\mu} \in \mathcal{P}$ is symmetric (invariant under all permutations of the variables).*

Proof. It suffices to show invariance under adjacent transpositions $(k\ k+1)$, since these generate S_N. For each k, the BK involution β_k gives a bijection on semistandard tableaux with $x^{\operatorname{cont}(\beta_k(T))} = \operatorname{rename}_{(k\ k+1)}(x^{\operatorname{cont}(T)})$. Reindexing the sum defining $s_{\lambda/\mu}$ by this bijection shows invariance. $\square$

Lemma 6.168 (Alternant times skew Schur expansion)**.** *For N-partitions λ, μ and any N-tuple α:*

$$a_\alpha \cdot s_{\lambda/\mu} = \sum_{T \in \operatorname{SSYT}(\lambda/\mu)} a_{\alpha + \operatorname{cont}(T)}.$$

Proof. Expand the alternant as $a_\alpha = \sum_\sigma (-1)^\sigma x^{\alpha\circ\sigma}$ and multiply by $s_{\lambda/\mu} = \sum_T x^{\operatorname{cont}(T)}$. Using the symmetry of $s_{\lambda/\mu}$ and regrouping terms gives the result. $\square$

6.4.10 Stembridge's involution infrastructure

Definition 6.169 (Violator columns and misstep indices)**.** Let T be a semistandard tableau and ν an N-tuple. A positive integer j is a *violator column* if $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$ is not an N-partition. An index k is a *misstep* for an N-tuple α if $k+1 < N$ and $\alpha_k < \alpha_{k+1}$ (i.e., the partition condition fails at k).

Lemma 6.170 (Non-Yamanouchi implies violator columns exist)**.** *If T is semistandard and not ν-Yamanouchi, then the set of violator columns is nonempty.*

Proof. By definition, T is not ν-Yamanouchi iff there exists $j > 0$ such that $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$ is not a partition. □

Lemma 6.171 (Non-partition implies misstep exists)**.** *An N-tuple α is not an N-partition if and only if it has a misstep. In particular, the misstep set is nonempty when α is not a partition.*

Proof. If α is not weakly decreasing, there exist $i < j$ with $\alpha_i < \alpha_j$. By well-founded induction on $j - i$, we find consecutive indices with this property. □

Definition 6.172 (Prefix-restricted Bender–Knuth)**.** The *prefix-restricted BK involution* $\beta_k^{<j}$ applies the parenthesis-matching swap of k and $k+1$ only to cells in columns $< j$. Cells in columns $\geq j$ are left unchanged:

$$\beta_k^{<j}(T)(c) = \begin{cases} k+1 & \text{if the column of } c \text{ is } < j \text{ and } c \text{ is unmatched free } k \text{ in prefix,} \\ k & \text{if the column of } c \text{ is } < j \text{ and } c \text{ is unmatched free } k{+}1 \text{ in prefix,} \\ T(c) & \text{otherwise.} \end{cases}$$

Lemma 6.173 (Prefix BK preserves suffix content)**.** *The prefix-restricted BK involution leaves columns $\geq j$ unchanged, so $\mathrm{cont}(\mathrm{col}_{\geq j}\, \beta_k^{<j}(T)) = \mathrm{cont}(\mathrm{col}_{\geq j} T)$.*

Proof. Cells in columns $\geq j$ are not modified by $\beta_k^{<j}$. □

Lemma 6.174 (Prefix BK preserves semistandardness (Stembridge context))**.** *Under the Stembridge context (where $\beta = \nu + \mathrm{cont}(\mathrm{col}_{\geq j+1} T)$ is a partition and k is a misstep for $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$), the prefix-restricted BK involution $\beta_k^{<j}(T)$ preserves semistandardness.*

Proof. Row-weak and column-strict ordering are verified by case analysis, similar to the full BK involution but restricted to the prefix. The Stembridge context hypotheses ensure that boundary cells (at column j) interact correctly with the modified prefix. □

Lemma 6.175 (Prefix BK involutivity in Stembridge context)**.** *Under the Stembridge context, the prefix-restricted BK involution is an involution: $\beta_k^{<j}(\beta_k^{<j}(T)) = T$. Moreover, the Stembridge context is preserved: $T' = \beta_k^{<j}(T)$ again satisfies the partition and misstep conditions needed for the next application.*

Proof. The suffix content is preserved (columns $\geq j$ are unchanged), so the violator column and misstep index remain the same. Applying $\beta_k^{<j}$ twice returns every prefix cell to its original value by the matching symmetry. □

Lemma 6.176 (Column j has no entry k at the max violator)**.** *Under the Stembridge context (where j is the max violator column and k is the misstep index), column j contains no entry equal to k. In particular, $\mathrm{cont}(\mathrm{col}_{\geq j} T)_k = \mathrm{cont}(\mathrm{col}_{\geq j+1} T)_k$.*

Proof. Since $\beta = \nu + \mathrm{cont}(\mathrm{col}_{\geq j+1} T)$ is a partition and $\alpha = \nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$ has a misstep at k, the difference $\alpha_k - \beta_k$ must be zero. Otherwise the misstep at k in α would contradict the partition property of β combined with the constraint $\alpha_{k+1} > \alpha_k$. □

6.4.11 Stembridge's involution and sign-reversing property

Definition 6.177 (Stembridge's involution)**.** For a semistandard tableau T that is not ν-Yamanouchi, define $T^* = \mathrm{stemb}_\nu(T)$ as follows:

1. Let j be the *largest* violator column (where $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$ is not a partition).
2. Let k be the *smallest* misstep index of $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$.
3. Apply the prefix-restricted BK involution: $T^* = \beta_k^{<j}(T)$.

Lemma 6.178 (Stembridge involution preserves semistandardness)**.** *If T is semistandard and not ν-Yamanouchi, then* $\mathrm{stemb}_\nu(T)$ *is semistandard.*

Proof. Follows from the involutivity theorem, which establishes semistandardness of T' as a prerequisite. ☐

Lemma 6.179 (Stembridge involution preserves non-Yamanouchi)**.** *If T is not ν-Yamanouchi, then* $\mathrm{stemb}_\nu(T)$ *is also not ν-Yamanouchi.*

Proof. The violator column j remains a violator for T' because $\beta_k^{<j}$ leaves columns $\geq j$ unchanged, so $\nu + \mathrm{cont}(\mathrm{col}_{\geq j} T') = \nu + \mathrm{cont}(\mathrm{col}_{\geq j} T)$, which is not a partition. ☐

Lemma 6.180 (Max violator column is preserved)**.** *Let j be the maximum violator column for T. Then $\nu + \mathrm{cont}(\mathrm{col}_{\geq j+1} T)$ is an N-partition. (This is because $j{+}1$ is* not *a violator column, being larger than the maximum.) If $j{+}1$ exceeds all column indices, the result reduces to ν being a partition.*

Proof. If $j{+}1 \leq \max_i \lambda_i + 1$, then $j{+}1$ is in the range of potential violator columns but is not a violator (since j is the max), so $\nu + \mathrm{cont}(\mathrm{col}_{\geq j+1} T)$ is a partition by definition. Otherwise, $\mathrm{cont}(\mathrm{col}_{\geq j+1} T) = \mathbf{0}$, and the result is ν, which is a partition by hypothesis. ☐

Theorem 6.181 (Stembridge involution is an involution)**.** *For N-partitions λ, μ, ν and a semistandard tableau T that is not ν-Yamanouchi: let $T' = \mathrm{stemb}_\nu(T)$. Then T' is semistandard, T' is not ν-Yamanouchi, and* $\mathrm{stemb}_\nu(T') = T$.

Proof. The key insight is that the parameters (j, k) are *the same* for T and T': the max violator column j' for T' equals j (since $\beta_k^{<j}$ leaves columns $\geq j$ unchanged, and columns $> j$ are not violators for either tableau). The misstep set is unchanged because $\mathrm{cont}(\mathrm{col}_{\geq j} T') = \mathrm{cont}(\mathrm{col}_{\geq j} T)$. Therefore $\mathrm{stemb}_\nu(T') = \beta_k^{<j}(\beta_k^{<j}(T)) = T$. ☐

Lemma 6.182 (Stembridge involution content transposition)**.** *Let $T' = \mathrm{stemb}_\nu(T)$. There exist indices $k \neq k'$ (namely k and $k{+}1$) such that*

$$\nu + \mathrm{cont}(T') + \rho = (\nu + \mathrm{cont}(T) + \rho) \circ (k\ k').$$

That is, the full exponent tuple is related by a transposition.

Proof. The proof uses the decomposition of content into prefix and suffix parts. The suffix is unchanged ($\mathrm{col}_{\geq j}$ is preserved). The prefix content swaps: the number of k's in the prefix of T' equals the number of $(k{+}1)$'s in the prefix of T, and vice versa. Combined with the misstep condition $\alpha_{k+1} = \alpha_k + 1$ and the rho shift $\rho_k - \rho_{k+1} = 1$, the full expressions at positions k and $k{+}1$ are exchanged. ☐

Lemma 6.183 (Prefix BK content swap). *For the prefix-restricted BK involution $T' = \beta_k^{<j}(T)$:*

$$\mathrm{cont}(T')_k + \mathrm{cont}(\mathrm{col}_{\geq j} T)_{k+1} = \mathrm{cont}(T)_{k+1} + \mathrm{cont}(\mathrm{col}_{\geq j} T)_k.$$

This identity captures how the BK swap in the prefix interacts with the unchanged suffix.

Proof. Decompose $\mathrm{cont}(T)_i = (\text{prefix } i \text{ count}) + \mathrm{cont}(\mathrm{col}_{\geq j} T)_i$. The suffix is unchanged, so it suffices to show: (prefix k count in T') = (prefix $(k+1)$ count in T). The prefix cells where $T'(c) = k$ are exactly the prefix cells that were k in T and stayed as k (forced k's and matched free k's), plus the unmatched free $(k+1)$'s (which became k). The prefix cells where $T(c) = k+1$ decompose into cells that stay as $(k+1)$ (forced and matched) and unmatched $(k+1)$'s (which become k). The unchanged cells contribute equally to both sides, and the unmatched $(k+1)$'s appear on both sides as well, yielding the identity. □

Lemma 6.184 (Stembridge involution is sign-reversing). *For a non-ν-Yamanouchi semistandard tableau T:*

$$a_{\nu+\mathrm{cont}(\mathrm{stemb}_\nu(T))+\rho} = -\, a_{\nu+\mathrm{cont}(T)+\rho}.$$

Proof. By Lemma 6.182, the exponent tuple is related by a transposition. By Lemma 6.152 **(b)**, swapping two entries of the exponent negates the alternant. □

Lemma 6.185 (Fixed points have zero alternant). *If $\mathrm{stemb}_\nu(T) = T$ (a fixed point), then $a_{\nu+\mathrm{cont}(T)+\rho} = 0$.*

Proof. If $T' = T$, then $\mathrm{cont}(T') = \mathrm{cont}(T)$, so the content transposition gives $f = f \circ (k\ k')$ with $k \neq k'$, where $f = \nu + \mathrm{cont}(T) + \rho$. This forces $f(k) = f(k')$, so f has repeated entries. By Lemma 6.152 **(a)**, $a_f = 0$. □

Lemma 6.186 (Non-Yamanouchi alternants cancel). *For N-partitions λ, μ, ν:*

$$\sum_{\substack{T \in \mathrm{SSYT}(\lambda/\mu) \\ T \textit{ not } \nu\textit{-Yamanouchi}}} a_{\nu+\mathrm{cont}(T)+\rho} = 0.$$

Proof. The Stembridge involution partitions the non-Yamanouchi tableaux into orbits of size 1 or 2. For each pair $\{T, T^*\}$ with $T^* \neq T$, the sign-reversing property gives $a_{\nu+\mathrm{cont}(T)+\rho} + a_{\nu+\mathrm{cont}(T^*)+\rho} = 0$. At fixed points (where $T^* = T$), the alternant is zero by Lemma 6.185. Hence, all terms cancel and the sum is zero. □

Lemma 6.187 (Alternant sum splits into Yamanouchi and non-Yamanouchi). *The sum of alternants over all semistandard tableaux equals the sum over Yamanouchi tableaux plus the sum over non-Yamanouchi tableaux:*

$$\sum_{T \in \mathrm{SSYT}} a_{\nu+\mathrm{cont}(T)+\rho} = \sum_{\substack{T \in \mathrm{SSYT} \\ \nu\textit{-Yam}}} a_{\nu+\mathrm{cont}(T)+\rho} + \sum_{\substack{T \in \mathrm{SSYT} \\ \textit{not } \nu\textit{-Yam}}} a_{\nu+\mathrm{cont}(T)+\rho}.$$

Proof. The semistandard tableaux partition into Yamanouchi and non-Yamanouchi subsets. The sum over a disjoint union equals the sum of the parts. □

6.4.12 Stembridge's Lemma

Lemma 6.188 (Stembridge's Lemma)**.** *Let λ, μ, ν be three N-partitions. Then,*

$$a_{\nu+\rho} \cdot s_{\lambda/\mu} = \sum_{\substack{T \text{ is a } \nu\text{-Yamanouchi} \\ \text{semistandard tableau} \\ \text{of shape } \lambda/\mu}} a_{\nu+\operatorname{cont} T+\rho}.$$

Proof. The proof proceeds in several steps.

Step 1. By Lemma 6.168 (which uses the symmetry of $s_{\lambda/\mu}$ from Lemma 6.167):

$$a_{\nu+\rho} \cdot s_{\lambda/\mu} = \sum_{T \in \operatorname{SSYT}(\lambda/\mu)} a_{\nu+\operatorname{cont} T+\rho}.$$

Step 2. By Lemma 6.187, split the sum into Yamanouchi and non-Yamanouchi tableaux.

Step 3. By Lemma 6.186, the non-Yamanouchi sum is zero (using the Stembridge involution from Definition 6.177, its sign-reversing property from Lemma 6.184, and its fixed-point behavior from Lemma 6.185). ☐

6.4.13 Consequences of Stembridge's Lemma

Lemma 6.189. *Let λ be any N-partition. Let T be a semistandard tableau of shape λ. Then, $T(i,j) \geq i$ for each $(i,j) \in Y(\lambda)$.*

Proof. This lemma is an easy consequence of the fact that the entries of a semistandard tableau increase strictly down each column. More precisely, the cells $(1,j), (2,j), \ldots, (i,j)$ all belong to $Y(\lambda)$ (since λ is weakly decreasing), and the entries $T(1,j) < T(2,j) < \cdots < T(i,j)$ are strictly increasing. Since $T(1,j) \geq 1$, we get $T(i,j) \geq i$ by induction. ☐

Lemma 6.190 (Minimalistic tableau is semistandard)**.** *The minimalistic tableau T_0 of shape λ (where $T_0(i,j) = i$ for all cells) is semistandard: entries are constant (hence weakly increasing) along rows, and strictly increasing down columns.*

Proof. Row-weak: $T_0(i,j_1) = i = T_0(i,j_2)$ for any $j_1 < j_2$. Column-strict: if $i_1 < i_2$, then $T_0(i_1,j) = i_1 < i_2 = T_0(i_2,j)$. ☐

Lemma 6.191 (Minimalistic tableau is Yamanouchi)**.** *Let λ be an N-partition. The minimalistic tableau T_0 of shape λ (whose entry at each cell (i,j) is i) is a $\mathbf{0}$-Yamanouchi semistandard tableau of shape $\lambda/\mathbf{0}$.*

Proof. The minimalistic tableau is semistandard (entries weakly increase along rows and strictly increase down columns). For the $\mathbf{0}$-Yamanouchi condition, the content of $\operatorname{col}_{\geq j} T_0$ at row i equals $\max\{\lambda_i - (j-1), 0\}$, which is weakly decreasing since λ is. ☐

Lemma 6.192 (Content of minimalistic tableau)**.** *Let λ be an N-partition. The content of the minimalistic tableau T_0 equals λ: $\operatorname{cont}(T_0) = \lambda$.*

Proof. The minimalistic tableau has all entries in row i equal to i, so the number of i's in T_0 is λ_i (the number of cells in row i). ☐

Lemma 6.193 (Yamanouchi rightmost entry equals row index)**.** *Let λ be an N-partition and T a $\mathbf{0}$-Yamanouchi tableau of shape $\lambda/\mathbf{0}$. For each row i with $\lambda_i > 0$, the rightmost cell (i, λ_i) satisfies $T(i, \lambda_i) = i$.*

Proof. By strong induction on i. Suppose $T(i, \lambda_i) = m \neq i$. By Lemma 6.189, $m \geq i$, and since $m \neq i$, we have $m > i$. From the $\mathbf{0}$-Yamanouchi condition, $\operatorname{cont}(\operatorname{col}_{\geq \lambda_i} T)$ is a partition, so $\operatorname{cont}(\operatorname{col}_{\geq \lambda_i} T)_i \geq \operatorname{cont}(\operatorname{col}_{\geq \lambda_i} T)_m \geq 1$. Hence row i has some entry i in columns $\geq \lambda_i$. But (i, λ_i) is the rightmost cell of row i, so $T(i, \lambda_i) \leq i$ by weak row increase from the cell where i appears. For rows $i' < i$, the inductive hypothesis gives $T(i', \lambda_{i'}) = i'$; the strict column increase then prevents any cell in the same column as (i, λ_i) from having value i in an earlier row. This contradicts $m > i$. $\square$

Lemma 6.194 (Uniqueness of $\mathbf{0}$-Yamanouchi tableau)**.** *Let λ be an N-partition. If T is a $\mathbf{0}$-Yamanouchi semistandard tableau of shape $\lambda/\mathbf{0}$, then T is the minimalistic tableau T_0.*

Proof. By Lemma 6.193, $T(i, \lambda_i) = i$ for each row. By Lemma 6.189, $T(i, j) \geq i$ for all (i, j). By weak row increase, $T(i, j) \leq T(i, \lambda_i) = i$ for all $j \leq \lambda_i$. Combining: $T(i, j) = i$ for all (i, j), so $T = T_0$. $\square$

Lemma 6.195 (Schur–alternant relation)**.** *Let λ be an N-partition. Then,*

$$a_{\lambda+\rho} = a_\rho \cdot s_\lambda.$$

(This is Theorem 6.107 ***(b)****.)*

Proof. Apply Lemma 6.188 with $\mu = \mathbf{0}$ and $\nu = \mathbf{0}$:

$$a_\rho \cdot s_\lambda = \sum_{\substack{T \text{ is a } \mathbf{0}\text{-Yamanouchi} \\ \text{semistandard tableau of shape } \lambda/\mathbf{0}}} a_{\operatorname{cont} T + \rho}.$$

By Lemma 6.191, the minimalistic tableau T_0 is $\mathbf{0}$-Yamanouchi, and by Lemma 6.194, it is the *only* such tableau. Therefore the sum has a single term. By Lemma 6.192, $\operatorname{cont}(T_0) = \lambda$, so we obtain $a_\rho \cdot s_\lambda = a_{\lambda+\rho}$. $\square$

6.4.14 The Littlewood–Richardson rule

Theorem 6.196 (Zelevinsky's generalized Littlewood–Richardson rule, in Yamanouchi form)**.** *Let λ, μ, ν be three N-partitions. Then,*

$$s_\nu \cdot s_{\lambda/\mu} = \sum_{\substack{T \text{ is a } \nu\text{-Yamanouchi} \\ \text{semistandard tableau} \\ \text{of shape } \lambda/\mu}} s_{\nu + \operatorname{cont} T}. \tag{6.8}$$

Proof. **Step 1.** By Lemma 6.195 (applied to ν instead of λ), $a_{\nu+\rho} = a_\rho \cdot s_\nu$.

Step 2. By Lemma 6.188,

$$a_{\nu+\rho} \cdot s_{\lambda/\mu} = \sum_{\substack{T \text{ is a } \nu\text{-Yamanouchi} \\ \text{semistandard tableau} \\ \text{of shape } \lambda/\mu}} a_{\nu + \operatorname{cont} T + \rho}.$$

Step 3. For each ν-Yamanouchi tableau T, the tuple $\nu + \operatorname{cont} T$ is an N-partition (by the $j = 1$ case of the Yamanouchi condition), so Lemma 6.195 yields $a_{\nu + \operatorname{cont} T + \rho} = a_\rho \cdot s_{\nu + \operatorname{cont} T}$.

Step 4. Substituting Steps 1 and 3 into Step 2:

$$a_\rho \cdot s_\nu \cdot s_{\lambda/\mu} = a_\rho \cdot \sum_{\substack{T \text{ is a } \nu\text{-Yamanouchi} \\ \text{semistandard tableau} \\ \text{of shape } \lambda/\mu}} s_{\nu + \operatorname{cont} T}.$$

Step 5. Since a_ρ is regular (Lemma 6.155), we can cancel it (Lemma 6.154) to obtain

$$s_\nu \cdot s_{\lambda/\mu} = \sum_{\substack{T \text{ is a } \nu\text{-Yamanouchi} \\ \text{semistandard tableau} \\ \text{of shape } \lambda/\mu}} s_{\nu+\operatorname{cont} T}.$$

□

6.5 The Pieri Rules and Jacobi–Trudi Identities

6.5.1 Strips and skew partitions

Definition 6.197. Let λ and μ be two N-partitions.

(a) We write λ/μ for the pair (μ, λ). Such a pair is called a *skew partition*.

(b) We say that λ/μ is a *horizontal strip* if we have $\mu \subseteq \lambda$ and the Young diagram $Y(\lambda/\mu)$ has no two boxes lying in the same column.

(c) We say that λ/μ is a *vertical strip* if we have $\mu \subseteq \lambda$ and the Young diagram $Y(\lambda/\mu)$ has no two boxes lying in the same row.

Now, let $n \in \mathbb{N}$.

(d) We say that λ/μ is a *horizontal n-strip* if λ/μ is a horizontal strip and satisfies $|Y(\lambda/\mu)| = n$.

(e) We say that λ/μ is a *vertical n-strip* if λ/μ is a vertical strip and satisfies $|Y(\lambda/\mu)| = n$.

Proposition 6.198. *Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_N)$ and $\mu = (\mu_1, \mu_2, \ldots, \mu_N)$ be two N-partitions.*

(a) *The skew partition λ/μ is a horizontal strip if and only if we have*

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \cdots \geq \lambda_N \geq \mu_N.$$

(b) *The skew partition λ/μ is a vertical strip if and only if we have*

$$\mu_i \leq \lambda_i \leq \mu_i + 1 \qquad \textit{for each } i \in [N].$$

Proof. **(a)** ($\Rightarrow$): A horizontal strip satisfies $\mu_i \geq \lambda_{i+1}$ for all valid i by definition, and $\lambda_i \geq \mu_i$ follows from the containment condition $\mu \subseteq \lambda$. ($\Leftarrow$): The interlacing condition $\mu_i \geq \lambda_{i+1}$ is exactly the horizontal strip condition.

(b) ($\Rightarrow$): A vertical strip satisfies $\lambda_i \leq \mu_i + 1$ by definition, and $\mu_i \leq \lambda_i$ follows from containment. ($\Leftarrow$): The condition $\lambda_i \leq \mu_i + 1$ is exactly the vertical strip condition. □

Strip characterization helpers

Lemma 6.199. *The definition of horizontal strip from Definition 6.197 is equivalent to the entry-wise characterization used in the enumeration of strip partitions.*

Proof. Immediate from the definitions. □

Lemma 6.200. *The definition of vertical strip from Definition 6.197 is equivalent to the entry-wise characterization used in the enumeration of strip partitions.*

Proof. Immediate from the definitions. □

Lemma 6.201. *If $\mu \subseteq \lambda$, λ/μ is a horizontal n-strip, and λ satisfies the upper bound constraints, then λ belongs to the finite set of horizontal n-strip partitions of μ. This provides a simplified membership criterion bridging the definition of horizontal n-strip with the finite set of strip partitions.*

Proof. Combine the horizontal n-strip hypothesis with the size-difference computation and apply the membership characterization. $\square$

Lemma 6.202. *If $\mu \subseteq \lambda$ and λ/μ is a vertical n-strip, then λ belongs to the finite set of vertical n-strip partitions of μ.*

Proof. Combine the vertical n-strip hypothesis with the size-difference computation and apply the membership characterization. $\square$

Partition transpose

Definition 6.203. The *transpose* (or *conjugate*) of an N-partition λ is the partition λ^t whose i-th part equals $|\{j : j < N,\ \lambda_j > i\}|$, i.e., the number of parts of λ that exceed i. Requires $N > 0$; the result is a partition of length λ_1.

Definition 6.204. Two functions $\lambda : \operatorname{Fin} N \to \mathbb{N}$ and $\lambda^t : \operatorname{Fin} M \to \mathbb{N}$ are *transposes* of each other if for all $i < N$ and $j < M$:

$$\lambda_i > j \iff \lambda^t_j > i.$$

This is the symmetric characterization of the transpose relation between Young diagrams.

Lemma 6.205. *If λ and λ^t are transposes, then there is a bijection between the boxes of the Young diagram of λ (pairs (i, j) with $j < \lambda_i$) and those of λ^t (pairs (j, i) with $i < \lambda^t_j$), given by swapping coordinates.*

Proof. The bijection maps (i, j) to (j, i). The transpose condition $\lambda_i > j \iff \lambda^t_j > i$ guarantees that this map is well-defined and bijective. $\square$

Lemma 6.206. *If λ and λ^t are transposes, then $\sum_i \lambda_i = \sum_j \lambda^t_j$. That is, the total number of boxes is preserved by transposition.*

Proof. Both sums count the cardinality of the respective Young diagram box sets. The bijection from Lemma 6.205 shows these sets have the same cardinality. $\square$

6.5.2 Schur and skew Schur polynomials

Definition 6.207. The *Schur polynomial* s_λ is defined as $s_\lambda = \sum_{T \in \mathrm{SSYT}(\lambda)} x_T$, where the sum is over all semistandard Young tableaux of shape λ with entries in $[N]$.

Lemma 6.208. *The Schur polynomial of the zero partition is 1: $s_0 = 1$.*

Proof. The only SSYT of shape 0 is the empty tableau, whose monomial is 1. $\square$

Definition 6.209. The *skew Schur polynomial* $s_{\lambda/\mu}$ is defined as $s_{\lambda/\mu} = \sum_{T \in \mathrm{SSYT}(\lambda/\mu)} x_T$, where the sum is over all semistandard Young tableaux of skew shape λ/μ.

Lemma 6.210. *For any N-partition λ, the skew Schur polynomial $s_{\lambda/0}$ equals the Schur polynomial s_λ.*

Proof. A skew partition $\lambda/0$ is just the partition λ, so $\mathrm{SSYT}(\lambda)$ and $\mathrm{SSYT}(\lambda/0)$ are in natural bijection preserving monomials. $\square$

Lemma 6.211. *For any N-partition λ, $s_\lambda = s_{\lambda/0}$.*

Proof. This is the reverse direction of Lemma 6.210. $\square$

6.5.3 Symmetric polynomials as Schur polynomials

Definition 6.212. The *row partition* $(n, 0, 0, \ldots, 0)$ with n boxes in the first row. This is the partition corresponding to h_n. Requires $N > 0$.

Definition 6.213. The *column partition* $(1, 1, \ldots, 1, 0, \ldots, 0)$ with n ones (i.e., n boxes in the first column). This is the partition corresponding to e_n. Requires $n \leq N$.

Theorem 6.214. *For $N > 0$, the complete homogeneous symmetric polynomial h_n equals the Schur polynomial indexed by the row partition:*

$$h_n = s_{(n,0,\ldots,0)}.$$

Proof. Both sides are sums of monomials: h_n sums over multisets of size n from $\mathrm{Fin}\, N$, while $s_{(n,0,\ldots,0)}$ sums over SSYT of row partition shape. There is a natural bijection identifying a multiset of size n with the unique SSYT whose only nonempty row (row 0) contains the sorted multiset entries. Weight preservation follows from the fact that both sides compute the product of x_j over the entries. □

Theorem 6.215. *For $n \leq N$, the elementary symmetric polynomial e_n equals the Schur polynomial indexed by the column partition:*

$$e_n = s_{(1^n, 0^{N-n})}.$$

Proof. Both sides are sums of monomials: e_n sums over subsets of $\mathrm{Fin}\, N$ of size n, while $s_{(1^n,0^{N-n})}$ sums over SSYT of column partition shape (single-column tableaux with n rows). There is a natural bijection identifying a size-n subset with the SSYT whose column entries are the sorted elements of the subset. Injectivity, surjectivity, and weight preservation are verified explicitly. □

Yamanouchi tableaux infrastructure for Pieri rules

Definition 6.216. A skew tableau T of shape λ/μ has *all entries zero* if every cell $(i, j) \in Y(\lambda/\mu)$ satisfies $T(i, j) = 0$. This is the key property characterizing Yamanouchi tableaux for the row partition $\nu = (n, 0, \ldots, 0)$.

Theorem 6.217. *If T is a semistandard tableau with all entries equal to 0, then T is ν-Yamanouchi for the row partition $\nu = (n, 0, \ldots, 0)$ for every n.*

This is the reverse direction of the Yamanouchi characterization for row partitions: the content at any position $k > 0$ vanishes when all entries are 0, so $\nu + \mathrm{cont}_{\geq j}(T)$ is automatically weakly decreasing.

Proof. When all entries are 0, the content $\mathrm{cont}_{\geq j}(T)$ at any position $k > 0$ is 0 (no cell has entry k). Therefore $\nu + \mathrm{cont}_{\geq j}(T) = (n + c_0, 0, \ldots, 0)$ for some $c_0 \geq 0$, which is weakly decreasing. □

Definition 6.218. Convert a multiset $s \in \mathrm{Sym}(\mathrm{Fin}\, N, n)$ to an SSYT of row partition shape $(n, 0, \ldots, 0)$: row 0 contains the sorted entries of s, and all other rows are empty.

Definition 6.219. Convert an SSYT of row partition shape $(n, 0, \ldots, 0)$ back to a multiset: extract the entries of row 0 (which form a weakly increasing sequence of length n).

Lemma 6.220. *The forward and inverse maps of the multiset–row SSYT bijection are inverses: converting a multiset to a row SSYT and back gives the original multiset.*

Proof. Sorting a multiset and extracting it gives back the original multiset. □

Definition 6.221. Convert a subset $s \subseteq \operatorname{Fin} N$ of size n to an SSYT of column partition shape $(1^n, 0^{N-n})$: column 0 contains the sorted elements of s in strictly increasing order.

Lemma 6.222. *The column entries of any SSYT of column partition shape are strictly increasing: if $k_1 < k_2 < n$, then $T(k_1, 0) < T(k_2, 0)$. This follows from the column-strict condition of semistandard tableaux.*

Proof. By induction on $k_2 - k_1$: the consecutive case uses column-strictness of SSYT directly; the general case follows by transitivity. □

6.5.4 The Pieri rules

Definition 6.223. The finite set of N-partitions λ such that λ/μ is a horizontal n-strip. This is computed by filtering all functions bounded by the horizontal strip constraints for weak decrease, horizontal strip condition, and size difference.

Definition 6.224. The finite set of N-partitions λ such that λ/μ is a vertical n-strip. This is computed by filtering all functions bounded by the vertical strip constraints.

Lemma 6.225. *An N-partition λ belongs to* horizontalNStripPartitions(μ, n) *if and only if $\mu \subseteq \lambda$, λ/μ is a horizontal strip, and $|\lambda| - |\mu| = n$.*

Proof. Unfold the filter conditions and relate them to the strip definitions. □

Lemma 6.226. *An N-partition λ belongs to* verticalNStripPartitions(μ, n) *if and only if $\mu \subseteq \lambda$, λ/μ is a vertical strip, and $|\lambda| - |\mu| = n$.*

Proof. Unfold the filter conditions and relate them to the strip definitions. □

Theorem 6.227 (Pieri rules)**.** *Let $n \in \mathbb{N}$. Let μ be an N-partition. Then:*

(a) *We have*

$$h_n s_\mu = \sum_{\substack{\lambda \text{ is an } N\text{-partition;} \\ \lambda/\mu \text{ is a horizontal } n\text{-strip}}} s_\lambda.$$

(b) *We have*

$$e_n s_\mu = \sum_{\substack{\lambda \text{ is an } N\text{-partition;} \\ \lambda/\mu \text{ is a vertical } n\text{-strip}}} s_\lambda.$$

Proof. **(a)** The proof follows from the Littlewood–Richardson rule (Theorem 6.196) by specializing to the case where one of the partitions is a single row $(n, 0, 0, \ldots, 0)$.

Alternatively, a direct combinatorial proof uses the RSK row insertion:

1. Expand $h_n = \sum_{s \in \operatorname{Sym}(\operatorname{Fin} N, n)} \prod_{j \in s} x_j$ and $s_\mu = \sum_{T \in \operatorname{SSYT}(\mu)} x_T$.
2. Distribute to get $h_n \cdot s_\mu = \sum_{s,T} \prod_{j \in s} x_j \cdot x_T$.
3. Apply the RSK row insertion bijection: $\{(s, T) : \operatorname{Sym}(\operatorname{Fin} N, n) \times \operatorname{SSYT}(\mu)\} \simeq \bigsqcup_{\lambda/\mu \text{ horiz } n\text{-strip}} \operatorname{SSYT}(\lambda)$. Given a sorted multiset $s = \{a_1, \ldots, a_n\}$ and tableau T of shape μ, row-insert $a_1, \ldots, a_n$ into T to get a tableau T' of shape λ where λ/μ is a horizontal n-strip.
4. Weight preservation: $\prod_{j \in s} x_j \cdot x_T = x_{T'}$.
5. Conclude $h_n \cdot s_\mu = \sum_\lambda s_\lambda$.

(b) Analogous using RSK column insertion, or via the ω-involution. □

6.5.5 The Jacobi–Trudi identities

Definition 6.228. The extended h_n: $h_n = 0$ for $n < 0$, $h_0 = 1$. This is needed since the Jacobi–Trudi formula may have negative indices.

Lemma 6.229. *There is a canonical equivalence*

$$\mathrm{LatticePath}(a, c) \simeq \mathrm{Sym}(\mathrm{Fin}\, N, (c-a)^+)$$

between lattice paths from $(a, 1)$ to (c, N) and multisets of size $(c-a)^+$ from $\mathrm{Fin}\, N$. The forward map extracts the multiset of east-step heights; the inverse sorts a multiset into a weakly increasing list.

Proof. Roundtrip properties follow from the fact that sorting a weakly increasing list is the identity, and sorting a multiset then extracting gives back the multiset. □

Definition 6.230. A *non-intersecting path tuple* (nipat) from $\mathbf{A} = (A_1, \ldots, A_N)$ to $\mathbf{B} = (B_1, \ldots, B_N)$ is an N-tuple of lattice paths $(p_1, \ldots, p_N)$ where p_i goes from $(\mu_i - i, 1)$ to $(\lambda_i - i, N)$, subject to a column-strictness condition: for $i < j$, if east steps k (in path i) and k' (in path j) correspond to the same tableau column ($\mu_i + k = \mu_j + k'$), then the height of path i at step k is strictly less than the height of path j at step k'.

The *weight* of a nipat is $\prod_i w(p_i)$.

Theorem 6.231 (Observation 1)**.** *Let a and c be two integers. Then,*

$$\sum_{\substack{p \text{ is a path} \\ \text{from } (a,1) \text{ to } (c,N)}} w(p) = h_{c-a}.$$

Proof. When $c - a \geq 0$, the bijection between lattice paths and $\mathrm{Sym}(\mathrm{Fin}\, N, (c-a))$ (via Lemma 6.229) gives:

$$\sum_p w(p) = \sum_{s \in \mathrm{Sym}(\mathrm{Fin}\, N, (c-a))} \prod_{j \in s} x_j = h_{c-a}.$$

When $c - a < 0$, both sides are 0. □

Nipat–SSYT bijection infrastructure

Definition 6.232. The forward map of the nipat–SSYT bijection: given a nipat $\mathbf{p} = (p_1, \ldots, p_N)$, construct the skew SSYT $T(\mathbf{p})$ of shape λ/μ whose i-th row contains the east-step heights of path p_i.

Row-weakness follows from the weakly increasing property of each path's east-step heights. Column-strictness follows from the column-strictness condition of the nipat.

Definition 6.233. The inverse map of the nipat–SSYT bijection: given a skew SSYT T of shape λ/μ, construct the nipat $\mathbf{p}(T)$ whose i-th path has east-step heights given by row i of T.

The weakly increasing property of each path follows from row-weakness of T. The column-strictness condition follows from column-strictness of non-adjacent rows in T.

Lemma 6.234. *Converting a nipat to a tableau and back gives the original nipat.*

Proof. The entries of each row of the constructed tableau are the east-step heights of the original nipat. Converting back produces the same east-step lists. □

Lemma 6.235. *Converting a tableau to a nipat and back gives the original tableau.*

Proof. The proof uses entry specification lemmas to show the entries agree pointwise. □

Lemma 6.236. *The weight of a nipat equals the monomial of the corresponding tableau:* $w(\mathbf{p}) = x_{T(\mathbf{p})}$ *for every nipat* $\mathbf{p}$.

Proof. The weight of a nipat is $\prod_i w(p_i)$ and the monomial of the SSYT is $\prod_i \prod_k x_{T(i,k)}$. Since the k-th entry of row i in the tableau equals the k-th east-step height of path p_i, these are equal. □

Lemma 6.237. *The forward and inverse maps of the nipat–SSYT bijection form an equivalence of types:*

$$\mathrm{Nipat}(\lambda, \mu) \simeq \mathrm{SkewSSYT}(\lambda/\mu).$$

Proof. Combine the two roundtrip lemmas. □

Lemma 6.238. *The sum of nipat weights equals the sum of SSYT monomials:*

$$\sum_{\mathbf{p} \in \mathrm{Nipat}(\lambda,\mu)} w(\mathbf{p}) = \sum_{T \in \mathrm{SkewSSYT}(\lambda/\mu)} x_T.$$

Proof. Transport the sum along the equivalence with weight preservation. □

Theorem 6.239 (Observation 2: Nipat–SSYT bijection)**.** *There is a bijection*

$$\{\textit{nipats from } \mathbf{A} \textit{ to } \mathbf{B}\} \xrightarrow{\sim} \mathrm{SSYT}(\lambda/\mu), \qquad \mathbf{p} \mapsto T(\mathbf{p}),$$

where for a nipat $\mathbf{p} = (p_1, \ldots, p_N)$*, the tableau* $T(\mathbf{p})$ *has its* i*-th row consisting of the heights of the east-steps of* p_i*.*

Moreover, $w(\mathbf{p}) = x_{T(\mathbf{p})}$ *for every nipat* $\mathbf{p}$.

Proof. The forward map sends a nipat $\mathbf{p}$ to the tableau $T(\mathbf{p})$; the inverse extracts the east-step heights from each row of a skew SSYT. The bijectivity is proved via:

- Injectivity: Lemma 6.234 shows that converting a nipat to a tableau and back gives the original nipat.
- Surjectivity: Lemma 6.235 shows that converting a tableau to a nipat and back gives the original tableau.

Weight preservation (Lemma 6.236) follows from the fact that the weight of each component path is the product of x_j over its east-step heights, which are exactly the row entries of the tableau. □

Lemma 6.240. *The sum of nipat weights equals the skew Schur polynomial:*

$$\sum_{\mathbf{p} \text{ nipat}} w(\mathbf{p}) = s_{\lambda/\mu}.$$

Proof. Follows from the nipat–SSYT bijection (Theorem 6.239) and the definition of $s_{\lambda/\mu}$ as a sum over SSYT monomials. The proof uses Lemma 6.238 followed by a bijective reindexing to relate the finset sum to the type sum. □

LGV connection for Jacobi–Trudi

Definition 6.241. The arc weight function for the Jacobi–Trudi proof on the integer lattice $\mathbb{Z}^2$: east-steps at height y (with $1 \le y \le N$) are weighted by x_{y-1}; north-steps are weighted by 1.

Definition 6.242. Define two N-vertices $\mathbf{A} = (A_1, \ldots, A_N)$ and $\mathbf{B} = (B_1, \ldots, B_N)$ by

$$A_i := (\mu_i - i,\ 1) \quad \text{and} \quad B_i := (\lambda_i - i,\ N) \quad \text{for each } i \in [N].$$

Lemma 6.243. *The source x-coordinates are weakly decreasing: $\mu_i - i \ge \mu_j - j$ when $i \le j$. This follows from the partition being weakly decreasing: $\mu_i \ge \mu_j$ and $i \le j$ imply $\mu_i - i \ge \mu_j - j$.*

Proof. Direct computation from the weakly decreasing property of partitions. □

Lemma 6.244. *The target x-coordinates are weakly decreasing: $\lambda_i - i \ge \lambda_j - j$ when $i \le j$.*

Proof. Analogous to Lemma 6.243. □

Definition 6.245. Convert an LGV path (in the integer lattice digraph) from $(a, 1)$ to (c, N) into a lattice path by extracting the y-coordinates at which east-steps occur and converting them to Fin N values (by subtracting 1).

Lemma 6.246. *The number of east-step y-coordinates extracted from an LGV path vertex list equals the x-displacement (final x minus initial x). This is a key property for the bijection between LGV paths and lattice paths.*

Proof. By induction on the vertex list: each east-step increases the count by 1 and the x-coordinate by 1; north-steps change neither. □

Lemma 6.247. *The east-step y-coordinates extracted from an LGV path are weakly increasing: since paths can only go north or east, y-coordinates can only increase or stay the same along the path.*

Proof. By induction on the vertex list. For an east-step from v_0 to v_1, the y-coordinates satisfy $y(v_0) = y(v_1)$, and the remaining heights are $\ge y(v_1)$ by the inductive hypothesis and y-monotonicity. □

Lemma 6.248. *Each east-step y-coordinate is bounded between the starting and ending y-coordinates of the path: if y is an east-step y-coordinate of a path from (a, y_0) to (c, y_1), then $y_0 \le y \le y_1$.*

Proof. Each east-step y-coordinate is the y-coordinate of some vertex of the path. The bound follows from y-monotonicity along the path. □

Theorem 6.249. *The LGV path weight sum from $(a, 1)$ to (c, N) using the Jacobi–Trudi arc weight equals h_{c-a}:*

$$\sum_{p:(a,1)\to(c,N)} w_{\mathrm{LGV}}(p) = h_{c-a}.$$

When $c - a \ge 0$, this is proved via a weight-preserving bijection between LGV paths and our lattice path type, composed with Theorem 6.231. When $c - a < 0$, both sides are 0.

Proof. For $c - a \ge 0$: compose the LGV-path-to-lattice-path bijection with the lattice path–multiset equivalence (Lemma 6.229), then apply Theorem 6.231. For $c - a < 0$: the set of LGV paths is empty (no path can have decreasing x-coordinate), and $h_{c-a} = 0$ by convention. □

Lemma 6.250. *The LGV arc-weight product of a path equals the lattice path weight of its conversion. This follows from the fact that the arc weight assigns x_{y-1} to east-steps at height y, which matches the $\prod x_j$ formula for lattice path weight.*

Proof. Reduce to showing that the product of x_{y-1} over east-step y-coordinates equals the product of x_h over the converted $\operatorname{Fin} N$ heights. ☐

Lemma 6.251. *The conversion from LGV path to lattice path is injective: two LGV paths with the same east-step y-coordinates have the same vertex sequence. This is because a path in the integer lattice is uniquely determined by its starting point and the sequence of east-step positions.*

Proof. Two paths with the same start, end, and east-step y-coordinates must have identical vertex lists, since at each step the direction (north or east) is determined by the remaining east-step y-coordinates and the current position. ☐

Lemma 6.252. *The Jacobi–Trudi matrix H equals the transpose of the LGV path weight matrix:*

$$H = M^T \quad \textit{where } M_{i,j} = \sum_{p : A_i \to B_j} w(p).$$

Proof. Each entry $H_{i,j} = h_{\lambda_i - \mu_j - i + j}$ equals the path weight sum from A_j to B_i by Theorem 6.249, since the path length is $(\lambda_i - i) - (\mu_j - j) = \lambda_i - \mu_j - i + j$. The transposition accounts for the index swap. ☐

Lemma 6.253. *The determinant of the Jacobi–Trudi matrix equals the determinant of the LGV path weight matrix:* $\det(H) = \det(M)$*. This follows from* $H = M^T$ *and* $\det(M^T) = \det(M)$*.*

Proof. Apply $\det(H) = \det(M^T) = \det(M)$ using Lemma 6.252 and the transpose-determinant identity. ☐

Lemma 6.254. *The LGV non-intersecting path tuple weight sum (using the LGV non-intersection condition) equals the nipat weight sum (using the column-strictness condition):*

$$\sum_{\mathbf{p}\ \textit{LGV-nipat}} w(\mathbf{p}) = \sum_{\mathbf{p}\ \textit{nipat}} w(\mathbf{p}).$$

Proof. This requires constructing an explicit weight-preserving bijection between LGV path tuples and nipats. The bijection extracts east-step y-coordinates from LGV paths and converts them to weakly increasing sequences in $\operatorname{Fin} N$. Non-intersection in the LGV sense corresponds to column-strictness in the nipat sense. Injectivity follows from Lemma 6.251; weight preservation from Lemma 6.250. ☐

Lemma 6.255. *If two non-intersecting lattice paths p and p' have starting y-coordinates satisfying $y_{\text{start}}(p) < y_{\text{start}}(p')$, and at some column x_0 the path p is weakly above p' (in y-coordinate), then p stays weakly above p' at all subsequent columns.*

This is the key geometric lemma ensuring that non-intersecting paths with ordered starting positions maintain their relative order.

Proof. By the integer lattice topology: if p is above p' at column x_0 but p' rises above p at some later column, then by the intermediate value property of lattice paths, the paths must share a vertex, contradicting non-intersection. ☐

Lemma 6.256. *The determinant of the Jacobi–Trudi matrix equals the sum of nipat weights:*

$$\det(H) = \sum_{\mathbf{p} \text{ nipat}} w(\mathbf{p}).$$

Proof. The proof proceeds in steps:

1. $\det(H) = \det(M)$ by Lemma 6.253.
2. Apply the LGV nonpermutable lemma to obtain $\det(M) = \sum_{\text{LGV-nipats}} w(\mathbf{p})$. The required sorting conditions on $\mathbf{A}$ and $\mathbf{B}$ follow from Lemmas 6.243 and 6.244.
3. Connect LGV non-intersecting path tuples to non-intersecting path tuples (Definition 6.230) via Lemma 6.254.

□

Theorem 6.257 (First Jacobi–Trudi formula)**.** *Let $M \in \mathbb{N}$. Let $\lambda = (\lambda_1, \lambda_2, \ldots, \lambda_M)$ and $\mu = (\mu_1, \mu_2, \ldots, \mu_M)$ be two M-partitions (i.e., weakly decreasing M-tuples of nonnegative integers). Then,*

$$s_{\lambda/\mu} = \det\left(\left(h_{\lambda_i - \mu_j - i + j}\right)_{1 \le i \le M,\ 1 \le j \le M}\right).$$

Proof. The proof combines two key lemmas. Lemma 6.240 yields

$$s_{\lambda/\mu} = \sum_{\mathbf{p} \text{ nipat}} w(\mathbf{p}),$$

and then Lemma 6.256 gives

$$\sum_{\mathbf{p} \text{ nipat}} w(\mathbf{p}) = \det\left(\left(h_{\lambda_i - \mu_j - i + j}\right)_{i,j}\right).$$

□

Theorem 6.258 (Second Jacobi–Trudi formula)**.** *Let λ and μ be two partitions. Let λ^t and μ^t be the transposes of λ and μ. Let $M \in \mathbb{N}$ be such that both λ^t and μ^t have length $\le M$. We extend the partitions λ^t and μ^t to M-tuples (by inserting zeroes at the end). Write these M-tuples λ^t and μ^t as $\lambda^t = (\lambda^t_1, \lambda^t_2, \ldots, \lambda^t_M)$ and $\mu^t = (\mu^t_1, \mu^t_2, \ldots, \mu^t_M)$. Then,*

$$s_{\lambda/\mu} = \det\left(\left(e_{\lambda^t_i - \mu^t_j - i + j}\right)_{1 \le i \le M,\ 1 \le j \le M}\right).$$

Proof. From the first Jacobi–Trudi formula (Theorem 6.257) applied to the transpose partitions λ^t and μ^t: $s_{\lambda^t/\mu^t} = \det(h_{(\lambda^t)_i - (\mu^t)_j - i + j})$. Applying the ω-involution ($\omega(h_n) = e_n$, $\omega(s_{\lambda/\mu}) = s_{\lambda^t/\mu^t}$) and using that $\omega^2 = \mathrm{id}$ gives $s_{\lambda/\mu} = \omega(s_{\lambda^t/\mu^t}) = \det(\omega(h_{(\lambda^t)_i - (\mu^t)_j - i + j})) = \det(e_{(\lambda^t)_i - (\mu^t)_j - i + j})$. □

6.5.6 Special cases

Theorem 6.259. *For non-skew Schur polynomials ($\mu = 0$):*

$$s_\lambda = \det\left(\left(h_{\lambda_i - i + j}\right)_{1 \le i,j \le N}\right).$$

Proof. Relate s_λ to $s_{\lambda/0}$ via Lemma 6.211, then apply the general Jacobi–Trudi formula (Theorem 6.257) with $\mu = 0$. □

6.5.7 Symmetry of Schur polynomials via Jacobi–Trudi

Lemma 6.260. *The extended complete homogeneous symmetric polynomial h_n (for any $n \in \mathbb{Z}$) is a symmetric polynomial: for any permutation σ of* Fin N*, we have $\sigma(h_n) = h_n$. When $n \geq 0$, this follows from the symmetry of h_n. When $n < 0$, $h_n = 0$ is trivially symmetric.*

Proof. By case analysis on n: the non-negative case uses the symmetry of h_n; the negative case is trivial. □

Lemma 6.261. *If all entries of a matrix M over $R[x_1, \ldots, x_N]$ are symmetric polynomials, then $\det(M)$ is a symmetric polynomial. This is because for any permutation σ, $\sigma(\det(M)) = \det(\sigma(M)) = \det(M)$, where the first equality uses that σ is a ring homomorphism and the second uses that σ fixes each entry.*

Proof. For any permutation σ of Fin N, the renaming by σ commutes with determinant (as a ring homomorphism preserves sums, products, and signs). Since each entry is fixed by the renaming, the determinant is also fixed. □

Lemma 6.262. *Every entry of the Jacobi–Trudi matrix H is a symmetric polynomial. This follows from the fact that each entry is $h_{\lambda_i - \mu_j - i + j}$, which is symmetric by Lemma 6.260.*

Proof. Each entry is an extended h-polynomial, which is symmetric. □

Theorem 6.263. *The determinant of the Jacobi–Trudi matrix $\det(H)$ is a symmetric polynomial.*

Proof. Apply Lemma 6.261 with the entry symmetry from Lemma 6.262. □

Theorem 6.264. *Skew Schur polynomials are symmetric: for any partitions $\lambda \supseteq \mu$, $s_{\lambda/\mu}$ is a symmetric polynomial.*

This provides a proof of symmetry via the Jacobi–Trudi formula, without relying on the Bender–Knuth involution.

Proof. By Theorem 6.257, $s_{\lambda/\mu} = \det(H)$. By Theorem 6.263, $\det(H)$ is symmetric. □

Theorem 6.265. *Schur polynomials are symmetric: for any partition λ, s_λ is a symmetric polynomial.*

This is a corollary of Theorem 6.264 with $\mu = 0$.

Proof. By Lemma 6.211, $s_\lambda = s_{\lambda/0}$. Apply Theorem 6.264. □

6.5.8 The ω-involution

The ω-involution is an algebra automorphism of the ring of symmetric functions that swaps the elementary and complete homogeneous symmetric polynomials: $\omega(e_n) = h_n$ and $\omega(h_n) = e_n$. It is essential for proving the second Jacobi–Trudi formula from the first.

Definition 6.266. The R-algebra homomorphism from $R[x_1, \ldots, x_n]$ to the symmetric subalgebra sending x_i to h_{i+1}. This is the analogous construction to the elementary symmetric algebra homomorphism (which sends x_i to e_{i+1}), using complete homogeneous instead of elementary symmetric polynomials.

Definition 6.267. The ω-*involution* on the symmetric subalgebra, defined as the composition of the complete homogeneous algebra homomorphism with the inverse of the elementary symmetric algebra equivalence. This maps $e_k \mapsto h_k$ for $1 \leq k \leq n$ and extends algebraically to all symmetric polynomials.

Theorem 6.268. *The ω-involution maps elementary to complete homogeneous: $\omega(e_{k+1}) = h_{k+1}$ for $k < n$.*

Proof. By definition of ω: since ω is the composition of the h-algebra homomorphism with the inverse of the e-algebra equivalence, applying it to e_{k+1} first sends e_{k+1} to the generator X_k, then maps X_k to h_{k+1}. $\square$

Theorem 6.269. *The symmetric Newton–Girard identity:*

$$\sum_{j=0}^{n}(-1)^j h_j\, e_{n-j} = 0 \quad \textit{for } n > 0.$$

This is the "h-first" version of the Newton–Girard relations, obtained from the standard "e-first" version by reindexing $j \mapsto n-j$ and using commutativity and the sign identity $(-1)^{n-j} \cdot (-1)^n = (-1)^j$.

Proof. Reindex the standard Newton–Girard identity $\sum_j (-1)^j e_j h_{n-j} = 0$ via $j \mapsto n-j$ to obtain $\sum_j (-1)^{n-j} h_{n-j} e_j = 0$. Then factor out $(-1)^n$ and use the sign identity $(-1)^{n-j} = (-1)^n \cdot (-1)^j$ (up to rearrangement and $(-1)^{2j} = 1$). $\square$

Lemma 6.270. *Recurrence for h_n from Newton–Girard: for $n > 0$,*

$$h_n = \sum_{j=1}^{n}(-1)^{j+1}\, e_j\, h_{n-j}.$$

This follows by isolating the $j = 0$ term in the Newton–Girard identity $\sum_{j=0}^{n}(-1)^j e_j h_{n-j} = 0$ and rearranging.

Proof. From $\sum_{j=0}^{n}(-1)^j e_j h_{n-j} = 0$, isolate $j = 0$: $h_n + \sum_{j=1}^{n}(-1)^j e_j h_{n-j} = 0$, hence $h_n = -\sum_{j=1}^{n}(-1)^j e_j h_{n-j} = \sum_{j=1}^{n}(-1)^{j+1} e_j h_{n-j}$. $\square$

Lemma 6.271. *Recurrence for e_n from the symmetric Newton–Girard: for $n > 0$,*

$$e_n = \sum_{j=1}^{n}(-1)^{j+1}\, h_j\, e_{n-j}.$$

This follows from $\sum_{j=0}^{n}(-1)^j h_j e_{n-j} = 0$ (Theorem 6.269) by isolating $j = 0$.

Proof. Analogous to Lemma 6.270. $\square$

Theorem 6.272. *The ω-involution maps complete homogeneous to elementary: $\omega(h_{k+1}) = e_{k+1}$ for $k < n$.*

Proof. By strong induction on k:

- Base case ($k = 0$): $h_1 = e_1$, so $\omega(h_1) = \omega(e_1) = h_1 = e_1$.
- Inductive step: Use the Newton–Girard recurrence (Lemma 6.270) to write $h_{k+1} = \sum_{j=1}^{k+1}(-1)^{j+1}\, e_j\, h_{k+1-j}$. Apply ω: by Theorem 6.268, $\omega(e_j) = h_j$, and by the IH, $\omega(h_{k+1-j}) = e_{k+1-j}$ for $k+1-j \leq k$. Hence $\omega(h_{k+1}) = \sum_{j=1}^{k+1}(-1)^{j+1}\, h_j\, e_{k+1-j}$, which equals e_{k+1} by the symmetric Newton–Girard recurrence (Lemma 6.271).

$\square$

Theorem 6.273. *The ω-involution is an involution:* $\omega \circ \omega = \mathrm{id}$.

Proof. To show $\omega \circ \omega = \mathrm{id}$, it suffices to check on generators. Every element of the symmetric subalgebra is a polynomial in $e_1, e_2, \ldots, e_n$ (fundamental theorem of symmetric polynomials). For each generator e_k: $\omega(\omega(e_k)) = \omega(h_k) = e_k$. Since ω is an algebra homomorphism, the identity extends to all elements by linearity and multiplicativity. $\square$

Theorem 6.274. *The ω-involution is bijective.*

Proof. Both injectivity and surjectivity follow from $\omega \circ \omega = \mathrm{id}$ (Theorem 6.273). $\square$

Definition 6.275. The ω-involution promoted to an algebra equivalence $\omega : \Lambda_R \xrightarrow{\sim} \Lambda_R$.

Lemma 6.276. *The ω-involution commutes with determinants: for any matrix M over the symmetric subalgebra, $\omega(\det(M)) = \det(\omega(M))$, where $\omega(M)$ applies ω entry-wise. This is because ω is an algebra homomorphism, and* det *is a polynomial expression in the entries (involving only addition, multiplication, and negation).*

Proof. This follows from the fact that any algebra homomorphism commutes with determinant. $\square$

Appendix A

Details

A.1 Details: Infinite products (part 1)

Fix a commutative ring K.

A.1.1 Essentially finite families

Definition A.1. A family $(k_i)_{i \in I}$ of natural numbers is *essentially finite* if all but finitely many entries equal 0, i.e., the set $\{i \in I : k_i \neq 0\}$ is finite. This corresponds to S^I_{fin} in the source.

Definition A.2. For a family $(p_{i,k})$ of FPSs, the *coefficient support set* T_m at degree m is the set $\{(i,k) : k \neq 0 \text{ and } [x^m]p_{i,k} \neq 0\}$.

Definition A.3. The *coefficient support union* $T'_n = T_0 \cup T_1 \cup \cdots \cup T_n$ is the union of coefficient support sets up to degree n.

Definition A.4. The *index set* I_n is the set of first components appearing in T'_n: $I_n = \{i \in I : \exists k,\ (i,k) \in T'_n\}$.

Definition A.5. The *value set* K_n is the set of second components appearing in T'_n: $K_n = \{k \in \mathbb{N} : \exists i,\ (i,k) \in T'_n\}$.

Definition A.6. The set S^I_{fin} of *essentially finite families* in $\prod_{i \in I} S_i$ is the set of all families $(k_i)_{i \in I}$ such that $k_i \in S_i$ for all i and the family (k_i) is essentially finite.

Definition A.7. Given a finite subset $I_n \subseteq I$ and a function $f : I_n \to \mathbb{N}$, *extend from finset* produces a function $I \to \mathbb{N}$ by setting $f(i)$ for $i \in I_n$ and 0 otherwise. This is used in the bijection between essentially finite families supported on I_n and functions $I_n \to S$.

Definition A.8. Given a finite subset $I_n \subseteq I$ and a function $k : I \to \mathbb{N}$, *restrict to finset* produces the restriction $k|_{I_n} : I_n \to \mathbb{N}$. This is the inverse direction of the bijection used in Claim 8.

A.1.2 Existence of approximators

Lemma A.9. *For a multipliable family $(\mathbf{a}_i)_{i \in I}$ and any $n \in \mathbb{N}$, there exists an x^n-approximator M for $(\mathbf{a}_i)_{i \in I}$. The proof takes the union of finite sets $M_0, M_1, \ldots, M_n$ where each M_k determines the x^k-coefficient.*

A.1.3 Multipliability criteria

Theorem A.10. *If $(f_i)_{i\in I}$ is a family of FPSs such that for each n, the set $\{i \in I : [x^n]f_i \neq 0\}$ is finite (i.e., the family is coefficient-wise summable), then the family $(1 + f_i)_{i\in I}$ is multipliable.*

The proof constructs an approximator M as the union of the finite sets $\{i : [x^m]f_i \neq 0\}$ for $m = 0, 1, \ldots, n$, and shows that multiplying by $(1 + f_i)$ for $i \notin M$ does not change coefficients up to degree n.

Theorem A.11. *If $(a_i)_{i\in I}$ is a family of FPSs such that $\{i \in I : a_i \neq 1\}$ is finite, then the family is multipliable. The finite support set itself serves as a determining set for every coefficient.*

Theorem A.12. *If $J \subseteq I$ and both the subfamily $(a_i)_{i\in J}$ and the subfamily $(a_i)_{i\in I\setminus J}$ are multipliable, then the full family $(a_i)_{i\in I}$ is multipliable. The proof combines x^n-approximators for the two subfamilies.*

A.1.4 Multipliable product and division rules (Lean formalization)

Theorem A.13. *If $(\mathbf{a}_i)_{i\in I}$ and $(\mathbf{b}_i)_{i\in I}$ are both multipliable, then the pointwise product family $(\mathbf{a}_i \cdot \mathbf{b}_i)_{i\in I}$ is multipliable. The proof takes $M = U \cup V$ where U and V are x^n-approximators for the two families.*

Theorem A.14. *Under the hypotheses of Theorem A.13, the infinite product of the pointwise product equals the product of the two infinite products:*

$$\prod_{i\in I}(\mathbf{a}_i \cdot \mathbf{b}_i) = \left(\prod_{i\in I}\mathbf{a}_i\right) \cdot \left(\prod_{i\in I}\mathbf{b}_i\right).$$

Theorem A.15. *If $(\mathbf{a}_i)_{i\in I}$ and $(\mathbf{b}_i)_{i\in I}$ are multipliable and each $\mathbf{b}_i$ has invertible constant term, then the family $(\mathbf{a}_i \cdot \mathbf{b}_i^{-1})_{i\in I}$ is multipliable.*

Theorem A.16. *Under the hypotheses of Theorem A.15,*

$$\prod_{i\in I}\frac{\mathbf{a}_i}{\mathbf{b}_i} = \frac{\prod_{i\in I}\mathbf{a}_i}{\prod_{i\in I}\mathbf{b}_i}.$$

A.1.5 Subfamily and reindexing lemmas

Lemma A.17. *If U is an x^n-approximator for a family $(\mathbf{a}_i)_{i\in I}$ of invertible FPSs and $J \subseteq I$, then $U \cap J$ (as a preimage under the subtype embedding) is an x^n-approximator for the subfamily $(\mathbf{a}_i)_{i\in J}$. This is the Lean formalization of Lemma A.28, using the invertibility hypothesis to cancel factors outside J.*

Theorem A.18. *If $(\mathbf{a}_i)_{i\in I}$ is multipliable and each $\mathbf{a}_i$ has invertible constant term, then for any $J \subseteq I$, the subfamily $(\mathbf{a}_i)_{i\in J}$ is multipliable.*

Theorem A.19. *If $f : S \to T$ is a bijection and $(\mathbf{a}_t)_{t\in T}$ is multipliable, then $(\mathbf{a}_{f(s)})_{s\in S}$ is multipliable.*

Theorem A.20. *Under the hypotheses of Theorem A.19, the reindexed infinite product equals the original: $\prod_{s\in S}\mathbf{a}_{f(s)} = \prod_{t\in T}\mathbf{a}_t$.*

A.1.6 Fiber product decomposition

Theorem A.21. *Let $f : S \to W$ be a map, and $(\mathbf{a}_s)_{s\in S}$ a multipliable family. Suppose that for each $w \in W$, the fiber subfamily $(\mathbf{a}_s)_{s\in f^{-1}(w)}$ is multipliable. Then the family of fiber products $\left(\prod_{s\in f^{-1}(w)} \mathbf{a}_s\right)_{w\in W}$ is multipliable. The proof uses $f(U)$ (the image of an x^n-approximator U) as the approximator for the outer product.*

Theorem A.22. *Under the hypotheses of Theorem A.21,*

$$\prod_{s\in S} \mathbf{a}_s = \prod_{w\in W} \Big(\prod_{s\in f^{-1}(w)} \mathbf{a}_s \Big).$$

*This is the Lean formalization of Proposition 1.372 (parts **(a)** and **(b)**).*

A.1.7 Fubini rules for infinite products

Theorem A.23. *Let $(\mathbf{a}_{(i,j)})_{(i,j)\in I\times J}$ be a multipliable family. Suppose that for each $i \in I$, the row family $(\mathbf{a}_{(i,j)})_{j\in J}$ is multipliable. Then the family of row products $\left(\prod_{j\in J} \mathbf{a}_{(i,j)}\right)_{i\in I}$ is multipliable.*

Theorem A.24. *Under the hypotheses of Theorem A.23,*

$$\prod_{(i,j)\in I\times J} \mathbf{a}_{(i,j)} = \prod_{i\in I} \Big(\prod_{j\in J} \mathbf{a}_{(i,j)} \Big).$$

This is the Fubini rule for infinite products of FPSs.

Theorem A.25. *If $(\mathbf{a}_{(i,j)})_{(i,j)\in I\times J}$ is a multipliable family of invertible FPSs (each has unit constant term), then both the row families $(\mathbf{a}_{(i,j)})_{j\in J}$ for each i and the column families $(\mathbf{a}_{(i,j)})_{i\in I}$ for each j are multipliable. No additional multipliability hypotheses are needed in the invertible case, because subfamilies of invertible multipliable families are automatically multipliable.*

A.1.8 Coefficient form of the generalized distributive law

Lemma A.26. *For any finite set $M \subseteq I$ and finite value sets $(S_i)_{i\in M}$,*

$$[x^n]\Big(\prod_{i\in M} \sum_{k\in S_i} p_{i,k} \Big) = [x^n]\Big(\sum_{(k_i)\in\prod_{i\in M} S_i} \prod_{i\in M} p_{i,k_i} \Big).$$

This is a direct corollary of Claim 11 (finite product-sum interchange).

A.1.9 Congruence lemma for division

Lemma A.27. *Let $a, b, c, d \in K[[x]]$ be four FPSs such that c and d are invertible. Let $n \in \mathbb{N}$. Assume that*

$$[x^m]\, a = [x^m]\, b \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Assume further that

$$[x^m]\, c = [x^m]\, d \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Then,

$$[x^m]\, \frac{a}{c} = [x^m]\, \frac{b}{d} \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

Proof. We have assumed that

$$[x^m] a = [x^m] b \qquad \text{for each } m \in \{0, 1, \ldots, n\}.$$

In other words, $a \overset{x^n}{\equiv} b$ (by the definition of the relation $\overset{x^n}{\equiv}$). Similarly, $c \overset{x^n}{\equiv} d$. Hence, by the compatibility of $\overset{x^n}{\equiv}$ with division (Theorem 1.328 **(e)**), we obtain $\dfrac{a}{c} \overset{x^n}{\equiv} \dfrac{b}{d}$. In other words,

$$[x^m] \frac{a}{c} = [x^m] \frac{b}{d} \qquad \text{for each } m \in \{0, 1, \ldots, n\}$$

(by the definition of the relation $\overset{x^n}{\equiv}$). This proves Lemma A.27. $\square$

A.1.10 Detailed proof of Proposition 1.367

Detailed proof of Proposition 1.367. This can be proved similarly to Proposition 1.366, with the obvious changes (replacing multiplication by division, and requiring each $\mathbf{b}_i$ to be invertible). Of course, instead of Lemma 1.346, we need to use Lemma A.27. $\square$

A.1.11 Approximator intersection lemma

Lemma A.28. *Let $(\mathbf{a}_i)_{i \in I} \in K[[x]]^I$ be a family of invertible FPSs. Let J be a subset of I. Let $n \in \mathbb{N}$. Let U be an x^n-approximator for $(\mathbf{a}_i)_{i \in I}$. Then, $U \cap J$ is an x^n-approximator for $(\mathbf{a}_i)_{i \in J}$.*

Proof. The set U is an x^n-approximator for $(\mathbf{a}_i)_{i \in I}$. In other words, U is a finite subset of I that determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i \in I}$ (by the definition of an x^n-approximator). Hence, in particular, U is finite. Moreover, $U \subseteq I$ (since U is a subset of I).

Let $M = U \cap J$. Thus, $M = U \cap J \subseteq U$, so that the set M is finite (since U is finite). Moreover, M is a subset of J (since $M = U \cap J \subseteq J$).

Now, let N be a finite subset of J satisfying $M \subseteq N \subseteq J$. We shall show that

$$\prod_{i \in N} \mathbf{a}_i \overset{x^n}{\equiv} \prod_{i \in M} \mathbf{a}_i.$$

Indeed, the set $N \cup U$ is finite (since N and U are finite) and is a subset of I (since $N \subseteq J \subseteq I$ and $U \subseteq I$); it also satisfies $U \subseteq N \cup U \subseteq I$. Now, we recall that U is an x^n-approximator for $(\mathbf{a}_i)_{i \in I}$. Hence, Proposition 1.364 **(a)** (applied to U and $N \cup U$ instead of M and J) yields

$$\prod_{i \in N \cup U} \mathbf{a}_i \overset{x^n}{\equiv} \prod_{i \in U} \mathbf{a}_i$$

(since $N \cup U$ is a finite subset of I satisfying $U \subseteq N \cup U \subseteq I$).

On the other hand, we have assumed that $(\mathbf{a}_i)_{i \in I}$ is a family of invertible FPSs. Thus, for each $i \in I$, the FPS $\mathbf{a}_i$ is invertible, so that its inverse $\mathbf{a}_i^{-1}$ is well-defined. We obviously have

$$\prod_{i \in U \setminus J} \mathbf{a}_i^{-1} \overset{x^n}{\equiv} \prod_{i \in U \setminus J} \mathbf{a}_i^{-1}$$

(since the relation $\overset{x^n}{\equiv}$ is reflexive).

Hence, by the compatibility of $\overset{x^n}{\equiv}$ with multiplication (Theorem 1.328 **(b)**), applied to $a = \prod_{i\in N\cup U} \mathbf{a}_i$ and $b = \prod_{i\in U} \mathbf{a}_i$ and $c = \prod_{i\in U\setminus J} \mathbf{a}_i^{-1}$ and $d = \prod_{i\in U\setminus J} \mathbf{a}_i^{-1}$, we obtain

$$\left(\prod_{i\in N\cup U} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right) \overset{x^n}{\equiv} \left(\prod_{i\in U} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right).$$

On the other hand, the sets N and $U\setminus J$ are disjoint (since $N\subseteq J$ and $U\setminus J\subseteq J^c$), and their union is $N\cup(U\setminus J) = N\cup U$ (since $U = (U\cap J)\cup(U\setminus J)$ and $U\cap J = M\subseteq N$). Hence, the set $N\cup U$ is the union of its two disjoint subsets N and $U\setminus J$. Thus, we can split the product $\prod_{i\in N\cup U} \mathbf{a}_i$ as follows:

$$\prod_{i\in N\cup U} \mathbf{a}_i = \left(\prod_{i\in N} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i\right).$$

Multiplying both sides of this equality by $\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}$, we obtain

$$\begin{aligned}
\left(\prod_{i\in N\cup U} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right) &= \left(\prod_{i\in N} \mathbf{a}_i\right)\underbrace{\left(\prod_{i\in U\setminus J} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right)}_{=\prod_{i\in U\setminus J}\left(\mathbf{a}_i\mathbf{a}_i^{-1}\right)} \\
&= \left(\prod_{i\in N} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \underbrace{\left(\mathbf{a}_i\mathbf{a}_i^{-1}\right)}_{=1}\right) = \left(\prod_{i\in N} \mathbf{a}_i\right)\underbrace{\left(\prod_{i\in U\setminus J} 1\right)}_{=1} \\
&= \prod_{i\in N} \mathbf{a}_i.
\end{aligned}$$

However, the set U is the union of its two disjoint subsets $U\cap J$ and $U\setminus J$. Thus, we can split the product $\prod_{i\in U} \mathbf{a}_i$ as follows:

$$\prod_{i\in U} \mathbf{a}_i = \left(\prod_{i\in U\cap J} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i\right).$$

Multiplying both sides by $\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}$, we obtain

$$\begin{aligned}
\left(\prod_{i\in U} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right) &= \underbrace{\left(\prod_{i\in U\cap J} \mathbf{a}_i\right)}_{=\prod_{i\in M} \mathbf{a}_i \text{ (since } U\cap J = M)}\underbrace{\left(\prod_{i\in U\setminus J} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \mathbf{a}_i^{-1}\right)}_{=\prod_{i\in U\setminus J}\left(\mathbf{a}_i\mathbf{a}_i^{-1}\right)} \\
&= \left(\prod_{i\in M} \mathbf{a}_i\right)\left(\prod_{i\in U\setminus J} \underbrace{\left(\mathbf{a}_i\mathbf{a}_i^{-1}\right)}_{=1}\right) = \left(\prod_{i\in M} \mathbf{a}_i\right)\underbrace{\left(\prod_{i\in U\setminus J} 1\right)}_{=1} \\
&= \prod_{i\in M} \mathbf{a}_i.
\end{aligned}$$

Substituting these two simplifications into the x^n-equivalence above, we obtain

$$\prod_{i\in N}\mathbf{a}_i \overset{x^n}{\equiv} \prod_{i\in M}\mathbf{a}_i.$$

In other words, each $m \in \{0, 1, \ldots, n\}$ satisfies

$$[x^m]\left(\prod_{i\in N}\mathbf{a}_i\right) = [x^m]\left(\prod_{i\in M}\mathbf{a}_i\right)$$

(by the definition of $\overset{x^n}{\equiv}$).

Forget that we fixed N. We have thus shown that every finite subset N of J satisfying $M \subseteq N \subseteq J$ satisfies the above equality for each $m \in \{0, 1, \ldots, n\}$.

Now, let $m \in \{0, 1, \ldots, n\}$. Then, every finite subset N of J satisfying $M \subseteq N \subseteq J$ satisfies

$$[x^m]\left(\prod_{i\in N}\mathbf{a}_i\right) = [x^m]\left(\prod_{i\in M}\mathbf{a}_i\right).$$

In other words, the set M determines the x^m-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ (by the definition of "determining the x^m-coefficient in a product").

Forget that we fixed m. We thus have shown that M determines the x^m-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ for each $m \in \{0, 1, \ldots, n\}$. In other words, M determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i\in J}$. In other words, M is an x^n-approximator for $(\mathbf{a}_i)_{i\in J}$ (by the definition of an "x^n-approximator", since M is a finite subset of J). In other words, $U \cap J$ is an x^n-approximator for $(\mathbf{a}_i)_{i\in J}$ (since $M = U \cap J$). This proves Lemma A.28. □

A.1.12 Detailed proof of Proposition 1.369

Detailed proof of Proposition 1.369. Let J be a subset of I. We shall show that the family $(\mathbf{a}_i)_{i\in J}$ is multipliable.

Fix $n \in \mathbb{N}$. We know that the family $(\mathbf{a}_i)_{i\in I}$ is multipliable. Hence, there exists an x^n-approximator U for $(\mathbf{a}_i)_{i\in I}$ (by Lemma 1.363). Consider this U.

Set $M = U \cap J$. Lemma A.28 yields that $U \cap J$ is an x^n-approximator for $(\mathbf{a}_i)_{i\in J}$. In other words, M is an x^n-approximator for $(\mathbf{a}_i)_{i\in J}$ (since $M = U \cap J$). In other words, M is a finite subset of J that determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i\in J}$ (by the definition of an x^n-approximator). Thus, the set M determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i\in J}$. Hence, in particular, this set M determines the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$. Therefore, the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ is finitely determined (by the definition of "finitely determined", since M is a finite subset of J).

Forget that we fixed n. We thus have proved that for each $n \in \mathbb{N}$, the x^n-coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ is finitely determined. In other words, each coefficient in the product of $(\mathbf{a}_i)_{i\in J}$ is finitely determined. In other words, the family $(\mathbf{a}_i)_{i\in J}$ is multipliable (by the definition of "multipliable").

Forget that we fixed J. We thus have shown that the family $(\mathbf{a}_i)_{i\in J}$ is multipliable whenever J is a subset of I. In other words, any subfamily of $(\mathbf{a}_i)_{i\in I}$ is multipliable. This proves Proposition 1.369. □

A.1.13 Finite products and x^n-equivalence

Lemma A.29. *Let $n \in \mathbb{N}$. Let V be a finite set. Let $(c_w)_{w\in V} \in K[[x]]^V$ and $(d_w)_{w\in V} \in K[[x]]^V$ be two families of FPSs such that*

$$\text{each } w \in V \text{ satisfies } c_w \overset{x^n}{\equiv} d_w.$$

Then, we have

$$\prod_{w\in V} c_w \overset{x^n}{\equiv} \prod_{w\in V} d_w.$$

Proof. This is just Theorem 1.328 **(f)** (the part about products), with the letters S, s, a_s, and b_s renamed as V, w, c_w, and d_w. □

A.1.14 Detailed proof of Proposition 1.372

Detailed proof of Proposition 1.372. We shall subdivide our proof into several claims.

Claim 1: Let $w \in W$. Then, the family $(\mathbf{a}_s)_{s\in S,\ f(s)=w}$ is multipliable.

[*Proof of Claim 1:* This was an assumption of Proposition 1.372.]

Let us set

$$\mathbf{b}_w := \prod_{\substack{s\in S;\\ f(s)=w}} \mathbf{a}_s \qquad \text{for each } w \in W.$$

This is well-defined, because for each $w \in W$, the product $\prod_{\substack{s\in S;\\ f(s)=w}} \mathbf{a}_s$ is well-defined (since Claim 1 shows that the family $(\mathbf{a}_s)_{s\in S,\ f(s)=w}$ is multipliable).

Now, let $n \in \mathbb{N}$. By Lemma 1.363 (applied to S and $(\mathbf{a}_s)_{s\in S}$ instead of I and $(\mathbf{a}_i)_{i\in I}$), there exists an x^n-approximator for $(\mathbf{a}_s)_{s\in S}$. Pick such an x^n-approximator, and call it U. Then, U is an x^n-approximator for $(\mathbf{a}_s)_{s\in S}$; in other words, U is a finite subset of S that determines the first $n+1$ coefficients in the product of $(\mathbf{a}_s)_{s\in S}$ (by the definition of an x^n-approximator).

The set U is finite. Thus, its image $f(U) = \{f(u) \mid u \in U\}$ is finite as well. Now, we claim the following:

Claim 2: For each $w \in W$, we have

$$\mathbf{b}_w \overset{x^n}{\equiv} \prod_{\substack{s\in U;\\ f(s)=w}} \mathbf{a}_s.$$

[*Proof of Claim 2:* Let $w \in W$. Let J be the subset $\{s \in S \mid f(s) = w\}$ of S. Then, the family $(\mathbf{a}_s)_{s\in J}$ is just the family $(\mathbf{a}_s)_{s\in S,\ f(s)=w}$, and thus is multipliable (by Claim 1). Furthermore, Lemma A.28 (applied to S and $(\mathbf{a}_s)_{s\in S}$ instead of I and $(\mathbf{a}_i)_{i\in I}$) yields that $U \cap J$ is an x^n-approximator for $(\mathbf{a}_s)_{s\in J}$ (since U is an x^n-approximator for $(\mathbf{a}_s)_{s\in S}$). Hence, Proposition 1.364 **(b)** (applied to J, $(\mathbf{a}_s)_{s\in J}$ and $U \cap J$ instead of I, $(\mathbf{a}_i)_{i\in I}$ and M) yields

$$\prod_{s\in J} \mathbf{a}_s \overset{x^n}{\equiv} \prod_{s\in U\cap J} \mathbf{a}_s.$$

However, we have $J = \{s \in S \mid f(s) = w\}$. Thus, $\prod_{s\in J} \mathbf{a}_s = \prod_{\substack{s\in S;\\ f(s)=w}} \mathbf{a}_s = \mathbf{b}_w$ and $U \cap J = \{s \in U \mid f(s) = w\}$. So the product sign "$\prod_{s\in U\cap J}$" is equivalent to "$\prod_{\substack{s\in U;\\ f(s)=w}}$". Thus,

$$\mathbf{b}_w \overset{x^n}{\equiv} \prod_{\substack{s\in U;\\ f(s)=w}} \mathbf{a}_s.$$

This proves Claim 2.]

Claim 3: The set $f(U)$ determines the x^n-coefficient in the product of $(\mathbf{b}_w)_{w\in W}$.

[*Proof of Claim 3:* Let T be a finite subset of W satisfying $f(U)\subseteq T$. We must show that $[x^n]\left(\prod_{w\in T}\mathbf{b}_w\right)=[x^n]\left(\prod_{w\in f(U)}\mathbf{b}_w\right)$.

By Claim 2, for each $w\in W$ we have $\mathbf{b}_w \overset{x^n}{\equiv} \prod_{\substack{s\in U;\\ f(s)=w}} \mathbf{a}_s$. Hence, by Lemma A.29, we get

$$\prod_{w\in T}\mathbf{b}_w \overset{x^n}{\equiv} \prod_{w\in T}\prod_{\substack{s\in U;\\ f(s)=w}}\mathbf{a}_s$$

and similarly

$$\prod_{w\in f(U)}\mathbf{b}_w \overset{x^n}{\equiv} \prod_{w\in f(U)}\prod_{\substack{s\in U;\\ f(s)=w}}\mathbf{a}_s.$$

Now, for $w\notin f(U)$, there are no $s\in U$ with $f(s)=w$, so the inner product is 1 (the empty product). Hence, for any finite set $V\supseteq f(U)$,

$$\prod_{w\in V}\prod_{\substack{s\in U;\\ f(s)=w}}\mathbf{a}_s=\prod_{s\in U}\mathbf{a}_s$$

(by the standard fiberwise reindexing of finite products). Applying this to both $V=T$ and $V=f(U)$, we conclude that both products equal $\prod_{s\in U}\mathbf{a}_s$. Therefore,

$$[x^n]\left(\prod_{w\in T}\mathbf{b}_w\right)=[x^n]\left(\prod_{s\in U}\mathbf{a}_s\right)=[x^n]\left(\prod_{w\in f(U)}\mathbf{b}_w\right).$$

This proves Claim 3.]

From Claim 3, the x^n-coefficient in the product of $(\mathbf{b}_w)_{w\in W}$ is finitely determined (since $f(U)$ is a finite subset of W). Since this holds for each $n\in\mathbb{N}$, the family $(\mathbf{b}_w)_{w\in W}$ is multipliable. This proves part **(a)** of Proposition 1.372.

For part **(b)**, we observe that by Proposition 1.364 **(b)**, the infinite product $\prod_{s\in S}\mathbf{a}_s$ is x^n-equivalent to $\prod_{s\in U}\mathbf{a}_s$. By the computation above, the infinite product $\prod_{w\in W}\mathbf{b}_w$ is also x^n-equivalent to $\prod_{s\in U}\mathbf{a}_s$. Since this holds for all n, the two infinite products are equal. $\square$

A.1.15 Detailed proof of Proposition 1.366

Detailed proof of Proposition 1.366. **(a)** Fix $n\in\mathbb{N}$. We know that the family $(\mathbf{a}_i)_{i\in I}$ is multipliable. Hence, there exists an x^n-approximator U for $(\mathbf{a}_i)_{i\in I}$ (by Lemma 1.363). Consider this U.

We also know that the family $(\mathbf{b}_i)_{i\in I}$ is multipliable. Hence, there exists an x^n-approximator V for $(\mathbf{b}_i)_{i\in I}$ (by Lemma 1.363, applied to $\mathbf{b}_i$ instead of $\mathbf{a}_i$). Consider this V.

We know that U is an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$. In other words, U is a finite subset of I that determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i\in I}$ (by the definition of an x^n-approximator). Hence, in particular, U is finite. Similarly, V is finite. Moreover, $U\subseteq I$; similarly, $V\subseteq I$.

Let $M=U\cup V$. This set M is finite (since U and V are finite). Moreover, $M=U\cup V\subseteq I\cup I=I$. Hence, M is a finite subset of I.

Now, let N be a finite subset of I satisfying $M \subseteq N \subseteq I$. We shall show that

$$[x^n] \left(\prod_{i \in N} (\mathbf{a}_i \mathbf{b}_i) \right) = [x^n] \left(\prod_{i \in M} (\mathbf{a}_i \mathbf{b}_i) \right).$$

Indeed, let $m \in \{0, 1, \ldots, n\}$. We have $U \subseteq U \cup V = M \subseteq N$. The set N is a finite subset of I and satisfies $U \subseteq N$. Now, recall that U determines the first $n+1$ coefficients in the product of $(\mathbf{a}_i)_{i \in I}$. Hence, U determines the x^m-coefficient in the product of $(\mathbf{a}_i)_{i \in I}$ (since $m \in \{0, 1, \ldots, n\}$). In other words, every finite subset T of I satisfying $U \subseteq T \subseteq I$ satisfies

$$[x^m] \left(\prod_{i \in T} \mathbf{a}_i \right) = [x^m] \left(\prod_{i \in U} \mathbf{a}_i \right).$$

We can apply this to $T = N$ (since N is a finite subset of I satisfying $U \subseteq N \subseteq I$), and thus obtain

$$[x^m] \left(\prod_{i \in N} \mathbf{a}_i \right) = [x^m] \left(\prod_{i \in U} \mathbf{a}_i \right).$$

However, we can also apply it to $T = M$ (since M is a finite subset of I satisfying $U \subseteq M \subseteq I$), and thus obtain

$$[x^m] \left(\prod_{i \in M} \mathbf{a}_i \right) = [x^m] \left(\prod_{i \in U} \mathbf{a}_i \right).$$

Comparing these two equalities, we obtain

$$[x^m] \left(\prod_{i \in N} \mathbf{a}_i \right) = [x^m] \left(\prod_{i \in M} \mathbf{a}_i \right).$$

The same argument (applied to $\mathbf{b}_i$ and V instead of $\mathbf{a}_i$ and U) yields

$$[x^m] \left(\prod_{i \in N} \mathbf{b}_i \right) = [x^m] \left(\prod_{i \in M} \mathbf{b}_i \right).$$

Forget that we fixed m. We thus have proved these equalities for each $m \in \{0, 1, \ldots, n\}$. Hence, Lemma 1.346 (applied to $a = \prod_{i \in N} \mathbf{a}_i$ and $b = \prod_{i \in M} \mathbf{a}_i$ and $c = \prod_{i \in N} \mathbf{b}_i$ and $d = \prod_{i \in M} \mathbf{b}_i$) yields that

$$[x^m] \left(\left(\prod_{i \in N} \mathbf{a}_i \right) \left(\prod_{i \in N} \mathbf{b}_i \right) \right) = [x^m] \left(\left(\prod_{i \in M} \mathbf{a}_i \right) \left(\prod_{i \in M} \mathbf{b}_i \right) \right)$$

for each $m \in \{0, 1, \ldots, n\}$. Applying this to $m = n$, we obtain

$$[x^n] \left(\left(\prod_{i \in N} \mathbf{a}_i \right) \left(\prod_{i \in N} \mathbf{b}_i \right) \right) = [x^n] \left(\left(\prod_{i \in M} \mathbf{a}_i \right) \left(\prod_{i \in M} \mathbf{b}_i \right) \right).$$

In view of

$$\prod_{i \in N} (\mathbf{a}_i \mathbf{b}_i) = \left(\prod_{i \in N} \mathbf{a}_i \right) \left(\prod_{i \in N} \mathbf{b}_i \right) \qquad \text{and} \qquad \prod_{i \in M} (\mathbf{a}_i \mathbf{b}_i) = \left(\prod_{i \in M} \mathbf{a}_i \right) \left(\prod_{i \in M} \mathbf{b}_i \right),$$

this rewrites as

$$[x^n]\left(\prod_{i\in N}(\mathbf{a}_i\mathbf{b}_i)\right)=[x^n]\left(\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i)\right).$$

Forget that we fixed N. We thus have shown that every finite subset N of I satisfying $M\subseteq N\subseteq I$ satisfies the above equality. In other words, M determines the x^n-coefficient in the product of $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ (by the definition of "determining the x^n-coefficient"). Hence, the x^n-coefficient in the product of $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ is finitely determined (since M is a finite subset of I).

Forget that we fixed n. We thus have proved that for each $n\in\mathbb{N}$, the x^n-coefficient in the product of $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ is finitely determined. In other words, the family $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ is multipliable. This proves Proposition 1.366 **(a)**.

(b) Proposition 1.366 **(a)** shows that the family $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$ is multipliable.

Let $n\in\mathbb{N}$. In our above proof of Proposition 1.366 **(a)**, we have seen the following:

- There exists an x^n-approximator U for $(\mathbf{a}_i)_{i\in I}$.
- There exists an x^n-approximator V for $(\mathbf{b}_i)_{i\in I}$.

Consider these U and V. Let $M=U\cup V$. Thus, $M=U\cup V\supseteq U$ and $M=U\cup V\supseteq V$. In our above proof, we have seen that M is a finite subset of I and M determines the x^n-coefficient in the product of $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$.

We recall that the relation $\overset{x^n}{\equiv}$ is an equivalence relation (by Theorem 1.328 **(a)**).

Now, the definition of the infinite product $\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i)$ (namely, Definition 1.343 **(b)**) yields that

$$[x^n]\left(\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i)\right)=[x^n]\left(\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i)\right)$$

(since M is a finite subset of I that determines the x^n-coefficient in the product of $(\mathbf{a}_i\mathbf{b}_i)_{i\in I}$). On the other hand, the set U is an x^n-approximator for $(\mathbf{a}_i)_{i\in I}$. Thus, Proposition 1.364 **(b)** (applied to U instead of M) yields

$$\prod_{i\in I}\mathbf{a}_i\overset{x^n}{\equiv}\prod_{i\in U}\mathbf{a}_i.$$

Since the relation $\overset{x^n}{\equiv}$ is symmetric, we thus obtain $\prod_{i\in U}\mathbf{a}_i\overset{x^n}{\equiv}\prod_{i\in I}\mathbf{a}_i$. Moreover, M is a finite subset of I satisfying $U\subseteq M$; hence, Proposition 1.364 **(a)** yields

$$\prod_{i\in M}\mathbf{a}_i\overset{x^n}{\equiv}\prod_{i\in U}\mathbf{a}_i\overset{x^n}{\equiv}\prod_{i\in I}\mathbf{a}_i.$$

Hence, $\prod_{i\in M}\mathbf{a}_i\overset{x^n}{\equiv}\prod_{i\in I}\mathbf{a}_i$ (since the relation $\overset{x^n}{\equiv}$ is transitive). The same argument (applied to $(\mathbf{b}_i)_{i\in I}$ and V instead of $(\mathbf{a}_i)_{i\in I}$ and U) yields $\prod_{i\in M}\mathbf{b}_i\overset{x^n}{\equiv}\prod_{i\in I}\mathbf{b}_i$. From these two x^n-equivalences, we obtain

$$\left(\prod_{i\in M}\mathbf{a}_i\right)\left(\prod_{i\in M}\mathbf{b}_i\right)\overset{x^n}{\equiv}\left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right)$$

(by Theorem 1.328 **(b)**). In view of

$$\left(\prod_{i\in M}\mathbf{a}_i\right)\left(\prod_{i\in M}\mathbf{b}_i\right)=\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i),$$

this rewrites as

$$\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i) \overset{x^n}{\equiv} \left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right).$$

In other words,

$$\text{each } m\in\{0,1,\ldots,n\} \text{ satisfies } [x^m]\left(\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i)\right) = [x^m]\left(\left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right)\right).$$

Applying this to $m=n$, we obtain

$$[x^n]\left(\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i)\right) = [x^n]\left(\left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right)\right).$$

In view of $[x^n]\left(\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i)\right) = [x^n]\left(\prod_{i\in M}(\mathbf{a}_i\mathbf{b}_i)\right)$, this rewrites as

$$[x^n]\left(\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i)\right) = [x^n]\left(\left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right)\right).$$

Now, forget that we fixed n. We thus have proved that each $n\in\mathbb{N}$ satisfies the above equality. In other words, each coefficient of the FPS $\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i)$ equals the corresponding coefficient of $\left(\prod_{i\in I}\mathbf{a}_i\right)\left(\prod_{i\in I}\mathbf{b}_i\right)$. Hence, we have

$$\prod_{i\in I}(\mathbf{a}_i\mathbf{b}_i) = \left(\prod_{i\in I}\mathbf{a}_i\right)\cdot\left(\prod_{i\in I}\mathbf{b}_i\right).$$

This proves Proposition 1.366 **(b)**. □

A.1.16 Product rule claims

Lemma A.30. ***(Claim 3.)*** *If $(i,k)\notin T'_n$ (the coefficient support union up to n) and $k\neq 0$, then $[x^m]p_{i,k}=0$ for all $m\in\{0,1,\ldots,n\}$.*

Proof. If $[x^m]p_{i,k}\neq 0$ for some $m\leq n$, then $(i,k)\in T_m\subseteq T'_n$, contradicting the hypothesis. □

Lemma A.31. ***(Claim 4.)*** *If $(k_i)_{i\in I}$ is essentially finite and $p_{i,0}=1$ for all i, then the family $(p_{i,k_i})_{i\in I}$ is multipliable.*

Proof. All but finitely many $i\in I$ satisfy $k_i=0$ (since $(k_i)_{i\in I}$ is essentially finite) and thus $p_{i,k_i}=p_{i,0}=1$. Hence, all but finitely many entries of the family $(p_{i,k_i})_{i\in I}$ equal 1. Thus, this family is multipliable (by Proposition 1.351). □

Lemma A.32. ***(Claim 5.)*** *If some $j\in I$ satisfies $k_j\neq 0$ and $(j,k_j)\notin T'_n$, then $[x^n](\prod_{i\in s}p_{i,k_i})=0$ for any finite $s\ni j$.*

Proof. The product $\prod_{i\in s}p_{i,k_i}$ is a multiple of p_{j,k_j} (since p_{j,k_j} is one of the factors of this product). Moreover, by Claim 3, we have $[x^m]p_{j,k_j}=0$ for each $m\in\{0,1,\ldots,n\}$. Hence, Lemma A.42 (applied to $u=p_{j,k_j}$ and $v=\prod_{i\in s}p_{i,k_i}$) shows that $[x^m](\prod_{i\in s}p_{i,k_i})=0$ for each $m\in\{0,1,\ldots,n\}$. Applying this to $m=n$ yields the claim. □

Lemma A.33. ***(Claim 6.)*** *If $j\notin I_n$ and $k_j\neq 0$, then $[x^n](\prod_{i\in s}p_{i,k_i})=0$ for any finite $s\ni j$.*

Proof. We have $k_j \in S_j$ and $k_j \neq 0$, so $(j, k_j) \in \overline{S}$. Moreover, $(j, k_j) \notin T'_n$ (because if we had $(j, k_j) \in T'_n$, then we would have $j \in I_n$ by the definition of I_n; but this would contradict $j \notin I_n$). Hence, $(j, k_j) \in \overline{S} \setminus T'_n$. Therefore, Claim 5 yields $[x^n](\prod_{i \in s} p_{i,k_i}) = 0$. $\square$

Lemma A.34. ***(Claim 7.)*** *The family $(\prod_{i \in I_n} p_{i,k_i})$ indexed by essentially finite families (k_i) supported on I_n is summable: for each n, only finitely many such families yield $[x^n](\prod_{i \in I_n} p_{i,k_i}) \neq 0$.*

Proof. Let $(k_i)_{i \in I} \in S^I_{\text{fin}}$ satisfy $[x^n](\prod_{i \in I} p_{i,k_i}) \neq 0$. Then it must satisfy two properties: (1) all $i \in I \setminus I_n$ satisfy $k_i = 0$ (otherwise Claim 6 would give a contradiction); (2) all $i \in I_n$ satisfy $k_i \in K_n \cup \{0\}$ (otherwise $(j, k_j) \in \overline{S} \setminus T'_n$ for some $j \in I_n$, and Claim 5 would give a contradiction). These two properties together restrict (k_i) to finitely many possibilities (since the sets I_n and $K_n \cup \{0\}$ are finite). Hence, all but finitely many families yield $[x^n](\prod_{i \in I} p_{i,k_i}) = 0$, proving summability. $\square$

Lemma A.35. ***(Claim 9.)*** *For $i \notin I_n$, we have $[x^m](\sum_{k \in S_i \setminus \{0\}} p_{i,k}) = 0$ for each $m \in \{0, 1, \ldots, n\}$.*

Proof. Each term $p_{i,k}$ with $k \neq 0$ has $[x^m] p_{i,k} = 0$ by Claim 3 (since $(i, k) \notin T'_n$ because $i \notin I_n$). $\square$

Lemma A.36. ***(Claim 10.)*** *For any finite set $J \supseteq I_n$, $[x^n](\prod_{i \in J} \sum_{k \in S_i} p_{i,k}) = [x^n](\prod_{i \in I_n} \sum_{k \in S_i} p_{i,k})$. The product stabilizes at I_n.*

Proof. For $i \in J \setminus I_n$, we have $\sum_{k \in S_i} p_{i,k} = 1 + \sum_{k \in S_i \setminus \{0\}} p_{i,k}$. By Claim 9, the remainder has zero coefficients up to degree n, so multiplying by it does not change the coefficient at degree n. $\square$

Lemma A.37. ***(Claim 11.)*** *For a finite index set I_n,*

$$\prod_{i \in I_n} \sum_{k \in S_i} p_{i,k} = \sum_{(k_i) \in S^{I_n}} \prod_{i \in I_n} p_{i,k_i}.$$

This is the standard finite distributive law (product of sums equals sum of products).

Proof. This is the standard finite distributive law (product of sums equals sum of products over all choice functions). See Proposition 1.383. $\square$

Lemma A.38. ***(Claim 8.)*** *The coefficient of the sum over essentially finite families equals the coefficient of the sum restricted to the finite index set I_n:*

$$[x^n]\Big(\sum_{(k_i) \in S^I_{\text{fin}}} \prod_{i \in I_n} p_{i,k_i}\Big) = [x^n]\Big(\sum_{(k_i) \in S^{I_n}} \prod_{i \in I_n} p_{i,k_i}\Big).$$

Proof. The proof establishes a bijection between essentially finite families supported on I_n and functions $I_n \to S$, via restriction and extension by 0. $\square$

A.1.17 Composition rules for infinite products

Lemma A.39. *If $f_1 \overset{x^n}{\equiv} f_2$ and $[x^0]g = 0$, then $f_1 \circ g \overset{x^n}{\equiv} f_2 \circ g$. That is, x^n-equivalence is preserved under composition with a fixed inner series.*

Proof. The coefficient $[x^k](f \circ g)$ involves a sum over d of $[x^d]f \cdot [x^k](g^d)$. For $d > k$, $[x^k](g^d) = 0$ since $\operatorname{ord}(g^d) \geq d > k$. For $d \leq k \leq n$, $[x^d]f_1 = [x^d]f_2$ by x^n-equivalence. □

Lemma A.40. *If M is an x^n-approximator for $(f_i)_{i \in I}$ (i.e., for all $k \leq n$ and $J \supseteq M$, $[x^k](\prod_{i \in J} f_i) = [x^k](\prod_{i \in M} f_i)$), and $[x^0]g = 0$, then for all $J \supseteq M$, $[x^n](\prod_{i \in J}(f_i \circ g)) = [x^n](\prod_{i \in M}(f_i \circ g))$.*

Proof. By Lemma A.43, $\prod_{i \in J}(f_i \circ g) = (\prod_{i \in J} f_i) \circ g$. The approximator condition gives $\prod_{i \in J} f_i \overset{x^n}{\equiv} \prod_{i \in M} f_i$. By Lemma A.39, composing with g preserves the equivalence. □

Proposition A.41. *If $(f_i)_{i \in I}$ is multipliable and $[x^0]g = 0$, then $(f_i \circ g)_{i \in I}$ is multipliable.*

Proof. For each n, combine the approximators for degrees $0, 1, \ldots, n$ into a single approximator M (their union), then apply Lemma A.40. □

A.1.18 Additional supporting lemmas

Lemma A.42. *If $u \mid v$ and $[x^m]u = 0$ for all $m \leq n$, then $[x^m]v = 0$ for all $m \leq n$.*

Proof. Write $v = u \cdot w$. Then $[x^m]v = \sum_{(i,j) \in \text{antidiag}(m)} [x^i]u \cdot [x^j]w$. Since $i \leq m \leq n$, we have $[x^i]u = 0$, so each term vanishes. □

A.2 Details: Infinite products (part 2) – Finite product composition rule

This chapter contains the detailed proof of the finite product composition rule for formal power series (Lemma A.43), which states that substitution (composition) distributes over finite products. This lemma is used in the proof of Proposition 1.399 (the infinite product composition rule), whose detailed proof is also given here.

Lemma A.43. *Let I be a* ***finite*** *set. If $(f_i)_{i \in I} \in K[[x]]^I$ is a family of FPSs, and if $g \in K[[x]]$ is an FPS satisfying $[x^0]\, g = 0$, then $\left(\prod_{i \in I} f_i\right) \circ g = \prod_{i \in I} (f_i \circ g)$.*

Proof. This follows by a straightforward induction on $|I|$.

Base case ($|I| = 0$): The empty product is $\underline{1}$ (the constant FPS with value 1), and we have $\underline{1} \circ g = \underline{1}$ for any $g \in K[[x]]$. Hence both sides equal $\underline{1}$.

Induction step: Assume the result holds for all sets of size $|I| - 1$. Pick some $a \in I$ and write $\prod_{i \in I} f_i = f_a \cdot \prod_{i \in I \setminus \{a\}} f_i$. By Proposition 1.181 **(b)** (the substitution rule for products: $(u \cdot v) \circ g = (u \circ g) \cdot (v \circ g)$), we have

$$\left(f_a \cdot \prod_{i \in I \setminus \{a\}} f_i\right) \circ g = (f_a \circ g) \cdot \left(\prod_{i \in I \setminus \{a\}} f_i\right) \circ g.$$

By the induction hypothesis (applied to the set $I \setminus \{a\}$, which has size $|I|-1$),

$$\left(\prod_{i\in I\setminus\{a\}} f_i\right) \circ g = \prod_{i\in I\setminus\{a\}} (f_i \circ g).$$

Combining, we obtain

$$\left(\prod_{i\in I} f_i\right) \circ g = (f_a \circ g) \cdot \prod_{i\in I\setminus\{a\}} (f_i \circ g) = \prod_{i\in I} (f_i \circ g).$$

□

A.2.1 Detailed proof of Proposition 1.399

Detailed proof of Proposition 1.399. Let $(f_i)_{i\in I} \in K[[x]]^I$ be a multipliable family of FPSs. Let $g \in K[[x]]$ be an FPS satisfying $[x^0]\, g = 0$.

We shall first show the following auxiliary claim:

Claim 1: Let M be an x^n-approximator for $(f_i)_{i\in I}$. Then, the set M determines the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$.

[*Proof of Claim 1:* The set M is an x^n-approximator for $(f_i)_{i\in I}$. In other words, M is a finite subset of I that determines the first $n+1$ coefficients in the product of $(f_i)_{i\in I}$ (by the definition of "x^n-approximator").

Let J be a finite subset of I satisfying $M \subseteq J \subseteq I$. Then, Lemma A.43 (applied to J instead of I) yields

$$\left(\prod_{i\in J} f_i\right) \circ g = \prod_{i\in J} (f_i \circ g). \tag{A.1}$$

Also, Lemma A.43 (applied to M instead of I) yields

$$\left(\prod_{i\in M} f_i\right) \circ g = \prod_{i\in M} (f_i \circ g). \tag{A.2}$$

However, Proposition 1.364 **(a)** (applied to $\mathbf{a}_i = f_i$) yields

$$\prod_{i\in J} f_i \overset{x^n}{\equiv} \prod_{i\in M} f_i.$$

We also have $g \overset{x^n}{\equiv} g$ (since the relation $\overset{x^n}{\equiv}$ is an equivalence relation). Hence, Proposition 1.338 (applied to $a = \prod_{i\in J} f_i$ and $b = \prod_{i\in M} f_i$ and $c = g$ and $d = g$) yields

$$\left(\prod_{i\in J} f_i\right) \circ g \overset{x^n}{\equiv} \left(\prod_{i\in M} f_i\right) \circ g.$$

In view of (A.1) and (A.2), this rewrites as

$$\prod_{i\in J} (f_i \circ g) \overset{x^n}{\equiv} \prod_{i\in M} (f_i \circ g).$$

In other words,

$$\text{each } m \in \{0,1,\ldots,n\} \text{ satisfies } [x^m]\left(\prod_{i\in J}(f_i \circ g)\right) = [x^m]\left(\prod_{i\in M}(f_i \circ g)\right)$$

(by the definition of the relation $\overset{x^n}{\equiv}$). Applying this to $m = n$, we obtain

$$[x^n]\left(\prod_{i\in J}(f_i \circ g)\right) = [x^n]\left(\prod_{i\in M}(f_i \circ g)\right).$$

Forget that we fixed J. We thus have shown that every finite subset J of I satisfying $M \subseteq J \subseteq I$ satisfies

$$[x^n]\left(\prod_{i\in J}(f_i \circ g)\right) = [x^n]\left(\prod_{i\in M}(f_i \circ g)\right).$$

In other words, the set M determines the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$ (by the definition of what it means to "determine the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$"). This proves Claim 1.]

Now, let $n \in \mathbb{N}$. Lemma 1.363 (applied to $\mathbf{a}_i = f_i$) shows that there exists an x^n-approximator for $(f_i)_{i\in I}$. Consider this x^n-approximator for $(f_i)_{i\in I}$, and denote it by M. Thus, M is an x^n-approximator for $(f_i)_{i\in I}$; in other words, M is a finite subset of I that determines the first $n+1$ coefficients in the product of $(f_i)_{i\in I}$ (by the definition of "x^n-approximator"). Claim 1 shows that the set M determines the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$. Hence, there is a finite subset of I that determines the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$ (namely, M). In other words, the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$ is finitely determined.

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$, the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$ is finitely determined. In other words, each coefficient in the product of $(f_i \circ g)_{i\in I}$ is finitely determined. In other words, the family $(f_i \circ g)_{i\in I}$ is multipliable (by the definition of "multipliable").

It remains to prove that $\left(\prod_{i\in I} f_i\right) \circ g = \prod_{i\in I}(f_i \circ g)$.

In order to do so, we again fix $n \in \mathbb{N}$. Lemma 1.363 (applied to $\mathbf{a}_i = f_i$) shows that there exists an x^n-approximator for $(f_i)_{i\in I}$. Consider this x^n-approximator for $(f_i)_{i\in I}$, and denote it by M. Thus, M is an x^n-approximator for $(f_i)_{i\in I}$; in other words, M is a finite subset of I that determines the first $n+1$ coefficients in the product of $(f_i)_{i\in I}$ (by the definition of "x^n-approximator"). Moreover, Proposition 1.364 **(b)** (applied to $\mathbf{a}_i = f_i$) yields

$$\prod_{i\in I} f_i \overset{x^n}{\equiv} \prod_{i\in M} f_i. \tag{A.3}$$

We also have $g \overset{x^n}{\equiv} g$ (since the relation $\overset{x^n}{\equiv}$ is an equivalence relation). Hence, Proposition 1.338 (applied to $a = \prod_{i\in I} f_i$ and $b = \prod_{i\in M} f_i$ and $c = g$ and $d = g$) yields

$$\left(\prod_{i\in I} f_i\right) \circ g \overset{x^n}{\equiv} \left(\prod_{i\in M} f_i\right) \circ g. \tag{A.4}$$

However, Lemma A.43 (applied to M instead of I) yields

$$\left(\prod_{i\in M} f_i\right) \circ g = \prod_{i\in M}(f_i \circ g).$$

In view of this, we can rewrite (A.4) as

$$\left(\prod_{i\in I} f_i\right) \circ g \overset{x^n}{\equiv} \prod_{i\in M} (f_i \circ g) .$$

In other words,

$$\text{each } m \in \{0, 1, \ldots, n\} \text{ satisfies } [x^m] \left(\left(\prod_{i\in I} f_i\right) \circ g\right) = [x^m] \left(\prod_{i\in M} (f_i \circ g)\right)$$

(by the definition of the relation $\overset{x^n}{\equiv}$). Applying this to $m = n$, we obtain

$$[x^n] \left(\left(\prod_{i\in I} f_i\right) \circ g\right) = [x^n] \left(\prod_{i\in M} (f_i \circ g)\right) . \tag{A.5}$$

However, Claim 1 shows that the set M determines the x^n-coefficient in the product of $(f_i \circ g)_{i\in I}$. Hence, the definition of the infinite product $\prod_{i\in I} (f_i \circ g)$ (specifically, Definition 1.343 **(b)**) yields

$$[x^n] \left(\prod_{i\in I} (f_i \circ g)\right) = [x^n] \left(\prod_{i\in M} (f_i \circ g)\right) .$$

Comparing this with (A.5), we obtain

$$[x^n] \left(\left(\prod_{i\in I} f_i\right) \circ g\right) = [x^n] \left(\prod_{i\in I} (f_i \circ g)\right) .$$

Forget that we fixed n. We thus have shown that each $n \in \mathbb{N}$ satisfies

$$[x^n] \left(\left(\prod_{i\in I} f_i\right) \circ g\right) = [x^n] \left(\prod_{i\in I} (f_i \circ g)\right) .$$

In other words, the FPSs $\left(\prod_{i\in I} f_i\right) \circ g$ and $\prod_{i\in I} (f_i \circ g)$ agree in all their coefficients. Hence, $\left(\prod_{i\in I} f_i\right) \circ g = \prod_{i\in I} (f_i \circ g)$. This completes our proof of Proposition 1.399. □

A.3 Domino tilings of height-3 rectangles

A.3.1 The tilings A_n, B_n, and C

Definition A.44. **(a)** For each even positive integer n, we let A_n be the domino tiling of $R_{n,3}$ consisting of the following dominos:

- the horizontal dominos $\{(2i-1,\ 1),\ (2i,\ 1)\}$ for all $i \in [n/2]$, which fill the bottom row of $R_{n,3}$, and which we call the **basement dominos**;
- the vertical domino $\{(1, 2),\ (1, 3)\}$ in the first column, which we call the **left wall**;
- the vertical domino $\{(n, 2),\ (n, 3)\}$ in the last column, which we call the **right wall**;
- the horizontal dominos $\{(2i,\ 2),\ (2i + 1,\ 2)\}$ for all $i \in [n/2 - 1]$, which fill the middle row of $R_{n,3}$ (except for the first and last columns), and which we call the **middle dominos**;

- the horizontal dominos $\{(2i,\ 3)\,,\ (2i+1,\ 3)\}$ for all $i \in [n/2-1]$, which fill the top row of $R_{n,3}$ (except for the first and last columns), and which we call the **top dominos**.

(b) For each even positive integer n, we let B_n be the domino tiling of $R_{n,3}$ obtained by reflecting A_n across the horizontal axis of symmetry of $R_{n,3}$ (swapping row 1 with row 3 and fixing row 2).

(c) We let C denote the domino tiling of $R_{2,3}$ consisting of three horizontal dominos: $\{(1,1),(2,1)\}$, $\{(1,2),(2,2)\}$, and $\{(1,3),(2,3)\}$.

A.3.2 Properties of A_n, B_n, and C

Lemma A.45. *For each even $n \geq 2$, the tiling A_n has a vertical domino in the top two squares of column 1, namely the left wall $\{(1,2),(1,3)\}$.*

Proof. The left wall $\{(1,2),(1,3)\}$ is explicitly among the dominos of A_n, is vertical, lies in column 1, and covers rows 2–3. □

Lemma A.46. *For each even $n \geq 2$, the tiling B_n has a vertical domino in the bottom two squares of column 1, namely the left wall $\{(1,1),(1,2)\}$ of B_n.*

Proof. The left wall of B_n is $\{(1,1),(1,2)\}$, which is vertical, lies in column 1, and covers rows 1–2. □

Lemma A.47. *The tiling C has no vertical domino in column 1. All three dominos of C are horizontal.*

Proof. C consists of three horizontal dominos $\{(1,y),(2,y)\}$ for $y=1,2,3$. None is vertical. □

A.3.3 Faultfreeness of A_n, B_n, and C

Lemma A.48. *For each even $n \geq 2$, the tiling A_n is faultfree. The basement dominos prevent faults between columns $2i-1$ and $2i$, while the middle and top dominos prevent faults between columns $2i$ and $2i+1$.*

Proof. Let k be a column with $1 \leq k < n$. If k is even, say $k=2j$, then the middle domino $\{(2j,2),(2j+1,2)\}$ spans columns $2j$ and $2j+1$, so there is no fault at column k. If k is odd, say $k=2j+1$, then the basement domino $\{(2j+1,1),(2j+2,1)\}$ spans columns $2j+1$ and $2j+2$, so there is no fault at column k. □

Lemma A.49. *For each even $n \geq 2$, the tiling B_n is faultfree.*

Proof. Since B_n is the reflection of A_n (Lemma A.56), and reflection preserves faultfreeness (Lemma A.53), B_n is faultfree because A_n is (Lemma A.48). □

Lemma A.50. *The tiling C is faultfree.*

Proof. The only potential fault in $R_{2,3}$ is at column 1. But the domino $\{(1,1),(2,1)\}$ spans columns 1 and 2, so there is no fault. □

A.3.4 Reflection lemmas

Definition A.51. Given a tiling T of $R_{n,3}$, the *reflection* of T is the tiling obtained by applying the cell map $(x, y) \mapsto (x, 4-y)$ to every cell of every domino. This swaps rows $1 \leftrightarrow 3$ and fixes row 2.

Proof. The reflected dominos are pairwise disjoint (since the cell map is injective) and their union covers $R_{n,3}$ (since the cell map is a bijection on $R_{n,3}$). $\square$

Lemma A.52. *The reflection of a tiling preserves the minimum and maximum column of each domino: for every domino d in T and its reflected image d', the minimum column of d' equals that of d, and the maximum column of d' equals that of d.*

Proof. The cell map $(x, y) \mapsto (x, 4-y)$ does not change the x-coordinate. Hence min and max of the x-coordinates of the cells are preserved. $\square$

Lemma A.53. *If T is a faultfree tiling of $R_{n,3}$, then its horizontal reflection is also faultfree.*

Proof. Reflection only changes row coordinates, not column coordinates. Therefore the minimum and maximum column of each domino are preserved (Lemma A.52), and faults (which depend only on column spans) are preserved. $\square$

Lemma A.54. *A tiling T of $R_{n,3}$ has a vertical domino in the top two squares of column c if and only if its reflection has a vertical domino in the bottom two squares of column c (and vice versa).*

Proof. The reflection swaps rows $1 \leftrightarrow 3$ and fixes row 2. A vertical domino covering rows 2–3 in column c is sent to one covering rows 1–2 in column c, and vice versa. $\square$

Lemma A.55. *Reflecting a tiling of $R_{n,3}$ twice yields the original tiling.*

Proof. The underlying cell reflection $(x, y) \mapsto (x, 4-y)$ is an involution on cells with $1 \le y \le 3$. $\square$

Lemma A.56. *For each even $n \ge 2$, B_n equals the reflection of A_n.*

Proof. By direct computation of the reflected domino sets. The basement dominos of A_n (in row 1) reflect to the "basement" dominos of B_n (in row 3), the left wall $\{(1, 2), (1, 3)\}$ reflects to $\{(1, 1), (1, 2)\}$, and similarly for the right wall, middle, and top/bottom dominos. $\square$

Lemma A.57. *If two tilings T_1 and T_2 of $R_{n,3}$ are equal, then their reflections are also equal.*

Proof. Reflection acts on the cells of each domino, so if T_1 and T_2 are equal, so are their reflections. $\square$

A.3.5 Auxiliary lemmas for the classification

Lemma A.58. *If $R_{n,3}$ admits a domino tiling, then n is even (or $n = 0$).*

Proof. $R_{n,3}$ has $3n$ cells, and each domino covers exactly 2 cells. Hence $3n$ must be even. Since 3 is odd, n must be even. $\square$

Lemma A.59. *If a tiling T of $R_{n,3}$ has a vertical domino in column 1, then either T has a vertical domino in the* ***top*** *two squares of column 1 (rows 2–3), or T has a vertical domino in the* ***bottom*** *two squares of column 1 (rows 1–2).*

Proof. A vertical domino in column 1 covers two adjacent rows. With only 3 rows available, the two adjacent rows must be $\{1, 2\}$ or $\{2, 3\}$. $\square$

Lemma A.60. *Let T be a tiling of $R_{n,3}$ with $n \geq 2$ that has a vertical domino in the top two squares of column 1. Then T contains the basement domino $\{(1, 1), (2, 1)\}$.*

Proof. The vertical domino covers $(1, 2)$ and $(1, 3)$. The remaining cell $(1, 1)$ in column 1 must be covered by some domino. It cannot be vertical (the cells above are taken). It cannot extend left (there is no column 0). So it must be horizontal extending right: $\{(1, 1), (2, 1)\}$. $\square$

Lemma A.61 (Claim 1)**.** *Let T be a faultfree tiling of $R_{n,3}$ that has a vertical domino in the top two squares of column 1. For each positive integer $i < n/2$, the tiling T contains:*

- *the basement domino $\{(2i - 1,\ 1),\ (2i,\ 1)\}$,*
- *the middle domino $\{(2i,\ 2),\ (2i + 1,\ 2)\}$,*
- *the top domino $\{(2i,\ 3),\ (2i + 1,\ 3)\}$.*

Proof. By strong induction on i.

Base case ($i = 1$): We must prove that T contains the basement domino $\{(1, 1)\,,\ (2, 1)\}$, the middle domino $\{(2, 2)\,,\ (3, 2)\}$ and the top domino $\{(2, 3)\,,\ (3, 3)\}$. For the basement domino, we have already proved this (Lemma A.60). For the middle and the top domino, we argue as follows: If T had a vertical domino in the 2-nd column, then this domino would cover the top two squares of that column (since the bottom square is already covered by the basement domino $\{(1, 1)\,,\ (2, 1)\}$), and thus T would have a fault between the 2-nd and 3-rd columns (since the basement domino $\{(1, 1)\,,\ (2, 1)\}$ also ends at the 2-nd column), which would contradict the faultfreeness of T. Hence, T has no vertical domino in the 2-nd column. Thus, the top two squares of the 2-nd column of T must be covered by horizontal dominos. These horizontal dominos must both protrude into the 3-rd column (since the corresponding squares in the 1-st column are already covered by the left wall), and thus must be the middle domino $\{(2, 2)\,,\ (3, 2)\}$ and the top domino $\{(2, 3)\,,\ (3, 3)\}$.

Induction step ($j \to j + 1$): By the induction hypothesis, T contains the basement domino $\{(2j - 1,\ 1)\,,\ (2j,\ 1)\}$, the middle domino $\{(2j,\ 2)\,,\ (2j + 1,\ 2)\}$ and the top domino $\{(2j,\ 3)\,,\ (2j + 1,\ 3)\}$. We must now show that T also contains the basement domino $\{(2j + 1,\ 1)\,,\ (2j + 2,\ 1)\}$, the middle domino $\{(2j + 2,\ 2)\,,\ (2j + 3,\ 2)\}$ and the top domino $\{(2j + 2,\ 3)\,,\ (2j + 3,\ 3)\}$.

The square $(2j + 1,\ 1)$ cannot be covered by a vertical domino in T, since this vertical domino would collide with the middle domino $\{(2j,\ 2)\,,\ (2j + 1,\ 2)\}$ (which we already know to belong to T). Thus, this square must be covered by a horizontal domino. This horizontal domino cannot protrude into the $(2j)$-th column (since it would then collide with the basement domino $\{(2j - 1,\ 1)\,,\ (2j,\ 1)\}$, which we already know to belong to T), and thus must be the basement domino $\{(2j + 1,\ 1)\,,\ (2j + 2,\ 1)\}$.

If T had a vertical domino in the $(2j + 2)$-nd column, then this domino would cover the top two squares of that column (since the bottom square is already covered by the basement domino $\{(2j + 1,\ 1)\,,\ (2j + 2,\ 1)\}$), and thus T would have a fault between the $(2j + 2)$-nd and $(2j + 3)$-rd columns (since the basement domino $\{(2j + 1,\ 1)\,,\ (2j + 2,\ 1)\}$ also ends at the $(2j + 2)$-nd column), which would contradict the faultfreeness of T. Hence, T has no vertical domino in the $(2j + 2)$-nd column. Thus, the top two squares of the $(2j + 2)$-nd column of T must be covered by horizontal dominos. These horizontal dominos must both protrude into the $(2j + 3)$-rd column (since the corresponding squares in the $(2j + 1)$-st column are already covered by the middle domino $\{(2j,\ 2)\,,\ (2j + 1,\ 2)\}$ and the top domino $\{(2j,\ 3)\,,\ (2j + 1,\ 3)\}$), and thus must be the middle domino $\{(2j + 2,\ 2)\,,\ (2j + 3,\ 2)\}$ and the top domino $\{(2j + 2,\ 3)\,,\ (2j + 3,\ 3)\}$. $\square$

Lemma A.62 (Claim 2)**.** *Let T be a faultfree tiling of $R_{n,3}$ that has a vertical domino in the top two squares of column 1. Then n is even.*

Proof. Since T is a tiling of $R_{n,3}$ by dominos, the total # of squares of $R_{n,3}$ must be even (since each domino covers exactly 2 squares). But this total # is $3n$. Thus, $3n$ must be even, so that n must be even.

(Alternatively, assuming n is odd, one can derive a contradiction using Claim 1 applied to $i = (n-1)/2$, which forces the cell $(n, 1)$ to be covered by a domino that collides with existing ones.) □

Lemma A.63. *If T is a faultfree tiling of $R_{n,3}$ with $n \geq 1$ and no vertical domino in column 1, then $n = 2$.*

Proof. Since the first column has no vertical domino, each of the three cells $(1, 1)$, $(1, 2)$, $(1, 3)$ must be covered by a horizontal domino extending into column 2. These three dominos cover all of column 2 as well. If $n > 2$, no domino would span columns 2 and 3, creating a fault at column 2, which contradicts faultfreeness. For $n = 1$, the rectangle $R_{1,3}$ has 3 cells, which is odd, so no domino tiling exists. □

A.3.6 The classification

Proposition A.64. *The faultfree domino tilings of height-3 rectangles are precisely the tilings*

$$A_2, A_4, A_6, A_8, \ldots, \qquad B_2, B_4, B_6, B_8, \ldots, \qquad C.$$

More concretely:

(a) *The faultfree domino tilings of a height-3 rectangle that contain a vertical domino in the **top** two squares of the first column are $A_2, A_4, A_6, A_8, \ldots$.*

(b) *The faultfree domino tilings of a height-3 rectangle that contain a vertical domino in the **bottom** two squares of the first column are $B_2, B_4, B_6, B_8, \ldots$.*

(c) *The only faultfree domino tiling of a height-3 rectangle that contains **no** vertical domino in the first column is C.*

Proof. It clearly suffices to prove parts **(a)**, **(b)** and **(c)**. We begin with the easiest part, which is **(c)**:

(c) Clearly, C is a faultfree domino tiling of $R_{2,3}$ that contains no vertical domino in the first column. It remains to show that C is the only such tiling.

Indeed, let T be any faultfree domino tiling of $R_{2,3}$ that contains no vertical domino in the first column. Thus, its first column must be filled with three horizontal dominos, which all must protrude into the second column and thus cover that second column as well. If T had any further column, then T would have a fault between its 2-nd and its 3-rd column, which is impossible for a faultfree tiling. Thus, T must consist only of the three horizontal dominos already mentioned. In other words, $T = C$.

(a) It is straightforward to check that the tilings $A_2, A_4, A_6, A_8, \ldots$ are faultfree (indeed, the basement dominos prevent faults between the $(2i-1)$-st and $(2i)$-th columns, whereas the top dominos prevent faults between the $(2i)$-th and $(2i+1)$-st columns). Thus, they are faultfree domino tilings of a height-3 rectangle that contain a vertical domino in the **top** two squares of the first column. It remains to show that they are the only such tilings.

Indeed, let T be any faultfree domino tiling of a height-3 rectangle that contains a vertical domino in the **top** two squares of the first column. Let n be the width of this rectangle (so that the rectangle is $R_{n,3}$). We shall show that n is even, and that $T = A_n$.

We know that T contains a vertical domino in the **top** two squares of the first column. In other words, T contains the left wall. The remaining square in the first column is the square $(1, 1)$, and it must thus be covered by the basement domino $\{(1, 1)\,,\ (2, 1)\}$ (since no other domino would fit). Hence, T contains the basement domino $\{(1, 1)\,,\ (2, 1)\}$.

Now, Claim 2 (Lemma A.62) shows that n is even, so that $n/2$ is a positive integer. Furthermore, we know that the tiling T contains the left wall, the first $n - 1$ basement dominos (by Claim 1, Lemma A.61), all the middle dominos and all the top dominos (also by Claim 1). This leaves only the four squares $(n - 1,\ 1)$, $(n,\ 1)$, $(n,\ 2)$ and $(n,\ 3)$ unaccounted for, but there is only one way to tile them: namely, with the last basement domino $\{(n - 1,\ 1)\,,\ (n,\ 1)\}$ and the right wall $\{(n, 2)\,,\ (n, 3)\}$. Thus, T must contain all the basement dominos, all the middle dominos, all the top dominos and both walls (left and right). Since these dominos cover all the squares of $R_{n,3}$, this entails that T consists precisely of these dominos. In other words, $T = A_n$.

(b) Proposition A.64 **(b)** is just Proposition A.64 **(a)** reflected across the horizontal axis of symmetry of $R_{n,3}$. More precisely, if T is a faultfree tiling with a bottom vertical in column 1, then its reflection T' is faultfree (Lemma A.53) with a top vertical in column 1 (Lemma A.54). By part **(a)**, $T' = A_n$ for some even $n \geq 2$. Since B_n is the reflection of A_n (Lemma A.56) and reflection preserves equality of tilings (Lemma A.57), reflecting back gives $T = B_n$.

The overall classification follows by Lemma A.59: any faultfree tiling of $R_{n,3}$ with $n \geq 1$ falls into exactly one of the three cases (top vertical, bottom vertical, or no vertical in column 1). □

Lemma A.65. *If T is a faultfree tiling of $R_{n,3}$ with a vertical domino in the top two squares of column 1, then n is even and $T = A_n$.*

Proof. By Claim 2, n is even and $n \geq 2$. By Claim 1 and the base case lemma, T contains all the dominos of A_n. Since both T and A_n tile the same rectangle, they have the same number of dominos, and so $T = A_n$. □

Lemma A.66. *If T is a faultfree tiling of $R_{n,3}$ with a vertical domino in the bottom two squares of column 1, then n is even and $T = B_n$.*

Proof. Reflect T to obtain T', which is faultfree with a top vertical in column 1. By part **(a)**, $T' = A_n$ for some even $n \geq 2$. Since reflection is an involution, T is the reflection of T', and hence T equals the reflection of A_n, which is B_n. □

Lemma A.67. *If T is a faultfree tiling of $R_{2,3}$ with no vertical domino in column 1, then $T = C$.*

Proof. In a 2×3 rectangle with no vertical domino in column 1, every domino must be horizontal (since a vertical domino in column 2 would force its column-1 neighbor to also be vertical in column 1, or would leave a cell that can only be covered by a vertical domino in column 1). Hence the tiling consists of three horizontal dominos $\{(1, y), (2, y)\}$ for $y = 1, 2, 3$, which is exactly C. □

A.3.7 Structural helpers for the classification

Lemma A.68. *If a tiling T of $R_{n,3}$ has a vertical domino in the top two squares of some column c, then $n \geq c$.*

Proof. The vertical domino has column coordinate c. Since all cells of a tiling lie in $R_{n,3}$, the column coordinate satisfies $c \le n$. $\square$

Lemma A.69. *If T is a faultfree tiling of $R_{n,3}$ with a vertical domino in the top two squares of column 1, then $n \ge 2$.*

Proof. By Claim 2 (Lemma A.62), n is even. Since T is nonempty (faultfree implies nonempty), $n \ge 1$, and being even forces $n \ge 2$. $\square$

A.3.8 Height-2 Fibonacci recurrence

These lemmas establish the Fibonacci recurrence $d_{n+2,2} = d_{n,2} + d_{n+1,2}$ using the natural-number-based domino tiling formalization.

Lemma A.70. *For any tiling T of $R_{n+1,2}$ (with $n+1 \ge 1$), there exists a domino $d \in T$ that covers the cell $(1,1)$.*

Proof. The cell $(1,1) \in R_{n+1,2}$, and by the coverage condition, some domino in T must contain it. $\square$

Lemma A.71. *If a domino d in a tiling of $R_{n+2,2}$ contains $(1,1)$, then either d is the vertical domino $\{(1,1),(1,2)\}$ or d is the horizontal domino $\{(1,1),(2,1)\}$.*

Proof. The domino is either horizontal or vertical. If horizontal, its second cell is $(2,1)$ (cannot be $(0,1)$ since $0 < 1$). If vertical, its second cell is $(1,2)$ (cannot be $(1,0)$ since $0 < 1$). $\square$

Lemma A.72. *If a tiling T of $R_{n+2,2}$ contains a horizontal domino d with $(1,1) \in d$, then T also contains a domino d' with $d' = \{(1,2),(2,2)\}$.*

Proof. The cell $(1,2)$ must be covered by some domino d'. This d' cannot overlap with d (which occupies $(1,1)$ and $(2,1)$). The only possibility for d' is $\{(1,2),(2,2)\}$. $\square$

Lemma A.73. *Every tiling T of $R_{n+2,2}$ either has a "vertical first column" (a domino with cells $\{(1,1),(1,2)\}$) or has a "horizontal pair" (two dominos with cells $\{(1,1),(2,1)\}$ and $\{(1,2),(2,2)\}$).*

Proof. By Lemma A.70, some domino covers $(1,1)$. By Lemma A.71, it is either vertical (giving the first case) or horizontal (giving the second case by Lemma A.72). $\square$

Lemma A.74. *Given a tiling T of $R_{n,2}$, prepending the vertical domino $\{(1,1),(1,2)\}$ and shifting all of T's dominos right by 1 gives a valid tiling of $R_{n+1,2}$.*

Proof. The prepended vertical covers column 1; the shifted dominos cover columns 2 through $n+1$. Pairwise disjointness holds by coordinate separation. $\square$

Lemma A.75. *Given a tiling T of $R_{n,2}$, prepending the two horizontal dominos $\{(1,1),(2,1)\}$ and $\{(1,2),(2,2)\}$ and shifting all of T's dominos right by 2 gives a valid tiling of $R_{n+2,2}$.*

Proof. The two horizontal dominos cover columns 1–2; the shifted dominos cover columns 3 through $n+2$. $\square$

Lemma A.76. *Given a tiling T of $R_{n+1,2}$ that contains the vertical domino $\{(1,1),(1,2)\}$, removing it and shifting all other dominos left by 1 gives a valid tiling of $R_{n,2}$.*

Proof. After removing the vertical domino, the remaining dominos all have column coordinates ≥ 2. Shifting left by 1 maps them to $R_{n,2}$. Disjointness and coverage are preserved. $\square$

Lemma A.77. *Given a tiling T of $R_{n+2,2}$ that contains both horizontal dominos $\{(1,1),(2,1)\}$ and $\{(1,2),(2,2)\}$, removing them and shifting all other dominos left by 2 gives a valid tiling of $R_{n,2}$.*

Proof. After removing the two horizontal dominos, the remaining dominos all have column coordinates ≥ 3. Shifting left by 2 maps them to $R_{n,2}$. ☐

Lemma A.78. *For every tiling T of $R_{n+2,2}$, classifying T by its first-column structure (vertical or horizontal pair), restricting, and then prepending recovers T.*

Proof. If T has a vertical first column, the restrict-then-prepend roundtrip produces a tiling whose dominos are the vertical domino plus all remaining dominos shifted right then left, recovering the original. The horizontal pair case is analogous. ☐

Lemma A.79. *The map from* $\mathrm{Tiling}(R_{n,2}) \sqcup \mathrm{Tiling}(R_{n+1,2})$ *to* $\mathrm{Tiling}(R_{n+2,2})$ *given by prepending the appropriate dominos is surjective: for every tiling T of $R_{n+2,2}$, there exists an element x of* $\mathrm{Tiling}(R_{n,2}) \sqcup \mathrm{Tiling}(R_{n+1,2})$ *whose prepended image equals T.*

Proof. Take x to be the classification of T (by its first-column structure). Then the prepended image equals T by Lemma A.78. ☐

Lemma A.80. *For every* $x \in \mathrm{Tiling}(R_{n,2}) \sqcup \mathrm{Tiling}(R_{n+1,2})$*, classifying the prepended tiling by its first-column structure and then restricting recovers x. This shows that the prepending map is injective and the classification map is surjective.*

Proof. If x corresponds to a tiling T of $R_{n+1,2}$: prepending a vertical domino produces a tiling whose first column is vertical, so the classification recovers T after restriction. The key is that restricting after prepending a vertical domino recovers T.

If x corresponds to a tiling T of $R_{n,2}$: prepending a horizontal pair produces a tiling whose first column is not vertical, so the classification recovers T after restriction. ☐

A.3.9 Bridge between $\mathbb{N}$-based and $\mathbb{Z}$-based domino tilings

Definition A.81. (a) There is a conversion from natural-number-based dominos (with natural number coordinates) to integer-based dominos (with integer coordinates) given by casting each coordinate.

(b) There is a conversion from integer-based dominos with positive coordinates back to natural-number-based dominos.

(c) There is a conversion from natural-number-based tilings of $R_{n,m}$ to integer-based tilings of $R_{n,m}$ obtained by converting each domino.

Proof. For the tiling conversion: the converted dominos are pairwise disjoint (since the coordinate casting preserves disjointness) and their union covers the integer-based rectangle (by the rectangle correspondence). ☐

Lemma A.82. *The $\mathbb{N}$-based rectangle $\{(i,j) \mid 1 \leq i \leq n, 1 \leq j \leq m\}$ and the $\mathbb{Z}$-based rectangle $\{(i,j) \in \mathbb{Z}^2 \mid 1 \leq i \leq n, 1 \leq j \leq m\}$ are in bijection via the coordinate casting map.*

Proof. The map is injective (since $\mathbb{N} \hookrightarrow \mathbb{Z}$ is injective) and surjective onto elements with positive coordinates. ☐

Lemma A.83. *If a set of natural-number-based dominos is pairwise disjoint, then their images under the coordinate casting are also pairwise disjoint.*

Proof. Since the coordinate casting preserves the cell set (up to the injective $\mathbb{N} \hookrightarrow \mathbb{Z}$ embedding), disjointness is preserved. $\square$

Lemma A.84. *If a set of natural-number-based dominos covers the natural-number-based rectangle $R_{n,m}$, then their images under the coordinate casting cover the $\mathbb{Z}$-based rectangle $R_{n,m}$.*

Proof. Every cell (i, j) in the $\mathbb{Z}$-based rectangle corresponds (via Lemma A.82) to a cell in the natural-number-based rectangle, which is covered by some domino. The image of that domino under the coordinate casting covers the original cell. $\square$

A.4 Details: Limits of FPSs

A.4.1 Detailed proof of Lemma 1.462

Detailed proof of Lemma 1.462. We have $\lim_{i\to\infty} f_i = f$. In other words, the sequence $(f_i)_{i\in\mathbb{N}}$ coefficientwise stabilizes to f. In other words, for each $n \in \mathbb{N}$,

$$\text{the sequence } \left([x^n] f_i\right)_{i\in\mathbb{N}} \text{ stabilizes to } [x^n] f \tag{A.6}$$

(by the definition of "coefficientwise stabilizing").

Now, let $n \in \mathbb{N}$. Furthermore, let $k \in \{0, 1, \ldots, n\}$. Then, the sequence $\left([x^k] f_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^k] f$ (by (A.6), applied to k instead of n). In other words, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^k] f_i = [x^k] f$$

(by the definition of "stabilizing"). Let us denote this N by N_k. Thus,

$$\text{all integers } i \geq N_k \text{ satisfy } [x^k] f_i = [x^k] f. \tag{A.7}$$

Forget that we fixed k. Thus, for each $k \in \{0, 1, \ldots, n\}$, we have defined an integer $N_k \in \mathbb{N}$ for which (A.7) holds. Altogether, we have thus defined $n+1$ integers $N_0, N_1, \ldots, N_n \in \mathbb{N}$.

Let us set $P := \max\{N_0, N_1, \ldots, N_n\}$. Then, of course, $P \in \mathbb{N}$.

Now, let $i \geq P$ be an integer. Then, for each $k \in \{0, 1, \ldots, n\}$, we have $i \geq P = \max\{N_0, N_1, \ldots, N_n\} \geq N_k$ (since $k \in \{0, 1, \ldots, n\}$) and therefore $[x^k] f_i = [x^k] f$ (by (A.7)). In other words, each $k \in \{0, 1, \ldots, n\}$ satisfies $[x^k] f_i = [x^k] f$. Renaming the variable k as m in this result, we obtain the following:

$$\text{Each } m \in \{0, 1, \ldots, n\} \text{ satisfies } [x^m] f_i = [x^m] f.$$

In other words, $f_i \overset{x^n}{\equiv} f$ (by the definition of x^n-equivalence).

Forget that we fixed i. We thus have shown that all integers $i \geq P$ satisfy $f_i \overset{x^n}{\equiv} f$. Hence, there exists some integer $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } f_i \overset{x^n}{\equiv} f$$

(namely, $N = P$). This proves Lemma 1.462. $\square$

A.4.2 Detailed proof of Proposition 1.463

Lemma A.85. *Let $(f_i)_{i\in\mathbb{N}}$ be a sequence of FPSs that coefficientwise stabilizes to an FPS f, and let $n \in \mathbb{N}$. Then there exists an integer $K \in \mathbb{N}$ such that*

$$\text{all integers } i \geq K \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Proof. This follows directly from Lemma 1.462 applied to the sequence (f_i) and limit f. □

Lemma A.86. *Let $(g_i)_{i\in\mathbb{N}}$ be a sequence of FPSs that coefficientwise stabilizes to an FPS g, and let $n \in \mathbb{N}$. Then there exists an integer $L \in \mathbb{N}$ such that*

$$\text{all integers } i \geq L \text{ satisfy } g_i \overset{x^n}{\equiv} g.$$

Proof. This is the exact same statement as Lemma A.85, but applied to the sequence (g_i) and limit g. It follows directly from Lemma 1.462. □

Lemma A.87. *Let $(f_i)_{i\in\mathbb{N}}$ and $(g_i)_{i\in\mathbb{N}}$ be sequences of FPSs that coefficientwise stabilize to f and g respectively, and let $n \in \mathbb{N}$. Then there exists an integer $P \in \mathbb{N}$ such that for all $i \geq P$,*

$$f_i \overset{x^n}{\equiv} f \qquad \text{and} \qquad g_i \overset{x^n}{\equiv} g.$$

Proof. By Lemma A.85, there exists K such that $f_i \overset{x^n}{\equiv} f$ for all $i \geq K$. By Lemma A.86, there exists L such that $g_i \overset{x^n}{\equiv} g$ for all $i \geq L$. Set $P := \max\{K, L\}$. Then for all $i \geq P$, both $i \geq K$ and $i \geq L$ hold, so both equivalences hold. □

Detailed proof of Proposition 1.463. Recall that $\lim_{i\to\infty} f_i = f$. Hence, Lemma 1.462 shows that there exists some integer $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Let us denote this N by K. Hence, all integers $i \geq K$ satisfy $f_i \overset{x^n}{\equiv} f$.

Thus, we have found an integer $K \in \mathbb{N}$ such that

$$\text{all integers } i \geq K \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Similarly, using $\lim_{i\to\infty} g_i = g$, we can find an integer $L \in \mathbb{N}$ such that

$$\text{all integers } i \geq L \text{ satisfy } g_i \overset{x^n}{\equiv} g.$$

Consider these K and L. Let us furthermore set $P := \max\{K, L\}$. Thus, $P \in \mathbb{N}$.

We shall now show that each integer $i \geq P$ satisfies $[x^n](f_i g_i) = [x^n](fg)$.

Indeed, let $i \geq P$ be an integer. Then, $f_i \overset{x^n}{\equiv} f$ (since $i \geq P = \max\{K, L\} \geq K$), whereas $g_i \overset{x^n}{\equiv} g$ (since $i \geq P = \max\{K, L\} \geq L$). Hence, we obtain

$$f_i g_i \overset{x^n}{\equiv} fg$$

(by the multiplicativity of x^n-equivalence, i.e., Theorem 1.328 **(b)**). In other words,

$$\text{each } m \in \{0, 1, \ldots, n\} \text{ satisfies } [x^m](f_i g_i) = [x^m](fg)$$

(by the definition of x^n-equivalence). Applying this to $m = n$, we find

$$[x^n](f_i g_i) = [x^n](fg).$$

Forget that we fixed i. We thus have shown that all integers $i \geq P$ satisfy $[x^n](f_i g_i) = [x^n](fg)$. Hence, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^n](f_i g_i) = [x^n](fg)$$

(namely, $N = P$). In other words,

$$\text{the sequence } \left([x^n](f_i g_i)\right)_{i \in \mathbb{N}} \text{ stabilizes to } [x^n](fg)$$

(by the definition of "stabilizes").

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$, the sequence $\left([x^n](f_i g_i)\right)_{i \in \mathbb{N}}$ stabilizes to $[x^n](fg)$. In other words, the sequence $(f_i g_i)_{i \in \mathbb{N}}$ coefficientwise stabilizes to fg (by the definition of "coefficientwise stabilizing"). In other words, $\lim_{i \to \infty}(f_i g_i) = fg$.

The same argument—but using the additivity of x^n-equivalence instead of the multiplicativity—shows that $\lim_{i \to \infty}(f_i + g_i) = f + g$. Thus, the proof of Proposition 1.463 is complete. ◻

Lemma A.88. *If (f_i) coefficientwise stabilizes to f and (g_i) coefficientwise stabilizes to g, then $(f_i + g_i)$ coefficientwise stabilizes to $f + g$.*

Proof. For each $n \in \mathbb{N}$:

1. By Lemma 1.462, obtain K_f such that $f_i \overset{x^n}{\equiv} f$ for $i \geq K_f$, and K_g such that $g_i \overset{x^n}{\equiv} g$ for $i \geq K_g$.
2. Set $P := \max\{K_f, K_g\}$.
3. For $i \geq P$, by the additivity of x^n-equivalence, $f_i + g_i \overset{x^n}{\equiv} f + g$.
4. Extracting the n-th coefficient: $[x^n](f_i + g_i) = [x^n](f + g)$.

◻

Lemma A.89. *If (f_i) coefficientwise stabilizes to f and (g_i) coefficientwise stabilizes to g, then $(f_i \cdot g_i)$ coefficientwise stabilizes to $f \cdot g$.*

Proof. For each $n \in \mathbb{N}$:

1. By Lemma 1.462, obtain K_f such that $f_i \overset{x^n}{\equiv} f$ for $i \geq K_f$, and K_g such that $g_i \overset{x^n}{\equiv} g$ for $i \geq K_g$.
2. Set $P := \max\{K_f, K_g\}$.
3. For $i \geq P$, by the multiplicativity of x^n-equivalence, $f_i \cdot g_i \overset{x^n}{\equiv} f \cdot g$.
4. Extracting the n-th coefficient: $[x^n](f_i \cdot g_i) = [x^n](f \cdot g)$.

◻

A.4.3 Helpers for Proposition 1.468

Lemma A.90. *Let $f, g \in K[[x]]$ with $f \overset{x^n}{\equiv} g$, and let u, v be units in K with $u = v$ as elements. Then the formal inverse of f with respect to u is x^n-equivalent to the formal inverse of g with respect to v.*

Lemma A.91. *Let $a \overset{x^n}{\equiv} b$ and $c \overset{x^n}{\equiv} d$, where c and d are invertible with constant terms u and v respectively. Then $a \cdot c^{-1} \overset{x^n}{\equiv} b \cdot d^{-1}$.*

Lemma A.92. *If (g_i) coefficientwise stabilizes to g and each g_i has invertible constant term, then eventually the unit associated to the constant term of g_i equals the unit associated to the constant term of g.*

Lemma A.93. *If (g_i) coefficientwise stabilizes to g and each g_i has invertible constant term, then the sequence of formal inverses $(g_i^{-1})_{i\in\mathbb{N}}$ coefficientwise stabilizes to g^{-1}. The proof proceeds by strong induction on the coefficient index, using the recursive formula for coefficients of the formal inverse.*

A.4.4 Detailed proof of Proposition 1.468

Detailed proof of Proposition 1.468. First, let us show that g is invertible:

Claim 1: *The FPS $g \in K[[x]]$ is invertible.*

Proof of Claim 1. We have assumed that $\lim_{i\to\infty} g_i = g$. In other words, the sequence $(g_i)_{i\in\mathbb{N}}$ coefficientwise stabilizes to g. In other words, for each $n \in \mathbb{N}$, the sequence $\left([x^n]g_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^n]g$ (by the definition of "coefficientwise stabilizing"). Applying this to $n = 0$, we see that the sequence $\left([x^0]g_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^0]g$. In other words, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^0]g_i = [x^0]g$$

(by the definition of "stabilizes"). Consider this N. Thus, all integers $i \geq N$ satisfy $[x^0]g_i = [x^0]g$. Applying this to $i = N$, we obtain $[x^0]g_N = [x^0]g$.

However, each FPS g_i is invertible (by assumption). Hence, in particular, the FPS g_N is invertible. By Proposition 1.111 (applied to $a = g_N$), this entails that its constant term $[x^0]g_N$ is invertible in K. In other words, $[x^0]g$ is invertible in K (since $[x^0]g_N = [x^0]g$).

But the FPS g is invertible if and only if its constant term $[x^0]g$ is invertible in K (by Proposition 1.111, applied to $a = g$). Hence, g is invertible (since its constant term $[x^0]g$ is invertible in K). This proves Claim 1. □

It remains to prove that $\lim_{i\to\infty} \frac{f_i}{g_i} = \frac{f}{g}$.

Recall that $\lim_{i\to\infty} f_i = f$. Hence, Lemma 1.462 shows that there exists some integer $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Let us denote this N by K. Hence, all integers $i \geq K$ satisfy $f_i \overset{x^n}{\equiv} f$.

Thus, we have found an integer $K \in \mathbb{N}$ such that

$$\text{all integers } i \geq K \text{ satisfy } f_i \overset{x^n}{\equiv} f.$$

Similarly, using $\lim_{i\to\infty} g_i = g$, we can find an integer $L \in \mathbb{N}$ such that

$$\text{all integers } i \geq L \text{ satisfy } g_i \overset{x^n}{\equiv} g.$$

Consider these K and L. Let us furthermore set $P := \max\{K, L\}$. Thus, $P \in \mathbb{N}$.

We shall now show that each integer $i \geq P$ satisfies $[x^n] \frac{f_i}{g_i} = [x^n] \frac{f}{g}$.

Indeed, let $i \geq P$ be an integer. Then, $f_i \overset{x^n}{\equiv} f$ (since $i \geq P = \max\{K, L\} \geq K$), whereas $g_i \overset{x^n}{\equiv} g$ (since $i \geq P = \max\{K, L\} \geq L$). Since both FPSs g_i and g are invertible (by Claim 1), we thus obtain

$$\frac{f_i}{g_i} \overset{x^n}{\equiv} \frac{f}{g}$$

(by Theorem 1.328 **(e)**, applied to $a = f_i$, $b = f$, $c = g_i$ and $d = g$). In other words,

$$\text{each } m \in \{0, 1, \ldots, n\} \text{ satisfies } [x^m] \frac{f_i}{g_i} = [x^m] \frac{f}{g}$$

(by the definition of x^n-equivalence). Applying this to $m = n$, we find

$$[x^n] \frac{f_i}{g_i} = [x^n] \frac{f}{g}.$$

Forget that we fixed i. We thus have shown that all integers $i \geq P$ satisfy $[x^n] \frac{f_i}{g_i} = [x^n] \frac{f}{g}$. Hence, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^n] \frac{f_i}{g_i} = [x^n] \frac{f}{g}$$

(namely, $N = P$). In other words,

$$\text{the sequence } \left([x^n] \frac{f_i}{g_i} \right)_{i \in \mathbb{N}} \text{ stabilizes to } [x^n] \frac{f}{g}$$

(by the definition of "stabilizes").

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$, the sequence $\left([x^n] \frac{f_i}{g_i} \right)_{i \in \mathbb{N}}$ stabilizes to $[x^n] \frac{f}{g}$. In other words, the sequence $\left(\frac{f_i}{g_i} \right)_{i \in \mathbb{N}}$ coefficientwise stabilizes to $\frac{f}{g}$ (by the definition of "coefficientwise stabilizing"). In other words, $\lim_{i \to \infty} \frac{f_i}{g_i} = \frac{f}{g}$. This proves Proposition 1.468 (since we already have shown that g is invertible). □

A.4.5 Detailed proof of Theorem 1.475

Proof of Theorem 1.475. The family $(f_n)_{n \in \mathbb{N}}$ is summable. In other words, the family $(f_k)_{k \in \mathbb{N}}$ is summable (since this is the same family as $(f_n)_{n \in \mathbb{N}}$). In other words, for each $n \in \mathbb{N}$,

$$\text{all but finitely many } k \in \mathbb{N} \text{ satisfy } [x^n] f_k = 0 \qquad \text{(A.8)}$$

(by the definition of "summable").

Let us define

$$g_i := \sum_{k=0}^{i} f_k \qquad \text{for each } i \in \mathbb{N}. \qquad \text{(A.9)}$$

Let us furthermore set

$$g := \sum_{k \in \mathbb{N}} f_k. \qquad \text{(A.10)}$$

Now, fix $n \in \mathbb{N}$. Then, all but finitely many $k \in \mathbb{N}$ satisfy $[x^n] f_k = 0$ (by (A.8)). In other words, there exists a finite subset J of $\mathbb{N}$ such that

$$\text{all } k \in \mathbb{N} \setminus J \text{ satisfy } [x^n] f_k = 0. \tag{A.11}$$

Consider this subset J. The set J is a finite set of nonnegative integers, and thus has an upper bound. In other words, there exists some $m \in \mathbb{N}$ such that

$$\text{all } k \in J \text{ satisfy } k \leq m. \tag{A.12}$$

Consider this m.

Let $k \in \mathbb{N}$ be such that $k \geq m+1$. If we had $k \in J$, then we would have $k \leq m$ (by (A.12)), which would contradict $k \geq m+1 > m$. Thus, we cannot have $k \in J$. Hence, $k \in \mathbb{N} \setminus J$ (since $k \in \mathbb{N}$ but not $k \in J$). Therefore, (A.11) yields $[x^n] f_k = 0$.

Forget that we fixed k. We thus have shown that

$$[x^n] f_k = 0 \qquad \text{for each } k \in \mathbb{N} \text{ satisfying } k \geq m+1. \tag{A.13}$$

Now, let i be an integer such that $i \geq m$. Then, $i \in \mathbb{N}$ (since $i \geq m \geq 0$) and $i+1 \geq m+1$ (since $i \geq m$). From $g = \sum_{k \in \mathbb{N}} f_k$, we obtain

$$\begin{aligned}
[x^n] g &= [x^n] \left(\sum_{k \in \mathbb{N}} f_k \right) = \sum_{k \in \mathbb{N}} [x^n] f_k \\
&= \sum_{k=0}^{i} [x^n] f_k + \sum_{k=i+1}^{\infty} \underbrace{[x^n] f_k}_{\substack{=0 \\ \text{(by (A.13)} \\ \text{since } k \geq i+1 \geq m+1)}} = \sum_{k=0}^{i} [x^n] f_k + \underbrace{\sum_{k=i+1}^{\infty} 0}_{=0} \\
&= \sum_{k=0}^{i} [x^n] f_k.
\end{aligned} \tag{A.14}$$

On the other hand, from $g_i = \sum_{k=0}^{i} f_k$, we obtain

$$[x^n] g_i = [x^n] \left(\sum_{k=0}^{i} f_k \right) = \sum_{k=0}^{i} [x^n] f_k.$$

Comparing this with (A.14), we obtain $[x^n] g_i = [x^n] g$.

Forget that we fixed i. We thus have shown that

$$\text{all integers } i \geq m \text{ satisfy } [x^n] g_i = [x^n] g.$$

Hence, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^n] g_i = [x^n] g$$

(namely, $N = m$). In other words, the sequence $\left([x^n] g_i\right)_{i \in \mathbb{N}}$ stabilizes to $[x^n] g$ (by the definition of "stabilizes").

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$,

$$\text{the sequence } \left([x^n] g_i\right)_{i \in \mathbb{N}} \text{ stabilizes to } [x^n] g.$$

In other words, the sequence $(g_i)_{i\in\mathbb{N}}$ coefficientwise stabilizes to g. In other words, $\lim_{i\to\infty} g_i = g$. In view of (A.9) and (A.10), we can rewrite this as

$$\lim_{i\to\infty} \sum_{k=0}^{i} f_k = \sum_{k\in\mathbb{N}} f_k.$$

Renaming the summation index k as n everywhere in this equality, we can rewrite it as

$$\lim_{i\to\infty} \sum_{n=0}^{i} f_n = \sum_{n\in\mathbb{N}} f_n.$$

This proves Theorem 1.475. □

A.4.6 Helpers for Theorem 1.476

Lemma A.94. *Let $f \in K[[x]]$ satisfy $[x^k]f = \delta_{k,0}$ for all $k \leq n$ (i.e., $f \equiv 1 \pmod{x^{n+1}}$). Then for any $g \in K[[x]]$ and any $k \leq n$,*

$$[x^k](g \cdot f) = [x^k]g.$$

The proof expands the product coefficient as a sum over the antidiagonal and observes that only the term with second component 0 survives, since $[x^j]f = 0$ for $0 < j \leq n$.

Lemma A.95. *If $f_m \equiv 1 \pmod{x^{n+1}}$, then for any $k \leq n$,*

$$[x^k]\prod_{j=0}^{m} f_j = [x^k]\prod_{j=0}^{m-1} f_j.$$

That is, extending a partial product by one more term that is $1 + O(x^{n+1})$ does not change the first $n+1$ coefficients.

Lemma A.96. *For a multipliable family (f_j), if $f_j \equiv 1 \pmod{x^{n+1}}$ for all $j \geq N$, then for all $i \geq N$ and $k \leq n$,*

$$[x^k]\prod_{j=0}^{i} f_j = [x^k]\prod_{j=0}^{N} f_j.$$

The proof proceeds by induction on i, repeatedly applying Lemma A.95.

Lemma A.97. *For a multipliable family (f_j), for each $n \in \mathbb{N}$, there exists N such that for all $i \geq N$,*

$$[x^n]\prod_{j=0}^{i} f_j = [x^n]\prod_{j\in\mathbb{N}} f_j.$$

This is the key stabilization lemma that directly implies Theorem 1.476.

A.4.7 Detailed proof of Theorem 1.476

Proof of Theorem 1.476. The family $(f_n)_{n\in\mathbb{N}}$ is multipliable. In other words, each coefficient in the product of this family is finitely determined (by the definition of "multipliable").

Let us define

$$g_i := \prod_{k=0}^{i} f_k \qquad \text{for each } i \in \mathbb{N}. \tag{A.15}$$

Let us furthermore set

$$g := \prod_{k\in\mathbb{N}} f_k. \tag{A.16}$$

Now, fix $n \in \mathbb{N}$. Then, the x^n-coefficient in the product of the family $(f_k)_{k\in\mathbb{N}}$ is finitely determined. In other words, there is a finite subset M of $\mathbb{N}$ that determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$. Consider this subset M.

The set M is a finite set of nonnegative integers, and thus has an upper bound. In other words, there exists some $m \in \mathbb{N}$ such that

$$\text{all } k \in M \text{ satisfy } k \leq m. \tag{A.17}$$

Consider this m.

Now, let i be an integer such that $i \geq m$. Then, $i \in \mathbb{N}$ (since $i \geq m \geq 0$) and $m \leq i$. From (A.17), we see that all $k \in M$ satisfy $k \leq m \leq i$ and therefore $k \in \{0, 1, \ldots, i\}$. In other words, $M \subseteq \{0, 1, \ldots, i\}$.

However, the set M determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$. In other words, every finite subset J of $\mathbb{N}$ satisfying $M \subseteq J \subseteq \mathbb{N}$ satisfies

$$[x^n]\left(\prod_{k\in J} f_k\right) = [x^n]\left(\prod_{k\in M} f_k\right)$$

(by the definition of "determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$"). Applying this to $J = \{0, 1, \ldots, i\}$, we obtain

$$[x^n]\left(\prod_{k\in\{0,1,\ldots,i\}} f_k\right) = [x^n]\left(\prod_{k\in M} f_k\right) \tag{A.18}$$

(since $\{0, 1, \ldots, i\}$ is a finite subset of $\mathbb{N}$ satisfying $M \subseteq \{0, 1, \ldots, i\} \subseteq \mathbb{N}$).

Moreover, the definition of the product of the multipliable family $(f_k)_{k\in\mathbb{N}}$ shows that

$$[x^n]\left(\prod_{k\in\mathbb{N}} f_k\right) = [x^n]\left(\prod_{k\in M} f_k\right)$$

(since M is a finite subset of $\mathbb{N}$ that determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$). Comparing this with (A.18), we find

$$[x^n]\left(\prod_{k\in\{0,1,\ldots,i\}} f_k\right) = [x^n]\left(\prod_{k\in\mathbb{N}} f_k\right).$$

In view of $\prod_{k\in\{0,1,\ldots,i\}} f_k = \prod_{k=0}^{i} f_k = g_i$ (by (A.15)) and $\prod_{k\in\mathbb{N}} f_k = g$ (by (A.16)), we can rewrite this as

$$[x^n]g_i = [x^n]g.$$

Forget that we fixed i. We thus have shown that

$$\text{all integers } i \geq m \text{ satisfy } [x^n]g_i = [x^n]g.$$

Hence, the sequence $\left([x^n]g_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^n]g$.

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$, the sequence $\left([x^n] g_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^n] g$. In other words, the sequence $(g_i)_{i\in\mathbb{N}}$ coefficientwise stabilizes to g. In other words, $\lim_{i\to\infty} g_i = g$. In view of (A.15) and (A.16), we can rewrite this as

$$\lim_{i\to\infty} \prod_{k=0}^{i} f_k = \prod_{k\in\mathbb{N}} f_k.$$

Renaming the product index k as n everywhere in this equality, we can rewrite it as

$$\lim_{i\to\infty} \prod_{n=0}^{i} f_n = \prod_{n\in\mathbb{N}} f_n.$$

This proves Theorem 1.476. $\square$

A.4.8 Detailed proofs of Corollary 1.477

Proof of Corollary 1.477. Let a FPS $a \in K[[x]]$ be written in the form $a = \sum_{n\in\mathbb{N}} a_n x^n$ with $a_n \in K$. Then, the family $(a_n x^n)_{n\in\mathbb{N}}$ is summable. Hence, Theorem 1.475 (applied to $f_n = a_n x^n$) yields

$$\lim_{i\to\infty} \sum_{n=0}^{i} a_n x^n = \sum_{n\in\mathbb{N}} a_n x^n = a.$$

In other words, $a = \lim_{i\to\infty} \sum_{n=0}^{i} a_n x^n$. This proves Corollary 1.477. $\square$

A.4.9 Detailed proof of Theorem 1.478

Proof of Theorem 1.478. Let us define

$$g_i := \sum_{k=0}^{i} f_k \qquad \text{for each } i \in \mathbb{N}. \tag{A.19}$$

We have assumed that the limit $\lim_{i\to\infty} \sum_{n=0}^{i} f_n$ exists. Let us denote this limit by g. Thus,

$$g = \lim_{i\to\infty} \sum_{n=0}^{i} f_n = \lim_{i\to\infty} g_i. \tag{A.20}$$

In other words, the sequence $(g_i)_{i\in\mathbb{N}}$ coefficientwise stabilizes to g. In other words, for each $n \in \mathbb{N}$,

$$\text{the sequence } \left([x^n] g_i\right)_{i\in\mathbb{N}} \text{ stabilizes to } [x^n] g. \tag{A.21}$$

Let $n \in \mathbb{N}$. Then, the sequence $\left([x^n] g_i\right)_{i\in\mathbb{N}}$ stabilizes to $[x^n] g$ (by (A.21)). In other words, there exists some $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } [x^n] g_i = [x^n] g. \tag{A.22}$$

Consider this N. Let M be the set $\{0, 1, \ldots, N\}$; this is a finite subset of $\mathbb{N}$.

Let $i \in \mathbb{N} \setminus M$. Thus,

$$i \in \mathbb{N} \setminus M = \mathbb{N} \setminus \{0, 1, \ldots, N\} = \{N+1, N+2, N+3, \ldots\}.$$

In other words, i is a nonnegative integer with $i \geq N+1$. Hence, $i-1 \geq N$. Therefore, (A.22) (applied to $i-1$ instead of i) yields $[x^n] g_{i-1} = [x^n] g$. But (A.22) also yields $[x^n] g_i = [x^n] g$ (since $i \geq N+1 \geq N$). Comparing these two equalities, we find

$$[x^n] g_{i-1} = [x^n] g_i. \tag{A.23}$$

However, (A.19) yields

$$g_i = \sum_{k=0}^{i} f_k = f_i + \sum_{k=0}^{i-1} f_k. \tag{A.24}$$

Moreover, (A.19) (applied to $i-1$ instead of i) yields $g_{i-1} = \sum_{k=0}^{i-1} f_k$. In view of this, we can rewrite (A.24) as

$$g_i = f_i + g_{i-1}.$$

Hence, $[x^n] g_i = [x^n](f_i + g_{i-1}) = [x^n] f_i + \underbrace{[x^n] g_{i-1}}_{\substack{=[x^n] g_i \\ \text{(by (A.23))}}} = [x^n] f_i + [x^n] g_i$. Subtracting $[x^n] g_i$ from both sides of this equality, we find $0 = [x^n] f_i$. In other words, $[x^n] f_i = 0$.

Forget that we fixed i. We thus have shown that all $i \in \mathbb{N} \setminus M$ satisfy $[x^n] f_i = 0$. Hence, all but finitely many $i \in \mathbb{N}$ satisfy $[x^n] f_i = 0$ (since all but finitely many $i \in \mathbb{N}$ belong to $\mathbb{N} \setminus M$ because the set M is finite). Renaming the index i as k in this statement, we obtain the following: All but finitely many $k \in \mathbb{N}$ satisfy $[x^n] f_k = 0$.

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$,

$$\text{all but finitely many } k \in \mathbb{N} \text{ satisfy } [x^n] f_k = 0.$$

In other words, the family $(f_n)_{n \in \mathbb{N}}$ is summable (by the definition of "summable").

Hence, Theorem 1.475 yields $\lim_{i \to \infty} \sum_{n=0}^{i} f_n = \sum_{n \in \mathbb{N}} f_n$. In other words, $\sum_{n \in \mathbb{N}} f_n = \lim_{i \to \infty} \sum_{n=0}^{i} f_n$. This completes the proof of Theorem 1.478. □

A.4.10 Helpers for Theorem 1.479

Lemma A.98. *If f_i has constant term 1 for every i in a finite set S, then $\prod_{i \in S} f_i$ also has constant term 1.*

A.4.11 Detailed proof of Theorem 1.479

Proof of Theorem 1.479. Let us define

$$g_i := \prod_{k=0}^{i} f_k \qquad \text{for each } i \in \mathbb{N}. \tag{A.25}$$

We have assumed that the limit $\lim_{i \to \infty} \prod_{n=0}^{i} f_n$ exists. Let us denote this limit by g. Thus,

$$g = \lim_{i \to \infty} \prod_{n=0}^{i} f_n = \lim_{i \to \infty} g_i. \tag{A.26}$$

Let $n \in \mathbb{N}$ be arbitrary. Lemma 1.462 (applied to g_i and g instead of f_i and f) shows that there exists some integer $N \in \mathbb{N}$ such that

$$\text{all integers } i \geq N \text{ satisfy } g_i \overset{x^n}{\equiv} g \tag{A.27}$$

(since $g = \lim_{i\to\infty} g_i$). Consider this N.

Let $M = \{0, 1, \ldots, N\}$. Thus, M is a finite subset of $\mathbb{N}$. We shall show that this subset M determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$.

For this purpose, we first observe that $M = \{0, 1, \ldots, N\}$, so that

$$\prod_{k\in M} f_k = \prod_{k\in\{0,1,\ldots,N\}} f_k = \prod_{k=0}^{N} f_k = g_N \tag{A.28}$$

(since g_N was defined to be $\prod_{k=0}^{N} f_k$). Applying (A.27) to $i = N$, we obtain $g_N \overset{x^n}{\equiv} g$ (since $N \geq N$). Therefore, $g \overset{x^n}{\equiv} g_N$ (since $\overset{x^n}{\equiv}$ is an equivalence relation and thus symmetric).

Next, we shall show two claims:

Claim 1: *For each integer $j > N$, we have $g \overset{x^n}{\equiv} gf_j$.*

Proof of Claim 1. Let $j > N$ be an integer. Thus, $j \geq N + 1$, so that $j - 1 \geq N$. Hence, (A.27) (applied to $i = j - 1$) yields $g_{j-1} \overset{x^n}{\equiv} g$. However, (A.27) (applied to $i = j$) yields $g_j \overset{x^n}{\equiv} g$ (since $j \geq N + 1 \geq N$). Thus, $g \overset{x^n}{\equiv} g_j$ (by symmetry).

Also, we obviously have $f_j \overset{x^n}{\equiv} f_j$ (by reflexivity).

However, the definition of g_{j-1} yields $g_{j-1} = \prod_{k=0}^{j-1} f_k$. Furthermore, the definition of g_j yields

$$g_j = \prod_{k=0}^{j} f_k = \left(\prod_{k=0}^{j-1} f_k\right) f_j = g_{j-1} f_j.$$

But recall that $g_{j-1} \overset{x^n}{\equiv} g$ and $f_j \overset{x^n}{\equiv} f_j$. Hence, $g_{j-1} f_j \overset{x^n}{\equiv} gf_j$ (by the multiplicativity of x^n-equivalence). In view of $g_j = g_{j-1} f_j$, we can rewrite this as $g_j \overset{x^n}{\equiv} gf_j$.

Now, recall that $g \overset{x^n}{\equiv} g_j$. Thus, $g \overset{x^n}{\equiv} g_j \overset{x^n}{\equiv} gf_j$, so that $g \overset{x^n}{\equiv} gf_j$ (by transitivity). This proves Claim 1. □

Claim 2: *If U is any finite subset of $\mathbb{N}$ satisfying $M \subseteq U$, then $g \overset{x^n}{\equiv} \prod_{k\in U} f_k$.*

Proof of Claim 2. We induct on the nonnegative integer $|U \setminus M|$:

Base case: Let us check that Claim 2 holds when $|U \setminus M| = 0$.

Indeed, let U be any finite subset of $\mathbb{N}$ satisfying $M \subseteq U$ and $|U \setminus M| = 0$. Then, $U \setminus M = \varnothing$ (since $|U \setminus M| = 0$), so that $U \subseteq M$. Combined with $M \subseteq U$, this yields $U = M$. Thus, $\prod_{k\in U} f_k = \prod_{k\in M} f_k = g_N$ (by (A.28)). But we have already proved that $g \overset{x^n}{\equiv} g_N$. In other words, $g \overset{x^n}{\equiv} \prod_{k\in U} f_k$ (since $\prod_{k\in U} f_k = g_N$). This completes the base case.

Induction step: Let $s \in \mathbb{N}$. Assume (as the induction hypothesis) that Claim 2 holds when $|U \setminus M| = s$. We must now prove that Claim 2 holds when $|U \setminus M| = s + 1$.

Indeed, let U be any finite subset of $\mathbb{N}$ satisfying $M \subseteq U$ and $|U \setminus M| = s + 1$. Then, the set $U \setminus M$ is nonempty (since $|U \setminus M| = s + 1 > 0$). Hence, there exists some $u \in U \setminus M$. Consider this u. By the induction hypothesis (applied to $U \setminus \{u\}$ instead of U), we obtain $g \overset{x^n}{\equiv} \prod_{k\in U\setminus\{u\}} f_k$.

However, $u \in U \setminus M$, so $u \notin M = \{0, 1, \ldots, N\}$, which gives $u > N$. Hence, Claim 1 (applied to $j = u$) yields $g \overset{x^n}{\equiv} gf_u$.

We have $g \overset{x^n}{\equiv} \prod_{k\in U\setminus\{u\}} f_k$ and $f_u \overset{x^n}{\equiv} f_u$. Hence, $gf_u \overset{x^n}{\equiv} \left(\prod_{k\in U\setminus\{u\}} f_k\right) f_u$ (by the multiplicativity of x^n-equivalence). In view of $\prod_{k\in U} f_k = \left(\prod_{k\in U\setminus\{u\}} f_k\right) f_u$, we can rewrite this as $gf_u \overset{x^n}{\equiv} \prod_{k\in U} f_k$.

Now, recall that $g \overset{x^n}{\equiv} gf_u$. Thus, $g \overset{x^n}{\equiv} gf_u \overset{x^n}{\equiv} \prod_{k\in U} f_k$, so $g \overset{x^n}{\equiv} \prod_{k\in U} f_k$ (by transitivity). This completes the induction step. Thus, Claim 2 is proved by induction. $\square$

Now, let J be a finite subset of $\mathbb{N}$ satisfying $M \subseteq J \subseteq \mathbb{N}$. Then, $g \overset{x^n}{\equiv} \prod_{k\in J} f_k$ (by Claim 2, applied to $U = J$). In other words,

$$\text{each } m \in \{0,1,\ldots,n\} \text{ satisfies } [x^m] g = [x^m] \left(\prod_{k\in J} f_k\right)$$

(by the definition of x^n-equivalence). Applying this to $m = n$, we find

$$[x^n] g = [x^n] \left(\prod_{k\in J} f_k\right).$$

The same argument can be applied to M instead of J (since $M \subseteq M \subseteq \mathbb{N}$), and thus yields

$$[x^n] g = [x^n] \left(\prod_{k\in M} f_k\right).$$

Comparing these two equalities, we find

$$[x^n] \left(\prod_{k\in J} f_k\right) = [x^n] \left(\prod_{k\in M} f_k\right).$$

Forget that we fixed J. We have now shown that every finite subset J of $\mathbb{N}$ satisfying $M \subseteq J \subseteq \mathbb{N}$ satisfies

$$[x^n] \left(\prod_{k\in J} f_k\right) = [x^n] \left(\prod_{k\in M} f_k\right).$$

In other words, the set M determines the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$. Hence, the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$ is finitely determined (since M is a finite subset of $\mathbb{N}$).

Forget that we fixed n. We thus have shown that for each $n \in \mathbb{N}$, the x^n-coefficient in the product of $(f_k)_{k\in\mathbb{N}}$ is finitely determined. In other words, each coefficient in the product of $(f_k)_{k\in\mathbb{N}}$ is finitely determined. In other words, the family $(f_k)_{k\in\mathbb{N}}$ is multipliable. In other words, the family $(f_n)_{n\in\mathbb{N}}$ is multipliable.

Hence, Theorem 1.476 yields $\lim_{i\to\infty} \prod_{n=0}^{i} f_n = \prod_{n\in\mathbb{N}} f_n$. In other words, $\prod_{n\in\mathbb{N}} f_n = \lim_{i\to\infty} \prod_{n=0}^{i} f_n$. This completes the proof of Theorem 1.479. $\square$

A.4.12 Helper lemmas from the Lean formalization

Lemma A.99. *If a sequence $(a_i)_{i\in\mathbb{N}}$ in K stabilizes to both ℓ_1 and ℓ_2, then $\ell_1 = \ell_2$.*

Proof. Let N_1 and N_2 be the stabilization indices for ℓ_1 and ℓ_2 respectively. Set $N := \max(N_1, N_2)$. Then $a_N = \ell_1$ and $a_N = \ell_2$, whence $\ell_1 = \ell_2$. $\square$

Lemma A.100. *If the family $(f_n)_{n\in\mathbb{N}}$ is summable, then the sequence of partial sums $\left(\sum_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to $\sum_{n\in\mathbb{N}} f_n$.*

Proof. This is a direct restatement of Theorem 1.475. □

Lemma A.101. *If the family $(f_n)_{n\in\mathbb{N}}$ is multipliable, then the sequence of partial products $\left(\prod_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to $\prod_{n\in\mathbb{N}} f_n$.*

Proof. This is a direct restatement of Theorem 1.476. □

Lemma A.102. *If the sequence of partial sums $\left(\sum_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to some limit, then the family $(f_n)_{n\in\mathbb{N}}$ is summable.*

Proof. This is a direct restatement of the summability part of Theorem 1.478. □

Lemma A.103. *If the sequence of partial sums $\left(\sum_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to some limit and the family $(f_n)_{n\in\mathbb{N}}$ is summable, then the infinite sum equals the limit.*

Proof. By Theorem 1.475, the partial sums converge to the infinite sum. By uniqueness of limits (Lemma 1.454), the limit equals the infinite sum. □

Lemma A.104. *If the sequence of partial products $\left(\prod_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to some limit, then the family $(f_n)_{n\in\mathbb{N}}$ is multipliable.*

Proof. This is a direct restatement of the multipliability part of Theorem 1.479. □

Lemma A.105. *If the sequence of partial products $\left(\prod_{j=0}^{i} f_j\right)_{i\in\mathbb{N}}$ coefficientwise stabilizes to some limit and the family $(f_n)_{n\in\mathbb{N}}$ is multipliable, then the infinite product equals the limit.*

Proof. By Theorem 1.476, the partial products converge to the infinite product. By uniqueness of limits (Lemma 1.454), the limit equals the infinite product. □

Lemma A.106. *If $f \overset{x^n}{\equiv} g$, then $f^d \overset{x^n}{\equiv} g^d$ for every $d \in \mathbb{N}$. The proof proceeds by induction on d, using the multiplicativity of x^n-equivalence at each step.*

A.5 Details on Laurent power series

Fix a commutative ring K.

A.5.1 Proof of Theorem 1.498 (sketched)

The set $K[x^{\pm}]$ is closed under addition, scaling, and the convolution product from Definition 1.497 (analogous to Theorem 1.148, replacing $\sum_{i=0}^{n}$ sums by $\sum_{i\in\mathbb{Z}}$ sums). The multiplication is associative, commutative, distributive, and K-bilinear (analogous to the corresponding parts of Theorem 1.76). The element $(\delta_{i,0})_{i\in\mathbb{Z}}$ is a neutral element for multiplication.

It remains to show that x is invertible. Set $\overline{x} := (\delta_{i,-1})_{i\in\mathbb{Z}}$. Then $x\overline{x} = 1$ and $\overline{x}x = 1$ by direct calculation. For example, from $x = (\delta_{i,1})_{i\in\mathbb{Z}}$ and $\overline{x} = (\delta_{i,-1})_{i\in\mathbb{Z}}$, we get

$$x\overline{x} = (c_n)_{n\in\mathbb{Z}}, \quad \text{where} \quad c_n = \sum_{i\in\mathbb{Z}} \delta_{i,1}\,\delta_{n-i,-1}$$

(by Definition 1.497). For each $n \in \mathbb{Z}$,

$$\begin{aligned} c_n &= \sum_{i \in \mathbb{Z}} \delta_{i,1}\, \delta_{n-i,-1} = \underbrace{\delta_{1,1}}_{=1}\, \delta_{n-1,-1} + \sum_{\substack{i \in \mathbb{Z} \\ i \neq 1}} \underbrace{\delta_{i,1}}_{=0 \text{ (since } i \neq 1)}\, \delta_{n-i,-1} \\ &= \delta_{n-1,-1} = \delta_{n,0}, \end{aligned}$$

since $n - 1 = -1$ if and only if $n = 0$. Hence $x\overline{x} = (\delta_{n,0})_{n \in \mathbb{Z}} = 1$. The proof of $\overline{x}x = 1$ is analogous, or follows by commutativity.

A.5.2 Helpers for Theorem 1.498

Lemma A.107 ($T(1) \cdot T(-1) = 1$). *In the Laurent polynomial ring, $T(1) \cdot T(-1) = 1$. This witnesses the right inverse of x.*

Proof. Follows from $T(a) \cdot T(b) = T(a + b)$ with $a = 1$, $b = -1$. □

Lemma A.108 ($T(-1) \cdot T(1) = 1$). *In the Laurent polynomial ring, $T(-1) \cdot T(1) = 1$. This witnesses the left inverse of x.*

Proof. Follows from $T(a) \cdot T(b) = T(a + b)$ with $a = -1$, $b = 1$. □

A.5.3 Multiplication by x shifts coefficients

Lemma A.109 (Multiplication by x and x^{-1} shifts coefficients). *Let $\mathbf{a} = (a_n)_{n \in \mathbb{Z}}$ be a Laurent polynomial in $K[x^{\pm}]$. Then,*

$$x \cdot \mathbf{a} = (a_{n-1})_{n \in \mathbb{Z}} \qquad \textit{and} \qquad x^{-1} \cdot \mathbf{a} = (a_{n+1})_{n \in \mathbb{Z}}.$$

Proof. From $x = (\delta_{i,1})_{i \in \mathbb{Z}}$ and $\mathbf{a} = (a_i)_{i \in \mathbb{Z}}$, we obtain

$$x \cdot \mathbf{a} = (\delta_{i,1})_{i \in \mathbb{Z}} \cdot (a_i)_{i \in \mathbb{Z}} = (c_n)_{n \in \mathbb{Z}}, \quad \text{where} \quad c_n = \sum_{i \in \mathbb{Z}} \delta_{i,1}\, a_{n-i}$$

(by Definition 1.497). For each $n \in \mathbb{Z}$,

$$\begin{aligned} c_n &= \sum_{i \in \mathbb{Z}} \delta_{i,1}\, a_{n-i} = \underbrace{\delta_{1,1}}_{=1}\, a_{n-1} + \sum_{\substack{i \in \mathbb{Z} \\ i \neq 1}} \underbrace{\delta_{i,1}}_{=0 \text{ (since } i \neq 1)}\, a_{n-i} \\ &= a_{n-1}. \end{aligned}$$

Thus, $x \cdot \mathbf{a} = (a_{n-1})_{n \in \mathbb{Z}}$.

It remains to prove that $x^{-1} \cdot \mathbf{a} = (a_{n+1})_{n \in \mathbb{Z}}$. Let $\mathbf{b} = (a_{n+1})_{n \in \mathbb{Z}}$; this is again a Laurent polynomial in $K[x^{\pm}]$. Applying the result just proved to $\mathbf{b}$ (with a_{n+1} in place of a_n) yields

$$x \cdot \mathbf{b} = \left(a_{(n-1)+1}\right)_{n \in \mathbb{Z}} = (a_n)_{n \in \mathbb{Z}} = \mathbf{a}.$$

Dividing by the invertible element x, we get $\mathbf{b} = x^{-1} \cdot \mathbf{a}$, i.e. $x^{-1} \cdot \mathbf{a} = (a_{n+1})_{n \in \mathbb{Z}}$. □

Lemma A.110 (Multiplication on the right by T shifts coefficients). *For any Laurent polynomial $\mathbf{a}$,*

$$\mathbf{a} \cdot x = (a_{n-1})_{n \in \mathbb{Z}} \qquad \textit{and} \qquad \mathbf{a} \cdot x^{-1} = (a_{n+1})_{n \in \mathbb{Z}}.$$

Proof. Follows from Lemma A.109 by commutativity of multiplication. □

A.5.4 Powers of x

Lemma A.111 ($T(m) \cdot T(n) = T(m+n)$)**.** *For all $m, n \in \mathbb{Z}$, we have $T(m) \cdot T(n) = T(m+n)$ in the Laurent polynomial ring. This is the fundamental multiplicative law for Laurent monomials.*

Proof. This is the group homomorphism property of T: $T(a) \cdot T(b) = T(a+b)$. □

Lemma A.112 ($T(1)^n = T(n)$ for natural numbers)**.** *For any natural number n, we have $T(1)^n = T(n)$ in the Laurent polynomial ring.*

Proof. Follows from $T(1)^n = T(1 \cdot n) = T(n)$ by the group homomorphism property. □

Proposition A.113 ($x^k = (\delta_{i,k})_{i \in \mathbb{Z}}$ for all $k \in \mathbb{Z}$)**.** *We have*

$$x^k = (\delta_{i,k})_{i \in \mathbb{Z}} \qquad \textit{for each } k \in \mathbb{Z}.$$

Proof. This is proved by two-sided induction on k (see [Grinbe15], §2.15 for a detailed explanation of two-sided induction).

Base case ($k = 0$): $x^0 = 1 = (\delta_{i,0})_{i \in \mathbb{Z}}$, which is the definition of the unity element.

Induction step from k to $k+1$: Assume $x^k = (\delta_{i,k})_{i \in \mathbb{Z}}$. By Lemma A.109 (the $x \cdot \mathbf{a} = (a_{n-1})_{n \in \mathbb{Z}}$ part),

$$x^{k+1} = x \cdot x^k = x \cdot (\delta_{i,k})_{i \in \mathbb{Z}} = (\delta_{n-1,k})_{n \in \mathbb{Z}} = (\delta_{n,k+1})_{n \in \mathbb{Z}},$$

since $n - 1 = k$ if and only if $n = k + 1$.

Induction step from k to $k-1$: Assume $x^k = (\delta_{i,k})_{i \in \mathbb{Z}}$. By Lemma A.109 (the $x^{-1} \cdot \mathbf{a} = (a_{n+1})_{n \in \mathbb{Z}}$ part),

$$x^{k-1} = x^{-1} \cdot x^k = x^{-1} \cdot (\delta_{i,k})_{i \in \mathbb{Z}} = (\delta_{n+1,k})_{n \in \mathbb{Z}} = (\delta_{n,k-1})_{n \in \mathbb{Z}},$$

since $n + 1 = k$ if and only if $n = k - 1$. □

Lemma A.114 (Coefficient of x^k)**.** *For all $k, i \in \mathbb{Z}$, the coefficient of x^k at position i is $\delta_{i,k}$.*

Proof. Immediate from Proposition A.113. □

A.5.5 Proof of Proposition 1.505 (sketched)

Definition A.115 (Summable family of monomials)**.** Given a Laurent series $\mathbf{a} = (a_n)_{n \in \mathbb{Z}}$, the family of monomials $\left(a_g \cdot x^g\right)_{g \in \operatorname{supp}(\mathbf{a})}$ forms a summable family. This construction is the key ingredient for expressing a Laurent series as a formal infinite sum of monomials.

Lemma A.116 (Coefficient of a Laurent series as a finitary sum)**.** *For any Laurent series $\mathbf{a} = (a_n)_{n \in \mathbb{Z}}$ and any $n \in \mathbb{Z}$, the n-th coefficient satisfies*

$$a_n = \sum_{i \in \operatorname{supp}(\mathbf{a})}^{\text{fin}} [x^n](a_i \cdot x^i).$$

Proof. Follows by rewriting the left-hand side as the sum of its monomial components and extracting the coefficient. □

Proposition 1.505 follows: any Laurent polynomial $\mathbf{a} = (a_i)_{i \in \mathbb{Z}}$ satisfies $\mathbf{a} = \sum_{i \in \mathbb{Z}} a_i x^i$, analogously to Corollary 1.85 but using Proposition A.113 instead of Proposition 1.84.

A.5.6 Proof of Theorem 1.509 (sketched)

The proof is analogous to Theorem 1.498. The only different piece is the proof that $K((x))$ is closed under multiplication.

Let $(a_n)_{n\in\mathbb{Z}}$ and $(b_n)_{n\in\mathbb{Z}}$ be two elements of $K((x))$. Their product is

$$(a_n)_{n\in\mathbb{Z}} \cdot (b_n)_{n\in\mathbb{Z}} = (c_n)_{n\in\mathbb{Z}}, \quad \text{where} \quad c_n = \sum_{i\in\mathbb{Z}} a_i\, b_{n-i}.$$

We must show that $(c_{-1}, c_{-2}, c_{-3}, \ldots)$ is essentially finite.

Since $(a_n) \in K((x))$, there exists a negative integer p such that $a_i = 0$ for all $i \le p$. Similarly, there exists a negative integer q such that $b_j = 0$ for all $j \le q$. Set $r := p + q$. For any integer $n \le r$:

- Each $i \ge p$ satisfies $n - i \le r - p = q$, hence $b_{n-i} = 0$.
- Each $i < p$ satisfies $a_i = 0$.

Therefore

$$c_n = \sum_{i\in\mathbb{Z}} a_i\, b_{n-i} = \sum_{\substack{i\in\mathbb{Z}\\ i<p}} \underbrace{a_i}_{=0}\, b_{n-i} + \sum_{\substack{i\in\mathbb{Z}\\ i\ge p}} a_i\, \underbrace{b_{n-i}}_{=0} = 0.$$

Hence all $n \le r$ satisfy $c_n = 0$, so the sequence $(c_{-1}, c_{-2}, \ldots)$ is essentially finite.

A.5.7 Helpers for the embeddings

Lemma A.117 (Power series coefficient preservation)**.** *The embedding $K[\![x]\!] \hookrightarrow K(\!(x)\!)$ preserves coefficients: for any power series f and any $n \in \mathbb{N}$, the n-th coefficient of the image equals the n-th coefficient of f.*

Proof. This is a standard result relating power series to Laurent series: the embedding preserves coefficients. □

Lemma A.118 (Laurent polynomial embedding is additive)**.** *The embedding $K[x^{\pm}] \hookrightarrow K(\!(x)\!)$ preserves addition: for Laurent polynomials p and q, the image of $p + q$ equals the sum of the images.*

Proof. By extensionality on coefficients: both sides have the same coefficient at each $n \in \mathbb{Z}$. □

Lemma A.119 (Laurent polynomial embedding preserves 1)**.** *The embedding $K[x^{\pm}] \hookrightarrow K(\!(x)\!)$ sends 1 to 1.*

Proof. By extensionality on coefficients. □

Lemma A.120 (Laurent polynomial embedding preserves coefficients)**.** *For any Laurent polynomial p and any $n \in \mathbb{Z}$, the n-th coefficient of the image equals $p(n)$.*

Proof. By definition of the embedding. □

A.5.8 Helpers for Theorem 1.483

Lemma A.121 (Existence of balanced ternary representation)**.** *For any integer n, there exists a balanced ternary representation of n. The proof finds k large enough that $|n| \le M_k$, then uses the k-bounded existence result (Lemma 1.485).*

Proof. Find k with $|n| \le M_k$ (using $|n| \le M_{|n|}$). Then $n \in [-M_k, M_k]$, so by Lemma 1.485 there exists a k-bounded representation, which is converted to a finitely supported function. $\square$

Lemma A.122 (Uniqueness of balanced ternary representation)**.** *If two balanced ternary representations have the same value n, then their digit functions are equal. This follows from Lemma 1.484 applied to the difference of the two digit sequences.*

Proof. Apply the uniqueness result for balanced ternary digits to the two digit functions, using that both have values in $\{-1, 0, 1\}$ and the same weighted sum. $\square$

Lemma A.123 (k-bounded values lie in the target interval)**.** *For any k-bounded balanced ternary representation $f \in \{-1, 0, 1\}^{k+1}$, the value $\sum_{i=0}^{k} f(i) \cdot 3^i$ lies in the interval $[-M_k, M_k]$ where $M_k = \sum_{i=0}^{k} 3^i$.*

Proof. Each digit $f(i) \in \{-1, 0, 1\}$ contributes at most $\pm 3^i$ to the sum. Summing gives the bounds $-\sum 3^i \le \text{value} \le \sum 3^i$. $\square$

Lemma A.124 (Balanced ternary injectivity)**.** *The evaluation map $f \mapsto \sum_{i=0}^{k} f(i) \cdot 3^i$ is injective on the set of k-bounded balanced ternary representations.*

Proof. By strong induction on k: the lowest digit is determined by the value modulo 3, and the remaining digits are determined by induction on the quotient. $\square$

Lemma A.125 (Digit differences are bounded)**.** *If $a, b \in \{-1, 0, 1\}$, then $|a - b| \le 2$. If additionally $a \ne b$, then $|a - b| \ge 1$.*

Proof. By case analysis on a and b. $\square$

Lemma A.126 (Equivalence of balanced ternary representation types)**.** *There is a type equivalence between the finitely-supported balanced ternary representation (using $\mathbb{N} \to \mathbb{Z}$ with digits in $\{-1, 0, 1\}$ and finite support) and the inductive-style balanced ternary representation. The conversions are inverse to each other.*

Proof. The forward direction maps each digit d to its balanced ternary digit, the inverse maps each balanced ternary digit to its integer value. Round-trip identities follow by checking both compositions are the identity. $\square$

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